

SMITHIAN FOLD THEORY V3

From Fold to Materials

An Exact, Parameter-Free and Machine-Closed Complete-Field Reconstruction of Materials
Science from Smithian Fold Theory

Maria Smith
Independent researcher and founder, Ernos Labs
Maria.Smith.Sftoe@gmail.com

Version 1.4.1
2 August 2026
DOI: 10.5281/zenodo.21761657
Paper: CC BY 4.0 - Code: Apache-2.0

Lean 4 whole-model report: PASS
SHA-256 6b6a5a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf

PUBLISHED OPEN ACCESS
ZENODO RECORD VERIFIED

Contents

Contents	2
An Exact, Parameter-Free and Machine-Closed Complete-Field Reconstruction of Materials Science from Smithian Fold Theory	14
Abstract	14
Successor headline findings	15
Results first: complete-field Materials reconstruction	15
Current status, evidence language and reader map	15
Corpus-wide terminology and evidence key	16
Three reading levels	16
Editorial change control	16
Public scientific mission and admission boundary	17
1. Publication, authorship and open-science boundary	17
2. Exact scope and meaning of closure	17
3. Constitutional arithmetic and prohibited premises	18
4. Dependency spine to the root theorem	18
5. Complete pre-source seal	18
6. Single fail-closed engine	18
7. Empirical method	18
8. Exact structural count certificates	18
8.1 Six simple-cubic neighbours	18
8.2 Rotation restriction	19
8.3 Seven crystal systems and fourteen Bravais classes	19
8.4 Three acoustic branches	19
9. V1/V2 reconciliation without answer import	19
10. Reading the exhaustive ledger	19
Family results, complete claim inventory and claim-level audit records	19
11. Measurement Identity	19
1. Material system and retained identity	19
2. Specimen and sampling relation	22
3. Material composition record	25
4. Material phase identity	28
5. Microstructure as mesoscale organisation	30
6. Material property relation	33
7. Materials uncertainty and metrological traceability	36
12. Crystal Quasicrystal	39
8. Generated lattice organisation	39
9. Primitive unit-cell identity	41
10. Crystalline translational order	44
11. Simple-cubic nearest-neighbour coordination	47
12. Crystallographic rotation restriction	50
13. Three-dimensional crystal-system classification	53

14. Three-dimensional Bravais classification	55
15. Reciprocal support and diffraction	58
16. Quasicrystalline aperiodic order	61
17. Acoustic lattice excitation branches	64
18. Exact quasicrystal substitution inflation	67
19. Acoustic, optical and thermal mode limits	70
13. Defects Microstructure	72
20. Point-defect identity	72
21. Vacancy as structural absence	75
22. Interstitial and substitutional defects	78
23. Dislocation line and slip	81
24. Grain and grain-boundary identity	83
25. Material interface law	86
26. Diffusive constituent transport	89
27. Nucleation and phase growth	92
14. Electronic Semiconductor	94
28. Electronic bands and structural gaps	95
29. Conductor, semiconductor and insulator distinction	97
30. Electron and hole carrier duality	100
31. Exclusion-bounded state occupation	103
32. Doping as controlled constituent substitution	106
33. p-type and n-type organisation	108
34. p-n junction and depletion organisation	111
35. Semiconductor transport relation	114
36. Interband optical transition	117
37. Exact p-n junction rectification	120
15. Superconducting Superfluid Topological	122
38. Coherent paired-carrier state	122
39. Dissipation-closed superconducting transport	125
40. Magnetic flux exclusion in the superconducting phase	128
41. Superconducting flux quantization	131
42. Weak-link coherent transfer	133
43. Dissipation-closed superfluid transport	136
44. Quantized superfluid circulation	139
45. Topological material invariant	142
46. Bulk-boundary protected transport	145
47. Superconducting isotope-response record	147
48. Integer and primary fractional Hall classes	150
49. Exact topological boundary-mode count	153
16. Mechanical	155
50. Stress and strain records	156
51. Elastic recovery	158
52. Plastic deformation	161

53. Dislocation-mediated slip	164
54. Elastic modulus as response ratio	167
55. Strength and hardness boundaries	170
56. Fracture and toughness	172
57. Fatigue and creep	175
17. Thermal Magnetic Optical	178
58. Heat capacity and excitation storage	178
59. Thermal conduction	181
60. Thermal expansion	184
61. Ferromagnetic order	187
62. Antiferromagnetic order	189
63. Dielectric polarization	192
64. Optical propagation and refractive response	195
65. Material phase transition	198
66. Ferrimagnetic sublattice order	201
18. Material Classes Bulk	203
67. Metallic material organisation	203
68. Ceramic material organisation	206
69. Polymeric material organisation	209
70. Composite material organisation	212
71. Glass and amorphous organisation	215
72. Porous material organisation	217
73. Molecular-to-bulk correspondence	220
74. Material anisotropy	223
75. Size and surface effects	226
76. Hydrogen-bonded water bulk-response ledger	229
19. Processing Degradation	232
77. Processing-structure-property path	232
78. Phase stability and coexistence organisation	235
79. Solidification	237
80. Sintering and densification	240
81. Heat treatment	243
82. Corrosion	246
83. Wear and tribological material transfer	249
84. Radiation damage	252
20. Advanced Functional Sustainable	255
85. Piezoelectric coupling	255
86. Ferroelectric order and switching	258
87. Thermoelectric coupling	260
88. Photonic material organisation	263
89. Functional magnetic material response	266
90. Nanoscale material boundary	269
91. Biomaterial interface boundary	272

92. Material lifecycle and recyclability	275
21. Integrated reconciliation table	278
22. External-source ledger	278
Complete-field family: Quantitative Crystallography Diffraction Disorder	282
93. Exact diffraction amplitude and intensity ledger	282
94. Finite structure-factor composition	285
95. Polycrystal texture and orientation distribution	287
96. Short-range order and diffuse-scattering relation	289
97. Stacking-fault sequence and diffraction consequence	292
98. Crystal twinning and domain relation	294
99. Incommensurate and modulated structure organisation	296
100. Total-scattering pair-distribution reconstruction	299
Complete-field family: Defects Microstructure Interfaces Multiscale	301
101. Defect population and site-fraction balance	301
102. Defect migration and retained path	303
103. Dislocation reaction, climb and cross-slip	306
104. Curvature-driven grain growth	308
105. Interface and grain-boundary segregation	311
106. Precipitation and coherent-incoherent inclusion boundary	313
107. Ostwald-type coarsening as exact carrier transfer	315
108. Interface migration and mobility record	318
109. Microstructure-to-bulk multiscale correspondence	320
Complete-field family: Phase Equilibria Transformations Metastability	322
110. Complete multiphase fraction ledger	322
111. Tie-line and lever-partition relation	325
112. Component potential and phase-coexistence handoff	327
113. Metastable-state retention boundary	330
114. Spinodal-equivalent instability organisation	332
115. Displacive and martensitic transformation	334
116. Reconstructive transformation path	337
117. Order-disorder transition	339
118. Glass-transition and kinetic-arrest boundary	341
119. Transformation kinetics and time-temperature path	344
Complete-field family: Mechanical Fracture Fatigue Creep Tribology Rheology	346
120. Tensor-resolved stress-strain response	346
121. Transverse-to-longitudinal strain relation	348
122. Viscoelastic memory and recovery	350
123. Viscoplastic flow relation	352
124. Yield-surface and loading-path boundary	354
125. Work hardening and retained deformation history	356
126. Fracture energy and toughness ledger	358
127. Stable and unstable crack-growth relation	361

128. Cyclic fatigue initiation and propagation	363
129. Creep mechanism and rupture-time custody	365
130. Impact and high-rate mechanical response	367
131. Friction and contact-state relation	369
132. Lubrication and tribofilm organisation	371
133. Rheological flow and relaxation classes	373
Complete-field family: Thermal Transport Storage Expansion Thermoelectric	375
134. Thermal diffusivity relation	375
135. Interfacial thermal-boundary resistance	377
136. Phonon scattering and mean-path ledger	380
137. Radiative thermal transport in materials	382
138. Thermoelectric coupled-response performance boundary	384
139. Phase-change thermal-storage ledger	386
140. Thermal-shock and thermal-fatigue boundary	389
Complete-field family: Electronic Dielectric Semiconductor Ionic Transport	391
141. Electrical conductivity and resistivity relation	391
142. Carrier mobility and concentration separation	393
143. Hall-response carrier ledger	395
144. Dielectric permittivity and loss relation	397
145. Ionic conductivity and transference	399
146. Mixed ionic-electronic transport	401
147. Tunnelling through a finite material barrier	403
148. Band-alignment and interface offset	405
149. Heterostructure carrier confinement	407
150. Defect and trap electronic states	410
151. Charge screening and depletion length	412
152. Electrochemical insertion and material-state response	414
Complete-field family: Magnetism Spin Superconductivity Superfluidity	416
153. Paramagnetic response	416
154. Diamagnetic response	418
155. Spin-glass freezing and history	420
156. Magnetic domains and walls	422
157. Magnetic hysteresis loop ledger	425
158. Magnetocrystalline anisotropy	427
159. Magnetoresistance and field-response relation	429
160. Spin transport and relaxation	431
161. Superconducting critical-field organisation	434
162. Superconducting vortex matter and pinning	436
163. Superconducting coherence-length boundary	438
164. Superfluid excitation and critical-flow ledger	440
Complete-field family: Optics Photonics Polarization Electromagnetic Materials	442
165. Optical absorption and extinction	442
166. Reflection and transmission ledger	445

167. Luminescence and quantum-yield custody	447
168. Elastic and inelastic light scattering	449
169. Birefringence and optical anisotropy	451
170. Nonlinear optical mixing response	454
171. Waveguide confinement and loss	456
172. Photonic-band-gap and defect mode	458
173. Plasmonic collective response	460
174. Exciton generation, transport and recombination	462
Complete-field family: Metals Alloys Ceramics Glasses Polymers Composites	465
175. Solid-solution and alloy phase organisation	465
176. Ordered intermetallic compound organisation	467
177. Compositionally complex alloy boundary	469
178. Refractory and ultra-high-temperature material class	472
179. Cementitious and concrete composite organisation	474
180. Fibre-reinforced composite load transfer	476
181. Particle-reinforced composite load transfer	479
182. Metallic-glass organisation	481
183. Structural and functional ceramic subclasses	483
184. Thermoplastic, thermoset and elastomer distinction	486
185. Gradient and functionally graded materials	488
186. Architected and cellular materials	490
Complete-field family: Soft Colloidal Liquid Crystalline Granular Materials	493
187. Colloidal stability and aggregation	493
188. Gelation and percolated soft network	495
189. Foam cell and drainage organisation	497
190. Liquid-crystal orientational order	499
191. Emulsion and multiphase droplet organisation	501
192. Membrane and thin-film soft matter	503
193. Granular packing and force-chain support	505
194. Jamming and unjamming boundary	507
195. Responsive and stimuli-sensitive soft materials	510
196. Active-material nonequilibrium organisation	512
Complete-field family: Biomaterials Biologically Derived Materials	514
197. Biocompatibility as a material-interface boundary	514
198. Bioresorption and degradation ledger	516
199. Tissue-scaffold porosity and connectivity	518
200. Cell-material adhesion handoff	520
201. Mechanical matching at a biological interface	522
202. Controlled-release material transport boundary	524
203. Mineralized biological material organisation	527
204. Biologically derived and biofabricated material identity	529
Complete-field family: Nanomaterials Two Dimensional Quantum Materials	531
205. Nanoparticle size and shape distribution	531

206. Nanowire and one-dimensional confinement	533
207. Two-dimensional layer and stacking organisation	535
208. Quantum-dot finite confinement	537
209. Surface-to-volume dominance relation	540
210. Nanoscale phase and melting boundary	542
211. Quantum-material collective-state classification	544
212. Moiré and twisted-layer superstructure	546
213. Nanocomposite interface density	548
214. Nanomaterial aggregation and dispersion custody	550
Complete-field family: Surfaces Coatings Adhesion Interfacial Response	553
215. Surface free-state and energy relation	553
216. Wetting and contact-angle custody	555
217. Adhesion and work-of-separation ledger	557
218. Coating layer and substrate organisation	559
219. Surface roughness and scale boundary	561
220. Surface reaction and catalysis handoff	563
221. Tribofilm formation and retention	565
222. Interface fracture and delamination	567
Complete-field family: Corrosion Oxidation Wear Radiation Ageing	569
223. Oxidation scale growth and transport	569
224. Corrosion-rate and electrochemical-path ledger	571
225. Passivation and film-breakdown boundary	573
226. Stress-corrosion cracking	575
227. Hydrogen uptake and embrittlement	577
228. Abrasive, adhesive and erosive wear distinction	579
229. Radiation-defect accumulation and recovery	581
230. Physical ageing and property drift	584
231. Environmental attack and weathering	586
232. Service-life and failure-time evidence boundary	588
Complete-field family: Processing Solidification Sintering Heat Treatment Additive	590
233. Casting and mould-filling material history	590
234. Thermomechanical forming and texture	592
235. Machining-induced surface and damage state	594
236. Additive layer-build and melt-pool history	596
237. Thin-film deposition and growth	598
238. Epitaxial growth and lattice matching	600
239. Welding, brazing and joining interface	602
240. Polymer processing and orientation history	604
241. Powder processing and compaction	606
242. Process-window provenance and reproducibility ledger	609
Complete-field family: Computational Materials Structure Property Inference	611
243. Exact material-structure data representation	611
244. Structure-property computation boundary	613

245. Finite numerical material simulation	615
246. Multiscale model composition	617
247. Numerical stability and error propagation in materials	619
248. Inverse materials problem	621
249. Machine-learning materials inference boundary	623
250. Materials database identity and provenance	625
251. Phase-field computational correspondence	627
252. Molecular-dynamics computational correspondence	629
253. Electronic-structure computational correspondence	631
254. Simulation-to-experiment validation ledger	633
Complete-field family: Materials Under Extreme Conditions	636
255. High-pressure material state	636
256. High-temperature material state	638
257. Cryogenic material response	640
258. High-electric-field material response	642
259. High-magnetic-field material response	644
260. High-strain-rate and shock response	646
261. Extreme-radiation material response	648
262. Combined-extreme condition and path custody	650
Complete-field family: Sustainable Material Cycles Lifecycle	652
263. Embodied material and energy ledger	652
264. Critical-material availability boundary	654
265. Material reuse and remanufacture	656
266. Recycling separation and recovery yield	658
267. Circular material-flow organisation	660
268. Durability and life-extension relation	662
269. Material toxicity and health handoff	664
270. Material substitution and function preservation	666
271. End-of-life fate and residual custody	668
Complete-field family: Complete Materials External Validation	670
272. Crystallography and diffraction validation vector	670
273. Defect and microstructure validation vector	673
274. Phase and transformation validation vector	676
275. Mechanical and tribological validation vector	678
276. Thermal and transport validation vector	681
277. Electronic, ionic and dielectric validation vector	684
278. Magnetic, superconducting and topological validation vector	687
279. Optical and photonic validation vector	690
280. Material-class and processing validation vector	693
281. Cross-source Materials reproducibility vector	697
282. Complete adverse, absent and out-of-bound Materials vector	706
283. Materials empirical Grand Lock	733
Complete-field family: Cross Branch Handoffs	760

284. Materials-to-Engineering one-owner handoff	760
285. Materials-to-Biology one-owner handoff	762
286. Materials-to-Medicine one-owner handoff	764
287. Materials-to-Earth and Environmental Science one-owner handoff	766
288. Materials-to-Astronomy one-owner handoff	769
289. Materials cross-branch one-owner completeness certificate	771
Reproducibility, falsification and extension boundary	773
Conclusion	774
Publication and repository links	774
Current authoritative statement and receipt reconciliation	774
SFT-MAT-MEAS-MATERIAL-001	774
SFT-MAT-MEAS-SPECIMEN-001	774
SFT-MAT-MEAS-COMPOSITION-001	775
SFT-MAT-MEAS-PHASE-001	775
SFT-MAT-MEAS-MICROSTRUCTURE-001	775
SFT-MAT-MEAS-PROPERTY-001	775
SFT-MAT-MEAS-TRACEABILITY-001	776
SFT-MAT-CRYST-LATTICE-001	776
SFT-MAT-CRYST-UNIT-CELL-001	776
SFT-MAT-CRYST-TRANSLATION-001	776
SFT-MAT-CRYST-CUBIC-COORDINATION-001	777
SFT-MAT-CRYST-ROTATION-RESTRICTION-001	777
SFT-MAT-CRYST-SYSTEMS-001	777
SFT-MAT-CRYST-BRAVAIS-001	777
SFT-MAT-CRYST-RECIPROCAL-001	778
SFT-MAT-CRYST-QUASICRYSTAL-001	778
SFT-MAT-CRYST-PHONON-001	778
SFT-MAT-DEFECT-POINT-001	778
SFT-MAT-DEFECT-VACANCY-001	779
SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001	779
SFT-MAT-DEFECT-DISLOCATION-001	779
SFT-MAT-MICRO-GRAIN-BOUNDARY-001	779
SFT-MAT-MICRO-INTERFACE-001	780
SFT-MAT-MICRO-DIFFUSION-001	780
SFT-MAT-MICRO-NUCLEATION-GROWTH-001	780
SFT-MAT-ELEC-BAND-GAP-001	780
SFT-MAT-ELEC-CONDUCTOR-CLASS-001	781
SFT-MAT-ELEC-CARRIER-DUALITY-001	781
SFT-MAT-ELEC-OCCUPATION-001	781
SFT-MAT-SEMI-DOPING-001	781
SFT-MAT-SEMI-PN-TYPE-001	782
SFT-MAT-SEMI-JUNCTION-001	782
SFT-MAT-SEMI-TRANSPORT-001	782

SFT-MAT-SEMI-OPTICAL-001	782
SFT-MAT-SC-PAIR-001	783
SFT-MAT-SC-ZERO-RESISTANCE-001	783
SFT-MAT-SC-MEISSNER-001	783
SFT-MAT-SC-FLUX-QUANTIZATION-001	783
SFT-MAT-SC-JOSEPHSON-001	784
SFT-MAT-SF-SUPERFLUID-001	784
SFT-MAT-SF-CIRCULATION-001	784
SFT-MAT-TOPO-INVARIANT-001	784
SFT-MAT-TOPO-BULK-BOUNDARY-001	785
SFT-MAT-MECH-STRESS-STRAIN-001	785
SFT-MAT-MECH-ELASTICITY-001	785
SFT-MAT-MECH-PLASTICITY-001	785
SFT-MAT-MECH-SLIP-001	786
SFT-MAT-MECH-MODULUS-001	786
SFT-MAT-MECH-STRENGTH-HARDNESS-001	786
SFT-MAT-MECH-FRACTURE-001	786
SFT-MAT-MECH-FATIGUE-CREEP-001	787
SFT-MAT-THERM-HEAT-CAPACITY-001	787
SFT-MAT-THERM-CONDUCTION-001	787
SFT-MAT-THERM-EXPANSION-001	787
SFT-MAT-MAG-FERROMAGNETISM-001	788
SFT-MAT-MAG-ANTIFERROMAGNETISM-001	788
SFT-MAT-DIEL-POLARIZATION-001	788
SFT-MAT-OPT-REFRACTIVE-001	788
SFT-MAT-PHASE-TRANSITION-001	789
SFT-MAT-CLASS-METAL-001	789
SFT-MAT-CLASS-CERAMIC-001	789
SFT-MAT-CLASS-POLYMER-001	789
SFT-MAT-CLASS-COMPOSITE-001	790
SFT-MAT-CLASS-GLASS-001	790
SFT-MAT-CLASS-POROUS-001	790
SFT-MAT-BULK-CORRESPONDENCE-001	790
SFT-MAT-BULK-ANISOTROPY-001	791
SFT-MAT-BULK-SIZE-SURFACE-001	791
SFT-MAT-PROC-PATH-001	791
SFT-MAT-PROC-PHASE-DIAGRAM-001	791
SFT-MAT-PROC-SOLIDIFICATION-001	792
SFT-MAT-PROC-SINTERING-001	792
SFT-MAT-PROC-HEAT-TREATMENT-001	792
SFT-MAT-DEGR-CORROSION-001	792
SFT-MAT-DEGR-WEAR-001	793
SFT-MAT-DEGR-RADIATION-001	793

SFT-MAT-FUNC-PIEZOELECTRIC-001	793
SFT-MAT-FUNC-FERROELECTRIC-001	793
SFT-MAT-FUNC-THERMOELECTRIC-001	793
SFT-MAT-FUNC-PHOTONIC-001	794
SFT-MAT-FUNC-MAGNETIC-001	794
SFT-MAT-FUNC-NANOMATERIAL-001	794
SFT-MAT-FUNC-BIOMATERIAL-001	794
SFT-MAT-SUST-LIFECYCLE-001	795
SFT-MAT-VALIDATION-CRYSTALLOGRAPHY-DIFFRACTION-VECTOR-001	795
SFT-MAT-VALIDATION-DEFECT-MICROSTRUCTURE-VECTOR-002	795
SFT-MAT-VALIDATION-PHASE-TRANSFORMATION-VECTOR-003	795
SFT-MAT-VALIDATION-MECHANICAL-TRIBOLOGICAL-VECTOR-004	796
SFT-MAT-VALIDATION-THERMAL-TRANSPORT-VECTOR-005	796
SFT-MAT-VALIDATION-ELECTRONIC-IONIC-DIELECTRIC-VECTOR-006	796
SFT-MAT-VALIDATION-MAGNETIC-SUPERCONDUCTING-TOPOLOGICAL-VECTOR-007	797
SFT-MAT-VALIDATION-OPTICAL-PHOTONIC-VECTOR-008	797
SFT-MAT-VALIDATION-MATERIAL-CLASS-PROCESSING-VECTOR-009	797
SFT-MAT-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010	797
SFT-MAT-VALIDATION-ADVERSE-ABSENT-OUT-OF-BOUND-VECTOR-011	798
SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012	798
Data and code availability	798
Shared claim-record clauses	799
MAT-S001 - applied at 8 claim locations	799
MAT-S002 - applied at 84 claim locations	799
MAT-S003 - applied at 92 claim locations	799
MAT-S004 - applied at 84 claim locations	799
MAT-S005 - applied at 87 claim locations	799
MAT-S006 - applied at 13 claim locations	799
MAT-S007 - applied at 8 claim locations	799
MAT-S008 - applied at 11 claim locations	799
MAT-S009 - applied at 197 claim locations	799
MAT-S010 - applied at 197 claim locations	799
MAT-S011 - applied at 8 claim locations	799
MAT-S012 - applied at 9 claim locations	800
MAT-S013 - applied at 10 claim locations	800
MAT-S014 - applied at 152 claim locations	800
MAT-S015 - applied at 12 claim locations	800
References	800
Complete current-census successor supplement	800
Preceding-version abstract preserved for chronology	800
Scope and ownership boundary	801
Mathematical, computational and empirical constitutions	801

Registered source surface	801
Limitations and open frontier	801
Successor machine-identity appendix	801
Lean 4 independent verification update	802
Verified result	802
Native theorem proof and artifact verification are different layers	802
Fail-closed custody event	802
What the result supports	802
Successor conclusion	803

An Exact, Parameter-Free and Machine-Closed Complete-Field Reconstruction of Materials Science from Smithian Fold Theory

Author: Maria Smith, independent researcher and founder, Ernos Labs

Publication authority: Maria Smith

Version: 1.4.1

Date: 2 August 2026

Status: Published open access; Zenodo record verified

Preceding version: 1.3.0, preserved unchanged at
publications/successors/materials/FROM_FOLD_TO_MATERIALS_PAPER_001_V1_3.md

Preceding DOI record(s): [10.5281/zenodo.21532481](https://doi.org/10.5281/zenodo.21532481), [10.5281/zenodo.21629306](https://doi.org/10.5281/zenodo.21629306)

Correction predecessor: Version 1.4.0 at
publications/successors/materials/FROM_FOLD_TO_MATERIALS_PAPER_001_V1_4.md

Correction predecessor DOI: [10.5281/zenodo.21760440](https://doi.org/10.5281/zenodo.21760440)

DOI: [10.5281/zenodo.21761657](https://doi.org/10.5281/zenodo.21761657)

Paper and documentation licence: CC BY 4.0

Repository code licence: Apache-2.0

Publication record. Maria Smith authorised this status-only correction.

It is published open access at [10.5281/zenodo.21761657](https://doi.org/10.5281/zenodo.21761657) through

Zenodo's new-version route after [10.5281/zenodo.21760440](https://doi.org/10.5281/zenodo.21760440)

in the same concept DOI lineage. The correction changes no scientific claim,

evidence classification, mathematical value, source identity or machine record.

Lean verification integration. The current SFT ordered census passed the

independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims,

898,902 candidate records and decisions, 11,108 controls, seventeen branches

and no reported issue. Lean natively proves the two-class operational root and

its unique survivor and constructs proof-bearing acceptance-gate certificates.

The remaining claim content is checked as the complete registered artifact

graph; that distinction is retained throughout this successor.

Abstract

This successor presents the complete current materials paper surface: 289 live model-admitted claims, 73,984 generated candidates, 289 unique survivors and 1,156 passed controls. It integrates 0 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 289 current claim(s) in its declared branch ownership boundary; 0 later claim section(s) are integrated in the successor supplement.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Results first: complete-field Materials reconstruction

Headline finding	Exact result	Empirical meaning
Complete dated field census	289/289 obligations; 0 open (display 0 denotes absence)	Every current obligation has its own receipt; future lawful additions remain permitted.
Exact enumeration	73,984 candidates and decisions; 289 sole survivors	Every law is selected by complete grammar enumeration, not fitting or consensus.
Falsification and reconstruction	1,156 adverse controls; 289 independent reconstructions	A missing distinction, changed source, altered artifact or boundary violation halts.
Quantitative structural classifications	six simple-cubic neighbours; orders 1, 2, 3, 4, 6; seven crystal systems; fourteen Bravais classes; three acoustic modes	These are exact forced classifications, not measured fits.
Full response surface	mechanics, transport, phase, electromagnetic, optical, collective, interfacial and lifecycle laws are separately retained	Specimen, method, scale, condition, uncertainty and adverse evidence remain attached.
Empirical Grand Lock	all 271 pre-validation Materials claims represented by 1,040 evidence lines and 334 source-identity occurrences	No favourable-only aggregation; adverse, absent, unavailable, unresolved and out-of-bound rows remain visible.
One-owner completeness	1,681 predecessor claims root-reachable through 28,437 dependency edges, including 19,852 cross-branch uses	Cross-field use is a directed dependency, never duplicate ownership.

Current status, evidence language and reader map

Status field	Current position
Branch and completion boundary	Materials Science: 289/289 frozen obligations; 289 live claims. This is dated, versioned completion and remains open to lawful extension.
Formal status	Every live claim represented here has a current model-admitted receipt. Formal admission does not by itself imply empirical confirmation.
Empirical status	Claim-specific. Blind, non-blind, development-observed, holdout, adverse, unresolved, unavailable and formal-only distinctions remain exactly as recorded in the claim ledger and evidence packages.
Chronology	Claim-specific registration, seal, custody and observation order governs. Later evidence does not retroactively become an earlier prediction.
Publication status	Version 1.4.1 is published open access at 10.5281/zenodo.21761657 in the existing Zenodo concept DOI lineage.
Existing record lineage	Current record DOI 10.5281/zenodo.21629306 ; concept DOI 10.5281/zenodo.21532481 . Publication is permitted only through the existing record's new-version route.
What is not claimed	Permanent closure of science, universal empirical confirmation, transfer of another branch's ownership or permission to replace an adverse or unresolved record.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the specified formal, computational or empirical test; its kind must be named.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications. None silently substitutes for another.
Foundational closure	Completion of the registered foundation only. It does not imply field-wide closure.
Field-wide closure	Completion of the frozen or registered dated field census named in this paper.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may be added by a later version without rewriting existing receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relationships; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The cited repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. Those files remain authoritative where prose abbreviates their display.

Editorial change control

This version may improve expression, order, typography and navigation. It may not change scientific meaning, scope, chronology, evidence class, ownership, claim status or machine identity. Any apparent conflict between authoritative records must be traced and either resolved by explicit supersession or recorded for Maria Smith; prose alone cannot manufacture agreement.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorise. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigour.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Materials Science, an attractive specimen, vendor datasheet or selected favourable measurement cannot stand for a material law. Carrier, composition, phase, microstructure, preparation, direction, scale, method and condition remain in the observation record. Exact structural counts are stated as exact; variable magnitudes remain bound to their measured specimen rather than being promoted into fictitious universal constants.

Maria Smith retains copyright and scientific authorship. Papers and documentation are licensed CC BY 4.0 and code is licensed Apache-2.0, permitting inspection, copying, criticism, reproduction, modification and redistribution with attribution. "Ernos Labs" is a separate, revocable standards-conformance designation. A reuse or fork may use the open work under its licences, but may describe itself as Ernos Labs only while it preserves the public empirical constitution, complete adverse evidence, unchanged admission route, critical review and community standards.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through <https://discord.gg/ucwGryVxGr>, or inspect the public project at <https://github.com/MettaMazza>.

1. Publication, authorship and open-science boundary

PUBLISHED OPEN-ACCESS PAPER. The current public version is 1.2 at DOI [10.5281/zenodo.21629306](https://doi.org/10.5281/zenodo.21629306). This version 1.3 manuscript is its complete-field successor and must remain in the same Zenodo concept lineage.

Maria Smith, independent researcher and founder of Ernos Labs. Contact: Maria.Smith.Sftoe@gmail.com. Reproducibility reports and submissions: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

The paper is prepared under CC BY 4.0 and code under Apache-2.0. Copyright preserves authorship while the licences preserve open inspection, reuse, criticism and independent replication. The Ernos Labs designation identifies conformance to the published methodological and community standards; copying alone does not confer that designation.

2. Exact scope and meaning of closure

Closure means every obligation in the frozen Materials question surface has one engine-admitted theorem within its exact generated grammar, full dependency trace, adverse-control boundary and registered empirical protocol. It does not mean that every possible material, specimen, processing history or future observation has been enumerated. Biology beyond the material interface, clinical outcomes, Earth-system fate and application engineering remain later or external validation domains.

3. Constitutional arithmetic and prohibited premises

Structural absence is empty One, never numerical zero. Counts are generated positive wholes; exact parts are positive held fractions. Opposition and direction are held fibres, never negative proof quantities. Irrational and imaginary proof values, floating proof arithmetic, a completed infinity, ungenerated continuum, axioms, free or fitted parameters, imported constitutive equations, pretrained models, consensus selection, target-derived rules and application results are prohibited. Host counters can enumerate artifacts but cannot act as scientific values.

4. Dependency spine to the root theorem

Each Materials registration has `root_theorems=(SFT-ROOT-THERE-IS-NO-NOTHING, )`, zero axioms and zero free parameters. Dependencies are accepted only when their immutable model-admission receipts already exist. A separate graph traversal over the materialized registrations confirms 289 of 289 paths reach the root. The path runs through the independently admitted Foundation, Mathematics, Information Science, Physics and Chemistry receipts named in each claim section; no branch name substitutes for an actual dependency identity.

5. Complete pre-source seal

Before NIST or BIPM source identities were chosen, V3 froze the original 84 obligations, exact counting module and target-blind blueprints. That seal records 21,504 candidates and remains

`sha256:da97a6cb6a001964a069b45a5a3698e7ea90f334a08d69c62bd09c46d8112035`. The later atomic V1/V2 audit exposed eight nonduplicative Materials-owned omissions. Before selecting any new source identity, V3 froze all eight successor obligations together, 2,048 candidates and their complete prediction set under `sha256:0e8d7f14a7389b7ec44a37205ce2c9074db65f7b6ed5a466b97f2a01418ef331`. Both seals expressly record that source identities were unselected and target content unopened.

6. Single fail-closed engine

For each claim, the engine verifies registration, source identity, admitted dependencies, exact candidate identity and cardinality, one decision per candidate, exactly one survivor, closure, minimality, named-shape uniqueness, four adverse controls and implementation-distinct reconstruction. Empirical evidence additionally requires an evaluator-verified derivation seal, capability isolation, target absence before seal, custody release only after a matching prediction seal, all-row retention, an explicit falsification condition and a tampered unfavourable control. A violation halts and preserves a rejection receipt; it cannot enter the model census.

7. Empirical method

Forty-two registered byte-frozen records from NIST and BIPM/JCGM supply the independent external boundary across the original and successor seals. Each claim requires at least two claim-specific discriminators. The predictor cannot read filesystem, network, clock, environment, subprocess, dynamic import or foreign functions. A distinct custodian opens source content only after sealing, verifies every snapshot hash and required feature, constructs the registered observation record and releases it through the portable custody exchange. External authority may falsify a correspondence but cannot alter the already sealed grammar or survivor.

This branch distinguishes exact structural counts from specimen-dependent magnitudes. Six neighbours, allowed rotation orders, seven systems, fourteen Bravais classes and three acoustic branches are exact count/classification claims. Elastic moduli, conductivities, strengths, transition temperatures and analogous quantities depend on material identity, preparation, direction, scale, method and condition; the Fold laws force those dependencies and ledgers, not one fictitious number for every material.

8. Exact structural count certificates

8.1 Six simple-cubic neighbours

Three independently admitted stable spatial generators each carry exactly two held Fold orientations. Their literal product contains six distinct nearest-adjacency positions. No signed coordinate or numerical zero is needed.

8.2 Rotation restriction

The rank-two integral rotation boundary requires Euler-totient degree no greater than two. The factor certificate proves depth independence: if odd prime p divides an order, $p-1$ divides its degree, leaving only odd prime three; the remaining powers satisfy $a \leq 2$, $b \leq 1$ and exclude simultaneous $a=2$, $b=1$. Orders one, two, three, four and six survive; five is the least excluded positive order.

8.3 Seven crystal systems and fourteen Bravais classes

The rank-three metric grammar enumerates three length-equivalence classes times four angle-equivalence classes and retains seven rotation-compatible quotient classes. Crossing those seven classes with five centering forms yields thirty-five candidates; basis and symmetry compatibility retain fourteen. The complete named survivors and exact claim receipts appear in their sections below.

8.4 Three acoustic branches

For one held propagation orientation in rank three, displacement has one propagation-aligned orientation and two independent transverse orientations. The complete displacement census therefore contains three acoustic branches.

9. V1/V2 reconciliation without answer import

The V2 corpus is used only as a question-level completeness obligation. Steps 47, 49, 52, 54, 72, 74, 75, 133, 137, 143, 193 and the Materials portion of 291 are mapped to newly executed V3 claims. None of their prior values, equations or certificates is exposed to the V3 derivation. Step 291's planetary and Tully-Fisher content is explicitly routed to Astronomy and Cosmology rather than hidden inside Materials closure.

10. Reading the exhaustive ledger

Every claim section below records the theorem, exact dependency identities, grammar and boundary, all eight binary axes, the 256-candidate census, sole survivor, base/successor certificate, witnesses, meaning, controls, independent implementation, post-seal source custody, falsifier and immutable evidence identities. The machine-readable JSON remains authoritative and is bundled with this paper.

Family results, complete claim inventory and claim-level audit records

The dependency-ordered ledger below is the complete human-readable Materials Science claim inventory for this version. Each claim section exposes its claim identity, family, formal and empirical status, exact statement, dependency route, carrier and boundary, candidate grammar and count, unique survivor, elimination logic, falsification condition, controls, source and custody records, chronology, scientific meaning and receipt identities. Where a generic clause is referenced, its full unchanged wording is retained in the shared claim-record appendix. The machine packages remain authoritative for complete candidates, decisions and executable traces.

11. Measurement Identity

Shared claim-record clause MAT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

1. Material system and retained identity

Claim identity: SFT-MAT-MEAS-MATERIAL-001

Question and exact theorem. A material is the least complete constituent-and-relation carrier whose composition, bonding organization and declared observation boundary remain distinguishable.

```
complete-constituent-carrier__composition-and-bonding-retained__material-boundary-held__declared-scale-identity__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-constituent-carrier, composition-and-bonding-retained, material-boundary-held and declared-scale-identity.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-constituent-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	composition-and-bonding-retained	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	material-boundary-held	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	declared-scale-identity	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-constituent-carrier__composition-and-bonding-retained__material-boundary-held__declared-scale-identity__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5fd1f0e5902535142cb31d865d5ea64e83b9e08cd7f630cac919c3a01e84d840.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:0bbc08685cb1726579e93191fb6f6bde56681445d2d229aa315724a6eb6ade94.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a401a612b8a1d870368aa2d6a487a31c012370a228b62c70915c9c6914706a23.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b2b3e2c6051ce0f6601c2466140f40ceeda0b4cb823d5e91c18daebcf3c5fa50.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:9a39682a714c1e73b5df88972c201028726440e1bf460169e557a45de1513cd5; certificate

sha256:bbe90ba7d0b746808b9a3dbe8e33c1c7ef995b34f1db2b994dcc8810c85f6ce; external-validation

hash sha256:f5880932fc39e96a61053ce252e6ffa011365bb09e677042321c321c9247ef32.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:683ba4d17cac39004f018001749fe7c52b9a376f80659e6bca9ecdabdeb44cf. Target content

opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-REFERENCE-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/reference-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-reference-materials.html;
sha256:2ff393988c4880068ca103e365b0f1e6aed5a02a493b33507f6684a2d9271db9; scope: material
identity, property characterization, uncertainty and traceability.

Comparison and custody records:

- sft-mat-meas-material-001-authority-correspondence: predicted complete-constituent-carrier__composition-and-bonding-retained__material-boundary-held__declared-scale-identity__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-constituent-carrier__composition-and-bonding-retained__material-boundary-held__declared-scale-identity__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced

- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered material system and retained identity observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-constituent-carrier, composition-and-bonding-retained, material-boundary-held and declared-scale-identity.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:3e6c63ac46ba6fa3a33796b77a14d8e56af8461367e178d8e4f7133e5691f15a; derivation seal sha256:03747f2235206b6276b4df2c863ec64edb6581fe3f24c6cd674312492061e806; engine receipt sha256:100e32f861c1188720ddb10a07f5fa28b9a94d7650c859382865519fb64e1be6 at receipts/engine/model_admitted/SFT-MAT-MEAS-MATERIAL-001-100e32f861c11887.json; empirical validation sha256:02058a82c7880fbce8194cc5b4834494351c29d565fe93525deabe152b873680; measurement receipt sha256:179da930e46c4b46067ebf11bd97d1a18acda644cb9cc6930c57c5e6e178f935; isolation sha256:88f0cd6f457584283c4106d0765538acfd07856df2cdba70cd8d43e5127c8da6; custody sha256:9586bdfec7797dd283c176a2c49ac47e128f15b36d4c3a52b4a31b1011477fa2.

2. Specimen and sampling relation

Claim identity: SFT-MAT-MEAS-SPECIMEN-001

Question and exact theorem. A specimen is a source-bound finite part of a material with an explicit sampling map, retained preparation history and declared representativeness boundary.

source-bound-specimen-carrier__specimen-to-material-provenance__sampling-boundary-retained__representativeness-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves source-bound-specimen-carrier, specimen-to-material-provenance, sampling-boundary-retained and representativeness-declared.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	source-bound-specimen-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	specimen-to-material-provenance	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	sampling-boundary-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	representativeness-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: source-bound-specimen-carrier__specimen-to-material-provenance__sampling-boundary-retained__representativeness-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9c35918d4fd63e38482a40eb699e350a2ee519aa20cf3e4da387a1c8afe55fd9.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:88c6c3ff183744cea93e433facfb571d89ebed74760801f5f079da131b8e6e3.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:63c1304a41770cfadab42af9fb67769ed75aaea6637ccef8bc2320976630e09b.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:90dbb259614c0b3e1e2f7c8918b80aaf091a937d31290d4872e84c9b43ee85ee.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:f58c2ea33d59fda86da1273df9a7eda41e54ca393804aa6349eb3f8e3140be34; certificate
sha256:23109cf272ed2894bb8ede6176ff82975653df8ff6a282aa3d2b31fedf0bdcad; external-validation
hash sha256:ea630c1ae5ba9b7c2b8b409f95a58ea69bcd82f8ea2dba1e62b98b4d681311aa.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:3a695fe79bc147ca2cae3cc48766472922fb8ad57766bc9f0d92fd979e449ac0. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TEXTURE-PHASE-FRACTION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot
experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html;
sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains,
phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-meas-specimen-001-authority-correspondence: predicted source-bound-specimen-carrier__specimen-to-material-provenance__sampling-boundary-retained__representativeness-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived source-bound-specimen-carrier__specimen-to-material-provenance__sampling-boundary-retained__representativeness-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered specimen and sampling relation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves source-bound-specimen-carrier, specimen-to-material-provenance, sampling-boundary-retained and representativeness-declared.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:eb6cb5d9d6699b6fa8c5d5a1b80d08a30e5340ee165f372adb13a378fc1c4c9d; derivation seal sha256:de5d12aa6b9d38df4dd41e838be989f5efdaa3c54b549b8eda98543e04d1c829; engine receipt sha256:64d737e3f4c0fc5a33a13a5af86282e4e74420b103aba81e067d3baalb55bbac at receipts/engine/model_admitted/SFT-MAT-MEAS-SPECIMEN-001-64d737e3f4c0fc5a.json; empirical validation sha256:fc00c8fadea60629d2db8350091da01088f9de13941e22ca9288749e6c0cb1bd; measurement receipt sha256:8b2eda391cb0daa95d3a916046d65defafa288867eda601ac9119b008d44878f; isolation sha256:6582228d88ed2f11aece7eb4830965c3a7cd5b5f166c50c7eaba4e4280125636; custody sha256:19398416e13e841a09e8a15c06fb484d7eb5911ea03afcef1ef03a01cd5e28aa.

3. Material composition record

Claim identity: SFT-MAT-MEAS-COMPOSITION-001

Question and exact theorem. Material composition is the exact part-to-whole record of every retained component on one declared amount or support basis.

complete-component-support__exact-part-to-whole-composition__component-identities-retained__basis-and-uncertainty-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Routed dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-component-support, exact-part-to-whole-composition, component-identities-retained and basis-and-uncertainty-declared.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-component-support	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	exact-part-to-whole-composition	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	component-identities-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	basis-and-uncertainty-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-component-support__exact-part-to-whole-composition__component-identities-retained__basis-and-uncertainty-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1495e5970a1ecdcd9d4372587b7b84cd29df2e68eb10bc9924511e2258d147b.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:8a163f92ab47dd17cee0a3c65a94c64a6eafced530cc6453a2784f2177cd17b6.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:09fd0a2fc18530f0598ae2dcbf496100a032e91d27f614edab80f0a72aac8539.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ee80dc3cb72bf4024bf186ddbc6b8334e46b57e03c145399957f2b9d82451f92.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
 sha256:1c370302c7b1df34e6027d61af197f02a43ec583c4b066fe47287bca0dedddfc; certificate
 sha256:ab5e53b673a0f24c340e78feed5d77ad4fdcc972a291abcc90cd54195a54b571; external-validation
 hash sha256:d7de9d562123cd8b519c68f77809dc68f4ce4fbd53724a2b4f4ed888b238f21.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
 sha256:ed1f6ea00fbb6f3c4efccac54ff08f15e6cf8eb6a86628d65ea245838980eb27. Target content
 opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-MATERIALS-DIVISION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/mml/materials-science-and-engineering-division>; snapshot
 experiments/external_sources/materials/snapshots/nist-materials-division.html;
 sha256:fac19ae0959946a8e73f970bf76c73325d93f4671ded97f8699dfcc5e1bf53e3; scope:
 materials measurement, composition, interfaces, processing, damage, classes and lifecycle.

Comparison and custody records:

- sft-mat-meas-composition-001-authority-correspondence: predicted complete-component-support__exact-part-to-whole-composition__component-identities-retained__basis-and-uncertainty-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-component-support__exact-part-to-whole-composition__component-identities-retained__basis-and-uncertainty-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered material composition record observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-component-support, exact-part-to-whole-composition, component-identities-retained and basis-and-uncertainty-declared.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
 sha256:cdcbc439c4cfe1e162e40a6b8bc72068658e552c4e4a03c16c6984d5dd8f8340; derivation seal
 sha256:5767520188e33d726e0ef14871b31adc74a3418ab648a59016e22946abc54465; engine receipt
 sha256:b172a8dfe74dc3bb3d5159178251f4b5a4c6de311ac14a084fe99ba9f87bf1a6 at
 receipts/engine/model_admitted/SFT-MAT-MEAS-COMPOSITION-001-b172a8dfe74dc3bb.json;
 empirical validation sha256:ff62f3576785762465d2a17af0abcf019afb29cf2e5e89f883176c02f23eb5f1;
 measurement receipt
 sha256:6879a614933015d04fd3da7337ddce7118ae2db28bb23b7ed439645af4394a62; isolation
 sha256:a133e07a5e3cb3a00fa8d464abe63e91a2c1dc18f20be9c5fd070e6cc73ef503; custody
 sha256:8de12da81639f5178c647e8da843491a125d1a038e8a92f7c88882f323409f6a.

4. Material phase identity

Claim identity: SFT-MAT-MEAS-PHASE-001

Question and exact theorem. A material phase is a maximal connected support whose constituents share one stable recurrence and symmetry observation class at declared conditions.

complete-phase-support__shared-recurrence-equivalence__phase-boundary-retained__condition-bounded-phase-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-phase-support, shared-recurrence-equivalence, phase-boundary-retained and condition-bounded-phase-class.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-phase-support	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	shared-recurrence-equivalence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	phase-boundary-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	condition-bounded-phase-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	authority-or-target-s elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in stance-lookup	positive-finite-successor-cl osure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or -extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-phase-support__shared-recurrence-equivalence__phase-boundary-retained__condition-bounded-phase-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:1ac7de1af45e524b8c9b0da7b3f8f877fba4a1b78e7a2f01c44861bb0d2901f3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:77b4f455bc737b32694908d129af28b0479a19481f4c990307d06cbc4a071c00`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:25fc762e494c9db8e616dca4b2a87b825f2d18221bb4b9de422f5e343895e0af`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:8fac8c656448533cc88dc6c3a67e5b772dd899d27934e94ed86ceda602fa3351`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

`sha256:dbb3ede679d2df917b3b1d3925870abcd640c7ac5b6bef94da3b36f5ad31bb09`; certificate
`sha256:9b3957f05da97dff4942b040c1abb90b95d8a5e7bc4a8d7656b4ccbfb48f3c04`; external-validation
hash `sha256:60dc5f40aab324386373a4beebe4ed741c3e27f21624dfe65830c09f7f607c2a`.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

`sha256:14b6f6f9b39bcb2995bc42d4164785b635c0a872b167d05174462ef7f62a4a06`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-MULTISCALE-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-meas-phase-001-authority-correspondence: predicted complete-phase-support__shared-recurrence-equivalence__phase-boundary-retained__condition-bounded-phase-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-phase-support__shared-recurrence-equivalence__phase-boundary-retained__condition-bounded-phase-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered material phase identity observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-phase-support, shared-recurrence-equivalence, phase-boundary-retained and condition-bounded-phase-class.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:8a93bd0b5a0225fb750f1e4251ae057e9ba3c649e1e767ba04f9b9ae00b3da99; derivation seal sha256:ad68cff33969540719db7d2b44560a371ddaa5bd4de6645407eaff0e0e21db79; engine receipt sha256:820c18b5745c2eb4d0f4dd9fb15f1c243d4f34d2c1c7b87d421316ff4426f1ee at receipts/engine/model_admitted/SFT-MAT-MEAS-PHASE-001-820c18b5745c2eb4.json; empirical validation sha256:3217c3fcbeef32eaab16508631f6a9200e0536e889546fb08ca0b278e7d225a9; measurement receipt sha256:40605d06c8c56d5d1cf3d24e347e2f925be621d0d0b1755d25c887234d0af836; isolation sha256:7f5971803416f24c7c627e35f7ad99fc03e6b838442f461e642f65437aea1736; custody sha256:68aa9c9ad49548681df4ac2ca134e5cae64aa76b8e217219cb3affd796e11a10.

5. Microstructure as mesoscale organisation

Claim identity: SFT-MAT-MEAS-MICROSTRUCTURE-001

Question and exact theorem. Microstructure is the complete declared-scale arrangement of phases, grains, interfaces and defects between constituent and bulk observation levels.

complete-mesoscale-carrier__constituent-arrangement-relation__interfaces-and-defects-retained__resolution-bounded-record__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-mesoscale-carrier, constituent-arrangement-relation, interfaces-and-defects-retained and resolution-bounded-record.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-mesoscale-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	constituent-arrangement-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	interfaces-and-defects-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	resolution-bounded-record	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-mesoscale-carrier__constituent-arrangement-relation__interfaces-and-defects-retained__resolution-bounded-record__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:4d9da1817be3190c573b9b4258210ab512e0d2c2e6fbc8e28e954970f97378ae.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:f9e6509348500baddf4417a261a58945012068b796a21f30110cb7079c17d2ce.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e7c3258831f4ada06f194a610d6b18187859f9ddb3d850ccf651c1241220632c.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:124c0a150b4c3c2041ea25e07f9a87131a8ee452c614b442f8bf200cf41b0fd9.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:5e3d4ac0287c0720de1b47e900c43f267b74a387e9ca2f75328eb09130bd251f; certificate
sha256:f076e7ddf7a8650c7b13642e5bae9cb217f53a23479887a911e2b00883c47195; external-validation
hash sha256:ee24cfe1b083142c03c82b0c868bf8c5112305f35839a1fff8a5f4204e76f52d.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:53a34e89965e7c97cfc0780ecddb9937112caa7dc849483c429122346ce0f93. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-MULTISCALE-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-multiscale-materials.html;
sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope:
multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- `sft-mat-meas-microstructure-001-authority-correspondence`: predicted complete-mesoscale-carrier__constituent-arrangement__relation__interfaces-and-defects-retained__resolution-bounded-record__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-mesoscale-carrier__constituent-arrangement__relation__interfaces-and-defects-retained__resolution-bounded-record__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered microstructure as mesoscale organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-mesoscale-carrier, constituent-arrangement-relation, interfaces-and-defects-retained and resolution-bounded-record.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:8e536fdd3d154712fc1b447a39f6c62acaa780b4680effe34a6c5b81c3de1e3c; derivation seal sha256:86d7179cfce073c0669747b856b118570b7f56bdb77f9049719a61d9479239b9; engine receipt sha256:2bd2dd39260bb2c978e97390f519d99570ec14a1ed71f2438040642a7a39fb2a at receipts/engine/model_admitted/SFT-MAT-MEAS-MICROSTRUCTURE-001-2bd2dd39260bb2c9.json; empirical validation sha256:d1d475bce79364060bd227a00440231974a97b9a698adcb5aa593114f145c1fe; measurement receipt sha256:687767a7a15260acce4b221cf3dcae4c7de94a93c6de310d64d9ea0763e65430; isolation sha256:86f3233608ea9a812c9e9b5f8ceee097f36d34afd46309e024c52f2c0f659472; custody sha256:51b87a705394f2363ee5beb0436787fdfaa257ee17792af99a438512bfb55e70.

6. Material property relation

Claim identity: SFT-MAT-MEAS-PROPERTY-001

Question and exact theorem. A material property is a reproducible relation between a declared material state and stimulus/response records, conditional on direction, scale, history and method.

material-state-carrier__stimulus-response-relation__condition-and-direction-retain
ed__method-bounded-property__complete-state-transition-boundary-trace__root-bound-
forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves material-state-carrier, stimulus-response-relation, condition-and-direction-retained and method-bounded-property.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	material-state-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	stimulus-response-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	condition-and-direction-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	method-bounded-property	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: material-state-carrier__stimulus-response-relation__condition-and-direction-retained__method-bounded-property__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9d5f10ef32de296ff8fd844e2d412fc5a4fa61e6b436547719a742ceae9254f6.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:eee60e6d8c42048b68a43ec3d3460d5ab672fe4c623828350fe25db95e1e6b0b.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5ba5573122548c6b1fbc93f566e3201eeb7d0a7862100fde5a2eb25342d30d5.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7a63bce6b4e5bcf0be1bec3e56ba88f15183e47d97e62c5be2ba087f4d9c2c66.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:c102a26be454aff89e7e294cb42102b49138996753da2059894e887ce197b1fd; certificate
sha256:a8cb28c31491b1f1a5a7f43ee96c82ef56e8f5ebf213bef1a308b35284d49599; external-validation
hash sha256:8d89cd0128a033bc11d35c7516a69ed90d3575999a75632bee9b7aed56a4b3bd.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:80cd9c287b41551e990818ae07a98fdf4cdd4528a814d66ff7bdd8f8ae2c4a0f. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-REFERENCE-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/reference-materials>; snapshot
[experiments/external_sources/materials/snapshots/nist-reference-materials.html](https://www.nist.gov/reference-materials/experiments/external_sources/materials/snapshots/nist-reference-materials.html);
sha256:2ff393988c4880068ca103e365b0f1e6aed5a02a493b33507f6684a2d9271db9; scope: material
identity, property characterization, uncertainty and traceability.

Comparison and custody records:

- sft-mat-meas-property-001-authority-correspondence: predicted material-state-carrier__stimulus-response-relation__condition-and-direction-retained__method-bounded-property__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-state-carrier__stimulus-response-relation__condition-and-direction-retained__method-bounded-property__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered material property relation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves material-state-carrier, stimulus-response-relation, condition-and-direction-retained and method-bounded-property.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:80fd697248d6321e6b90d9794452f6b9fd387ea961db765471df7548a94761d7; derivation seal sha256:1d1f8fd9378247d9dfefb92613f2b5fec6cb4c9c3909dc46815a7f416d4db2bdc; engine receipt sha256:d13a1540d3b9d11dd631f188e97f4096b58dcc42f19ad0dae0b4d7ad1eb60857 at receipts/engine/model_admitted/SFT-MAT-MEAS-PROPERTY-001-d13a1540d3b9d11d.json; empirical validation sha256:13e08bdc1143aa9d50e2532125113b4afabc13db1c34e8d4079f2f91bb00e27c; measurement receipt sha256:7b6727c92a907c19971097f1e19c34a2064840a58b1892285497d238cffffc014; isolation sha256:fadc55d82e80bdb6922f6cf2b8e4dc9e0b233a27f6e4324a1db1b348cba2c95d; custody sha256:a75d0c32265d651a4565cb6f302dae96060190de2899773c0b4fcb191e5981ee.

7. Materials uncertainty and metrological traceability

Claim identity: SFT-MAT-MEAS-TRACEABILITY-001

Question and exact theorem. A materials measurement is admissible only with a complete result, uncertainty, method, specimen, condition, calibration and unbroken reference chain.

complete-measurement-record__result-to-reference-chain__uncertainty-components-retained__calibration-and-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-measurement-record, result-to-reference-chain, uncertainty-components-retained and calibration-and-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-measurement-record	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	result-to-reference-chain	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	uncertainty-components-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	calibration-and-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-measurement-record__result-to-reference-chain__uncertainty-components-retained__calibration-and-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:06fa01f03b954ef5285336e526a90aed8ebffbc55a443245ef359e1ca0562e4.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:00c29832548d03a45275175072a6636fe6db09d58277db2f6b593eb666bb770b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:664397bfce016fac6e18934501599f9cd4d638f486cf68fccd6a6e85a98e4e09.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:816bbf074bc4def5e8a11de819f574d9f945a4197e01ebce28f46f3b6c1fa869.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
 sha256:f70bd0aa9ed0e1a4e81927798902ca76757e566963e1a41c71ec14bf9b9f06d8; certificate
 sha256:93eb0026b754f4235aed15c8d33ccada191ea0ddfe7879abed1bee7aa0b05f57; external-validation
 hash sha256:67f1a8e737f8f5da46378dcebd60f9ee9adce75c99f638dc9252cc1ee114c5b4.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
 sha256:da9c91f2ca355e816db0feb24791e6b9e2f07fd54d1ae885376692c784507db3. Target content
 opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- BIPM-JCGM-VIM-TRACEABILITY-2026-07-24 - Joint Committee for Guides in Metrology / Bureau International des Poids et Mesures; <https://jcg.m.bipm.org/vim/en/2.41.html>; snapshot
 experiments/external_sources/materials/snapshots/bipm-vim-traceability.html;
 sha256:3c06dca2ea8ad71516b027b9979e7a8381edd1576606ad10a379591c4cb6dba4; scope:
 measurement result, reference chain, calibration and uncertainty.

Comparison and custody records:

- sft-mat-meas-traceability-001-authority-correspondence: predicted complete-measurement-record__result-to-reference-chain__uncertainty-components-retained__calibration-and-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-measurement-record__result-to-reference-chain__uncertainty-components-retained__calibration-and-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered materials uncertainty and metrological traceability observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-measurement-record, result-to-reference-chain, uncertainty-components-retained and calibration-and-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
 sha256:0331cf713780ef603d89ab128524abbac7f2778801318dd0b8fd437d65b9ca0e; derivation seal
 sha256:827cfa9bc16633bc9e45d63d74209619bcfd40942f632c5bcd6d0cf83da8098; engine receipt
 sha256:b0bf205091f5c263309fad96e87e51572c4a3f60f40d61263205778c4509f299 at
 receipts/engine/model_admitted/SFT-MAT-MEAS-TRACEABILITY-001-b0bf205091f5c263.json;
 empirical validation sha256:7929a09e55e51aa0aa0d30d435bfa2a2755c8f23d5951673931c13c94d98e889;
 measurement receipt
 sha256:075d6446b70e416f2a1961f712d1b26ac009326b0eec2664ac4e1edb2398ad06; isolation
 sha256:e79a2ff20201393fcff90b3a3fa417752ae743648e9b0cc3f7eb73989252a169; custody
 sha256:16ebf8cce951c7bf7d037eaa5be0825ab8703ad75184e4291b569d394fb4dff3.

12. Crystal Quasicrystal

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

8. Generated lattice organisation

Claim identity: SFT-MAT-CRYST-LATTICE-001

Question and exact theorem. A lattice is a completely generated site-and-adjacency network whose local word recurs under retained translations.

```
complete-site-carrier__translation-recurrence-relation__adjacency-word-repeated__finite-support-lattice-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-site-carrier, translation-recurrence-relation, adjacency-word-repeated and finite-support-lattice-class.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-site-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	translation-recurrence-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	adjacency-word-repeated	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	finite-support-lattice-class	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-site-carrier__translation-recurrence-relation__adjacency-word-repeated__finite-support-lattice-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c9646ff6ef73ed550d9dd57d3bd3376e1f2b6d1505670ff5b473a2ecc1f16dc5.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:4b1c36d0e527ba983cfb859c1ee7030c18d678f7766b0b00850691b28a340ce7.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d9773d30b11ed06149d52d2aaade731e6df807698242f853bb43b56cad410efe.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3fffc29ee144f3e4f558b5a20eed2e6c152a3d985a503c009f484019144df44f.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:45dc456bf89de22438417c63625a628d65c2cfed0f0a944c50fb3e1b2f26e5cb; certificate
sha256:4daea4fea105991ed38bba1541939504c775085e2bc923e88d0c73fbdc4374dc; external-validation
hash sha256:b25db7f0facb055cb4017842334dd5509e5a9064e847f2b0cafb42796cf683ec.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:79e502524e622bf1c982a0b73cbb777eb604a59fdc448a82abdc3696c033cabf. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24 - National Institute of Standards and Technology; https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html; sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- sft-mat-cryst-lattice-001-authority-correspondence: predicted complete-site-carrier__translation-recurrence-relation__adjacency-word-repeated__finite-support-lattice-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-site-carrier__translation-recurrence-relation__adjacency-word-repeated__finite-support-lattice-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered generated lattice organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-site-carrier, translation-recurrence-relation, adjacency-word-repeated and finite-support-lattice-class.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:953d8aba886fe20066df18e9a13f9d6984043da3ff9379cd8346719abac8a594; derivation seal sha256:8de7be19a12b78faf87780e5d8b279fb8c8e7ba7e9a087d8adb7f85a14bde14a; engine receipt sha256:29a30c7d21227319c53ac130261c6ef9c7619d92e841bef00c3bc3f2f8813209 at receipts/engine/model_admitted/SFT-MAT-CRYST-LATTICE-001-29a30c7d21227319.json; empirical validation sha256:53c0b57c7770e4ba2e65a1ded9835b42fcf0fa452df42f4323d3f4cabac1b40b; measurement receipt sha256:5f15d36c9fc8bcd15bcaddbcaa0a7f8868ebf2f9bcee0598548c35f646f00bd3; isolation sha256:76de27215d27a3fdf1c148dce465d257b19dc89c1c9cae51905839acfd2d905b; custody sha256:ffa931b5491e2c28acb540ecd2204c2262a022b75ecf8e633df32ad8b3791436.

9. Primitive unit-cell identity

Claim identity: SFT-MAT-CRYST-UNIT-CELL-001

Question and exact theorem. A primitive unit cell is the least complete labelled cell whose generated translations reproduce every lattice site and relation once.

minimal-cell-carrier__translation-generates-whole__no-redundant-site-copy__basis-choice-canonicalized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves minimal-cell-carrier, translation-generates-whole, no-redundant-site-copy and basis-choice-canonicalized.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	minimal-cell-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	translation-generates-whole	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	no-redundant-site-copy	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	basis-choice-canonicalized	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: minimal-cell-carrier__translation-generates-whole__no-redundant-site-copy__basis-choice-canonicalized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:95bee5e1bc790838af7571ea763454bb76b0b4cfd29afc969b46e5b894b1773a.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:655991b393c1177811a27031f45194e75585bd304199e7024f505539a7895dd8.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:52f038e5dccc305e1b474d16809934ad563804788d0130f0902e8985630f440e.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a0d6eaaecef72a08b0d991621e1f772a0c5b2da721de6f518654924f3b606102.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:0154a35feeac91950959dc4366d8f1f7fbc32bdd85bba4a14373ce7828cc2de1; certificate
sha256:07c7eb750e1bfdc7075611d7232bff0c62de4bedbe700286e4d62936adec502a; external-validation
hash sha256:85798ec472c45e4f89fdcf19152f5e5835f6dbbbd8af31fdda44cfc5c1ec8d11.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:7df3d5e1106dd9be2853c3ceff4a96c592603507e4909374b63428425490fbd0. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24 - National Institute of Standards and Technology;
https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html;
sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- sft-mat-cryst-unit-cell-001-authority-correspondence: predicted minimal-cell-carrier__translation-generates-whole__no-redundant-site-copy__basis-choice-canonicalized__complete-state-transition-boundary-trace__root-bound-forward-forcing__

positive-finite-successor-closure__no-extra-rule; source-derived minimal-cell-carrier__translation-generates-whole__no-redundant-site-copy__basis-choice-canonicalized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered primitive unit-cell identity observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves minimal-cell-carrier, translation-generates-whole, no-redundant-site-copy and basis-choice-canonicalized.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:0ee1191c7afc84e64e0772d8b89185e5f90e00eae010409b4705ada433fe51f1; derivation seal sha256:1d98de21f5d7e99681d01977448efd664603dde6233b1a0a289b6339db5dae3c; engine receipt sha256:062606db8c6509ec2a58c31f865d99c78fe46b8a407e9a0bf823b7abbc04e868 at receipts/engine/model_admitted/SFT-MAT-CRYST-UNIT-CELL-001-062606db8c6509ec.json; empirical validation sha256:0f5045547d470ffff7cff7a76b331bcc87321b131e410d01dfedc202fef8b7fd; measurement receipt sha256:370d8291daf74de34741f463bb14759e3ae77c4e70542c12bb03fce0e4694a0c; isolation sha256:b54849491c03c4b7826fcc79b350bbff9e6969437a8a7f6ebd296f1e84f0de55; custody sha256:6b4d5550b99b1ece2327e422d03af3ab252fd537e9bcd3b86868b07fdce0cfb7.

10. Crystalline translational order

Claim identity: SFT-MAT-CRYST-TRANSLATION-001

Question and exact theorem. Crystalline translational order is invariance of the complete local constituent word and adjacency class under a generated integer translation basis.

complete-periodic-support__integer-translation-action__local-word-and-adjacency-invariant__observation-depth-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-periodic-support, integer-translation-action, local-word-and-adjacency-invariant and observation-depth-declared.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-periodic-support	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	integer-translation-action	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	local-word-and-adjacency-invariant	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	observation-depth-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-periodic-support__integer-translation-action__local-word-and-adjacency-invariant__observation-depth-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:2dc5baa4e9c0b21b287632d0b7879ad79b6a20ac0f10e9043a7a9fc7c91eeaec.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:699c736da5c750925561e80681e07c690f72b1d453701d89b034b39beec00b55.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c59d80682ced4dc231ab5a5e2eee49f48e53e6d78753baad047450f99ab1a36b.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f7bde3998b13bbb464ee34cba1ecf270b66e6a5fe4a30c001e0b4b0784a4e4b7.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:cf1267c3e93cd30aadb138996e6f53303d50d7bb730c304a24acfd22e2b4d913; certificate
sha256:31470cfc67cc99e83e12e9f967814d55ed8aeb6725c114e1be33cd919c8ee3a9; external-validation
hash sha256:0232399b0b20900fae5f945a17c32ff14a146310471f3d8e1013d98dc9fdd112.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:9fc8e3d73650812afd6ca2ff83c2faaf5fedd96177422de98734a64c4236f2aa. Target content
opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24 - National Institute of Standards and Technology;
https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html;
sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- `sft-mat-cryst-translation-001-authority-correspondence`: predicted complete-periodic-support__integer-translation-action__local-word-and-adjacency-invariant__observation-depth-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-periodic-support__integer-translation-action__local-word-and-adjacency-invariant__observation-depth-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered crystalline translational order observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar

- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-periodic-support, integer-translation-action, local-word-and-adjacency-invariant and observation-depth-declared.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:82c21d5ad24be00ddd2907d027deee7e0332eef8d95834009a8623769d1e6667; derivation seal sha256:ed8de31a2f6de9736c9f8dde2f1add2bd74b1cd305d41a3f2252e2f2b3e3f759; engine receipt sha256:bf36a3534aab9194516ce002531273c8bd47093f8db5304d723bcb5179c8af3e at receipts/engine/model_admitted/SFT-MAT-CRYST-TRANSLATION-001-bf36a3534aab9194.json; empirical validation sha256:d5820432483ce1f6db98ac5243feb7d9739305010874fb8312bedf5b1efa5fb2; measurement receipt sha256:d863a33e6f7b1f9df5b48c1ca7be3000834fe51921330c7db33edd8a06780fed; isolation sha256:4a2fcc8f791a7da72410ba1c5ab79d7ea3b06949abc307bcb34e0032e28236e3; custody sha256:31b05b66c269dfbe59d8109ccf02e598b00d239c8876be66387f24d6d6765ad0.

11. Simple-cubic nearest-neighbour coordination

Claim identity: SFT-MAT-CRYST-CUBIC-COORDINATION-001

Question and exact theorem. Three stable spatial generators with two held orientations per generator force six nearest neighbours for the simple-cubic adjacency class.

three-axis-site-carrier__two-orientations-per-axis__nearest-adjacency-complete__coordination-count-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves three-axis-site-carrier, two-orientations-per-axis, nearest-adjacency-complete and coordination-count-six.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	three-axis-site-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	two-orientations-per-axis	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	nearest-adjacency-complete	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	coordination-count-six	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: three-axis-site-carrier__two-orientations-per-axis__nearest-adjacency-complete__coordination-count-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.
- coordination-six: three directions times two held orientations generate six distinct neighbours; passed true.

Detailed mathematical and physical meaning. The exact product is three independently admitted spatial generators times the two Fold fibre orientations. The six generated neighbours are: axis-one/first-fibre, axis-one/second-fibre, axis-two/first-fibre, axis-two/second-fibre, axis-three/first-fibre, axis-three/second-fibre. No coordinate subtraction is used; opposed directions are held labels.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a557cd3bae435643b540f71a220d86dead2a0af33a4b9db33127a736db1746d4.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

sha256:ffdd89973296587921dcf25a21228bea827ef70e3557d3b263e39c5b3c5ecf3d.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8836eb5156029bb0d0d5e87199dbd03ea5ca0cc32842231958a80aca8dd63da1.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:559485317a0696c17c8bee2f0a302354a5738b655be68072aa8f84e6516ff1ea.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:20be4530b1fbc3b79cce20b0c8a94ff0794a8ac42dc8d633c2228dd92dbe0d82; certificate
sha256:7705717fd0f2eb7831c6a4f828389f697c3b1c0a157da3d9d59907a04de53b48; external-validation
hash sha256:882c896ef2f8a38ebf582d61bf71dd229923e81adfc46d9474398cb812a2caa3.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:fa547f7fa1727d64e737ac3f8ebc4a041bd00f93f18621aebd87ac98e47da155. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SIX-NEIGHBOUR-MESH-2026-07-24 - National Institute of Standards and Technology;
https://math.nist.gov/oommf/doc/userguide20a0/userguide/Standard_Oxs_Ext_Child_Clas.html; snapshot
experiments/external_sources/materials/snapshots/nist-six-neighbour-mesh.html;
sha256:505ca000d91c5ef134836cbc75307df1584750dfbfb2faf5c4e55756d054f5c2; scope: six
nearest neighbours as forward/backward positions on three coordinate axes.

Comparison and custody records:

- sft-mat-cryst-cubic-coordination-001-authority-correspondence: predicted three-axis-site-carrier__two-orientations-per-axis__nearest-adjacency-complete__coordination-count-six__complete-state-transition-boundary-trace__root-bound-forward-f orcing__positive-finite-successor-closure__no-extra-rule; source-derived three-axis-site-carrier__two-orientations-per-axis__nearest-adjacency-complete__coordination-count-six__complete-state-transition-boundary-trace__root-bound-forward-f orcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered simple-cubic nearest-neighbour coordination observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves three-axis-site-carrier, two-orientations-per-axis, nearest-adjacency-complete and coordination-count-six.

Immutable evidence identities. Pre-source branch seal
experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:d9f52d70cc6c34ca39fc40035d128e1c8aa5334910900b03aab1998a85286082; derivation seal

sha256:e42124c8f179997887b6b5e7ab012c11153998f111f7ba98c1ec85a2d82d83b; engine receipt
 sha256:b63b31c5bf7aea6cde9b7495c3aaed6f34fac4baf513c2824a18d4c42102155a at receipts/engine/model_admitted/SFT-MAT-CRYST-CUBIC-COORDINATION-001-b63b31c5bf7aea6c.json; empirical
 validation sha256:8a82df98e9284baf366a3492530356dfb56430eafa1cfd56fb362f46856df7e0; measurement receipt
 sha256:be7fafca4dbef1c845769908689a1b966df05d448dd42a7b6f91a704d1258123; isolation
 sha256:1b7a43467ab4c4a739ade2cb74f2bb3d1aeda8ad1fbb25ed3adfe8a9cd5c1d10; custody
 sha256:5608949988ae3687ef60d5f1df5ed7441439153ecac0d84f2fb6d20b19b26046.

12. Crystallographic rotation restriction

Claim identity: SFT-MAT-CRYST-ROTATION-RESTRICTION-001

Question and exact theorem. A finite rotation preserves a periodic lattice only for generated orders 1, 2, 3, 4 and 6; five is the least excluded positive order.

integer-lattice-action-carrier__finite-order-rotation-relation__translation-compatibility-retained__orders-one-two-three-four-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves integer-lattice-action-carrier, finite-order-rotation-relation, translation-compatibility-retained and orders-one-two-three-four-six.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	integer-lattice-action-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	relation-imported-fitted-or-erased	finite-order-rotation-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	translation-compatibility-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	orders-one-two-three-four-six	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: integer-lattice-action-carrier__finite-order-rotation-relation__translation-compatibility-retained__orders-one-two-three-four-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.
- rotation-orders: the depth-independent factor certificate retains exactly orders 1, 2, 3, 4 and 6; passed true.

Detailed mathematical and physical meaning. The rank-two integral rotation boundary requires totient degree at most two. If an odd prime p divides the order, $p-1$ divides the degree, so only odd prime three survives. Every order is therefore 2^a times 3^b with $a \leq 2$, $b \leq 1$, not ($a=2$ and $b=1$). Exhaustion leaves (1, 2, 3, 4, 6); five is the least excluded positive order. This is a depth-independent factor certificate, not a depth-seven lookup and not a trigonometric approximation.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8e118cd050ba1f91713c4bac85af97b055953558ef2a08ad26c17387a3abde50.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:903699ea736f5b69ac09a10ca848e32b3b7bf4ee50f11d913c7d630147c8f3a7.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a9244da0719ec18d4622cf446c89292a3c2267c7bb08fa457861ad57cf3f2c31.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:f6f523f473b879608080940789f14fa8b6f533bc3ca361aafce15dfffb28763b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:0a7974f67520983c1fdd51fccf1df5b8e5802ebc0562dd9a4e97fc4778af4b0f; certificate sha256:ce3cc18e06f56be003ec62dea474e6214221da3577fe714dc39437d1d192a4e3; external-validation hash sha256:2231c2f7810cf10930fb666398885e031d743432e234512ad4452968d2844c38.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:5bbabc66fa0c30a2122ad479107cf8fdbb52650090cfff138262db47417d2bda1. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SHECHTMAN-QUASICRYSTALS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/nist-and-nobel/dan-shechtman>; snapshot experiments/external_sources/materials/snapshots/nist-shechtman-quasicrystals.html; sha256:911c06e466a95d58749422c56cd4e33fa1fbee04ec06977af428d7dac23ae0e; scope: periodic rotation orders and fivefold quasicrystal observation.

Comparison and custody records:

- sft-mat-cryst-rotation-restriction-001-authority-correspondence: predicted integer-lattice-action-carrier__finite-order-rotation-relation__translation-compatibility-retained__orders-one-two-three-four-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived integer-lattice-action-carrier__finite-order-rotation-relation__translation-compatibility-retained__orders-one-two-three-four-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered crystallographic rotation restriction observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves integer-lattice-action-carrier, finite-order-rotation-relation, translation-compatibility-retained and orders-one-two-three-four-six.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:14f138cda0fee15f74ec859ff51be2b51c67c810be00b72df700c3d8a14211ca; derivation seal sha256:4731ceaa58a721703393d44192a3a9c00d6b8a04efad601d81c99abff6efaab7; engine receipt sha256:8fe85fc4bd955efdb5fcc37b159fec38f73a027159ead37660749a04313cd465 at receipts/engine/model_admitted/SFT-MAT-CRYST-ROTATION-RESTRICTION-001-8fe85fc4bd955efd.json; empirical validation sha256:e93f338ce62e1bdbbba29721ad38fa839cd63180536515cf90c50df1a4e6b463; measurement receipt sha256:2ff74beb5fbbcbd98e4efb667c9107057fe01c33a592299c0ed2a9b5e4604d9d; isolation

sha256:e6db0b5eb3ece993947ca064c20b18abdb90df5cc6c021dc2e2281cc38d72220; custody
 sha256:290d1315e496177b5646f2fbc830723be3df434fc37fc87de423ee348b62e35f.

13. Three-dimensional crystal-system classification

Claim identity: SFT-MAT-CRYST-SYSTEMS-001

Question and exact theorem. Complete rank-three metric and crystallographic-rotation equivalence partitions periodic crystals into seven crystal-system classes.

```
rank-three-metric-carrier__axis-length-angle-equivalence__rotation-compatible-classes__seven-system-partition__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-metric-carrier, axis-length-angle-equivalence, rotation-compatible-classes and seven-system-partition.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	rank-three-metric-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	axis-length-angle-equivalence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	rotation-compatible-classes	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	seven-system-partition	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: rank-three-metric-carrier__axis-length-angle-equivalence__rotation-compatible-classes__seven-system-partition__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.
- seven-systems: the complete rank-three length/angle quotient has seven survivors; passed true.

Detailed mathematical and physical meaning. The complete rank-three metric grammar is the product of three length-equivalence classes and four angle-equivalence classes. Quotienting by axis relabelling and rotation compatibility retains seven forms: cubic (all-equal; all-right), trigonal (all-equal; all-equal-nonright), tetragonal (two-equal; all-right), hexagonal (two-equal; two-right), orthorhombic (all-distinct; all-right), monoclinic (all-distinct; two-right), triclinic (all-distinct; all-distinct).

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d6d53473525caba97aac0eef120adbe564356c71e4097f541c0d987a977f95f7.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d5ecbdab14d6f4451eabd1972a86974801daa1b3f29fbc0d095e536484148aa9.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:89c2d1f3852452f2909a9e5f7bd2121e7df76e5501c07b86b1cb3253df66abf7.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ef648732ef475e15f7f44a87514589d7ad95c001bf0beb0fb102ffd1422f6f1e.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:d5514861f898ded8174cc4056d7825fd0e5b938747e48538eaebed3cc84f8f5b8; certificate

sha256:0e7973bb51c318e20750b3f30a48728851edd1fc9f4b8a4303aae2f3d8841560; external-validation hash sha256:e71174f60ede1e2827735dbd37e20c4ab1938398f2be76698070fa1426b66624.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:d7a99045f74e35aa49659bcc6ecb8293ac686764c1a2e6a3f153593a1df6f002. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24 - National Institute of Standards and Technology; https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html; sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- sft-mat-cryst-systems-001-authority-correspondence: predicted rank-three-metric-carrier__axis-length-angle-equivalence__rotation-compatible-classes__seven-system-partition__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived rank-three-metric-carrier__axis-length-angle-equivalence__rotation-compatible-classes__seven-system-partition__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered three-dimensional crystal-system classification observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-metric-carrier, axis-length-angle-equivalence, rotation-compatible-classes and seven-system-partition.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:e2966380d9a9fda51fc07e0b73d5afee4d49e91d5a0af0869a4cb930524697b9; derivation seal sha256:05c0736603eda6753dcaed3534e5a3f3e54e3ae4d3a62489c4fcaa0dcf19f335; engine receipt sha256:e09270361d9fcad5e3e2e58018dd2c46dace3a54af702d711d8b22b0e34919dd at receipts/engine/model_admitted/SFT-MAT-CRYST-SYSTEMS-001-e09270361d9fcad5.json; empirical validation sha256:c4b816b9a81da48b1bc8d1a3cc70e41847ad2f22c86313a5106036f69a7322e7; measurement receipt sha256:bd488ae3983b3fa598b381a978e307b19aa3263489e578414e7e71d91ec6a9b9; isolation sha256:a068a2cff6c89dff9921d705b6a08c6eac4f097a5ba44a238ab72455d3a278be; custody sha256:27e2d5486e52bf5f47c1ae8bcad33818e99eea50d8b1efd4791f031a0c6e82c4.

14. Three-dimensional Bravais classification

Claim identity: SFT-MAT-CRYST-BRAVAIS-001

Question and exact theorem. Complete rank-three translation lattices modulo basis and centering equivalence form fourteen Bravais classes.

rank-three-translation-carrier__centering-equivalence-relation__primitive-lattice-classes__fourteen-bravais-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-translation-carrier, centering-equivalence-relation, primitive-lattice-classes and fourteen-bravais-classes.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	rank-three-translation-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	centering-equivalence-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	primitive-lattice-classes	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	fourteen-bravais-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-specimen-or-in-stance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: rank-three-translation-carrier__centering-equivalence-relation__primitive-lattice-classes__fourteen-bravais-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.
- fourteen-bravais: the complete system/centering quotient has fourteen survivors; passed true.

Detailed mathematical and physical meaning. The system/centering grammar contains seven system classes times five centering classes, or thirty-five candidates. Exact basis/centering compatibility retains fourteen: cubic/primitive, cubic/body, cubic/face, trigonal/rhombohedral, tetragonal/primitive, tetragonal/body, hexagonal/primitive, orthorhombic/primitive, orthorhombic/base, orthorhombic/body, orthorhombic/face, monoclinic/primitive, monoclinic/base, triclinic/primitive. The paper claims this quotient only inside the registered rank-three translation and centering grammar.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:78101747be3a1f60452a2a41fef4bbceacc6e5337dcbc7573bf342ffcd82ed15.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1ae5e1a2f713f1df2305360821d929b44dc0ff0a5d58e620083b46859e5a6c77.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a636fbeb7e7ea076b33edcc43949081d4b538dfbf0e6a76d84730ca300815c6.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e348fb47b5b11603b3bef45c7ed6a17b63d22d9d6ef59d63964b9ab0dbd010fb.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:5255faf88e98383237d2363fc087370a4d87195e1b2a7960052221bcf3b45d69; certificate
sha256:2ca709648229e6c57487ee5fa522f233b1084b25fdce360491a2b8656d3f7032; external-validation
hash sha256:0247320bda7feded0c92b38377a9b25f7c6959cf48051ef443a41a804b8a6237.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is
sha256:6232a8f37319ed3012b7aa4f9826ecf52c3b38617950fa4facfc4637cfadd968. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24 - National Institute of Standards and Technology; https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html; sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.
- NIST-BRAVAIS-CLASSIFICATION-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification>; snapshot experiments/external_sources/materials/snapshots/nist-bravais-classification.html; sha256:82fc7e011a992352d5200131028339a83eca0938c7462131ada2dcc059f38c91; scope: diffraction classification across all fourteen Bravais lattices.

Comparison and custody records:

- sft-mat-cryst-bravais-001-authority-correspondence: predicted rank-three-translation-carrier__centering-equivalence-relation__primitive-lattice-classes__fourteen-bravais-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived rank-three-translation-carrier__centering-equivalence-relation__primitive-lattice-classes__fourteen-bravais-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered three-dimensional bravais classification observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-translation-carrier, centering-equivalence-relation, primitive-lattice-classes and fourteen-bravais-classes.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:1ba2a2e7694b4e6f545b596b6a15b62501b2a48434945e1c12cd508bc407a61e; derivation seal sha256:da91f3783dc2f1e6d6e61900a6e577dc770edb550cbf5e3eef3b8144d95a5a50; engine receipt sha256:402977c8ad6256e131a459efd0762f6f572b919f6a35e60619b4acaf66ad684 at receipts/engine/model_admitted/SFT-MAT-CRYST-BRAVAIS-001-402977c8ad6256e1.json; empirical validation sha256:333e6b1b653c99e953f630f7eb8804adce7dbdd7d3cf7d3f79293700b0f60d16; measurement receipt sha256:003021a8d559eb1b26af67c2f11c9ed26cf12f49becff5a2006773efb7a33977; isolation sha256:527f1213f3d7c8a54c383d938c9a3d8232e0db8136f64e8aed5995f316676a4a; custody sha256:567f6142968e0c21f7265d5e24402a84854192f9c201dae1a7319410381357bb.

15. Reciprocal support and diffraction

Claim identity: SFT-MAT-CRYST-RECIPROCAL-001

Question and exact theorem. Reciprocal support is the exact class of probe path labels whose phase returns coherently over the complete generated material organization; retained classes form diffraction peaks.

```
complete-scatterer-carrier__path-phase-return-relation__translation-dual-support__
peak-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__
positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-scatterer-carrier, path-phase-return-relation, translation-dual-support and peak-resolution-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-scatterer-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	path-phase-return-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	translation-dual-support	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	peak-resolution-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-specimen-or-in-stance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-scatterer-carrier__path-phase-return-relation__translation-dual-support__peak-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:dbbd7f07ef61e5278af026e0de60234376a22109306b91f4d1bf17bf3fc92db7.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:c3cecb0b4bea526fd329b1d5f1ddb4790afef854b36f7c80a3b54d70d679c209.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6e2d17ef89a20df4e99c1242813468012f08aabc1d0f82b92ef260085e73c59b.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:c03422bc2ec42444b078c5e063341eaaaf36057cc69ec87b869a1fba8c8c0f016.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:7fde687d50176b337eb1d74f974b35828f044bea87496e0145f9ed715e7a66c7; certificate
sha256:b79a9101b1cd1d53c7602769ae73503014124fe572a19913e5e2d32930989b98; external-validation
hash sha256:d01489f20f1663afdd82a9d4e83200885117dfa5c34feeaa4a85ea9f15449730.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:1f65c1f62878d5c4562fa2d3b4250a9d4266167ee478f1b4b1631a12b10e9daf. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-BRAVAIS-CLASSIFICATION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification>; snapshot

experiments/external_sources/materials/snapshots/nist-bravais-classification.html; sha256:82fc7e011a992352d5200131028339a83eca0938c7462131ada2dcc059f38c91; scope: diffraction classification across all fourteen Bravais lattices.

Comparison and custody records:

- sft-mat-cryst-reciprocal-001-authority-correspondence: predicted complete-scatterer-carrier__path-phase-return-relation__translation-dual-support__peak-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-scatterer-carrier__path-phase-return-relation__translation-dual-support__peak-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered reciprocal support and diffraction observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-scatterer-carrier, path-phase-return-relation, translation-dual-support and peak-resolution-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:7fe0ba8c722df50c77b1ae34c1574c76db1044d3115cd0de6c04b557de63cfac; derivation seal sha256:2cf710a5f2a9502019e2d3816d744823c545bd807f7de28a0f80368e5d6d9856; engine receipt sha256:bb69b8443063baf90849c3e3fbb54fe746bf27a39cbb5892e3b7988d38e35d78 at receipts/engine/model_admitted/SFT-MAT-CRYST-RECIPROCAL-001-bb69b8443063baf9.json; empirical validation sha256:71cca974d8629c82315c237652911b0b5577f25513e2fa24cdfec188ef578ea0; measurement receipt sha256:b4975788f2336635d491d04088bc4f366ecdc98223a688a48adabce8c40f16f8; isolation sha256:7ce6b243c09044f0a801fd8ae4c47c89f24bec45f413bb579ba34fb0b8e017e3; custody sha256:2f6fc08ab9a531f9412ae510e8bf4f74b19507eb4c3f421d074c2ff4a0a5b0e8.

16. Quasicrystalline aperiodic order

Claim identity: SFT-MAT-CRYST-QUASICRYSTAL-001

Question and exact theorem. Quasicrystalline order is long-range coherent recurrence with sharp reciprocal support and no finite translation period; fivefold order is thereby admitted only outside periodic-lattice closure.

complete-aperiodic-carrier__long-range-recurrence-without-translation__forbidden-periodic-order-retained__sharp-nonperiodic-diffraction__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-aperiodic-carrier, long-range-recurrence-without-translation, forbidden-periodic-order-retained and sharp-nonperiodic-diffraction.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-aperiodic-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	long-range-recurrence-without-translation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	forbidden-periodic-order-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	sharp-nonperiodic-diffraction	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-aperiodic-carrier__long-range-recurrence-without-translation__forbidden-periodic-order-retained__sharp-nonperiodic-diffraction__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: f482fe7557c1f3d9096abc4c5b6ab86ff984ddeb47b893b09a2df71291ecca34.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256: dce768e3d1858fac538cdf97980ba03c7c7a351a5d3243eae790fb0d6fe1b82.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: b018cbd8640c5d2740f37cd8fb376d36267f8057724af2a3ad80b4bf6ef62ec7.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: e752ea45c00eb03832377dbbdd9c0bae8c0df9aa99f886ff58cb8cc499898244.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 8713bd0cc130a811d5e81c9332eb037108cbdcecb35a4ea3afd962489cb870f0; certificate

sha256: 6315b1810b5134e92f29441df2c6d737beebb9e3277bcb4032a37e8d06f5c938; external-validation

hash sha256: 25e7cbc99190b490ca1de85a180d406e5f69d1060cc67b8232a4b4d1cdfc488f.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256: c5eb79da6aa59b45f7955fda6e77963fcbcc4468da12df970f9ce6c9df522f7c. Target content

opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot experiments/external_sources/materials/snapshots/nist-quasicrystal-measurement.html;
sha256: alb7cc0a5882502bb31b9706225b926c9ffb1e8d93ea5c0043249376c4d6ff18; scope: microscope confirmation, nonrepeating order and fivefold symmetry.

Comparison and custody records:

- sft-mat-cryst-quasicrystal-001-authority-correspondence: predicted complete-aperiodic-carrier__long-range-recurrence-without-translation__forbidden-periodic-order-retained__sharp-nonperiodic-diffraction__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-aperiodic-carrier__long-range-recurrence-without-translation__forbidden-periodic-order-retained__sharp-nonperiodic-diffraction__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered quasicrystalline aperiodic order observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-aperiodic-carrier, long-range-recurrence-without-translation, forbidden-periodic-order-retained and sharp-nonperiodic-diffraction.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:7b1915442585a900282440427561c6f7b4df02ce32449a92d2ea47ff898edc5d; derivation seal sha256:6a4386e007b87fab581b9006924276970082bcd9cd7259a4384243c894a9a07; engine receipt sha256:77617aba1e766abd8b64273551afa2ca856c9f67a68f869d86448ef1c5266e21 at receipts/engine/model_admitted/SFT-MAT-CRYST-QUASICRYSTAL-001-77617aba1e766abd.json; empirical validation sha256:977cb5c6105037ae75d97e2312a249489f97a3fa3fe524912be1b7a5fc79bb29; measurement receipt sha256:0ad97e77ff4a78474e22cdfc34fa63f25835bd04395e2ce541ef8d2e2fec0bf9; isolation sha256:cf3c21b892268b1d10e85c3b0d1f1f505778ea51b5f4939b71f2288730876745; custody sha256:d89c32434f6f04e88eb371e35d05e15a90f83cf32e23b31d9f17ec49afb58967.

17. Acoustic lattice excitation branches

Claim identity: SFT-MAT-CRYST-PHONON-001

Question and exact theorem. A rank-three lattice has exactly one longitudinal and two transverse displacement orientations, forcing three acoustic excitation branches.

rank-three-displacement-carrier__one-longitudinal-two-transverse__collective-excitation-recurrence__three-acoustic-branches__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-displacement-carrier, one-longitudinal-two-transverse, collective-excitation-recurrence and three-acoustic-branches.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	rank-three-displacement-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	one-longitudinal-two-transverse	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	collective-excitation-recurrence	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	three-acoustic-branches	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: rank-three-displacement-carrier__one-longitudinal-two-transverse__collective-excitation-recurrence__three-acoustic-branches__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.
- three-acoustic-branches: one longitudinal plus two transverse rank-three orientations are retained; passed true.

Detailed mathematical and physical meaning. A propagation direction consumes one rank-three orientation. The remaining two independent cross-axis orientations are transverse. The complete census is: longitudinal/propagation-axis, transverse/first-cross-axis, transverse/second-cross-axis, yielding three acoustic branches without an irrational normalization or continuous-mode premise.

Shared claim-record clause MAT-S006 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d2e24b132e0f5c972671ff849c506788f98e5f442ca5e40467459c911b8e4636.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:274bdf879f97e05579ec9587adb448fbfa54aaae500b5c950c6b1702e2192c84.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:82599c5b8d2e6423bcd6b667f6a82fc0e740bed1f5baf566351026c7353152f6.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1a63a5afef7b1226f407ae481433648a075f0dfe892a946fd213606d40485070.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:32104da4daf1fd380faf112d1ca088d4bffc61a352e61ffa4c246635b881e4d3; certificate
sha256:e712e3cecd112890f70c47e00d2985ebb8941fb0a3933a188b1522998c457edc; external-validation
hash sha256:e6f329a8b9200d04c64b33e216bae238ce8bb9fcd662f88aea11ae209aa5a186.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is
sha256:6913f74d3c5f5fbb3d58697fbb956c68bb97d0cd5240834d73dd97e6109226ff. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716cccf34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.
- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- `sft-mat-cryst-phonon-001-authority-correspondence`: predicted rank-three-displacement-carrier__one-longitudinal-two-transverse__collective-excitation-recurrence__three-acoustic-branches__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived rank-three-displacement-carrier__one-longitudinal-two-transverse__collective-excitation-recurrence__three-acoustic-branches__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered acoustic lattice excitation branches observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-displacement-carrier, one-longitudinal-two-transverse, collective-excitation-recurrence and three-acoustic-branches.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:d76418e36f165132c2d2757608ccfd33cbf2caabbf7849321efeb8a6ceec504c; derivation seal sha256:91b3320ea3273de7f8f7250145371464466d13e21357218c3c7688707b79a25a; engine receipt sha256:8c3472817c5da55f7509b6085db046570a7109ea880dd7e38a086f72c07b216c at receipts/engine/model_admitted/SFT-MAT-CRYST-PHONON-001-8c3472817c5da55f.json; empirical validation sha256:79983d2c6c9ccc400c7d5b9a724ca15387801673a1881287815dbdefe12706ca; measurement receipt sha256:ae8c2aac5b764cf95d5996bbecc43548237405912e7bca76b77a1f77161bb877; isolation sha256:088998a56bf790b485497a4f0d902cf7c2cb52cba726aac1e09c34716acfbfe5; custody sha256:dac4e779449deddb57373829413ebff00af16ad5936d9cf9b3f2b72ec1a778d3.

18. Exact quasicrystal substitution inflation

Claim identity: SFT-MAT-CRYST-QUASICRYSTAL-INFLATION-002

Question and exact theorem. The two-label substitution A to AB and B to A generates an exact positive finite successor chain whose population pair never admits a positive rational fixed scale satisfying the recurrence.

two-labelled-aperiodic-word-carrier__first-label-to-pair-second-label-to-first-substitution__finite-word-provenance-and-nonperiodic-population-retained__positive-successor-and-no-rational-fixed-scale__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001
- SFT-MAT-CRYST-QUASICRYSTAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor crystal_quasicrystal grammar preserving two-labelled-aperiodic-word-carrier, first-label-to-pair-second-label-to-first-substitution, finite-word-provenance-and-nonperiodic-population-retained and positive-successor-and-no-rational-fixed-scale.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	two-labelled-aperiodic-word-carrier	The material carrier cannot be reconstructed. The complete generated carrier is retained.
relation	relation-imported-fitted-or-erased	first-label-to-pair-second-label-to-first-substitution	An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.
organization	organization-collapse	finite-word-provenance-and-nonperiodic-population-retained	Required material distinctions are merged. Every required organisation distinction remains held.
observation	observation-boundary-unrecorded	positive-successor-and-no-rational-fixed-scale	The observation class is unidentified. The declared observation boundary is retained.
record	result-without-transition-trace	complete-state-transition-boundary-trace	An answer without its transition trace is not auditable. The complete trace is retained.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.
generality	finite-answer-lookup	positive-finite-successor-closure	A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.

Unique survivor, base and successor. Sole survivor: two-labelled-aperiodic-word-carrier__first-label-to-pair-second-label-to-first-substitution__finite-word-provenance-and-nonperiodic-population-retained__positive-successor-and-no-rational-fixed-scale__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: The first complete carrier retains its relation, organisation, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- substitution: six positive successors reproduce the declared recurrence; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:6baf2e887f2ad563e892b4b309c33d7c64d01903860f8c4c44e0b15bd9637512.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

sha256:653aca8f3f9546781c62f5e9b6bdb94ddf0b6462f460e38d59775e088eaada7a.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:6b6ba95cbbf2e29b9240cab9960e0b6d05e9c8038e9a8abbad4e99f30b2194bd.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:47aeae71239f94c8706673ec051643230a1b6874b53c5a45a27082ae544a1e54.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:8cbdc5aaab90614797e081352f548ef5022193a74c0e1082d3136687b0d68ba1; certificate

sha256:6e2476ed2a1c3d4c619f2883987c2e27002bdbcab0035705a2fe538e3d5c3bb6; external-validation

hash sha256:6e2476ed2a1c3d4c619f2883987c2e27002bdbcab0035705a2fe538e3d5c3bb6.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:6a0c71797dcd96a3ef4ba7ac7e435a44f46230005a964544d63b2492d860b4a0. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-QUASICRYSTAL-APERIODIC-2026-07-27 - National Institute of Standards and Technology; <https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot experiments/external_sources/materials/snapshots/nist-quasicrystal-measurement.html; sha256:a1b7cc0a5882502bb31b9706225b926c9ffb1e8d93ea5c0043249376c4d6ff18; scope: post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Shared claim-record clause MAT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor crystal_quasicrystal grammar preserving two-labelled-aperiodic-word-carrier, first-label-to-pair-second-label-to-first-substitution, finite-word-provenance-and-nonperiodic-population-retained and positive-successor-and-no-rational-fixed-scale.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_v1_v2_successor_complete_pre_source.json;

source manifest sha256:1a0b9bcf3caf449f37b9bb0b2a6064bbdea94e1cf1cdfaa7d9a9a7ac77b946da;

derivation seal sha256:e238ce3615d8bddc05e3ed617bf643de8a5f67a3674b79a8eb9bfc90203e53a3;

engine receipt sha256:ac6f202a08f452225c0fe8a378d95f216de0c49b147d5a3b1c11a1a20fe242f0 at re

ceipts/engine/model_admitted/SFT-MAT-CRYST-QUASICRYSTAL-INFLATION-002-ac6f202a08f45222.json; empirical validation sha256:3ab04ed10e478a9583e79b4a72699385f23bc90781885ed16c4700407ea2063a; measurement receipt sha256:4555a6c602a633cac8b36021f8bf6425de5b9395cf7df626871c70857622d512; isolation sha256:0c86b7ea5498aa0a2c3c5ee338b79c3ab58bb855e834d9db19f6f67433aed2e7; custody sha256:b078b06c9bbd3d8961e0a6b25b7c066fc4ab55cf744f3a20b10a7c57a0e71e32.

19. Acoustic, optical and thermal mode limits

Claim identity: SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002

Question and exact theorem. Uniform shared displacement supplies a gapless acoustic class, a retained two-constituent basis supplies an opposed optical class, and positive rank-three support contains the exact cube count of low-support modes and three displacement modes per site at complete support.

rank-three-labelled-displacement-carrier__shared-and-opposed-basis-displacement-relation__acoustic-optical-and-complete-mode-support-retained__gapless-shared-mode-and-rank-three-cube-count__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001
- SFT-MAT-CRYST-PHONON-001
- SFT-MAT-THERM-HEAT-CAPACITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor crystal_quasicrystal grammar preserving rank-three-labelled-displacement-carrier, shared-and-opposed-basis-displacement-relation, acoustic-optical-and-complete-mode-support-retained and gapless-shared-mode-and-rank-three-cube-count.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	rank-three-labelled-displacement-carrier	The material carrier cannot be reconstructed. The complete generated carrier is retained.
relation	relation-imported-fitted-or-erased	shared-and-opposed-basis-displacement-relation	An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.
organization	organization-collapse	acoustic-optical-and-complete-mode-support-retained	Required material distinctions are merged. Every required organisation distinction remains held.
observation	observation-boundary-unrecorded	gapless-shared-mode-and-rank-three-cube-count	The observation class is unidentified. The declared observation boundary is retained.
record	result-without-transition-trace	complete-state-transition-boundary-trace	An answer without its transition trace is not auditable. The complete trace is retained.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.
generality	finite-answer-lookup	positive-finite-successor-closure	A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.

Unique survivor, base and successor. Sole survivor: rank-three-labelled-displacement-carrier__shared-and-opposed-basis-displacement-relation__acoustic-optical-and-complete-mode-support-retained__gapless-shared-mode-and-rank-three-cube-count__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: The first complete carrier retains its relation, organisation, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- mode-support: shared/opposed classes, three directions and cube counts reconstruct exactly; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f4c3423093e584fe900a9f0ba616ca878be81ab2e31eda877a510579f98aa307.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:5c544f51ddbc572dac1e1294db6634e15c7d7547762aaa2bbce56021664a9874.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a7c1d1e7be607560a5b99d1e133480c90affe9f3f97f82df128d29deb1ea712d.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1513991d2d07cb7a0c1c673c20824ab45793e31e0934a37eeeb7d60b4d0286c8.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:15f5be0593f058d90ae68a29ae9542eab4bd640398e8297b12ec62a3419adc571; certificate
sha256:8ff5ab0de67996fa2a97558cbb4a86897a2caf4cb8e6439a3f0300125807403d; external-validation
hash sha256:8ff5ab0de67996fa2a97558cbb4a86897a2caf4cb8e6439a3f0300125807403d.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:edb33397f9512d80b106927d5839c7601a60d4b000a10c53c0d49649fd226967. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-PHONON-THERMAL-LIMITS-2026-07-27 - National Institute of Standards and Technology; <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication960-11.pdf>; snapshot experiments/external_sources/materials/snapshots/nist-phonon-thermal-limits-2026-07-27.txt; sha256:208d3fff48584c6548ccb00d3b1559ba63d66b1fabe027af9bb499504d5cad64b; scope: post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Shared claim-record clause MAT-S007 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor crystal_quasicrystal grammar preserving rank-three-labelled-displacement-carrier, shared-and-opposed-basis-displacement-relation, acoustic-optical-and-complete-mode-support-retained and gapless-shared-mode-and-rank-three-cube-count.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_v1_v2_successor_complete_pre_source.json; source manifest sha256:747cf3a2b40171ad8987eb1f6b539fc7e6019fe27ab1871647c424352a9d019e; derivation seal sha256:887b47334d338944c8d3b4f6fb0819ba228ecad61ea6986dcaad9912333beddd; engine receipt sha256:658c8a9808a71ff33f048f529b88e8189db27a9d4321aad19f9944df4249eab6 at receipts/engine/model_admitted/SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002-658c8a9808a71ff3.json; empirical validation sha256:ae7de6e618edc99685896d9578e59cf835e95576599bd3ac5f2f7cab13c7a453; measurement receipt sha256:9ef211f19dc0d100f0f5a22d6e11df0097d11dc72f300711601678c045298e76; isolation sha256:e6e27b5f28a15fe84d3930fdf06bcb2a87956db1d52ce50d77e43b4721d1e2c8; custody sha256:d78acee38aef99baedf775f7acf5ab233d2f3137e25a8f63032d9d4beeedad7b.

13. Defects Microstructure

Defects are retained differences from a declared reference organisation; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

20. Point-defect identity

Claim identity: SFT-MAT-DEFECT-POINT-001

Question and exact theorem. A point defect is the least localized retained difference between an actual site record and its declared reference-lattice word.

```
site-and-occupant-carrier__local-reference-difference__defect-provenance-retained__
resolution-localized__complete-state-transition-boundary-trace__root-bound-forward-forcing__
positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves site-and-occupant-carrier, local-reference-difference, defect-provenance-retained and resolution-localized.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	site-and-occupant-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	local-reference-difference	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	defect-provenance-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	resolution-localized	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: site-and-occupant-carrier__local-reference-difference__defect-provenance-retained__resolution-localized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Defects are retained differences from a declared reference organisation; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d5cca37e5943297a8a5b2d3d85d2a054989f1a133065f6d35ed160baaa65cfd9.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:f68f961248b9bc315028a34f7cabd89e0b1f40a716840e7fae8f3a0bf451eac3.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3f18a95e86b4d63ffc8ae28d1018e1b039db7a8de3cd6e79b9e1cfe87a40d635.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7dd783b6de1acf2c3cf3ee8c73e69fd6b520ec7fb967d22b238900a5d0b23b8c.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:3059cc726018f803ac57d5d892c51d517508bfa3be460c81d502fdc7d2c68244; certificate

sha256:c759d08df8a75a473cd5e9dec1433b27cd200bb7d4cca6abb72cabb0f4d37eae; external-validation

hash sha256:3c2b96d2f15fc90e4531999b446db3da556e38c8b51ef1660d68d2855aef2cf3.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:7988e4e687fecb304714c434cb7545ce9d70fa8f823acf23c085c562186f7272. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POINT-DEFECTS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot
experiments/external_sources/materials/snapshots/nist-point-defects.html;
sha256:b885250bd645ed8eae55a29d99ea4436b691f77c17c74a457acb6acf88335a; scope:
vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-defect-point-001-authority-correspondence: predicted site-and-occupant-carrier__local-reference-difference__defect-provenance-retained__resolution-localized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived site-and-occupant-carrier__local-reference-difference__defect-provenance-retained__resolution-localized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered point-defect identity observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves site-and-occupant-carrier, local-reference-difference, defect-provenance-retained and resolution-localized.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:ce5dba442d917fdfaad1110e0264bcecad74c6a4c9be9ef58ada814cbe2d7cf1; derivation seal sha256:25283d53bed00346043921382e190f5b91210459c356f8bfbf58a89ea2aa456; engine receipt sha256:b1b3d3707d89ec075a1c23ad71477dc4af6341260e33d7e712aea0d4c35d4b22 at receipts/engine/model_admitted/SFT-MAT-DEFECT-POINT-001-b1b3d3707d89ec07.json; empirical validation sha256:a94db58787562293e8bb51a26af1d3f6966955e3d91a4255ca338c8c4eb7daea; measurement receipt sha256:38aecf7fa7d9859f15d89dc3a4baf8b8d0b60043dcc2052a2a76b90835f55a43; isolation sha256:a019338b83f6ca0ef9f51f77476ad7d44a4673ef585264cc7173c3bddcde9759; custody sha256:d2acebdeb9581146a364ab315d10012ef51ac622920719654b9919208b89890e.

21. Vacancy as structural absence

Claim identity: SFT-MAT-DEFECT-VACANCY-001

Question and exact theorem. A vacancy is a retained reference site carrying structural occupant absence, never deletion of the site or a numerical-zero material amount.

reference-site-carrier__occupant-absence-held-label__site-identity-preserved__empty-one-not-numerical-zero__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves reference-site-carrier, occupant-absence-held-label, site-identity-preserved and empty-one-not-numerical-zero.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	reference-site-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	occupant-absence-held-label	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	site-identity-preserved	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	empty-one-not-numerical-zero	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: reference-site-carrier__occupant-absence-held-label__site-identity-preserved__empty-one-not-numerical-zero__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Defects are retained differences from a declared reference organisation; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:34bcd6eefbd08e1ae00fee7c919cbaa8d558a45544826b10a57c2c7bd698f9de.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:5a8c40e74cfe96c496d6ff7e40fa33f620678a3a3a5afa2f3ba76056e2d6c7e7.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:0e53cc842ec91c069e757ab5bb956cb974cf58b4cbfe3f5b671dbb3b85de8551.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2a6374a8e168c792adec3263140527f62b65afebcd1c2a85a14fe573562262ee.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:505a9fdb929e4fe4912e574520857cc16b9843a3599ba905f951ef28e415e1a; certificate
sha256:072c8a2d131081cd814ed1c6d543f5a27b00b28d1743049df66be684c1a19b1e; external-validation
hash sha256:1dc1c03612c8e602721414875702b896f302df78952ae98f40bee3f3bb53a80f.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:d66f5170de57eaea5c0a8c4e930eb3e728aaeed6a13417345f946147d5fe3d83. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POINT-DEFECTS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot
experiments/external_sources/materials/snapshots/nist-point-defects.html;
sha256:b885250bd645ed8eae5a29d99ea4436b691f77c17c74a457acb6acf88335a; scope:
vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- `sft-mat-defect-vacancy-001-authority-correspondence`: predicted reference-site-carrier__occupant-absence-held-label__site-identity-preserved__empty-one-not-numerical-zero__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived reference-site-carrier__occupant-absence-held-label__site-identity-preserved__empty-one-not-numerical-zero__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered vacancy as structural absence observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar

- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves reference-site-carrier, occupant-absence-held-label, site-identity-preserved and empty-one-not-numerical-zero.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:8b6b1466e80a30239147f070f141ead32c10c87ce24d155d975af5d548b78089; derivation seal sha256:b4b7599659a0361406a02a2850911db2a54c6f046756a9562f280abd80656e6b; engine receipt sha256:e380e7e72cf8fb701cc5cf6bd37e711db7fa7cc9a9fef348aebb81fec9d73c53 at receipts/engine/model_admitted/SFT-MAT-DEFECT-VACANCY-001-e380e7e72cf8fb70.json; empirical validation sha256:6b86a2e46c5ebfa7a9310081ee9972a6f67bc2d21d4684747b684a2a26521959; measurement receipt sha256:2d2091fe1ae56f8660db3b54d2cb6b8747cfcef0a10d58857c60c5a4c9cf857c; isolation sha256:ae1764141e70fda66147266e60303f406510c93e4f2b9099554fb0c5c4c0ad18; custody sha256:2a8ecf59f12ea4aa2c6785dc707dd2b1052495407984155615a5a2ba719e8082.

22. Interstitial and substitutional defects

Claim identity: SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001

Question and exact theorem. Interstitial and substitutional defects are distinguished respectively by an added occupied site relation and a changed occupant label at a retained reference site.

complete-local-site-carrier__occupant-to-reference-comparison__extra-site-versus-replaced-label__defect-kind-distinguished__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves complete-local-site-carrier, occupant-to-reference-comparison, extra-site-versus-replaced-label and defect-kind-distinguished.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-local-site-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	occupant-to-reference-comparison	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	extra-site-versus-replaced-label	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	defect-kind-distinguished	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-local-site-carrier__occupant-to-reference-comparison__extra-site-versus-replaced-label__defect-kind-distinguished__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Defects are retained differences from a declared reference organisation; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f3948b872639a13d3ee135782f30c8aff9a7773c9fdf26149a4866148efaa330.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:9adeb2d2247f93e2869d911066bf4021bd6108567408c919a97b0c473380fe15.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ba6464150d8d66f60708c21cf335227879f90b82d93bd8b62b1ef6011aaea5e1.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3b95d8ae638b3a67b12120e44708eadb6d42a6a0db6034c93420958988deb289.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:d26a7f8e2a347a232ba359abd7b605a579728ef6581446703faf206ce7b37f9d; certificate
sha256:b7a03f2805f3eeab439d48fef49cd06f8670e0d7637638a18a4edd759c7b01ee; external-validation
hash sha256:337ae07bd1c202cf5affc6fa9cf3f4c6045a8d83a55056f625a351ad6fd99929.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:2b552685e6c97371ba6be2557169b0890be1cc01817fc2a39de49338915614c1. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POINT-DEFECTS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot
experiments/external_sources/materials/snapshots/nist-point-defects.html;
sha256:b885250bd645ed8eae55a29d99ea4436b691f77c17c74a457acb6acf88335a; scope:
vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-defect-interstitial-substitution-001-authority-correspondence: predicted complete-local-site-carrier__occupant-to-reference-comparison__extra-site-versus-replaced-label__defect-kind-distinguished__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-local-site-carrier__occupant-to-reference-comparison__extra-site-versus-replaced-label__defect-kind-distinguished__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered interstitial and substitutional defects observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves complete-local-site-carrier, occupant-to-reference-comparison, extra-site-versus-replaced-label and defect-kind-distinguished.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:b8b0fce4b02e0890ef18ec6beb51e1064210749c16b5dcf939dee868c30a8a4e; derivation seal
sha256:28dea2afe3f1fdcf79309764e992de951478be45317f96c14af4e0e839b8640a; engine receipt
sha256:47a87b557fa200d5e2177b408e83e080eb7d358c52c10f61c3817b60b34707c7 at receipts/engine/model_admitted/SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001-47a87b557fa200d5.json
; empirical validation

sha256:4e6ee278b992402fd86a3894cf2d3ce3b6e29c3cb964073d732f8b1e6f66b673; measurement receipt sha256:e78d1384af71ffceb827e8a592a7cd061f1534e806e90b511867da52567835fd; isolation sha256:5c8ac345a9fea15a0989b063f23ceb749d6a7e46b80c9b45cc28951c28f4f23c; custody sha256:8f9a5e072757dc231f8a117bb38e7f9d6c5dc43a4051c82e0019ab109d7a988b.

23. Dislocation line and slip

Claim identity: SFT-MAT-DEFECT-DISLOCATION-001

Question and exact theorem. A dislocation is a line-supported lattice mismatch whose complete circuit retains a nonclosing held translation and supplies a lawful slip path.

line-supported-mismatch-carrier__closure-failure-around-loop__burgers-path-record-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves line-supported-mismatch-carrier, closure-failure-around-loop, burgers-path-record-retained and slip-system-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	line-supported-mismatch-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	closure-failure-around-loop	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	burgers-path-record-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	slip-system-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `line-supported-mismatch-carrier__closure-failure-around-loop__burgers-path-record-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Defects are retained differences from a declared reference organisation; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d59f3ceee9a9ea4c5fedc027dca3afe12a11b3071f83718e717f5ea4d7962b94.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:7d81ee24b84c6e40d34cd4b294fb323ec65c1baf32a075bbba85a363860296e0.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6f2b30a9d67d8c00448c0baccd1113371c830b432e882418e1938dc936c54ed0.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:870afb6bf3c0fa3e16cb9c36ffa1fffb021c357a0cb1f7df6080efb52a9d7288.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:ad2fbb72e7f7bffff1236de535b9cc4ddf9618d802e99ba9238b109b1e619136c; certificate
sha256:eba1b94470fbb42e12dfdbb14fb076926ce197f9965de57ce9e47b36f17a0746; external-validation
hash sha256:039ab49886efc164de273e4dd73105148fbcc04ed8e98851087a40d5474150fe.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:e7f7cc953536991061b1f57ee711cea8d62059945102b7d6622beb403907358a. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SUPERCONDUCTING-GRAIN-BOUNDARIES-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature-snapshot-experiments/external_sources/materials/snapshots/nist-superconducting-grain-boundaries.html; sha256:1309d6599f6894e517dac8f3adc0b6b8506842cdcccc680329007351cde6abf2; scope: grain boundaries, strain, dislocations, current flow and superconducting response.

Comparison and custody records:

- sft-mat-defect-dislocation-001-authority-correspondence: predicted line-supported-mismatch-carrier__closure-failure-around-loop__burgers-path-record-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived line-supported-mismatch-carrier__closure-failure-around-loop__burgers-path-record-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered dislocation line and slip observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves line-supported-mismatch-carrier, closure-failure-around-loop, burgers-path-record-retained and slip-system-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:6188c63c7eca703914838536c9b19141566ce78b61fe8a0b217cd8c483e9dac3; derivation seal sha256:956b315813b9db004e985d97806bcf3ad06e798b1c442da065d7a1737b696f2f; engine receipt sha256:ff273682b7e9bf3e07a0ebe94391f6a7f6046677fe786751d96667017dbdaa50 at receipts/engine/model_admitted/SFT-MAT-DEFECT-DISLOCATION-001-ff273682b7e9bf3e.json; empirical validation sha256:74c2249f8ad197da4a480dfc1c7f5a336bd21868f2b600eb0388eff6b8d0c0d9; measurement receipt sha256:9ddbale33caf86d459b837718673e37a49246d7fed354968f71ce5b33376e7cd; isolation sha256:0e4803cccc3ea43aa1cf21b67ac24af49e54f17f8fa33c9d452ccbf535666406; custody sha256:d8faf7892749b1143d64114e17b66476b9329ff8a8919918c02542daa4f249ea.

24. Grain and grain-boundary identity

Claim identity: SFT-MAT-MICRO-GRAIN-BOUNDARY-001

Question and exact theorem. A grain is a maximal connected orientation-coherent crystal region; its boundary is the retained adjacency where two such orientation classes meet.

orientation-coherent-region__adjacent-orientation-mismatch__interface-network-retained__grain-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves orientation-coherent-region, adjacent-orientation-mismatch, interface-network-retained and grain-scale-declared.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	orientation-coherent-region	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	adjacent-orientation-mismatch	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	interface-network-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	grain-scale-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: orientation-coherent-region__adjacent-orientation-mismatch__interface-network-retained__grain-scale-declared__complete-state-transiti

on-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Defects are retained differences from a declared reference organisation; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c942c9634762db6520d32ff1ac8e7d7a1c63e39d2c641022dcd7138ef3598216.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:220e22dfdd09c0f9d947632d1b79d338308a1b410141a93266443951748ef628.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ddf2ad7f91bf18ad74cad7113fd23d4b572fc232ae3f3dacd4a2505c0501db95.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4bed4e2d09f13bdd5c2a7afc570796ce789516387035f49315e09e75991a704b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:349f54df0d5774c84cd33aa94c0bd9445971a549f8d5040319e8c1bb49c31cda; certificate

sha256:77a0e2b7f0df63949e54011d392bbc08d9bd34adb66f336af40b02a4ae3368e7; external-validation

hash sha256:834416d05874d2011eb1ffd3f172e77f75eb414b9ab7d09635a4ac2362a0ea45.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:17e0d614290974fd2dc557705e24b055f6725241f36b930e4e3fed566c413d49. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TEXTURE-PHASE-FRACTION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot
experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html;
sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains,
phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-micro-grain-boundary-001-authority-correspondence: predicted orientation-coherent-region__adjacent-orientation-mismatch__interface-network-retained__grain-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived orientation-coherent-region__adjacent-orientation-mismatch__interface-network-retained__grain-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule;

ntation-mismatch__interface-network-retained__grain-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered grain and grain-boundary identity observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves orientation-coherent-region, adjacent-orientation-mismatch, interface-network-retained and grain-scale-declared.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:98bcef7ff326b2a30a2c04f4526b617cc2e6b5a08adb067693acf16d4d1d1f67; derivation seal sha256:e514b24f767337ca53e38189e80c4ba8a8acb74613b672b8a5fcfdd042ab9144; engine receipt sha256:ac7a66efd3aedc5341c9fbbd7d72a278c5f2e9e3420f3ac6c5d121b9af1505fb at receipts/engine/model_admitted/SFT-MAT-MICRO-GRAIN-BOUNDARY-001-ac7a66efd3aedc53.json; empirical validation sha256:db61a7d6a0f2d4f932107505d4ce4551d79e5139bf4aabb45252e82265b5f1cf; measurement receipt sha256:28bae4d3ddb832f965a34d4a5f50d64ca9a7d2818cc04cfe3eced3f53bc3c7d; isolation sha256:b23111358cbbd16ddffebadea0f98f4d4710932725c2d69776d218c8afa597b0; custody sha256:b7738555c09fa628fe8a367ef171ca1c2e41c56a620fbc9f304df2afc63d428.

25. Material interface law

Claim identity: SFT-MAT-MICRO-INTERFACE-001

Question and exact theorem. A material interface is the complete boundary relation joining distinct material or phase carriers with explicit structural, content and transfer compatibility records.

two-material-boundary-carrier__cross-boundary-compatibility__traction-content-and-structure-ledger__interface-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves two-material-boundary-carrier, cross-boundary-compatibility, traction-content-and-structure-ledger and interface-scale-declared.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	two-material-boundary-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	cross-boundary-compatibility	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	traction-content-and-structure-ledger	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	interface-scale-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: two-material-boundary-carrier__cross-boundary-compatibility__traction-content-and-structure-ledger__interface-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Defects are retained differences from a declared reference organisation; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_promise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c1e4dfd258e36706fa84e3cd75aaca7071913c560ebf12f38ff39bbcdf3d222e.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:38e2f1f70bfd4f0bc36468142b4370421240bc7e98c6c54d8d0436482af344e6.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c1db60d562c2617ef8de5671afd5d54569ef6985af0e15a0014cc202d4747fca.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3439e9c2f088c1bb6c5ce841494806e4d057028abcf2413c50f6c51c1a129bdf.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:057c33aa17088d597f45fd732daa9b8051863699e686069cfa4abb4d15629506; certificate
sha256:644657ee1c81408bd1464213a5340603387c8ba06f5db9339efcd369df0fdbf0; external-validation
hash sha256:f722c72b28540d303f2c1be3ed5525483c8f7771260a1132e5fc02a2517180fb.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:9f783a523f300809cccd2615a4c90f84a593fd2fbbdb5267829fe4312999299b5. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POLYMER-INTERFACE-CONSORTIUM-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/el/mssd/polymer-surfaceinterface-consortium>; snapshot experiments/external_sources/materials/snapshots/nist-polymer-interface-consortium.html;
sha256:2d48772b1b901e657fa86e4ee28fad15e176dd675911ce8322b54cc718c32635; scope: surface damage, interfacial adhesion and interface-property measurement.

Comparison and custody records:

- `sft-mat-micro-interface-001-authority-correspondence`: predicted two-material-boundary-carrier__cross-boundary-compatibility__traction-content-and-structure-ledger__interface-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived two-material-boundary-carrier__cross-boundary-compatibility__traction-content-and-structure-ledger__interface-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered material interface law observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule

- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves two-material-boundary-carrier, cross-boundary-compatibility, traction-content-and-structure-ledger and interface-scale-declared.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:a20fe778e5b1b9c80a7b99792163778ffe132c5828b30a105a4f8dcf48e297e1; derivation seal sha256:52261a548994417867de3243a46b64609a8e73134b022c544173fcc2d5b3f129; engine receipt sha256:a75d45c8a0bd91e7f206b23cc49ca7c97649ccc12672945d9699ea6e1c8eb1bd at receipts/engine/model_admitted/SFT-MAT-MICRO-INTERFACE-001-a75d45c8a0bd91e7.json; empirical validation sha256:d7d15758d7c7347d521260dd8c03fdd20873caee2dccc34297ab1ee1994a911b; measurement receipt sha256:2789b2f3f5f5d2da7dabda5141971f793ac0c3aa17426e32a28f0a1dd66fe03c; isolation sha256:62c4e87fefaa3b164744c4ab25db1267d0676a9fa2507c9733253ee4d2c723e7; custody sha256:ac3e49780e5c320a0c9b97e62913dbc25e34efa901a369b2f41be3150e774490.

26. Diffusive constituent transport

Claim identity: SFT-MAT-MICRO-DIFFUSION-001

Question and exact theorem. Diffusion is counted adjacent-site redistribution of retained constituent labels with complete global content conservation and declared support/time recurrence.

labelled-constituent-carrier__adjacent-site-transition-count__global-content-conserved__time-and-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves labelled-constituent-carrier, adjacent-site-transition-count, global-content-conserved and time-and-support-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	labelled-constituent-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	adjacent-site-transition-count	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	global-content-conserved	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	time-and-support-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: labelled-constituent-carrier__adjacent-site-transition-count__global-content-conserved__time-and-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Defects are retained differences from a declared reference organisation; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1d11be3a1ad882f5cd29b311457e62302459d6833393f74d4e942187679fae66.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:2ed5233a8868de6e7a8153374b3de47540029249169de8412bcef7af92ebfb5d.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:0a97789d5adcb49aef646568c5f015237315bb8fa83088af4563c2e6bd4bc54b.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 48a8135d5b012c395466cb8e4986b7799d63e98542cd8ca4fa9d311f782ba773.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 9c59f387133a29636589558b2530ee2118991e6afe52a24776db39acffb14f43; certificate
sha256: 5e1381704dc4bb016cca880cdf834a2838db3c214f1ecbd148234faf2c29a052; external-validation
hash sha256: 64cf3e5bd60590e55b421e0fbf9f707a7adb5412ef80884d4445ab68dfd899a6.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256: 06c38d5113645138ec19c40e94ce2a9cdb3ace8703038b4f73d9197bfbf084c5. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POINT-DEFECTS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot
experiments/external_sources/materials/snapshots/nist-point-defects.html;
sha256: b885250bd645ed8eae55a29d99ea4436b691f77c17c74a457acb6acf88335a; scope:
vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-micro-diffusion-001-authority-correspondence: predicted labelled-constituent-carrier__adjacent-site-transition-count__global-content-conserved__time-and-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived labelled-constituent-carrier__adjacent-site-transition-count__global-content-conserved__time-and-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered diffusive constituent transport observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves labelled-constituent-carrier, adjacent-site-transition-count, global-content-conserved and time-and-support-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256: 40f7cd3e3ea9de556a959e4850c76bd1d2b8cb4801a32bcd9ebb56d720beeb74; derivation seal
sha256: fe6aabdae82bc2c12a9539bf9af505c44f86e4fc12a9f8e911c8aea9cf8c2469; engine receipt
sha256: 3e6618450ee819a245f475c45283d761896e350e8c0e1462eb2dabc10c400881 at
receipts/engine/model_admitted/SFT-MAT-MICRO-DIFFUSION-001-3e6618450ee819a2.json;
empirical validation sha256: 3a20a291f5964b0d603cb10e0cb85139d358db3dba65b48bc57dc3ce8996b56e;
measurement receipt

sha256:d5d1645eced54a2feb7096397771785e507c1d1fe702218f2732de6c9e0ad0d4; isolation
 sha256:a3eab68585733f829113f1f4cc7e7fe19431f1984508c8ed8967fde128196956; custody
 sha256:8da37d5bfaf2abcc2106df7c0398aa92c2f6ef38027ea47d4c74c4f90cecd65b.

27. Nucleation and phase growth

Claim identity: SFT-MAT-MICRO-NUCLEATION-GROWTH-001

Question and exact theorem. A nucleus is the least generated cluster whose interior recurrence persists against its complete boundary relation; growth appends compatible cells while retaining phase and transfer provenance.

candidate-phase-cluster-carrier__boundary-versus-interior-recurrence__critical-persistence-and-growth__condition-and-size-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves candidate-phase-cluster-carrier, boundary-versus-interior-recurrence, critical-persistence-and-growth and condition-and-size-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	candidate-phase-cluster-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	boundary-versus-interior-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	critical-persistence-and-growth	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	condition-and-size-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: candidate-phase-cluster-carrier__boundary-versus-interior-recurrence__critical-persistence-and-growth__condition-and-size-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Defects are retained differences from a declared reference organisation; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:02e1b41356ebd232f4e1ffce226a6e14589a289969615744ce95a85aad756a91.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:85311f60dd07cce5138f6bd6e045ddff85be231164ee115db8a452a521f1fee8.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:56ef43ac04bcd811cb45b4c7c6fdbbcff6ccdb17e704f75735d8372bd501a556a.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:dd63ed8dc2853cf4afc78234cae08767527ac748ef137aa0f4a17520975b4ce6.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:9683ac6de970ada2d2d72b0be6c6d861cc8dddd1b21b397d0d9b60b6defbedbc; certificate

sha256:1dbae78f92c78f583b62df211ba0173ab8f2c89a7c42f608e32c4b285d976cc8; external-validation

hash sha256:c3b79b0253401ef2d732672ae16b5c5ea73b0c67e363e3f8a7a2f5247d212895.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is `sha256:1c85d46f9369e6f5de9f63e0cbd6cbb7f02c8739fb3b08ad26d55f7238bbc31e`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SOLIDIFICATION-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/publications/solidification-0>; snapshot `experiments/external_sources/materials/snapshots/nist-solidification.html`; `sha256:d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c`; scope: solidification transport, nucleation, growth, interfaces, segregation and processing.

Comparison and custody records:

- `sft-mat-micro-nucleation-growth-001-authority-correspondence: predicted candidate-phase-cluster-carrier__boundary-versus-interior-recurrence__critical-persistence-and-growth__condition-and-size-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`; source-derived candidate-phase-cluster-carrier\_\_boundary-versus-interior-recurrence\_\_critical-persistence-and-growth\_\_condition-and-size-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered nucleation and phase growth observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects `_microstructure` grammar that preserves candidate-phase-cluster-carrier, boundary-versus-interior-recurrence, critical-persistence-and-growth and condition-and-size-bounded.

Immutable evidence identities. Pre-source branch seal

`experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:5a3a4319df079e15b89fcc578f2f7ad41536a4d633b5fcea4746de982194b30c`; derivation seal `sha256:00b4f5e631130452dec26095c7e8906936b03dd6d06a591f3a22989b1a66c1df`; engine receipt `sha256:a9245f3d9e9386799ada6b1cc5f501b011bfa74c4b21132e17030eaed3e79bdd` at `receipts/engine/model_admitted/SFT-MAT-MICRO-NUCLEATION-GROWTH-001-a9245f3d9e938679.json`; empirical validation `sha256:2a1889999b40d5869f6b9b7ff919265c7bcfd95f77b61d002f917d8d71b7ba49`; measurement receipt `sha256:c2a6cc6f62a2898a9dfc49b69b928f8aea8f9d17b2110bda7b6eb6eba6342a3d`; isolation `sha256:b8d3766cf94a9c7e8e9e750abb2bf1a0feef8f57dba717ddb3236c425ac6764b`; custody `sha256:90ae992b4e7826a09190aad26ac3d29e2df6487b1e1529c8ef4b21b5f718c1c4`.

14. Electronic Semiconductor

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

28. Electronic bands and structural gaps

Claim identity: SFT-MAT-ELEC-BAND-GAP-001

Question and exact theorem. Electronic bands are complete phase-compatible recurrence classes on a material network; a band gap is a declared interval with no compatible generated state path.

```
complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-
support-retained__energy-observation-bounded__complete-state-transition-boundary-t
race__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves complete-lattice-state-support, phase-compatible-path-classes, allowed-and-absent-support-retained and energy-observation-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-lattice-state-support	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	phase-compatible-path-classes	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	allowed-and-absent-support-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	energy-observation-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	authority-or-target-s elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in stance-lookup	positive-finite-successor-cl osure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or -extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-lattice-state-support__phase-compatible__path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:e7658e1f06537e680b3ac39b7f4cc55681b1f55427f53eac74a237a7914ca6ac`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:2430cf6e628f3745292c24ba39678422bf65e31c5bafae7c88e050479ec28fb5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:ccb0fb8846d5f8b815561ff55751d3207480100b14da0039bbbdd6dd480954b2`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:bfde1e54c6b003d0a0606d7d52bfc1046a37dd8f1357f843a5fc77d332188f2a`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
`sha256:1e5c2bccf261f989160ab67cea576566eb955e4ef4e8e027f5cef5f3b063634c`; certificate
`sha256:21963d4131f0a68880134816c335271db01e439c0b6a68b848658dcbb318d1bd`; external-validation
hash `sha256:c100ff6484b4b8bf70291a8f5b401b7de9a126a6cac9f6a74ad43142ec5a41ae`.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
`sha256:68ff9628a37dce468af12c6384628b91da884cec8991a7cc7e4e45528a7c86d1`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TOPOLOGICAL-INSULATORS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/topological-insulators>; snapshot `experiments/external_sources/materials/snapshots/nist-topological-insulators.html`; sha256:4cbe823dfffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449; scope: topological invariants, insulating bulk, band gap and metallic boundaries.

Comparison and custody records:

- sft-mat-elec-band-gap-001-authority-correspondence: predicted complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered electronic bands and structural gaps observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves complete-lattice-state-support, phase-compatible-path-classes, allowed-and-absent-support-retained and energy-observation-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:44addc87284413f5cb47df08df006e5926357c33b694da2ae9da9862470876c6; derivation seal sha256:175a8e789f8b27dd035c876d34d6ad6c7f59c521f797fd73ebde832f3f81d385; engine receipt sha256:7b3d73980e43e5e27c0796cda5840f72f2139cd68462f5de28c32b6fd289f085 at receipts/engine/model_admitted/SFT-MAT-ELEC-BAND-GAP-001-7b3d73980e43e5e2.json; empirical validation sha256:7940d848f5f262579b0b16624dc6ba450141f83939e11ec7508f2c14b97aea6e; measurement receipt sha256:87710ab5dcb4ae5cc803063185f54704445502dbebf37fb7affafa72b7f0d29; isolation sha256:28fc22f923f5c9d892f7e2e01dcccfdde00144bae2fced6a676eef828729958f4; custody sha256:5d21c6ba0ac6008740250a9ad2c4d5b32112de387241c602fd7c36e703c3128a.

29. Conductor, semiconductor and insulator distinction

Claim identity: SFT-MAT-ELEC-CONDUCTOR-CLASS-001

Question and exact theorem. Conductor, semiconductor and insulator classes are forced by whether occupied carrier support connects to accessible states directly, across a finite activation gap, or not within the declared conditions.

carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves carrier-access-support, occupied-to-accessible-state-relation, gap-and-filling-retained and condition-bounded-transport-class.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	carrier-access-support	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	occupied-to-accessible-state-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	gap-and-filling-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	condition-bounded-transport-class	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 6b5fc1f6d32f1642235870027f3f6d09f09a59e0640b2c8e0ceed9372a3354a0.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256: 8bc23cae5864aba8f8171dd7eabe70270b7240ec4dd08547ac1f140c0eab3df7.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 6a3880e037cb135ac340b234e00c15070e7aa00ca0d95ab77f580baab73b002d.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 22f9b7b6ae28eef1be31d92d3cc3dea0076e729e6321efcec9f9b1998e18189f.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: e51bfd79933191b3db914530ce1a3f2fe977a4e15d0a373a00bfd0620f7dca2b; certificate

sha256: 40be778743ea52098d5752b1130728f7507bc2b71ea29381c51cfba7898bd47c; external-validation

hash sha256: 1caf6cf61b772309acea478f82ffc1af61f3837d67f720225570c7b1955149c2.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256: 9c9409a2e572bb2d0836dca51f779846076dceabda444c74ff1840b1ae8b95a6. Target content

opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TOPOLOGICAL-INSULATORS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/topological-insulators>; snapshot
experiments/external_sources/materials/snapshots/nist-topological-insulators.html;
sha256: 4cbe823dffeabd728844a897aalc5a222dfc3aef465263a06501974e194df449; scope:
topological invariants, insulating bulk, band gap and metallic boundaries.
- NIST-SEMICONDUCTORS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot
experiments/external_sources/materials/snapshots/nist-semiconductors.html;
sha256: ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope:
semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- `sft-mat-elec-conductor-class-001-authority-correspondence`: predicted carrier-access-support__occupied-to-accessible-state
-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-b

ound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered conductor, semiconductor and insulator distinction observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves carrier-access-support, occupied-to-accessible-state-relation, gap-and-filling-retained and condition-bounded-transport-class.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:3a0672b7756ff9d49ab3b20a85f2c1915685f491d53339920343d42872af33af; derivation seal sha256:82986bf06b4f76150a9f44e3485710dde93593cb487b90aff3d4518af3f5b29b; engine receipt sha256:e66114ebeeef098723db07db0f8fcf46b242222fa43dfc62cafd7fe664832ca2 at receipts/engine/model_admitted/SFT-MAT-ELEC-CONDUCTOR-CLASS-001-e66114ebeeef0987.json; empirical validation sha256:f4a79e3f2a36290a2a22ba128e7587792a20995bcb71d7eed2ab65c2cbbf0dba; measurement receipt sha256:5cdfb489014fb0be924fde87b453d556866f08a9f42d39b4a2de8460e51b467c; isolation sha256:31ff4726366c6ee968c9e2a86e07d4f4481376fd4c23fc08e32844ea5fd39622; custody sha256:dff32d01f05e8099cb18a076661a0d64c6a7fe2d9bc75d4612b2c046c2a7b273.

30. Electron and hole carrier duality

Claim identity: SFT-MAT-ELEC-CARRIER-DUALITY-001

Question and exact theorem. A binary exclusion-bounded state yields two transport descriptions: retained occupation and its held local absence, corresponding after sealing to electron and hole carriers.

binary-occupation-carrier__occupied-and-held-absence-pair__charge-orientation-retained__two-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves binary-occupation-carrier, occupied-and-held-absence-pair, charge-orientation-retained and two-carrier-classes.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	binary-occupation-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	occupied-and-held-absence-pair	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	charge-orientation-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	two-carrier-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: binary-occupation-carrier__occupied-and-held-absence-pair__charge-orientation-retained__two-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_promise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6913b47e4395df4244aac3f1636c31e345f1650091defd63a5a55ad53215427c.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:a06921b3e66f349f59a1691bdf6941ce10a2bb46d011ff17d496c3e9879cb00b.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:elf4603d1ca5bf94be4382a5ff0171fb92d97f2ed2dac597bdcdcff1969cd9e2.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8488d36cd152a36b3fca8f7c07b9aee54038f04802540d3e561994b31f723093.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:2c00743cc5db244ed341a55c207f31476829df1719b36131ab05a2e9bdd1d2a7; certificate
sha256:d95c160677403b50e44e149e3775ceab1a29413ec3de7c3840d02d669e98325d; external-validation
hash sha256:fee3f69c239830d86d49cf5c348de9f74b0ba9301d4365196d7a093b2bcb179c.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:ba6eaeeda55aaefbe09d5d56edd2f15d1f36e1cd04ad57756489bc2a67100d93. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SEMICONDUCTOR-CARRIERS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/news-events/news/2017/03/testing-performance-semiconductors-light>; snapshot
experiments/external_sources/materials/snapshots/nist-semiconductor-carriers.html;
sha256:05c774e4a30e6009f031b3064b16319d181ddac92e6d088a01bb6fe362172cc5; scope: electron
and hole carrier signs, doping, transport and Hall measurement.
- NIST-OPTICAL-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-optical-materials.html;
sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of
refraction, absorption, reflectance, dielectric response and photonic crystals.

Comparison and custody records:

- `sft-mat-elec-carrier-duality-001-authority-correspondence`: predicted binary-occupation-carrier__occupied-and-held-absence-pair__charge-orientation-retained__two-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived binary-occupation-carrier__occupied-and-held-absence-pair__charge-orientation-retained__two-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered electron and hole carrier duality observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves binary-occupation-carrier, occupied-and-held-absence-pair, charge-orientation-retained and two-carrier-classes.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:487f354580f1b942ef88107d8a7e67acaeb612e4800a2d3db21957d7393baa69; derivation seal sha256:02e62b0bb16a65e63274071f63abc8695d4dc712151cc16d37c39dea4b57dcee; engine receipt sha256:fd598a7661fd3dad3da24311efdcba303cbcee8dad7ddcd14e32c6e69afa7a0a at receipts/engine/model_admitted/SFT-MAT-ELEC-CARRIER-DUALITY-001-fd598a7661fd3dad.json; empirical validation sha256:32242bbc2ab119d300a72f453ece708ab5c3c3431cf5d19c4d742a0d2a85b81a; measurement receipt sha256:fdc881b8019988cb2fa9ffcaf6c97a23f6e9abe4541a24471546cd30626b139c; isolation sha256:dbdf98301457a308b3ac52745fe174bc74d6241df9a017d8fe11b0655ffa36b0; custody sha256:529bbb4a33e307039b20ec33f0fc389b025fcd8244504553e92cbc392540b299.

31. Exclusion-bounded state occupation

Claim identity: SFT-MAT-ELEC-OCCUPATION-001

Question and exact theorem. An exclusion-labelled canonical material mode admits structural absence or one retained indistinguishable occupation; a duplicate occupation is rejected.

canonical-mode-carrier__empty-or-singly-held-support__indistinguishable-duplicate-occupation-rejected__occupation-bound-one__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves canonical-mode-carrier, empty-or-singly-held-support, indistinguishable-duplication-rejected and occupation-bound-one.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	canonical-mode-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	empty-or-singly-held-support	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	indistinguishable-duplication-rejected	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	occupation-bound-one	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `canonical-mode-carrier__empty-or-singly-held-support__indistinguishable-duplication-rejected__occupation-bound-one__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9e095d2545b97492889f9c036a377954bc6c30b71ea4df6cf590c33a7ff73e15.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:489cedb9542a3ae1a73c6c0b0cfcae0a35d5b408bb26dcc08cbbac120e976a04.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:450062506d9f4b952b6cfc258ddee57e835465c3ba6069f3fa260372f2bd7be5.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:02cac4eb83231697c6648ffde545483083d1b1beb4f661539dde2c0d7ade8718.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:68927cecb586cf05cfe48019cdd0639a70cf2b46c72e876832380c4b24a11265; certificate
sha256:e5f507b0a7e69aa7d8ad60023d63d64a9590dee11e5f7b8667d863fb2f58209a; external-validation
hash sha256:c996a7a959521342393af3d9614300fa090a7d658dc386a86baffb1f4268760a.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:586a491bdc820fe537ca3c2684b67f21a9afdd214deb310fdc5f78d2ecc7daf. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html;
sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- `sft-mat-elec-occupation-001-authority-correspondence`: predicted canonical-mode-carrier__empty-or-singly-held-support__indistinguishable-duplication-rejected__occupation-bound-one__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived canonical-mode-carrier__empty-or-singly-held-support__indistinguishable-duplication-rejected__occupation-bound-one__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered exclusion-bounded state occupation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves canonical-mode-carrier, empty-or-singly-held-support, indistinguishable-duplication-rejected and occupation-bound-one.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:2d0fe62702f566ed470e063bb5e5b66dd78ec57a1ecf5952711e588f64e6f0a0; derivation seal
sha256:a0a7516941112ac9fc011c492b523eedf7b8aee7a793fc958fc72c9dacc71d9; engine receipt
sha256:506e8ef22d4cde99c31da1531b362820047c16b74619ef23b75785d57ef7cc68 at

```
receipts/engine/model_admitted/SFT-MAT-ELEC-OCCUPATION-001-506e8ef22d4cde99.json;
empirical validation sha256:ee35ddf7d203268b4935a82979d01e280f6475b227d378ec8b8eff821b9381df;
measurement receipt
sha256:084ad1d875948d1719f83b9b6a861e91642abbbaa1885ef0ce125aa3a60706a7; isolation
sha256:938d56f9988c6c5ec659b1c08f398e04a05f1f4b0b848dcf4430a32356ad6967; custody
sha256:02e2391d12546aa8a4d5e40c33b22264075664191749b1011b94efbf0606c846.
```

32. Doping as controlled constituent substitution

Claim identity: SFT-MAT-SEMI-DOPING-001

Question and exact theorem. Doping is a source-bound substitution or addition that retains host organization while introducing an exact carrier-support difference.

```
host-lattice-and-dopant-carrier__valence-support-difference__host-structure-and-provenance-retained__concentration-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves host-lattice-and-dopant-carrier, valence-support-difference, host-structure-and-provenance-retained and concentration-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	host-lattice-and-dopant-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	valence-support-difference	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	host-structure-and-provenance-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	concentration-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in-stance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `host-lattice-and-dopant-carrier__valence-support-difference__host-structure-and-provenance-retained__concentration-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:56562612ce8c35623d0f0eccc35d803dedbbb5024e8a9b6ddd9af1ea5273c1dc.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:e0b7aa75ce3cc58f9f38cccc833e5fffb828ed6a780b76c27e72333ae94cc79d.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cce931a0a9fc2d4bf92c7abfb6695c3c7131c95cf90e190f274acdf42801dd19.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:bb0257ed565ef5277fab5ae273fec802528089e7c14e6b69b5869605228db7ce.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:1b2754be6101b1b3994aa165f56cdc9719922bda78bb8a3e7b49541a130c5cec; certificate

sha256:bb931bf8b8986710e3d9c87afc938b21108e51f8711df610b710f2d2d37447c36; external-validation

hash sha256:af6af3fcedf777a897ccb954d3a94939fda8a9a940635212ca5f0c7ff90910d.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:122bb299288127d263f0e8816b29197cafb3b27fd5ab9450db9e64b49090bfc3. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-semi-doping-001-authority-correspondence: predicted host-lattice-and-dopant-carrier__valence-support-difference__host-structure-and-provenance-retained__concentration-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived host-lattice-and-dopant-carrier__valence-support-difference__host-structure-and-provenance-retained__concentration-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered doping as controlled constituent substitution observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves host-lattice-and-dopant-carrier, valence-support-difference, host-structure-and-provenance-retained and concentration-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:7d6ba4126f6021c12701e140f9662df0dcac166c04f0411174670731bcacf36f2; derivation seal sha256:2c55ece8edc00804dddf74490222c7b077a2b1fb138d08925bacd67c16fa258d; engine receipt sha256:2dac934d9906877ee45538a866d4b4e5d01bca7db0fb4b83d1d13f6f3dfe5bc0 at receipts/engine/model_admitted/SFT-MAT-SEMI-DOPING-001-2dac934d9906877e.json; empirical validation sha256:4b2237c7244ac6417cf900786078c5e3e67a6690e0716fa7cd6ce20c2bd1e4b9; measurement receipt sha256:76ac92ae9fbcfb18e6d6249bc79a99402af54af309c24eeb36531ec6dc8d07b8; isolation sha256:65640601d6884e72972e3ccfd89d29f60ac5ac53b9d2f2642fb10fc37a5041e7; custody sha256:75feef131ff8f5a61484f23bc35e133e4d4dcf9aa845ebd919d5e0050a69d51b.

33. p-type and n-type organisation

Claim identity: SFT-MAT-SEMI-PN-TYPE-001

Question and exact theorem. The two held orientations of a doped carrier imbalance force occupation-majority and absence-majority semiconductor classes.

doped-support-carrier__excess-occupation-or-absence__opposed-charge-orientations__
two-majority-carrier-classes__complete-state-transition-boundary-trace__root-bound
-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves doped-support-carrier, excess-occupation-or-absence, opposed-charge-orientations and two-majority-carrier-classes.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	doped-support-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	excess-occupation-or-absence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	opposed-charge-orientations	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	two-majority-carrier-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: doped-support-carrier__excess-occupation-or-absence__opposed-charge-orientations__two-majority-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:56ed005fb5cdc0097997f439b277475193ecbc6b55d69b1a477b88ee060926ec.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:cb92a6f95ff3cdea45d2e28b701382cb2a7f52dae78b06701b10a88ec5f994ec.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4b980f33219ddfc1ad61861742c6de890fe9d280fdaa67a1f556462fcbeaa528.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:d8e94a97ea900fc4dfdec8667a3e360ee94746b72f6b7e344184e234fc506842.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:e452a66255f1a519151033fcce9e6faa6547ddefcecff55bca8aa84762cbfd35; certificate

sha256:30d82bd025a5aef37135c2532e86294b973a60acc75b57d4c7832a10a1aeb2e; external-validation

hash sha256:008460b8937e1e40da3576894f4ebb00fdd06ac578102266d2755dfd33919f57.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 3 claim-specific source discriminators were required; their ordered identity is

sha256:c9d08fdf23cf1d1c4a0cbd021530dcec9fe237faec27dcda06c3bc8c6ba06828. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SEMICONDUCTOR-CARRIERS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/news-events/news/2017/03/testing-performance-semiconductors-light>; snapshot experiments/external_sources/materials/snapshots/nist-semiconductor-carriers.html; sha256:05c774e4a30e6009f031b3064b16319d181ddac92e6d088a01bb6fe362172cc5; scope: electron and hole carrier signs, doping, transport and Hall measurement.

Comparison and custody records:

- sft-mat-semi-pn-type-001-authority-correspondence: predicted doped-support-carrier__excess-occupation-or-absence__opposed-charge-orientations__two-majority-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived doped-support-carrier__excess-occupation-or-absence__opposed-charge-orientations__two-majority-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered p-type and n-type organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves doped-support-carrier, excess-occupation-or-absence, opposed-charge-orientations and two-majority-carrier-classes.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:c315630088c00248b7a14519816565651cba129871a017b8b196d44835b21999; derivation seal sha256:508e9b18d4e217dbe7bb9880439e5ad62146420eedddbd5f4d38ab155c89bbc1; engine receipt sha256:09af6c90f13d03c26aa83c7e45157858c3cc873c105d60d24b5d8ecf5416bb56 at receipts/engine/model_admitted/SFT-MAT-SEMI-PN-TYPE-001-09af6c90f13d03c2.json; empirical validation sha256:de8958fcad4ba8ece21a19486d0c82b04ba99b43f1aefa231a80e2a69b25851d; measurement receipt sha256:4401c9d604bc206b01f0b0240170a20ab4dbcdbac070de79b10ebfa528e1ffaa; isolation sha256:d097f756290910c99b0b5274c7508ea5752523009ebbbae60bc424a248147a40; custody sha256:f248e50cffb18f18a9831313e6d9d11fb1281018d9cf6fe04bfe88b3687c4896.

34. p-n junction and depletion organisation

Claim identity: SFT-MAT-SEMI-JUNCTION-001

Question and exact theorem. Joining opposed doped regions forces interfacial carrier transfer until a retained depletion/space-charge organization balances transport and field response at declared conditions.

opposed-doped-region-carrier__cross-interface-carrier-transfer__space-charge-and-field-retained__equilibrium-and-bias-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves opposed-doped-region-carrier, cross-interface-carrier-transfer, space-charge-and-field-retained and equilibrium-and-bias-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	opposed-doped-region-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	cross-interface-carrier-transfer	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	space-charge-and-field-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	equilibrium-and-bias-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: opposed-doped-region-carrier__cross-interface-carrier-transfer__space-charge-and-field-retained__equilibrium-and-bias-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.

- **observation**: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- **false_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:04fea8715f50e7048d9416d376c53e3450f90c1c8994797f356f59fc28bb26fd.
- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b60cab4caeac9244a50a12bfbcdaba9f6971a485a5c9dca8d0dc056c9e1461973.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:341cc639ba0af26441cfc73b6163439bb85143317cd5e75e78911ae88dc2e7b5.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:bed3d1635d1a890677f6efc509da67b859b05cda4243428f4bfa2cb10c603d12.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:e6eb0781d6783a76498a288507518312a16cd86e3b3062787095f1b02f171edd; certificate
sha256:e9b13a5eb19c146ded679178d0be32f2a73e257475860e846b49a09f9cb13d16; external-validation
hash sha256:d4b7e5fdc300feba712eba719d137d821e3bf69c7e235835a0281a7832a17eea.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is
sha256:28acbaad472972858e5363c8410375f0fe98fecce35a163c6af00e8e96a968c8. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-JOSEPHSON-STANDARD-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/news-events/events/josephson-volt-nists-first-quantum-electrical-standard>; snapshot
experiments/external_sources/materials/snapshots/nist-josephson.html;
sha256:371bb54f09d8e66d74cda08e66518b44ebf752445ee2a049b26a76fd669487e0; scope:
superconductive tunnel junctions and quantized Josephson voltage.
- NIST-SEMICONDUCTORS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot
experiments/external_sources/materials/snapshots/nist-semiconductors.html;
sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope:
semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-semi-junction-001-authority-correspondence: predicted opposed-doped-region-carrier__cross-interface-carrier-transfer__space-charge-and-field-retained__equilibrium-and-bias-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived opposed-doped-region-carrier__cross-interface-carrier-transfer__space-charge-and-field-retained__equilibrium-and-bias-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered p-n junction and depletion organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves opposed-doped-region-carrier, cross-interface-carrier-transfer, space-charge-and-field-retained and equilibrium-and-bias-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:4d54dd75b86ef466bb3ae5cbc87a0ca7f701301e43197d0ab078c61c49750fc3; derivation seal sha256:555154742ac70136d63a345d0f10ab9c0c161554e1e8014f4b9ba7d52847822c; engine receipt sha256:779d6b98fcddc660d82e4349746ac115bf9463663074b27bac1a52fba754d2e0 at receipts/engine/model_admitted/SFT-MAT-SEMI-JUNCTION-001-779d6b98fcddc660.json; empirical validation sha256:bcae9c1e255550e984a5e79458201c89f70b0258eba44b7b1903d1bba3cf78d3; measurement receipt sha256:35ceda6bfff041209f8db359a7dfcba8f80dafb726fdce296417f61bf22051e3; isolation sha256:76cfb9f4f250904e62748743c6525a455e0f0750a8adcbabd145a21d2c142a99; custody sha256:533feb47e28eae5a49bc786c08228949054a2dffe1623d130133575a3ec9d5a6.

35. Semiconductor transport relation

Claim identity: SFT-MAT-SEMI-TRANSPORT-001

Question and exact theorem. Semiconductor transport is the complete counted motion of occupied and held-absence carriers through accessible support under declared field, gradient and scattering conditions.

labelled-mobile-carrier-support__field-and-gradient-transition-count__scattering-provenance-retained__mobility-condition-bounded__complete-state-transition-boundary__trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves labelled-mobile-carrier-support, field-and-gradient-transition-count, scattering-provenance-retained and mobility-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	labelled-mobile-carrier-support	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	field-and-gradient-transition-count	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	scattering-provenance-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	mobility-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: labelled-mobile-carrier-support__field-and-gradient-transition-count__scattering-provenance-retained__mobility-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0e7b64dccdebd3f781a411c669794f0bb3a3e5bdc46971765b794957d212a9.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:013b382715fc6ec176e0dc3408ede980c818daf766daf96c476885b998c34ddd.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:126ab210a36b80039e752758c90fa83b171da3f4688ecbfab8fb9478a547d6f8.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:bbf482bcded2d0df85a2fcb371710d3e58ddec033cd86e9729e94016097fd96b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:aedaad9f2cdfe6887baf53d955eae7a0fb8b34bef6cbbb1ef936381ff1b3def2; certificate
sha256:1f9ea281dc6bffeca2b178a18d709033fcb47d9f2666f9691f5c20615e0f8693; external-validation
hash sha256:07cc521721d714dd3db3311e20236df3c3daf59ff0064523b530398b1e8f4db1.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:416db3b9005e4976355fa7996612862d673d7aced19bdae0fe27a041a619ee80. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716cccf34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope:
thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- sft-mat-semi-transport-001-authority-correspondence: predicted labelled-mobile-carrier-support__field-and-gradient-transition-count__scattering-provenance-retained__mobility-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived labelled-mobile-carrier-support__field-and-gradient-transition-count__scattering-provenance-retained__mobility-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered semiconductor transport relation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves labelled-mobile-carrier-support, field-and-gradient-transition-count, scattering-provenance-retained and mobility-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:3af983283357631283d3947def3fd55314859ce664d93c96ae2d9a4aa1d2e69c; derivation seal sha256:fcd13ccc157a4adf9d8bd511d17f75266e4d4514d3c5a8008bb2c4a9fe2e2a2c; engine receipt sha256:7a9e1a097ec9c3c48321750ba1f42b728a9bec278024122acdb49c71d1bfecb at receipts/engine/model_admitted/SFT-MAT-SEMI-TRANSPORT-001-7a9e1a097ec9c3c4.json; empirical validation sha256:6b0761d0b8abf30d604d8cd0c0d8fc0bca015f1b61338f43fd2910a8b178aa52; measurement receipt sha256:8b454ccd9aee42a77be377617c292db6ea57bd02cbebae033513adf4e802cf3f; isolation sha256:9a8ad3fe9172c8975e080fdd5a07d52725f60710ce518c29d8a6ea02fa14784b; custody sha256:db2e9f0628622dc8e4a0e9ae541b76556fe17d30ad12204a0b24021289e87f25.

36. Interband optical transition

Claim identity: SFT-MAT-SEMI-OPTICAL-001

Question and exact theorem. An interband optical response exists only when a probe supplies a complete compatible transition between retained occupied and accessible band states under the material selection rules.

initial-final-band-carrier__probe-energy-compatible-transition__occupation-gap-and-momentum-retained__selection-and-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves initial-final-band-carrier, probe-energy-compatible-transition, occupation-gap-and-momentum-retained and selection-and-resolution-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	initial-final-band-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	probe-energy-compatible-transition	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	occupation-gap-and-momentum-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	selection-and-resolution-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: initial-final-band-carrier__probe-energy-compatible-transition__occupation-gap-and-momentum-retained__selection-and-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c936b0d5bb1ae8aec13da9742921982d562b55f6d4774bd1aab6e334a1e5b753.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:8701918a766daef17bc285510ac5721438c7e56272c88217cd085a9d3e145154.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:bc252118ced3be1bd50755e418df34a442529528051ab45a471d57b3cfcfb8f6.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:ceec9c1581794a42484f4f193268ea44aecb938181070c46a7f3f65721a2f46a.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:a3afed5ac8947640ccdafaf35625a2212acb73b39ad06c991752279f3cad2019; certificate

sha256:8076ebc7ac57b55ce725e87f69dcce17a303f9329a50b04a951c98f157d8053e; external-validation hash sha256:f68ffc662b3b7c7128a78f5abc04b5a0e0eaa09941e02795cea9ced09ac95ef0.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:278dc1caac24eb6c922c3a6bf6e833a99a93040d869c9a79be2ae4104dbf0bf3. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-OPTICAL-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-optical-materials.html;
sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of
refraction, absorption, reflectance, dielectric response and photonic crystals.
- NIST-SEMICONDUCTORS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot
experiments/external_sources/materials/snapshots/nist-semiconductors.html;
sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope:
semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-semi-optical-001-authority-correspondence: predicted initial-final-band-carrier__probe-energy-compatible-transition__occupation-gap-and-momentum-retained__selection-and-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived initial-final-band-carrier__probe-energy-compatible-transition__occupation-gap-and-momentum-retained__selection-and-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered interband optical transition observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves initial-final-band-carrier, probe-energy-compatible-transition, occupation-gap-and-momentum-retained and selection-and-resolution-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest

sha256:8d8279fa3008b12b0b1051877f0af738d00569ad9ba188399d1eba5df11f24f8; derivation seal

sha256:8e4a5b2c675b5e98ec5e642838d2228f57b6aee5ee1ac95b8b21082fbcf31b8a; engine receipt
 sha256:809590cfd0ac95b5b110d63963478dbeff718642ced0621bb7bd593ce0f09e93 at
 receipts/engine/model_admitted/SFT-MAT-SEMI-OPTICAL-001-809590cfd0ac95b5.json; empirical
 validation sha256:d223c4d8de7571cd7987e65eab435a979ca11071a431f928801ba7ff40fc18f7;
 measurement receipt
 sha256:bbdfdb43fb4478403d0f7fbdebc1396367ef0eb93db04319e389aad5fb1104d3; isolation
 sha256:bef5fea3faf3ab08ccdddef2b1a5053453c9a471eee182464a6cd2ce6a342af1; custody
 sha256:2a01435955d2824cb67bf66500b68c1a34f4933eeb7ecee81a2e04ca4db6e43.

37. Exact p-n junction rectification

Claim identity: SFT-MAT-SEMI-RECTIFICATION-002

Question and exact theorem. The two held bias orientations of a retained p-n junction force distinct forward barrier-opening and reverse barrier-strengthening transition classes without signed proof quantities.

junction-barrier-and-held-bias-carrier__orientation-conditioned-barrier-transition
 __forward-opening-and-reverse-strengthening-retained__complete-two-orientation-tra
 nsport-census__complete-state-transition-boundary-trace__root-bound-forward-forcin
 g__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-MAT-SEMI-JUNCTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor electronic_semiconductor grammar preserving junction-barrier-and-held-bias-carrier, orientation-conditioned-barrier-transition, forward-opening-and-reverse-strengthening-retained and complete-two-orientation-transport-census.

Shared claim-record clause MAT-S003 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	junction-barrier-and-held-bias-carrier	The material carrier cannot be reconstructed. The complete generated carrier is retained.
relation	relation-imported-fitted-or-erased	orientation-conditioned-barrier-transition	An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.
organization	organization-collapse	forward-opening-and-reverse-strengthening-retained	Required material distinctions are merged. Every required organisation distinction remains held.
observation	observation-boundary-unrecorded	complete-two-orientation-transport-census	The observation class is unidentified. The declared observation boundary is retained.
record	result-without-transition-trace	complete-state-transition-boundary-trace	An answer without its transition trace is not auditable. The complete trace is retained.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.
generality	finite-answer-lookup	positive-finite-successor-closure	A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.

Unique survivor, base and successor. Sole survivor: `junction-barrier-and-held-bias-carrier__orientation-conditioned-barrier-transition__forward-opening-and-reverse-strengthening-retained__complete-two-orientation-transport-census__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: The first complete carrier retains its relation, organisation, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `bias-census`: forward and reverse held orientations yield distinct barrier states; passed `true`.

Shared claim-record clause MAT-S005 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:72091144fd5e5e2f17fb0af1236f878771e05c91a88f2d97aa1b5e2187a75b40.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:9929efcd682e523925a4517b24689bc98c434b00c3121fd02619458c61f8c0bc.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1fd0a27d51a77ca70c0055f7e0b90401b16bb9eddda5a4c552a9733f42ea9c2c.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1ddd52f23e78c87eb0542e05a59b000a8285827ad99039256e09441808f242fb.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:274dadcd795cbad20b0196e6035f10b5d8be04f5669237100a5e40812bacb1d9; certificate
sha256:336a6381c77c7668726d3aff4dc72e9d7f6b8f8edaf623d2703d3f950dd3290e; external-validation
hash sha256:336a6381c77c7668726d3aff4dc72e9d7f6b8f8edaf623d2703d3f950dd3290e.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:5f25f10dcea8438b7df990add66ee5fbf6485ee5cb207a08aa417df77a3419b2. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-PN-RECTIFICATION-2026-07-27 - National Institute of Standards and Technology; <https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir4414.pdf>; snapshot experiments/external_sources/materials/snapshots/nist-pn-rectification-2026-07-27.txt; sha256:ca301b8fc47ba509909982ca2b077c238b92f3ff4978ed7377d086071c519ba4; scope: post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Shared claim-record clause MAT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor electronic_semiconductor grammar preserving junction-barrier-and-held-bias-carrier, orientation-conditioned-barrier-transition, forward-opening-and-reverse-strengthening-retained and complete-two-orientation-transport-census.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_v1_v2_successor_complete_pre_source.json; source manifest sha256:3808d1a1c973341ac5988f70736ad6acf2671f1d6de4e44ca6a6c76f39e9a696; derivation seal sha256:4235421a299754c0130b0539c84de12ccc13f420cb83c72cd00d73bc69c6ee34; engine receipt sha256:e71ca636880f795b1e68037e7ad6f62ce1388b4a8a8d6c6a05399847beaa836c at receipts/engine/model_admitted/SFT-MAT-SEMI-RECTIFICATION-002-e71ca636880f795b.json; empirical validation sha256:c9e1c1cda237826ef15921efe0dd41a8bc39c8cd60cd3357c3670a45d14f70f7; measurement receipt sha256:06c4e475020998315f98335528e79fcb642bc1176030bca0f7f729022ac026b9; isolation sha256:65b5f7ae534d603f11f917cc1061ba772035490b6a36f4e51b868a596df904b4; custody sha256:e970053159f8be987db9dd34005e22f72951151146d54ed1a9a82ab1bf89168a.

15. Superconducting Superfluid Topological

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

38. Coherent paired-carrier state

Claim identity: SFT-MAT-SC-PAIR-001

Question and exact theorem. The least exclusion-compatible collective carrier joins two opposed fermionic labels into one integer-composite recurrence class.

two-fermion-joint-carrier__opposed-label-pairing__integer-composite-recurrence__pair-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-folding__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves two-fermion-joint-carrier, opposed-label-pairing, integer-composite-recurrence and pair-support-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	two-fermion-joint-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	opposed-label-pairing	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	integer-composite-recurrence	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	pair-support-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: two-fermion-joint-carrier__opposed-label-pairing__integer-composite-recurrence__pair-support-bounded__complete-state-transition-bound

ary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:588d928b1e15481455a5bc3aed20e5ecc939285f701ef568ab8854c37a456594.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1fcelab61ac5cdfd56f21d9a82a0f122803c48d6893b8182a0ec054e44125969.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:06212265fed2d4576c34bb29a7fa1db2397ec20e3ecdda764ca853158bb137eb.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:d85c54fbbab7de7503b8d8e13076bebd80a9d80d2bf9d3463f2ba5d851c6bf6e.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:ccdabd49d56b90757f60cc9f3d1de2c6b1e70499d23109147029465b0c1a2078; certificate

sha256:4425dcd748752cd4959cb4846753a73d83912db2c333b36d3295d3212ba1f9a1; external-validation

hash sha256:cd91799f0836165d952f55222fdc867f9baf72874a1e3c76abb6f573efb6766f.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:6c15a4430a3b63ae8b2d9b46dc34a353ae01acb710cd8481e153960a6e86cc26. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html;
sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-sc-pair-001-authority-correspondence: predicted two-fermion-joint-carrier__opposed-label-pairing__integer-composite-recurrence__pair-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived two-fermion-joint-carrier__opposed-label-pairing__integer-comp

osite-recurrence__pair-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positiv
e-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered coherent paired-carrier state observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves two-fermion-joint-carrier, opposed-label-pairing, integer-composite-recurrence and pair-support-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:d93b4d5e2b622347f1676960ee60f23945f6a7ced001e0126d43a7c8b8ef3a8e; derivation seal
sha256:a1338a1e0276c18373474215b722f972da5d845bbbd492ccb324d205c46553b5; engine receipt
sha256:f65c0f0c9bd7124faa85bdb47c873a6abc42e8e8708b6d1e1a59820b33115855 at
receipts/engine/model_admitted/SFT-MAT-SC-PAIR-001-f65c0f0c9bd7124f.json; empirical
validation sha256:713c8da110495dfd47aaea6dd319378273a24160916d2c4318d89f2f2fd1d278;
measurement receipt
sha256:61a58faeb48a968867e3b01590511b20540c5a14b184e41518714cf58b250eba; isolation
sha256:7785d4374ef3eff679275b146a79ab9f5de3cea69341d630072727b1c8d59a9f; custody
sha256:9c164976a0cf9ee4d1d0855105a2d8a6351a716ee3bee38ae12ec035fb2c8d5c.

39. Dissipation-closed superconducting transport

Claim identity: SFT-MAT-SC-ZERO-RESISTANCE-001

Question and exact theorem. Superconducting zero-resistance transport is phase-locked pair recurrence on complete connected support with no retained label-compatible dissipative transition.

phase-locked-pair-network__collective-translation-recurrence__momentum-randomizing
-paths-absent__critical-condition-bounded__complete-state-transition-boundary-trac
e__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves phase-locked-pair-network, collective-translation-recurrence, momentum-randomizing-paths-absent and critical-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	phase-locked-pair-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	collective-translation-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	momentum-randomizing-paths-absent	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	critical-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: phase-locked-pair-network__collective-translation-recurrence__momentum-randomizing-paths-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8d318b6cd754552f1635d2d3376fb19a7943dac948ddf64cdb2c469dd2d57981.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:2d3c2643f67651e4ca0ce67d52da102053664b47fd07c1aec55f2625f3b5820d.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4d3bbf5b42e26656f92e162c71c1b9b62f1636bacdd84ef0eade4dc87e68aa53.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:51e5ed61aacce99183b4b16d20ea877065858d4c167fda8309eca0ac820dd396.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:b1b9adcd5b385285baebd3bcc32081e7bcb07bc9a7423d1d0d8f4b24623b8b64; certificate
sha256:e767c0f4c1ff932262442a7d1c6b5ae435d89b7adfc2ede2eb66da249e8a5872; external-validation
hash sha256:bc66886b883b3b04e280b2c157934a163799a8e3abcd37cc54ea9170516dc3b5.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:c5d02ac8fd4e460114d7267f20dc67b2baff4b6cb76be84227c69b692b6557a0. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SUPERCONDUCTING-FLUX-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot
experiments/external_sources/materials/snapshots/nist-superconducting-flux.html;
sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; scope: zero
resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- `sft-mat-sc-zero-resistance-001-authority-correspondence`: predicted phase-locked-pair-network__collective-translation-recurrence__momentum-randomizing-paths-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived phase-locked-pair-network__collective-translation-recurrence__momentum-randomizing-paths-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered dissipation-closed superconducting transport observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar

- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves phase-locked-pair-network, collective-translation-recurrence, momentum-randomizing-paths-absent and critical-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:b9e796760dea97e1e9f71baab1a6c8bfd297371b6fbdd5aa16ee8c36edf8c366; derivation seal sha256:edbbea4729b78617df2c8f1ecdb55716528ac140a38827276580103223e2b971; engine receipt sha256:12eb3dc5c5b6718a083650f79097da45e15e1c0f9a7d6d962fb3afdf672576ea at receipts/engine/model_admitted/SFT-MAT-SC-ZERO-RESISTANCE-001-12eb3dc5c5b6718a.json; empirical validation sha256:4463105306d4f42745346031c4a24c8910c65ae30aaf2ea125d0108575905655; measurement receipt sha256:1bebb8a089facd8820e247e0c2b56def398143bf54d99b01e82b17c475edc90b; isolation sha256:3a31a682ebb7914cb6dfe9a572940f32f1a21e3af617a0e474f367180900ffee; custody sha256:4b18283de437e6a909184edac099a512c370d092730bd30c5ef62b3927e90677.

40. Magnetic flux exclusion in the superconducting phase

Claim identity: SFT-MAT-SC-MEISSNER-001

Question and exact theorem. A phase-locked charged carrier network generates a boundary response whose complete bulk recurrence closes an incompatible applied magnetic distinction, yielding flux exclusion apart from the declared penetration boundary.

coherent-boundary-current-carrier__applied-field-response-relation__bulk-phase-loc
k-retained__penetration-boundary-declared__complete-state-transition-boundary-trac
e__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves coherent-boundary-current-carrier, applied-field-response-relation, bulk-phase-lock-retained and penetration-boundary-declared.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	coherent-boundary-current-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	applied-field-response-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	bulk-phase-lock-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	penetration-boundary-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: coherent-boundary-current-carrier__applied-field-response-relation__bulk-phase-lock-retained__penetration-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:41fdd77f83e8d59c2bfcf7bfb501d37db3c95f5bc0b83d16b44837f40d88f75.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:e4bd038bb4b94d51173bdfb9230bddd24ccc1144277314f85858688c05c4e10d.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b0b3d381e964f5faaa7119aeaac1005b9b4d579b20faed23038adc1dd012e4cc.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e1e636a30e4b8e61b27e684806bc21fbf45696298651b7107e96b7f089e0a892.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:928c10c30b33c7559a3dbbc0c913420ed2ab81b7bc1d15273d91edeb97eee1d0; certificate
sha256:06e8deb4ea6d9b5aa05a4dfec72f2b735bfa417cc18126dfb1493310c209e47; external-validation
hash sha256:b50c818dc8d34d3a26d99222c94ed78c6d5e46fe14e683dadfd25d9519ca71ce.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:0c547443d909099a887a74f3853eb1e7fc46b03829baa87ede9b6fae8b103170. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SUPERCONDUCTING-FLUX-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot
experiments/external_sources/materials/snapshots/nist-superconducting-flux.html;
sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; scope: zero
resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- `sft-mat-sc-meissner-001-authority-correspondence`: predicted coherent-boundary-current-carrier__applied-field-response-relation__bulk-phase-lock-retained__penetration-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived coherent-boundary-current-carrier__applied-field-response-relation__bulk-phase-lock-retained__penetration-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered magnetic flux exclusion in the superconducting phase observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting__superfluid__topological grammar that preserves coherent-boundary-current-carrier, applied-field-response-relation, bulk-phase-lock-retained and penetration-boundary-declared.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:b59e8e71bf983f7a08fd389c4b3c03789f7d0f5a8bd9b4c5f8864fbfb35608e8; derivation seal

```
sha256:f9146fca79784576aa3a1589231dca173d7efe82cd3885b978b60ee3e13ef2bf; engine receipt
sha256:2cd04f09d3f7371d20ce1b7209cc1dc496ea19835d019733d1fda439b0195fce at
receipts/engine/model_admitted/SFT-MAT-SC-MEISSNER-001-2cd04f09d3f7371d.json; empirical
validation sha256:ed1c150af3c3925bcb4915bb464cdcb311c930e5f83b00b8d6175f225bb0f044;
measurement receipt
sha256:3f7d9d6861cb68227b9a4061d8ab60bd253c3e2a7a84b9134c79a13fa58ec14e; isolation
sha256:f141e62bf73a7f12e615ada65bbff9248644c0f2540ddc848f69ebca25a2b0b4; custody
sha256:713d272d7326c231a0ba2e21288be842ea24652fdb5f251e11611b0f3113d7ae.
```

41. Superconducting flux quantization

Claim identity: SFT-MAT-SC-FLUX-QUANTIZATION-001

Question and exact theorem. Single-valued return of a paired-carrier phase around a closed path forces whole winding classes and therefore discrete flux support.

```
closed-phase-loop-carrier__whole-return-winding-relation__paired-charge-label-retained__integer-flux-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves closed-phase-loop-carrier, whole-return-winding-relation, paired-charge-label-retained and integer-flux-classes.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	closed-phase-loop-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	whole-return-winding-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapse d	paired-charge-label-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	integer-flux-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `closed-phase-loop-carrier__whole-return-winding-relation__paired-charge-label-retained__integer-flux-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f0e9a5316c0cbe435499c31ffe40b86071cc48d07b4534c765299a5a35bf0f9d.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d7f3655cbbeb3b9850a02dc2d27cfd0a9c033adbd9caa63cba999c19e29c974.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c69e26fe4417441d2552b06d56b73199016f8853baa5f0e2a5f34bcac21d8bcd.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:64af51fc38ea344b01e2131b58ecba3a1728ed24158136831a4a38b3507f95cb.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:ba1d06e0c591660c25b9d494d140c2eb9c5b5ad7a60667e026d6c7c3f845b47d; certificate
sha256:7cf210102f6487dc10ff5b4ceb32d3bf816945f961e947c3b153c669c4d4c632; external-validation
hash sha256:76c78e4fa8a0436139f72942375cfe051a5b9bb6dd6496e8e17cbb720971c2f6.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:a59499d54f66da57c505ea1a9006d13183cf8c79f97b6694d0562a7789a16489. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SUPERCONDUCTING-FLUX-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-flux.html; sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; scope: zero resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- sft-mat-sc-flux-quantization-001-authority-correspondence: predicted closed-phase-loop-carrier__whole-return-winding-relation__paired-charge-label-retained__integer-flux-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived closed-phase-loop-carrier__whole-return-winding-relation__paired-charge-label-retained__integer-flux-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered superconducting flux quantization observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves closed-phase-loop-carrier, whole-return-winding-relation, paired-charge-label-retained and integer-flux-classes.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:c8b02fef0c388c57e46670d576003802b43b86270c1afaff3f410a87423f6911; derivation seal sha256:85f5457eddebd585da75c677d2b205089e95a4558c7cc6f9af8783fe82ea8e0e; engine receipt sha256:5222d9301d5077db8f7fe1adccb876b76af5caeec23508a322fc8393ac62e22b at receipts/engine/model_admitted/SFT-MAT-SC-FLUX-QUANTIZATION-001-5222d9301d5077db.json; empirical validation sha256:0b4b3abdbac20b0e33a7ee4a688a80a5a4090133d5ad8bb7d756508ea41e47aa; measurement receipt sha256:ee23932a5a16f9a89e0b3dfd7839be86e66580e95a8c9a7621fef766dd025324; isolation sha256:b80a2d163e7842712cce6b99df6f2c0337bcf60b91f002180b285ac4d1ab8ed6; custody sha256:9d61fe50609999ec7322ca722ebf3962578a79550c05102d4c8a2a06e82e31dc.

42. Weak-link coherent transfer

Claim identity: SFT-MAT-SC-JOSEPHSON-001

Question and exact theorem. A weak link between phase-locked pair supports carries a coherent transfer recurrence determined by their retained phase distinction without dissolving pair identity.

```
two-condensate-and-link-carrier__phase-difference-transfer-recurrence__pair-identity-retained-across-link__junction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves two-condensate-and-link-carrier, phase-difference-transfer-recurrence, pair-identity-retained-across-link and junction-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	two-condensate-and-link-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	phase-difference-transfer-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	pair-identity-retained-across-link	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	junction-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-specimen-or-in-stance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: two-condensate-and-link-carrier__phase-difference-transfer-recurrence__pair-identity-retained-across-link__junction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:399845f807dcd160de6e469aad258bfa25f83f72dc4570e597b8b67651d0aab2.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:c0e22c5b1eff9f4e8d5756445e63ce5b47873c214a9e6225659f30250ae45e44.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7b8e05bab5f8a0f80750a7c96e67151bc76e63513889500c048513a184222c3.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f9a41df391a6e3a93cb1c58b912920fb74a9f3d3cfe4d8231ca7cded8f9f5816.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:4d646761ef86e203d4dc92fc307815d351f2f955293517f5549e5dcc1e949b39; certificate
sha256:28ea20d0b7536a61f4df9e90490a956ebd4d4d57e33dbf816d7a7011b0260a1b; external-validation
hash sha256:74274c0b5926ea435009d31d46f20c137898814c025f01bac6ccec42b1905e3c.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:ee4e8e99314f54e4c03074f8570f0e1abe2959be6265b7bcf80807a92d298924. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-JOSEPHSON-STANDARD-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/news-events/events/josephson-volt-nists-first-quantum-electrical-standard>; snapshot experiments/external_sources/materials/snapshots/nist-josephson.html; sha256:371bb54f09d8e66d74cda08e66518b44ebf752445ee2a049b26a76fd669487e0; scope: superconductive tunnel junctions and quantized Josephson voltage.

Comparison and custody records:

- sft-mat-sc-josephson-001-authority-correspondence: predicted two-condensate-and-link-carrier__phase-difference-transfer-recurrence__pair-identity-retained-across-link__junction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived two-condensate-and-link-carrier__phase-difference-transfer-recurrence__pair-identity-retained-across-link__junction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered weak-link coherent transfer observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves two-condensate-and-link-carrier, phase-difference-transfer-recurrence, pair-identity-retained-across-link and junction-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:de9469a89d2867722b9eaed6fa3760649daaa5553d70c469eb7cf83e15e2525a; derivation seal sha256:3bd854b2b132fd641e4e5f5955b5ffce6518e0c7ac28bcdd65660d450b12d489; engine receipt sha256:7081c6da685356122c147a5935e3b767d81a0c0c6ce6d5e0a7b75f6c0b4da6d0 at receipts/engine/model_admitted/SFT-MAT-SC-JOSEPHSON-001-7081c6da68535612.json; empirical validation sha256:d871b3b487789842475feacb4d5f08c14ca3faa00faab4518a71d1f3cf2ceb03; measurement receipt sha256:2faf6ca3e067d3c131f05245b0149df1d19104e157da47b7accab873cd704cda; isolation sha256:7bc6387d84c77a4c1527b40b42bd1d5b808abd5603f45727497e2621e7d90a08; custody sha256:48ac2f8f2ee7cacdc2cdd908a0e4e30408ceced8311b194c717c481488f74827.

43. Dissipation-closed superfluid transport

Claim identity: SFT-MAT-SF-SUPERFLUID-001

Question and exact theorem. Superfluid flow is shared-phase recurrence of a neutral collective carrier when the complete accessible transition census contains no allowed dissipative path below its first retained excitation boundary.

neutral-coherent-fluid-carrier__shared-phase-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves neutral-coherent-fluid-carrier, shared-phase-flow-recurrence, loss-paths-below-gap-absent and critical-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	neutral-coherent-fluid-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	shared-phase-flow-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	loss-paths-below-gap-absent	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	critical-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: neutral-coherent-fluid-carrier__shared-phase-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state

-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:53aeb9bd83d18de3cd8d06065a5ef322fc212de21e769e8ca85c254f3a235c1e.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:0dd29512db07cf4c534254d6059cf520fc5edc37285f2abc39c5a4c037549c9b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f6e09bc043b7b7342f350da3404ad33d5781dccc6d6a9ee2adb0e1d3c39d5b21.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:502b285ce7563d3d21f1c79789e6ae682a6022b3e36db0a107366d25c3c7fa2b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:f56b139525ec8db1263aebf85b98f0d1417c8c2ef70d4f4ecef2abb94bc22a8e; certificate

sha256:737b7188dd1647b964cd6655764a2a2db0aea236180922a145ffbddfb1ffa9ad; external-validation

hash sha256:dbdd999b36e350be4f7fc5b44e9ef260cea53c782eccf9d0cb8e356c5eb0cc93.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:00b0b7c16330fdcb90a5ea51473a63afe04ee29fbb606a9c345ba91717e8eee. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SUPERFLUID-CIRCULATION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/news-events/news/2012/11/first-controllable-atom-squid>; snapshot
experiments/external_sources/materials/snapshots/nist-superfluid-circulation.html;
sha256:49a5edb2dc307d0849d5fb5a245be46166dbffe2c50f2e9422a78f139aa5f8a0; scope:
superfluid flow, vortices and discrete quantized circulation.

Comparison and custody records:

- sft-mat-sf-superfluid-001-authority-correspondence: predicted neutral-coherent-fluid-carrier__shared-phase-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived neutral-coherent-fluid-carrier__shared-phase-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

se-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered dissipation-closed superfluid transport observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves neutral-coherent-fluid-carrier, shared-phase-flow-recurrence, loss-paths-below-gap-absent and critical-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:cb2ec54d212e954c015f00c0dac3b0b91ad8433dddd8dfb5e842f81a20a3158; derivation seal sha256:c4fa97a72987df4fb8e8db6b3e67e94202e5bd15d51c09701501a72e26228d5f; engine receipt sha256:f97e7d21474da7f1f117d9fdf57a2a714f0844f88ece7bf4d90daf91e8e79b3b at receipts/engine/model_admitted/SFT-MAT-SF-SUPERFLUID-001-f97e7d21474da7f1.json; empirical validation sha256:e10012064a7012eaa97c30443e19ed8db341b8c01a039ad34cec73910fc9ebd; measurement receipt sha256:f5e2e939e5d8bbbed493cbca552c915b723dbef99a7a0602b8a960eac7d1a8b2d; isolation sha256:5c0d5f2cea3b7d6c7242bf7c842f4a0713086806e9d89c4909cff5b976545779; custody sha256:65f0b4b610b2e75226f3cae0b0a7eb10f9d9093837f3fc6cb940dde6bbfb33e8.

44. Quantized superfluid circulation

Claim identity: SFT-MAT-SF-CIRCULATION-001

Question and exact theorem. Single-valued return of a superfluid phase around a closed path forces whole winding and discrete circulation classes with a retained vortex-core boundary.

closed-superflow-loop-carrier__whole-phase-return-winding__vortex-core-boundary-retained__integer-circulation-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves closed-superflow-loop-carrier, whole-phase-return-winding, vortex-core-boundary-retained and integer-circulation-classes.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	closed-superflow-loop-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	whole-phase-return-winding	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	vortex-core-boundary-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	integer-circulation-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: closed-superflow-loop-carrier__whole-phase-return-winding__vortex-core-boundary-retained__integer-circulation-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f02eec7391d7631590d2bdf547034d8deb9704bd1216a21b9f245de6f8279715.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:47e3e962a411f836a67dcfcfb25ec672026bec43ec887acbf46cdd0e5d2fd37.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4e454943e4f2ef6f47e060c907ae345a5b14d1aa029de2e0f12f423693052ff7.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:65529a60c0bbbd22624bcc4ef2744b5a19a59898aaea44440259859ce3e8d166.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:8309220cfb4f71ac91bb6c78e1b8e93d5fd7d982a212003a218b11b52185129b; certificate
sha256:c2182c823e85f1379edf6bbcae692fb0a0597df0bf835c2b4a567d343189c797; external-validation
hash sha256:c4741e9aec860d4696db44b536e7ed688067581da702492261813d3450aa487d.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:3b4dda8db9ff0cf748bd3069c4aa37b649f64bf399b70947c75841856eaf0257. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SUPERFLUID-CIRCULATION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/news-events/news/2012/11/first-controllable-atom-squid>; snapshot
experiments/external_sources/materials/snapshots/nist-superfluid-circulation.html;
sha256:49a5edb2dc307d0849d5fb5a245be46166dbfffe2c50f2e9422a78f139aa5f8a0; scope:
superfluid flow, vortices and discrete quantized circulation.

Comparison and custody records:

- `sft-mat-sf-circulation-001-authority-correspondence`: predicted closed-superflow-loop-carrier__whole-phase-return-winding__vortex-core-boundary-retained__integer-circulation-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived closed-superflow-loop-carrier__whole-phase-return-winding__vortex-core-boundary-retained__integer-circulation-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered quantized superfluid circulation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar

- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves closed-superflow-loop-carrier, whole-phase-return-winding, vortex-core-boundary-retained and integer-circulation-classes.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:f16ac71b88734483350d41ca16b2b40556ee06e0987c89a426fb1985a4b5907a; derivation seal sha256:b372c91d91c4134850d6c8fb14dca1272e0daaaacce8ca48d23bd0f66a3498b; engine receipt sha256:d7416c15ecaa36ba63cd90357f14a677eba102cc7a50158fc5fda860d820aebd at receipts/engine/model_admitted/SFT-MAT-SF-CIRCULATION-001-d7416c15ecaa36ba.json; empirical validation sha256:d0a3ab6803a908c6d5278dadb5d3099c6ef00e2f4b1c721818568eaa604bcb2d; measurement receipt sha256:01272ae44e4f4e7d9ba40480b6afd4158ecdb7a6bdd572fceb256005fe603f75; isolation sha256:a6bf12743e23eb7e27f83abfac2fab464763d8dcc4f16491218c03255d3b3e3a; custody sha256:b9112c2aaa8190490bcec24ceccf35ccc1ef0b1a0a82050c5612ebe0472af434.

45. Topological material invariant

Claim identity: SFT-MAT-TOPO-INVARIANT-001

Question and exact theorem. A material invariant is topological when its complete global path class is unchanged by every generated local deformation that retains connectivity and boundary records.

complete-global-path-carrier__connectivity-preserving-equivalence__local-deformations-quotiented__integer-path-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves complete-global-path-carrier, connectivity-preserving-equivalence, local-deformations-quotiented and integer-path-class.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-global-path-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	connectivity-preserving-equivalence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	local-deformations-quotiented	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	integer-path-class	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-global-path-carrier__connectivity-preserving-equivalence__local-deformations-quotiented__integer-path-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:13efa656a19aecf32a12b6813904ce4d2fcde0814a104238b0f2940d99e2c78e.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:e181bd02e17925831b4a6b22c0d3ae5c3578c1a6ce53367eeabced88df33d8a6.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8905e525f2d1831fae398bb4d42ebe70cd9d0eb4243660c738c48c791372c83e.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2b0c618640e78c9e6a61d0cd591b39a31111d3a4bcb515fa4230310307ad5b0f.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:52be2a7f7455b7444b99c8ee1fb48683ea99fdce6dd35ca5f94546db602cac5a; certificate
sha256:324729b6e2e4e99286bacf7e230b6054ad0584c770944fde8bac4f7d7e3ca46f; external-validation
hash sha256:d5da1e69cff9663a4438aa6a46b03c0d80ee306d6917c10d13e544d6c476d28d.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:e25423817b22188b5ad5603572fe72556d6d623de47ea8d79cac1a22ac17e8e8. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TOPOLOGICAL-INSULATORS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/topological-insulators>; snapshot
experiments/external_sources/materials/snapshots/nist-topological-insulators.html;
sha256:4cbe823dffeabd728844a897aalc5a222dfc3aef465263a06501974e194df449; scope:
topological invariants, insulating bulk, band gap and metallic boundaries.

Comparison and custody records:

- `sft-mat-topo-invariant-001-authority-correspondence`: predicted complete-global-path-carrier__connectivity-preserving-equivalence__local-deformations-quotiented__integer-path-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-global-path-carrier__connectivity-preserving-equivalence__local-deformations-quotiented__integer-path-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered topological material invariant observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves complete-global-path-carrier, connectivity-preserving-equivalence, local-deformations-quotiented and integer-path-class.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:39b15b0d045f1b1d1b7ff8ad2ee76ac387e0f1a555cdf04c6cfe0a7e5c93e16a; derivation seal
sha256:355e1bda262e641217018e5bfd12142fb515f53642ce68873f49b667d47e9dbb; engine receipt

sha256:0aa06ea813c11acea4d063c83c11e606519be491423b0806a196e2876fe9391d at
 receipts/engine/model_admitted/SFT-MAT-TOPO-INVARIANT-001-0aa06ea813c11ace.json;
 empirical validation sha256:e2546e5a5c2cffbad153836b434eee2a7d90ffd41f42024f1d71c7fbfa5e9c84;
 measurement receipt
 sha256:c2ed694d7588b3efc388b89c3d8225bce6a6b223050cfa18c3666f6752c8d69b; isolation
 sha256:d877347a8eb8de659a662a2d14c4f0f00357e99386626f2e72ebd2da2b6a733e; custody
 sha256:1ec350459cdf675e1b52403b1065a89816d482f69231144349ef769376bfac96.

46. Bulk-boundary protected transport

Claim identity: SFT-MAT-TOPO-BULK-BOUNDARY-001

Question and exact theorem. A nontrivial bulk path class forces a boundary recurrence whose removal would require closing the retained bulk distinction; compatible local backscattering paths are therefore absent within the declared gap.

bulk-and-boundary-joint-carrier__global-class-boundary-obstruction__local-backscatter-paths-excluded__gap-and-boundary-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves bulk-and-boundary-joint-carrier, global-class-boundary-obstruction, local-backscatter-paths-excluded and gap-and-boundary-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	bulk-and-boundary-joint-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	global-class-boundary-obstruction	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapse d	local-backscatter-paths-excl uded	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	gap-and-boundary-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-bo undary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-s elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in stance-lookup	positive-finite-successor-cl osure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or -extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: bulk-and-boundary-joint-carrier__global-class-bo
undary-obstruction__local-backscatter-paths-excluded__gap-and-boundary-bounded__compl
ete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-succ
essor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f204c5e7146eb04130ad4807231ad0637e358eeb604715a0c22917bb8e16da4a.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:8ed623d8b4faf8ed6f5d8eabc89a41c340ddf240b0b45ecc9e15bd39089bc68f.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ad1aad827f8955b0b0041cf25546c6841d11c1b396331884019eeb73aab28e00.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4d467a33e09e676142c88ff0e26ce9d516bfa0201a789ecef92a812414280b8f.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:c1ad057b49304b4368ba4fad93e4f0ee05abf12ee690c16fd5c5d7c3a8eda48b; certificate
sha256:b0899dd406db35b3d8d9d3b00a5254964967f26d618d1a345bb4f7a4f3537c19; external-validation
hash sha256:5b8e12fedd650e5769d332dbaf6afda85a21d1f6ab5c33aede7b21fedad07d32.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is `sha256:1d4a59568502972f6927fa8cb2944be5ea47b884c616a7a56c4d71c0d8fede16`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TOPOLOGICAL-INSULATORS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/topological-insulators>; snapshot `experiments/external_sources/materials/snapshots/nist-topological-insulators.html`; `sha256:4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449`; scope: topological invariants, insulating bulk, band gap and metallic boundaries.

Comparison and custody records:

- `sft-mat-topo-bulk-boundary-001-authority-correspondence: predicted bulk-and-boundary-joint-carrier__global-class-boundary-obstruction__local-backscatter-paths-excluded__gap-and-boundary-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`; source-derived `bulk-and-boundary-joint-carrier__global-class-boundary-obstruction__local-backscatter-paths-excluded__gap-and-boundary-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered bulk-boundary protected transport observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves `bulk-and-boundary-joint-carrier`, `global-class-boundary-obstruction`, `local-backscatter-paths-excluded` and `gap-and-boundary-bounded`.

Immutable evidence identities. Pre-source branch seal

`experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:624e50eb6c37a7cb6aa302bb729524f97d404589da3d4dacfb4b391c2f0fd9ee`; derivation seal `sha256:786171e535a2997aaaa15642857733ac66028d16eab119c2349d6501ea056489`; engine receipt `sha256:abb28732eccdc748940dce9e3629db011dc41a205ef3b968b06d5cf1c0fb4f1b` at `receipts/engine/model_admitted/SFT-MAT-TOPO-BULK-BOUNDARY-001-abb28732eccdc748.json`; empirical validation `sha256:74b655977a85543c9d7d4722f055436f75559358e69d248ace5caaa8253b1b88`; measurement receipt `sha256:6fe03baefebf6c5f8b5194955b9d1132490c993a55fc531f5407d89ae26235b7`; isolation `sha256:6b2e7114ed5081ebd7faf0314e4d1e0aa595766da8b28fd6ea5bbb6f8dc75bc0`; custody `sha256:03b7784db09f06251f28e2779209ce357bee47b308dce3fabf2cb5625b6e2dc5`.

47. Superconducting isotope-response record

Claim identity: `SFT-MAT-SC-ISOTOPE-RESPONSE-002`

Question and exact theorem. A superconducting isotope comparison is admissible only when isotope identity, constituent-mass distinction and transition response are retained together while the observation remains incapable of selecting the pairing law.

paired-material-and-isotope-identity-carrier__isotope-substitution-to-transition-response-relation__pairing-law-independent-provenance-retained__specimen-method-condition-bounded-response__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-MAT-SC-PAIR-001
- SFT-MAT-PHASE-TRANSITION-001
- SFT-CHEM-ELEM-ISOTOPE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor superconducting_superfluid_topological grammar preserving paired-material-and-isotope-identity-carrier, isotope-substitution-to-transition-response-relation, pairing-law-independent-provenance-retained and specimen-method-condition-bounded-response.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answered-only	paired-material-and-isotope-identity-carrier	The material carrier cannot be reconstructed. The complete generated carrier is retained.
relation	relation-imported-fitted-or-erased	isotope-substitution-to-transition-response-relation	An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.
organization	organization-collapse	pairing-law-independent-provenance-retained	Required material distinctions are merged. Every required organisation distinction remains held.
observation	observation-boundary-unrecorded	specimen-method-condition-bounded-response	The observation class is unidentified. The declared observation boundary is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	result-without-transition-trace	complete-state-transition-boundary-trace	An answer without its transition trace is not auditable. The complete trace is retained.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.
generality	finite-answer-lookup	positive-finite-successor-closure	A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.

Unique survivor, base and successor. Sole survivor: paired-material-and-isotope-identity-carrier__isotope-substitution-to-transition-response-relation__pairing-law-independent-provenance-retained__specimen-method-condition-bounded-response__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: The first complete carrier retains its relation, organisation, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- record-boundary: isotope identity and transition response are retained without a universal exponent; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:94fd92ce0920aed5e96f9ac44e642ab502467bb08a12d724a44c2c2d21c9398f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:f21319ea8ef1ac3766fecbd963eacd861eb50473dcf9179c57e4935140c92658.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e2db6f23721473fd237164511f5c59498218b933478ddaf811445464216cdd7f.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:c16ace5c2e9c0f270d5d3ae27ef67924b682328cbcdc69b58d803fe0d0c60029.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:e4da47dc86627db27b968f024b231d219853ae5f4e3b645c14a7bc0e6093ea84; certificate
sha256:bd352525e4d60943a208319d14de9b2a49f8fd6ac05f10ea747d2f0e801405e3; external-validation
hash sha256:bd352525e4d60943a208319d14de9b2a49f8fd6ac05f10ea747d2f0e801405e3.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:f7fe74a4077c16022bd5816fa81794f75fb1b7a9be1aedb285474284704e694a. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SUPERCONDUCTING-ISOTOPE-2026-07-27 - National Institute of Standards and Technology;
https://nvlpubs.nist.gov/nistpubs/jres/094/jresv94n3p147_A1b.pdf; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-isotope-2026-07-27.txt;
sha256:48fa89f9f63d467e9259e766f81fd8f854924ad479ec0006d14075cfe48d3ae3; scope:
post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Shared claim-record clause MAT-S007 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor superconducting_superfluid_topological grammar preserving paired-material-and-isotope-identity-carrier, isotope-substitution-to-transition-response-relation, pairing-law-independent-provenance-retained and specimen-method-condition-bounded-response.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_v1_v2_successor_complete_pre_source.json;
source manifest sha256:9cc83826c3a6be6add3771eb5e8946b13bfdebd21a8e4c7858277c74ac0ca0a9;
derivation seal sha256:71ac33cc8762d33fd9f32a0930ae2c7e9fc20b15fbfec30946d731a6ffbc5efc;
engine receipt sha256:a55394460feb15801fb15a215f8e6c6ab5b9fb62523db33307c4f721c15dc1f6 at
receipts/engine/model_admitted/SFT-MAT-SC-ISOTOPE-RESPONSE-002-a55394460feb1580.json;
empirical validation sha256:f2d5a758829778c576c552b919b86b71464aaa09f2e926e02e9dff32d034e0bf;
measurement receipt
sha256:3d1c99fa2505455dd15bedca5b4a2d02c7fffafebd31cfa330f7dc7686ecfeeb; isolation
sha256:505e2002a7536d3a6689bc18f93811c6179c22bc8f52754b75580209c58c542b; custody
sha256:cfad539f900cf896749cefd972a21ffcfdd0bbe1e91cefd8a8fb4d1137a4a51.

48. Integer and primary fractional Hall classes

Claim identity: SFT-MAT-HALL-QUANTIZATION-002

Question and exact theorem. A gapped oriented two-generator material carrier retains whole winding classes and a separately bounded primary hierarchy of reduced positive odd-denominator parts, invariant under gap-preserving deformation.

gapped-oriented-two-generator-material-carrier__whole-winding-and-reduced-odd-part
-relation__gap-preserving-transport-classes-retained__integer-and-primary-odd-deno
minator-filling-census__complete-state-transition-boundary-trace__root-bound-forwa
rd-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-MAT-TOPO-INVARIANT-001
- SFT-MAT-ELEC-CARRIER-DUALITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor superconducting_superfluid_topological grammar preserving gapped-oriented-two-generator-material-carrier, whole-winding-and-reduced-odd-part-relation, gap-preserving-transport-classes-retained and integer-and-primary-odd-denominator-filling-census.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	gapped-oriented-two-generator-material-carrier	The material carrier cannot be reconstructed. The complete generated carrier is retained.
relation	relation-imported-fitted-or-erased	whole-winding-and-reduced-odd-part-relation	An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.
organization	organization-collapse	gap-preserving-transport-classes-retained	Required material distinctions are merged. Every required organisation distinction remains held.
observation	observation-boundary-unrecorded	integer-and-primary-odd-denominator-filling-census	The observation class is unidentified. The declared observation boundary is retained.
record	result-without-transition-trace	complete-state-transition-boundary-trace	An answer without its transition trace is not auditable. The complete trace is retained.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.
generality	finite-answer-lookup	positive-finite-successor-closure	A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.

Unique survivor, base and successor. Sole survivor: gapped-oriented-two-generator-material-carrier__whole-winding-and-reduced-odd-part-relation__gap-preserving-transport-classes-retained__integer-and-primary-odd-denominator-filling-census__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: The first complete carrier retains its relation, organisation, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- hall-census: the bounded primary hierarchy contains only reduced positive odd-denominator parts; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:efdabcd3c4418e00dcb59d89337e8c44bcaf27ba5c4436d0aa642fc45cc492f9.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:fb1c4724e66f9325dc667f98edbfef7eea78a89066058c5b775b2bfb88ec2b9.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a4e744f2ecce5e97eb0f6e47dd983fb87b6c837ab7f935d7f03b51c0268749cc.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8f0c7e1c3bb60b2866b5329f881997595405b99cf2520e4cf080fe63800f0982.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:3dc4f426d44e402de2b381525079f7b2f183435006a131c8efc1ee2b0877862f; certificate

sha256:fa11d3b9672c0cb258bf232d11fd91bf5d31df65359fc501fa8af42599dbf2e0; external-validation

hash sha256:fa11d3b9672c0cb258bf232d11fd91bf5d31df65359fc501fa8af42599dbf2e0.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 3 claim-specific source discriminators were required; their ordered identity is

sha256:2551ebd4dc35246430f129face2ce828648425f4c82491bf347ce4c37d907108. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-HALL-QUANTIZATION-2026-07-27 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/integer-and-fractional-quantum-hall-effect>; snapshot experiments/external_sources/materials/snapshots/nist-hall-quantization-2026-07-27.html;
sha256:4536e34ab401e72216f8b9010bafc9f1cacdbbfc47ca8fc998b781fa9125994c; scope:
post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Shared claim-record clause MAT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed

- Every positive finite generated carrier in the registered successor superconducting_superfluid_topological grammar preserving gapped-oriented-two-generator-material-carrier, whole-winding-and-reduced-odd-part-relation, gap-preserving-transport-classes-retained and integer-and-primary-odd-denominator-filling-census.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_v1_v2_successor_complete_pre_source.json;
 source manifest sha256:d9290b7cb8c1eba4c906fce12feb75d9af06433ecf31acef8ed28f8815231988;
 derivation seal sha256:d20afc47519f5439e90081f10d46ee922c9ade6f7c477841dc669c59e5677379;
 engine receipt sha256:8bf14f74a78ae92fd3ff5878df726d92cbac72870b8af7f44601acf2b01397f5 at
 receipts/engine/model_admitted/SFT-MAT-HALL-QUANTIZATION-002-8bf14f74a78ae92f.json;
 empirical validation sha256:f53d1d556220384209dc27cc0306054c51b5de98895eb8d0695e3c3cadece320;
 measurement receipt
 sha256:3a469de7be7ae142f5d62b887d5328d625080a048be09342e457bbcc5d86bb66; isolation
 sha256:928f854b15080f487374967b75ab4e6ba0890c1d04168758285303f4e70f0cfc; custody
 sha256:be256a30cf9a0ea3ad6a654e2f75833deaddc556892def83888c95ae99d8f0fb.

49. Exact topological boundary-mode count

Claim identity: SFT-MAT-TOPO-EDGE-COUNT-002

Question and exact theorem. For two distinct adjacent whole bulk winding classes, the exact protected boundary recurrence count is their positive whole gap with orientation retained as a label rather than a signed number.

adjacent-bulk-winding-pair-carrier__positive-whole-winding-gap-relation__bulk-orie
 ntation-and-boundary-recurrence-retained__exact-protected-edge-class-count__comple
 te-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-su
 ccessor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-MAT-HALL-QUANTIZATION-002
- SFT-MAT-TOPO-BULK-BOUNDARY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor superconducting_superfluid_topological grammar preserving adjacent-bulk-winding-pair-carrier, positive-whole-winding-gap-relation, bulk-orientation-and-boundary-recurrence-retained and exact-protected-edge-class-count.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	adjacent-bulk-winding-pair-carrier	The material carrier cannot be reconstructed. The complete generated carrier is retained.
relation	relation-imported-fitted-or-erased	positive-whole-winding-gap-relation	An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.
organization	organization-collapsed	bulk-orientation-and-boundary-recurrence-retained	Required material distinctions are merged. Every required organisation distinction remains held.
observation	observation-boundary-unrecorded	exact-protected-edge-class-count	The observation class is unidentified. The declared observation boundary is retained.
record	result-without-transition-trace	complete-state-transition-boundary-trace	An answer without its transition trace is not auditable. The complete trace is retained.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.
generality	finite-answer-lookup	positive-finite-successor-closure	A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.

Unique survivor, base and successor. Sole survivor: adjacent-bulk-winding-pair-carrier__positive-whole-winding-gap-relation__bulk-orientation-and-boundary-recurrence-retained__exact-protected-edge-class-count__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: The first complete carrier retains its relation, organisation, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- edge-gap: bulk classes five and two force three boundary classes; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Collective recurrence, whole return classes and global path obstruction organise superconducting, superfluid and topological material response.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b0f8a1570544fe9e553b882d2599e7eb2c174700b4ba3a54af58951a5eab4531.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:0435e9e62c8543856b653f65753315b2720a09a2a8c63d8a4b17e43d328c5c6b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6a85996d29e292434ba2ad82021ff6c843a64e1cf1ccf64adalefc2b7d39f816.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4e2210aa723d5207ae3e372cf3b8277d4fe7023863ccab3eef2b4ce9e3132a2d.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:68e58fe1d0215233990f1fac100ed7b05b73412a0fd8e54d63e8c235d3ae8a11; certificate sha256:bab99db1cc19f41183013609ec5b8ff2a0384c102f8296c8625320820eacee25; external-validation hash sha256:bab99db1cc19f41183013609ec5b8ff2a0384c102f8296c8625320820eacee25.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:6faca2d50bb9d04fa3b6f0418840d7433420dd842e94e195b0e330114bc786a2. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-HALL-QUANTIZATION-2026-07-27 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/integer-and-fractional-quantum-hall-effect>; snapshot experiments/external_sources/materials/snapshots/nist-hall-quantization-2026-07-27.html; sha256:4536e34ab401e72216f8b9010bafc9f1cacdbbfc47ca8fc998b781fa9125994c; scope: post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Shared claim-record clause MAT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor superconducting_superfluid_topological grammar preserving adjacent-bulk-winding-pair-carrier, positive-whole-winding-gap-relation, bulk-orientation-and-boundary-recurrence-retained and exact-protected-edge-class-count.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_v1_v2_successor_complete_pre_source.json; source manifest sha256:4f4001fc31772c1638559bd8221ac0bce34d58e0c13b761118a5d68f32af7b7f; derivation seal sha256:1b1b1e75f7751982016da82cf278cc89f9c5ff8e8eba04fbfd9302a46a5f0005; engine receipt sha256:5b742a33b9d5b5822b2ff2c8db6375cbe0b9f3e605643f187fee05f43a921c57 at receipts/engine/model_admitted/SFT-MAT-TOPO-EDGE-COUNT-002-5b742a33b9d5b582.json; empirical validation sha256:5cac72ae62bf75b24d851e95ace663da5c23e2af0aab8998a300fa047eea0012; measurement receipt sha256:81460434a3dacd79b30d5e6ed2a7e9e1f717b3ec5ebef047eff1afb352d2f3cf; isolation sha256:e63e4d1570fddfad718942053f80e4f9bfecdc47fc34fd58412f66b0bb9febc2; custody sha256:9dfcc9d49aacd64cf256ffa9f50b353709b943c016f4fac3881755be01c05e31.

16. Mechanical

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

50. Stress and strain records

Claim identity: SFT-MAT-MECH-STRESS-STRAIN-001

Question and exact theorem. Stress is oriented load transfer per generated boundary support; strain is the exact retained change of internal separation relative to the same reference material state.

material-region-and-boundary-carrier__load-transfer-and-shape-change__direction-support-and-reference-retained__method-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-region-and-boundary-carrier, load-transfer-and-shape-change, direction-support-and-reference-retained and method-and-scale-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	material-region-and-boundary-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	load-transfer-and-shape-change	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	direction-support-and-reference-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	method-and-scale-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	authority-or-target-s elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in stance-lookup	positive-finite-successor-cl osure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or -extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: material-region-and-boundary-carrier__load-transfer-and-shape-change__direction-support-and-reference-retained__method-and-scale-bound__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:adb5d05a21d454b6bf5a2017f8be3f371d55fbd9a0b7ad25b947caa373ad534f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:109a21a2e46dcd23f5466f23be5156d57b500eff7391011d89ecc112be5e6872.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6391dbcd50225bace528b7afbb330ee156b47e635d52cdd610f75541e72b83b.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:04e5725ac444cc83fe68338efe05502691abd15b8808ea95150d1c8bb7212fe6.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:60cf9be4f23f11d4bef081dbc622511a6b00ba9a95d7c50d982b096248f1af5e; certificate
sha256:166b798626def8349847d2bbf1f68dca63d749c0a498661c151eb032cc0da216; external-validation
hash sha256:64538778c885dec7c2c0b9e159dc79e019cbcc17cace487879a2ca3a86e5b28f.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:58cbbf4e31983da152d2e9df9813867504aea0339fd3dd88255134648592ea34. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FATIGUE-FRACTURE-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html; sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; scope: processing-structure-property-performance, defects, fatigue and fracture.
- NIST-SUPERCONDUCTING-GRAIN-BOUNDARIES-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature>; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-grain-boundaries.html; sha256:1309d6599f6894e517dac8f3adc0b6b8506842cdcccc680329007351cde6abf2; scope: grain boundaries, strain, dislocations, current flow and superconducting response.

Comparison and custody records:

- sft-mat-mech-stress-strain-001-authority-correspondence: predicted material-region-and-boundary-carrier__load-transfer-and-shape-change__direction-support-and-reference-retained__method-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-region-and-boundary-carrier__load-transfer-and-shape-change__direction-support-and-reference-retained__method-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered stress and strain records observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-region-and-boundary-carrier, load-transfer-and-shape-change, direction-support-and-reference-retained and method-and-scale-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:72ec1a5c263bd16dcc14f88e80cde6ddbce93c49574494ea6d506ef427d69a6b; derivation seal sha256:d3b9e2aa529d399c2335b68a60fb2429b910b89d6a5c9463d99af63692002ffa; engine receipt sha256:85482dc44ea7d4480cdd7474370679a5ef92c2b374ade6acd5604e5da740e766 at receipts/engine/model_admitted/SFT-MAT-MECH-STRESS-STRAIN-001-85482dc44ea7d448.json; empirical validation sha256:b255306584d1626058d63a03e6c05d19c2039983e3939726f8e7e9d1feae2f65; measurement receipt sha256:905fe24fd20a32060c8e829ca181725b21b382c21b9580ce56ac405699ed9f02; isolation sha256:4eb94e72da4ebae40c69edca01de9e83b273d60729611f3ac44fa71099fd27e2; custody sha256:f680002c5dbcf1e4fa1ab346fedf28362a1c9de3b8c2c7e067d99fe7112d62b.

51. Elastic recovery

Claim identity: SFT-MAT-MECH-ELASTICITY-001

Question and exact theorem. Elastic response is a material transition path whose load release returns every retained constituent and adjacency relation to the original canonical state.

bond-network-state-carrier__reversible-displacement-path__original-adjacency-class-restored__elastic-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-network-state-carrier, reversible-displacement-path, original-adjacency-class-restored and elastic-boundary-declared.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	bond-network-state-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	reversible-displacement-path	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	original-adjacency-class-restored	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	elastic-boundary-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: bond-network-state-carrier__reversible-displacement-path__original-adjacency-class-restored__elastic-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d965c15a37557dcfc3605cc273e1a1880c9e41083dbcf9d3a708f979636fd61.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:cd6eb0e5456c7e6e8a8e5f7fd98dfc5b0a7eae9a32d7ecce593b09f29c908bae.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ac6d6673b36ac8725d7bfe8f9d12e9924b9024d0edd255e90883210b8efb8fe9.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:47aaed4f4a0ea0724465aceb9bf4aba1ef44ce469cad8d3d06fa3e0f47f21236.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:b0a0f914de0a4b46b3d7efd5c82bdd05cd890bec50d642f36863a2e98494e644; certificate

sha256:9d1e0777b17da1c0589983ef1da7f0c3ec0ba76e129d86df0a1b1a58f37c3c93; external-validation

hash sha256:a0374dd3a632fe0a266ad5bf3f615aac82e7a7b413a508f0b7c44bcd0e433ad.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:1f5231592213d1cb22ac6eee1db2e48d66131790c95c6401a6712d5e12f4bc25. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SEMICONDUCTORS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot
experiments/external_sources/materials/snapshots/nist-semiconductors.html;
sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope:

semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-mech-elasticity-001-authority-correspondence: predicted bond-network-state-carrier__reversible-displacement-path__original-adjacency-class-restored__elastic-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bond-network-state-carrier__reversible-displacement-path__original-adjacency-class-restored__elastic-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered elastic recovery observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-network-state-carrier, reversible-displacement-path, original-adjacency-class-restored and elastic-boundary-declared.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:b346b8fee12d428022c4ae51969a24edb3f509f9de381374c5735ea1eb0d92a5; derivation seal sha256:260626b207e5763c1058d0fc4ec4281cb5ce31d71314b505f4027525115ab722; engine receipt sha256:b490d2bc4771b5f91d7458764359748fcb9fd73ecb665b5f66637be92c9d9b17 at receipts/engine/model_admitted/SFT-MAT-MECH-ELASTICITY-001-b490d2bc4771b5f9.json; empirical validation sha256:91e76ec99100ab2b4bcb9de89117654abe30c261b652a123188a14eccb7d5e2e; measurement receipt sha256:0fc70106063eb948741b12ccaf1d72685502d7185847683b38e43c272ba1b467; isolation sha256:5f44ba6f53e33a525c41ec5a037db2da522c7f9ab958f32f1cc61771aa19cae6; custody sha256:4746d3544ccf6e014f821317b7525878c79c6de8fa91be7cb75968f039c58fc0.

52. Plastic deformation

Claim identity: SFT-MAT-MECH-PLASTICITY-001

Question and exact theorem. Plastic response retains material content and phase identity while a defect-mediated transition changes the canonical adjacency organization after load release.

bond-and-defect-network-carrier__irreversible-same-material-reorganization__content-retained-adjacency-changed__yield-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-and-defect-network-carrier, irreversible-same-material-reorganization, content-retained-adjacency-changed and yield-boundary-declared.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	bond-and-defect-network-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	irreversible-same-material-reorganization	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	content-retained-adjacency-changed	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	yield-boundary-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: bond-and-defect-network-carrier__irreversible-same-material-reorganization__content-retained-adjacency-changed__yield-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organisation are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9150c7c34bf6ef1d0009e84afa667c913cb1142a022c9425f3fc6a140ffd3b9d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:a7d6901fc35b585477d6b688a690b3c162e2e621559cad8b4d96464a4ec0170f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:844e5f851097100443c4f3d0dfa284fbae50e59dff142d5d5c9b8fdddf629d3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:316c01f0fe9eb2fb72cde97a9c3e7af4199b41650b03f8a77be1d286fd0b766f.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:f15e6957edff21a3f20f016b86d2c1eec8d49f37c03d565312ea934715ed36d1; certificate
sha256:db298f3a0e0fc3ab181511dd1bea67a8856c29679fb939576d2814ae6dc44876; external-validation
hash sha256:20e178859f18007683e55801f7a6cb2240a269f32c4aa11dd20fa58e04218056.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:c36087458657bcd08909aee6600aa4bcabf26dce1f9f96112864f0e05ec356d9. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TEXTURE-PHASE-FRACTION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot
experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html;
sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains,
phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-mech-plasticity-001-authority-correspondence: predicted bond-and-defect-network-carrier__irreversible-same-material-reorganization__content-retained-adjacency-changed__yield-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bond-and-defect-network-carrier__irreversible-same-material-reorganization__content-retained-adjacency-changed__yield-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered plastic deformation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-and-defect-network-carrier, irreversible-same-material-reorganization, content-retained-adjacency-changed and yield-boundary-declared.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:3006a582b76abd466eb5e052f89c2465497a3c270e20fe6db6dbb339179a8ac2; derivation seal sha256:12ebaca61ddae97d7c9eb477672cc0de736f0ab3c7f9af57c6d2e80f45d61158; engine receipt sha256:77ce9abbd0c078fd565e3a48d5a5b2f65b128d91f104012cc32b4c4a02eb7d2f1 at receipts/engine/model_admitted/SFT-MAT-MECH-PLASTICITY-001-77ce9abbd0c078fd5.json; empirical validation sha256:b3532d1895dd1296d276e1735e26099cf264926c90c65d74200d5e625debdfe7; measurement receipt sha256:391ccc840a880b49ecdd2940ed67b3596c473564b0d566fe1b63c106bffed26a; isolation sha256:d873591d0a8d9bb028422f4b911183a711f9c215eb9344d67d2f95798c1a8f7e; custody sha256:efd5e346f7ffd5349aa7bb765688da67e26c84fda13093741e87732a585a073b.

53. Dislocation-mediated slip

Claim identity: SFT-MAT-MECH-SLIP-001

Question and exact theorem. Slip is a counted whole-lattice translation propagated by a retained dislocation along a compatible plane/direction without requiring simultaneous rupture of the complete plane.

dislocation-and-slip-plane-carrier__whole-translation-advance__burgers-label-and-content-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001

- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves dislocation-and-slip-plane-carrier, whole-translation-advance, burgers-label-and-content-retained and slip-system-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	dislocation-and-slip-plane-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	whole-translation-advance	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	burgers-label-and-content-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	slip-system-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: dislocation-and-slip-plane-carrier__whole-translation-advance__burgers-label-and-content-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:89538877c541c0e3b80b081a953e40d057a01f631b47a0e8282b46c7a892f43f.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:0151dd250ba88c10559598d8778c71c3c11e36e9edcd1034f8ee2fc45cbfb0c5.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f077732e53b69bf50948fe5266a30e63f5f80a7293c8bc4ae7d2a46e0d248e76.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:16203aa2aa450696e0421d92c1b3cc0a003c3fd86ca1413a987ca923e90c1af7.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:ce30006112583d5a105cd9e64a8deec6bdef9c9a58e2ccafa5b65005bd351415; certificate
sha256:c9a1f12156a69f5368449a3cc781bbff9923903bd31e5d7f38c8d32b9beec75; external-validation
hash sha256:1ca21fd13495c4d6ec9d83b742a4d8006a197f875c48dccdd1140cb435248696.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:a5c1c011368a43bfcccc04cf8c412ee846c22c570ed3a97aa228d2280274c1f1. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot experiments/external_sources/materials/snapshots/nist-quasicrystal-measurement.html;
sha256:a1b7cc0a5882502bb31b9706225b926c9ffb1e8d93ea5c0043249376c4d6ff18; scope:
microscope confirmation, nonrepeating order and fivefold symmetry.

Comparison and custody records:

- `sft-mat-mech-slip-001-authority-correspondence`: predicted dislocation-and-slip-plane-carrier__whole-translation-advance__burgers-label-and-content-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived dislocation-and-slip-plane-carrier__whole-translation-advance__burgers-label-and-content-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered dislocation-mediated slip observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves dislocation-and-slip-plane-carrier, whole-translation-advance, burgers-label-and-content-retained and slip-system-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:1e7c4abd9525c4f5aa9107b87bff628f69fee285b2d2a20cf4c4b5ddb071f60a; derivation seal sha256:5c9bc421d35c9d3b9ed2d15aacce9dda69982b549984f8c43b1d769f9d16501e; engine receipt sha256:ca7d3c3104a0dfdf5dde9127e49e575edfbbba784c1c2d58859ab2f22267eb1e3 at receipts/engine/model_admitted/SFT-MAT-MECH-SLIP-001-ca7d3c3104a0dfdf.json; empirical validation sha256:262f49da288b06c0eadc6b88cf734f56ef39fe5c5f9ca0093b53805735c49a9b; measurement receipt sha256:612c261a80046a405268cd495f42234bf4b68f6b2b4e82c49826f0939b1e8b52; isolation sha256:7274e98371ac622e94f1b2247c5c951a7650f0e5270ff34fc798957594909bbd; custody sha256:d3ef1d91e8259ecd6710a62a7ebf7ee7c3741be9f8d9e6b5668b3591fa03c339.

54. Elastic modulus as response ratio

Claim identity: SFT-MAT-MECH-MODULUS-001

Question and exact theorem. An elastic modulus is the exact stress/strain response relation for one declared material direction, condition, reference state and reversible response window.

elastic-state-pair-carrier__stress-to-strain-exact-relation__direction-condition-and-range-retained__linear-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves elastic-state-pair-carrier, stress-to-strain-exact-relation, direction-condition-and-range-retained and linear-window-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	elastic-state-pair-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	relation-imported-fitted-or-erased	stress-to-strain-exact-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	direction-condition-and-range-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	linear-window-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `elastic-state-pair-carrier__stress-to-strain-exact-relation__direction-condition-and-range-retained__linear-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:7d6248f498d15a2edbf0aea112d99e5a2f1daf19a804222723337b55c86c4ace`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:06222eed10c6b068ebb6c92d7456fde86013ee12119f30ebb92fe895b8f0fffc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:2bbd140e1af9ff6e2ba9427d90da1fbe2b21c9c8a379fecc1902d3f131edf130`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:9d740d421a442846bdb6849544d4930fc9837d3a2f4ef6f6a838b16d1b9541f1`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:18bf6748d138471a11dcd6ba96c75fe25842d895dd65d2f2ced937d8eca68784; certificate
 sha256:36744a94fc531a579c5f99ec93300f4892e6f6e9dd73cd52fe56f5d70fba9de6; external-validation
 hash sha256:111c32e35dd4f098864a80c781a12cc6cf9e944f89cbae2640eb53a44f1a4854.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
 sha256:85c03f12333c98d0238b67dfce07d56b00dbd561755af223f44fbc04919ff7e9. Target content
 opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SEMICONDUCTORS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot
 experiments/external_sources/materials/snapshots/nist-semiconductors.html;
 sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope:
 semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-mech-modulus-001-authority-correspondence: predicted elastic-state-pair-carrier__stress-to-strain-exact-relation__direction-condition-and-range-retained__linear-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived elastic-state-pair-carrier__stress-to-strain-exact-relation__direction-condition-and-range-retained__linear-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered elastic modulus as response ratio observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves elastic-state-pair-carrier, stress-to-strain-exact-relation, direction-condition-and-range-retained and linear-window-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
 sha256:1153be6043acfefa94d2197d88eb7ba687bea8d2e7db45b379d2b884ec2fe603; derivation seal
 sha256:e9f4cc9b2d4c4888f1326180b6143f911b195fa9076791fc8bb4e403fb286e78; engine receipt
 sha256:fd39c697c448c56d56a6d63557ec19a7b249f2ca0ff72bae1f895ee7fc77fd88 at
 receipts/engine/model_admitted/SFT-MAT-MECH-MODULUS-001-fd39c697c448c56d.json; empirical
 validation sha256:841d85ef828079b201e69f2e3cb0bfc8c17992ebc430516174765f5a3641725a;
 measurement receipt
 sha256:e195b8c1f1a29cab72fdd2ba5198b76a7c3eaf63c8afce760acfd8ab09ba87bf; isolation
 sha256:9281062f09a9ad8c389d3cc4cd86a231f35c1720f0bb32a9799f953a5285a3da; custody
 sha256:0269158f503aae4190601227607fce846f38422cd5356c8b20326c355ff2d890.

55. Strength and hardness boundaries

Claim identity: SFT-MAT-MECH-STRENGTH-HARDNESS-001

Question and exact theorem. Strength and hardness are method-bounded thresholds at which a declared loading path first produces retained plastic, indentation or failure organization.

material-and-test-carrier__load-to-persistent-change-threshold__method-geometry-and-scale-retained__threshold-not-universal-constant__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-and-test-carrier, load-to-persistent-change-threshold, method-geometry-and-scale-retained and threshold-not-universal-constant.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	material-and-test-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	load-to-persistent-change-threshold	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	method-geometry-and-scale-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	threshold-not-universal-constant	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in-stance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: material-and-test-carrier__load-to-persistent-change-threshold__method-geometry-and-scale-retained__threshold-not-universal-constant__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:862336e016548ff4c2c82fd3ac77ca0323be9c77556e971a6aebbacd0c76fb5.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:bc14a47bca29bd87794372fa028e2771a41e0b2d2e2b3f1ec19666f2e362e0c7.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4be80645c59da9b639daaaf753ec6ce5c2df9dec4c06899ac073ceb7572efc38.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:c2c68a86de634604b832653e2398fde3c925a952212ee0b01e72d81390035d1d.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:bd81bef0333505d69e1bb6b71f2ea590586d3910f93d18e8c9770aa2d979ea68; certificate
sha256:6e74e10acadf20f87e2b884dbb28242db27c0b03c2917387319e000cb73953a8; external-validation
hash sha256:05602a8fb8c3d7a23827d52a2847b910135f7aa50c9622b0198f2e704299548c.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:306bd49533d46cb19fcf0627564636eda8ee8cc1a15790255742306b9d9a8c67. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POLYMER-COMPOSITES-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/polymer-composites>; snapshot
experiments/external_sources/materials/snapshots/nist-polymer-composites.html;
sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer
matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-mech-strength-hardness-001-authority-correspondence: predicted material-and-test-carrier__load-to-persistent-change-threshold__method-geometry-and-scale-retained__threshold-not-universal-constant__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-and-test-carrier__load-to-persistent-change-threshold__method-geometry-and-scale-retained__threshold-not-universal-constant__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered strength and hardness boundaries observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-and-test-carrier, load-to-persistent-change-threshold, method-geometry-and-scale-retained and threshold-not-universal-constant.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:8c586386b9a6b3354402a0c4f30ff03c2f27d866530dcb39d08757be0caad2e2; derivation seal
sha256:74b8b045099cf7851df7063a11da3c2a0e7870c7148bf04fb703f638f213e800; engine receipt
sha256:2db59637af7814e2711eba819d183bafc63ff7b259573bed972188dd7aaf2a08 at receipts/engine/model_admitted/SFT-MAT-MECH-STRENGTH-HARDNESS-001-2db59637af7814e2.json; empirical
validation sha256:0e7d03af5c7af07d3e7714193bd5d51bb707adabef48ef5c6d82ed7100c6176d;
measurement receipt
sha256:43ad4c5ed82a361d710e614767541948b7402e476eac6a6b773b321d8ee0ec09; isolation
sha256:cb858f2adbe1786193d85075804549f15206839b715fad6d7a87296ed263bf6c; custody
sha256:7605813ca3176eda76d4d8ce6a2512d1a68bb9f25cf017efef4d138523c4197f.

56. Fracture and toughness

Claim identity: SFT-MAT-MECH-FRACTURE-001

Question and exact theorem. Fracture is connected bond-path separation creating new retained boundary support; toughness is the declared resistance relation for crack initiation or advance with a complete work ledger.

```
crack-and-bond-network-carrier__advance-versus-resistance-relation__new-boundary-and-energy-ledger-retained__geometry-and-rate-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves crack-and-bond-network-carrier, advance-versus-resistance-relation, new-boundary-and-energy-ledger-retained and geometry-and-rate-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	crack-and-bond-network-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	advance-versus-resistance-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	new-boundary-and-energy-ledger-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	geometry-and-rate-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: crack-and-bond-network-carrier__advance-versus-resistance-relation__new-boundary-and-energy-ledger-retained__geometry-and-rate-bounded

d__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finalite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a5733f9eb6f9903bd7d45570055f974ecaedde3f68a6a62ed63ebde4607095d2.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:6a5d0a7bb0a8e2f66788209faf7630ad1686264d0c376bc6aa1c3c9c934ce1e4.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f27d462d208ffce1fd4f85a280d48e599aefab736bdf059196e25dc0714e5118.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6509c61f547b936474888b2417bcd4a5c72cfd3a0ae057432717257c1443a7e0.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:d6bc4e4588e8d5185f39670215ad2f0e681c18f43d64af3415da20f97af3e995; certificate

sha256:f13fcb4efc42df89e9e318ffd78f831ed7652201114029d301e13d7ace8e69e6; external-validation

hash sha256:a99696668e62b4f2d3dce17f8752af6a911915de7dffa7eb432db42293f12515.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:01b75c79f54df1e6372093d29ee1c45470f56ff393dbed86e238d8a2c2e91a74. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FATIGUE-FRACTURE-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot
experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html;
sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; scope:
processing-structure-property-performance, defects, fatigue and fracture.
- NIST-POLYMER-COMPOSITES-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/polymer-composites>; snapshot
experiments/external_sources/materials/snapshots/nist-polymer-composites.html;
sha256:0ca34811fd8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer
matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-mech-fracture-001-authority-correspondence: predicted crack-and-bond-network-carrier__advance-versus-resistance-relation__new-boundary-and-energy-ledger-retained__geometry-and-rate-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived crack-and-bond-network-carrier__advance-versus-resistance-relation__new-boundary-and-energy-ledger-retained__geometry-and-rate-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered fracture and toughness observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves crack-and-bond-network-carrier, advance-versus-resistance-relation, new-boundary-and-energy-ledger-retained and geometry-and-rate-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:b8c88ba37928e3ac9514e4b72395806d1d0a2cc60f604090a4fabd73f7b3bdbbc; derivation seal sha256:8b5df3b57c3581998968ea6f3337fb3d1ade12a540eaae72c4dc0b7e3bf1a88b; engine receipt sha256:66cd7e1bb5e969fb2cf9920411459685ec4b19cce6ddc54415e5bf1e449497cf at receipts/engine/model_admitted/SFT-MAT-MECH-FRACTURE-001-66cd7e1bb5e969fb.json; empirical validation sha256:31ff5bbedcc5575ff326542e7720d80181ab36505299b0828dd244e41ab283b0; measurement receipt sha256:f5fcd156b6b0806c04a181af2e91fb90172cfe31ca081e8d9fb7b3f2e123eabb; isolation sha256:90c60d61e5bac21634efcd59ce8c04fb5e939f5118bbfad4c168a6032a6678e1; custody sha256:1f63d814bc73955153dabccf74d08cbe10b6657f10c76bb37a2872dff423ae46.

57. Fatigue and creep

Claim identity: SFT-MAT-MECH-FATIGUE-CREEP-001

Question and exact theorem. Fatigue and creep are distinct history-dependent material processes: recurrence-counted damage under cyclic load and retained deformation under sustained load and time/temperature conditions.

history-labelled-material-carrier__cyclic-or-sustained-transition-accumulation__damage-defect-and-time-provenance__load-temperature-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves history-labelled-material-carrier, cyclic-or-sustained-transition-accumulation, damage-defect-and-time-provenance and load-temperature-time-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	history-labelled-material-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	cyclic-or-sustained-transition-accumulation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	damage-defect-and-time-provenance	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	load-temperature-time-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: history-labelled-material-carrier__cyclic-or-sustained-transition-accumulation__damage-defect-and-time-provenance__load-temperature-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed `true`.
- relation: the generated relation and organisation are separately retained; passed `true`.
- observation: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:138e417ec271e2d16689255fb807b76d4e11a863a53f75a4be29e4664f40edaa.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:f4a38be94309bdd54a09b8c098de0bf328e971aaf99abb5afda17dce0565aaa3.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5964c78e400b2cccbe8f604adf24d54cb4afaadca4cf3524164fcc0b10c9811e.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:96533e94cd35ecc95ced0615590e5f0915afcc9ae1e73b589bc9ad5feb6ff99.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:50718c3f7afe9f2f21ff7e49da32cb9694d31dceb34873026cf5ab117835fb04; certificate
sha256:b9dc76d661417daa718f8f2c9614ab248424d3786dd7fb08cfc03b9d7680ae6; external-validation
hash sha256:22152c9fcd29cc509c7d11b289060e20c3b52f613b2a4721bd6678fd3f58976a.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:30bae96d9d87e88765e2e13ea6f741387b1141dca008910c29f8318ec151b536. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FATIGUE-FRACTURE-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot
experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html;
sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; scope:
processing-structure-property-performance, defects, fatigue and fracture.
- NIST-POLYMER-COMPOSITES-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/polymer-composites>; snapshot
experiments/external_sources/materials/snapshots/nist-polymer-composites.html;
sha256:0ca34811fd8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer
matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-mech-fatigue-creep-001-authority-correspondence: predicted history-labelled-material-carrier__cyclic-or-sustained-transition-accumulation__damage-defect-and-time-provenance__load-temperature-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived history-labelled-material-carrier__cyclic-or-sustained-transition-accumulation__damage-defect-and-time-provenance__load-temperature-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-cl

osure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered fatigue and creep observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves history-labelled-material-carrier, cyclic-or-sustained-transition-accumulation, damage-defect-and-time-provenance and load-temperature-time-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:6259261c2f433280ac5b338213ad30a639c165bf120e029689cd7cc6b0f61e4b; derivation seal sha256:161e752c1fbcc5111fa1dca42857a94f4d8918ae581e3e23f81e30df3059a745; engine receipt sha256:0533e9af7333170230d829ce66828f6bc1982defb9c8fb311cf01e53196f13e8 at receipts/engine/model_admitted/SFT-MAT-MECH-FATIGUE-CREEP-001-0533e9af73331702.json; empirical validation sha256:b2728846e5e1ddf985715934a704a0d3e072542786645908f030c00d21547ca7a; measurement receipt sha256:645d29b7f2bb33fcad245033ea69bd2534dba99f4df7b9a74d6b153c77f1df80; isolation sha256:fc65a26867d6ef55044a2f9e33afc6460d9a90381f191f27c46fe0a05893a6f1; custody sha256:a2f03701274180403fafd17c220b8ad21dad330cab01ff5cb46bd025c103928a.

17. Thermal Magnetic Optical

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

58. Heat capacity and excitation storage

Claim identity: SFT-MAT-THERM-HEAT-CAPACITY-001

Question and exact theorem. Material heat capacity is the exact relation between supplied thermal carrier change and the resulting accessible excitation-support change at declared conditions.

material-excitation-support__energy-to-state-count-change__condition-and-phase-retained__temperature-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves material-excitation-support, energy-to-state-count-change, condition-and-phase-retained and temperature-window-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	material-excitation-support	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	energy-to-state-count-change	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	condition-and-phase-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	temperature-window-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: material-excitation-support__energy-to-state-count-change__condition-and-phase-retained__temperature-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.

- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9b1c92f07930722df9def1146b48afb9053a331fe13649f8dbc811937f62bda5.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:f5a5463e58ef4db286448b3ccf1df2c4a0f2f2c5934e286ad3c5c109878ac038.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a6b3bf01252235db636981ba4ca9b96662f720f8b2f46ea7bfc07d9beb3bdc9.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:09b10d18c04b2ebd6f0139afdc0e36b99291ceb2381d3b1485aa6dbff83644ee.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:c85a958591bc27b8bea1709497f13281fc169c363ae08e5f0d3c2d231bc34257; certificate
sha256:3b8c30953cea0c023f330da888aa7ee92a4162a93e01cd00fa210aa29c8a15ad; external-validation
hash sha256:609c83bbdd41d06080381f919f9aac4e58eb32bf48a496e68e78b41ed213737c.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:962425c0c7e89286623b46cb21a73a7bfd40a4e3549f825ec5937dd421beca5d. Target content
opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.htm
1; sha256:4db74747d716cccf34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope:
thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- `sft-mat-therm-heat-capacity-001-authority-correspondence`: predicted material-excitation-support__energy-to-state-count-change__condition-and-phase-retained__temperature-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-excitation-support__energy-to-state-count-change__condition-and-phase-retained__temperature-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered heat capacity and excitation storage observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor

- no numerical zero, negative, irrational, imaginary or floating point quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves material-excitation-support, energy-to-state-count-change, condition-and-phase-retained and temperature-window-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:e42ae574ea210e58e7db4db2b4dea63b4c2a952a2c3003d1494486b831a51cce; derivation seal sha256:7abc09c00bae8245182e239280d27c4e0fcd5752a0f60ddf6adc1531bfefbc332; engine receipt sha256:9a919059d9d03fcdc366ad97dd95eed602d6df50b8db4d49e3acb1d492dcf1d8 at receipts/engine/model_admitted/SFT-MAT-THERM-HEAT-CAPACITY-001-9a919059d9d03fcd.json; empirical validation sha256:6e1bf64a4d135d263d2f6e1a0a7ce530df19e721915617a6ed4e4cc56060a319; measurement receipt sha256:5c348ebe8e6ee46f9ac074932fb49e4af0501d044da5826744228e4d7d8afc9e; isolation sha256:e3b1e98bbbbb1fa295e5694748d0a260d5e9516397fdad151680d860c0d743008; custody sha256:cb2428cec3ca8bfac3b1cd7e5da284e0d4f80482444e02eb6f6b01292f22deac.

59. Thermal conduction

Claim identity: SFT-MAT-THERM-CONDUCTION-001

Question and exact theorem. Thermal conduction is counted adjacent transfer of retained energy labels through connected material support with complete source, boundary and scattering provenance.

energy-labelled-cell-network__adjacent-energy-transfer-count__total-energy-ledger-retained__gradient-time-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001

- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves energy-labelled-cell-network, adjacent-energy-transfer-count, total-energy-ledger-retained and gradient-time-support-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	energy-labelled-cell-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	adjacent-energy-transfer-count	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	total-energy-ledger-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	gradient-time-support-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: energy-labelled-cell-network__adjacent-energy-transfer-count__total-energy-ledger-retained__gradient-time-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:19069236c3c4b3a7dfaaab7b12bbef6c542b7fa84585b15712bd64dabb87e580.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

sha256:e830330106bc942c9651b68e4685a201d4266513d442e71e66fd2321a9463b3e.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:14fe4fe4b473610bb862eb6beff2dfd07c036ad39ee6a12848a26d4e24a8af58.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7a817c1177ec039fa173c7e2f4b2458c3bc9a478948cec1e1e7b45539ae54486.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:e6979253f0608812554abf41f81ef0099f77c5cc4c214da12304a1b55437d891; certificate
sha256:8e9fd289da0b25f75129f0b35c2ace40504a81b64a66962a0b25cfa6e7386d8e; external-validation
hash sha256:31ba40a4f94cd387d02921b23d30d4c580558c96bdbfe0ef2e03fce22fcc583a.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:f0aa7c18ee1acfc7349c5163388a2ca04b1c881e3486769bc42c642df190e9803. Target content
opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.htm
l; sha256:4db74747d716ccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope:
thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- sft-mat-therm-conduction-001-authority-correspondence: predicted energy-labelled-cell-network__adjacent-energy-transfer-count__total-energy-ledger-retained__gradient-time-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived energy-labelled-cell-network__adjacent-energy-transfer-count__total-energy-ledger-retained__gradient-time-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered thermal conduction observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves energy-labelled-cell-network, adjacent-energy-transfer-count, total-energy-ledger-retained and gradient-time-support-bounded.

Immutable evidence identities. Pre-source branch seal
experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest

sha256:e897e584b3dd80bc4168883855f14430c86cd1a0d62bda4ad2814b4c6ebf8554; derivation seal
 sha256:1fdc4a1db8b717fa6573da8c1c79cf3d21145d4cc07736eb12ed7fd13e1e50c5; engine receipt
 sha256:1bc2b97d196c9750d41d454a0aa47fa7b36d142e108da02aeba0e7fed8134e33 at
 receipts/engine/model_admitted/SFT-MAT-THERM-CONDUCTION-001-1bc2b97d196c9750.json;
 empirical validation sha256:e0c3e7778a79d6e1256434b13cb64fcc8c903bf2cadef8c62fc60890f47bd1d1;
 measurement receipt
 sha256:afa71e5c2000da0945d1c1cc12bcaa435c411f13ac286cfadcfb847b0cc71a2a; isolation
 sha256:e0ae991f27d27091ef5c4f2d234f76b510a90c4d02cd87451b0cc9a7af342af7; custody
 sha256:be4f7ae87af0abe604d3a65bbcb6ace9caf68fc7941833d5e2d5b4d44144779d.

60. Thermal expansion

Claim identity: SFT-MAT-THERM-EXPANSION-001

Question and exact theorem. Thermal expansion is the condition-bounded change of recurrent constituent separation with excitation while material identity and complete adjacency provenance remain retained.

bond-separation-network__excitation-conditioned-separation-change__material-identity-retained__direction-and-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bond-separation-network, excitation-conditioned-separation-change, material-identity-retained and direction-and-temperature-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	bond-separation-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	excitation-conditioned-separation-change	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	material-identity-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	direction-and-temperature-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `bond-separation-network__excitation-conditioned-separation-change__material-identity-retained__direction-and-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:0e53cee45fef353af8c1fc3a048db16697e59a4d1945eea47b3e9a9ae4ae9c0c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:28f81c0bd83923a6393f7cf96abef9aa601ace13fbe9c2e13d8a95e762b6775e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:35b43f42468a1c1e771a8aa75e61ee96f3241e1249bfb8bfde861a74df429ab0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:016edca3decdf384605400b1fcfa0e140add94b17145adf5269e163906f9311f`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:2430b2dfd7b2daa237185f98bc1c08842018b04eedfc432b4aca14a108bc9c25; certificate sha256:12cf1d124a03f9181174b282315f745c3854f4129d0540c209fd60966356acc3; external-validation hash sha256:f43b131aa11d8a1e83be60efe50ca5486c5336a52fb45526c6d32755c4bf3a72.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:51a71476779069d408c1bdc3413b8c879c8a17f2e40b35bbd6caca235d49ee36. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-MULTISCALE-MATERIALS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-therm-expansion-001-authority-correspondence: predicted bond-separation-network__excitation-conditioned-separation-change__material-identity-retained__direction-and-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bond-separation-network__excitation-conditioned-separation-change__material-identity-retained__direction-and-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered thermal expansion observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bond-separation-network, excitation-conditioned-separation-change, material-identity-retained and direction-and-temperature-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:d08e166df831402c873d497a2b6ebe99c4557b02f7a6686f6bbcf42d229df9c8; derivation seal sha256:0214f5934c9a967a9aa0ac4ddcde7068fc7560b1c26f9ea683d9c5f23a5cc25a; engine receipt sha256:e43221596aae9f61e140436942ee5dbb1625f0bfa5a520f34f217d780558cbdb at receipts/engine/model_admitted/SFT-MAT-THERM-EXPANSION-001-e43221596aae9f61.json; empirical validation sha256:be6cdeb54ff17afaa198f0414e8ea3404907e36e1b79a041e23f13f399cec260; measurement receipt sha256:b871d56ca42a2531ba7e55209ef6306482b6aeb06c6825d06320164217ab5345; isolation

sha256:a299c268357a8d8afccf08b9285f38ec3440b0a33f6dc2497ec2a36ad2b1d44f; custody
 sha256:3f767b087bfbcc3375d204a69b2fba54d2c14a7282e893c1d63f95f35b76e021.

61. Ferromagnetic order

Claim identity: SFT-MAT-MAG-FERROMAGNETISM-001

Question and exact theorem. Ferromagnetism is connected recurrence of aligned local held-moment orientations producing a retained macroscopic orientation class.

local-moment-network__aligned-shared-orientation-recurrence__connected-net-moment-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves local-moment-network, aligned-shared-orientation-recurrence, connected-net-moment-retained and ordering-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	local-moment-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	aligned-shared-orientation-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	connected-net-moment-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	ordering-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: local-moment-network__aligned-shared-orientation-recurrence__connected-net-moment-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:18c6e7d85f2e9146578d2645a8213fdbf355c75e79e05273dcfc9a89bdc6c8f7.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:09dd1edb95f74fb641872c23fef4e458b8b03d180b58c83bcf96284e7c2e84d9.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:22995d5074cfb5ecda427c7205df471e65beb5216cf1f63a3073a4d46746ab5b.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:290a1b4e2cb6f280907d51338d128896ce8ed6e11c6cd49d332f1bbbccb0340c.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:d0db4c43c76befd8e81f178bb7d60c32918904c030aec387054fd7cac0d279d3; certificate
sha256:b6ccb098a72046791e3b63c766f6baafd7006b7b10f8a30543b374f3f2a1b383; external-validation
hash sha256:dd32b24dd4b45270d8a255cfbbcbd691505465416487c457ea4a0d8c47875be8.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:8f81d89cf2d73a6809d674ceca2c208912e0f9998eaa5252837c8d3ce4fd2a3e. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-mag-ferromagnetism-001-authority-correspondence: predicted local-moment-network__aligned-shared-orientation-recurrence__connected-net-moment-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived local-moment-network__aligned-shared-orientation-recurrence__connected-net-moment-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered ferromagnetic order observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves local-moment-network, aligned-shared-orientation-recurrence, connected-net-moment-retained and ordering-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:874d335c33a53a6df0256d9f5e149acc452d0743a4f57132fbbad96958d83b63; derivation seal sha256:ef659703b1b78755c811fadd47b2d4d01767e0be0db9aa4ed6357fb2e6738878; engine receipt sha256:8702b789cbad4cb166e5003ed111ac105b9908c3b921c7694876a7542a47c9ab at receipts/engine/model_admitted/SFT-MAT-MAG-FERROMAGNETISM-001-8702b789cbad4cb1.json; empirical validation sha256:81ddf6d5cd8f35488a8fd9a766f2aa59bd53d2cfce1aef09c382fea4d32521be; measurement receipt sha256:711bb821fc4020f451a199fa3d1e50db79292aec3b2096e80b7f84bc7be5ba0c; isolation sha256:706c62f32b92289aebef28bb5c38f54efd56a53938d566f43efca10e89f4062; custody sha256:e06f5ee0a0adc7afe327d4983e262f49654c4374c76cc69517f35ffb41a20a5f.

62. Antiferromagnetic order

Claim identity: SFT-MAT-MAG-ANTIFERROMAGNETISM-001

Question and exact theorem. Antiferromagnetism is stable opposed recurrence on distinguishable sublattices that retains local order while the complete macroscopic moment distinction closes.

bipartite-moment-network__opposed-sublattice-recurrence__local-order-global-closure-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bipartite-moment-network, opposed-sublattice-recurrence, local-order-global-closure-retained and ordering-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answered-only	bipartite-moment-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	opposed-sublattice-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	local-order-global-closure-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	ordering-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: bipartite-moment-network__opposed-sublattice-recurrence__local-order-global-closure-retained__ordering-condition-bounded__complete-st

ate-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:3a97cd3e05edbee53ee54b4478bab16cf055ce518e721868e07f6c11502ba8cd.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d91fbf826b29dfc56d95eaf525c93c5a097a496f66fd54bdc07270e4d97c1709.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:21fc3784752313c95d471cc7f4c37eba8dc481651af93a44b97ae34e2574b7f9.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:00d0d75dd210b3b851ee3af7c6b1b5e9382df37025806f700f9e3c4a914ab1b2.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:554a049abd839fa53066a0d078721f9d32d1e2da2ab50d6bad079166de236149; certificate

sha256:5bcf84137ee43fbc7eaa1e6c5cb69622b45205f73a6e2c53f144f0c95bcab8ce; external-validation

hash sha256:a46609ffce4544872f8f33cad521d02fd03fb08c46966d098291c136698c7eca.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 3 claim-specific source discriminators were required; their ordered identity is

sha256:e52715234c5d0275b83377bae9ec6c31e571fc23fe49d47cd26aac633767439c. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-ANTIFERROMAGNETIC-COUPLING-2026-07-24 - National Institute of Standards and Technology;
https://www.nist.gov/ncnr/acns-2020-tutorial-ii-practical-approach-fitting-neutron-reflectometry-data/understanding_snapshot_experiments/external_sources/materials/snapshots/nist-antiferromagnetic-coupling.html; sha256:b8ce5105622c02b84bffe643693c05c199df57beae79bd9c5e29f26c4360e85e;
scope: polarized-neutron measurement of magnetic structure and antiferromagnetic coupling.

Comparison and custody records:

- sft-mat-mag-antiferromagnetism-001-authority-correspondence: predicted bipartite-moment-network__opposed-sublattice-ecurrence__local-order-global-closure-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bipartite-moment-network

`__opposed-sublattice-recurrence__local-order-global-closure-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True`

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered antiferromagnetic order observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bipartite-moment-network, opposed-sublattice-recurrence, local-order-global-closure-retained and ordering-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:3df466cadcac208382ade4c4777ac52b418e0c3e01286abe21cd23a00899994d; derivation seal sha256:66b2d9f5998a96ae9e7a4b0f607b4b3701336de1c18415dd2cfa78eaff6de448; engine receipt sha256:4e024776e6572a09e72fd47738e804b731ccffff0685efb0993023aecafc15c7 at receipts/engine/model_admitted/SFT-MAT-MAG-ANTIFERROMAGNETISM-001-4e024776e6572a09.json; empirical validation sha256:9c06b2a0f59c2933efdaa85f3a3f7107cdfb23da035952b54ebbb1571c31b065; measurement receipt sha256:1144769f267e47cca17f0dc4ddd226220357394daaece0f0da693ac5622311f1; isolation sha256:324097c4049c4005726db64e2a93937542c96b9be63d8c95bea908389c63d4e8; custody sha256:c52d305b11f8dbf80db49ca64d8ce900ce41725947b43d8721e896ad58ea6163.

63. Dielectric polarization

Claim identity: SFT-MAT-DIEL-POLARIZATION-001

Question and exact theorem. Dielectric polarization is a retained field-conditioned redistribution or orientation of bound held-charge labels with complete conservation and relaxation records.

`bound-charge-orientation-carrier__applied-field-displacement-relation__charge-conservation-and-relaxation-retained__frequency-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bound-charge-orientation-carrier, applied-field-displacement-relation, charge-conservation-and-relaxation-retained and frequency-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	bound-charge-orientation-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	applied-field-displacement-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	charge-conservation-and-relaxation-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	frequency-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: bound-charge-orientation-carrier__applied-field-displacement-relation__charge-conservation-and-relaxation-retained__frequency-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.

- **observation**: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- **false_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ab4c971eddd4a0592c52799f300cbb68ccdd19e0d17929704c030e932e3147ef.
- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:f811d9b97206cb817d188a47fc41c6ee4857a456316be6a1d8a144544071bcd3.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e5b70b12ddb119fd8aab8971c8adb87e5830aadd0fa9b644bfc648aa356cbdf8.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:419fd084389f321aa44b601dd00e318fb0647694ec700f5298800692c0e802c1.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:00a9ba701d96966b56fd31c7b4d206d442ec6d652fd7453700c985196d1c92b6; certificate

sha256:59d7757274ab8ecbe48f4c00bc9b612f3d32ff9fa583f3cd2803d0fed7eb7482; external-validation hash sha256:53450986c1b2ebc93c03ac06621f5d9b2598b3048bfeec6730e0dda3f02dcadf.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:a96625233799c6676585da9b9389b00e18e4f90a9ec7b3b86d06d2b1c40ced6b. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-diel-polarization-001-authority-correspondence: predicted bound-charge-orientation-carrier__applied-field-displacement-relation__charge-conservation-and-relaxation-retained__frequency-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bound-charge-orientation-carrier__applied-field-displacement-relation__charge-conservation-and-relaxation-retained__frequency-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered dielectric polarization observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor

- no numerical zero, negative, irrational, imaginary or floating point quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bound-charge-orientation-carrier, applied-field-displacement-relation, charge-conservation-and-relaxation-retained and frequency-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:1498859c348dca8e370798bb01a8832aea7e4e8d376b158d788bd6156378dc0e; derivation seal sha256:0a118061761c83630046d49ac13b24f8877554e4f646101aa11e5dba5b8f1420; engine receipt sha256:f432cb7b60b503a89a22333e6942725b659572fd12da81a606267858f3ac637d at receipts/engine/model_admitted/SFT-MAT-DIEL-POLARIZATION-001-f432cb7b60b503a8.json; empirical validation sha256:606adb7ecb391af5f1f37d0122a6ca840f6a4b532ea334cd0fb1c4205cad2aa; measurement receipt sha256:e912dcb76211979f7aca50f6f3348c37518499fc73f6cd86ec3c8aad480477c; isolation sha256:3abe9bbad740c8ab4999df5dbec74fe762361aee816a6c135941212a29e44b2c; custody sha256:8344d6447b2b2cb16be34b7690434539f57962edbde04bd804c4b811d17e8539.

64. Optical propagation and refractive response

Claim identity: SFT-MAT-OPT-REFRACTIVE-001

Question and exact theorem. Material refractive response is the source-bound phase/delay relation produced by coherent probe coupling to accessible material excitations while propagation, loss and dispersion records remain separate.

probe-and-material-excitation-carrier__coherent-delay-and-transfer-relation__frequency-phase-and-loss-retained__spectral-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001

- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves probe-and-material-excitation-carrier, coherent-delay-and-transfer-relation, frequency-phase-and-loss-retained and spectral-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	probe-and-material-excitation-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	coherent-delay-and-transfer-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	frequency-phase-and-loss-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	spectral-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: probe-and-material-excitation-carrier__coherent-delay-and-transfer-relation__frequency-phase-and-loss-retained__spectral-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9c630b0915ecb27ecd4a8e9e3a7467474e98d990d0618c7a85c74363f19b6bc7.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:9ae669c1738af532adaf47a7e7ad4f3d7043304c6043f5d358a4a2940af80005.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:73d971daac95bf3c58df6f424c3181e678ef094a14cfca717ca19c769d266212.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b5e103cfd88598d9b7655978391197ab5224bcc9d005792573d8a058c49d84c4.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:65dc955c5bc0b935e97be9bc2733c36b012a32a1a1e56f9c7c9d67b631ab63ef; certificate
sha256:191167b5a78855912ddc9e723c619c26dce388f41e3e0573f98e36c081b53714; external-validation
hash sha256:7338f1b3d4505446488721394e707d1a2d078e07f0c0fd0dcf8a817f8ba2905d.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:d8b7289d06e63b982d20adbe30509ea26d9351d73e84dbb6f49daa777954288e. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-OPTICAL-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-optical-materials.html;
sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of
refraction, absorption, reflectance, dielectric response and photonic crystals.

Comparison and custody records:

- `sft-mat-opt-refractive-001-authority-correspondence`: predicted probe-and-material-excitation-carrier__coherent-delay-and-transfer-relation__frequency-phase-and-loss-retained__spectral-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived probe-and-material-excitation-carrier__coherent-delay-and-transfer-relation__frequency-phase-and-loss-retained__spectral-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered optical propagation and refractive response observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves probe-and-material-excitation-carrier, coherent-delay-and-transfer-relation, frequency-phase-and-loss-retained and spectral-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:6338ee48ea6cf26874c7b32eb792944b85ed28d91cbcf78412f4f07738f8a2f3; derivation seal sha256:1b93339c2b96508f0faecd14fa3af55d3369e93d5a2cf3410c3b8cffeecf3243a; engine receipt sha256:82335b4d53a29fa158d4484e54572994b56b5dda246c8654034cbdc01f485d99 at receipts/engine/model_admitted/SFT-MAT-OPT-REFRACTIVE-001-82335b4d53a29fa1.json; empirical validation sha256:a09d090deef51116d29b164006bf17e5a792eac1e4af65e9ca63528a3ab350e1; measurement receipt sha256:f2305f1463ef3636441aa52e0a33cc956f137b10e879102d6f956a3ca8c464ec; isolation sha256:029f56e5c4af77cb4977858abb7ad0a734f216f21e388c340a7c094da910c654; custody sha256:ddcc328651656989f385b50503615b950a835d1186cc1970f2953b7db45a12e7.

65. Material phase transition

Claim identity: SFT-MAT-PHASE-TRANSITION-001

Question and exact theorem. A material phase transition is a provenance-retaining change between stable macro-recurrence and symmetry classes with explicit conditions, path and hysteresis boundary.

initial-final-phase-carrier__stable-macro-fibre-change__symmetry-content-and-path-
ledger__condition-and-hysteresis-bounded__complete-state-transition-boundary-trace
__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves initial-final-phase-carrier, stable-macro-fibre-change, symmetry-content-and-path-ledger and condition-and-hysteresis-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	initial-final-phase-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	stable-macro-fibre-change	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	symmetry-content-and-path-ledger	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	condition-and-hysteresis-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `initial-final-phase-carrier__stable-macro-fibre-change__symmetry-content-and-path-ledger__condition-and-hysteresis-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:699bb0538a254745ef67cc361ac8f4844e8c1fa2538837ac5938e52053255541.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:3192cf970f56809f280e80c2c9820e5498273e7e42e1999057f39d8970014427.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e183813b92b1fe15c3e6eccee915107ea2ab934872dcf80d50f64eea683505b9.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:406f3613f24df373d9f299a2659d9ad0e2bcf07a282d0ba2cebccb1230a0620b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:a5971eb324e6f0a7959b7a16ebae120a132038c909c3c221b6c9a779eef1c7c; certificate sha256:c7ca23959f208aede92422515466358ab2b11bb5ecbefd73d808513ca2d5dc7a; external-validation hash sha256:cafe572aaadaa8b0658384852fee24d1d662945aa2b0d6c7931694fe40e7de38.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256:e20ae266b0ec1dd474ba078d282cf54d525ff318993fa9787c5b9fd426c2c8a6. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-MULTISCALE-MATERIALS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.
- NIST-SUPERCONDUCTING-FLUX-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-flux.html; sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; scope: zero resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- sft-mat-phase-transition-001-authority-correspondence: predicted initial-final-phase-carrier__stable-macro-fibre-change__symmetry-content-and-path-ledger__condition-and-hysteresis-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived initial-final-phase-carrier__stable-macro-fibre-change__symmetry-content-and-path-ledger__condition-and-hysteresis-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered material phase transition observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves initial-final-phase-carrier, stable-macro-fibre-change, symmetry-content-and-path-ledger and condition-and-hysteresis-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:0297cd0782de90adb0d271e4db03d88413a6636799f16e3fbdcf3b5d84617c4; derivation seal sha256:c219d7cd41090ebb065ba94181b6f6cdea11e507feac6b44e0424b3af3e26e8b; engine receipt sha256:c40ad4320e96682f79e7b338df47728fd53e8df1455276c8a6673294e60ea745 at

receipts/engine/model_admitted/SFT-MAT-PHASE-TRANSITION-001-c40ad4320e96682f.json;
 empirical validation sha256:5cefdebc7a29e14018d3791b7023149b6671cfa8c81121bee0a4e4667f743cd9;
 measurement receipt
 sha256:f97aa41aceeec4f79873517d7940d2804059220eee142435cde997ab2b1fe951; isolation
 sha256:fca5b15e153357ae3fcc8f912e9b78e0da3ec9e98cfefb1944b5528738c27536c; custody
 sha256:fbfbbaa82bf8ce93b146cb1a97a3e85e00bc21ab4c4f20e2516ee0ccdf94433a.

66. Ferrimagnetic sublattice order

Claim identity: SFT-MAT-MAG-FERRIMAGNETISM-002

Question and exact theorem. Opposed distinguishable magnetic sublattices with unequal positive counted moment support force a nonempty partial bulk moment class distinct from aligned and equal-opposed order.

two-distinguishable-magnetic-sublattice-carrier__opposed-unequal-positive-moment-r
 elation__local-opposition-and-nonempty-bulk-gap-retained__partial-net-moment-order
 -class__complete-state-transition-boundary-trace__root-bound-forward-forcing__posi
 tive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor thermal_magnetic_optical grammar preserving two-distinguishable-magnetic-sublattice-carrier, opposed-unequal-positive-moment-relation, local-opposition-and-nonempty-bulk-gap-retained and partial-net-moment-order-class.

Shared claim-record clause MAT-S003 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	two-distinguishable-magnetic-sublattice-carrier	The material carrier cannot be reconstructed. The complete generated carrier is retained.
relation	relation-imported-fitted-or-erased	opposed-unequal-positive-moment-relation	An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.
organization	organization-collapse	local-opposition-and-nonempty-bulk-gap-retained	Required material distinctions are merged. Every required organisation distinction remains held.
observation	observation-boundary-unrecorded	partial-net-moment-order-class	The observation class is unidentified. The declared observation boundary is retained.
record	result-without-transition-trace	complete-state-transition-boundary-trace	An answer without its transition trace is not auditable. The complete trace is retained.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.
generality	finite-answer-lookup	positive-finite-successor-closure	A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.

Unique survivor, base and successor. Sole survivor: two-distinguishable-magnetic-sublattice-carrier__opposed-unequal-positive-moment-relation__local-opposition-and-nonempty-bulk-gap-retained__partial-net-moment-order-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: The first complete carrier retains its relation, organisation, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- unequal-opposition: five opposed to three leaves exactly two in a held orientation; passed true.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:7a1f40900a717d2d47abefffd7fa3f938bd727cf128299586801e6773007f9fd.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:f969091576a41a221b052aa295f4b3fd36204a3833515368554ec03bffdabc86f.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:10a38363fea4202d9bea4baacaale3fae35f20c66884ab7e5ea925f361662f5f.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:56c4c2eaa33d2a9aa6276bc9282f40ae508181277a751a908423ef0d6319be2f.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:62401af2379325f04b5f3a09888974f68d2f7a8fabd1d5c319f04b0774216657; certificate
sha256:c6e26e75582fe70ec9aae72fb463dce2692d02ccef11d704ccfd7263bb8723cb; external-validation
hash sha256:c6e26e75582fe70ec9aae72fb463dce2692d02ccef11d704ccfd7263bb8723cb.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:591cbb5129d22ba5ed720fac188ac2eb94e59e4b1a97b279c6f0e541ecea06e2. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FERRIMAGNETIC-SUBLATTICES-2026-07-27 - National Institute of Standards and Technology; [https://www.nist.gov/publications/thickness-dependent-magnetism-variation-ferrimagnetic-rare-earthtransition-metal-fe1;snapshot experiments/external_sources/materials/snapshots/nist-ferrimagnetism-2026-07-27.html](https://www.nist.gov/publications/thickness-dependent-magnetism-variation-ferrimagnetic-rare-earthtransition-metal-fe1;snapshot%20experiments/external_sources/materials/snapshots/nist-ferrimagnetism-2026-07-27.html); sha256:685872920e5663b2ec71636b953d0e7ddec3d1cb679fb4f17bb94b6d6d811795; scope: post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Shared claim-record clause MAT-S007 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor thermal_magnetic_optical grammar preserving two-distinguishable-magnetic-sublattice-carrier, opposed-unequal-positive-moment-relation, local-opposition-and-nonempty-bulk-gap-retained and partial-net-moment-order-class.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_v1_v2_successor_complete_pre_source.json;
source manifest sha256:181be2cb81d02e04eff8c616cafaa4f2344670bc2097a787aa7f2e0d52281817;
derivation seal sha256:cf87fe52b9ef52f74ba1c74afee83891689fbfa06d7655dc65e6287aa5658602;
engine receipt sha256:42f94c9f6bfbf33931ff8fdc7603dde8134e2b6afb67e2a0b22d96942f0a29a7 at
receipts/engine/model_admitted/SFT-MAT-MAG-FERRIMAGNETISM-002-42f94c9f6bfbf339.json;
empirical validation sha256:43fc76865edc7dcf64c02c3f89d6a339fb9b19b282f6ec6e0f9ba982122cf8eb;
measurement receipt
sha256:0726f5ccf3befd94cc3579c24513bc880f1b92956c909944b8b101170f6d6d05; isolation
sha256:fc95b2f9f5452cf386454313a923348361f4da6a3d6efe5293c26ded3d395984; custody
sha256:933c669680d48554da6586e68932bbc12e149d53dbaf06e28bbbdcca913901ef.

18. Material Classes Bulk

Shared claim-record clause MAT-S008 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

67. Metallic material organisation

Claim identity: SFT-MAT-CLASS-METAL-001

Question and exact theorem. A metallic material is a connected atomic organization with delocalized carrier access across multiple constituent cells and no declared transport gap at its operating conditions.

connected-atomic-network__delocalized-carrier-access__translation-and-bonding-retained__condition-bounded-metal-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves connected-atomic-network, delocalized-carrier-access, translation-and-bonding-retained and condition-bounded-metal-class.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	connected-atomic-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	delocalized-carrier-access	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	translation-and-bonding-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	condition-bounded-metal-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `connected-atomic-network__delocalized-carrier-access__translation-and-bonding-retained__condition-bounded-metal-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 75257e1c7d9eaa9bff969b6d2330e1265d2a0f9bba87c9aece8e402510ab3988.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256: 1b001bf09b7d2091d3d1b8bbeb596db998ff93fd4e7203badeebe99ab867c5b4.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: c7b74d8a2285a0aab500e235bd3898e00a6c65155c152646c59586906e97f402.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: cc6ff66be0902ac0ba61c87beee88d6283725c77049ca60aecc1a9358568fb3b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: bb4ef02ce0fb08afdd526d8b4a3fc086373575480832e1d9cfb5b462bb5f6bff; certificate

sha256: 2458dd1b4f8560378cf749959b1476d3cc8b0e8a8e003582c2fbb109136fbb6e; external-validation hash sha256: 61ea3121a344907c6519355dde4ec7510dcf6362f3a0fda5897da2741b33cb77.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256: 87fd1a3ef3fb7ef8871ddb7ecd0d074dda38732b6f59dc8a0acfa4dafb814bd1. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SOLIDIFICATION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/publications/solidification-0>; snapshot
experiments/external_sources/materials/snapshots/nist-solidification.html;
sha256: d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c; scope:
solidification transport, nucleation, growth, interfaces, segregation and processing.

Comparison and custody records:

- `sft-mat-class-metal-001-authority-correspondence`: predicted `connected-atomic-network__delocalized-carrier-access__translation-and-bonding-retained__condition-bounded-metal-class__complete-state-transition-boundary-trace__root-bound-for`

ward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived connected-atomic-network__delocalized-carrier-access__translation-and-bonding-retained__condition-bounded-metal-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered metallic material organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves connected-atomic-network, delocalized-carrier-access, translation-and-bonding-retained and condition-bounded-metal-class.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:5f02bd1b0f32f5693c1791781165d3bcb38fc8c49ff0ec01254dc518ced91df6; derivation seal sha256:af475c9901883694ec27d4a4693e3ca1eb56373d75c42e6c0b073c66e977bced; engine receipt sha256:ff88492d77ed22ed6106d1f981e4d70142431b1bfe9aeaa7cc5a6f26966951e2 at receipts/engine/model_admitted/SFT-MAT-CLASS-METAL-001-ff88492d77ed22ed.json; empirical validation sha256:c5422a5c61e616c84db792a559545f816ea6a3d6dfce8456af9a560e2d6c8588; measurement receipt sha256:84f09dc3af6ef8df70c69226be6eec901246c4e203c597a51b981db60890fc1f; isolation sha256:3ec6b3346deb1532d82db336b639a7be66c91b7b1c4b4146bcf4d40108b37540; custody sha256:eb8356b4e42e5e256d654f54b378fdeed0f9fb5059c09c23e394e2e3d4102391.

68. Ceramic material organisation

Claim identity: SFT-MAT-CLASS-CERAMIC-001

Question and exact theorem. A ceramic is an inorganic nonmetallic material whose bulk carrier is a localized ionic/covalent network with retained phase, porosity and grain/interface organization.

inorganic-network-carrier__localized-ionic-covalent-joining__phase-and-grain-structure-retained__processing-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves inorganic-network-carrier, localized-ionic-covalent-joining, phase-and-grain-structure-retained and processing-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	inorganic-network-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	localized-ionic-covalent-joining	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	phase-and-grain-structure-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	processing-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: inorganic-network-carrier__localized-ionic-covalent-joining__phase-and-grain-structure-retained__processing-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.

- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:76deb008a987f16ea7cf2d23813da1ce5b61f9665f182459ad93958acd125655.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:62d814937e216c438b3031aa1287c27a1c099fc17fb38a779a32a4c0abda3867.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:52134d6b9c5c728526c88b12b66102baf449f59f6b8deab50eda64a5d8430d18.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b1df93ebe9661b7a1504c2c444af7ff8ce24fd932d309719a4a077d0effcf610.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:48db99ea691a8154b6bd4b946ecd20cd7d61481745e9e73e611241b91eb35d41; certificate

sha256:9872a8af3e2b80be8d45b13f6f0d4175c23baf4f58eacdedd1afdee08f3fa44e; external-validation hash sha256:fdb0f31db398bd99134453418f3830811ffc0ef4924dd1e4b3e2d9d5d24cb566.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:01eb6862c745c23944528b05975766210e00b897c0ef858c5cccb000421b24cf. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-CERAMIC-ADDITIVE-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/additive-manufacturing-ceramics>; snapshot experiments/external_sources/materials/snapshots/nist-ceramic-additive.html; sha256:540ef912a827c1834a113352ac0305c7e8d25a76283dcc1bcbab48f0b826a3e9; scope: ceramic feedstock, defects, sintering, densification, phases and microstructure.

Comparison and custody records:

- `sft-mat-class-ceramic-001-authority-correspondence`: predicted inorganic-network-carrier__localized-ionic-covalent-joining__phase-and-grain-structure-retained__processing-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived inorganic-network-carrier__localized-ionic-covalent-joining__phase-and-grain-structure-retained__processing-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered ceramic material organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor

- no numerical zero, negative, irrational, imaginary or floating point quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves inorganic-network-carrier, localized-ionic-covalent-joining, phase-and-grain-structure-retained and processing-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:36cccb1dd2e5e1628e3512f5fa6df179b27bb82e1499c77dcc1baca99f0d6238; derivation seal sha256:369e843e6bc60b72015a21594789a04f78935ae133eed27afa02dad43c23c0ea; engine receipt sha256:c8d2a975402210330273a38078b9cb98953aa151f947aef809e2a6877aa82749 at receipts/engine/model_admitted/SFT-MAT-CLASS-CERAMIC-001-c8d2a97540221033.json; empirical validation sha256:05b56e875a7c653dbdeb13e30878e702946a5d6c85e3e47da657abf31287437e; measurement receipt sha256:d5391cef95ca7f7c7ab19f64f4b6eab3ddcd8ade6de2050843457b42c1f1bb85; isolation sha256:34ae11265cd7a19fde0a5f163f473131a9d88c056516a17c01e0a67aa9493a6f; custody sha256:c65e9e11483fa4575b83fe65d209fea3569ede6cbad5ccf6048ed0aed39d9905.

69. Polymeric material organisation

Claim identity: SFT-MAT-CLASS-POLYMER-001

Question and exact theorem. A polymeric material is a population of covalently recurrent chain carriers with explicit repeat identity, sequence, length/branching distribution and interchain organization.

repeated-unit-chain-carrier__covalent-backbone-recurrence__sequence-length-branching-and-entanglement__distribution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001

- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves repeated-unit-chain-carrier, covalent-backbone-recurrence, sequence-length-branching-and-entanglement and distribution-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	repeated-unit-chain-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	covalent-backbone-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	sequence-length-branching-and-entanglement	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	distribution-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: repeated-unit-chain-carrier__covalent-backbone-recurrence__sequence-length-branching-and-entanglement__distribution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:19eea086be5305ab4d9ffbe6bbf3f5a9286aec315bf29b7d4403a02f9aeb76c3.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:7e11f2ef9579ffc07192430aeb6c40fa225c2ac06d7e114724ecc9219834280a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:11aaf3801d8105f70582cb88872a4c36298334d504540dafcbb43a38041c1ed7.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:881a7d9a75b566364a003b319abaeca54bdf2cd3263ffbd9ab1aeb7813f0df7.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:c1061dd2fce337b40a2e45f157414d832614f3037b6f908a09deb0ef7f1e04d4; certificate
sha256:e2dc53b388ebdf459e8929547810cf4c3d778209fea737eb1bf27af6795d1b08; external-validation
hash sha256:7c8e5ef81f03d00e04ef56779653dd0d5cffe973bca994f80b6ae1c9ed47974.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:b4d03886dc24480e527ac4522ef8eb3213f4940617f016588852655c95ef6d8a. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POLYMER-PROCESSING-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/polymer-advanced-manufacturing-and-rheology>; snapshot
experiments/external_sources/materials/snapshots/nist-polymer-processing.html;
sha256:11279429040fb5daff9108435f0c510fc12d89e09710e799b1a6182f63a63a18; scope: polymer
processing, crystallization, rheology, kinetics and recycling.

Comparison and custody records:

- `sft-mat-class-polymer-001-authority-correspondence`: predicted repeated-unit-chain-carrier__covalent-backbone-recurrence__sequence-length-branching-and-entanglement__distribution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived repeated-unit-chain-carrier__covalent-backbone-recurrence__sequence-length-branching-and-entanglement__distribution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered polymeric material organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered `material_classes_bulk` grammar that preserves repeated-unit-chain-carrier, covalent-backbone-recurrence, sequence-length-branching-and-entanglement and distribution-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:6a1b61cf7b27fd57917e294a49338c0d42f55298366417e4c1c0e6ee8680e27e; derivation seal sha256:f3bf4d70c0a28bed53287d8770575102bf8e2585e68ae7310e414b9dc218ef3c; engine receipt sha256:a99eafa3c8ace458d61210eebad617b1826e865d7025839889d21e69e28f6ff6 at receipts/engine/model_admitted/SFT-MAT-CLASS-POLYMER-001-a99eafa3c8ace458.json; empirical validation sha256:5fd119728138b9711ab1e4dc0020f406fcd5489ae5d9dda98813de8936b3983e; measurement receipt sha256:cd90b19a87eb545697cc05bd0370d1b1ac3f616e0baa870bf5a41fe44481e356; isolation sha256:11825e09dd0ac943dd33d6d22bdbbd9ff7ba42a8c0c68a92d5c72c8e5ccd07e6; custody sha256:ffa4c921c9ac95448b5471c3dfe8126bbce6d40721a51c4ba3744e339a384af7.

70. Composite material organisation

Claim identity: SFT-MAT-CLASS-COMPOSITE-001

Question and exact theorem. A composite is a joint material whose distinguishable constituent carriers persist while interfaces and architecture compose their load, transport or barrier responses.

multiple-constituent-carriers__interface-mediated-load-transfer__constituent-identities-retained__architecture-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves multiple-constituent-carriers, interface-mediated-load-transfer, constituent-identities-retained and architecture-and-scale-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	multiple-constituent-carriers	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	interface-mediated-load-transfer	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	constituent-identities-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	architecture-and-scale-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `multiple-constituent-carriers__interface-mediate-d-load-transfer__constituent-identities-retained__architecture-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c6c46419af214daaf146350e487c95de5bc30d082ae54dad12b5579ad28b1633.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:ecccf75b15e102f438af9f7c385a5cc558ed848ca7c8313848451f843f4856a9.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ce4a29905dae4aa34196f4664e2f919f02c6b3eab2d8cd11401b744b38a79e1b.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e02674fffcbbf54bd9a50f41cef4217f5351d525fb98fc7931dae2a22496d021.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:8d4c7ec3138919dd7154707c09abc73b7d0fcecfebf5de1ef5330929a19b6ef56; certificate sha256:e0737335b366ab6f6be9f854b63f51a38c01a6727f89f1e92a8958ab93e87def; external-validation hash sha256:ab91a5e87b1f7ead61352f386d150be5fec82f2d722d41a666c2f3e18d595300.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:99c767633d82ee90ffa49c7f507f599fa6fd42c297c565e2ce1dc9e4a339472c. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POLYMER-COMPOSITES-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/polymer-composites>; snapshot experiments/external_sources/materials/snapshots/nist-polymer-composites.html; sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-class-composite-001-authority-correspondence: predicted multiple-constituent-carriers__interface-mediated-load-transfer__constituent-identities-retained__architecture-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived multiple-constituent-carriers__in interface-mediated-load-transfer__constituent-identities-retained__architecture-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered composite material organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves multiple-constituent-carriers, interface-mediated-load-transfer, constituent-identities-retained and architecture-and-scale-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:0c4d6765e00200da4b3797781c97c267cd941b131622a573de77c323c27824ab; derivation seal sha256:c8d209ef1e580b1bae19f6f4a3c8c816ecac58ba3747d66810463b448cf61240; engine receipt sha256:e068e79f1b5490cd352d10a8669b04741ce9fedbf15e89115fbfa514487b6983 at receipts/engine/model_admitted/SFT-MAT-CLASS-COMPOSITE-001-e068e79f1b5490cd.json; empirical validation sha256:a8e4192ff987d1ec55606bf440fe850195cf076253bef45c4287a44b7dd1ceec; measurement receipt sha256:04a5264efc4594b50729f4788e24b5b702585fd07b9127b6840eb7df565b6d49; isolation sha256:1451cd65a42e6b1336107ce0653b3d3a2c9742c625bcc8d8af0e498b61c5cfc0; custody sha256:b55a8ef809423e1b68ca5143fc757b398f522cd4b4ec94fdad9342c02fa767b5.

71. Glass and amorphous organisation

Claim identity: SFT-MAT-CLASS-GLASS-001

Question and exact theorem. A glass is a connected material with retained local structural recurrence but no crystalline translation class, whose macrostate depends on preparation and observation-time relaxation.

connected-nonperiodic-network__local-order-without-translation__history-and-relaxation-retained__observation-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves connected-nonperiodic-network, local-order-without-translation, history-and-relaxation-retained and observation-time-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	connected-nonperiodic-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	local-order-without-translation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	history-and-relaxation-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	observation-time-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `connected-nonperiodic-network__local-order-without-translation__history-and-relaxation-retained__observation-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c8117ff6f4fb77e532becbb51883f31b96b64fc2032238b47ee0761bfd73ce3c.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:48ccd96f2cef517058dcc908ea88079c276754135e112e1ce69ebf05da1070a2.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:dc7537443e2a29e6ec64d48fdd218123e16753cbd276ee7cbcd8adee543f54a.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e4c5829e88d7ee7cf3bd6e46e6305425f6c0aae67dfa329036fbe4f8ab143be4.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:7e5fd03d9ff4c3789544066770b9bce294e351f8cc31f47b82515957acbebd5e; certificate

sha256:633a1eca778acd3c503f297e2512b44ae93ce6dde346cdcb88dde7ef694d8f51; external-validation hash sha256:bcd2bad747a845796be107bb29cfd82be8b8a5f595131d1dadaaf578261e267d.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 3 claim-specific source discriminators were required; their ordered identity is
sha256:bca2ffd249200181edd6ffefb138fc5f42e3029ab07afc7b74f34735ef7970d2. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-GLASS-TRANSITION-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/publications/glass-transition-its-measurement-and-underlying-physics>; snapshot experiments/external_sources/materials/snapshots/nist-glass-transition.html; sha256:2d6b99e46aae9d3c6a503494eedd17db45d4b48a633bb254cd3a6a5195440777; scope: thermodynamic and kinetic glass-transition measurement in glass-forming materials.

Comparison and custody records:

- sft-mat-class-glass-001-authority-correspondence: predicted connected-nonperiodic-network__local-order-without-translation__history-and-relaxation-retained__observation-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived connected-nonperiodic-network__local-order-without-translation__history-and-relaxation-retained__observation-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered glass and amorphous organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves connected-nonperiodic-network, local-order-without-translation, history-and-relaxation-retained and observation-time-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:ead873ba4e541f4086b9cf10d55335b648a49936c705d212d34a67042618ba67; derivation seal sha256:20b63fcf87d6da5eff04eb20dda0fe29cf66a0f8502357ec603f104e95066877; engine receipt sha256:48121416b6e96a7ca43dfe1cd3cd81cb6122dca865c2d5757249708911c2d8bb at receipts/engine/model_admitted/SFT-MAT-CLASS-GLASS-001-48121416b6e96a7c.json; empirical validation sha256:82fa9de65ba0566f78de7820f4837b4fa446ff6d8564270c79fd9abe6b2d82b6; measurement receipt sha256:963696e40f2dd738d98875acbeb98ee72b964edfd075b4ea88f9ca04de50fa1a; isolation sha256:2fb7bfab1a75e763ff1da7ffbc65554ff91371b787f01d199b07624b7c7f2dac; custody sha256:5b5891f1bf93f5e9ce4ef68c1d14aafb3e29547eebf393d707bd525b16a298d.

72. Porous material organisation

Claim identity: SFT-MAT-CLASS-POROUS-001

Question and exact theorem. A porous material is a joint solid/structural-absence support whose pore connectivity, interface and accessibility are retained at a declared resolution.

solid-and-empty-support-carrier__connected-phase-and-pore-network__interface-topology-retained__resolution-and-access-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves solid-and-empty-support-carrier, connected-phase-and-pore-network, interface-topology-retained and resolution-and-access-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	solid-and-empty-support-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	connected-phase-and-pore-network	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	interface-topology-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	resolution-and-access-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `solid-and-empty-support-carrier__connected-phase-and-pore-network__interface-topology-retained__resolution-and-access-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:86474f877d49101e7e3ace7b1ea74f2d9a093c01fe7680f3b1271e4489adc258.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:aa09a3db065810ef015d5e9c1c6c88deee9b27a1f27fe6dbf4b0cf92e5801cc6.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:de7c2b045d20ac78f59d8ec13cc4ff932cc6c3c6f7cd7c236358dbf0836e8d40.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:79a409094422197b89e8c3e50fdf69c57e508575d8db62334202a4d2f429b0c0.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:9df78b11cbc6e6c95b2d9a9e0a9f99e1c79ff7e0c979633cebef497f96e33a74; certificate

sha256:a3e148d5b2218edb7365c024ec306f7d09058931911d8cc7a88ee26c66aa1af0; external-validation hash sha256:7d669197ff1fddc9a4452d46a59fde18bc7e56e1d41e894b145b83fd6603ac8c.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:13583e6afba8522a3fd7dd9d3c8fd22796763215d55db5dfbd11a552b2b30528. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-MULTISCALE-MATERIALS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- `sft-mat-class-porous-001-authority-correspondence`: predicted `solid-and-empty-support-carrier__connected-phase-and-pore-network__interface-topology-retained__resolution-and-access-bounded__complete-state-transition-boundary-trace__root-`

bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived solid-and-empty-support-carrier__connected-phase-and-pore-network__interface-topology-retained__resolution-and-access-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered porous material organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves solid-and-empty-support-carrier, connected-phase-and-pore-network, interface-topology-retained and resolution-and-access-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:69f8356c8af5397580d2a11cf32c7e72fd277e6e8bed3270680e722abc3cefbe; derivation seal sha256:c54ad9ff0af549841d8deaf72c13a3262cecdelf9c6babf627c99aa57579d918; engine receipt sha256:38df22075765cbfddde541b7c7c5909a5657a9a5d88c17badc55cf9e1691d896f at receipts/engine/model_admitted/SFT-MAT-CLASS-POROUS-001-38df22075765cbfd.json; empirical validation sha256:450e94e6f317469bb6b703f57926d750ce630510f52b2625b57fb1b86b746636; measurement receipt sha256:7ba33b9a0f10949d4026a021623572caf081489251cb9fd806df2a9ca191e427; isolation sha256:21dd1f085996b65e7f1dd705fab7ba3a6e648ad01ebaa1cfb389d1ecd6e2366; custody sha256:6b400b1a8dfc8c669425cc3c4d17624d8621bf1ca1828ad820b17ab91ea2d375.

73. Molecular-to-bulk correspondence

Claim identity: SFT-MAT-BULK-CORRESPONDENCE-001

Question and exact theorem. A bulk material law is admitted only when the complete constituent/interaction network composes to a scale-stable observation while fluctuations, interfaces and preparation history remain explicitly bounded.

complete-constituent-ensemble__local-relation-to-network-composition__fluctuation-interface-and-history-retained__representative-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves complete-constituent-ensemble, local-relation-to-network-composition, fluctuation-interface-and-history-retained and representative-scale-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-constituent-ensemble	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	local-relation-to-network-composition	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	fluctuation-interface-and-history-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	representative-scale-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: complete-constituent-ensemble__local-relation-to-network-composition__fluctuation-interface-and-history-retained__representative-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organisation are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8aaa2be2cb1aaf75093de967bfa9bb883f60e3bfb38d10c1501e9e67043ca4e0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:aflae25c467d916ea85f9b6610c5ba82923fef5eece84c75ea61a4f185806e16.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1aa726163603796dc22e356d791678eb295c11adc115abd3c4435a34c6960e8e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:64c386dbbe1c3aa910b83a1516e7b297cbebdb40d4ca072c8985bbf628c260b1.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:2730e1897d8309a7942b0a6250869b93142c515f1222b012fab05c66195db660; certificate

sha256:dc8f66c5f350939ed51ffd3201e28c851664e6533f6eeefa0bb451050bad8ee9b; external-validation

hash sha256:d0944a5281e0ad757098205af86c358371b8c15f418d733c9961798ef1fafa56.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:95c6dd46934ae777e83fddc8889a1df0d2b9d0e0f586a7c703736741414c412f. Target content

opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POLYMER-COMPOSITES-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/polymer-composites>; snapshot
experiments/external_sources/materials/snapshots/nist-polymer-composites.html;
sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer
matrix composites, interfaces, strength, toughness, fatigue and recycling.
- NIST-MULTISCALE-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-multiscale-materials.html;
sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope:
multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-bulk-correspondence-001-authority-correspondence: predicted complete-constituent-ensemble__local-relation-to-network-composition__fluctuation-interface-and-history-retained__representative-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-constituent-ensemble__local-relation-to-network-composition__fluctuation-interface-and-history-retained__representative-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced

- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered molecular-to-bulk correspondence observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves complete-constituent-ensemble, local-relation-to-network-composition, fluctuation-interface-and-history-retained and representative-scale-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:b7c45e293722535b1fee18b96470c7d74e777249add1635ab885cfb643012bce; derivation seal sha256:18006e7469ea4d53ae3e2432398c8677dce5f5cd10dd2be3fa2658c714da22ad; engine receipt sha256:447912e5ddb8853c1ee41231f0d106962ca3940a118a3fd420384a77f06509b4 at receipts/engine/model_admitted/SFT-MAT-BULK-CORRESPONDENCE-001-447912e5ddb8853c.json; empirical validation sha256:37e602662e5ef301a345ca2dcaa89ced48957308154af015e5d5a8e23fb71221; measurement receipt sha256:3844092558e29c9ff4835b0c2c4df9bc85fdb6b9418f73032b67837012d39920; isolation sha256:3953777fe17d106465d44bfe5c9158304296a968e2186bf3fbe0ba5891ce252d; custody sha256:feadd5dd045f6ba5e2e5f2c2688ab882456e1efeb3630d53f2e0ca0ab2066b0b.

74. Material anisotropy

Claim identity: SFT-MAT-BULK-ANISOTROPY-001

Question and exact theorem. Anisotropy is inequality of response relations across retained material directions after specimen, condition and method identities are held fixed.

direction-labelled-structure-carrier__orientation-dependent-response__same-material-identity-retained__coordinate-and-symmetry-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves direction-labelled-structure-carrier, orientation-dependent-response, same-material-identity-retained and coordinate-and-symmetry-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	direction-labelled-structure-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	orientation-dependent-response	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	same-material-identity-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	coordinate-and-symmetry-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: direction-labelled-structure-carrier__orientation-dependent-response__same-material-identity-retained__coordinate-and-symmetry-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6cbc7c78de53a91c49c276e589ee1970b58c34859a374b9e536116f635df6977.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:32e5a3fe10672f05fcc0c334670205b39f340f5e57431dd46f3b50d3742f3fa3.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:175cd6c6266b529caa8f9d2671fa7ae2b95dd96739c241f9fbdef8a23d8e613a.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:d1c1e14e228849b8d8b6e38554430a79cd3247aadccdb0a8de858714adf77819.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:b984029abd34332b72aa4acbe09bba4fa14c8c7e328e04abe4a2d287d0d8ec73; certificate
sha256:d3277213f7cc1a63985c32223dc1176a3a60269bd1d12589f6b5becf88e0d77; external-validation
hash sha256:f31046c84f6ca5ab88a49c834753bfd3a8c8fb142ffb4cb2b1148a7cd13517c1.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:5415b2701a13d019225ae29acb353bfe745f8505d009fe5b4176d0a86c279069. Target content
opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TEXTURE-PHASE-FRACTION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot
experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html;
sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains,
phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- `sft-mat-bulk-anisotropy-001-authority-correspondence`: predicted direction-labelled-structure-carrier__orientation-dependent-response__same-material-identity-retained__coordinate-and-symmetry-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived direction-labelled-structure-carrier__orientation-dependent-response__same-material-identity-retained__coordinate-and-symmetry-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered material anisotropy observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule

- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves direction-labelled-structure-carrier, orientation-dependent-response, same-material-identity-retained and coordinate-and-symmetry-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:72f09f5534c17c27099b1d5257c3786dbac85ef798492155e0d32d5a6a10e3e3; derivation seal sha256:fbdd70413189e72b586102c3e9366e8bcb48fc378e7e134073135a4e0529a77b; engine receipt sha256:c3c22fd6b9e4de8ba6af34698e7d7dba28cf604ee5d62cef540b8459446182b6 at receipts/engine/model_admitted/SFT-MAT-BULK-ANISOTROPY-001-c3c22fd6b9e4de8b.json; empirical validation sha256:551c9e104169aae809fc0394df484bdc23a543b85a8346628b6209e3723d3b3; measurement receipt sha256:1a455d7914018cf69f07f94afb5e7a4abf16bf467f77e237738d2008f9cff2c2; isolation sha256:e984687ace695a4b2491c37d9f3cfe69d4dbb7cbb9d5f60cb8b992d004c67869; custody sha256:61e237fb7d23a7c17168cc1fb16bc841b1020e9a1d5e5df1860c43e029becd5a.

75. Size and surface effects

Claim identity: SFT-MAT-BULK-SIZE-SURFACE-001

Question and exact theorem. A material size effect occurs when the exact boundary/interior support relation changes the accessible state, transfer or mechanical census; it is never a free scale correction.

finite-bulk-and-boundary-carrier__boundary-to-interior-support-ratio__surface-state-and-curvature-retained__size-shape-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves finite-bulk-and-boundary-carrier, boundary-to-interior-support-ratio, surface-state-and-curvature-retained and size-shape-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	finite-bulk-and-boundary-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	boundary-to-interior-support-ratio	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	surface-state-and-curvature-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	size-shape-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: finite-bulk-and-boundary-carrier__boundary-to-interior-support-ratio__surface-state-and-curvature-retained__size-shape-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a71adddee9b5e490d396e46c2d4ce8e7d6df9fd103a368f317d5404c794f5962.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1cb16132da27181a2ea269de5aeccca7fd5c6fb9308e530c9eb4762bdd7c6cd9.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:16e259f649e3a6a6d3f8aaf73de1ae125511d669c7b238cab1230047fa53e70b.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0abbadc72e2bb18b61d548347a7fc809891b92f314caf46fd012efe82f96adb7.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:8b01df9a6447d154451b82d24d6fbe0a6cd3d71941e037c9c9c056cc28fdf109; certificate
sha256:71bf3b0b6332824a4a93e13e69b343a7f22245cac791167a29f3c097e5d0f4c3; external-validation
hash sha256:5b57f4f341beaf218e7fdd9becd318ff208a58060bf09221cbfdda95b8aea683.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is
sha256:96c7a8b7a44f93e71147fa23ce577cc4434028fcf92b7d9f39cab20f952bc8b5. Target content
opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SURFACE-DAMAGE-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project>; snapshot
experiments/external_sources/materials/snapshots/nist-surface-damage.html;
sha256:7f0a61e225462f1514a9b2b9e459f5a0727ef2dc291bbcf5e83626f52359169d; scope:
surface/bulk distinction, degradation, radiation, moisture, cracking and nanofiller release.
- NIST-NANO-PROTOCOLS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/mml/nano-measurement-protocols>; snapshot
experiments/external_sources/materials/snapshots/nist-nano-protocols.html;
sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; scope:
nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

Comparison and custody records:

- sft-mat-bulk-size-surface-001-authority-correspondence: predicted finite-bulk-and-boundary-carrier__boundary-to-interior-support-ratio__surface-state-and-curvature-retained__size-shape-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived finite-bulk-and-boundary-carrier__boundary-to-interior-support-ratio__surface-state-and-curvature-retained__size-shape-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered size and surface effects observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves finite-bulk-and-boundary-carrier, boundary-to-interior-support-ratio, surface-state-and-curvature-retained and size-shape-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:95bea428b8ee5e8f23690a77f8b094becf3ba585242b61cb740f778c8cc4f8; derivation seal sha256:5f43cb2e35c18fe49c584059308b7af0afc7c6a72bad42a13c824446b7e64489; engine receipt sha256:d42d7f0a8dfb8cd74cb9b7857420beef13a926f5ec874c61e613976f557dac14 at receipts/engine/model_admitted/SFT-MAT-BULK-SIZE-SURFACE-001-d42d7f0a8dfb8cd7.json; empirical validation sha256:46a2c11f07c97d5bca30736ccaac4a8677cea56d58998eaf25795921cd21e80c; measurement receipt sha256:22c019851ed0dcc75e91525362b462fb42429f0006f626a2460bac869fe59b66; isolation sha256:6a00639920ccb0d0be88becd640b378050d2bd32f0cd668d7a8022d2e2aab7fb; custody sha256:96a1bc3e068eea0c455024d13b01d9ecf1d2b794de18c5c4783991da5e4fc51c.

76. Hydrogen-bonded water bulk-response ledger

Claim identity: SFT-MAT-BULK-WATER-RESPONSE-002

Question and exact theorem. A water bulk-response claim must retain the molecular hydrogen-bond network, specimen and phase identity, boiling, solid/liquid density and heat-capacity observations with their conditions, methods and uncertainty boundaries.

molecular-network-and-bulk-specimen-carrier__intermolecular-organization-to-bulk-response-relation__phase-density-boiling-and-heat-capacity-records-retained__identity-condition-method-and-uncertainty-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-INTERMOLECULAR-BINDING-011
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-PHASE-TRANSITION-001

- SFT-MAT-THERM-HEAT-CAPACITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor material_classes_bulk grammar preserving molecular-network-and-bulk-specimen-carrier, intermolecular-organisation-to-bulk-response-relation, phase-density-boiling-and-heat-capacity-records-retained and identity-condition-method-and-uncertainty-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	molecular-network-and-bulk-specimen-carrier	The material carrier cannot be reconstructed. The complete generated carrier is retained.
relation	relation-imported-fitted-or-erased	intermolecular-organization-to-bulk-response-relation	An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.
organization	organization-collapsed	phase-density-boiling-and-heat-capacity-records-retained	Required material distinctions are merged. Every required organisation distinction remains held.
observation	observation-boundary-unrecorded	identity-condition-method-and-uncertainty-bounded	The observation class is unidentified. The declared observation boundary is retained.
record	result-without-transition-trace	complete-state-transition-boundary-trace	An answer without its transition trace is not auditable. The complete trace is retained.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.
generality	finite-answer-lookup	positive-finite-successor-closure	A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.

Unique survivor, base and successor. Sole survivor: molecular-network-and-bulk-specimen-carrier__intermolecular-organization-to-bulk-response-relation__phase-density-boiling-and-heat-capacity-records-retained__identity-condition-method-and-uncertainty-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: The first complete carrier retains its relation, organisation, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- bulk-ledger: all nine required molecular-to-bulk evidence fields are retained; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause MAT-S008 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:18b0d69913ad5bc692e76806db29181e55a3393a181f9f7d7525f066e86c78dd.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:03ddd75b6d96b73054222c608d9a9de3960543fa91c25ecb965e03e9905724c5.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:93be1ca1dd2bfd31a348aa94e09bcac089cb5a27baf8cc4d06188fd1fd1b7b1.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8470568cf156f40d4f230a507c433c53b7ef0c2608ca12e252ce1c3b5f3f34a5.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:740847c6a0eb2ab7caafe3433bdc61a3264c9e1ac5f79bd7664bcb4bb557e55; certificate
sha256:adb5f7d3a68e71521a0d0aa531d4600ab203beedc87c96451fb9aafb6138880c; external-validation
hash sha256:adb5f7d3a68e71521a0d0aa531d4600ab203beedc87c96451fb9aafb6138880c.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:dde23657166381ae8b42a76512717a54870f1fc924267c72b3e0618487891a25. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-WATER-THERMOPHYSICAL-2026-07-27 - National Institute of Standards and Technology;
<https://webbook.nist.gov/cgi/cbook.cgi?ID=C7732185&Mask=FFF&Units=CAL>; snapshot experiments/external_sources/materials/snapshots/nist-water-complete-2026-07-27.html;
sha256:ba9fc5225bc8d0d89cbe15703bff2d9dc06ef64244598c420c7666931f1befbf; scope:
post-seal claim-specific authoritative comparison.
- NIST-LIQUID-WATER-DENSITY-2026-07-27 - National Institute of Standards and Technology;
<https://www.nist.gov/document/jpcrd38200921ppdf>; snapshot experiments/external_sources/materials/snapshots/nist-liquid-water-properties-2026-07-27.txt;
sha256:b3af256bcffdf7ff87fbf2aa00f545910a531d5a1b285fa9bb9f926f7afad9e24; scope:
post-seal claim-specific authoritative comparison.
- NIST-ICE-DENSITY-2026-07-27 - National Institute of Standards and Technology;
https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=926353; snapshot experiments/external_sources/materials/snapshots/nist-ice-properties-2026-07-27.txt;
sha256:7073b61255cf73163595c9ca25646dcf2c323dd2aca942d797f9f0e4dd6c178e; scope:
post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Shared claim-record clause MAT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor material_classes_bulk grammar preserving molecular-network-and-bulk-specimen-carrier, intermolecular-organisation-to-bulk-response-relation, phase-density-boiling-and-heat-capacity-records-retained and identity-condition-method-and-uncertainty-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_v1_v2_successor_complete_pre_source.json;
source manifest sha256:e181e7b9d0b370a10d0fdda4e459caf66dbc40665ab7cae962f3e73b1d218d67;

```

derivation seal sha256:f74d7a789438f6e069854e4651e70699007c810fe1a232a9e5ad63f4d1549679;
engine receipt sha256:bbc61e7fbba2d3cf5ce25e2015c3aa1713e03d741067f422760e797ab6a42774 at
receipts/engine/model_admitted/SFT-MAT-BULK-WATER-RESPONSE-002-bbc61e7fbba2d3cf.json;
empirical validation sha256:0ea1185d4ba1ffe80e5c49895cb65c9f6bdde00361f4c14fd0ad8124e77c6239;
measurement receipt
sha256:c60a627a4bb3afc89140f7d6e4e88493a2ead819f8c5ac7c8221f3b4aad224bb; isolation
sha256:4b02bcb242983a2408823c7bab95f0b878716b006850a00f86f1643c68a2cfdc; custody
sha256:b97e1b5ece66313d657ab0e2dc92b92e34bea313db4f1da2009c649d06b07c29.

```

19. Processing Degradation

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

77. Processing-structure-property path

Claim identity: SFT-MAT-PROC-PATH-001

Question and exact theorem. A materials process is a complete ordered path from input carrier through retained structural states to property relations; endpoint labels alone cannot erase the path.

```

history-labelled-material-carrier__process-to-structure-to-response-map__intermedi
ate-states-retained__route-condition-bounded__complete-state-transition-boundary-t
race__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves history-labelled-material-carrier, process-to-structure-to-response-map, intermediate-states-retained and route-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	history-labelled-material-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	process-to-structure-to-response-map	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	intermediate-states-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	route-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `history-labelled-material-carrier__process-to-structure-to-response-map__intermediate-states-retained__route-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:de1ccf6eb7be4b98104a537ca43370f1260822868e975e19263742b742042635.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:2a2242612fcf027a838edcc8079ef2b4a0a8103741fd3d4dac0fb3faa9783e70.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:361024627d19d9bae6cc5a148da978c4f72f58f46563b8d250daba4ecff56ba3.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:c570df606d3d54c7acc82474a18b688a269c52278fa3ed9e40cf3fb952620731.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
 sha256:3de27f586e0a54a133982ba8f398e3da38fac53b3510ae159a9d1aa2f07a33c0; certificate
 sha256:03e7e4f1fc0dc184d8177651acc4a903420c58e4b0d458d84f3f307ecb17b308; external-validation
 hash sha256:7f4acaefbfb04449e58fbb65dc23f59ec79c48a4162681365f5b4d931ef7faaa.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
 sha256:7cbdc5dc38d614e32c43f7f765797cd184d0e4187360667ac81f809ede32d647. Target content
 opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FATIGUE-FRACTURE-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot
 experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html;
 sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; scope:
 processing-structure-property-performance, defects, fatigue and fracture.

Comparison and custody records:

- sft-mat-proc-path-001-authority-correspondence: predicted history-labelled-material-carrier__process-to-structure-to-response-map__intermediate-states-retained__route-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived history-labelled-material-carrier__process-to-structure-to-response-map__intermediate-states-retained__route-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered processing-structure-property path observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves history-labelled-material-carrier, process-to-structure-to-response-map, intermediate-states-retained and route-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
 sha256:1f883e2d57a7b61755f14fc762c6739e9ce7beb81863b37d8e95f4ad26909eeb; derivation seal
 sha256:702060f13eedb527da17fb13321451beb4f372bf20b7eba39441e125afe297d1; engine receipt
 sha256:44a5d1efe1f1541917ea77e5eed4af2fdadb277107194470e0cd69a279a2bf6 at
 receipts/engine/model_admitted/SFT-MAT-PROC-PATH-001-44a5d1efe1f15419.json; empirical
 validation sha256:96ece9f7a71e3deb2748bec2f31385040cbee3d68883d1f5bcf6a94d67a8cd6d;
 measurement receipt
 sha256:cbc82b899d3bc7f483ec6881a345df54813c7cbdd9a4ab034da56209bb2f053d; isolation
 sha256:87b63a192292d43bf9591f2f3dfa1354236cc5b24b75811f1ba2e0each225d5d; custody
 sha256:c441dd3a1ce057baa9ad7680adcd01a5ba977076a692f9633121dd6ae9017865.

78. Phase stability and coexistence organisation

Claim identity: SFT-MAT-PROC-PHASE-DIAGRAM-001

Question and exact theorem. A phase map partitions generated composition/condition support by the stable material recurrence class, retaining coexistence, fraction and transition boundaries without fitted extrapolation beyond its domain.

composition-condition-support__stable-phase-minimality-relation__coexisting-fractions-and-boundaries__declared-domain-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves composition-condition-support, stable-phase-minimality-relation, coexisting-fractions-and-boundaries and declared-domain-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	composition-condition-support	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	stable-phase-minimality-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	coexisting-fractions-and-boundaries	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	declared-domain-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	authority-or-target-s elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in stance-lookup	positive-finite-successor-cl osure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or -extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `composition-condition-support__stable-phase-minimality-relation__coexisting-fractions-and-boundaries__declared-domain-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:9054bd12994d5f41bcd6177b6fed1a34c64e07f8bc3fa509b475f602c4ea5c2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:40353a579820b968d584b0a61eba404cd3123713f03e571bd52c017083c2f5cc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:b3bf6511512a363058e595b8843d98098ef829b121614b2c706fac58032f2bf3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:2f7e1698d0a482289fd10c54b7e271b8255002aff2c60c9580d49d52be03b00b`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
`sha256:5b0d3302fca9faacc388d07b983abf4e483d21355e3e376bafddfaeb2611318b`; certificate
`sha256:f57fb9ac93816e6dfe7d21e87d4a975ff683877da72cf233739c2ef07f175a5a`; external-validation
hash `sha256:e19d6dc3ca0588280f74e04f9517cf7ac7566243ff1d1a70e74bb3323873b877`.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is
`sha256:cac509177fb4e7db7eaf36ef805abfc1b25c482a863f8a04e987b8d380a19de2`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716cccf34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.
- NIST-TEXTURE-PHASE-FRACTION-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html; sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains, phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-proc-phase-diagram-001-authority-correspondence: predicted composition-condition-support__stable-phase-minimality-relation__coexisting-fractions-and-boundaries__declared-domain-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived composition-condition-support__stable-phase-minimality-relation__coexisting-fractions-and-boundaries__declared-domain-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered phase stability and coexistence organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves composition-condition-support, stable-phase-minimality-relation, coexisting-fractions-and-boundaries and declared-domain-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:2697defe1bb570f0e0a431bf86071a61097ef77fa6a094e7bed3496b89580dd0; derivation seal sha256:e0c1577c109e318217fbd8353609550171447bf5838f1c229e48b62946ff50ac; engine receipt sha256:b2ab10b5f18cb90daa694ebe0ed304d3a153b06b007290c11a4ef97f4e157d0d at receipts/engine/model_admitted/SFT-MAT-PROC-PHASE-DIAGRAM-001-b2ab10b5f18cb90d.json; empirical validation sha256:ef30582b4adcc27a061c4c1f53dafc306fd144b7f3d0da2fdda718fbee0ef391; measurement receipt sha256:b6121ca9be3efcfdea534009debc7982d58154d474e27907ba17eac057b4615; isolation sha256:9b3a1a5224c4dacf767cf438ecbae4065682fefe679f87ef9cdda13a118c6e3d; custody sha256:e89d44de4b24180fffe9afd082483ad17c54feee960bc30fad6ece7eda5a639a.

79. Solidification

Claim identity: SFT-MAT-PROC-SOLIDIFICATION-001

Question and exact theorem. Solidification is the ordered nucleation and growth path from mobile to position-retaining organization with complete energy, constituent rejection and interface provenance.

```
liquid-and-solid-phase-carrier__nucleation-growth-and-rejection-path__latent-trans
fer-and-composition-ledger__rate-gradient-bounded__complete-state-transition-bound
ary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra
-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves liquid-and-solid-phase-carrier, nucleation-growth-and-rejection-path, latent-transfer-and-composition-ledger and rate-gradient-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	liquid-and-solid-phase-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	nucleation-growth-and-rejection-path	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	latent-transfer-and-composition-ledger	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	rate-gradient-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	authority-or-target-s elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in stance-lookup	positive-finite-successor-cl osure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or -extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `liquid-and-solid-phase-carrier__nucleation-growt
h-and-rejection-path__latent-transfer-and-composition-ledger__rate-gradient-bounded__
complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite
-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:28b0e07297d86f968e940b1b2ac43f3094e57aeb51ca1cf3957be002d4d3c900`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:1d9abadcefe83f3d19ed07b99fe3c194097758fd2c19ca09640c2d36d329b587`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:25a0e1d82af8173155367a44fd6a6c03a8a457875f6d6237beb39a838f16497`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:97e19a857b778351e4607a1e2b02197c20cd922c00b6ab7a535f726ea471fec4`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

`sha256:9008ceb7d860deb437d0278efb1db506b4282e8994c079cb8310927d30804be8`; certificate
`sha256:ca3d3ad1afbbf9780ad9132a1260095507c75475a0390dfe859deddde190e2b3`; external-validation
hash `sha256:8442dec51ae104811a14c830ef71370b48550b273d30be09822d7dcd9ec6d08f`.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

`sha256:0c9f870f7f230ffeb3890f9e71ae58c89c7463037ae121df10cc17d648146081`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SOLIDIFICATION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/publications/solidification-0>; snapshot
experiments/external_sources/materials/snapshots/nist-solidification.html;
sha256:d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c; scope:
solidification transport, nucleation, growth, interfaces, segregation and processing.

Comparison and custody records:

- sft-mat-proc-solidification-001-authority-correspondence: predicted liquid-and-solid-phase-carrier__nucleation-growth-and-rejection-path__latent-transfer-and-composition-ledger__rate-gradient-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived liquid-and-solid-phase-carrier__nucleation-growth-and-rejection-path__latent-transfer-and-composition-ledger__rate-gradient-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered solidification observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves liquid-and-solid-phase-carrier, nucleation-growth-and-rejection-path, latent-transfer-and-composition-ledger and rate-gradient-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:cd4d0b64eac45c1d3a28447a50aa223d74e3c5531439f94b4aafb0ad44ec43ff; derivation seal
sha256:44d972ac53f7700b71e42f3b296ec7072c34c45c9f864c2c029cbe4af795bad3; engine receipt
sha256:882454d5441763ff1ecfcd81b6298141741d84056879c62ad3fa9b1cd7c3bf85 at
receipts/engine/model_admitted/SFT-MAT-PROC-SOLIDIFICATION-001-882454d5441763ff.json;
empirical validation sha256:8f82f3fcd57f8f8f6dcbd0b69e42bc940cd8c94021d24df8dafcd8f016a00f4f;
measurement receipt
sha256:e07ec032b3d672f10f84002045b8112a885dc28567393f3f27097c22abaa2d6c; isolation
sha256:d2bc013d9aadeadd5d8593a793caf61908d3490425424619a807233e8dfa3da7; custody
sha256:549b964885d9add4a9fb9c1b65671c42f0e813321766e22d95b1f4d9e922ea1e.

80. Sintering and densification

Claim identity: SFT-MAT-PROC-SINTERING-001

Question and exact theorem. Sintering is interface-mediated constituent transport that joins particles and reorganizes structural absence while conserving material content and retaining the processing path.

particle-and-pore-network__interface-driven-material-transfer__content-conserved-pore-reorganized__temperature-time-load-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves particle-and-pore-network, interface-driven-material-transfer, content-conserved-pore-reorganized and temperature-time-load-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	particle-and-pore-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	interface-driven-material-transfer	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	content-conserved-pore-reorganized	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	temperature-time-load-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `particle-and-pore-network__interface-driven-material-transfer__content-conserved-pore-reorganized__temperature-time-load-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:427fba8bb3e13d9e946485513932e9addb8efa06915387bb6f29c16955dd431c.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:a95765c073bde4df0dde77cbd0bba9d791e2da4a9df0f3b85ffba97e4c9a869.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a86e1fcd6ea91d3d19e8c4ba54fe00c6aed81ef80e74c371cb01a3cf5d220e80.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:973a0bc896c8d4274acedad03ad2cbdb968478e6872d029074dcbe70d0885cb8.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:6b45355ff97dda505fb391570767e63172d3e98a3db5bc1a18561bc19b1ee990; certificate

sha256:e171069695aad4a42995390f5a4d33efa475bcf7ab4de7d4e49caa9013ac2cfa; external-validation

hash sha256:8ad24f2106da658a5eaf4ae15e3958ee5bbdccfd4fe1e6c0753e5c4770761a6a.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:7892b6461c84d32bd9487a05f441b0e38c9d4640a478edd10b49c14a6cb2ace7. Target content

opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-CERAMIC-ADDITIVE-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/additive-manufacturing-ceramics>; snapshot
experiments/external_sources/materials/snapshots/nist-ceramic-additive.html;
sha256:540ef912a827c1834a113352ac0305c7e8d25a76283dcc1bcbab48f0b826a3e9; scope: ceramic
feedstock, defects, sintering, densification, phases and microstructure.

Comparison and custody records:

- sft-mat-proc-sintering-001-authority-correspondence: predicted particle-and-pore-network__interface-driven-material-transfer__content-conserved-pore-reorganized__temperature-time-load-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived particle-and-pore-network__interface-driven-material-transfer__content-conserved-pore-reorganized__temperature-time-load-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered sintering and densification observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves particle-and-pore-network, interface-driven-material-transfer, content-conserved-pore-reorganized and temperature-time-load-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:3a48aa91150d2edb674a0ea59f30e9d6dcff091178031b3568c20f1bfe7608f4; derivation seal sha256:6dcdbf2eb9b8b5012bad74a050454b19eca9f1a608ddefc30e364e667083445; engine receipt sha256:2405176e959c34190b77cd5b5acfb77dedd92245de91133d652c24ae73e097da at receipts/engine/model_admitted/SFT-MAT-PROC-SINTERING-001-2405176e959c3419.json; empirical validation sha256:5954fe3b5f12ec2660cfb6021c0d0517f40add8351bdeb0d8a388492b1e1028c; measurement receipt sha256:6db1b0cc6f2ba17d3ce2852251728a164746c4d146429002f978129e6e66979a; isolation sha256:a13f7d3ad965c5aa3dcaaf81c675099da3d320214db342f785bf416fcf7506eb; custody sha256:06ec2311f8604a307fcde247724ac7551f9fa3cb88f4b52cd156ca063fe0f795.

81. Heat treatment

Claim identity: SFT-MAT-PROC-HEAT-TREATMENT-001

Question and exact theorem. Heat treatment is a declared thermal/time path whose phase, defect and microstructure transitions are retained and whose property consequence is path-bound.

```
history-labelled-phase-network__thermal-path-to-microstructure-map__time-temperature-transition-record__cooling-and-specimen-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves history-labelled-phase-network, thermal-path-to-microstructure-map, time-temperature-transition-record and cooling-and-specimen-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	history-labelled-phase-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	thermal-path-to-microstructure-map	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	time-temperature-transition-record	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	cooling-and-specimen-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: history-labelled-phase-network__thermal-path-to-microstructure-map__time-temperature-transition-record__cooling-and-specimen-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed `true`.
- relation: the generated relation and organisation are separately retained; passed `true`.
- observation: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b9bccf53662d1f29124adbd3570acc88fe4777fd0a652a1e1919e532d8ed6c2e.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:54c522c6df12bc9faa22bd4f2e7cb3e5466453da6c0e58ea1133735b6fbc9b0.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:955e631bff387114e5252f19d68542f07a446c13563573d355fc803804d8d9d4.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e69374df1be210f621e0f3e134dc23dc86ba34914fdd90ad287d39530861f637.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:d5c8b9fd2403519162e8c42d68270b6519c09b147cb35d04e29d194cbd536948; certificate
sha256:2009f709ca9587b3ee9090d9fe6638b2040f5762ac7c979215d8cb8ec2821dce; external-validation
hash sha256:18935d07883ce48e19d9973db302e753092903a41dd892d13d8a9f6384a3fe4b.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:f4f5c6df4ecec53d108f79157d9e39adee7ab3f6e995d38609ad491640f1c285b. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-MULTISCALE-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.
- NIST-METAL-ADDITIVE-CORROSION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/additive-manufacturing-metals>; snapshot experiments/external_source s/materials/snapshots/nist-metal-additive-corrosion.html; sha256:albb9a6bc22eb85fb7a1b3c7d0ac4bc0837d68832b7b7c6f357a68ea6cde0322; scope: phase transformation, heat treatment, corrosion and environmental cracking.

Comparison and custody records:

- sft-mat-proc-heat-treatment-001-authority-correspondence: predicted history-labelled-phase-network__thermal-path-to-microstructure-map__time-temperature-transition-record__cooling-and-specimen-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived history-labelled-phase-network__thermal-path-to-microstructure-map__time-temperature-transition-record__cooling-and-specimen-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rul

e; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered heat treatment observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves history-labelled-phase-network, thermal-path-to-microstructure-map, time-temperature-transition-record and cooling-and-specimen-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:137c5dcf16efb098c8ba33efc12ca790c51997d030f971ed5ee21538263c3e0e; derivation seal sha256:b48749c0b7e8c1220f3d810e0fe1c7315f214c907330a1c10b73fa00ced47eab; engine receipt sha256:71c2de533e261b753919a23d2dda9c99717dc6b0893a99806067e734e9ef5081 at receipts/engine/model_admitted/SFT-MAT-PROC-HEAT-TREATMENT-001-71c2de533e261b75.json; empirical validation sha256:48550300b7b1bc18d7b5a89db1d438340b5f3332fa47c775af37b4a7e93a9529; measurement receipt sha256:ea6182376624969f7747697ca792534ee7b5b7583dc9a85812fd0f0bcfeb5341; isolation sha256:3990d3430a57e033ce1931a8557cb3f20880d2e069e3da1e186471eebee159f6; custody sha256:5136a32f8fdbfbbdbel1df064160bc81eb12efd63dc487f609789aaaecff8d.

82. Corrosion

Claim identity: SFT-MAT-DEGR-CORROSION-001

Question and exact theorem. Corrosion is an interfacial reaction/transport process that transforms a material and environment jointly while retaining complete constituent, charge, product and boundary provenance.

material-environment-joint-carrier__interfacial-redox-and-transport-path__mass-charge-and-product-ledger__environment-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves material-environment-joint-carrier, interfacial-redox-and-transport-path, mass-charge-and-product-ledger and environment-time-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	material-environment-joint-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	interfacial-redox-and-transport-path	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	mass-charge-and-product-ledger	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	environment-time-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: material-environment-joint-carrier__interfacial-redox-and-transport-path__mass-charge-and-product-ledger__environment-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9d15174b2bb1a9c07c26c8ba5711f9cdd9bb2f8a11a30aa320ce3f46bf1fe60d.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:66247b21764c92c8fd72734d44cdcdaba2d4db4e84b4faaa730823ee7a5db3327.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6c13de46c871b855dc4deb0ad522cb2e7c51962bf71905b9717e9d62e1decd18.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7cf442abda319bc3a9c74bd4d226b43396730c350c7f6f90566a15f15c27c7db.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:bc99ee0b1e7dd4c872735e6e261e18d1b1a6c045e6165dba52feb87aba82c3b0; certificate
sha256:322b3fc3c7a431f278d9a3f9d46a2322315693e645c3c955a86a8a9358fa0a8e; external-validation
hash sha256:c1b0a450d3020e0da5e0c078254ae93c3e1f4b760c4df000c6e1da4857be82d7.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:12b98b8c7184be61af10b2461cf587f811865f7d0f89a232f61fe44e983ae817. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-METAL-ADDITIVE-CORROSION-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/additive-manufacturing-metals>; snapshot experiments/external_source
s/materials/snapshots/nist-metal-additive-corrosion.html;
sha256:a1bb9a6bc22eb85fb7a1b3c7d0ac4bc0837d68832b7b7c6f357a68ea6cde0322; scope: phase
transformation, heat treatment, corrosion and environmental cracking.

Comparison and custody records:

- sft-mat-degr-corrosion-001-authority-correspondence: predicted material-environment-joint-carrier__interfacial-redox-and-transport-path__mass-charge-and-product-ledger__environment-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-environment-joint-carrier__interfacial-redox-and-transport-path__mass-charge-and-product-ledger__environment-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered corrosion observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves material-environment-joint-carrier, interfacial-redox-and-transport-path, mass-charge-and-product-ledger and environment-time-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:cbe464ce8433a734a64bd9b7432682d3246b2a27c2e1dfdf8fa429a58d8d2513; derivation seal sha256:dd05f442eff002b7acbae6f66aed2befa009ad9fc2ef5adada01cdaa20bf3838; engine receipt sha256:433102762a4f8027717c9429dddf7428947bf0bb850b88acb4929dd5d6b5078c at receipts/engine/model_admitted/SFT-MAT-DEGR-CORROSION-001-433102762a4f8027.json; empirical validation sha256:7397c246db07236a4bebe94e39a5390cbe1a75c43d58addd67e2e87aca429bfff; measurement receipt sha256:bdb80356d1b616ae5ea6fb262d58658987549ab20ca590623ae3d8c4b2906183; isolation sha256:e79ebce2c1fddc766f90acb9028cfed588fe6beb296c5bfc4cd88d717a42323; custody sha256:72c2ef2583c1c8b4bbe902480e18bf0cb56cca88d710ecb969a1f8347cf9f995.

83. Wear and tribological material transfer

Claim identity: SFT-MAT-DEGR-WEAR-001

Question and exact theorem. Wear is retained material transfer or surface reorganization produced by a declared contact/load/motion path; frictional response and material loss remain separately recorded.

contacting-surface-carriers__load-motion-interface-transition__debris-transfer-and-surface-ledger__contact-history-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves contacting-surface-carriers, load-motion-interface-transition, debris-transfer-and-surface-ledger and contact-history-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	contacting-surface-carriers	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	load-motion-interface-transition	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	debris-transfer-and-surface-ledger	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	contact-history-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `contacting-surface-carriers__load-motion-interface-transition__debris-transfer-and-surface-ledger__contact-history-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0efa487d64b1eee6cd7ed3a8527673f988a9ec16ecd45ae8220ccc0757354b1f.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:8229d0af59b1f56d1ea82fa4cdbbbd8e7ede783433fe74e9419d3f2bb8f61fca.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4347b34ae82985dfadd12d02f667736bf31f2ea92d6347f510d65e4f5162297f.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4b02a5f8e1ee1b658b888159069c30bfcca22de1c25516ba7278faf493cbb6d3.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:247df125f5c11760b072521e72106849cde1ca0ddf036aed5c47f6b5100eb78f; certificate
sha256:46c5627398f978342d464c3c65e7c094a0b903406e797f56d2b374c14effe281; external-validation
hash sha256:a2426bbe0d4a4c3e36b80076f60ad0181b8c4e09889767de1e1dbd2bc78b985f.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:6b5d358984148115f3731a4f470093239e7a3bde5e29e0180f773b33385efa39. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-NANOTRIBOLOGY-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/nanotribology-nanomanufacturing-archived>; snapshot
experiments/external_sources/materials/snapshots/nist-nanotribology.html;
sha256:3823f313343779c2a4a74a3e251a620319c89fef2142f2b1e58fa03250fec523; scope:
interacting surfaces, relative motion, friction, adhesion, lubrication and wear.

Comparison and custody records:

- `sft-mat-degr-wear-001-authority-correspondence`: predicted contacting-surface-carriers__load-motion-interface-transition__debris-transfer-and-surface-ledger__contact-history-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived contacting-surface-carriers__load-motion-interface-transition__debris-transfer-and-surface-ledger__contact-history-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered wear and tribological material transfer observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves contacting-surface-carriers, load-motion-interface-transition, debris-transfer-and-surface-ledger and contact-history-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:99ac90f50feb5f2af8a5030ff952e87e3438f02da297fb6bee8c61f4489fb672; derivation seal
sha256:2bc4d5b3e790c3d5ea3c51b06600f9aa2470632ff440c74aacd6b3be89fa1f11; engine receipt

sha256:92c996a0f5bfa825783a55376275b47e3635f67bc54552647a07899572b75f73 at
 receipts/engine/model_admitted/SFT-MAT-DEGR-WEAR-001-92c996a0f5bfa825.json; empirical
 validation sha256:b834c07064aee6699c3fcdc85825d0c731978cca680844c88aa92ea00f0d4ab5;
 measurement receipt
 sha256:6a0798c98fc0d49a3d347d1f9652267352ca1c9c64617f6609f52c626d2f9f82; isolation
 sha256:857edb5a43adcea4878ef787f307a6ffd5010ed00e07ec12712474e3c5f8c7a1; custody
 sha256:48e96ae55cb515efe98c53c5d32c53794e7a66ffa1bb72dd0a6e1b02e1f06021.

84. Radiation damage

Claim identity: SFT-MAT-DEGR-RADIATION-001

Question and exact theorem. Radiation damage is the complete source-bound path from probe energy transfer to retained displacement, excitation or transmutation defects and their subsequent migration/recovery.

material-and-probe-carrier__energy-transfer-defect-generation__displacement-transmutation-and-recovery__dose-rate-spectrum-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves material-and-probe-carrier, energy-transfer-defect-generation, displacement-transmutation-and-recovery and dose-rate-spectrum-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	material-and-probe-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	relation-imported-fitted-or-erased	energy-transfer-defect-generation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	displacement-transmutation-and-recovery	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	dose-rate-spectrum-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: material-and-probe-carrier__energy-transfer-defect-generation__displacement-transmutation-and-recovery__dose-rate-spectrum-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9038baa8e438b1ac2c472866e42bf6bf72685d5b5fccccafff338d1022d68b1fc.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:41a6bac5d899d6b3dc3443ea2f8ad03f200a69dc058c1c064bb1692905e50d94.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c5a8b28f1713b2c745128747d78b44c51d07371449023bddd94d0cb6f48d0518.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0eae8804b4960020b1078c44aeb79a2bcf0b3b24bd45b92fee471ed6121b6014.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:4f95022daaa462fc02c4d5629e2dce03074ed2a6402f7db3fe88c45f732ebe14; certificate
 sha256:2a3ea836c86f60a0cd147b73185a1047da7c534efca61fa7978d850f566f80b3; external-validation
 hash sha256:8e49be83d0c7ef29a09413543caed1843c57a721e1441a22efdf106c9a749ebb.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is
 sha256:f6da78675cb2d1530cb7a3007c0e23d3a8e7ad00b3a3bd09478b36b62572f8ae. Target content
 opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SURFACE-DAMAGE-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project>; snapshot
 experiments/external_sources/materials/snapshots/nist-surface-damage.html;
 sha256:7f0a61e225462f1514a9b2b9e459f5a0727ef2dc291bbcf5e83626f52359169d; scope:
 surface/bulk distinction, degradation, radiation, moisture, cracking and nanofiller release.
- NIST-MULTISCALE-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot
 experiments/external_sources/materials/snapshots/nist-multiscale-materials.html;
 sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope:
 multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-degr-radiation-001-authority-correspondence: predicted material-and-probe-carrier__energy-transfer-defect-generation__displacement-transmutation-and-recovery__dose-rate-spectrum-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-and-probe-carrier__energy-transfer-defect-generation__displacement-transmutation-and-recovery__dose-rate-spectrum-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered radiation damage observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves material-and-probe-carrier, energy-transfer-defect-generation, displacement-transmutation-and-recovery and dose-rate-spectrum-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
 sha256:6cd407968904d1edcba930dfcd606074c06ccc12d24464f820a04bf357931e8d; derivation seal
 sha256:698efffcc699b74190a413a43a06ba91aa6b1d1dd5c490766e0a6d22b501ca42; engine receipt
 sha256:e6e36bc645b877f36b9c969dd7bcbdda391a9f6e765e806c80d5fc11e214de42 at
 receipts/engine/model_admitted/SFT-MAT-DEGR-RADIATION-001-e6e36bc645b877f3.json;

empirical validation sha256:ee06521307cfbf5c880200c3b4c53e271fa83da81341ee32e8eb79fba8b1e43c;
 measurement receipt
 sha256:8b764f65f5eb032dc6e38b45a153ac5030f775a4900fc2c5164e8c19bcc87faa; isolation
 sha256:72088d354ffef387f18fefdddef742e2a4eae7b9cac2a1b92dd65903dd7638b17; custody
 sha256:e03d344139946255498886f2ef0be18c7ccf242d053080f8aea0062802a9381a.

20. Advanced Functional Sustainable

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

85. Piezoelectric coupling

Claim identity: SFT-MAT-FUNC-PIEZOELECTRIC-001

Question and exact theorem. Piezoelectricity is a reversible cross-relation between mechanical deformation and held electric polarization permitted only when the retained material symmetry does not close the orientation distinction.

noncentrosymmetric-structure-carrier__mechanical-electric-cross-response__orientat
 ion-and-charge-ledger__field-stress-condition-bounded__complete-state-transition-b
 oundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-e
 xtra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves noncentrosymmetric-structure-carrier, mechanical-electric-cross-response, orientation-and-charge-ledger and field-stress-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	noncentrosymmetric-structure-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	mechanical-electric-cross-response	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	orientation-and-charge-ledger	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	field-stress-condition-bound	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `noncentrosymmetric-structure-carrier__mechanical-electric-cross-response__orientation-and-charge-ledger__field-stress-condition-bound__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8ab926031b7ddd6f9ae6723610d89598c05082b6a6aae33124e1485e2155a550.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:46b8080073955bdaff587c54bb623525c892ff9137218f8adc2ce6d36b3c9278.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:769c539ede54297da90196ce94b20c57bbbfa5cea6148c8d4b022abc7c36e8a9.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:178a33c9bd0f4b6d3e6f7842ba574a87781b2a3174b4f3c77ea99f725c82198f.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:0edb98b42db8eb35abf9129bf24b204bde6a7b1f7723f398d61319711f842fec; certificate sha256:0eb2407afb1deb0d42da9134ce6101bc3eb518446e460f33ed4948cea367c142; external-validation hash sha256:7ea355f4b6500f3743e60bfa36f6f943be05e13ae2c759ce168cc4085e9b94ae.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:0199fe3b4329ace3bce2d717af5ca80589160d0648df85bf0b5b56d1a2e622b0. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-func-piezoelectric-001-authority-correspondence: predicted noncentrosymmetric-structure-carrier__mechanical-electric-cross-response__orientation-and-charge-ledger__field-stress-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived noncentrosymmetric-structure-carrier__mechanical-electric-cross-response__orientation-and-charge-ledger__field-stress-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered piezoelectric coupling observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves noncentrosymmetric-structure-carrier, mechanical-electric-cross-response, orientation-and-charge-ledger and field-stress-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:cdfe88675f774917ec9e902154853bc89336ea93db2feda54d87a9ac4e61fb8d; derivation seal sha256:4f680e305121b953aa576a0d8655400d3fff5a059be9a6c72c1b8035da333bc0; engine receipt sha256:7f640ec82c5c73c47f5749e38393747843704417ad8bd9279f671aab42365d26 at receipts/engine/model_admitted/SFT-MAT-FUNC-PIEZOELECTRIC-001-7f640ec82c5c73c4.json; empirical validation sha256:37594ac98d49f7968d62437760964d340829aacbf3c9f628d17f905f8c83d; measurement receipt sha256:87d6d92549f3c144716efa44e1286f3c5f188352f6208e71998bf895a634c516; isolation

sha256:ef1082ab87a9d58062773893911c04c5a836d5e6cc67300969b03531e0478173; custody
 sha256:e55d04e7bca4b730de4be5419299976028216ef5ec38114ceadd743681279539.

86. Ferroelectric order and switching

Claim identity: SFT-MAT-FUNC-FERROELECTRIC-001

Question and exact theorem. Ferroelectricity is a material phase with stable opposed polarization recurrence classes and a retained field-driven switching path between them.

polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves polar-domain-network, stable-opposed-polarization-classes, field-driven-domain-path-retained and hysteresis-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	polar-domain-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	stable-opposed-polarization-classes	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	field-driven-domain-path-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	hysteresis-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c8c27f9a90d8b0fdf120aa63794dad5be1b2554f0f0e6cd36b647d26c59abc87.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:8162c61883e1b3fe67fef308ebacf2e9ed54670ae52ee7067edf8d80ca467aa7.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d63ee927baa6c8404ff75eb6407622cdfafe2ad7d894e0556ae9b7572cd4bab7.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1f47c545d9bf91517601315bcd5136409a1a228df3c7edab9920e71bc1afb39d.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:5467315283f5f64b807d962309ddcf27cd5f1c65a26683e6d316faf565008499; certificate

sha256:e8f91344b1dc05fb3c8e81712d23b08b06cb1612c30f64a3a83a49a66b4e595f; external-validation

hash sha256:f6cc2ab5195e78512f9c6a49d2fc79945b2b79a484890d38ade80e7a93576c82.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:950537b9a096411ed993b95bfaeb1b77308569ce1b70e1834e2c82c09f6a60c2. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-func-ferroelectric-001-authority-correspondence: predicted polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered ferroelectric order and switching observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves polar-domain-network, stable-opposed-polarization-classes, field-driven-domain-path-retained and hysteresis-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:da506701f72bb70612bca1f2478c1b91e30dbc5c121e71ccb91e601b080acb0d; derivation seal sha256:8611bbfd8a418797896a462e54d5cfd398aa65d0e3bd39effa88daa38add9c80; engine receipt sha256:36580c9520ef5c2f5e57824b854c8e16716aacb4b2165fe37a23e6a3d14fbd55 at receipts/engine/model_admitted/SFT-MAT-FUNC-FERROELECTRIC-001-36580c9520ef5c2f.json; empirical validation sha256:74498fd9ace1ce0b625e2b44cc82a6c1ebd9faec3e735e2f19fca20778d3a743; measurement receipt sha256:d6c8aaadf1d41547c53d2d042b751edc04af4f75c71c14f1bb8dd0b08fed2817; isolation sha256:1d266587e8d483f234e7469691b423578536738ae724f1795770a9a17601a8cf; custody sha256:4ae27b63ebc319dcd721ece0a9d989885ce280368273a53f8e1a402036bc0ad9.

87. Thermoelectric coupling

Claim identity: SFT-MAT-FUNC-THERMOELECTRIC-001

Question and exact theorem. Thermoelectric response is the joint conserved relation between energy-gradient transport and held-charge transport on the same material network.

charge-and-energy-carrier-network__temperature-gradient-charge-transfer__joint-energy-charge-ledger__direction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

le

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves charge-and-energy-carrier-network, temperature-gradient-charge-transfer, joint-energy-charge-ledger and direction-condition-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	charge-and-energy-carrier-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	temperature-gradient-charge-transfer	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	joint-energy-charge-ledger	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	direction-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: charge-and-energy-carrier-network__temperature-gradient-charge-transfer__joint-energy-charge-ledger__direction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5b2e498cf9e75a44a5a5fbc0345af47f4deb173f28fe5dc65f0ae665ee2fe21.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d7fd528b32ee51d1a154d5a48b34ed9216d7534090f6e967ecc4f8a632ad9f69.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1b3c38b099cd9b9ec5998f66e220e1435bbe784b5c2e30a5f0d2835f7280ed36.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7c8fad072572679744edac6081724e4c31ee8f824747cf90d31041aa70856b65.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:52d94c68bc0c817a04d1992ddf14e2f8e7371008376f88e58b6b2212c7c9659c; certificate

sha256:e2af0809eb20acbef176b000e81b81f08d1d24274e74136cc94a9af4ceb83c5d; external-validation hash sha256:3a69e07d1138a3073f22b46120a349145fdd7034830be55e85a4b4271c1f9a61.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:49468d618ab6361c0f75798af8c71ae5516ac5325bf6b6c287bd154616fb8059. Target content

opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716ccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- sft-mat-func-thermoelectric-001-authority-correspondence: predicted charge-and-energy-carrier-network__temperature-gradient-charge-transfer__joint-energy-charge-ledger__direction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived charge-and-energy-carrier-network__temperature-gradient-charge-transfer__joint-energy-charge-ledger__direction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered thermoelectric coupling observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves charge-and-energy-carrier-network, temperature-gradient-charge-transfer, joint-energy-charge-ledger and direction-condition-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:95f83d012fde1ad0ca6942fe7be129412ba9e0e36dbbeb433f5c4d778a9daec4; derivation seal sha256:1c5893f35485b76b913f5de741de5cf87349e4cb7eb89f62b4687a7902a62dc7; engine receipt sha256:c861a777de27b860ffce46411542047fe44e88ddd68fcae21cc9d48d658184cd at receipts/engine/model_admitted/SFT-MAT-FUNC-THERMOELECTRIC-001-c861a777de27b860.json; empirical validation sha256:6c525fdf2f2a8591cbb0fe07899e059cc14b5446ff5d88a1b60538aa0f4d771b; measurement receipt sha256:827f8e8fa21a88f168df8e3788a9549165f2e074652d33693846f2b23bb5e6df; isolation sha256:81aad8c3ba087051b0eca6244280ea87263116cda7f57708077f8efdcc15aba9; custody sha256:edaa008f449455c72fd96ef412f4bab31a0314fc502f8bea8e75a80e3280d417.

88. Photonic material organisation

Claim identity: SFT-MAT-FUNC-PHOTONIC-001

Question and exact theorem. A photonic material organizes electromagnetic recurrence support so that allowed propagation classes and structural gaps follow from complete generated geometry and boundary relations.

electromagnetic-cell-network__wave-recurrence-and-gap-support__geometry-frequency-and-boundary-retained__spectral-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves electromagnetic-cell-network, wave-recurrence-and-gap-support, geometry-frequency-and-boundary-retained and spectral-scale-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	electromagnetic-cell-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	wave-recurrence-and-gap-support	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	geometry-frequency-and-boundary-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	spectral-scale-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: electromagnetic-cell-network__wave-recurrence-and-gap-support__geometry-frequency-and-boundary-retained__spectral-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed `true`.
- relation: the generated relation and organisation are separately retained; passed `true`.
- observation: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c32943d095b89193d400a5b2631eca6463a05ce83ff023c092e18e7ba3433f6f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:50acfe0be40722d399943981a25721636a151897789b475bafcd1015ad34e87.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:61b3cdac655eb4a710cc30fadb2ad4d1adc5a38bc277fcf964e0cb41b326b6f.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ee7373f233a08e9471fb6686dcb5b89984f07cb5f74d32b13cd25379e37c4c9b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:22bfd80dd14b7366e546e51abc799a76a5ab463562e49218bb9c373fce71cd15; certificate
sha256:f38b1b5ef15f21b39d7a46ca08c82d87709a63df3623fc8d55d110c32d94ea34; external-validation
hash sha256:65befe7fa71ec4b61efbb9c9025b153fa0b8b56543006349263735440208c414.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:8f8bf66262959ed2118bcb56bb0a0fa706206788b7efc3da91b17c7d057f1c92. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-OPTICAL-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot
experiments/external_sources/materials/snapshots/nist-optical-materials.html;
sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of
refraction, absorption, reflectance, dielectric response and photonic crystals.

Comparison and custody records:

- sft-mat-func-photonic-001-authority-correspondence: predicted electromagnetic-cell-network__wave-recurrence-and-gap-support__geometry-frequency-and-boundary-retained__spectral-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived electromagnetic-cell-network__wave-recurrence-and-gap-support__geometry-frequency-and-boundary-retained__spectral-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered photonic material organisation observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves electromagnetic-cell-network, wave-recurrence-and-gap-support, geometry-frequency-and-boundary-retained and spectral-scale-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:b8359125c415f70252bad5e8662f32a4a0411fd4ecab90b210f238e5d4eff3f7; derivation seal sha256:d5719938752bfff0b363ba54eb5c0394c4b5fa74ca27361fee3d5493208f569e9; engine receipt sha256:b63114cadba712eeb0a65b991315d9035f61024e44e181fc9dbb26f93b2947c7 at receipts/engine/model_admitted/SFT-MAT-FUNC-PHOTONIC-001-b63114cadba712ee.json; empirical validation sha256:d70953f2830575c85ac5367761e35a0a7daa230e39683dbb25e951db3f69a00b; measurement receipt sha256:076b47a085e881f1daba570530905b12afa40b4cfd62cb8dd1c26da66228211a; isolation sha256:95a9d5ee8ac6b7e943fc57db2431acee31ac9e4115a590b4eae9b3b301dc8a94; custody sha256:6a0be96588221e8b3bbc5e6b3e84565643db68aecf32f4b3ee6a428c2135b7bb.

89. Functional magnetic material response

Claim identity: SFT-MAT-FUNC-MAGNETIC-001

Question and exact theorem. A functional magnetic response is the complete relation among applied field, domain transition path, retained orientation, loss and remanence at declared frequency and temperature.

moment-domain-and-field-carrier__field-history-response-map__domain-wall-loss-and-remanence__frequency-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves moment-domain-and-field-carrier, field-history-response-map, domain-wall-loss-and-remanence and frequency-temperature-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	moment-domain-and-field-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	field-history-response-map	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	domain-wall-loss-and-remanence	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	frequency-temperature-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: moment-domain-and-field-carrier__field-history-response-map__domain-wall-loss-and-remanence__frequency-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:db19f1a2cc9d5f45d18837e5d0172d17a4f6e0b27b08b0096dc0622a8dfec16.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:5ee5c3258afeb367bb0cd4365b9db18fdb8207765c46dad0c76f440d2f2a0614.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ae310bd447bd0da6fef8f21bcd1ce0c0df36b3523c38cc3df2dd5e5be48d62f.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:334673e317625efa1bbb433ce8d7cdec742cf726cb5311cb9c5b0b1bbd9bd85f.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:cd8911fe8913dbfca7c344e88e4917190bbc07e7d04e1403ca7f513d06fce2f7; certificate
sha256:a0b1af464cab9fc6344d8193cfe38a2651617f9c887720c78ad0eae55b49a065; external-validation
hash sha256:d28ec86f9c2ea983d46ed613c328135ead66046bc3f0baf7bb15769aeafd3b8.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:8ab5d335d373302646984a94eb1f45dccc0a137eea7206abf17029cac4627ff1. Target content
opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources
/materials/snapshots/nist-functional-electronic-materials.html;
sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants,
defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-func-magnetic-001-authority-correspondence: predicted moment-domain-and-field-carrier__field-history-response-map__domain-wall-loss-and-remanence__frequency-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived moment-domain-and-field-carrier__field-history-response-map__domain-wall-loss-and-remanence__frequency-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered functional magnetic material response observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves moment-domain-and-field-carrier, field-history-response-map, domain-wall-loss-and-remanence and frequency-temperature-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:6d22b16de46e8b5d63aa64b2a61e80240e18216a5feb89964e7099a2854d017c; derivation seal sha256:395244dc506ff152b9c09887ab33883310f04880d049893f0e8d4cd625914bed; engine receipt sha256:1ed29896c9091b1948b2d855f0c0d51738df688e864db22fcde9f344da2dfca4 at receipts/engine/model_admitted/SFT-MAT-FUNC-MAGNETIC-001-1ed29896c9091b19.json; empirical validation sha256:b117f217af70207aaacd9b09bb785b027e22a059780fa54a291a85793545b6cc; measurement receipt sha256:5d4519d974b7c1ab4790a14e7d29d2343daa1b3b423846555bf212c4f5e7716a; isolation sha256:58661800655851db8c9f5dc1c1717efb27bcdcf0624e4d19967f68b2400e53f2; custody sha256:2b92d674f030c9acc3cb3072a05f64cf6665bb62637a7c91f52e7aad01229652.

90. Nanoscale material boundary

Claim identity: SFT-MAT-FUNC-NANOMATERIAL-001

Question and exact theorem. A nanomaterial regime begins when boundary and finite-support distinctions remain in the material state/property census and cannot be closed by the bulk observation quotient.

finite-constituent-boundary-carrier__boundary-support-comparable-to-interior__discrete-state-and-interface-retained__size-shape-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves finite-constituent-boundary-carrier, boundary-support-comparable-to-interior, discrete-state-and-interface-retained and size-shape-resolution-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	finite-constituent-boundary-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	boundary-support-comparable-to-interior	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	discrete-state-and-interface-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	size-shape-resolution-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: finite-constituent-boundary-carrier__boundary-support-comparable-to-interior__discrete-state-and-interface-retained__size-shape-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:31c76c3f386bed3a498183b0f457baf72566e4999e3bbfde6be9cf01415a227d.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:dc40a0ec44efafc6e16bc5818d4e5876e2169c4a4d15e75a7c44b057a6232ce2.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:42da32bcf34d016e4fb04da37d8e1a77bed1c2942845d0bbdb58abb5d23c657f.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e95636207c13c2d934d5cc7503c8af418875ef0267cb94376f56979200802167.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:69caf8202ee5680eb52cb0ab81104a75f12ccbafe87e08f079578e9953f1ba93e; certificate
sha256:77995cabcd5179d50722c7318e402d37e31b3cbe2ac7ad4afa6a53c415d9547; external-validation
hash sha256:5a9764c3798b5bc1028e1d681d140e937971b971a0207684cd14a01d4c057987.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is
sha256:453e167dbba6de5524a9410a9889cf4b02c39e7b2c962716c0a287338c24ee3e. Target content
opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-NANO-PROTOCOLS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/mml/nano-measurement-protocols>; snapshot
experiments/external_sources/materials/snapshots/nist-nano-protocols.html;
sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; scope:
nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

Comparison and custody records:

- sft-mat-func-nanomaterial-001-authority-correspondence: predicted finite-constituent-boundary-carrier__boundary-support-comparable-to-interior__discrete-state-and-interface-retained__size-shape-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived finite-constituent-boundary-carrier__boundary-support-comparable-to-interior__discrete-state-and-interface-retained__size-shape-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered nanoscale material boundary observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves finite-constituent-boundary-carrier, boundary-support-comparable-to-interior, discrete-state-and-interface-retained and size-shape-resolution-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:a772561a12e9f1e33384ed3b074205b589ac789c8348c602e27c320ce8c2b925; derivation seal
sha256:2cc9bc4362cbc4dee60c592336a2df482696e5330953579100fc5e1f15ceb9b1; engine receipt
sha256:a815d0fcf264919778c652d76e2b46dd60f5f75df6e79c325e98bc7192cc85b3 at

receipts/engine/model_admitted/SFT-MAT-FUNC-NANOMATERIAL-001-a815d0fcf2649197.json;
 empirical validation sha256:092002f7236e19aaf27f182f5f1ae15d177a80a287affd058a15195835790be3;
 measurement receipt
 sha256:3d6f316b710b89615ba3a69f4c6b1bfff427dcfdbf675756a2f8359d144c7f311; isolation
 sha256:35d270615bc1e69314b08304d5ffda3838eaf6b7e856a0a86c3d8d89b2c1b02c; custody
 sha256:7d55c2da7e6abd7970c8a0ba99fd51def81599e28c42cc318be5f4e2b0bcaa3a.

91. Biomaterial interface boundary

Claim identity: SFT-MAT-FUNC-BIOMATERIAL-001

Question and exact theorem. A biomaterial claim is a joint material/biological interface relation with explicit exposure, transfer, response and function boundaries; biological outcomes cannot select the material law.

material-biological-interface-carrier__bidirectional-response-and-transfer__material-and-biological-identities-retained__exposure-function-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves material-biological-interface-carrier, bidirectional-response-and-transfer, material-and-biological-identities-retained and exposure-function-bounded.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	material-biological-interface-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	bidirectional-response-and-transfer	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapse d	material-and-biological-iden- tities-retained	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary- unrecorded	exposure-function-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state- transition-record	complete-state-transition-bo- undary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-s elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in- stance-lookup	positive-finite-successor-cl osure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or- -extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: material-biological-interface-carrier__bidirectional-response-and-transfer__material-and-biological-identities-retained__exposure-function-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organisation are separately retained; passed true.
- observation: the observation boundary is distinct from the material organisation; passed true.

Shared claim-record clause MAT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c77ecdcc4cee7390b374f11dac9a22f81f990b7789c59a924c07fe55f79a889e.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:cb1ea4682cb262aad2796a49a64cedd62acb16f9534428567b7cd8dc5a118235.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1d101e88432f57335048d4d9f0a2b09fe305b1cf62dd6fb95fe211598e2553a4.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:da528078953f494ae3b594ae054f42104f1e21c78ec26b6bae2a906ca4dc1689.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash
sha256:01f66c16a5ba692961fbabd31a4335924f06b60bb42767fddf09c7682a57a104; certificate
sha256:ea0c22c327dabffe543439a4565a94849959aebfa0e4fe2e9c48504b4de0085; external-validation
hash sha256:4f1fda699d5d3bb69d2a10eac5fcdeb43aa1257bb54bf96f88d3f002556b7209.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256:bf5574013b2ace3aec4970b1b058bd142529651a68815439d9d26d155166b6f2. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-NANO-PROTOCOLS-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/mml/nano-measurement-protocols>; snapshot
experiments/external_sources/materials/snapshots/nist-nano-protocols.html;
sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; scope:
nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.
- NIST-POLYMER-COMPOSITES-2026-07-24 - National Institute of Standards and Technology;
<https://www.nist.gov/programs-projects/polymer-composites>; snapshot
experiments/external_sources/materials/snapshots/nist-polymer-composites.html;
sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer
matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-func-biomaterial-001-authority-correspondence: predicted material-biological-interface-carrier__bidirectional-response-and-transfer__material-and-biological-identities-retained__exposure-function-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-biological-interface-carrier__bidirectional-response-and-transfer__material-and-biological-identities-retained__exposure-function-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered biomaterial interface boundary observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves material-biological-interface-carrier, bidirectional-response-and-transfer, material-and-biological-identities-retained and exposure-function-bounded.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest
sha256:1f6cf609fdd14e3ef9884853b588680c81246b326cb6628497bb7524f680b695; derivation seal
sha256:e04744de5df085dc908548847c8701d22343f9735c7e28a67250206beeddc22f; engine receipt
sha256:5ed0cf28440ad78632017718d57debfcdda937fdc7acb83e53a782aefad24d3 at
receipts/engine/model_admitted/SFT-MAT-FUNC-BIOMATERIAL-001-5ed0cf28440ad786.json;
empirical validation sha256:9810925c7e30dd79c38a3577eca33c9fbd681aded4f9d2526db42d8ebbf8359c;
measurement receipt
sha256:b61bf887f8ab6b53a09748fbffeff3dad5477ea07a50a081cc25e46e1899ac14; isolation

sha256:d4685e1d38fb5855d3c4519cb602ff7498c3fc36794645bd703176a74392c664; custody
 sha256:c8acfc9ecf4861523b5dcfff4402a370a9b10fde0758517ffbf0bd42fbd9ed50.

92. Material lifecycle and recyclability

Claim identity: SFT-MAT-SUST-LIFECYCLE-001

Question and exact theorem. A material lifecycle is the complete source-to-processing-to-use-to-recovery flow ledger; recyclability requires an explicit identity-preserving recovery path and records every transformation, loss and quality boundary.

```
source-process-use-recovery-carrier__complete-material-flow-path__loss-transformat
ion-and-quality-ledger__system-boundary-declared__complete-state-transition-bounda
ry-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-
rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Shared claim-record clause MAT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves source-process-use-recovery-carrier, complete-material-flow-path, loss-transformation-and-quality-ledger and system-boundary-declared.

Shared claim-record clause MAT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	source-process-use-recovery-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	complete-material-flow-path	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapse	loss-transformation-and-quality-ledger	Collapsing organisation merges materials states that the question requires to remain distinct. Every organisation, interface, history or boundary distinction required by the claim remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	system-boundary-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-elected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-in-stance-lookup	positive-finite-successor-closure	One favourable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `source-process-use-recovery-carrier__complete-material-flow-path__loss-transformation-and-quality-ledger__system-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organisation, observation boundary and proof record.

Shared claim-record clause MAT-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organisation are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organisation; passed `true`.

Shared claim-record clause MAT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:42532e43e708387eab368ef316263c2ed870195ef765bab6a76e7b16dd4a334b.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1360d7af1f0267dfc482885e6b6289c9a952d85d977db53826e457f0fe3d337a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:599f2eca8225f1d225f32ee678bb4237582333e452455db62034e12fdbfc6981.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:29bef3fbd7454e3857b3b0be39cd76cdc6725365e836fe943c4c620e0cc2da2d.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:8c2a8e8674efd78cc9aeb6e6ff178591c51fdae5c0f7573769bd763ad4c69bca; certificate
sha256:7271f451ff5b28ae59d344e9b829463616b6e8225c85562520df817c592ef606; external-validation
hash sha256:a34eca90ac5590c6a6444768b73d44224b164109b923c033cce4da264b285775.

Post-seal empirical correspondence. This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256:d1aa7f6fa9c4241ced6e2ef6f2f5900b28d79e8a963460e50d55a0fabda4f634. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-MATERIALS-DIVISION-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/mml/materials-science-and-engineering-division>; snapshot experiments/external_sources/materials/snapshots/nist-materials-division.html; sha256:fac19ae0959946a8e73f970bf76c73325d93f4671ded97f8699dfcc5e1bf53e3; scope: materials measurement, composition, interfaces, processing, damage, classes and lifecycle.
- NIST-NANO-PROTOCOLS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/mml/nano-measurement-protocols>; snapshot experiments/external_sources/materials/snapshots/nist-nano-protocols.html; sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; scope: nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

Comparison and custody records:

- sft-mat-sust-lifecycle-001-authority-correspondence: predicted source-process-use-recovery-carrier__complete-material-flow-path__loss-transformation-and-quality-ledger__system-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived source-process-use-recovery-carrier__complete-material-flow-path__loss-transformation-and-quality-ledger__system-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavourable control rejected

Falsification boundary: The claim fails if a complete registered material lifecycle and recyclability observation lacks any sealed carrier, relation, organisation or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favourable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves source-process-use-recovery-carrier, complete-material-flow-path, loss-transformation-and-quality-ledger and system-boundary-declared.

Immutable evidence identities. Pre-source branch seal

experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:ae6eca4c43b073a3ed7eda20aa9836c009c9bb3b05a6f443ab2d24738fd48e57; derivation seal sha256:3e9a7308036c4ef8d2b7fc56f68e6b33cff28eab1ed004ae9f243609dc70434e; engine receipt sha256:0046611012fa957e6166fedel24a4a43e24de052957566bb7411afb54ed2bc43 at receipts/engine/model_admitted/SFT-MAT-SUST-LIFECYCLE-001-0046611012fa957e.json; empirical validation sha256:06b1e6c1f4a57c40bb00be257aff0e6c57a0e88208a7fb723ec9fb26800cfbc4; measurement receipt sha256:4feb710f4285d4f1a27cabaeca9e8b34fe808c3da8ccd7e9033c74c077225620; isolation sha256:b568d984120dc112f697754f4fb3c44d6f111e8acca6c1dcdc953243ff61c6f1; custody sha256:33b0dc3bcf6a48ff0c1dd145f2729a0d5e168d02649d0ced23e6809078094545.

21. Integrated reconciliation table

- Step 47: cubic lattice and six nearest neighbours -> SFT-MAT-CRYST-CUBIC-COORDINATION-001
- Step 49: crystalline order and crystallographic restriction -> SFT-MAT-CRYST-TRANSLATION-001, SFT-MAT-CRYST-ROTATION-RESTRICTION-001, SFT-MAT-CRYST-SYSTEMS-001, SFT-MAT-CRYST-BRAVAIS-001
- Step 52: paired-carrier superconducting response -> SFT-MAT-SC-PAIR-001, SFT-MAT-SC-ZERO-RESISTANCE-001, SFT-MAT-SC-MEISSNER-001, SFT-MAT-SC-FLUX-QUANTIZATION-001, SFT-MAT-SC-JOSEPHSON-001
- Step 54: electronic bands, gaps and transport classes -> SFT-MAT-ELEC-BAND-GAP-001, SFT-MAT-ELEC-CONDUCTOR-CLASS-001
- Step 72: three acoustic lattice-excitation branches -> SFT-MAT-CRYST-PHONON-001
- Step 74: aligned and opposed magnetic order -> SFT-MAT-MAG-FERROMAGNETISM-001, SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- Step 75: semiconductor carrier types, doping and junction balance -> SFT-MAT-ELEC-CARRIER-DUALITY-001, SFT-MAT-SEMI-DOPING-001, SFT-MAT-SEMI-PN-TYPE-001, SFT-MAT-SEMI-JUNCTION-001, SFT-MAT-SEMI-TRANSPORT-001
- Step 133: quasicrystalline order without periodic translation -> SFT-MAT-CRYST-ROTATION-RESTRICTION-001, SFT-MAT-CRYST-RECIPROCAL-001, SFT-MAT-CRYST-QUASICRYSTAL-001
- Step 137: superfluid flow and quantized circulation -> SFT-MAT-SF-SUPERFLUID-001, SFT-MAT-SF-CIRCULATION-001
- Step 143: topological class and bulk-boundary protection -> SFT-MAT-TOPO-INVARIANT-001, SFT-MAT-TOPO-BULK-BOUNDARY-001
- Step 193: elastic, plastic, slip, strength, fracture, fatigue and creep response -> SFT-MAT-MECH-STRESS-STRAIN-001, SFT-MAT-MECH-ELASTICITY-001, SFT-MAT-MECH-PLASTICITY-001, SFT-MAT-MECH-SLIP-001, SFT-MAT-MECH-MODULUS-001, SFT-MAT-MECH-STRENGTH-HARDNESS-001, SFT-MAT-MECH-FRACTURE-001, SFT-MAT-MECH-FATIGUE-CREEP-001
- Step 291: quasicrystal component of the mixed materials-and-astrophysics step -> SFT-MAT-CRYST-QUASICRYSTAL-001; routed remainder: Planetary and Tully-Fisher claims are not Materials results and remain assigned to Astronomy and Cosmology.

22. External-source ledger

- BIPM-JCGM-VIM-TRACEABILITY-2026-07-24 - <https://jcg.m.bipm.org/vim/en/2.41.html>; snapshot experiments/external_sources/materials/snapshots/bipm-vim-traceability.html; sha256:3c06dca2ea8ad71516b027b9979e7a8381edd1576606ad10a379591c4cb6dba4; measurement result, reference chain, calibration and uncertainty.
- NIST-REFERENCE-MATERIALS-2026-07-24 - <https://www.nist.gov/reference-materials>; snapshot experiments/external_sources/materials/snapshots/nist-reference-materials.html; sha256:2ff393988c4880068ca103e365b0f1e6aed5a02a493b33507f6684a2d9271db9; material identity, property characterization, uncertainty and traceability.
- NIST-MATERIALS-DIVISION-2026-07-24 - <https://www.nist.gov/mml/materials-science-and-engineering-division>; snapshot experiments/external_sources/materials/snapshots/nist-materials-division.html; sha256:fac19ae0959946a8e73f970bf76c73325d93f4671ded97f8699dfcc5e1bf53e3; materials measurement, composition, interfaces, processing, damage, classes and lifecycle.
- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24 - https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html; sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; lattice

generation, rotation, seven crystal systems and fourteen Bravais classes.

- NIST-BRAVAIS-CLASSIFICATION-2026-07-24 - <https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification>; snapshot experiments/external_sources/materials/snapshots/nist-bravais-classification.html; sha256:82fc7e011a992352d5200131028339a83eca0938c7462131ada2dcc059f38c91; diffraction classification across all fourteen Bravais lattices.
- NIST-SHECHTMAN-QUASICRYSTALS-2026-07-24 - <https://www.nist.gov/nist-and-nobel/dan-shechtman>; snapshot experiments/external_sources/materials/snapshots/nist-shechtman-quasicrystals.html; sha256:911c06e466a95d58749422c56cd4e33fa1fbee04ec06977af428d7dac23ae0e; periodic rotation orders and fivefold quasicrystal observation.
- NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24 - <https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot experiments/external_sources/materials/snapshots/nist-quasicrystal-measurement.html; sha256:a1b7cc0a5882502bb31b9706225b926c9ffbf1e8d93ea5c0043249376c4d6ff18; microscope confirmation, nonrepeating order and fivefold symmetry.
- NIST-POINT-DEFECTS-2026-07-24 - <https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot experiments/external_sources/materials/snapshots/nist-point-defects.html; sha256:b885250bd645ed8eae5a29d99ea4436b691f77c17c74a457acb6acf88335a; vacancies, interstitials, substitutions, diffusion, composition and defect metrology.
- NIST-TEXTURE-PHASE-FRACTION-2026-07-24 - <https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html; sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; grains, phase fractions, texture, deformation and uncertainty.
- NIST-MULTISCALE-MATERIALS-2026-07-24 - <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.
- NIST-SEMICONDUCTORS-2026-07-24 - <https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot experiments/external_sources/materials/snapshots/nist-semiconductors.html; sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; semiconductor structural, nanomechanical, thermal and spectroscopic measurement.
- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.
- NIST-SUPERCONDUCTING-GRAIN-BOUNDARIES-2026-07-24 - <https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature>; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-grain-boundaries.html; sha256:1309d6599f6894e517dac8f3adc0b6b8506842cdcccc680329007351cde6abf2; grain boundaries, strain, dislocations, current flow and superconducting response.
- NIST-SUPERCONDUCTING-FLUX-2026-07-24 - <https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-flux.html; sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; zero resistance, Meissner expulsion, quantized flux and vortices.

- NIST-JOSEPHSON-STANDARD-2026-07-24 - <https://www.nist.gov/news-events/events/josephson-volt-nists-first-quantum-electrical-standard>; snapshot experiments/external_sources/materials/snapshots/nist-josephson.html; sha256:371bb54f09d8e66d74cda08e66518b44ebf752445ee2a049b26a76fd669487e0; superconductive tunnel junctions and quantized Josephson voltage.
- NIST-SUPERFLUID-CIRCULATION-2026-07-24 - <https://www.nist.gov/news-events/news/2012/11/first-controllable-atom-squid>; snapshot experiments/external_sources/materials/snapshots/nist-superfluid-circulation.html; sha256:49a5edb2dc307d0849d5fb5a245be46166dbffe2c50f2e9422a78f139aa5f8a0; superfluid flow, vortices and discrete quantized circulation.
- NIST-TOPOLOGICAL-INSULATORS-2026-07-24 - <https://www.nist.gov/programs-projects/topological-insulators>; snapshot experiments/external_sources/materials/snapshots/nist-topological-insulators.html; sha256:4cbe823dffeabd728844a897aalc5a222dfc3aef465263a06501974e194df449; topological invariants, insulating bulk, band gap and metallic boundaries.
- NIST-FATIGUE-FRACTURE-2026-07-24 - <https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html; sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; processing-structure-property-performance, defects, fatigue and fracture.
- NIST-POLYMER-COMPOSITES-2026-07-24 - <https://www.nist.gov/programs-projects/polymer-composites>; snapshot experiments/external_sources/materials/snapshots/nist-polymer-composites.html; sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.
- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - <https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716ccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.
- NIST-CERAMIC-ADDITIVE-2026-07-24 - <https://www.nist.gov/programs-projects/additive-manufacturing-ceramics>; snapshot experiments/external_sources/materials/snapshots/nist-ceramic-additive.html; sha256:540ef912a827c1834a113352ac0305c7e8d25a76283dcc1bcbab48f0b826a3e9; ceramic feedstock, defects, sintering, densification, phases and microstructure.
- NIST-POLYMER-PROCESSING-2026-07-24 - <https://www.nist.gov/programs-projects/polymer-advanced-manufacturing-and-rheology>; snapshot experiments/external_sources/materials/snapshots/nist-polymer-processing.html; sha256:11279429040fb5daff9108435f0c510fc12d89e09710e799b1a6182f63a63a18; polymer processing, crystallization, rheology, kinetics and recycling.
- NIST-SURFACE-DAMAGE-2026-07-24 - <https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project>; snapshot experiments/external_sources/materials/snapshots/nist-surface-damage.html; sha256:7f0a61e225462f1514a9b2b9e459f5a0727ef2dc291bbcf5e83626f52359169d; surface/bulk distinction, degradation, radiation, moisture, cracking and nanofiller release.
- NIST-NANO-PROTOCOLS-2026-07-24 - <https://www.nist.gov/mml/nano-measurement-protocols>; snapshot experiments/external_sources/materials/snapshots/nist-nano-protocols.html; sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.
- NIST-SOLIDIFICATION-2026-07-24 - <https://www.nist.gov/publications/solidification-0>; snapshot experiments/external_sources/materials/snapshots/nist-solidification.html; sha256:d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c; solidification

transport, nucleation, growth, interfaces, segregation and processing.

- NIST-METAL-ADDITIVE-CORROSION-2026-07-24 -
<https://www.nist.gov/programs-projects/additive-manufacturing-metals>; snapshot experiments/external_sources/materials/snapshots/nist-metal-additive-corrosion.html; sha256:a1bb9a6bc22eb85fb7a1b3c7d0ac4bc0837d68832b7b7c6f357a68ea6cde0322; phase transformation, heat treatment, corrosion and environmental cracking.
- NIST-NANOTRIBOLOGY-2026-07-24 -
<https://www.nist.gov/programs-projects/nanotribology-nanomanufacturing-archived>; snapshot experiments/external_sources/materials/snapshots/nist-nanotribology.html; sha256:3823f313343779c2a4a74a3e251a620319c89fef2142f2b1e58fa03250fec523; interacting surfaces, relative motion, friction, adhesion, lubrication and wear.
- NIST-OPTICAL-MATERIALS-2026-07-24 -
<https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot experiments/external_sources/materials/snapshots/nist-optical-materials.html; sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; index of refraction, absorption, reflectance, dielectric response and photonic crystals.
- NIST-SIX-NEIGHBOUR-MESH-2026-07-24 -
https://math.nist.gov/oommf/doc/userguide20a0/userguide/Standard_Oxs_Ext_Child_Clas.html; snapshot experiments/external_sources/materials/snapshots/nist-six-neighbour-mesh.html; sha256:505ca000d91c5ef134836cbc75307df1584750dfbfb2faf5c4e55756d054f5c2; six nearest neighbours as forward/backward positions on three coordinate axes.
- NIST-POLYMER-INTERFACE-CONSORTIUM-2026-07-24 -
<https://www.nist.gov/el/mssd/polymer-surfaceinterface-consortium>; snapshot experiments/external_sources/materials/snapshots/nist-polymer-interface-consortium.html; sha256:2d48772b1b901e657fa86e4ee28fad15e176dd675911ce8322b54cc718c32635; surface damage, interfacial adhesion and interface-property measurement.
- NIST-SEMICONDUCTOR-CARRIERS-2026-07-24 -
<https://www.nist.gov/news-events/news/2017/03/testing-performance-semiconductors-light>; snapshot experiments/external_sources/materials/snapshots/nist-semiconductor-carriers.html; sha256:05c774e4a30e6009f031b3064b16319d181ddac92e6d088a01bb6fe362172cc5; electron and hole carrier signs, doping, transport and Hall measurement.
- NIST-ANTIFERROMAGNETIC-COUPPLING-2026-07-24 -
<https://www.nist.gov/ncnr/acns-2020-tutorial-ii-practical-approach-fitting-neutron-reflectometry-data/understanding>; snapshot experiments/external_sources/materials/snapshots/nist-antiferromagnetic-coupling.html; sha256:b8ce5105622c02b84bffe643693c05c199df57beae79bd9c5e29f26c4360e85e; polarized-neutron measurement of magnetic structure and antiferromagnetic coupling.
- NIST-GLASS-TRANSITION-2026-07-24 -
<https://www.nist.gov/publications/glass-transition-its-measurement-and-underlying-physics>; snapshot experiments/external_sources/materials/snapshots/nist-glass-transition.html; sha256:2d6b99e46aae9d3c6a503494eedd17db45d4b48a633bb254cd3a6a5195440777; thermodynamic and kinetic glass-transition measurement in glass-forming materials.
- NIST-QUASICRYSTAL-APERIODIC-2026-07-27 -
<https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot experiments/external_sources/materials/snapshots/nist-quasicrystal-measurement.html; sha256:a1b7cc0a5882502bb31b9706225b926c9fffb1e8d93ea5c0043249376c4d6ff18; post-seal claim-specific authoritative comparison.
- NIST-PHONON-THERMAL-LIMITS-2026-07-27 -
<https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication960-11.pdf>; snapshot experiments/external_sources/materials/snapshots/nist-phonon-thermal-limits-2026-07-27.txt; sha256:208d3ff48584c6548ccb00d3b1559ba63d66b1fabe027af9bb499504d5cad64b; post-seal claim-specific authoritative comparison.

- NIST-PN-RECTIFICATION-2026-07-27 - <https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir4414.pdf>; snapshot experiments/external_sources/materials/snapshots/nist-pn-rectification-2026-07-27.txt; sha256:ca301b8fc47ba509909982ca2b077c238b92f3ff4978ed7377d086071c519ba4; post-seal claim-specific authoritative comparison.
- NIST-SUPERCONDUCTING-ISOTOPE-2026-07-27 - https://nvlpubs.nist.gov/nistpubs/jres/094/jresv94n3p147_A1b.pdf; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-isotope-2026-07-27.txt; sha256:48fa89f9f63d467e9259e766f81fd8f854924ad479ec0006d14075cfe48d3ae3; post-seal claim-specific authoritative comparison.
- NIST-FERRIMAGNETIC-SUBLATTICES-2026-07-27 - <https://www.nist.gov/publications/thickness-dependent-magnetism-variation-ferrimagnetic-rare-earth-transition-metal-fe1>; snapshot experiments/external_sources/materials/snapshots/nist-ferrimagnetism-2026-07-27.html; sha256:685872920e5663b2ec71636b953d0e7ddec3d1cb679fb4f17bb94b6d6d811795; post-seal claim-specific authoritative comparison.
- NIST-HALL-QUANTIZATION-2026-07-27 - <https://www.nist.gov/programs-projects/integer-and-fractional-quantum-hall-effect>; snapshot experiments/external_sources/materials/snapshots/nist-hall-quantization-2026-07-27.html; sha256:4536e34ab401e72216f8b9010bafc9f1cacdbbfc47ca8fc998b781fa9125994c; post-seal claim-specific authoritative comparison.
- NIST-WATER-THERMOPHYSICAL-2026-07-27 - <https://webbook.nist.gov/cgi/cbook.cgi?ID=C7732185&Mask=FFF&Units=CAL>; snapshot experiments/external_sources/materials/snapshots/nist-water-complete-2026-07-27.html; sha256:ba9fc5225bc8d0d89cbe15703bff2d9dc06ef64244598c420c7666931f1befbf; post-seal claim-specific authoritative comparison.
- NIST-LIQUID-WATER-DENSITY-2026-07-27 - <https://www.nist.gov/document/jpcrd38200921ppdf>; snapshot experiments/external_sources/materials/snapshots/nist-liquid-water-properties-2026-07-27.txt; sha256:b3af256bcffdf7ff87fbf2aa00f545910a531d5a1b285fa9bb9f926f7afad9e24; post-seal claim-specific authoritative comparison.
- NIST-ICE-DENSITY-2026-07-27 - https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=926353; snapshot experiments/external_sources/materials/snapshots/nist-ice-properties-2026-07-27.txt; sha256:7073b61255cf73163595c9ca25646dcf2c323dd2aca942d797f9f0e4dd6c178e; post-seal claim-specific authoritative comparison.

Complete-field family: Quantitative Crystallography Diffraction Disorder

93. Exact diffraction amplitude and intensity ledger

Field: quantitative_crystallography_diffraction_disorder

Obligation: SFT-MAT-OBL-CRYS-001

Claim: SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001

Question, law and forced result. Every labelled scattering path enters one exact phase class; opposed classes cancel only into structural absence, while retained coherent path pairs force the exact intensity ledger.

CRYS-001 uniquely retains period-four-phase-cancellation-and-coherent-pair-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered CRY5-001 preservation axes before external target release.

Boundary: Every positive finite generated material carrier in CRY5-001, retaining its exact local, reciprocal, specimen, method, condition, uncertainty and proof-trace distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- period-four-phase-cancellation-and-coherent-pair-ledger
- complete-local-and-reciprocal-organization
- probe-condition-scale-and-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-material-carrier__period-four-phase-cancellation-and-coherent-pair-ledger__complete-local-and-reciprocal-organization__probe-condition-scale-and-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.`

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof sha256:6e537f274ee9db6367e1b5655c3e63189d63d1ae3ba654d075786dc3166ed8cc; generality sha256:8be3080a8f7f68cb693e975383d2ccbe361eb8c5fccc984785c531553c9c59c.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:4be4b13eec5ad3539966212b556a5ae83398962ca29eaba943f19dc03240f0f9.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:7312a19238bdc9342f9684b433f23e6eab149d04d4fc448ed2b6eb924caf43aa.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:ed36bce31b9f3ab8ba51e8c2595bece9772c69ee26c142b6cfeddf586f4fb1ec.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:8702bc1de8d732a993f8abefe48111a3616aa5e800f6e0f2d8751c24d0d86177.

Independent reconstruction and engine admission. Independent implementation

sha256:fdc016dfb48b1783f69764b017b86891c4ecd97cf9adc8fe2659ba65b524ef4b regenerated the declared census and sole survivor; independent certificate

sha256:43ec51859fa7fe34f5625a94fc2a850e8ae84e9487fe1326f9d8ba2d41b5a74d. Derivation seal

sha256:e059b84535239df7214e0305d2eae681bd1adc08259a149b9857ac68bc573cc7. Source manifest

sha256:1d969bc2fb8dd266df6bd325e65242143c606db2ee126e2dc620a466e5a4a0c2. Engine receipt

sha256:66b25d6e80686a090e0f5bb0e9c95cde9568726a998edcf6389ce22abfce5739 at receipts/engine/model_admitted/SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001-66b25d6e80686a09.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-TOTAL-SCATTERING-PDF-2014
- IUCR-TWINNED-DIFFRACTION-DATA-2022

Measured, observed, adverse and boundary records:

- CRY-001: 2 distinct registered comparisons preserved
- complete family custody: 8 captured and 2 explicitly unavailable sources
- capability-closed exact correspondence: True; omission control rejected: True
- NIST-TOTAL-SCATTERING-PDF-2014: captured; total scattering function; related to interatomic distances; favourable-use=True
- IUCR-TWINNED-DIFFRACTION-DATA-2022: captured; twin domains scatter coherently; bragg reflections and diffuse streaks; favourable-use=True

Shared claim-record clause MAT-S011 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:2b9590a0b1447f152c9760d7db8d8147d036d290901dfffb3bb863ba7d39a7c00; empirical validation

sha256:fe70c74a7dfeeee632c745ccbd95151e48b63d6af8d1a79a7a0395114e39241c; external validation

sha256:c9b42efe293f12e19b4987c12432f18e970c1f8ae33a0e609ad91b643eb5e760; isolation

sha256:586fb52eea0294a7fe3cda6d0930dbb1d55c8997c74ba248a0a1dbafa2097746; custody

sha256:cedff55ad05f782ebaa00c0cbc958f1d4927d48027c21eae2352fb48bf259b24.

Prohibited inputs.

- no V1/V2 proof artifact, conventional complex amplitude, continuum field or named equation as a premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and phase opposition are held forms rather than prohibited scalars
- no target value, official source, measured diffraction pattern or favourable specimen selects a survivor

- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary row
- no failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

94. Finite structure-factor composition

Field: quantitative_crystallography_diffraction_disorder

Obligation: SFT-MAT-OBL-CRYS-002

Claim: SFT-MAT-CRYS-STRUCTURE-FACTOR-002

Question, law and forced result. A finite structure factor is the permutation-invariant Fold composition of every scatterer weight, exact lattice position, reciprocal path and retained phase relation.

CRYS-002 uniquely retains finite-scatterer-position-phase-composition with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered CRYS-002 preservation axes before external target release.

Boundary: Every positive finite generated material carrier in CRYS-002, retaining its exact local, reciprocal, specimen, method, condition, uncertainty and proof-trace distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- finite-scatterer-position-phase-composition

- complete-local-and-reciprocal-organization
- probe-condition-scale-and-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__finite-scatterer-position-phase-composition__complete-local-and-reciprocal-organization__probe-condition-scale-and-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:184c53a6fad1987acc6556ed47dc53f71b16ad585fcd88e8848cb456aa237906; generality
sha256:408762e1248902823f24d4c4a9708faeedee12e9162a9e768b96d5a23e78804f.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:b1d50a02f871775ea161303d416a32a6d3f8937531c44f5ec4112d27cbdd2637.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:596b77f60443ba6c1868975ad042e63e08f908dac6fd7b557242a9182342ad77.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:c501cab38deb0ac6340835cace6d4bb699eefde671b31766858289765e994c22.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:eece2e80b4f708549f6e3b27db2dcc3caa778943bcc812f20663b8476c998880.

Independent reconstruction and engine admission. Independent implementation

sha256:fdc016dfb48b1783f69764b017b86891c4ecd97cf9adc8fe2659ba65b524ef4b regenerated the declared census and sole survivor; independent certificate

sha256:4d7d46f1b3674498ce4e1232456bea63bfe65152fdf580ff11bd162648017c60. Derivation seal

sha256:a2d897122201ba111efac6a752f4bc9ce1ad5a53b10eaa4e1b3029691b7dea99. Source manifest

sha256:437b3a8756bc97af522d1f37a49d5b516d6b339268f3e45dda9b031d001884c7. Engine receipt

sha256:a5389501fdf3f002a63f0852699d9aa5eaf2def492ebb793a555af20d788f7a3 at receipts/engine/model_admitted/SFT-MAT-CRYS-STRUCTURE-FACTOR-002-a5389501fdf3f002.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SP846-POWDER-DIFFRACTION-1992
- NIST-TOTAL-SCATTERING-PDF-2014

Measured, observed, adverse and boundary records:

- CRYS-002: 2 distinct registered comparisons preserved
- complete family custody: 8 captured and 2 explicitly unavailable sources
- capability-closed exact correspondence: True; omission control rejected: True

- NIST-SP846-POWDER-DIFFRACTION-1992: captured; structure factor for reflection; relates the intensity to the crystal structure parameters; favourable-use=True
- NIST-TOTAL-SCATTERING-PDF-2014: captured; atomic scattering factors; scattering vector magnitude; favourable-use=True

Shared claim-record clause MAT-S011 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:34f12bc639f822a3bc895da28a9a28558430b6cfce085a3af2c352ff6c9cbd20; empirical validation
 sha256:634c51f960ad70d2aa1ca0e0572b3e4f231a95d30015eb5e1d5ce93e66a7995f; external validation
 sha256:9c1cdacd2a9195d5fdfe59942dc910685128cda09baedca04ff33257ba39f9da; isolation
 sha256:9c483c82f2c6390366fe1c726805f86b7533dff995f4d9eb6005191418d901e8; custody
 sha256:4dc9b7a746f686c8e9f31b11b70918cab23c018d98287c4159c393e0461a7f76.

Prohibited inputs.

- no V1/V2 proof artifact, conventional complex amplitude, continuum field or named equation as a premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and phase opposition are held forms rather than prohibited scalars
- no target value, official source, measured diffraction pattern or favourable specimen selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary row
- no failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

95. Polycrystal texture and orientation distribution

Field: quantitative_crystallography_diffraction_disorder

Obligation: SFT-MAT-OBL-CRYS-003

Claim: SFT-MAT-CRYS-TEXTURE-ORIENTATION-003

Question, law and forced result. Polycrystal texture is the complete exact rational distribution of retained grain-orientation labels, phase labels, specimen identity and measurement boundary.

CRYS-003 uniquely retains exact-grain-orientation-rational-distribution with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001

- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-MAT-CRYS-STRUCTURE-FACTOR-002

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered CRY5-003 preservation axes before external target release.

Boundary: Every positive finite generated material carrier in CRY5-003, retaining its exact local, reciprocal, specimen, method, condition, uncertainty and proof-trace distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- exact-grain-orientation-rational-distribution
- complete-local-and-reciprocal-organization
- probe-condition-scale-and-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__exact-grain-orientation-rational-distribution__complete-local-and-reciprocal-organization__probe-condition-scale-and-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d6af3fdae60e22c4d2b6d7795fdc02bcf6b3a4a711d0f20bb7c0e87fd07e6442; generality

sha256:39a3fa0175568cd87d65c04a8952fe50e487dd3c7415a5b85da85336164a86b2.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:4c4e5ab8a633ac8a193ade9dab6ad553fa3b543df3f3abcf27e3803f880f261c.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:2d15dd6d81e5e87d7b5b6b7a766ee8e4d10eedbba2b78b77a19a01e76972a7bd.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:5c2cc4164a02b60d6d788101845ab3ad683f95a5c70b51c59db21606ccd533ba.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:dd261dc898b217229572ac3eeb1fa2d51cf9e8258b7c5ad0cabf99eef0cdf141.

Independent reconstruction and engine admission. Independent implementation

sha256:fdc016dfb48b1783f69764b017b86891c4ecd97cf9adc8fe2659ba65b524ef4b regenerated the

declared census and sole survivor; independent certificate

sha256:27e2c06c797c7b6763b87257f6c734ff74b02e1557c555ee57dd0e35a7c92b7b. Derivation seal
 sha256:08de036dc57e14b910274ad200f97ab38391f440f68bdee47b2dc9cd933b593f. Source manifest
 sha256:d9350a3274d5c68aa4cd592e5f8cdcc9600a97b3f691f8d15d14488bf927a1d7. Engine receipt
 sha256:ccc88e8824914311822eb8eae03ea58811d1fd7bf93fea9f91a810f096a2a8ef at receipts/engine/model_admitted/SFT-MAT-CRYS-TEXTURE-ORIENTATION-003-ccc88e8824914311.json. Closure
 scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-NCAL-TEXTURE-PHASE-FRACTION-2026

Measured, observed, adverse and boundary records:

- CRYS-003: 2 distinct registered comparisons preserved
- complete family custody: 8 captured and 2 explicitly unavailable sources
- capability-closed exact correspondence: True; omission control rejected: True
- NIST-NCAL-TEXTURE-PHASE-FRACTION-2026: captured; grains sharing a common crystal structure but having different crystallographic orientations; cross comparison between measurement techniques such as neutron diffraction; favourable-use=True
- NIST-NCAL-TEXTURE-PHASE-FRACTION-2026: captured; uncertainty metrics are also typically unquantified; phase fraction measurements may exhibit bias; favourable-use=True

Shared claim-record clause MAT-S011 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:d47712ab03f41a787b324ca79d679ba49ea4f82d0b1be7c29d7ccec41697f52d; empirical validation
 sha256:c20abbaaa92ff9418ab8ba437716d8a179dd86a46bbbbb13b5e2df8c166f825f; external validation
 sha256:60e91cbfe4e785746f2b69a903d676d498d4d76742e0940233b0929e26802efc; isolation
 sha256:ec9c34df452bbe121fe02894bf37abd8972d83d0d45c896db1d60733749a1375; custody
 sha256:6c43dd8fad12fad41ec9c3ffeee9a68a8ca324480ad97ab861bccef69e1db1f6.

Prohibited inputs.

- no V1/V2 proof artifact, conventional complex amplitude, continuum field or named equation as a premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and phase opposition are held forms rather than prohibited scalars
- no target value, official source, measured diffraction pattern or favourable specimen selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary row
- no failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

96. Short-range order and diffuse-scattering relation

Field: quantitative_crystallography_diffraction_disorder

Obligation: SFT-MAT-OBL-CRYS-004

Claim: SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004

Question, law and forced result. Short-range order is the exact lag-labelled pair distribution of a generated material word; noncollapsed pair classes form the complete diffuse-support ledger.

CRYS-004 uniquely retains lag-labelled-local-pair-and-diffuse-support-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-MAT-CRYS-STRUCTURE-FACTOR-002

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered CRYST-004 preservation axes before external target release.

Boundary: Every positive finite generated material carrier in CRYST-004, retaining its exact local, reciprocal, specimen, method, condition, uncertainty and proof-trace distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- lag-labelled-local-pair-and-diffuse-support-ledger
- complete-local-and-reciprocal-organization
- probe-condition-scale-and-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-material-carrier__lag-labelled-local-pair-and-diffuse-support-ledger__complete-local-and-reciprocal-organization__probe-condition-scale-and-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof sha256:c097000e096e2977d8b95fd012cc5812b6e514e08a460ac7f80129313017e9cc; generality sha256:59a46b303cd3b2f9a038fe6b238bc9eebe8439206cd82cc6ce7a104777e883f1`.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:f04f881bc1a956e795e4e73fa91ca6f2c7dd04c4aeaffeb8d12d06f8972271d4.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:cd64deace565b9b68e81659f50ce4bdc1858d94bf39829c24c2e1da473d714c1.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:ddc422d3b55487ebfe63690f499eed501a5391e17e1112927b9c803f29b77cd2.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:4952e0501d20b720beb5666b1eb8eecddef71e6ad3fae2e3f1f9226a064aa45f.

Independent reconstruction and engine admission. Independent implementation

sha256:fdc016dfb48b1783f69764b017b86891c4ecd97cf9adc8fe2659ba65b524ef4b regenerated the declared census and sole survivor; independent certificate

sha256:f3de94934c20730c11fbbcb4dbae2b73cacc50aedac3a994648a76dd93062f71. Derivation seal

sha256:674ead9ca80601aa8891c0f57d43fb30b5be703bb57784acb0cf47876016a970. Source manifest

sha256:94e3e0bf5d0976ba5f64f7cb541f4b7174e2adb37dd9568c63d647a20484912d. Engine receipt

sha256:c567f9ebd825af83bca30b576eb97adc7b6cee2e9c032e48afc8ca6e818b57be at receipts/engine/model_admitted/SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004-c567f9ebd825af83.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- IUCR-MODULATION-WAVE-DIFFUSE-2015
- IUCR-STACKING-DIFFUSE-2023

Measured, observed, adverse and boundary records:

- CRYS-004: 2 distinct registered comparisons preserved
- complete family custody: 8 captured and 2 explicitly unavailable sources
- capability-closed exact correspondence: True; omission control rejected: True
- IUCR-MODULATION-WAVE-DIFFUSE-2015: captured; both longer range order and truly short-range order are simultaneously encoded; highly structured diffuse intensity distributions; favourable-use=True
- IUCR-STACKING-DIFFUSE-2023: captured; any faulted sequence of layers produces streaks of diffuse scattering; preferred local structure; favourable-use=True

Shared claim-record clause MAT-S011 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:cbf4e027c8ba08d664b97babccb061f4d2a866b9d12378c1690b61e587c30bdf; empirical validation

sha256:8aa5ecf4f052443c3d5f41b88fb19386ac1fe57db1e437f5297cbee347f7a770; external validation

sha256:552a870ca1954ca9db819ddb3241285b179475120c1878bd2fff907b544eb51d; isolation

sha256:98e528e8c83db8d68f261e91655b0a63d0a3da2f09e8cc52acb811ae1b2e34e4; custody

sha256:6344791a6b442f8f052cf5bca69e53e2c4c6511501c606d912e3b54e4ab191eb.

Prohibited inputs.

- no V1/V2 proof artifact, conventional complex amplitude, continuum field or named equation as a premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude

- structural absence and phase opposition are held forms rather than prohibited scalars
- no target value, official source, measured diffraction pattern or favourable specimen selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary row
- no failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

97. Stacking-fault sequence and diffraction consequence

Field: quantitative_crystallography_diffraction_disorder

Obligation: SFT-MAT-OBL-CRYS-005

Claim: SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005

Question, law and forced result. A stacking fault is an exact departure from the unique cyclic successor in a close-packed A/B/C word, with every changed phase-support and diffraction class retained.

CRYS-005 uniquely retains cyclic-layer-successor-fault-and-diffraction-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered CRYS-005 preservation axes before external target release.

Boundary: Every positive finite generated material carrier in CRYS-005, retaining its exact local, reciprocal, specimen, method, condition, uncertainty and proof-trace distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- cyclic-layer-successor-fault-and-diffraction-ledger
- complete-local-and-reciprocal-organization
- probe-condition-scale-and-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__cyclic-layer-successor-fault-and-diffraction-ledger__complete-local-and-reciprocal-organization__probe-condition-scale-and-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:cc5ef77b402693399118a1d9a7e69156f33f2be974760e9c27442e9495039910; generality

sha256:42bd32adeaec2bb2f3033029990693da66ab97e9ef503a3dce21b43180b1a979.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:068bd1c1b4916709fe9ae1bbf4404979606ee0b5ceb99cc37dc541c59cd0463c.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:eee27b5b26055ad0e32f36e7b514fef1b25fe0bdc0b5c7764fc8285a088e86f9.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:6ffbea93a74c5960a4cddb5afb7799799c02faf09c2cbc38343c46a27404b3a2.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:5fffd46c869667fe1448df6eb20c9f91df2079f229d6a8c6d0fb562eb4610eca.

Independent reconstruction and engine admission. Independent implementation

sha256:fdc016dfb48b1783f69764b017b86891c4ecd97cf9adc8fe2659ba65b524ef4b regenerated the declared census and sole survivor; independent certificate

sha256:245782adcc258e5098edba4663a86499ff75ba45bf059ffff694962f58217096. Derivation seal

sha256:59fea8c9fa1861efcfb118e1f4f3997870617b293d4921c85fc082a2104efd8d. Source manifest

sha256:f75c5846ee3749e3efd34835496d4cdf39b1d458f37a6f586e59401550b9152. Engine receipt

sha256:e240c408ba701401889e6d105471e556c07b63778a70c74bc8ec6d5660177167 at receipts/engine/model_admitted/SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005-e240c408ba701401.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- IUCR-STACKING-FAULT-LDH-2020

Measured, observed, adverse and boundary records:

- CRY5-005: 2 distinct registered comparisons preserved
- complete family custody: 8 captured and 2 explicitly unavailable sources
- capability-closed exact correspondence: True; omission control rejected: True

- IUCR-STACKING-FAULT-LDH-2020: captured; recursive routine for generating and averaging supercells; measured diffraction data; favourable-use=True
- IUCR-STACKING-FAULT-LDH-2020: captured; occurrence of stacking faults causes diffuse scattering; degree and type of faulting; favourable-use=True

Shared claim-record clause MAT-S011 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:378b2724d95e770873dd1933f6932d2ae88ce59a521ed40e3904983ffb517d8d; empirical validation
 sha256:9efc29a32d14e23416b581d85de57f01ee78138268f965bd568c41b7f066fc57; external validation
 sha256:03891bf74d9b5ff298e22c79a39e32e3388c28d0a5adfb67804841d20be2899f; isolation
 sha256:7cf394907cc8172e13500ddb0966657eb36b9c05ef141ec4cfba0134e34c6c54; custody
 sha256:f65924399692b331653da3fc623de98d07f8f22cfd5ef934f87dba889f93cd72.

Prohibited inputs.

- no V1/V2 proof artifact, conventional complex amplitude, continuum field or named equation as a premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and phase opposition are held forms rather than prohibited scalars
- no target value, official source, measured diffraction pattern or favourable specimen selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary row
- no failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

98. Crystal twinning and domain relation

Field: quantitative_crystallography_diffraction_disorder

Obligation: SFT-MAT-OBL-CRYS-006

Claim: SFT-MAT-CRYS-TWIN-DOMAIN-006

Question, law and forced result. A twin is a distinct domain generated by an exact nonidentity involution of the crystal carrier; shared and domain-specific diffraction support remain separately reconstructible.

CRYS-006 uniquely retains nonidentity-involution-and-two-domain-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001

- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered CRY5-006 preservation axes before external target release.

Boundary: Every positive finite generated material carrier in CRY5-006, retaining its exact local, reciprocal, specimen, method, condition, uncertainty and proof-trace distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- nonidentity-involution-and-two-domain-ledger
- complete-local-and-reciprocal-organization
- probe-condition-scale-and-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__nonidentity-involution-and-two-domain-ledger__complete-local-and-reciprocal-organization__probe-condition-scale-and-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:e7976fd5355bb06220f9ffc1d80aa45e185a6b9b0c40b2bf4601dcbad45af1b3; generality
 sha256:060c4d038e1a67e5494f1edd4cc094345a6bdce065ff2406de98456d3cbde47a.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:cf06ac54eea20566e69e011c796338650632c119a48611effeae01fb0a2b94ab.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:23c6739f2dec3f7f8c811eb3c95e273e18a8a0b31059ea1bade4365a5bf98c78.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:70a3bbc2f9f8acc6c49e9caae1658fe9e5c28daacde957dcb1278e822ebfb81b.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:ce23df3fa494292dcad7e87d3c5eb04a0ee6444b1eef8c154f81e48dbda3229d.

Independent reconstruction and engine admission. Independent implementation

sha256:fdc016dfb48b1783f69764b017b86891c4ecd97cf9adc8fe2659ba65b524ef4b regenerated the

declared census and sole survivor; independent certificate

sha256:8a90b68b00418c8cb6fb24c4322df0aa2c636cf69437b9465a1166cfd3130ca5. Derivation seal
 sha256:7f46a1dd14791e1bb356ba0d4d7e4a03186806441036dfe359a7ecf1cc5ae770. Source manifest
 sha256:fea627ec346677b7ede6121f7d1e249e4d6e7271345db6ae0b4b1ccf6dc5f8f4. Engine receipt
 sha256:a400404257782ed601e29462864cff39aa505b407a7327bf1f16f37e65865b09 at
 receipts/engine/model_admitted/SFT-MAT-CRYS-TWIN-DOMAIN-006-a400404257782ed6.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- IUCR-TWINNED-DIFFRACTION-DATA-2022
- IUCR-TWIN-DICTIONARY-2026

Measured, observed, adverse and boundary records:

- CRYS-006: 2 distinct registered comparisons preserved
- complete family custody: 8 captured and 2 explicitly unavailable sources
- capability-closed exact correspondence: True; omission control rejected: True
- IUCR-TWINNED-DIFFRACTION-DATA-2022: captured; non-merohedrally twinned with a twofold rotation; twin domains scatter coherently; favourable-use=True
- IUCR-TWIN-DICTIONARY-2026: source_unavailable_403_preserved; source unavailable 403 preserved; not used for favourable comparison; favourable-use=False

Shared claim-record clause MAT-S011 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:bdd5c1b4c1876d687100de6ed2cef15390367f4facbae2135be5044f1e8917b; empirical validation
 sha256:0440e756c45bf9d9665c620668b1d2a011a772f0a4be43921bc75343f8536629; external validation
 sha256:39ebe93526490be36a137a0ff4a1d7cbd21990d64196b95e210206e1adf8650a; isolation
 sha256:9a68980a8f04e88c011baae3d82d6b4e56dacffd725478b7b5dca42053b5109c; custody
 sha256:f331897a0e13dde425b3b757cc93511a532d2d1467191ce31b88c320c97c1870.

Prohibited inputs.

- no V1/V2 proof artifact, conventional complex amplitude, continuum field or named equation as a premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and phase opposition are held forms rather than prohibited scalars
- no target value, official source, measured diffraction pattern or favourable specimen selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary row
- no failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

99. Incommensurate and modulated structure organisation

Field: quantitative_crystallography_diffraction_disorder

Obligation: SFT-MAT-OBL-CRYS-007

Claim: SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007

Question, law and forced result. An incommensurate modulation is an independent nonclosing successor fibre over the base lattice, forcing compound main/satellite indices without irrational proof coordinates.

CRYS-007 uniquely retains independent-modulation-successor-and-compound-index-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-MAT-CRYST-QUASICRYSTAL-001
- SFT-MAT-CRYS-TWIN-DOMAIN-006

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered CRYST-007 preservation axes before external target release.

Boundary: Every positive finite generated material carrier in CRYST-007, retaining its exact local, reciprocal, specimen, method, condition, uncertainty and proof-trace distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- independent-modulation-successor-and-compound-index-ledger
- complete-local-and-reciprocal-organization
- probe-condition-scale-and-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-material-carrier__independent-modulation-successor-and-compound-index-ledger__complete-local-and-reciprocal-organization__probe-condition-sc`

ale-and-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:547a0d35319f86f4f3ac33f988628d5206960d88977af8aeae38bd2bee8597b7; generality

sha256:81e4eb1dfbffd1cfdd650a03c676c48f2f311035d70549db81ecbb7e35aabdb81.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:571694e656374fa4e5c04ba7c9bcb0a39479222d22bc5b28f445252ef8780e3.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:f312a7ffe7b248a87e1e76eed20efda574189934e4bd7f03e917603efc45c823.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:6ea718d6463baca65ceff9ba1df1184f8e18ba161b0cdbc265be12ade7b5a0c2.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ed26eb46924ab2d9efe7bf40cf9114f6d6c06d6d8acac4fba7c993d1c42c147d.

Independent reconstruction and engine admission. Independent implementation

sha256:fdc016dfb48b1783f69764b017b86891c4ecd97cf9adc8fe2659ba65b524ef4b regenerated the declared census and sole survivor; independent certificate

sha256:42c4cb14adc471dc8cfd3ae737db8b72f590ae2309e6d8c6a776cd8383e1fc56. Derivation seal

sha256:51a2878cd3e555ef5d05ad15b2401caa0072ed3976c86b6e76d04adec42d6428. Source manifest

sha256:9727a3945b14ac18d23d86b08e6bf3b01f49a80bbfa4c44bb4e834a3e4b5269b. Engine receipt

sha256:313ced02f25e6e053754c7fcd26e889ec8babb7640db65351c95fd2f1c1cdab2 at receipts/engine/model_admitted/SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007-313ced02f25e6e05.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- IUCR-MODULATED-STRUCTURES-2009
- IUCR-INCOMMENSURATE-DICTIONARY-2026

Measured, observed, adverse and boundary records:

- CRYS-007: 3 distinct registered comparisons preserved
- complete family custody: 8 captured and 2 explicitly unavailable sources
- capability-closed exact correspondence: True; omission control rejected: True
- IUCR-MODULATED-STRUCTURES-2009: captured; additional bragg reflections; integer indexing with three indices hkl is not possible; favourable-use=True
- IUCR-MODULATED-STRUCTURES-2009: captured; satellite reflections; loss of periodicity in three dimensions; favourable-use=True
- IUCR-INCOMMENSURATE-DICTIONARY-2026: source_unavailable_403_preserved; source unavailable 403 preserved; not used for favourable comparison; favourable-use=False

Shared claim-record clause MAT-S011 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:b2b7ea37e73abcb5b3c3d9b46c01c3fe8e5e744c2c30064f0ff9ba6a4ee6df09; empirical validation

sha256:dc88f0084d8e7983765cc359a60cec26849c27324ff8537b7da86ca08b628da2; external validation

sha256:047d4980924554a975c732c252295de9203e006a5d30c8d9d5ca434b83689152; isolation
 sha256:073e6389ee85dcd5172a035891bd67c8a082d97afe4c73c07b57e625242a6f42; custody
 sha256:f12f32ec9dede5be7fd06d0fa95c4e388484cb3cb49483dd3e670482ec09e890.

Prohibited inputs.

- no V1/V2 proof artifact, conventional complex amplitude, continuum field or named equation as a premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and phase opposition are held forms rather than prohibited scalars
- no target value, official source, measured diffraction pattern or favourable specimen selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary row
- no failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

100. Total-scattering pair-distribution reconstruction

Field: quantitative_crystallography_diffraction_disorder

Obligation: SFT-MAT-OBL-CRYS-008

Claim: SFT-MAT-CRYS-PAIR-DISTRIBUTION-008

Question, law and forced result. The real-space pair distribution is the complete exact rational multiplicity ledger of every positive constituent separation retained by total scattering.

CRYS-008 uniquely retains complete-positive-pair-separation-multiplicity-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered CRYIS-008 preservation axes before external target release.

Boundary: Every positive finite generated material carrier in CRYIS-008, retaining its exact local, reciprocal, specimen, method, condition, uncertainty and proof-trace distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- complete-positive-pair-separation-multiplicity-ledger
- complete-local-and-reciprocal-organization
- probe-condition-scale-and-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__complete-positive-pair-separation-multiplicity-ledger__complete-local-and-reciprocal-organization__probe-condition-scale-and-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:b1edd7ca337eda9a4ca55ccff533292d6482e4ebd7d30bb24011346532de2fec; generality sha256:bb6e7844dbb3dfef1b11038e91fa505a7f2102274aa0cf82578e5c383dbd962d.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:e9b48d0b14cfba2acd4f7194003dc6aa29420fe13301007cfd6d9d9013307b96.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:1b5195c7daa0672e3f0c63af425223c4e69548c1a83e67961a22b61700f8d1f5.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:b6ee57c3880d54ac8395dfa6a998f100105f292190cad68fe6a24d9d3dd8d2bb.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:bee409391dd274035ac37e1405e63c02b1a480e43be7df4e66dd568dce22e6db.

Independent reconstruction and engine admission. Independent implementation

sha256:fdc016dfb48b1783f69764b017b86891c4ecd97cf9adc8fe2659ba65b524ef4b regenerated the declared census and sole survivor; independent certificate sha256:6fcf4edd6f0342b8a66455c8e11d54cf8845a7f0f5093b9052256950e8aea4cf. Derivation seal sha256:bcdcdcc5dbd269cf6c6c54762b952feaf56b4fc4eb8b32ab9040fe523032fde0. Source manifest sha256:46929b486f5544b59baa705a6c7589161d209f8865d5614d8672c3cce087b526. Engine receipt sha256:5af3a5acad0cf0ecf3c601c2db4549f59423fdd26cd799ede24ad34664753557 at receipts/engine/model_admitted/SFT-MAT-CRYS-PAIR-DISTRIBUTION-008-5af3a5acad0cf0ec.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-TOTAL-SCATTERING-PDF-2014

Measured, observed, adverse and boundary records:

- CRY5-008: 2 distinct registered comparisons preserved
- complete family custody: 8 captured and 2 explicitly unavailable sources
- capability-closed exact correspondence: True; omission control rejected: True
- NIST-TOTAL-SCATTERING-PDF-2014: captured; fourier transform of the was data; pair-distribution function; favourable-use=True
- NIST-TOTAL-SCATTERING-PDF-2014: captured; total scattering function; related to interatomic distances; favourable-use=True

Shared claim-record clause MAT-S011 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:6832678cfff7059a433eca8527e66ea278005ffcc03b01592f142bda4a149d18a; empirical validation
 sha256:0efb35a25dbfa1b5446cc0d907fee9ffdc8eb313d3417cae095b6f99711a22b1; external validation
 sha256:bd64fb89fcc4f0500692940a3df3da5df83665aefce5f176a4766642e00db11a; isolation
 sha256:00cb7ea9da8a958394858bcc32464f9a0a00243d44f063a34248eee23bf01fe7; custody
 sha256:7f90cd2fbdb0ffe1fea315c2f6639c3d5d8045c8675fec8ba1be88d26724af6b.

Prohibited inputs.

- no V1/V2 proof artifact, conventional complex amplitude, continuum field or named equation as a premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and phase opposition are held forms rather than prohibited scalars
- no target value, official source, measured diffraction pattern or favourable specimen selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary row
- no failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

Complete-field family: Defects Microstructure Interfaces Multiscale

101. Defect population and site-fraction balance

Field: defects_microstructure_interfaces_multiscale

Obligation: SFT-MAT-OBL-MICRO-001

Claim: SFT-MAT-MICRO-DEFECT-POPULATION-001

Question, law and forced result. A defect population is the complete exact partition of a finite site carrier into held defect and host classes, whose rational parts recompose the whole carrier.

MICRO-001 uniquely retains complete-site-partition-and-rational-fraction-balance with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered MICRO-001 preservation axes before external target release.

Boundary: Every positive finite generated microstructure carrier in MICRO-001, with complete site, path, interface, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-microstructure-carrier
- complete-site-partition-and-rational-fraction-balance
- complete-site-path-interface-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-microstructure-carrier__complete-site-partition-and-rational-fraction-balance__complete-site-path-interface-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof sha256:47a635633addd782c4a1dce9cbcdab936c5663bfcf07293287d7648625d14030; generality sha256:2b935bd58fe2cd4b45a8935e1aa9fb8f5588343d847a4d8329f8a4f1c94f86cd`.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt

sha256:cfdd8f6013644e8ebf2ea90936b858174e893d6b310b7f9a9f984bce235eaa96.

- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:13ce7abb61091067a8009237013a01ffe84e79e976f9cf5a20a0fdc3e9500f35.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:b43f314f7b2129f309cd807faaac683bcf015f601d70e8455ff31db8be4d22d1.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:c195763952dfc9f6254d66a80e311ba84917077deb6308ffb25577d9b5c942a.

Independent reconstruction and engine admission. Independent implementation

sha256:09f2ca2e37e2e5caa120823c6594cff6841df612380d4ab55ae7d8cb4b516727 regenerated the declared census and sole survivor; independent certificate

sha256:8fbec5a662076c9e1e80b257fc6b8a90735056d442489fbc96b5811958fc0c77. Derivation seal

sha256:c96e346a546733931ad621281d1a6ef47f46cb85fc0960c64f4bba17a2db7d1b. Source manifest

sha256:f1bd2c83a8e5f30525ace1e69cf4cfd4259fc45b1f075694a173b03c34bf73f7. Engine receipt

sha256:902c083fc7c51628739a79389dae656223864ca0966bb5dd7f28a6ff7e66cff7 at receipts/engine/model_admitted/SFT-MAT-MICRO-DEFECT-POPULATION-001-902c083fc7c51628.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-POINT-DEFECTS-2026

Measured, observed, adverse and boundary records:

- MICRO-001: 1 distinct comparisons retained
- family custody: 9 usable and 1 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-POINT-DEFECTS-2026: captured_preexisting_official_snapshot; dilute concentrations of point defects; measured defect chemistry; favourable-use=True

Shared claim-record clause MAT-S012 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:7582e77d997e4909b89a595b51f74f6caaf88eb6fbc91088b22f1d86b48ba45; empirical validation

sha256:72942447e2681507be5b26f251439e429ea8a188d250a2015f392b48692e0351; external validation

sha256:fc33fbcf8b759193735f11560aca165d34706b936b3b2bd74f40c353cd083bd5; isolation

sha256:b5e0c0dc3f355c1556ea66e4d75d74be3667cc2a6c3ce1906702adb09f923e3; custody

sha256:a713c52cbc96556329fb4c0fecedac3ef1256edba9a690a1ffecc6f87b832549.

Prohibited inputs.

- no V1/V2 proof artifact, conventional continuum law, fitted constitutive equation or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and opposed direction remain held labels
- no target value, official source, measured micrograph, selected specimen or favourable outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary evidence row
- no first failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

102. Defect migration and retained path

Field: defects_microstructure_interfaces_multiscale

Obligation: SFT-MAT-OBL-MICRO-002

Claim: SFT-MAT-MICRO-DEFECT-MIGRATION-002

Question, law and forced result. Defect migration is a complete adjacent-site transition word retaining initial site, every axis-oriented step, terminal site, time order and carrier identity.

MICRO-002 uniquely retains adjacent-site-axis-oriented-complete-migration-word with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-DEFECT-POPULATION-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered MICRO-002 preservation axes before external target release.

Boundary: Every positive finite generated microstructure carrier in MICRO-002, with complete site, path, interface, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-microstructure-carrier
- adjacent-site-axis-oriented-complete-migration-word
- complete-site-path-interface-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace

- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-microstructure-carrier__adjacent-site-axis-oriented-complete-migration-word__complete-site-path-interface-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f690e43ed63b14ab3ca84550679671040963d6cc84dadd2752a48fae0dba5d59; generality

sha256:3f8a80a2da7f165974223ab1f0ec51e1fb6964551f8787f117059a36b2369ffe.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:e7c2f26b8b58995acefd4f8c857efce430c1b7ecc3fff8184c821f518824c119.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:68e5acb857199ca3f8492984c83ce1a01db6c7ccfd4fc5de9716db6ecffa2624.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:673fd12a549509e89dcfc1cfc140ff8a8cfa4c6a7830c09f1484963e9b04d4d6.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:a0fc65e157e9e8fd55cd0e29d3820d44b2d4447be7a5f2cd7eac5c8d7b64df7b.

Independent reconstruction and engine admission. Independent implementation

sha256:09f2ca2e37e2e5caa120823c6594cff6841df612380d4ab55ae7d8cb4b516727 regenerated the declared census and sole survivor; independent certificate

sha256:a4cb945f8d8ef08683e45332a16f9d4eba839c3b4a7e88bf088216dc4bddd4af. Derivation seal

sha256:cb1e1ffc59df5bbd49ba30aa3aac7fda333287b1797a83581daa9a88333c087d. Source manifest

sha256:0f7dff2d97bd92bd18902b1b7ff3e2d107b03edf1db28a01ac300eb77d2f4fba. Engine receipt

sha256:313402a897492a0bc284213931b53f8418db5afd04b5e943dbc27a94e6c11e58 at receipts/engine/model_admitted/SFT-MAT-MICRO-DEFECT-MIGRATION-002-313402a897492a0b.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-POINT-DEFECTS-2026
- NIST-DISLOCATION-DYNAMICS-2021

Measured, observed, adverse and boundary records:

- MICRO-002: 2 distinct comparisons retained
- family custody: 9 usable and 1 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-POINT-DEFECTS-2026: captured_preexisting_official_snapshot; vacancy diffusion; mobility and local interaction of ionic charge carriers; favourable-use=True
- NIST-DISLOCATION-DYNAMICS-2021: captured; evolution of dislocation structures; vacancy concentration; favourable-use=True

Shared claim-record clause MAT-S012 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c1e1ea25de13bd67599583f5e7f7d885fa0d096bda9d4096dcc13bd75c2b3e0b; empirical validation
 sha256:41585da40048d048898807266bbd7859f597a2de4c780b6521ccb5284b708e18; external validation
 sha256:c4db1c45c9ceab52922ef7a64ac1618b6c64d52116682b7f27436f8283e1badd; isolation
 sha256:5ad2caa5f50f3fdd03c71c8a25f07a30a5ea22acf5364b1a1b105b0a1901c7c6; custody
 sha256:d18bc7468d5154adb75126a3ae70375e53cd3aa754546344ca3b6e76d27373fd.

Prohibited inputs.

- no V1/V2 proof artifact, conventional continuum law, fitted constitutive equation or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and opposed direction remain held labels
- no target value, official source, measured micrograph, selected specimen or favourable outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary evidence row
- no first failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

103. Dislocation reaction, climb and cross-slip

Field: defects_microstructure_interfaces_multiscale

Obligation: SFT-MAT-OBL-MICRO-003

Claim: SFT-MAT-MICRO-DISLOCATION-REACTION-003

Question, law and forced result. Dislocation reactions compose exact line magnitudes with held orientations; climb retains vacancy-mediated plane change and cross-slip retains line identity across a distinct plane.

MICRO-003 uniquely retains held-orientation-line-reaction-climb-cross-slip-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001

- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-DEFECT-MIGRATION-002

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered MICRO-003 preservation axes before external target release.

Boundary: Every positive finite generated microstructure carrier in MICRO-003, with complete site, path, interface, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-microstructure-carrier
- held-orientation-line-reaction-climb-cross-slip-ledger
- complete-site-path-interface-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-microstructure-carrier__held-orientation-line-reaction-climb-cross-slip-ledger__complete-site-path-interface-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:fc9fd58480b1df08b46639c5d00942c56227334986c0ceffcb5079fdbea2950b; generality`

`sha256:408fa3784fa8cc4e54e4d5f571e3512b8144ed4b0bd1341028cab7f0e66f5a3a.`

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:c10b358d4895da2def97a015ca90aa13c0501e3b23052df64805c7575cad89e6.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:07b51d18c7b02cb341a5dbad90f3c5d252313389229a7dd32c2d39eb120b7666.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:8bb49b29fba0095fec99730c4ad8043a7f4d7fafb8e6787786e20bc7931a6680.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:a5bb790145c344406470058c79ec433b1d77f5e2c69a3ebbd3aabf6f1bdfd98.`

Independent reconstruction and engine admission. Independent implementation

`sha256:09f2ca2e37e2e5caa120823c6594cff6841df612380d4ab55ae7d8cb4b516727` regenerated the declared census and sole survivor; independent certificate

`sha256:f19012ae6361b1b8a367f5323a1d77415a9ffcb7573e64059e4914e236add967.` Derivation seal

`sha256:47864384145f5719b1a4408ee6c60b4bb1f2b66e240ec91c41d2e3180a0872ec.` Source manifest

sha256:3ab132de78a16a085e4b53075d3e4289c3021019f4070426724fffd3441adf8e. Engine receipt
 sha256:89293aa1943dddf99f7c67de12a3d8e3391cd0f6303d7333338148ef8551c3d0 at receipts/engine/model_admitted/SFT-MAT-MICRO-DISLOCATION-REACTION-003-89293aa1943dddf9.json. Closure
 scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-DISLOCATION-CLIMB-MONOGRAPH-59
- NIST-DISLOCATION-DYNAMICS-2021

Measured, observed, adverse and boundary records:

- MICRO-003: 2 distinct comparisons retained
- family custody: 9 usable and 1 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-DISLOCATION-CLIMB-MONOGRAPH-59: captured; dislocation climb is necessary before unlike dislocations can unite; lattice vacancies must deposit all along the dislocation line; favourable-use=True
- NIST-DISLOCATION-DYNAMICS-2021: captured; evolution of dislocation structures; vacancy concentration; favourable-use=True

Shared claim-record clause MAT-S012 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:632239a2b49dc87256347109fdf65e21122dfaaede7d114c35c37911c83b8cfd; empirical validation
 sha256:a0a72743e3ec117d99387f78acbea625e4e3bb340e76571103516c068edd9d90; external validation
 sha256:c98e2d666c2515b03d134a6bdcdaf91871717a8cd1c39c56c68ef31d959c0fe1; isolation
 sha256:0188347c0f9993de046aaaca9a9a756b4578b0f0c8dc58766d9052d8b940f25f; custody
 sha256:27282f7c9b7b00ce1e66c34d2788ae43824b8714a5f7fb00524b3cf04fe89468.

Prohibited inputs.

- no V1/V2 proof artifact, conventional continuum law, fitted constitutive equation or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and opposed direction remain held labels
- no target value, official source, measured micrograph, selected specimen or favourable outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary evidence row
- no first failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

104. Curvature-driven grain growth

Field: defects_microstructure_interfaces_multiscale

Obligation: SFT-MAT-OBL-MICRO-004

Claim: SFT-MAT-MICRO-GRAIN-GROWTH-004

Question, law and forced result. Discrete curvature is the exact boundary-cell to grain-cell part; one lawful growth step transfers a cell from greater to lesser curvature while preserving the complete material carrier.

MICRO-004 uniquely retains boundary-cell-per-grain-cell-curvature-transfer with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-DISLOCATION-REACTION-003

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered MICRO-004 preservation axes before external target release.

Boundary: Every positive finite generated microstructure carrier in MICRO-004, with complete site, path, interface, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-microstructure-carrier
- boundary-cell-per-grain-cell-curvature-transfer
- complete-site-path-interface-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-microstructure-carrier__boundary-cell-per-grain-cell-curvature-transfer__complete-site-path-interface-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.`

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof sha256:b13cdea9d1694c576f92a8528b44fd9b3b7bcb2198f2b753644fd4944282ac14; generality sha256:9268b5c82a44ee78d5289dd2f3f25a23d5ac204f44d4537e53ae3d3a07e6ece9.`

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:dd7bbf25db4771bb5415dc3641f11528344d1bf16d099e214902f4acd84b35df.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:b02fe5dc4a3622ef306e0b2d892a45f1a9d9b84b849f56f1e3ae5e9c027ec420.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:609c0156e85f7cd412d56b835b6f345d6c80aa56573fff00d79ed9bd58de866f.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:3701828f31339cf4e02b7fc048c667f925fbc2a805211d6c03a250ac27b73c55.

Independent reconstruction and engine admission. Independent implementation

sha256:09f2ca2e37e2e5caa120823c6594cff6841df612380d4ab55ae7d8cb4b516727 regenerated the declared census and sole survivor; independent certificate
sha256:cfcf69568f2138389df07acfc3d239b4480872825403884db9e1db38c1703c24. Derivation seal
sha256:2f83e6e159aaa2c702f667277612c726d3510f8341000ca3de7f92b4bb6d92c0. Source manifest
sha256:c32d1ffc891eacbf54adaceed4b42eb7ebd71a70e301d6e49b26d0382e92ab2b. Engine receipt
sha256:dfc9ebf5aaa6044a4121ec7b94cd92ea9c79f1f47d6453258b637d3a5efdfec6 at
receipts/engine/model_admitted/SFT-MAT-MICRO-GRAIN-GROWTH-004-dfc9ebf5aaa6044a.json.
Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SHARP-INTERFACE-GRAINS-2001

Measured, observed, adverse and boundary records:

- MICRO-004: 1 distinct comparisons retained
- family custody: 9 usable and 1 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-SHARP-INTERFACE-GRAINS-2001: captured; normal velocity of the boundary is proportional to both its curvature; interfacial energy and mobility; favourable-use=True

Shared claim-record clause MAT-S012 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:fb5db7d602b306229fcb8dfd7368c4b76c939813bac6c549c3956fea302c8709; empirical validation
sha256:289fb215f545c7d5be3d8c4bd0bcc3b04ef9451c45e4335be77c94ade946684b; external validation
sha256:3cafbb54e6cf5fcd1824ce1ba12be66aabc0c5a9d86f8b9cb1347f4e0bfe6f42; isolation
sha256:04e2870642ecd68872e58731f3bc8c3a6bf3f3e342bc3bbea142f5fc19c7e7d1; custody
sha256:2d41873e20f886c663a2ee3bed7f8440b63b017357e62ee58bb801f1b58e177f.

Prohibited inputs.

- no V1/V2 proof artifact, conventional continuum law, fitted constitutive equation or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and opposed direction remain held labels
- no target value, official source, measured micrograph, selected specimen or favourable outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary evidence row

- no first failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

105. Interface and grain-boundary segregation

Field: defects_microstructure_interfaces_multiscale

Obligation: SFT-MAT-OBL-MICRO-005

Claim: SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005

Question, law and forced result. Boundary segregation is the complete species-by-species comparison of exact bulk and boundary parts, retaining enrichment, depletion, equality and structural absence without signed concentration.

MICRO-005 uniquely retains species-wise-bulk-boundary-part-comparison with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-GRAIN-GROWTH-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered MICRO-005 preservation axes before external target release.

Boundary: Every positive finite generated microstructure carrier in MICRO-005, with complete site, path, interface, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-microstructure-carrier
- species-wise-bulk-boundary-part-comparison

- complete-site-path-interface-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-microstructure-carrier__species-wise-bulk-boundary-part-comparison__complete-site-path-interface-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:43a81fe7bac6f53a67b4ef3ce77b8688c276e3f301dbe7ff1411972ad32460e2; generality

sha256:f9e3b5d304daef4b33b1878d11a4c00d881cf21c731c1dcb12d831c6ad7cf9f0.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:6c24f81296aee9b5af5345db78152a2f1b23b7d3a865ed1b02d48cb372458606.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:47ecbf85fc23571877bd4244d90ee2db88a22b12c1eb677eff258657007bfb1c.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:3510037983578147e3323a36384eb3c655742fb99ef2b4b4f2d36b3a18a65eb0.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:9e33ec7214394d24fe89e756eb438bb608c4c8fbe552a94a268705878a9caeb4.

Independent reconstruction and engine admission. Independent implementation

sha256:09f2ca2e37e2e5caa120823c6594cff6841df612380d4ab55ae7d8cb4b516727 regenerated the declared census and sole survivor; independent certificate

sha256:99a2c4a1e6a4e37b005eaaafb66699f34a08e203d6a82ffd9faf04ce54c23edc5. Derivation seal

sha256:920519a00f693c8f9abf5c03968d39096fd369b11ca10f9a1a06181ce796ef46. Source manifest

sha256:5fc20f2275f6b13aeafefc74b73601b37e93dad49da56486777d742149880a01. Engine receipt

sha256:ealfd4a4308216474a7b3bafb4c82b8b26e08ade0458f2fb9f5c29842ed00134 at receipts/engine/model_admitted/SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005-ealfd4a430821647.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SEGREGATION-PRECIPITATION-2021

Measured, observed, adverse and boundary records:

- MICRO-005: 1 distinct comparisons retained
- family custody: 9 usable and 1 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-SEGREGATION-PRECIPITATION-2021: captured; grain boundary segregation; composition; favourable-use=True

Shared claim-record clause MAT-S012 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:beabe638bf8f63089fc42d9ce16ab84148ab20a14e5f488035e20dd9cabdab2a; empirical validation
 sha256:be148cee350fd04e3832177d3f572b4c0ac4458c10a1862630440188fd4b8610; external validation
 sha256:9613c96e33aed4b6e68689ef44b8dfca76b5e7fcd3a030abcca564f4937c8cfd; isolation
 sha256:9d40d8a33c800b00c40b28b4ee4f0e2c89de86e35e09c653a4a83b3a4da706cd; custody
 sha256:08fa4d15daadfae09dd2d8f63234fbb6b63638e565864a5b1873b99cee29df12.

Prohibited inputs.

- no V1/V2 proof artifact, conventional continuum law, fitted constitutive equation or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and opposed direction remain held labels
- no target value, official source, measured micrograph, selected specimen or favourable outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary evidence row
- no first failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

106. Precipitation and coherent-incoherent inclusion boundary

Field: defects_microstructure_interfaces_multiscale

Obligation: SFT-MAT-OBL-MICRO-006

Claim: SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006

Question, law and forced result. A precipitate inclusion retains matrix and inclusion recurrences; equality is coherent, a finite common return is semicoherent, and no return within the declared observation span is incoherent at that boundary.

MICRO-006 uniquely retains matrix-inclusion-recurrence-and-common-return-boundary with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-INTERFACE-001

- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered MICRO-006 preservation axes before external target release.

Boundary: Every positive finite generated microstructure carrier in MICRO-006, with complete site, path, interface, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-microstructure-carrier
- matrix-inclusion-recurrence-and-common-return-boundary
- complete-site-path-interface-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-microstructure-carrier__matrix-inclusion-recurrence-and-common-return-boundary__complete-site-path-interface-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a1a4738a945967e6c342dc8d8affdab2f84bcc857e42d0f63c5f67e2f16aa253; generality

sha256:10832ccc85e345f31e1f93bbf657090672ac11a98cdda41f7701739d1a78c1cc.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:e6d839e510dd29301635cc13685f81af003b05dbf4b88896fbcbbfac751b7233.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:d3689b76c299068995ead6a8d40a0ba805e5f37f1ea9d4d5454cf791362fbb41.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:cd6fe368b3503f2bb769391f3d13f05b61f580b4777cf44f95c155b21ebbc6bb.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:1e1b99c9b2004255cd3edc4e90cb17e16fdc3823f7d368d50ba9fe8fdcf125fd.

Independent reconstruction and engine admission. Independent implementation

sha256:09f2ca2e37e2e5caa120823c6594cff6841df612380d4ab55ae7d8cb4b516727 regenerated the declared census and sole survivor; independent certificate

sha256:ef9c43483debcf57e5c1659cb7e7df63d376bf9f0507588c9507fb06cba96e6e. Derivation seal

sha256:32ed3deacf2dc96ba541af8d1610a991c8a50c95763284c2360b460e8e12e7e1. Source manifest

sha256:2278c8c47471f5453b858521d02e5199cb023cd302333ce0962bc9783bb7478b. Engine receipt

sha256:c1165612f8dec6355601012603eff1c402cfc1d793ae01fd5c3d850bd0c72b1b at receipts/eng

ine/model_admitted/SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006-c1165612f8dec635.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SEGREGATION-PRECIPITATION-2021
- NIST-STRUCTURES-PRECIPITATION-HANDBOOK

Measured, observed, adverse and boundary records:

- MICRO-006: 2 distinct comparisons retained
- family custody: 9 usable and 1 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-SEGREGATION-PRECIPITATION-2021: captured; precipitation of a secondary phase; partially or completely incoherent with the host lattice; favourable-use=True
- NIST-STRUCTURES-PRECIPITATION-HANDBOOK: source_unavailable_http_404_preserved; source unavailable 404 preserved; not used for favourable comparison; favourable-use=False

Shared claim-record clause MAT-S012 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:29d20e1e80a6d299146106c05e27a835e1bdd58c2d68359eb59f948f5941efdf; empirical validation
 sha256:b00048d5a22e6909e7c9d12cc40bb4f3b42598c448631d5858b1f51a7ff839e7; external validation
 sha256:938a65c4d0605a555c77979275f52795a39369e3627f2c57323ab029c63ebf69; isolation
 sha256:db9a1792a63dae0100e0e1a1c0acdef4c5a6654ad12ead83f5cd1a40461488f8; custody
 sha256:50bb934d6edc428cf8f8deaebd485b2dd7653c1f2e63367161f82e4f56612ef9.

Prohibited inputs.

- no V1/V2 proof artifact, conventional continuum law, fitted constitutive equation or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and opposed direction remain held labels
- no target value, official source, measured micrograph, selected specimen or favourable outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary evidence row
- no first failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

107. Ostwald-type coarsening as exact carrier transfer

Field: defects_microstructure_interfaces_multiscale

Obligation: SFT-MAT-OBL-MICRO-007

Claim: SFT-MAT-MICRO-COARSENING-TRANSFER-007

Question, law and forced result. Coarsening is conserved exact carrier transfer from a declared donor particle to a receiver particle, with donor disappearance held as structural absence and no unaccounted material gain.

MICRO-007 uniquely retains conserved-particle-carrier-transfer-and-absence with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered MICRO-007 preservation axes before external target release.

Boundary: Every positive finite generated microstructure carrier in MICRO-007, with complete site, path, interface, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-microstructure-carrier
- conserved-particle-carrier-transfer-and-absence
- complete-site-path-interface-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-microstructure-carrier__conserved-particle-carrier-transfer-and-absence__complete-site-path-interface-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:883cfe3b68ea31a2eb67cbfa8c84f1cde2dbd5f4d2f685e339bde47ae24b843f; generality sha256:eb400630315906389c08546bccaf1660f1d5df10cc107a5e4b348a228e8f609.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:451e5041cbfc0ccf4f71c3ecba0693de2e1da672f322bb811c70de7724230a0a.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:86647a9f582dc9dd7126f7fec88d83f923787db5399301fa6668cd17288dc374.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:038a9f819b98a423f27c362fe8bd21b8e8fac79603e038814869df4631636c46.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:8f24e3c92f58e1ebb84f04106a27c9b11173a5effc55f976ccb8e0b3837ed4f7.

Independent reconstruction and engine admission. Independent implementation

sha256:09f2ca2e37e2e5caa120823c6594cff6841df612380d4ab55ae7d8cb4b516727 regenerated the declared census and sole survivor; independent certificate

sha256:af5ec7dd6edbe0a0fe1c19203458c78569515336da1a8e7432c0f278a271e276. Derivation seal

sha256:559a40a8d1321d7a2834c558bf0cc60918b12a456c80286540cf4eae25850e3f. Source manifest

sha256:c878d36a1fd912c15c634f23f729cf08c64e008d4be498148e9b48b771e4d954. Engine receipt

sha256:1c4b55d223e02a856d7ec3484f9db556792b27b8b6c9e36e1db8c1efdd4049ec at receipts/engine/model_admitted/SFT-MAT-MICRO-COARSENING-TRANSFER-007-1c4b55d223e02a85.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-BENCHMARK-COARSENING-2017
- NIST-STRUCTURES-PRECIPITATION-HANDBOOK

Measured, observed, adverse and boundary records:

- MICRO-007: 2 distinct comparisons retained
- family custody: 9 usable and 1 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-BENCHMARK-COARSENING-2017: captured; growth and coarsening of a second phase; ostwald ripening problem; favourable-use=True
- NIST-STRUCTURES-PRECIPITATION-HANDBOOK: source_unavailable_http_404_preserved; source unavailable 404 preserved; replacement addendum retained; favourable-use=False

Shared claim-record clause MAT-S012 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:4425a149b7c34a3fb7df00445ba126e97bb3ef0b78aa1847a20f54d4e8b0f972; empirical validation

sha256:49cfa5d4b4fa68716f5b60c2c68d74b899aec30c5894376a50138fdb9a89e78a; external validation

sha256:9bcd465f9671885fe57ab15f0017269871a897171e7236c6c51b5ff28b61e91; isolation

sha256:9bb3ac3b3a71d708fa40d7afe2c263e6c7af3c8bd36676c8bf36d10aba995c5e; custody

sha256:e19728449632c37400368bb65fbd907d9eeb9fe47c686cfae21bf002cab5c56d.

Prohibited inputs.

- no V1/V2 proof artifact, conventional continuum law, fitted constitutive equation or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and opposed direction remain held labels
- no target value, official source, measured micrograph, selected specimen or favourable outcome selects a survivor

- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary evidence row
- no first failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

108. Interface migration and mobility record

Field: defects_microstructure_interfaces_multiscale

Obligation: SFT-MAT-OBL-MICRO-008

Claim: SFT-MAT-MICRO-INTERFACE-MOBILITY-008

Question, law and forced result. Interface mobility is the exact oriented displacement part per retained driving count, inseparable from the complete position-time path and observation boundary.

MICRO-008 uniquely retains oriented-interface-path-velocity-and-mobility-parts with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-COARSENING-TRANSFER-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered MICRO-008 preservation axes before external target release.

Boundary: Every positive finite generated microstructure carrier in MICRO-008, with complete site, path, interface, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-microstructure-carrier

- oriented-interface-path-velocity-and-mobility-parts
- complete-site-path-interface-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-microstructure-carrier__oriented-interface-path-velocity-and-mobility-parts__complete-site-path-interface-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:42ccdce0c36b289abce808a7b2798bce9141cf39d99f9b9a46e9ccb9cdc293a7; generality

sha256:9de254499377e16467b4853386d7932c94aaa08330fb9c0bd7676ae2dc880148.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:d11bb142c5f07748541be5db5a6c625a9703c1efda76b4639b69b7ad96013941.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:ecd377074285543227f1e72ca54cb603974a741d44e2cf25e452641551a8a0ef.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:738c0b35f7fa78a272501116b756bfae3949a5d2d9fc4f14b55b2f928f64a65c.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:cc0ac8b3731e664f6365761237f9b59437a067a351c423911816b9f0b7928e78.

Independent reconstruction and engine admission. Independent implementation

sha256:09f2ca2e37e2e5caa120823c6594cff6841df612380d4ab55ae7d8cb4b516727 regenerated the declared census and sole survivor; independent certificate

sha256:9acd32865c287c909fcd3cd0c485644555b0132591333d283d2041d04b41a6c7. Derivation seal

sha256:119c0198a996bef0d5ca8af90c821941275eb6fb33ac22fb184d4f3b71535b2e. Source manifest

sha256:f4728e65656404c4643b7ff163eb59fe310ca34ce4b2d0195bc6d9e8097aa16a. Engine receipt

sha256:ab5d106f5833f889f7251c0d603ff9f2616b6250b2c7f181a1a8553305576d5d at receipts/engine/model_admitted/SFT-MAT-MICRO-INTERFACE-MOBILITY-008-ab5d106f5833f889.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SHARP-INTERFACE-GRAINS-2001
- NIST-UNIFIED-GRAIN-BOUNDARY-MOTION-2008

Measured, observed, adverse and boundary records:

- MICRO-008: 2 distinct comparisons retained
- family custody: 9 usable and 1 unavailable sources
- exact correspondence: True; omission rejected: True

- NIST-SHARP-INTERFACE-GRAINS-2001: captured; normal velocity of the boundary is proportional to both its curvature; interfacial energy and mobility; favourable-use=True
- NIST-UNIFIED-GRAIN-BOUNDARY-MOTION-2008: captured; normal motion of a grain boundary; proportional to the normal motion of the interface; favourable-use=True

Shared claim-record clause MAT-S012 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:1942e0ce1fdb9c103fc85f48d45f517e47e5feff6eef78dccc162df68010cf9f; empirical validation
 sha256:1b1c1f89b4fc57354149f6e2bdd7eb417fefcee7c1cc36d5ad66734f83a37b13; external validation
 sha256:fb9c3594fcf1104098ff1cf4afb476c05c43fe735930ed5cd0458671d167f61c; isolation
 sha256:07e88f4b033b8f71ac1840baddb84b3836e447b97f1c8ea6001bfe4063585e83; custody
 sha256:0fbd3ba1e63cd77c32e90a7fa8ea3eb07687f8b460d120deace7627335b8616d.

Prohibited inputs.

- no V1/V2 proof artifact, conventional continuum law, fitted constitutive equation or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and opposed direction remain held labels
- no target value, official source, measured micrograph, selected specimen or favourable outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary evidence row
- no first failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

109. Microstructure-to-bulk multiscale correspondence

Field: defects_microstructure_interfaces_multiscale

Obligation: SFT-MAT-OBL-MICRO-009

Claim: SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009

Question, law and forced result. A bulk material response is the exact site-weighted composition of every retained microstructural feature response, with the complete feature-to-bulk reconstruction ledger preserved.

MICRO-009 uniquely retains complete-site-weighted-feature-to-bulk-composition with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001

- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-INTERFACE-MOBILITY-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered MICRO-009 preservation axes before external target release.

Boundary: Every positive finite generated microstructure carrier in MICRO-009, with complete site, path, interface, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-microstructure-carrier
- complete-site-weighted-feature-to-bulk-composition
- complete-site-path-interface-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-microstructure-carrier__complete-site-weighted-feature-to-bulk-composition__complete-site-path-interface-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:7f8ac13e6be3521e0c45dc893e8970ad68eb4f58cf4c38800fed54f5abcd3e67; generality

sha256:f4089131f4797671276e951c21bcb86fea0117b9c9b0436e8a441fbe3f297ede.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:7141915714b3bb630721000d2200a72fef5fd11aef2d0597af05d64b22721f81.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:ec77e6a73711b3f81aef80501a2131de84c72bb93e09fbf7136ee8def7d104dc.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:bfd7dfb3be08773d6199c57b3c3fc3192f2eaa05ad075e59c8cca348e5a4d014.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ba38f4310f82303b76fe4dc2cabe9b89fbd56d1dbcc019b53aa9e1080394627b.

Independent reconstruction and engine admission. Independent implementation

sha256:09f2ca2e37e2e5caa120823c6594cff6841df612380d4ab55ae7d8cb4b516727 regenerated the declared census and sole survivor; independent certificate
 sha256:04a4f75dbfde26ba520ea2bc0acce01f9c5a5971c30d6a2048893cb56bbd6301. Derivation seal
 sha256:7cd699585d77a1b8176a55dd5b95a66c7b115e99ba138461a65593f2c8b7bd61. Source manifest
 sha256:c164a089689dc7cc234a7365458321660555a86d2816b0a63e5cb8ffef90a3c. Engine receipt
 sha256:ceeac26aabb0b5e76fef5e2e0f234d22ac74a228da008d6a685d43163623c7da at receipts/engine/model_admitted/SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009-ceeac26aabb0b5e7.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-MULTISCALE-MATERIALS-2026
- NIST-MICROSTRUCTURE-PROPERTY-TOOLS-2026

Measured, observed, adverse and boundary records:

- MICRO-009: 2 distinct comparisons retained
- family custody: 9 usable and 1 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-MULTISCALE-MATERIALS-2026: captured_preexisting_official_snapshot; over multiple length scales; local microstructural arrangements; favourable-use=True
- NIST-MICROSTRUCTURE-PROPERTY-TOOLS-2026: captured; macroscopic properties from images of real or simulated microstructures; structure-property linkages at the microstructure scale; favourable-use=True

Shared claim-record clause MAT-S012 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:f8a88fc74c75634e8784d4e702a5375557244676f6a2c89db6f987e7389c99e5; empirical validation
 sha256:ec122a79bb8a48e544ba5e61f7b51143effe38b436f4520a5e69425733276d48; external validation
 sha256:fe74bcc81ccca41d082b598fe9a3c3f8936c36a339a15376faf6817051706151; isolation
 sha256:3ac39681a4e2a91d8c2c7d117becd456cf592174c54f46a73ba7dca487ce7172; custody
 sha256:eb21a442631ec50cde3be180b2fa19c204a299fbf44def20335371a15436b986.

Prohibited inputs.

- no V1/V2 proof artifact, conventional continuum law, fitted constitutive equation or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and opposed direction remain held labels
- no target value, official source, measured micrograph, selected specimen or favourable outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or scope-boundary evidence row
- no first failed route retires an obligation and no engine, verifier, receipt or admitted certificate change

Complete-field family: Phase Equilibria Transformations Metastability

110. Complete multiphase fraction ledger

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-001

Claim: SFT-MAT-PHASE-FRACTION-LEDGER-001

Question, law and forced result. A multiphase material state is the complete partition of one finite specimen carrier into held phase classes; every exact phase part is retained and the parts recompose the One.

PHASE-001 uniquely retains complete-held-phase-partition-and-one-recomposition with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-001 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-001 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- complete-held-phase-partition-and-one-recomposition
- complete-phase-component-path-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-specimen-and-phase-carrier__complete-held-phase-partition-and-one-recomposition__complete-phase-component-path-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d64af0cd3a6a40c9cabfe3e17e2a3d75017cc1ccc2a395396057beca9982c5e1; generality

sha256:197ad4d9de038c0663f44eba22abca75a2744463860372dd59d2344c538caf89.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:c275fef6b780ee27821e2d34e547d3be1555bcefc897a51c0ae62444f19e2d2.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:185f6589e1304e9a9ed2adb951d4a2dd861f0d0af84497f016029a68d1291f6b.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:84674513df450f75659aa7a77fd797f782d06c02d0b71a128768e3b0d9cba1a9.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:89581574f3e5d6ac123c80e6cabf09faf301829b340213dea2afe154ff5cefa0.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate

sha256:fb6bc1056ac45720c3374ce4f67b1b5b5b966892b63e3588dad59fcb1ee005ac. Derivation seal

sha256:1025f35b9f1fbf9df6585c0b682409d8a613dd730ac17b73461d5083a29ff30b. Source manifest

sha256:52efe5904d2814e582e5c644d91b74ebbc3be8d0736627b8e08a71648c392cf8. Engine receipt

sha256:fdb02cf19624df051b3716fc053c3bc646376a2afbddd8b09d6dc3415421c1 at receipts/engine/model_admitted/SFT-MAT-PHASE-FRACTION-LEDGER-001-fdb02cf19624df05.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-TEXTURE-PHASE-FRACTION
- NIST-LEVER-RULE-SOLIDIFICATION

Measured, observed, adverse and boundary records:

- PHASE-001: 2 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-TEXTURE-PHASE-FRACTION: captured_preexisting_official_snapshot; Accurate quantification of the crystallographic texture; phase fraction measurements; first=True; second=True
- NIST-LEVER-RULE-SOLIDIFICATION: captured_post_registry; phase compositions; phase fractions and enthalpies; first=True; second=True

Shared claim-record clause MAT-S013 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e8a19e481509b325f98a1d2d5e3cc249c3c7f3e2051c29d7d747393d39858ebd; empirical validation

sha256:c49848bd5dde8caa68fd302af29cf4f13d02d468cbcc40f287e83bf901d61fd5; external validation

sha256:efebed4d525c78c2de2cc17d1555d1732bbe6784721fae756a96aa451384b3c2; isolation
 sha256:c6016e10a727ee076610c982f07f1d08fe55cb9509e324305e417d0ef8cb855f; custody
 sha256:9ac77d8a1ef80c7d4e936426e865362c1f1466e6fb5f61a60384f360631ca3f4.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

111. Tie-line and lever-partition relation

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-002

Claim: SFT-MAT-PHASE-TIE-LINE-LEVER-002

Question, law and forced result. A coexistence tie-line retains both endpoint compositions and the bulk coordinate; the two exact opposite-span parts uniquely reconstruct the bulk and recompose the One.

PHASE-002 uniquely retains coexistence-endpoint-tie-line-and-opposite-span-partition with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009

- SFT-MAT-PHASE-FRACTION-LEDGER-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-002 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-002 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- coexistence-endpoint-tie-line-and-opposite-span-partition
- complete-phase-component-path-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-specimen-and-phase-carrier__coexistence-endpoint-tie-line-and-opposite-span-partition__complete-phase-component-path-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:6cef2cdb1ba498d5ce30112f76b8f1c16701c675601cf0f022daf7f88116961f; generality

sha256:a86afa863ca10af3baf50b5cb9dce7713dbf568c256389c66bb7e0acc2626f2c.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:03b3ef6fb953df6111238ce2e08a242bcd6a222c67245e03f4a06e4702b288d9.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:4779e452da8ad1e123aa5c987d94967c3b4757c11b9c5d1c6b4a94f4eb4371e7.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:5050773ff3fbe38714a32c1c10e1be6b5b32a09c4cade7926acd78787992cea1.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:aba3c559860c7fb9ca71dee3f55964c40d572d5fdfebcb87019d98de2ad88a801.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate

sha256:6b6b55c3c9548710e1d3b9047e0b8911545552e538344c1d284d74b19efc7b2a. Derivation seal

sha256:2428c529688e17a16fa71f532498b3a8b0a91c57ded5617e9c54aa6757b371b2. Source manifest

sha256:e9313fbc8014acd386fc75a117325e52da1cb5f23e43b1a354485560dda0977b. Engine receipt

sha256:76bf325991a7c5071cc2e173d9e6b9a523f8e95cedc5fd3e13d9bdbbed103850 at receipts/engine/model_admitted/SFT-MAT-PHASE-TIE-LINE-LEVER-002-76bf325991a7c507.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-LEVER-RULE-SOLIDIFICATION
- NIST-LLE-TERNARY-TIE-LINES

Measured, observed, adverse and boundary records:

- PHASE-002: 2 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-LEVER-RULE-SOLIDIFICATION: captured_post_registry; single point equilibrium calculation; phase compositions, phase fractions and enthalpies; first=True; second=True
- NIST-LLE-TERNARY-TIE-LINES: captured_post_registry; compositions of co-existing phases; Tie-line data are represented as two data sets; first=True; second=True

Shared claim-record clause MAT-S013 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:67b297bc933663bcb6b1ca7ba38d9fd364dd6347bcddd593138f274781a4afb0; empirical validation
 sha256:96413f1519d6f211f13a1d46f224fbdf049d827adbf68728b0eebf11fba47d3b; external validation
 sha256:a72362fd977bf4c637d0ba26f3f7a57ce4986f80810ed73a823772d0c518105e; isolation
 sha256:8d5c537dc76a7c68b60934ea8d2eb1988669ae387d6e4c76e83bb6c9f452652c; custody
 sha256:d6921747736814cad939434aaf9eeac24938af68937d5311e20f99b61ee61eab.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

112. Component potential and phase-coexistence handoff

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-003

Claim: SFT-MAT-PHASE-COMPONENT-HANDOFF-003

Question, law and forced result. Phase coexistence requires every held component identity to present the same exact handoff token across all coexisting phase words while phase identities remain distinct.

PHASE-003 uniquely retains component-wise-equal-handoff-across-distinct-phase-words with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-PHASE-TIE-LINE-LEVER-002

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-003 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-003 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- component-wise-equal-handoff-across-distinct-phase-words
- complete-phase-component-path-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-specimen-and-phase-carrier__component-wise-equal-handoff-across-distinct-phase-words__complete-phase-component-path-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof sha256:868ea649de183e4adb0e2123e9a8cc26c75f5ebce14339cc2954fc34ab0752ee; generality sha256:d2b8c7a6c3bec5f14ada423673a8360976d6c9e822bf277de00d2c518d689448`.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:0c6d5859ae9c86a97228422d39935b430a88eeb3c0bf6a2f418fb97eaf60d7ae.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:2fd46ed63c18ff9322e3b09bcc472fbc04b06e872492b6f346965ac91ad2edad.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:b09215c3e9b0851cc94cd3297874ccc44adaac1e289b783101b69a84b535c4b2.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ee8e0cbc9b8c8d2f7c74be48d121b445ef94b971a71d3de618062c8194a2e4a9.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate

sha256:60d01b3b733429b6b8f23a7fcd4e5152b2f643eaeb831c717a284c79dc32da6b. Derivation seal

sha256:bdb7b43ea8261261e0de86296fc9544eea2da68c55bf945eebffc35340d1b8ba. Source manifest

sha256:a1c9d56eb285edef342aca399c469d09fc4d64b3b9ad2d82ac3f3c3aba1468c8. Engine receipt

sha256:02c5a4d91cad16088babbf0a32333ab8c4950ff6b2f519d16a3cdd31d34eb2ab at receipts/engine/model_admitted/SFT-MAT-PHASE-COMPONENT-HANDOFF-003-02c5a4d91cad1608.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SAT-TMMC-COEXISTENCE

Measured, observed, adverse and boundary records:

- PHASE-003: 1 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-SAT-TMMC-COEXISTENCE: captured_post_registry; activity (chemical potential; pressures of the liquid and vapor phases were equal; first=True; second=True

Shared claim-record clause MAT-S013 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:136a5b0530bfd7a392432e6d7fba50df590579e4a569900fb4d7509c6c4fda1d; empirical validation

sha256:16c4eab01916804cfe44b9cc1cbf7f51cc46cd4dcfa9b328b756c3bc5b52b4fc; external validation

sha256:db4a6660a2cfd0a0b325942c99eedc682dd7ae5bde6e70ff24406533643e42b; isolation

sha256:55faf2df01dbde0159d45fa1defa4a1c85cc5c3f68f5baf63dde0a35d0558599; custody

sha256:40a3e71a227f89495c83ce16e1adbb07ed16659b7a9972e101f03ddbde5a09c5.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

113. Metastable-state retention boundary

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-004

Claim: SFT-MAT-PHASE-METASTABLE-RETENTION-004

Question, law and forced result. A metastable state is a nonterminal held state retained through a positive observation interval on a complete path before an exact escape transition; its history and boundary cannot be erased.

PHASE-004 uniquely retains nonterminal-state-retention-until-exact-escape-transition with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-PHASE-COMPONENT-HANDOFF-003

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-004 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-004 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- nonterminal-state-retention-until-exact-escape-transition
- complete-phase-component-path-organization

- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-specimen-and-phase-carrier__nonterminal-state-retention-until-exact-escape-transition__complete-phase-component-path-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f81304b99c4bafeca84c65e71021b47d832547edf342f9a5d952a926ac08cadd; generality

sha256:b5abc2237b38aa65b95579c19b3a9cadef6246393062bc5989a16eeb2b53293c.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:2c951b93c1bd8e54a29899497850b9d4c333c7f6131cfa860b3bec30fc6120a0.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:7a6faa6c8ef182ff2a525581820c83edf81956f56a5eb26f98ee0c337a15041f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:168ab2174e15e0479fdb917c55453ec20a2c8743e4c696425d9ef70f6d018c95.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:f21dde540a7cbb5307156975e14fb219e01e0ff128af092c91f1a7863d3a5cb5.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate

sha256:031bb9386a4bb4b1160c6822926cd219bc2d573d22701599826a2bd15d2206d7. Derivation seal

sha256:0679e4c2f8af2d11755c91d77458324c1d0bdb0849f99595d465f70e68c6d8cb. Source manifest

sha256:7475f62e3beb3cc2f7ba72a2a2e5d5d523d3e9950850fde621dcb1d7d1228a4c. Engine receipt

sha256:7c50a76f079d3017e36ac5f0b281cb70a3a3359fcb64e6259f0f0b52ba6c5226 at receipts/engine/model_admitted/SFT-MAT-PHASE-METASTABLE-RETENTION-004-7c50a76f079d3017.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-LIQUID-WATER-METASTABLE
- NIST-BINARY-HALIDE-TRANSFORMATIONS

Measured, observed, adverse and boundary records:

- PHASE-004: 2 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-LIQUID-WATER-METASTABLE: captured_preexisting_official_snapshot; some temperatures where liquid water is metastable; metastable liquid phase; first=True; second=True

- NIST-BINARY-HALIDE-TRANSFORMATIONS: captured_post_registry; Irreversible crystal structure transformations; function of time; first=True; second=True

Shared claim-record clause MAT-S013 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:5e114ed809d533ff73fb28973a9e92faf6f776252c643f584cdcbf8f11e373c6; empirical validation
 sha256:014855975131639c87ca7bc6376c96f16375be7b4952cd36575c728fb4ff055f; external validation
 sha256:24cb51a06e7a132929fcb98e76b67cf46c3c61e8892c72acc8b4092dcff299e3; isolation
 sha256:4164a7dc17809624cb59fbdc0192a471ad96e8061caabc3f31ed9ce4ad9e4316; custody
 sha256:22b1fc48714b90d7cc45c559146340461aa3867e06a5a2845ed67a2a94373efc.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

114. Spinodal-equivalent instability organisation

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-005

Claim: SFT-MAT-PHASE-SPINODAL-INSTABILITY-005

Question, law and forced result. A finite composition neighbourhood is separation-amplifying exactly when the doubled centre energy count exceeds the two-neighbour sum; the opposed orientation and equality boundary remain labels, never negative curvature.

PHASE-005 uniquely retains three-site-separation-amplifying-or-restoring-instability-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012

- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-PHASE-METASTABLE-RETENTION-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-005 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-005 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- three-site-separation-amplifying-or-restoring-instability-ledger
- complete-phase-component-path-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-specimen-and-phase-carrier__three-site-separation-amplifying-or-restoring-instability-ledger__complete-phase-component-path-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:3c87802e68e4ce44d8b2a79f06f2d5e0541f168363b7c7f96a0030d7bfe56835; generality
sha256:88584f944cfd79c958433112da694e3696049e4299116833336ab17db06ca7b3.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:71ce225893d413f9d96ee9f91483e1f3aa3dc086c708a4d6c81632c423074aae.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:43df58bc3e34e29cceb073d3875d514b924e4972ccd63375d2101ab35b275e0f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:7d0f2be7e95343274e735cfaf45618a7be7e927ba7b9b87bb617229b8820e5c1.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:c3cd6265c4861ac39b17c3f02e2b674db9147d9e5e30b26118363495a7a76af6.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate
 sha256:d59d602da42ce0df71a412bc602f0a55efc77b194ddf1c389ad749f77649efdc. Derivation seal
 sha256:1d77498cee046e6f1df735f7ddbda4435178bb4f8f13740007a2bc4a3b8aea0f. Source manifest
 sha256:6d63f60c385ffedeba12289058e1bf6274c281270e01d8762e860fbd00d87985. Engine receipt
 sha256:8c984394fff7d2c1c89630eb0e8a4be3ebe1699700699e20aee68de755386788 at receipts/engine/model_admitted/SFT-MAT-PHASE-SPINODAL-INSTABILITY-005-8c984394fff7d2c1.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-LIQUID-WATER-METASTABLE
- NIST-ORDER-DISORDER-SEPARATION

Measured, observed, adverse and boundary records:

- PHASE-005: 2 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-LIQUID-WATER-METASTABLE: captured_preexisting_official_snapshot; liquid-vapor spinodal; re-entrant spinodal; first=True; second=True
- NIST-ORDER-DISORDER-SEPARATION: captured_post_registry; Order-disorder and phase separation; driven by energetics of opposite sign; first=True; second=True

Shared claim-record clause MAT-S013 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e6615d0d0af1c461294909a251e83aa17de60f52f730494c262dda8488e77d30; empirical validation
 sha256:1962d5489a1405ed2526dd265d2b19c711d546b88844bfed53997bb0a6eecdab; external validation
 sha256:55e78fd91bc6c14d96aab21b3eae9e57d4249b04d8020a78e850e5e404afd0f6; isolation
 sha256:30048f8986ce6419f33d53f242a06cbe564e0ed77c7e859727e366e623214098; custody
 sha256:77aaf922c634c18b7ebbf4dab42f1b440ff7defd4d238d5804d81c9018a4fe1d.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

115. Displacive and martensitic transformation

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-006

Claim: SFT-MAT-PHASE-MARTENSITIC-006

Question, law and forced result. A displacive transformation is a cooperative held-direction mapping of every atom identity to a displaced structural state without a diffusive identity handoff.

PHASE-006 uniquely retains atom-bijective-held-direction-cooperative-displacive-map with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-PHASE-SPINODAL-INSTABILITY-005

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-006 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-006 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- atom-bijective-held-direction-cooperative-displacive-map
- complete-phase-component-path-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-specimen-and-phase-carrier__atom-bijective-held-directi
on-cooperative-displacive-map__complete-phase-component-path-organization__specimen-m
ethod-condition-scale-uncertainty-held__complete-state-transition-resource-trace__roo
t-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:4d271bd98e82022492c4014169fa717846b130dd19a0db973e8eeae734184d70; generality

sha256:43cc22ca30170e6654fb27f0852e37f2178781792bf6faa5d0451efa5e979fce.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:80caccf3138ae3b6de84bd4b5d9eec49855893a22af63b596222919847f995f8.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:f05af558063103b7fa6c4bc6856c3390827f3c770c0c60dc068ea3d7334d227e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:33fb13fed3022c55649e4d625c8e1d44f1355a6dde8eb1ef78079574faf63ba8.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:f86dc958675e7e9c28f2221eac48fdd84569a4b9bb96a324a4482a354ff701fb.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate

sha256:4228ef388a27a4534bf3c9b689c29422641c97da338df520f86c9c9b9da28b72. Derivation seal

sha256:e16860462af4eb435e3060175a0975469fcb64fc9258af963f547b68c77060c4. Source manifest

sha256:242b85a33a7fb9f8ba0949d1de22184d0f811bf598ad2a2626707ec7662cb281. Engine receipt

sha256:2b3d48180790e9a789f5bc4b71f17578b0931d32264772538c413bc40ef5a2ee at

receipts/engine/model_admitted/SFT-MAT-PHASE-MARTENSITIC-006-2b3d48180790e9a7.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MARTENSITIC-MATERIALS-STUDY

Measured, observed, adverse and boundary records:

- PHASE-006: 1 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-MARTENSITIC-MATERIALS-STUDY: captured_post_registry; virtually diffusionless structural change; associated shape change; first=True; second=True

Shared claim-record clause MAT-S013 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ff42cb7b08e8c076e984f177e4d96cd2e4329c151cebfa00729d5c3ab169a455; empirical validation

sha256:ef6825b8299f9dd291aa3fa9fcef8a6961fd8367ddfc64f33eccdd61fe31286c; external validation

sha256:55e7c459e6737e3ee8044b896e3a5266708be63cf3b8c762c2a94e2a66618ca2; isolation

sha256:4bd1f57f822d262aac2a69f3a2deb9b6bf66453043d1d96e270120f66d9b3dd2; custody

sha256:c1b358dc053dc657e5f3f7d5ab0ca7bf87bf2468379f0318f7f1170d739a197b.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

116. Reconstructive transformation path

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-007

Claim: SFT-MAT-PHASE-RECONSTRUCTIVE-007

Question, law and forced result. A reconstructive transformation preserves the complete atom carrier while breaking and forming exact bonds so that the connectivity topology changes and the full path remains reconstructible.

PHASE-007 uniquely retains carrier-conserving-bond-breaking-and-forming-topology-change with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-PHASE-MARTENSITIC-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-007 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-007 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- carrier-conserving-bond-breaking-and-forming-topology-change
- complete-phase-component-path-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-specimen-and-phase-carrier__carrier-conserving-bond-breaking-and-forming-topology-change__complete-phase-component-path-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:b88c027dbab3c2a8c3dea833dac8c4952a75742847c8f4396068017868b120f3; generality

sha256:a55a7b678af93e19d0121ae29e9bdf13a37deb9f279cf07e66a5760c88103f.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:498d4923fcd67be82b3c6f83521fd56a130b59c765798d05b0ddc1a2f73311c0.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:694947b6f4bdb71e9f48b6a45772eb15020b34e43054b9cb157e9a7eec5c26bc.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:80ab986edb57fa0fedeb820a39c672d32701622fa01710a29f3f3b7bdb3e88c2.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:d26ad7817cb70d7500d8d1c9087cc01600b309401b01677527a40ef276e09abc.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate

sha256:a9c8f4b05acfbceb3f3f53e75861ee3659234b3b9bbc52b36cb290ea56a44d1f. Derivation seal

sha256:74747abf40879622e70e8b40ace7589b90e1d78a2dc6dabc803bc1741ef6ff92. Source manifest

sha256:96d4708761a2e45ba8c6543e8fb5d2f841bf777751f1af8622c321c505ad9aab. Engine receipt

sha256:d50bf0ad012aabb49da0e88e5c0b82cd35e9e4d71e2897974a34a653d9776611 at receipts/engine/model_admitted/SFT-MAT-PHASE-RECONSTRUCTIVE-007-d50bf0ad012aabb4.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-BINARY-HALIDE-TRANSFORMATIONS

Measured, observed, adverse and boundary records:

- PHASE-007: 1 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-BINARY-HALIDE-TRANSFORMATIONS: captured_post_registry; Irreversible crystal structure transformations; remains in the new transformed structure even after cooling; first=True; second=True

Shared claim-record clause MAT-S013 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:cddd29cbd79095983ca4fdae4b72cf0daa60deb1126b9920828b7d096ca0abb0; empirical validation
 sha256:05ce710acf4e9052177b48acde340d657a3b2624f5f6695c7fa517c89c65ac; external validation
 sha256:5fd89b75382b2286b26af94be30791148146af1d62f98667eaf98c6611544ec9; isolation
 sha256:2c11abb7c8bdfdfccc0b4d58ad7d50c5900dd63b6c6f7d33806e2bb5539c87957; custody
 sha256:14b86b1e3a53fbfb546e184e6b9c792ffe643b79a82f006a052194946e1ea324.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

117. Order-disorder transition

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-008

Claim: SFT-MAT-PHASE-ORDER-DISORDER-008

Question, law and forced result. Order and disorder are exact site-by-site relations between two words over the same conserved constituent carrier, retaining matching and reassigned parts without a fitted order parameter.

PHASE-008 uniquely retains same-carrier-site-match-and-reassignment-order-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-PHASE-RECONSTRUCTIVE-007

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-008 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-008 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- same-carrier-site-match-and-reassignment-order-ledger
- complete-phase-component-path-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-specimen-and-phase-carrier__same-carrier-site-match-and-reassignment-order-ledger__complete-phase-component-path-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:d6deddc400ec19f42c110da7f9d08c8219635e3bd5ce4a84e98a4384cba7599b; generality`

`sha256:40f06d3b706aede6cd78655a9e6f105e5c29f311a6286cc02c76c293b73c7480.`

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:b85f72062cc85e82a841e68fa0b3160117ee1503f5a5c7067fe60b4c02d0405f.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:3cef4befe78bb1332724c34fdf347fe7a582041f318cde623b9043d6dcd6a74a.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt

sha256:2e56e6bce4a5a8d92617cc8b032d64bcfe2431fcea8846e9d7926d891edc430a.

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:0d9a4b67c08f73b2aebf06fdd46fe3317e7a4c972c57769e3c7d1722ad263a06.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate

sha256:c9d22f86833af7dd2f06f9146ce74aec1c4fd8f312c609613e4e4a912f3e27ed. Derivation seal

sha256:366bf3b0600c3bcb816fec0bcf293a94ced846d6b31b8d7b70a964d873339345. Source manifest

sha256:438dc2dce8d1b8a6cf16b660c6a9015b52f6f5c66e77a654381200627ecc1b65. Engine receipt

sha256:6370db6cbde2d909482bdedef0ebda821fe9caf134555297275a058acdf653c7 at receipts/engine/model_admitted/SFT-MAT-PHASE-ORDER-DISORDER-008-6370db6cbde2d909.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-ORDER-DISORDER-SEPARATION
- NIST-BINARY-HALIDE-TRANSFORMATIONS

Measured, observed, adverse and boundary records:

- PHASE-008: 2 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-ORDER-DISORDER-SEPARATION: captured_post_registry; condensed solutions of two or more components; disordered state has greater configurational entropy; first=True; second=True
- NIST-BINARY-HALIDE-TRANSFORMATIONS: captured_post_registry; increase in disorder with rising temperature; order-disorder change; first=True; second=True

Shared claim-record clause MAT-S013 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:b1894181edac990778cecd94c0903a5a383199547ca6bb99565456e9a3b777c8; empirical validation

sha256:cfc051a6cf3cd5e4df1e1907ec765296a971ef4c7d4a926acdb0878f14f7e383; external validation

sha256:74b6d9d866f11ffe4d1ccac3796ae5abf21c9540afdeae1207ebdb0587bf0606; isolation

sha256:373908fa3307243303e3142d7f132553ef9d37deafcc0467a8bdb0bb4dd62f73; custody

sha256:83907ff5eefce1a3297a24943253538d8f830a34f5a3538946cab9a0e7d80f46.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

118. Glass-transition and kinetic-arrest boundary

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-009

Claim: SFT-MAT-PHASE-GLASS-ARREST-009

Question, law and forced result. Kinetic arrest occurs when the retained observation word closes before the positive relaxation recurrence; equality and observed relaxation are separate exact boundary classes.

PHASE-009 uniquely retains observation-to-relaxation-recurrence-arrest-boundary with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-PHASE-ORDER-DISORDER-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-009 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-009 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- observation-to-relaxation-recurrence-arrest-boundary
- complete-phase-component-path-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing

- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-specimen-and-phase-carrier__observation-to-relaxation-recurrence-arrest-boundary__complete-phase-component-path-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:e6b19ceadf72453278234c55adaadaf90ce005a26dd4bac9e3d8a7431d468976; generality sha256:50275f09842415a21d2e1123b358876291673c6226b16fe95c406595ea6ca383.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:116cf49d45294794d9dc16f437fbd6159cbdc37f304a2dc6d2c1d37a1f3497bb.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:ef191397d2426a45e271df09f4663e539a0cd0ab5e2e0a99a41e9407b1a1c144.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:48d459fe621c7d9986c2507e57f150bb0281f678a1b57f2621b49e169cd04f9a.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:bca22ae8aa93e43a95b342881b1436551ea27659f866d6c5c7297506d9d57c00.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate sha256:a90b39eeb8251ac1841b73d3e93a50497a34ebd4794eca782f4ecdd43cdfaa1c. Derivation seal sha256:8b3651105d831a6554c42092dc533c268d80e4e8a5f9c80da3d5b26a4dec9c25. Source manifest sha256:f6f5ea556350bafc719cab47693962eef5297a85b2c38f53f9aad5ee3b9d27d0. Engine receipt sha256:144c6de994caf84feffc382ac0402dfa0bf200d0d0fb1e7172ec02354e2712d8 at receipts/engine/model_admitted/SFT-MAT-PHASE-GLASS-ARREST-009-144c6de994caf84f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-GLASS-TRANSITION

Measured, observed, adverse and boundary records:

- PHASE-009: 1 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-GLASS-TRANSITION: captured_preexisting_official_snapshot; thermodynamic and kinetic aspects of the glass transition event; techniques used to measure the glass transition temperature; first=True; second=True

Shared claim-record clause MAT-S013 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:23efc535395c0f21063245a0e56796af74fc7878f17f23f81ffe18a5ea76ac8; empirical validation sha256:97c37194ceacf95ed7792e5c58d46ed9b03630d007ee8a60044ffda3c280ccd4; external validation sha256:44bf7f1712ac1f1e075e93c9896d3a9383bdb886031701f2957e638d5ead500c; isolation

sha256:8df660d3653d9be8ae30aac93640d842607aa24932840e2b7e470363f6a38964; custody
sha256:0312d7754c9d03e3b111b8fd30e57f360cf87635134e87b8d39ae802079a6d07.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

119. Transformation kinetics and time-temperature path

Field: phase_equilibria_transformations_metastability

Obligation: SFT-MAT-OBL-PHASE-010

Claim: SFT-MAT-PHASE-TIME-TEMPERATURE-010

Question, law and forced result. Transformation kinetics is the complete ordered set of exact time, temperature, transformed-carrier and total-carrier records; structural absence is labelled and every nonempty fraction is rational.

PHASE-010 uniquely retains complete-ordered-time-temperature-transformed-carrier-path with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-PHASE-GLASS-ARREST-009

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete literal product of the eight registered PHASE-010 preservation axes before external target release.

Boundary: Every positive finite generated PHASE-010 carrier, with complete constituent, phase, site, path, specimen, method, condition, scale, uncertainty and proof distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-specimen-and-phase-carrier
- complete-ordered-time-temperature-transformed-carrier-path
- complete-phase-component-path-organization
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-specimen-and-phase-carrier__complete-ordered-time-temperature-transformed-carrier-path__complete-phase-component-path-organization__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:7c169f93fde5f0213e43969654a72f2ad916ae6ae794ee38cabd77a831196b24; generality
 sha256:ad46b26a5a60ca75151ac152a9a7128dd9697daccbfff98796ede076f4949e.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:e0e7fb853c975806873eb7cd03a75edc94d06971df7c27623d44990c7f3e7964.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:7f486f6618c1526d9ab2c6faa9e9bbb9f01b67446e56be248a4723ba2ea6823f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:63517c19722c78703bb6f9b2ef4e9a91d1d13573cffe9cd075694095a317af9b.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:8588b9aceac124c3cf2ed9d0cfa87f7a1af84e9e83e2ef5931550bc7f486948d.

Independent reconstruction and engine admission. Independent implementation

sha256:6db7a6170f4a02ac96311fa86aa5fd224b3a5f7b7b6ae27b2de7c05182991802 regenerated the declared census and sole survivor; independent certificate
 sha256:74507603c1514b2b4936173ea956eb63d7b19c33f8d655bc9746853bf06dd940. Derivation seal
 sha256:8cd59423c0082437aa7463dca509eef25d4ee8ef13df1538bda8fcc526dc7f01. Source manifest
 sha256:469b44dbb0c13178c909a10d7bdb1c89ee7d1045abe0d4f73553dd88e5fc6707. Engine receipt
 sha256:71edbf8a3e8928061e68a084735b8b1584ca55d77bfcd1924b466206d6ed2e23 at receipts/engine/model_admitted/SFT-MAT-PHASE-TIME-TEMPERATURE-010-71edbf8a3e892806.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SOLIDIFICATION
- NIST-PHASE-TRANSITION-TEMPERATURES

Measured, observed, adverse and boundary records:

- PHASE-010: 2 authoritative comparisons retained
- family custody: 11 usable and 0 unavailable sources
- exact correspondence: True; omission rejected: True
- NIST-SOLIDIFICATION: captured_preexisting_official_snapshot; nucleation, growth kinetics; Polyphase solidification; first=True; second=True
- NIST-PHASE-TRANSITION-TEMPERATURES: captured_post_registry; compiled and evaluated phase transition temperatures; recommended phase transitions; first=True; second=True

Shared claim-record clause MAT-S013 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:574469a9e22e4e5441dafd719b1dfba6da76ae1c179d067f9aaeb86e33ce6137; empirical validation
 sha256:40e2a3bcadee32eb74e32c16f3a0e5ff6b416ca9f9fe87eb2b8e356c63059d0b; external validation
 sha256:29bd29d931209e64729c8c5a5ff202d169ae88eb433b8b00be7f1c766c13227e; isolation
 sha256:638c657dd18ce3f06f2a44cc9bdd4fd85d334cf6601e158754a2f7d11f1ba335; custody
 sha256:6d59faa875ac5d664724a5f356b6a6d5013c6829d77aeb91006191b29bad7fd6.

Prohibited inputs.

- no V1/V2 proof artifact, continuum thermodynamic potential, named phase equation, fitted constitutive law or consensus classification as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, opposed orientation and boundary equality remain held labels
- no target value, source fragment, measured transition, favourable specimen or database outcome selects a survivor
- no omitted adverse, absent, unavailable, unresolved, tampered or condition-bound evidence row
- no failed attempt retires an obligation and no engine, verifier, receipt or admitted certificate changes

Complete-field family: Mechanical Fracture Fatigue Creep Tribology Rheology

120. Tensor-resolved stress-strain response

Field:mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-001

Claim: SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001

Question, law and forced result. Mechanical response is a complete exact axis-by-axis ledger of positive load, extension and held orientation; no scalar average erases tensor distinctions.

MECH-001 uniquely retains axis-resolved-held-orientation-load-extension-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-001 axes before target release.

Boundary: Every positive finite MECH-001 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- axis-resolved-held-orientation-load-extension-ledger
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-mechanical-carrier__axis-resolved-held-orientation-load-extension-ledger__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.`

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:c038117b5a81c1e9be903336c0c18737907721ed4f3fc559c696fba7558f4f1f; generality`

`sha256:4f55f3223610fe5962ddc38e3581efe0e9e5daeff2dd4ea93707af0cbd0ba4bb.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:61669ea5951edc28006a6631aa9f2eb73840de2a4a0ea1f8426a88aa8d9ce9c8.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:7b6156c4e4e85f06490ce526d7da8b6ecefef419002bbaf93b9a46103d65d3fcb.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:a05e91b96bb66f2da2a8cf0c95ee0949813d5435682de5739480e3d4f72d1334.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:c6804c6b5db1d56366060b7e142d5f97a8f8d6f2734b3b58287edc58bd077a79.`

Independent reconstruction and engine admission. Independent implementation

`sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613` regenerated the

declared census and sole survivor; independent certificate

sha256:16c1ec537dd90a8e891412468d7846010eff9fa3b467b5107191526319d4221a. Derivation seal
 sha256:a0ee3f814e0eaed582563a5d7867100aabde5d04aa9a255642c70491cf86a8f4. Source manifest
 sha256:2c62f0eca86aab1d68cb60861fb9b23ad2769adc66f5711553ec24776d2f18f2. Engine receipt
 sha256:75ee953ca94f9c3df746a63cd62d94ec034df39948be0e474fe8666f61cfc4ae at receipts/engine/model_admitted/SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001-75ee953ca94f9c3d.json. Closure
 scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-MARTENSITIC-MATERIALS-STUDY

Measured, observed, adverse and boundary records:

- MECH-001: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-MARTENSITIC-MATERIALS-STUDY: stress-strain curves; elastic constants

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e3c53230e159aed7c2330763bd1fc17366c28dbf5498ae42744bc1ac9b067027; empirical validation
 sha256:5547f99bf426929f50901d644a1e243a493cc106311c46cd8b19a80195d1b124; external validation
 sha256:64e44154f605eb0278cc50dfd710462f176f7cb114f37724102a1ae8862c45e6; isolation
 sha256:2a40cf93a00b4dabc99f0511c7ff9736c879c202da89d849ee43a431f3fcec84; custody
 sha256:a5c05d9d8ee6be261088fcc0a2ec6cb8b883fcf9d5354268a30e7a2bc80db6ea.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

121. Transverse-to-longitudinal strain relation

Field:mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-002

Claim: SFT-MAT-MECH-TRANSVERSE-STRAIN-002

Question, law and forced result. Transverse response is the exact positive magnitude part of longitudinal extension with contraction or expansion retained as a label, never a signed quantity.

MECH-002 uniquely retains transverse-longitudinal-positive-part-and-held-direction with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-002 axes before target release.

Boundary: Every positive finite MECH-002 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- transverse-longitudinal-positive-part-and-held-direction
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-mechanical-carrier__transverse-longitudinal-positive-part-and-held-direction__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:6b9774dba264904b2bb059057352986d62e3ff0da3275e00efa7aa7fd1c464f3; generality`

`sha256:41519ce0db3f799093000933ae1530be6819e00344953d8517c9a053436282ad.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:e73d6aaa9a2f32f924fb0d0a19355b14623c29f69da3f30791f030bd918786df.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:b6d4cfed63bfa8d08680939b4e143883d78a75f72076b2276b673824b7b46364.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:576bd096bca7768e942325b0c0b0e8eff6238d5eaedd8fb44d24d25682476660.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:8bc014e6b85edf467cd41dd20432292e88074dba58b7e06f0992c6d01407649f.`

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate
 sha256:2b7bdd711295a1c4f1982a9ab3a063a83f22ed5766cb8563fa64011a69470697. Derivation seal
 sha256:81a19fd74f637270ac9f7478e5a0f284a7a823296ae29486490194c80fbf7205. Source manifest
 sha256:320beb3f07c080ff16665561c5230e6816561a5587ad8db353613e2150c460fd. Engine receipt
 sha256:6da865c282e26407adf565ed67d845b00a50f9cdfb654764e90974f230c1e96 at receipts/engine/model_admitted/SFT-MAT-MECH-TRANSVERSE-STRAIN-002-6da865c282e26407.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MARTENSITIC-MATERIALS-STUDY

Measured, observed, adverse and boundary records:

- MECH-002: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-MARTENSITIC-MATERIALS-STUDY: stress-strain curves; elastic properties

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:4a55ad8800a803dbe5c817b13e416d5b02387fd1360dc84cd8ff8edc788e49fe; empirical validation
 sha256:ded0b55501fba64d82f47c196a1225c37459ea112377c5ab7a765436862a2f57; external validation
 sha256:a2873df73a008e0a62ff1aa118d972e42def28522332c0ed9b16e0d744ba3a5b; isolation
 sha256:69831e2862a3d5388ab9371905ab6f6ecb8280ef1a22df051c8137cebc1ba2dc; custody
 sha256:43bddc8d7f6f1da82204f3727b7d858244788f5282da544df29ddd06888759e7.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

122. Viscoelastic memory and recovery

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-003

Claim: SFT-MAT-MECH-VISCOELASTIC-MEMORY-003

Question, law and forced result. Viscoelasticity retains the complete loading word, positive recovery and any unrecovered remainder as structural absence or positive count.

MECH-003 uniquely retains complete-loading-memory-recovery-remainder-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-TRANSVERSE-STRAIN-002

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight MECH-003 axes before target release.

Boundary: Every positive finite MECH-003 carrier with complete path and observation distinctions.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- complete-loading-memory-recovery-remainder-ledger
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-mechanical-carrier__complete-loading-memory-recovery-remainder-ledger__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:40c21841e6cb4cf2e741a183a2b39a06d4f5eeefbb06f83ea2db8cdb9d77c3cc; generality`

`sha256:064c071d7709b1fa278d072f366337d67f4057ed33e29f227a097c8a2052df44.`

Shared claim-record clause `MAT-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:89105f2ee757983fbaae174f70972efa185d64de1988db1b0e6bc4fb602cdb23.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:6cb0bfb137e444fc9e6cf801c5d52d4141ea37c569a9288ac68aa9b4eeb65299.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:aba234f52dcc46ea8629c4017934bd1984803caef6d4e9e127a9cef9373be5e3.`

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:1543f3dff640e36dc0d26e53b9f79333036df78454744c8fe6ad8a2780f75403.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate
sha256:28d70fcaf65d1418b8f90aff4f5609086ed08e5f3085f1f0efc54d706e83440d. Derivation seal
sha256:04d9b835049e5e1be139831f02404ac9fb3e1ff3af4e58b17ea383dbb35d92a4. Source manifest
sha256:0c4ac9c69387c007247dfefc7228a4993c46e29c3d5ac27dd9c9cc116747b54a. Engine receipt
sha256:553f84e6d310b623a0c7a4ba93e55634866255fba42e2e3c7e9986dd2803e3c9 at receipts/engine/model_admitted/SFT-MAT-MECH-VISCOELASTIC-MEMORY-003-553f84e6d310b623.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-VISCOELASTIC-SEALANT

Measured, observed, adverse and boundary records:

- MECH-003: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-VISCOELASTIC-SEALANT: non-linear viscoelastic properties; stress-relaxation experiments

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:9d86ef160654270e355aaadeec355bdcd4da6579f57489a8162624170438c218; empirical validation
sha256:1f4cc749fbf035edc5900b4038c7ad2727ee320c45bc6a008167d4c062708a5c; external validation
sha256:008e0a6b0d5d30d1fee7306d9e6abc50e11b5654d308f899af9ec38b83ce3abe; isolation
sha256:a81f48eedda4d673cc1662263b08d8382fe30d582e0c912d5d5287e5f81419c2; custody
sha256:82df9cd2363118308793e7c4f8fd74a538ba06cc9bd437e272af80eda56a3a59.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

123. Viscoplastic flow relation

Field:mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-004

Claim: SFT-MAT-MECH-VISCOPLASTIC-FLOW-004

Question, law and forced result. Viscoplastic flow is exact applied deformation partitioned into recovered and permanently retained parts after the observation word closes.

MECH-004 uniquely retains applied-recovered-permanent-deformation-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-VISCOELASTIC-MEMORY-003

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-004 axes before target release.

Boundary: Every positive finite MECH-004 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- applied-recovered-permanent-deformation-partition
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-mechanical-carrier__applied-recovered-permanent-deformation-partition__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f79a8f87cb40e52f79fda20510553230c5579ebefe801ab2f4577f6c07e149af; generality

sha256:f7c745a7576a83d9348a96c3189befa09d1982bbf044b8e964d150e0542d9491.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:d3b580cdc628a324c37aca6a9999fefb4e0566f67c34b872f07a3b379ecdb7eb.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:2e4ab0415f045735e30ac0acba3011a75a0d25077f75ae9a740bd7e20c261727.

- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:86f9545ee7cc33d83f440dd5df8f6710bd6144b8a39f04aaf29bbc2207a2ac4b.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:0cc48ab6efeb86490e408d5f910fc52cf80fee5a8ee683ff11e33e8a2991fbf7.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate
sha256:ef784270e95cbcee88d251933466962b5d4e22fec4a92836c459c3cfeebd2bf3. Derivation seal
sha256:2564ce362a15432bedd648136f1d1ddf9f438e04320efde2c4383f75cc19bd35. Source manifest
sha256:4df5760989aa5080d7ab701bbf97cd9842de2e12db27141577f07023f9444493. Engine receipt
sha256:799d82588b4d636407becfcbd95c697a1a68ea15178be65b30bb5b5517984437 at receipts/engine/model_admitted/SFT-MAT-MECH-VISCOPLASTIC-FLOW-004-799d82588b4d6364.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-RHEOLOGY

Measured, observed, adverse and boundary records:

- MECH-004: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-RHEOLOGY: resistance to flow; function of the shear rate

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:6eedc2ce69a1d0ff8d6a88d82113e4cacb0f1afc7fdf2814bc7d2fe20d7b7d5b; empirical validation
sha256:a881df1fc17d9a461c7d0dfb159d1915514a46afca871ab5ef9fd30bcedd15ef; external validation
sha256:0816ac794fe774781480757d9013d614b2632850882187940c728fe4160c2b07; isolation
sha256:2e58d1167c0b23922d706a838e5498085e8f361b274164e96ebe8383140e97a5; custody
sha256:a8cb46e131cea19a213cde8da6adcf8de56d5185729f513e1c52beee83476166.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

124. Yield-surface and loading-path boundary

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-005

Claim: SFT-MAT-MECH-YIELD-PATH-005

Question, law and forced result. Yield is the first position on a retained loading path at which deformation changes from recoverable to retained.

MECH-005 uniquely retains first-retained-state-on-complete-loading-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-VISCOPLASTIC-FLOW-004

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-005 axes before target release.

Boundary: Every positive finite MECH-005 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- first-retained-state-on-complete-loading-path
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-mechanical-carrier__first-retained-state-on-complete-loading-path__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.`

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:e80a62ce7b2d775cb6db9ad1bf745cebe434d2ea1d17f9f09ad19a8a790083fb; generality`

`sha256:872655748c178608dcb2c29adbb75e0a78e28b1f8cb7d0de9b84c5489f2e0509.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:5ff26249ec2655aa5d46f5fbde588a20743b5d801dfd134a90e8c3b1efe77d50.`

- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:035c3287589212fffe822d7193f716a8a3dbb542f852c91f1906c10ca663bf5a.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:079c430bd619d33282d594a7b0f0cb6a3ffa392380a52d2b92efc75cf2546c12.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:759c37515efa675dc460969fc4962e7c6048c06915cf06f04966a662874ad64b.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate

sha256:341787def0dd8cf0245921e081152ac2608a94eac1f06b647601cfa556527adf. Derivation seal

sha256:2c803c37db4bb9d2e4bbb694d69adc9fbe42d23bd0c6508a9175a2f6e46dfeb3. Source manifest

sha256:53125923e57e26f77662c7d0973b0895389d733c83c5acf3010d07d3869a9d94. Engine receipt

sha256:5f252fe1aa5ac3debb6d6f9e63d8738e4736715e5dd458ed4ad3a0cc2ff6f06a at

receipts/engine/model_admitted/SFT-MAT-MECH-YIELD-PATH-005-5f252fe1aa5ac3de.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-RHEOLOGY

Measured, observed, adverse and boundary records:

- MECH-005: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-RHEOLOGY: viscosity and yield stress; force needed to initiate flow

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:308d16ef05176a0e2697e8e140da71d7fe7c6986e1f473d3c11a8b1c4185d491; empirical validation

sha256:414534b9dcbd9eec22e51a57a6f1239c3ec488793d539932dedd7fac02aaf975; external validation

sha256:faa14b73942e16b376db0f42c851763f7d4489e8db158ea57a575d684a5e53c6; isolation

sha256:ea9766501eff225450e9ddd147e0db75d221aa0dc9fdf5c4e5917ec5d600d5f7; custody

sha256:45d247a856e2b7ef6366cf20739f442f9e84690b0f409c3d64ffa8a18f00ea91.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

125. Work hardening and retained deformation history

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-006

Claim: SFT-MAT-MECH-WORK-HARDENING-006

Question, law and forced result. Work hardening is a later exact yield count greater than the first, inseparable from the retained deformation history that distinguishes the states.

MECH-006 uniquely retains history-bound-increased-later-yield-count with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-YIELD-PATH-005

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-006 axes before target release.

Boundary: Every positive finite MECH-006 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- history-bound-increased-later-yield-count
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-mechanical-carrier__history-bound-increased-later-yield-count__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof sha256:27e1b85c467cbd268300a84b10f2acf3e650d6ded242d4bcb8990ec912c1f8ef; generality sha256:a1e1fc5434503915cf414429fbca7aa760e5fdc3b5bd655f8e63bf69ff9792c4`.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:89f280f9d5588c1118ba3fa01f73e78b5626a2be6c2e4aacfc5c38491ceb47c2.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:bee073301dd0167de2011821f1959095671f13a8c0e0e33375ecdc91760f6860.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:fb63db258201e8e3f746f742fa90003cea5ef64766b7dbb297efacbac5fa6656.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:4a9e9fc9c8de3d4a939079e568ff364d873a4282420ac3158fdef05f173e998c.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate
sha256:fcf51250761f122fdfa6e784bd48f58df56ed54be671bd8a744eb34dec3807c0. Derivation seal
sha256:9d86ee18838b532cc384b308c146e63783f03f12b03915a60cbcd7db90091722. Source manifest
sha256:436fa75d62fe0fe1ea9014ec2a3eefae74759877d686cdbc8551d1e2626f07d. Engine receipt
sha256:9ac5c14209b96d41254394f844de68749fb9ac484d9d5657ad7182a9bd77f827 at
receipts/engine/model_admitted/SFT-MAT-MECH-WORK-HARDENING-006-9ac5c14209b96d41.json.
Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MARTENSITIC-MATERIALS-STUDY

Measured, observed, adverse and boundary records:

- MECH-006: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-MARTENSITIC-MATERIALS-STUDY: degree of work hardening; flow strength

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c4ea7950e0af6797847e3cb9d29de25fa325199555bd8c0ee3dab5f806bab418; empirical validation
sha256:bb997c28fd6c24553d62ed2058879e7f15ce882b2928a71e7b4e88ba19655dde; external validation
sha256:6d5abb0d61fd706f5fb409feb548097eed8a2816d6bc5ff1c164c95a53827b11; isolation
sha256:499eae3a439fe6c03506551d6d995d4ea86b702c42a8e7a8bd1cfc15d2d71c1b; custody
sha256:b520b920f90ecd2113cf2eb08a4e159025d04e85c80b660809c21a302b7d1fb6.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

126. Fracture energy and toughness ledger

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-007

Claim: SFT-MAT-MECH-FRACTURE-ENERGY-007

Question, law and forced result. Fracture work is the exact positive work carrier per newly formed surface carrier with complete specimen and crack-boundary custody.

MECH-007 uniquely retains exact-work-per-new-surface-fracture-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-WORK-HARDENING-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-007 axes before target release.

Boundary: Every positive finite MECH-007 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- exact-work-per-new-surface-fracture-ledger
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-mechanical-carrier__exact-work-per-new-surface-fracture-ledger__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:2c713596033e03562b4ee18370a7c1dd13345e382547bf34cbcdc173833242db; generality

sha256:6a2d19255d9e368b6a772368db7b0fdde8f5f0da0f0d1d93bd05267755a519e0.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:765150cfa51df6c6b3c542c521a29c837408c9288c5898b73d6616ba76cec57f.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:6c2dd18c1f227964cba58f2df75c8bd4ae7529f212f04980422d4ccb25730eb0.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:e8b747f73338bc3a307156f216a5f7c052c840a409e1bdbb5f96ced89916d8ea.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:a039145c9f07b3127cc23064c34d31bcd6f477a85cd0795f4596218e8b3e3abe.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate
sha256:bf2a3e181fc388fdda6dc63d9c32eb8e40b6ef2954a518aa59ff02844b4d92c0. Derivation seal
sha256:c7e31960202a99b4387da64153a449f645fe2d6e6b59e0594fab478cd5f8d189. Source manifest
sha256:42ad725796695acb6ee4fe7754547179f30f1acb26049c0a01637a97c0156f9a. Engine receipt
sha256:cb8423938cc668b23e699e5659bbfa492ffd835fd7cda1da0ec58c43d40258bd at receipts/engine/model_admitted/SFT-MAT-MECH-FRACTURE-ENERGY-007-cb8423938cc668b2.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FAILURE-PROPERTY-TESTS

Measured, observed, adverse and boundary records:

- MECH-007: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-FAILURE-PROPERTY-TESTS: fracture toughness; resist this fracture

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:8db429ab69cdc6eaf9b16092e333c23f63fe01b10d7d2cd9ab0fcc5cf9caab74; empirical validation
sha256:95d18f84e7137dab5d82a2e1e3000d3da78aa6425c53a0fb27c464081e52ce55; external validation
sha256:d5f8e11fc6c3a8077ff4352038dab8a768882fcd1ddf193e4a81a88c30bb5d15; isolation
sha256:e438bdb4ee17faecc62d7b0d143da7c4cc49b0c17892ce19c2d08f08811c3b5b; custody
sha256:42060a4f5b43e1ea6839a7cacbb28ec014ef333c0cba0a086d0e5be315035b3c.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

127. Stable and unstable crack-growth relation

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-008

Claim: SFT-MAT-MECH-CRACK-GROWTH-008

Question, law and forced result. Crack growth is a complete nondecreasing length word whose terminal state is held below or at the registered instability boundary.

MECH-008 uniquely retains nondecreasing-crack-word-and-instability-boundary with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FRACTURE-ENERGY-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-008 axes before target release.

Boundary: Every positive finite MECH-008 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- nondecreasing-crack-word-and-instability-boundary
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-mechanical-carrier__nondecreasing-crack-word-and-instability-boundary__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:9eee0b4d62012fdd1103922a14e470cd68f54921fd93ebb48da2c7e0abde4542; generality
 sha256:9d20ba764520e9e6c411d90e6d846340a8b2787e5155f40560b81383b6c78cde.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:bb9845c6efaed5175fa34056f5bc9575d97754fff6be8474d6770540c5a00076.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:96c04b0cb6020c7cdd3c92e9d60cb796981d1c1f38f77d291f68d0edd73c4399.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:01748b184f70517e3df7fe2b1a57e64e356adbff3c82f4ce54f09fc42018e304.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:f8c8e2821fdd52e77030845a2dc4dbae795e3c0af0aa0bdc6793e3b4b48011c7.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate
 sha256:4ea29fe1ad5441ebcf0a6e2161027841aede4d26986b11a19885ac51f9ccc955. Derivation seal
 sha256:10b976d191cd41d01dd64be5f74fe44eb4e4a9d3bf21b49b5f8b0f5ff58961df. Source manifest
 sha256:f2a8fe6623575b531e9f581d21104a728a24d836d0e82adad2c914f075145338. Engine receipt
 sha256:f12803c9e26564d123a26760f7250e727637707dcccdaaf0b86caef3261a9efa6 at
 receipts/engine/model_admitted/SFT-MAT-MECH-CRACK-GROWTH-008-f12803c9e26564d1.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-FAILURE-PROPERTY-TESTS
- NIST-FATIGUE-FRACTURE

Measured, observed, adverse and boundary records:

- MECH-008: 2 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-FAILURE-PROPERTY-TESTS: crack growth tests; rate of crack growth
- NIST-FATIGUE-FRACTURE: fatigue crack initiation; fracture surface

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:7170db503ad53f5959797a8a961ea57a3d07837c82cda3d66eab6e33ef68c059; empirical validation
 sha256:ac21c7b3ff9d959aee5dad99f1e5e9707039615fa1a6b643f776b9fa160b5150; external validation
 sha256:614b89bdc54ec0e5cedacc2471278814f2257d0cbb5660cec45a7e879612597e; isolation
 sha256:607aa34901ef6dcd8530b283c4265bbf36e9df0755c6bfb0a4e4ae0a0320a019; custody
 sha256:2ee0ba16f08a61305faacc3d63150bea6c3867126167dc5bd31f67d67dbbd3b4.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise

- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

128. Cyclic fatigue initiation and propagation

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-009

Claim: SFT-MAT-MECH-FATIGUE-009

Question, law and forced result. Fatigue retains every counted cycle, the first initiation cycle and every later crack record; no cycle-average erases propagation history.

MECH-009 uniquely retains cycle-initiation-propagation-complete-history with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-CRACK-GROWTH-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-009 axes before target release.

Boundary: Every positive finite MECH-009 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- cycle-initiation-propagation-complete-history
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure

- no-extra-rule

Unique survivor: complete-positive-mechanical-carrier__cycle-initiation-propagation-complete-history__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:83369af3d3d642376877d5441b14ac75e0c34f986acc16ab45889e1d8ceadcba; generality

sha256:3c3f65b2a6d2eea8381ea4a62d3821bc7da5e89e4b78cee079b4cd5c78e34d39.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:6b13895bc616793f4949312fc2162c3e6d304464a85ca35b949d9db299396412.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:3e9855995e9571c85d93ddfb2f218bfbaee26476997c29c1375f93575a9272fa.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:1644a823680abcbf909e3ddab58a95426cc9e5e31e513797c5dc5008f13f46ad.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:9cde2e06c4991ca55cd393290635869e29c7a0b88f9e7772825c563bac62c662.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate

sha256:c9452de33a0d3d50dac71e5880ebdd893ee197d1ff42e0ce2452aaf45c91e071. Derivation seal

sha256:828ab5c34634fdfb27c67c7df2b80bd0f159825f79b3124b4cd6266a4e0a5cd8. Source manifest

sha256:e2d61502186cfa309b1f62106cf6bd91134a5b1793832c80ff7fa06ca01dfefc. Engine receipt

sha256:1d81cfe5aebce710cd17d91ced8742f0ce3e7596d21dec2880daa1916d4f037b at

receipts/engine/model_admitted/SFT-MAT-MECH-FATIGUE-009-1d81cfe5aebce710.json. Closure

scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FAILURE-PROPERTY-TESTS
- NIST-FATIGUE-FRACTURE

Measured, observed, adverse and boundary records:

- MECH-009: 2 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-FAILURE-PROPERTY-TESTS: fatigue cycles; crack growth rates
- NIST-FATIGUE-FRACTURE: high-cycle fatigue fracture surfaces; fatigue crack initiation

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:6e09181b0466571f3167604c44e11b4d832a3ac9d62e8e70ff5c9a0ba8e4e435; empirical validation

sha256:3ed27c1d3db2404e125c7860d4c0d97ac1fb30128b90398d0971fc9e530ca55f; external validation

sha256:c12ab2765f2b176019c4177d7cd458e6cd874a6f60231f4b1759aa2e7244f8b6; isolation

sha256:e8a2c7ffbe02067b5c9da7d6a92bf84e1a8fbb42d976346380fa3af559ed5fa5; custody
 sha256:a1c808b36c953714f27d7dd415ee06338ea16a5c6701f4e28b28d09b287a252a.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

129. Creep mechanism and rupture-time custody

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-010

Claim: SFT-MAT-MECH-CREEP-RUPTURE-010

Question, law and forced result. Creep is a time-ordered deformation path under retained conditions terminating at an exact observed rupture time.

MECH-010 uniquely retains time-ordered-creep-path-and-rupture-custody with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FATIGUE-009

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-010 axes before target release.

Boundary: Every positive finite MECH-010 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- time-ordered-creep-path-and-rupture-custody
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held

- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-mechanical-carrier__time-ordered-creep-path-and-rupture-custody__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:4b726c18dff223ec043500035a5bfabcfee223b925881ea9c405512173a1aa59; generality sha256:10ca95a6132c7df2644bcaad09251fa3897996742ee9ffed9abff3aef7b87978.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:0e3bd367897856f43b1bee4341ef7dd1d997fa4ad2f71f6b7978e0e1dd2e1784.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:9a795a04d8d3ad211cb7f3c3670c31a0d14ff9607a0c8b31902245bd3c22708c.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:db42e786b5efb7163e61b4b252fccdeb8605dae26f08772ba2d1c9f5452e7ec2.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:826667f9b5fadd6cc27733bfbf698e6acfaef420241fa8cc516e9ed32f9aa3bc.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate sha256:bd2d31589b7799f556311f0ddce263e7d51ca533172a950c69196d504a62286d. Derivation seal sha256:3e8eb1a0e6a8de557f0f7ce04493276c463557476bc37b494cb71f364461bc47. Source manifest sha256:be820eb978394cec4d9235a711cb127cb45d81ea8b876a2c4d7607932eddaa55. Engine receipt sha256:fde173039ba0f1d8f74bb5cba12c8b774943cfff9ee14be7bec852e41e1f3e94 at receipts/engine/model_admitted/SFT-MAT-MECH-CREEP-RUPTURE-010-fde173039ba0f1d8.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FAILURE-PROPERTY-TESTS

Measured, observed, adverse and boundary records:

- MECH-010: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-FAILURE-PROPERTY-TESTS: Creep Crack Growth Rate; creep deformation

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0e762772ba311b411fccd099dae71551b642d68a238302ee1b570b7451479fef; empirical validation

sha256:632ee075965dc439f5203bd750b03de0afb1040465ea8c067759a9acf696b9ef; external validation
 sha256:62bb3f8365a643727cde8bc1c3cea0f9443c5789a5ceebfe7b2832c3e1269308; isolation
 sha256:d601d5c05890af6bfde072359a1d042cca788d55dd9a133fe432ced6c52bae0c; custody
 sha256:2892fdee764450d4fc7fc24570bff4460a0dee7f25236ea7200b50cce93c8b20.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

130. Impact and high-rate mechanical response

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-011

Claim: SFT-MAT-MECH-IMPACT-011

Question, law and forced result. Impact response is the complete exact partition of one input carrier among absorbed, rebounded and damage channels.

MECH-011 uniquely retains input-absorption-rebound-damage-one-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-CREEP-RUPTURE-010

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-011 axes before target release.

Boundary: Every positive finite MECH-011 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- input-absorption-rebound-damage-one-partition

- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-mechanical-carrier__input-absorption-rebound-damage-one-partition__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:1472a503d251b6deb8234d835bf7ead41ba08443ad525a1fbd168384784af97b; generality
sha256:926fccd31db20de700790991a99295d7d78ea6b9e9cdfa6add7877b638c5ef63.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:10934047348740fb81c528ffd90470e9533f98df9cbbd701505dabae7d460fd4.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:149186209bbcfbc757846760f34b40fe9d708a8891376b2e27b8a3ce32a8cf.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:d7fdab2547a011c3b25ca1b34e3d9b953b3683be2f9d0b403357a3f01a964a43.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ac511ac3b67d2b0eba8386acf5e86c69a4adc0f2c4e96108f60f8948dfaeb005.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate

sha256:0f155b5a2445d71ff4023c5efa90b5f41ea4fd04bbf9233270b6237a615a3979. Derivation seal
sha256:8ee72a11c6f9865678f71f77e0ab8ae3ed9d9bec9b796eb50df10ef012ae8402. Source manifest
sha256:c9c3484c0857096ff02f643a28650815936493bfddd40c4f1940e488e99eb17e. Engine receipt
sha256:8a8153930d7dd7d03553d9964c49e9f2ad78645d461525bd73f75bd9399b7fb4 at
receipts/engine/model_admitted/SFT-MAT-MECH-IMPACT-011-8a8153930d7dd7d0.json. Closure
scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FAILURE-PROPERTY-TESTS

Measured, observed, adverse and boundary records:

- MECH-011: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-FAILURE-PROPERTY-TESTS: Absorbed Impact Energy; notched metal

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c101b74137a3d34691ef1f6279d0ba45e6e9de4877e66fa6403da28324fed7e6; empirical validation
 sha256:1aff3346556e9055a9a8882f2b97a184019373e7e6435473d87bc5ef9268408f; external validation
 sha256:a50f11625b834115aadfb72356fee6a6d20cfaea75a50cbcd141c8baa435db2b; isolation
 sha256:e43b26a864199216ca8f35b7c0af6ace989c2f2cdfa519a63e26c312c93f0aac; custody
 sha256:96acb0431cce6b08a449bc5ddd411b738c0ad2f82ea81b5aa3ae24fea397bcaa.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

131. Friction and contact-state relation

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-012

Claim: SFT-MAT-MECH-FRICTION-CONTACT-012

Question, law and forced result. Friction retains exact tangential and normal carriers, their rational part and the complete contact population and condition.

MECH-012 uniquely retains tangential-normal-contact-population-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-IMPACT-011

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-012 axes before target release.

Boundary: Every positive finite MECH-012 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier

- tangential-normal-contact-population-ledger
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-mechanical-carrier__tangential-normal-contact-population-ledger__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:aee19c549b6c7d9286c49ae1269cf65789c911318ab8fad794cc55ebf26fffc5; generality

sha256:0a02338084e858a84facf382bf65fe7ebea4282f71db55a424b124340ddccdc6.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:b6df230f5a86574795d9ec1a134268f905d32e6541aa0c83a80c9cbe1d41efee.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:57b03c1abb7dd74c36597f9b865d620c4b8366ae58bc7056872b35e2cb0dff3f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:350c640f923a2d6488de7a3275f81b21dfe87bed1083b49776183babe9850f15.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:65c5a097a89f3c626d34b943b28e212034a9e0c0c0e7313ddb493a5ccab36733.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate

sha256:88ad68c83ebe938cabca4fba5cf6422551dae69223fe13d5e475b57db4b5a741. Derivation seal

sha256:b33b5006d5b82441af87d2823c30322a81e7d58411b626b1e170f9f28d1a1833. Source manifest

sha256:8960f6fa0ee43176609c1dad7e52a9a6218c5eca01731d2a33ed193e0be3e48a. Engine receipt

sha256:23017fea0eff13d900b3690c75560f00929caf5cf6910cec388d88b177045b91 at receipts/engine/model_admitted/SFT-MAT-MECH-FRICTION-CONTACT-012-23017fea0eff13d9.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-NANOTRIBOLOGY
- NIST-LUBRICATION-HANDBOOK

Measured, observed, adverse and boundary records:

- MECH-012: 2 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True

- NIST-NANOTRIBOLOGY: interacting surfaces in relative motion; friction, adhesion, lubrication, and wear
- NIST-LUBRICATION-HANDBOOK: reduce friction or resistance to motion; prevented from making contact by a film

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e43bc36c6be898f9b347617e9e74e8017c8e09b0b6a9ec6768be3617b719e683; empirical validation
 sha256:07280ebca920f964e3156f0fd4ce878332575da0dc6787273bb322bdb52a582e; external validation
 sha256:9232cc13fa66a5e5b4a04112bd7cc7976a1a40b0eaf769fd948b476375dc2a3d; isolation
 sha256:1f83fd88ad56a802897cb19c13bb9ca0098dca0d38f14eacf8f445033c578a47; custody
 sha256:2a034240b431ba58f8eda8953effe324fea8c3e59142c9a8ca331822b1629e2f.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

132. Lubrication and tribofilm organisation

Field: mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-013

Claim: SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013

Question, law and forced result. Lubrication is a complete transition word among bare contact, held film and broken film states, retaining formation and loss rather than averaging them away.

MECH-013 uniquely retains bare-film-held-film-broken-transition-word with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FRICTION-CONTACT-012

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-013 axes before target release.

Boundary: Every positive finite MECH-013 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- bare-film-held-film-broken-transition-word
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-mechanical-carrier__bare-film-held-film-broken-transition-word__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:cc67f349f25ee92ec8e556587d07d8ebf085cb33e885397f1f955a9eb158baf7; generality

sha256:d7f27678e85382e4267ea781dac49c6e16d0dbaec7c17ba21dbb7928dcc839f7.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:37fee2ce6cf2c22ccd656616525d77d44184ad50fad392d08f35030183ccf0b1.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:64bbab189e75e421596f7e126d95d42d23fdde46ad71fbc84bca1536e29cb72d.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:6754538e10f2db208f2827165e26bc0a308d92f143aa81e6b5ff3579a5eb607f.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:d3f9d3468e89c0b32ef22f6e9bc64255fa6252227d201fb49294f21a72522983.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate

sha256:539a75cd2b95708451dd0bb1188744cea4f346f0098ba2737ddba3942d99c065. Derivation seal

sha256:e40eddf9f116a8971d09da81edafba4cf7147eb4ed15a5170b1059eb3141fa23. Source manifest

sha256:f75e378121a2e839aff140b39e7290504b09a1067e5e8a86c4a34115bb4229fc. Engine receipt

sha256:be276516d621b0d48b943e10c062d1f583a7106c302c1ae63ac99c1d5c04315a at receipts/engine/model_admitted/SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013-be276516d621b0d4.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-LUBRICANT-FILM
- NIST-LUBRICATION-HANDBOOK

Measured, observed, adverse and boundary records:

- MECH-013: 2 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-LUBRICANT-FILM: monomolecular film during burnishing; tribological environments
- NIST-LUBRICATION-HANDBOOK: forming a plastic lubricating film; fluid lubricant

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:3fc9d8ff0c02116cfca1ccfeb907ccb9af33bc39a5632ceecd5b8bf730af90aa; empirical validation
 sha256:f0432b895a46d269c17e033ea6de19861e04828f39c8d66ccb476b75910731e4; external validation
 sha256:922999ccf21072ba8a51a8e9d915772d95f8166f2bb05b66d089dd811151005f; isolation
 sha256:67fca83e930f2bbe11b7d14440e923b36e15fa37dd3ab26d6697323d1d9bebf; custody
 sha256:3bba3e925a1c8e51472cdb02a5d349c65cd1a8f665ebef485a8f58511860e695.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

133. Rheological flow and relaxation classes

Field:mechanical_fracture_fatigue_creep_tribology_rheology

Obligation: SFT-MAT-OBL-MECH-014

Claim: SFT-MAT-MECH-RHEOLOGY-014

Question, law and forced result. Rheology is the exact relation of applied carrier to flow carrier plus the retained relaxation recurrence and material condition.

MECH-014 uniquely retains applied-flow-resistance-part-and-relaxation-class with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010

- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MECH-014 axes before target release.

Boundary: Every positive finite MECH-014 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-mechanical-carrier
- applied-flow-resistance-part-and-relaxation-class
- complete-loading-time-contact-history
- specimen-method-condition-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-mechanical-carrier__applied-flow-resistance-part-and-relaxation-class__complete-loading-time-contact-history__specimen-method-condition-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a5dc4a67bddd7f2404fd35dcdcf8d6c8ba3b4d3c1411aae55db0fab333dd45c55; generality

sha256:1567d32d7068741ee2491dd4287f4248dbabafa1ac281ba707c52913190f17eb.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:66f4f5a92c0d81c614e54a808d272e7db3e389f37d33cb09c88310a8dfaa2366.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:046cc41522d26cebb0856a55764b8e1769ac992ea6f73f1f04993e52e62ba9dc.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:5ff5cdf48727d416c4c89b2e1e9b06476357ba131467331b8a1ecea180c90ea8.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:7b889ab48a310ccd5f578df5f4aca9eaccc3ba14280877e8860fded0708b84652.

Independent reconstruction and engine admission. Independent implementation

sha256:f8c3545aee7537cd2bad0e823a04ba6f9e72abfc7b98f02c31b1f7d280a6c613 regenerated the declared census and sole survivor; independent certificate

sha256:c7fe0fdd9edd0753ac1041e434233df20cc29a6acee8021538895ab08e77316b. Derivation seal

sha256:c8fac109ebfc4d02a591b7760bc50ecaeb358ff4fbbedb631e5c0d9293802758. Source manifest

sha256:c5aa330dc4e3c5c6f94d10716e81e05b35da7b6f8f732215e6475959b511105b. Engine receipt

sha256:b38317d23bcc37b7ffe7dbf46629754d041fa15a27dc33f70f774cbefeeef9b6 at

receipts/engine/model_admitted/SFT-MAT-MECH-RHEOLOGY-014-b38317d23bcc37b7.json. Closure

scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.

Comparison passed: true.

Source identities:

- NIST-RHEOLOGY
- NIST-VISCOELASTIC-SEALANT

Measured, observed, adverse and boundary records:

- MECH-014: 2 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-RHEOLOGY: Rheology is the study of the flow properties of liquids; viscosity and yield stress
- NIST-VISCOELASTIC-SEALANT: time dependence and strain dependence; stress-relaxation experiments

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0e7e267f15f7169f66242631faa0a9aa8bebd8975d728dcb8a5e8de80f2638ce; empirical validation
 sha256:2671e0e8392c860d795b43ab88453379aa315758e504ff240505ec7d7bcbb933; external validation
 sha256:454c1e1ddce08807d75397a5c2a927b409cb97658474e90182a25f3a7c84e1cf; isolation
 sha256:ec3d5023a20bc849d1a7c95414c578732f8e49920264552a752bf0f548976f8e; custody
 sha256:917d1d1d3ce87bd0804e0eacc97bda7a81f19e014d4fbb2a24282169dc6052bd.

Prohibited inputs.

- no imported continuum law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and opposed direction remain held labels
- no external outcome selects a survivor
- all adverse and boundary rows retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Thermal Transport Storage Expansion Thermoelectric

134. Thermal diffusivity relation

Field: thermal_transport_storage_expansion_thermoelectric

Obligation: SFT-MAT-OBL-THERM-001

Claim: SFT-MAT-THERM-DIFFUSIVITY-001

Question, law and forced result. Thermal diffusivity is the exact rational part of the conductivity carrier over the complete density-times-specific-heat storage carrier, with all specimen conditions retained.

THERM-001 uniquely retains conductivity-over-density-specific-heat-exact-diffusivity-part with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-HEAT-CAPACITY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-MAT-THERM-EXPANSION-001
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FATIGUE-009
- SFT-MAT-MECH-FRACTURE-ENERGY-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight THERM-001 binary axes before target release.

Boundary: Every positive finite THERM-001 carrier with complete condition, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-thermal-carrier
- conductivity-over-density-specific-heat-exact-diffusivity-part
- complete-interface-scattering-phase-cycle-path
- specimen-method-temperature-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-thermal-carrier__conductivity-over-density-specific-heat-exact-diffusivity-part__complete-interface-scattering-phase-cycle-path__specimen-method-temperature-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:404870fb5e0c5d676bbab63cf35ef2653a91e16a5117c714160f41aa91db4c87; generality
sha256:38db08355eb427a9495cc92bfbfd53233b6119bc552b102294c2a3f90f05b78a4.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:8a95cdc5d5a68f059c1ff2f0f629f7a02d69fca0b5dc87576f531eddc8aff4e.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:b4e830390bacd3a52fe4d201ddcec72b80aedb58c378f6e57bd951344e6a407a.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:c3765c6fbd789bac174653639ff88bbaf80ccf18260273d83e10738d6fb6e2d3.

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:bc20f48f488b2909ebb07ed6c81aae42259570cdbcc33c926976b31bffd617ee.

Independent reconstruction and engine admission. Independent implementation

sha256:2fc6203babd13114223350c5993531b6c4abb4cc4073d8101459d69aed4f353e regenerated the declared census and sole survivor; independent certificate
sha256:35cd3ad26d4cf85a392891fc8d77c35eb1a95eb6c7a941798783647eae0b4a45. Derivation seal
sha256:3b8252191b70101b383daace33052e9dd65f93728e2d90b3048ea9fea6f28e44. Source manifest
sha256:4c290067aeca356e5f5299b00aa86827b8d16deea26b10e35cf56760ee73158. Engine receipt
sha256:203cd80e00d1c50650d093112e8231834b7fe241a34552f88d05c3402f2f625a at
receipts/engine/model_admitted/SFT-MAT-THERM-DIFFUSIVITY-001-203cd80e00d1c506.json.
Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
Comparison passed: true.

Source identities:

- NIST-MATERIALS-DATA-GUIDE
- NIST-THERMAL-DIFFUSIVITY-AM

Measured, observed, adverse and boundary records:

- THERM-001: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-MATERIALS-DATA-GUIDE: K=pCpD; thermal conductivity and thermal diffusivity should be correlated
- NIST-THERMAL-DIFFUSIVITY-AM: thermal diffusivity; thermal conductivity

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:63c3ca600535618d446b3c604e37a79e18b070f7d0b357da3c7c485ef92d4e1f; empirical validation
sha256:fcbb27e2aadcca358568302e89cb62360fba4739e3dcb0dfd71a537ee4913b396; external validation
sha256:d0bb5730b2c5c331e362f568b8f09d6db56bca645cd919bb78b66f0c4001fb93; isolation
sha256:70a163a49ec8572be7f69fc44f1b59f7b4dc25748c541faf7fdb7ae5ac558d00; custody
sha256:bb408de6ebe86a7a647e782e366cea99ed61cc557a364e35dc4ddd86ee638098.

Prohibited inputs.

- no imported continuum transport law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and thermal orientation remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and boundary rows remain retained
- no failed attempt retires an obligation or changes protected authority

135. Interfacial thermal-boundary resistance

Field: thermal_transport_storage_expansion_thermoelectric

Obligation: SFT-MAT-OBL-THERM-002

Claim: SFT-MAT-THERM-BOUNDARY-RESISTANCE-002

Question, law and forced result. Interfacial thermal resistance is the exact positive temperature-drop-times-area carrier per heat-flow carrier, with the interface identity and reciprocal conductance retained.

THERM-002 uniquely retains interface-held-temperature-drop-area-per-heat-flow-reciprocal-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-HEAT-CAPACITY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-MAT-THERM-EXPANSION-001
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FATIGUE-009
- SFT-MAT-MECH-FRACTURE-ENERGY-007
- SFT-MAT-THERM-DIFFUSIVITY-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight THERM-002 binary axes before target release.

Boundary: Every positive finite THERM-002 carrier with complete condition, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-thermal-carrier
- interface-held-temperature-drop-area-per-heat-flow-reciprocal-ledger
- complete-interface-scattering-phase-cycle-path
- specimen-method-temperature-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-thermal-carrier__interface-held-temperature-drop-area-per-heat-flow-reciprocal-ledger__complete-interface-scattering-phase-cycle-path__specimen-method-temperature-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:e7f67aab4a9855f128baab664ade69c1dcceb220f3f62466815295ab4ac04c0e; generality
 sha256:9e7685249875fdd362ead9a294b2ec454b2a345b5efe7f15e635552eb2cc9665.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:10c867cbe2ddc982fe448e863a12a1c1f6b0bba2407351e0799f0c6c4509fa62.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:840e90def92ea59207a2b7fc9a80890f62a98a7b429041d3b5f645196c67183b.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:07a8734a85983357079eea056c9ecc9598ad523602cce41922e564a5bbc8850a.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:4df3ca55898cd95f20adcc51152a01e3f96d21ddca828a7610c18ba3c1110821.

Independent reconstruction and engine admission. Independent implementation

sha256:2fc6203babd13114223350c5993531b6c4abb4cc4073d8101459d69aed4f353e regenerated the declared census and sole survivor; independent certificate
 sha256:7ae77f1416aa99ee19737c38ec22a931256c395d83ed3954748cc126a679ff13. Derivation seal
 sha256:f0e3c13447cdf68f158d91c2845200dc70a93113a0d933f275872e602224311b. Source manifest
 sha256:b5e680e529f7b2dde9289bb2eee123e5322009e6441d3fecc2cf3ada2f0963c5. Engine receipt
 sha256:8892aa549888cf5c8e45886e5b893074c167bf4480ebe3313adf30ae90bd7e71 at receipts/engine/model_admitted/SFT-MAT-THERM-BOUNDARY-RESISTANCE-002-8892aa549888cf5c.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FDTR-TRANSPORT

Measured, observed, adverse and boundary records:

- THERM-002: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-FDTR-TRANSPORT: thermal interface resistance; multilayered materials

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:00916aaa566a050e815e1f9f0127e0cdcb0f9bed7f4e0707da322d45a1fe4bc7; empirical validation
 sha256:d86b7ce75565e607b6b3946b64bcf0c87c52eedb0cd60342d83dddca11ac05e; external validation
 sha256:c4742acb0ce731b54afd6362c4a677675c3e451e0db70c2fa2b074617019c63f; isolation
 sha256:bcbdd361705dc534f9cf62809fb859b787fdf98dc08daa514da2e58f868496c31; custody
 sha256:befaabdd860df3fdd2f0384aba7318b13291ef56a50407b5711e3c6272af78dd.

Prohibited inputs.

- no imported continuum transport law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and thermal orientation remain held labels

- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and boundary rows remain retained
- no failed attempt retires an obligation or changes protected authority

136. Phonon scattering and mean-path ledger

Field: thermal_transport_storage_expansion_thermoelectric

Obligation: SFT-MAT-OBL-THERM-003

Claim: SFT-MAT-THERM-PHONON-MEAN-PATH-003

Question, law and forced result. Phonon transport is a complete counted path word whose every positive segment is paired with a held scattering event; mean path is their exact rational total per event.

THERM-003 uniquely retains complete-phonon-path-segment-scattering-event-mean-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-HEAT-CAPACITY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-MAT-THERM-EXPANSION-001
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FATIGUE-009
- SFT-MAT-MECH-FRACTURE-ENERGY-007
- SFT-MAT-THERM-BOUNDARY-RESISTANCE-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight THERM-003 binary axes before target release.

Boundary: Every positive finite THERM-003 carrier with complete condition, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-thermal-carrier
- complete-phonon-path-segment-scattering-event-mean-ledger
- complete-interface-scattering-phase-cycle-path

- specimen-method-temperature-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-thermal-carrier__complete-phonon-path-segment-scattering-event-mean-ledger__complete-interface-scattering-phase-cycle-path__specimen-method-temperature-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f9d5da91718316eb924457bc5f46ccbab7008d8331f218336e57cba72146eecb; generality

sha256:c2131887fb8b8b783b886de5b3ae6a5721fdd9b10efe570c6666e6237fb1aeac.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:28c9d1ba3ca1823ff7b48fef026142609714ec24f22aa06f6d758686acf7a556.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:361cbc325c31bdc9dc5ea86acdf3b1d28782db73167fa6a5ecf4fe04b839fe1.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:1042aa67b0490841ce8da2b6bbece7ad678a043c5fb669f81335e0d682f6cd9a.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:17c33769ca0727f63be62395f67da882153a36c3eae83393e618599c0e02d020.

Independent reconstruction and engine admission. Independent implementation

sha256:2fc6203babd13114223350c5993531b6c4abb4cc4073d8101459d69aed4f353e regenerated the declared census and sole survivor; independent certificate

sha256:9c897e51a6a4345cd917c189224a9fa2f832ffa837f05261dddc3c7c41668789. Derivation seal

sha256:88d0b2077bc833d464b0465838e7346071f16b408e51575b934d8c08ea1f792a. Source manifest

sha256:ca01828cda7e46c8ef9a01466929eab7cae8054a28dd83c314fb59f2c0bdab27. Engine receipt

sha256:76a926b4634a25262369ade1a043e5dfbb4abc62139b3f5fc7d05fe1d5326895 at receipts/engine/model_admitted/SFT-MAT-THERM-PHONON-MEAN-PATH-003-76a926b4634a2526.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-INTERFACE-SCATTERING
- NIST-PHONON-THERMAL-LIMITS

Measured, observed, adverse and boundary records:

- THERM-003: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-INTERFACE-SCATTERING: phonon interface scattering; mean free path
- NIST-PHONON-THERMAL-LIMITS: collective modes of these vibrations called phonons; thermal energy

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e5864180d6ec697d39917555809e4c1668ce02a77f6eccf990532a6b55944e6e; empirical validation
 sha256:d35b1066b9fb9fb4a089742f8517f881d1f17e275e2b560d92a5fe16f1c886c2; external validation
 sha256:49c3d58b792abce3f1e4357066a5e95c39af020dac97bedbc233f18493d89d30; isolation
 sha256:e4a22d8887f360f0c24781f7d9cd6f1a8c428d26ba97418a91c8959679fb1a06; custody
 sha256:51f134744cc90e15a39bf7f6a62d36b159876ea45f938f7d5b164cac30a824ac.

Prohibited inputs.

- no imported continuum transport law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and thermal orientation remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and boundary rows remain retained
- no failed attempt retires an obligation or changes protected authority

137. Radiative thermal transport in materials

Field: thermal_transport_storage_expansion_thermoelectric

Obligation: SFT-MAT-OBL-THERM-004

Claim: SFT-MAT-THERM-RADIATIVE-TRANSPORT-004

Question, law and forced result. Radiative transport is an exact spectral and directional partition of one incident carrier into reflected, transmitted and absorbed carriers without continuum or signed proof magnitudes.

THERM-004 uniquely retains spectral-directional-incident-reflected-transmitted-absorbed-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-HEAT-CAPACITY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-MAT-THERM-EXPANSION-001
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FATIGUE-009

- SFT-MAT-MECH-FRACTURE-ENERGY-007
- SFT-MAT-THERM-PHONON-MEAN-PATH-003

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight THERM-004 binary axes before target release.

Boundary: Every positive finite THERM-004 carrier with complete condition, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-thermal-carrier
- spectral-directional-incident-reflected-transmitted-absorbed-partition
- complete-interface-scattering-phase-cycle-path
- specimen-method-temperature-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-thermal-carrier__spectral-directional-incident-reflected-transmitted-absorbed-partition__complete-interface-scattering-phase-cycle-path__specimen-method-temperature-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:406cc72c09abe8a5ea62a0022ddadc82436104ec52f7786694230473223f0736; generality
sha256:1d1e54af46fc83866f1062e0dae9a9b334852e5099c87735dc9a242e003fe803.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:af620239c22732699642d3df34824f8074bfad2363d6427badeb46b840f8cee9.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:0d447374db696e41af1d1130d496f87bbb1f4528b9430f513b21b5b66e634a89.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:80d16a397c26c01a94d3b6cb2dceb9d54d53a998626ca16dbc297fb9e64f3d29.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:4d212bd7caed6ffffdb5f241901b16ad4f5c1697f276b7b0d2c18d123207737fb.

Independent reconstruction and engine admission. Independent implementation

sha256:2fc6203babd13114223350c5993531b6c4abb4cc4073d8101459d69aed4f353e regenerated the declared census and sole survivor; independent certificate

sha256:ea754f9402f1fa084496e0fd35f033f56899e69d2ba3c49487dd774600a4b58a. Derivation seal

sha256:fba1746289b54723ae066ffc236249287ec16174dc7e171a488a43d78e23d536. Source manifest

sha256:289ef0f2b2d0adf698c955f0046221b7fe837907d28c86f281fac80f3e6eaa57. Engine receipt

sha256:0c4494cc76fe6ed5ed829a4f289158669fac95e271e47bd3d97db42ce4b770c4 at receipts/engine/model_admitted/SFT-MAT-THERM-RADIATIVE-TRANSPORT-004-0c4494cc76fe6ed5.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-INFRARED-OPTICAL-PROPERTIES

Measured, observed, adverse and boundary records:

- THERM-004: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-INFRARED-OPTICAL-PROPERTIES: reflectance, transmittance and emittance; radiative heat transfer measurements

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a339c5b9bd5cdf3b375104574e10479fef2a91b422ba4e51d438c3d83d89101a; empirical validation
sha256:4f50c05fb0d3ece302077e007a9cd1293ae0921d98b18707d230bcf012f673ac; external validation
sha256:1f0d62f5b1c8dd4d47051df78f7ab8de8f7674bc301db2038279a7cc2c02fd3a; isolation
sha256:cf70b510d0cab4f4f0f779c0615ffcd8ef7b57951bf8270a02d4ef7f7bdf0a8; custody
sha256:eedec28cf4fb9b729008984049c982d90041ca1c20355fa30575ab7a9a9b9fb3.

Prohibited inputs.

- no imported continuum transport law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and thermal orientation remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and boundary rows remain retained
- no failed attempt retires an obligation or changes protected authority

138. Thermoelectric coupled-response performance boundary

Field: thermal_transport_storage_expansion_thermoelectric

Obligation: SFT-MAT-OBL-THERM-005

Claim: SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005

Question, law and forced result. Thermoelectric performance is the exact coupled rational relation of the Seebeck carrier squared and temperature carrier to retained resistivity and thermal-conductivity carriers.

THERM-005 uniquely retains seebeck-squared-temperature-over-resistivity-conductivity-boundary with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001

- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-HEAT-CAPACITY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-MAT-THERM-EXPANSION-001
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FATIGUE-009
- SFT-MAT-MECH-FRACTURE-ENERGY-007
- SFT-MAT-THERM-RADIATIVE-TRANSPORT-004

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight THERM-005 binary axes before target release.

Boundary: Every positive finite THERM-005 carrier with complete condition, path and observation distinctions.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-thermal-carrier
- seebeck-squared-temperature-over-resistivity-conductivity-boundary
- complete-interface-scattering-phase-cycle-path
- specimen-method-temperature-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-thermal-carrier__seebeck-squared-temperature-over-resistivity-conductivity-boundary__complete-interface-scattering-phase-cycle-path__specimen-method-temperature-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent`; `minimality true`; `named-shape uniqueness true`; `proof`

`sha256:e34acb8d70fc5e57ca763b743f4c76d0632370aae77ee75d6dbe384f56b79faf`; `generality`

`sha256:cb9d11f0a74594a53492a50e6bd6a31b11b848e5d54d6c090cc036bfd3f70468`.

Shared claim-record clause `MAT-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:b079c5d84c7eef2c781eb7211ffdc14db67273a0c5143217c87a5c7cbfbde054`.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:accc4307c4e207d42db41d3489261af92d877181e2f41430a50fb6515356840e`.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:e05531e8a51d7f08b498e0ce350406eddb184865597e8744830b4ddae277a4af`.

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:8b205d7e5cbb930c28a70a5071cd862a9e077cb63a061835a3d956bd64b89822.

Independent reconstruction and engine admission. Independent implementation

sha256:2fc6203babd13114223350c5993531b6c4abb4cc4073d8101459d69aed4f353e regenerated the declared census and sole survivor; independent certificate

sha256:d906643cc9c88e41597a3bf9f05ea7f0723eaa448d0291672d1bb778a41854de. Derivation seal

sha256:b8abbcca8e19ff252885dadfa158ae5f3fb65e0dd7142a5e12fdfcc702ec248a. Source manifest

sha256:d6fc54f672f2e67e1dbceb0e4c48d3d74e17a62019e25ceddefed6a1d02be142. Engine receipt

sha256:f448e0cc638e3d7fd5ad7f72245002d2215a386f08e74b71c34ad5ba3d2f5cb6 at receipts/engine/model_admitted/SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005-f448e0cc638e3d7f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-THERMOELECTRIC-MEASUREMENTS
- NIST-TRANSPORT-THERMOELECTRIC

Measured, observed, adverse and boundary records:

- THERM-005: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-THERMOELECTRIC-MEASUREMENTS: figure of merit zT ; Seebeck coefficient
- NIST-TRANSPORT-THERMOELECTRIC: bulk and thin film thermoelectric materials; High-Temperature Seebeck Coefficient Standard

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:08846f1c3811839b5eba7343bbaaf2150a76537a6ff418b9677942ebd527006f; empirical validation

sha256:177dfafc5548c5a601bd7ef721e03bbaf161d4f7097ce5399877c2e27f9fba82; external validation

sha256:3e5cf2c2e946904e4fb826ece04dd57c64559b3b3b75f73c1cd99e19cf2b7086; isolation

sha256:63f87cb1068aa3f6f46461d8bcecb8ad7dfcef8f49d305001cff0f562de8295d; custody

sha256:2612fa164f0c21918a4cd185b51aaae757a0ce098f1654f48dbb7d94572caf8c.

Prohibited inputs.

- no imported continuum transport law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and thermal orientation remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and boundary rows remain retained
- no failed attempt retires an obligation or changes protected authority

139. Phase-change thermal-storage ledger

Field: thermal_transport_storage_expansion_thermoelectric

Obligation: SFT-MAT-OBL-THERM-006

Claim: SFT-MAT-THERM-PHASE-STORAGE-006

Question, law and forced result. Phase-change storage is the exact sum of pre-transition sensible, latent and post-transition sensible carriers with the complete phase path retained.

THERM-006 uniquely retains sensible-latent-sensible-complete-phase-storage-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-HEAT-CAPACITY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-MAT-THERM-EXPANSION-001
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FATIGUE-009
- SFT-MAT-MECH-FRACTURE-ENERGY-007
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight THERM-006 binary axes before target release.

Boundary: Every positive finite THERM-006 carrier with complete condition, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-thermal-carrier
- sensible-latent-sensible-complete-phase-storage-ledger
- complete-interface-scattering-phase-cycle-path
- specimen-method-temperature-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-thermal-carrier__sensible-latent-sensible-complete-phase-storage-ledger__complete-interface-scattering-phase-cycle-path__specimen-method-temperature-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:506a7074b5da57c756508c3199515cac8f0921857fbd3f254aed917b13fa5878; generality
 sha256:3ad60e1d509898d68022c24e9a2e411935d55906aa6b7f5a5e9242ffb4d56775.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:855455ea96496ca275000dd6e7e696123981be39138ff7fd241b95651be.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:34b3071c5757fc60d7c2bd02344fd9ddcc3c284298453cb4579a3a24f3ac0f46.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:ad6d3818ee46e1d15bf3633d6924e33f97bd046d88d1819d027330aa92b6c6bd.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:54a7308fd86bd533d1c7ba620c7e76ae8fef49bdeddaf62b063d0be8b2773519.

Independent reconstruction and engine admission. Independent implementation

sha256:2fc6203babd13114223350c5993531b6c4abb4cc4073d8101459d69aed4f353e regenerated the declared census and sole survivor; independent certificate
 sha256:2bce7abc20d0eb8b953015beb8dcf8df1430e05641e961f77d5e11036f0f9fea. Derivation seal
 sha256:7fc37ad855a1529b27dd59e00ed3ad6e8cffbc3912b30528747b93d08bb055d2. Source manifest
 sha256:63d83fc5c1b8ba0ba7169510bf945693185b779ce64d93b04920987cba7bec58. Engine receipt
 sha256:db2cc00f176e3b82dd10915ca448451fdb759fa1897e514b16678b193df63179 at
 receipts/engine/model_admitted/SFT-MAT-THERM-PHASE-STORAGE-006-db2cc00f176e3b82.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-PHASE-CHANGE-STORAGE
- NIST-NANOCALORIMETRY

Measured, observed, adverse and boundary records:

- THERM-006: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-PHASE-CHANGE-STORAGE: thermal energy storage via phase-change materials; thermal resistance
- NIST-NANOCALORIMETRY: phase transitions; heat capacity

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ee7d7fd4da130602ecfd2dee60e755f0ff74c397e474c7adefc360b4411e97ca; empirical validation
 sha256:049e531b4d32b45466f68541024cbc9ddf7d946ee7e1bcdbfd9543b606d026d7; external validation
 sha256:b8771c14af5ca9ae91163665c8d1c9064a3d35558900ec7bd1cd79d7fd819cc; isolation
 sha256:964b7c8dd226fd66d7bbea03e7c7c2eded2d47a212b20388adacfc0c174dd6ef; custody
 sha256:1c32f290a73de7fbe940af55c5da17c8fe2da8479e2c58ff2bc25e3c03f2c13c.

Prohibited inputs.

- no imported continuum transport law, fitted constitutive equation, named mechanism or prior proof as premise

- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and thermal orientation remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and boundary rows remain retained
- no failed attempt retires an obligation or changes protected authority

140. Thermal-shock and thermal-fatigue boundary

Field: thermal_transport_storage_expansion_thermoelectric

Obligation: SFT-MAT-OBL-THERM-007

Claim: SFT-MAT-THERM-SHOCK-FATIGUE-007

Question, law and forced result. Thermal shock and fatigue retain every positive temperature-step magnitude, heating or cooling orientation, cycle and crack record through the first critical boundary.

THERM-007 uniquely retains oriented-temperature-step-cycle-crack-critical-boundary-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-HEAT-CAPACITY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-MAT-THERM-EXPANSION-001
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-MECH-FATIGUE-009
- SFT-MAT-MECH-FRACTURE-ENERGY-007
- SFT-MAT-THERM-PHASE-STORAGE-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight THERM-007 binary axes before target release.

Boundary: Every positive finite THERM-007 carrier with complete condition, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-thermal-carrier

- oriented-temperature-step-cycle-crack-critical-boundary-ledger
- complete-interface-scattering-phase-cycle-path
- specimen-method-temperature-scale-uncertainty-held
- complete-state-transition-resource-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-thermal-carrier__oriented-temperature-step-cycle-crack-critical-boundary-ledger__complete-interface-scattering-phase-cycle-path__specimen-method-temperature-scale-uncertainty-held__complete-state-transition-resource-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:5f9669871642b8a03c3133b6af93e3a0bfe112e2434c974465dee0b0593e4f1f; generality

sha256:9b004f289606c496cfac4b3ae23bccb2428e5f11569906c60476a4163d6a4a47.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:c2c7a1a4642d7f2db5e64260045aa35a8101a9bcfd061f9a4741761cb220807e.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:11ecb5f2e47c4d87c725e3316d1f81a7f601df68af3b87ef4130d058f4332e3d.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:7bc8f5dbb5e2824dc654d8b06fd7f9558df0d05c73c98a833355d2f5e0370885.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:d2c63d9e3d930ff2b3b7149104fe753a1e7e96faf122d63ce96b5727dc8353dc.

Independent reconstruction and engine admission. Independent implementation

sha256:2fc6203babd13114223350c5993531b6c4abb4cc4073d8101459d69aed4f353e regenerated the declared census and sole survivor; independent certificate

sha256:fe7231ca4fab63f1596ff5541c168fb1729f2ead31d1c1a6e1520b89ef696620. Derivation seal

sha256:85c0d614082db64b5c102007d4837d4cdad2cad88117f86f89b57a5e1d0c5c0a. Source manifest

sha256:9dabba3fda8d46ac294e206ae4534cab4b8a2fa39b6a84a756073e04845d6351. Engine receipt

sha256:384a57b2c1e8032b7ee57c398fbbae7cbdc24b349d7781ab2eabe9724fb7568f at

receipts/engine/model_admitted/SFT-MAT-THERM-SHOCK-FATIGUE-007-384a57b2c1e8032b.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.

Comparison passed: true.

Source identities:

- NIST-FRACTOGRAPHY-THERMAL-SHOCK
- NIST-THERMAL-SHOCK-SILICON-NITRIDE

Measured, observed, adverse and boundary records:

- THERM-007: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True

- NIST-FRACTOGRAPHY-THERMAL-SHOCK: Thermal stresses and strains occur; cyclic fatigue
- NIST-THERMAL-SHOCK-SILICON-NITRIDE: critical quench temperature for failure; extension of Vickers radial cracks

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:7af1729a56a4d706e505cb9d0d6638c883d6831f4eeb9392b3870e9d2ad5b4f1; empirical validation
 sha256:6e4a2ff52f8560effb02974230ff10ccbd48dab13dd9766c720b816fb400319c; external validation
 sha256:ab511999f024ad83e8eaa4e21049859445415c9cc728bf03abad7b960c4cda85; isolation
 sha256:2327b3edafe0c30411168ff69256a007fd22d1c097643d7530e1b1f398d60e3b; custody
 sha256:3d91b20f53cc2326a7f9498c558f06ea043f224bc6217e37dbd2bb742d3de546.

Prohibited inputs.

- no imported continuum transport law, fitted constitutive equation, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and thermal orientation remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and boundary rows remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Electronic Dielectric Semiconductor Ionic Transport

141. Electrical conductivity and resistivity relation

Field: electronic_dielectric_semiconductor_ionic_transport

Obligation: SFT-MAT-OBL-ELEC-001

Claim: SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001

Question, law and forced result. Conductivity and resistivity are exact reciprocal positive rational relations with specimen geometry retained.

ELEC-001 uniquely retains reciprocal-conductivity-resistivity-specimen-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight ELEC-001 axes before target release.

Boundary: Every positive finite ELEC-001 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-electrical-carrier
- reciprocal-conductivity-resistivity-specimen-ledger
- complete-interface-field-cycle-state-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-electrical-carrier__reciprocal-conductivity-resistivity-specimen-ledger__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:547ea6c379eb4903911c7263a268839c6e5fdd633d5a578aa10db3145c9c5a59; generality

sha256:b3e39a39d9cfc45a7bcb8dc084203c442aa6b50ed7ac01d5239fcdf6bf9b812f.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:2bf143ec27e893ce534b5821a597fca148709a3c2f1439b2c8f7ca87915db73a.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:7f2a6bdb06ebfd9e9656e08bee2dda22a788de9c9be10f8e67f6da5fd039047a.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:522a7807af289eb6601932ae33f5448537aa73b79cc210c3cc42caa8a972b913.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:922eb1308648f9275e87df94ca486882d5bf3807e6ade548b43206f0bc990a2a.

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the declared census and sole survivor; independent certificate

sha256:6f1d77d441713a7cab5bfe05633617d2a34adbea3e2bcc1c8abe072190ab88ca. Derivation seal

sha256:91606eaa595ecc755255bafbe62f317cfd810be6b77df01c88e603c01478c402. Source manifest

sha256:b3f9465a6538e829b68f9b5789d9e592c644c16fcd3e61d452c8d608b0352765. Engine receipt

sha256:f36daae94e69e01b10297efefdd4da77b2d0431de6b372029d751d0197229759 at receipts/engine/model_admitted/SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001-f36daae94e69e01b.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HALL-EFFECT

Measured, observed, adverse and boundary records:

- ELEC-001: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-HALL-EFFECT: resistivity (or conductivity); current-carrying capability

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:bbe187d25be4f0e6f5a729f4fcf30422151c6ee37f5038ea333d4bc194cdb018; empirical validation
 sha256:700d188a98bd903dbd1ad484588a24c005d423de1ab6b1f165b39264adacdafb; external validation
 sha256:65c426925055d43834f2fd036e31af1640f7b0250deb740feff648ae2b1390c1; isolation
 sha256:308ccfbc1a31be9e37527986903de654625b4fb7336c6336889e6f681a9b6718; custody
 sha256:4b2219243d108a82914ca46754bfa5857b31f29faa2d2ee5cedb0ee437f45bf8.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

142. Carrier mobility and concentration separation

Field: electronic_dielectric_semiconductor_ionic_transport

Obligation: SFT-MAT-OBL-ELEC-002

Claim: SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002

Question, law and forced result. Carrier flow is separated exactly into retained concentration and mobility carriers rather than conflated into one response.

ELEC-002 uniquely retains concentration-mobility-separated-carrier-flow with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001

- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight ELEC-002 axes before target release.

Boundary: Every positive finite ELEC-002 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-electrical-carrier
- concentration-mobility-separated-carrier-flow
- complete-interface-field-cycle-state-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-electrical-carrier__concentration-mobility-separated-carrier-flow__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:0a58ba021f0c30d5f3e88abc9e8391ee370c18ac5b94bb3ff150fa6cf69d966d; generality
sha256:01bd1e597b783bd98fd2dd4270673666d7989597b44b7cb2ed67e7764a752430.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:1945318327b6618b459e9012b06a53c88b1c979283bcd07d9a1b553140dde9fb.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:274224a4d5504e65f31a89db9c9367d40d294abfacbf3024009f154c009bccd9.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:0a47d5168c028af1115364f51e1f8634d745e1e7fe53c86a5b656d5c5f0f36e3.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:5d10fbd8acfb3082d6b0d5e802cc519ea0c766e4645a3841815d9ee7cbe2c336.

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the declared census and sole survivor; independent certificate

sha256:474231fce8a92b24d9ece5f4c40a37ea5eff0f4db6b79eb9a45d407ab46cb2c7. Derivation seal

sha256:3e28130a82d8d0a3df07f2df50b1a56e7e46431e6948aa8624c414807702bf6d. Source manifest

sha256:5941e86244c27f7ee3270b2c0b365ff1c887fa2559d65cc5360f30531a5dc082. Engine receipt

sha256:af184916c5fa1e1d858363839ad505f93d0034903d7bbecca988845562fa85e10 at receipts/engine/model_admitted/SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002-af184916c5fa1e1d.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HALL-EFFECT

Measured, observed, adverse and boundary records:

- ELEC-002: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-HALL-EFFECT: carrier density; mobility

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:61dbbc9ae6b02b62cc94e684438ac602042dec37bd28e31999ef8b74ea81da44; empirical validation
sha256:c2c6ce99155ba794431bffc4a25e44e6f4a639a3ffe0eba9b5125678e536eea1; external validation
sha256:69f482339ca234b8eb2c17b8edc91fda16891aa2c45502f98b6fe3132ff21387; isolation
sha256:91f6b193b99900fbbb4c3079e632366958f93e9abe2e580b6a9be20c3be41555; custody
sha256:e6c74553ae60e8e0c42b9b46e06cf6fd491c899a3f9091395ef5acb62e0985b9.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

143. Hall-response carrier ledger

Field: electronic_dielectric_semiconductor_ionic_transport

Obligation: SFT-MAT-OBL-ELEC-003

Claim: SFT-MAT-ELEC-HALL-RESPONSE-003

Question, law and forced result. Hall response retains current, field, charge, thickness, carrier density and electron-or-hole orientation as one exact ledger.

ELEC-003 uniquely retains hall-current-field-density-thickness-orientation-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001

- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight ELEC-003 axes before target release.

Boundary: Every positive finite ELEC-003 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-electrical-carrier
- hall-current-field-density-thickness-orientation-ledger
- complete-interface-field-cycle-state-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-electrical-carrier__hall-current-field-density-thickness-orientation-ledger__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:5121b179ef0976470b248f703c4d1054126c8203c020bb71458237e0d99df4f0; generality`

`sha256:3d52e18faf83afa2a06460432c55af3e2607b06a71d3faa16ae212a558821e05.`

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:446288c8a45752c8b597aa168c0cb07598219f44d47cd360180db28ab677e4cd.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:6e3d02f23a2ca6fa60ff75347f098048f18a0e83bdedd1393ae9acb57a6ad1b5.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:906fa2c216ae5c2d5703a92ca4fec21283dcabad87f2b17810bfc417e11ff1c.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:0a33524c8e674681cc8e8e13f63b30d7ae7af65aeb31623bb1afeeae6cb6bc53.`

Independent reconstruction and engine admission. Independent implementation

`sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83` regenerated the declared census and sole survivor; independent certificate

`sha256:542cb954ec8937d8d9e02a79a1f1b0acbbbb2d3069e9df26d4a6c95c118cd928.` Derivation seal

`sha256:044748b4fd2349fb72502554dec583141b9187c355bcbef8631f76b5677e32fd.` Source manifest

`sha256:e9cd10247fd258d483c2105e9bb6d1772da3beeb10ca52b7700a2467340a6d3f.` Engine receipt

`sha256:c6147980136418e1e89200bae0e4574a2ac2d5f10db011ec4b7e69e701377ac3` at

receipts/engine/model_admitted/SFT-MAT-ELEC-HALL-RESPONSE-003-c6147980136418e1.json.
Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
Comparison passed: true.

Source identities:

- NIST-RESISTIVITY-HALL

Measured, observed, adverse and boundary records:

- ELEC-003: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-RESISTIVITY-HALL: Hall voltage; sheet carrier density

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:40a9221eb7848c27eaf7c6f53c50973b9dcdbd050d58bd37ba5a50e89aa7ed55e; empirical validation
sha256:e491ac8cd49dbe6c5bb556968685fddcc56888417088f4880ece00c15875f3c9; external validation
sha256:a167d3c0938ca18fb9f3a1a16801b73d0be8641837f640ecbf3561eaa1692679; isolation
sha256:2fc5bf01e176a823f17c63c9f7c86b40e4a5c94556f3bc9cf594270f1c3ea1bd; custody
sha256:a2c9f72283bc626ab24e92acdfcfb498a5b36288ca5ba8df85e01581952c41fe.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

144. Dielectric permittivity and loss relation

Field: electronic_dielectric_semiconductor_ionic_transport

Obligation: SFT-MAT-OBL-ELEC-004

Claim: SFT-MAT-ELEC-DIELECTRIC-LOSS-004

Question, law and forced result. Dielectric response retains exact stored and lost carriers, cycle conditions and their rational loss relation.

ELEC-004 uniquely retains dielectric-stored-lost-cycle-response with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-ELEC-HALL-RESPONSE-003

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight ELEC-004 axes before target release.

Boundary: Every positive finite ELEC-004 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-electrical-carrier
- dielectric-stored-lost-cycle-response
- complete-interface-field-cycle-state-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-electrical-carrier__dielectric-stored-lost-cycle-response__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:da7c90122b34535ec29f57a95fcb653999ccf2f7791931321e7109cfcff4776f; generality`

`sha256:ab002b5ade0831a8bcfa7f8beb370d40fb9b803364a9b44c4a9c31510bd71064.`

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:3e629a6a022e052c27c322e29768dac054eaba9b93f8f1366f3649f8620de9f9.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:d94acf0fb8f8cf88068447296f1edcc1d7b43d367e8038bbb89ca08b92ff7aa4.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:0352cc248b6301867e0046b60a2354c9fc5ca298c58d84926a3f7cd14bd135c5.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:07053c60a29729887765be6fc013fdeec8be6d468257e599b95027e4c10dabb2.`

Independent reconstruction and engine admission. Independent implementation

`sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83` regenerated the declared census and sole survivor; independent certificate

`sha256:d9c7efc5d89a041e2cbe3e0314798202ca4d821af31ffb937751060df4d536d7.` Derivation seal

sha256:9709df19a1ee4d268b8e62ee8a3d1411a9ff6f49cc1779cd8811ce3b46e4129f. Source manifest sha256:e3b558657d64a0377766b8967e9853f5d2fa629a490ec4132f6d342fd14826b7. Engine receipt sha256:4d86ad5d9cc531086b23f5b8db6c82191dcfdb230d639b9aa242dda8aeb9fca at receipts/engine/model_admitted/SFT-MAT-ELEC-DIELECTRIC-LOSS-004-4d86ad5d9cc53108.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-DIELECTRIC-PERMITTIVITY-LOSS

Measured, observed, adverse and boundary records:

- ELEC-004: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-DIELECTRIC-PERMITTIVITY-LOSS: relative permittivity; loss tangent

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:81aa095a5d6df83abb0cd0e231237dfbecbe13e9180cba6d5e9ecfc1a6e50e27; empirical validation sha256:867c4ae49d128a8c5019f338825302f10b65e4150713b5751520a5cafa6b2990; external validation sha256:8bd878c658bb1bfa3da671c97fala559a4e2c3d01e3235ecaa556c6dfa18cf21; isolation sha256:96bf742d0e14e274bbc135fd9ce15365bfec19ad0d0fb7b73b39e5c08424e0bf; custody sha256:d495d43602689ebdf02fd8e7d5f1070acc8aa8a7c54aa2bd4b2d54eb9dc709c0.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

145. Ionic conductivity and transference

Field:electronic_dielectric_semiconductor_ionic_transport

Obligation: SFT-MAT-OBL-ELEC-005

Claim: SFT-MAT-ELEC-IONIC-TRANSFERENCE-005

Question, law and forced result. Ionic transport is the complete species-labelled partition of the total mobile carrier with exact transference parts.

ELEC-005 uniquely retains species-labelled-ionic-transference-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-ELEC-DIELECTRIC-LOSS-004

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight ELEC-005 axes before target release.

Boundary: Every positive finite ELEC-005 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-electrical-carrier
- species-labelled-ionic-transference-partition
- complete-interface-field-cycle-state-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-electrical-carrier__species-labelled-ionic-transference-partition__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:100f0832b6291f3e1c2bf1de5f5fda4fd330217d4d8921ff0f64b6e743bb792c; generality

sha256:348977bd83da26d86da26e1506a2ae8fbc8b89542f899ff2d66f5ccdc47fba7.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:6f5e891b3c94eff1619a438a3d40056717aee257411bc0b1a6af7af831a9fd26.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:cf0bff4b2994132a79bcbca38583426ceb6f7907f9c928548c0bd1e5553112cd.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:854472c67a6a1121e8bf5069b4db3763939656fb2032dad0ff83e548583a3c47.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:06731bba3e061c669276769a305c3c0ed7ed9fb379bd9e46ae851d637a662abe.

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the

declared census and sole survivor; independent certificate

sha256:d3bfe38784c8e50a3bb05049068b2ab0ed557d22fca1022ad51eb6d5fa69d684. Derivation seal
 sha256:4f59843ebe202ca39ecd59377ca48aae82bc2793771ca6c4f42cf9d4094d1751. Source manifest
 sha256:58159e92256402b2ea2739ea41e0085531ecddef82ccf7895f90a0c3b497377c. Engine receipt
 sha256:cfc2c55ab5621ddc6c4fd2d6b9dce3f44984316a49a2cfc156e5cd520740257 at receipts/engine/model_admitted/SFT-MAT-ELEC-IONIC-TRANSFERENCE-005-cfc2c55ab5621ddc.json. Closure
 scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-IONIC-CONDUCTIVITY

Measured, observed, adverse and boundary records:

- ELEC-005: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-IONIC-CONDUCTIVITY: oxygen vacancy ionic hopping; impedance spectroscopy

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ba2366edeb36af4b3395de986cba2a8a04646aad7ca6c7cb7e30b28ab611416b; empirical validation
 sha256:2c4e0dca1173f344ff1fa549e434d5c0ef0061ff1e02e8a3f8cda7ad513c4823; external validation
 sha256:a7aefae7d83a2b06692a09fafdb05a3d3a03f5d035572a5721ae9778fa5a91b9; isolation
 sha256:152d4749b9300ee80a02f482619a96e8fbf467a0cdcd3f0a0cdf0197c6e6cd28; custody
 sha256:5befac4131304f5c7e187b44a67609d03e8096482d6c6164f466c0c04ae3f0b6.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

146. Mixed ionic-electronic transport

Field: electronic_dielectric_semiconductor_ionic_transport

Obligation: SFT-MAT-OBL-ELEC-006

Claim: SFT-MAT-ELEC-MIXED-TRANSPORT-006

Question, law and forced result. Mixed transport retains simultaneous ionic and electronic carriers as exact complementary parts without erasing either channel.

ELEC-006 uniquely retains simultaneous-ionic-electronic-complementary-transport with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-ELEC-IONIC-TRANSFERENCE-005

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight ELEC-006 axes before target release.

Boundary: Every positive finite ELEC-006 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-electrical-carrier
- simultaneous-ionic-electronic-complementary-transport
- complete-interface-field-cycle-state-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-electrical-carrier__simultaneous-ionic-electronic-complementary-transport__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:46ca214f00eeaf737ca6e6d29b1197db6acbda8da478941f346b16afe17c088; generality`

`sha256:8675a3ed4b1bb9c651e800713767e3d68874db34ca8fc54ee250bea89b627bef.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:e2548aa5371ee11a72a50369cc0614a8d3716e3266a3b70a3a869575d7d61947.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:3acb9df78c879b0b10ecc6f67f1ee69d7068523ac7fe0423c2e67d00ff5a85f6.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:b0ab462cd4464c0038d5821794561adeb1d26bf73465536d142f34ba4b324dff.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:02aec6a61187e6e4dc6c21959d5b816fe343898e57c3936655222aca35613f49.`

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the declared census and sole survivor; independent certificate
 sha256:696b9756656cd6baeac6afb087f247943835bc793ac8e6a9a19ec7aeab1327f7. Derivation seal
 sha256:4c0d9be1eef74bc2cf6be25b942d439acd1792560fa64b59e846cac5845281b8. Source manifest
 sha256:e0983e234214eb5f447ee052f32b28840ee482bbd4b4474fafdcf1e0ca27c523. Engine receipt
 sha256:75d2701859b7a6b04ba078ad5baa2cb0d9c519d675f0f4bd74362995d2988215 at receipts/engine/model_admitted/SFT-MAT-ELEC-MIXED-TRANSPORT-006-75d2701859b7a6b0.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MIXED-IONIC-ELECTRONIC

Measured, observed, adverse and boundary records:

- ELEC-006: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-MIXED-IONIC-ELECTRONIC: mobile electronic charge carriers; simultaneously supporting ionic transport

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:777814d14de9b750175e757f0ed761d32f1ead4b219475edaeaa3ec959db86c; empirical validation
 sha256:00f43cf1498679462ce8fdf84cae30f70cda3113b4488c0fe22b11bace2436a9; external validation
 sha256:7ecfb40769e3f4adedc2f836bddc64ef048a9f21a334e40d289f4d1cfb9f11a5; isolation
 sha256:aa0ac5bb2c83540f2ac05cf8150df707b052d53a464417104fd07a7a1e8b14cb; custody
 sha256:2b80fd90dbe8e3accce33852c1822a575e0a227b49849a7758e33bc1d797249c.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

147. Tunnelling through a finite material barrier

Field: electronic_dielectric_semiconductor_ionic_transport

Obligation: SFT-MAT-OBL-ELEC-007

Claim: SFT-MAT-ELEC-FINITE-BARRIER-TUNNELLING-007

Question, law and forced result. Finite-barrier transport is an exact incident partition into transmitted and reflected carriers with barrier width retained.

ELEC-007 uniquely retains finite-barrier-incident-transmitted-reflected-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-ELEC-MIXED-TRANSPORT-006

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight ELEC-007 axes before target release.

Boundary: Every positive finite ELEC-007 carrier with complete path and observation distinctions.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-electrical-carrier
- finite-barrier-incident-transmitted-reflected-partition
- complete-interface-field-cycle-state-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-electrical-carrier__finite-barrier-incident-transmitted-reflected-partition__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:efb13d4a8cd17d81e0c7f0e37e5f28ef010f5a857d0c8ceb5bbb7ef2134a583c; generality`

`sha256:cd618ef4d61abca7e0893793c859475e04b1a1888318b481060eb7fa89d4981d.`

Shared claim-record clause `MAT-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:b293b587e7e928876beb0401b603614b5c1a449bc23ad8de3180eddb7bf41b7.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:a0c01456c9ad22fb3df37b8579d1b669ec8b5fb7a8c384777f8e928c2b6b3ac5.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:1696ae1cd3b97a052fbe7153da9bd600c3f8b91664cfa51203d485ae344cc99b.`

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:c7444fd8234cfec486681f2af20699afbfb228d7e594f017496eb67619922a140.

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the declared census and sole survivor; independent certificate
sha256:20abd95e98a5079d781e97d1388fc4edcea1a1a5b4e605718f693107cffe7822. Derivation seal
sha256:61234b2e41680f80e7ab10ba0dee4b7f737d0a9ea82fd83ba4f5fb605e40e887. Source manifest
sha256:9f0088471b30dc4a6e1485eef058044ea9fb4559ce45b54a6d78776d038c63fe. Engine receipt
sha256:223f4c84743d38002d7031222be1d66f5892b50f1119bc655902a8468b155c74 at receipts/engine/model_admitted/SFT-MAT-ELEC-FINITE-BARRIER-TUNNELING-007-223f4c84743d3800.json.
Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
Comparison passed: true.

Source identities:

- NIST-TUNNEL-BAND-OFFSET

Measured, observed, adverse and boundary records:

- ELEC-007: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-TUNNEL-BAND-OFFSET: barrier heights; quantum tunneling efficiency

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:f25e15eeb7cab7eb5356e0c3fe8e0bc0421f89a438ae2ab979711a1d0c2b5c5b; empirical validation
sha256:2e2f79b20618941069fd4a803982eb803598cd663cfbd65f3795f138b6886bb8; external validation
sha256:67a913fc2025a182c794fedd1c43253a069fc64b9351d0bef4dad99123d0d57d; isolation
sha256:17e0a188a352aada508804cda347ba47cc0622a1f1a3fa3f11554d57fe89b846; custody
sha256:ab819a8c299d4a836b57351d5c1f6082e04d5d647af48bee92383c90a79ab476.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

148. Band-alignment and interface offset

Field: electronic_dielectric_semiconductor_ionic_transport

Obligation: SFT-MAT-OBL-ELEC-008

Claim: SFT-MAT-ELEC-BAND-ALIGNMENT-008

Question, law and forced result. Band alignment retains both positive levels, exact offset magnitude, held ordering and interface identity without signed proof magnitudes.

ELEC-008 uniquely retains two-level-offset-held-order-interface-alignment with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-ELEC-FINITE-BARRIER-TUNNELLING-007

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight ELEC-008 axes before target release.

Boundary: Every positive finite ELEC-008 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-electrical-carrier
- two-level-offset-held-order-interface-alignment
- complete-interface-field-cycle-state-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-electrical-carrier__two-level-offset-held-order-interface-alignment__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d5f5a07c3b3e52e162a9194d9d244ea5ac00c4c0201a4c6fbbb00beebcce699b; generality

sha256:2e42ee32a68e63bf10182f45db36f30620fd435f3ba16db0e2b01582e3e7f117.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:40f8f28c2ee660ab4e25a4116f137f92256287bfb74058d4ec9c03a2a37e7233.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:477fbff2e89d50da5b3cbad31fee6212304870927ee0c6594c19383bd3f331a3.

- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:5f8341ff1b651627e1824ce141c03a95a92154393515f89d07e211795e85df50.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:b6af52fde433e4c118a287b8b57a547c1f929c5920e22156d5e70acff13fd0a4.

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the declared census and sole survivor; independent certificate
sha256:1041452155e4f6307844f128a278eb8c5183b8909c9dae26f646fa453dcbf4f0. Derivation seal
sha256:dc93fbbb8ea736350ef680c08a60ef17f3d03ce82d996beec94cd448814e3c16. Source manifest
sha256:968763331d5163dff4a0127ccde1d31101a4a79024ad4e20faf0d1335a61ff09. Engine receipt
sha256:9b3afc5ae9b18eaf8dd83976578802ae51f2d67b80f3cf0e170228bc5511d4bc at
receipts/engine/model_admitted/SFT-MAT-ELEC-BAND-ALIGNMENT-008-9b3afc5ae9b18eaf.json.
Closure scope `depth_independent`. External status `empirically_tested_and_independently_replicated`.

Post-seal external comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`.
Comparison passed: `true`.

Source identities:

- `NIST-TUNNEL-BAND-OFFSET`
- `NIST-OPTOELECTRONIC-BAND-METROLOGY`

Measured, observed, adverse and boundary records:

- `ELEC-008`: 2 comparisons
- family sources: 10
- exact correspondence: `True`; omission rejected: `True`
- `NIST-TUNNEL-BAND-OFFSET`: complete energy band alignment; band offset
- `NIST-OPTOELECTRONIC-BAND-METROLOGY`: band alignment/offset; heterojunction interfaces

Shared claim-record clause `MAT-S014` **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0dfe355afd72e26121f98232a3e63e4730b044dfc3ea93250803e44d3fdaefb6; empirical validation
sha256:d7de0bc81c4e0503930e9c8a65a33d51a704e0a0d35dde5b427eaadd7f246672; external validation
sha256:6fcfb29257bc2a330923081553684a99362d7405a8fbfeb60e83362fdbde35e4; isolation
sha256:37ddf331b996293104d1cf8ff2f6f09ef7ab184fe78cae9a09ccb00dfd0a9cb7; custody
sha256:5b145e4b42db1b7fc1b8cff481612306b3e1f58868e639dfb6549a3f743f29db.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

149. Heterostructure carrier confinement

Field: `electronic_dielectric_semiconductor_ionic_transport`

Obligation: `SFT-MAT-OBL-ELEC-009`

Claim: `SFT-MAT-ELEC-CARRIER-CONFINEMENT-009`

Question, law and forced result. Carrier confinement is the exact retained depth of a positive level below a held interface barrier in a named layer.

ELEC-009 uniquely retains below-barrier-layer-held-carrier-confinement with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-ELEC-BAND-ALIGNMENT-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight ELEC-009 axes before target release.

Boundary: Every positive finite ELEC-009 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-electrical-carrier
- below-barrier-layer-held-carrier-confinement
- complete-interface-field-cycle-state-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-electrical-carrier__below-barrier-layer-held-carrier-confinement__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:ce4a379aaee188d684aecb85ec4fb4a2f9a23c4815d49d6f00294a4301746244; generality sha256:903989a1aaff65de88cff5a98fdffc73c4e39f6be70c504f80e7f49235b65f68.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:247177d5a14e8b8a21ba15e1301cf48cd30dcc32dfb145a8427dd592aa84dc7d.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:b8c3e4c9be372fbcf9d69c61e50a19070b71c543fbb979be2016217c45ef298f.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:23bac81e4ff5d22f31ad06057fe4535fcc2f64abedf36b0d525b4dc8b313d3a0.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:93a064abfb6160d74365451b5bad32619cc24e38b4d214afb53bd0dfb84e1435.

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the declared census and sole survivor; independent certificate

sha256:4e8bf2de9666382b6a668e2662f9b136547128c76d349ed70968ef1a10c9dc2a. Derivation seal

sha256:b0b6e58fb9909bf784dd2771832af8decfd260f5440179167c33460c3e802697. Source manifest

sha256:e622f660ff7967fe158613636fb21f614f238f67dda646ccb26271f800a4f68f. Engine receipt

sha256:0afccf622730db4176a14b347eee8d4e453e1eb52fb78bc04835549031d00805 at receipts/engine/model_admitted/SFT-MAT-ELEC-CARRIER-CONFINEMENT-009-0afccf622730db41.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-TUNNEL-BAND-OFFSET
- NIST-OPTOELECTRONIC-BAND-METROLOGY

Measured, observed, adverse and boundary records:

- ELEC-009: 2 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-TUNNEL-BAND-OFFSET: semiconductor heterojunction interface; electron and hole barrier heights
- NIST-OPTOELECTRONIC-BAND-METROLOGY: 2-D heterostructure semiconductors; multiple layer structures

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0fc43b432baa61c906cfbf764d6c3403f90bbcc50dfa9ca36dbf827326343ae6; empirical validation

sha256:c6d2eaea387cdcb501f1c68f63ca23b2f856f9b77ffb1d3e07b28d1bbcc9d348b; external validation

sha256:3cdb3f5ca6dec6fc7f17cf78e06b0cdeaa64357ac9662234c2e01f4abe234b27; isolation

sha256:10890fc30578d73a7153da43ccf6e23eea4446c92362531a21ca93f6f001c229; custody

sha256:cd363ff3d1780ab1f41821ee13c2a090679ef7a03318ed6bcf644540013e90f3.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

150. Defect and trap electronic states

Field: `electronic_dielectric_semiconductor_ionic_transport`

Obligation: `SFT-MAT-OBL-ELEC-010`

Claim: `SFT-MAT-ELEC-DEFECT-TRAP-STATES-010`

Question, law and forced result. Defect states retain the complete defect population, occupied traps and structural absence or positive unoccupied remainder.

ELEC-010 uniquely retains defect-occupied-unoccupied-trap-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MAT-MEAS-MATERIAL-001`
- `SFT-MAT-MEAS-SPECIMEN-001`
- `SFT-MAT-MEAS-PROPERTY-001`
- `SFT-MAT-MEAS-TRACEABILITY-001`
- `SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005`
- `SFT-MAT-ELEC-CARRIER-CONFINEMENT-009`

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight ELEC-010 axes before target release.

Boundary: Every positive finite ELEC-010 carrier with complete path and observation distinctions.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- `complete-positive-electrical-carrier`
- `defect-occupied-unoccupied-trap-ledger`
- `complete-interface-field-cycle-state-path`
- `specimen-method-field-temperature-uncertainty-held`
- `complete-trace`
- `root-bound-forward-forcing`
- `positive-finite-successor-closure`
- `no-extra-rule`

Unique survivor: `complete-positive-electrical-carrier__defect-occupied-unoccupied-trap-ledger__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:d95010453cd585b67db1b13a897d8a2fd7f716d8858e167554526b2b8cbe4cc7; generality
 sha256:aef8c746c424687b021e7d9f0ed23361ecc04206fc0d9d6210291617f238bd9a.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:2aaa8539be46fa9341eb24e44406a3990ae44fb9d7a03514d8adf78edc500416.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:ad37202821a9eae062435b792fb5f29e3f106a78c34c9f9592e97e111176c027.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:3c23cbab921977653490ecc2ef36bde8d5a92d4830c5661bf49b7984f8b76e6c.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:b1a56d1cddd9b141aff84a4e4e8eaf69988123ee24e8bc47d01f2a4b9c9b9070.

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the declared census and sole survivor; independent certificate
 sha256:834b80f360b32e2bbef75cafddc13c9dcf1c937ecd6d93d6acca383c9385707. Derivation seal
 sha256:9433149002b242403916290c916cff4645c0b9a5655dc3e49b69a5b78513da9a. Source manifest
 sha256:74dc57e441d18285b7d44a38654192a93ff5a7c560cd52153d0350a7b4d49e98. Engine receipt
 sha256:0a9b20a06741a4d825d5b6b8eae58e1df3cd01650b72bbfeb6a913d214d5efdc at receipts/engine/model_admitted/SFT-MAT-ELEC-DEFECT-TRAP-STATES-010-0a9b20a06741a4d8.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-POINT-DEFECT-CHEMISTRY

Measured, observed, adverse and boundary records:

- ELEC-010: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-POINT-DEFECT-CHEMISTRY: point defects; valence state

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:f2989a88f1d4d903b33603e8824957078a8253844b856b49f323d14bd6d6e3d0; empirical validation
 sha256:f13ed32eb0cc11bfa14074e63c9ae7deaf1f8e51f5163e7bd16ab800df9bcbb1; external validation
 sha256:71b6c1fe7d5db567c7a76b21624bf8851e1bd17f7bc8b0be0dd356e860159455; isolation
 sha256:25922e7cf54539319c2642829e8bdd7d1febcbfed663fa5fd3e7607d19cc995e; custody
 sha256:59e9dc32eef975b5e94537a5b59a46e43c4b8035776625524eccceee87bf6d4f.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels

- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

151. Charge screening and depletion length

Field: `electronic_dielectric_semiconductor_ionic_transport`

Obligation: `SFT-MAT-OBL-ELEC-011`

Claim: `SFT-MAT-ELEC-SCREENING-DEPLETION-011`

Question, law and forced result. Screening retains the exact accumulation, depletion and inversion partition and the observed regime label.

ELEC-011 uniquely retains accumulation-depletion-inversion-screening-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MAT-MEAS-MATERIAL-001`
- `SFT-MAT-MEAS-SPECIMEN-001`
- `SFT-MAT-MEAS-PROPERTY-001`
- `SFT-MAT-MEAS-TRACEABILITY-001`
- `SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005`
- `SFT-MAT-ELEC-DEFECT-TRAP-STATES-010`

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight ELEC-011 axes before target release.

Boundary: Every positive finite ELEC-011 carrier with complete path and observation distinctions.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- `complete-positive-electrical-carrier`
- `accumulation-depletion-inversion-screening-partition`
- `complete-interface-field-cycle-state-path`
- `specimen-method-field-temperature-uncertainty-held`
- `complete-trace`
- `root-bound-forward-forcing`
- `positive-finite-successor-closure`
- `no-extra-rule`

Unique survivor: `complete-positive-electrical-carrier__accumulation-depletion-inversion-screening-partition__complete-interface-field-cycle-state-path__specimen-method-field-t`

emperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:49f4c5f8e22fb228cc709315c8d82bede2538eb94b07a7f54a8b2c55b35ec0fd; generality

sha256:c989d14a01bf4d556bfa75575362137754eafd19758d00007dff06939d06260b.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:40b4952815401aee46948deb298fa3e6287b5be8bb9d7dbff388ca4a7edac45e.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:2abc45e7b03a0aa2fac321b890e2d197368a17ec1d78b877f0d0554121a85c19.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:b10147d4b5c1e89334a62954425f77e3a6fdcde898b4efed60aa571b0a8110c1.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ae2b1df8da5514a896bd8229176da7d42b3ae616308d360914240c20b051e85.

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the declared census and sole survivor; independent certificate

sha256:e13e68cbe664e0656c77aa17df510e1d7d00f61897507d9452004264f4b1d440. Derivation seal

sha256:add8627017640fda86606e5d361923e3b870641dfe6d853133b5bbf843bbae26. Source manifest

sha256:ebb8d33a40f6a6553fd1d9454d049032a78d397b2484609559da3330fe50aa4a. Engine receipt

sha256:7f783979da059859e675242ec28ed6a5da90c706599e0163fe1b7f3d229cf3ef at receipts/engine/model_admitted/SFT-MAT-ELEC-SCREENING-DEPLETION-011-7f783979da059859.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-DIELECTRIC-DEPLETION

Measured, observed, adverse and boundary records:

- ELEC-011: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-DIELECTRIC-DEPLETION: accumulation, depletion and inversion; trapped charge density

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0730b4682d32d1be1d35438b4d2102ea9a03e3257215b79e06c4df17b1e5579b; empirical validation

sha256:59c7b5f3e0404f567defd8bad51f6e643385037c6cdb1b575adb89671e4ad7dc; external validation

sha256:7311057b5dc8b1862c8b4502e35c33f0783db6b488d0f829c428e33fb66b66d5; isolation

sha256:11a1ec2b889e3f99d00ed0eada956f92667175c3073a17d54d382b71b54a9433; custody

sha256:c8697560a6bef9dd7ff0c89bc74f74f57920d3cdadc8dedb6a2b0d4e69f18175.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude

- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

152. Electrochemical insertion and material-state response

Field: `electronic_dielectric_semiconductor_ionic_transport`

Obligation: `SFT-MAT-OBL-ELEC-012`

Claim: `SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012`

Question, law and forced result. Electrochemical insertion retains host capacity, inserted and removed carriers and the complete operando material-state path.

ELEC-012 uniquely retains host-inserted-removed-operando-state-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MAT-MEAS-MATERIAL-001`
- `SFT-MAT-MEAS-SPECIMEN-001`
- `SFT-MAT-MEAS-PROPERTY-001`
- `SFT-MAT-MEAS-TRACEABILITY-001`
- `SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005`
- `SFT-MAT-ELEC-SCREENING-DEPLETION-011`

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight ELEC-012 axes before target release.

Boundary: Every positive finite ELEC-012 carrier with complete path and observation distinctions.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- `complete-positive-electrical-carrier`
- `host-inserted-removed-operando-state-path`
- `complete-interface-field-cycle-state-path`
- `specimen-method-field-temperature-uncertainty-held`
- `complete-trace`
- `root-bound-forward-forcing`
- `positive-finite-successor-closure`
- `no-extra-rule`

Unique survivor: complete-positive-electrical-carrier__host-inserted-removed-operando-state-path__complete-interface-field-cycle-state-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-success-or-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:e8643d1e0864df795c54083537423da24b504d67f893a5ec7e30e16936435bf2; generality
 sha256:b0bab61d2adb651dce9e83a020799370691e09853b3811c9dd33e0df22a6f05d.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:3719e751f66a1be6645ce1154a3de78739b6f5082eaf00c8ef7ec152b12b9754.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:471021fb68d762f601a82771c04a642a47104f6e384aa762134435a57346c988.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:022b710aefddf47c6ec3438538df0088923d4c467508dd6a837570d2b33a4b48.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:cf94e2ca1c109ba1ce5899e437c36c54453c1d9f92437c949810815f05e0da9e.

Independent reconstruction and engine admission. Independent implementation

sha256:107cdea449594c2a94274bd2959f2914a517da4c8d0c7e66a021d588420e1d83 regenerated the declared census and sole survivor; independent certificate
 sha256:ea6054ef105764ff32becdc894eeee4446bc7253b25ced51cf197bfe86eee520. Derivation seal
 sha256:e5c9d92e6a278c2d02ca7260725e45357feef6703b7dd7fb335e6c32b4c77e30. Source manifest
 sha256:ea0adee0c799bc5ea4f8b47d549eabfa0b7e7652926e65729307342b949e5f38. Engine receipt
 sha256:39e1ed13fdf4eda1887c6c54107dbeb519158eae317691102248899dc4553333 at receipts/engine/model_admitted/SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012-39e1ed13fdf4eda1.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-OPERANDO-ELECTROCHEMICAL

Measured, observed, adverse and boundary records:

- ELEC-012: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-OPERANDO-ELECTROCHEMICAL: battery charging and discharging; measure the charge transfer processes while they happen

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:d97f12cefe11cfb6c2392b4f61ed14c4c191721f5e37f249abb9e71385aeab76; empirical validation
 sha256:6fbd7ca74dfffbe7d869ec55646cfd1db6a4d038af819ac96e0384c56841869; external validation
 sha256:f174aa52887f064dac2482ab789aac4d963ff41e89506896ede2d6fcc742a7cd; isolation
 sha256:840551264e94a32c62bfa9106413dc174b9509db91fdd7f57b8a9b333b5a9073; custody
 sha256:ac36d83756776f355988f6c494dc192d84e4783a9982e56c96ec4a83c74faefd.

Prohibited inputs.

- no imported transport equation, continuum band model, fitted constitutive law or named mechanism as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- absence and carrier orientation remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Magnetism Spin Superconductivity Superfluidity

153. Paramagnetic response

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-001

Claim: SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001

Question, law and forced result. Paramagnetic response retains positive moment and field carriers with parallel orientation as a held label and exact rational susceptibility magnitude.

MAGSC-001 uniquely retains parallel-moment-field-susceptibility-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-001 axes before target release.

Boundary: Every positive finite MAGSC-001 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- parallel-moment-field-susceptibility-ledger
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__parallel-moment-field-susceptibility-ledger__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:587aae3abb4e72405d20ad42220cf4d2cce8613eb3758e2c6e49948047eeb044; generality

sha256:21a8aeb3fee27631bdf7a4477bdec6f36c486a487ace6b3f4dc13351e2756ee7.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:5a9c3020cbfedbed8414fd6766a7d96641aaeb5f9a98ad7e26f0404cc9351d0f.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:ee069daca07c4a4e19de1cdcd5a140aa646cb37df32cd3bbad79921c8a26494.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:174e5ca8818dd934b194a069424948e5d0464ef1a4d9ba772370e9afd0f048d4.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:7760e954412eb7a7fc87e3275cf626eb1eca0b58dd3fc4250f3e2aa0b148289e.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eefb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate

sha256:e899c2ddd7f107dac4932fbcca243ff4d0179fcbd39bd77b94876e1b296098f7. Derivation seal
sha256:17e59ef321aeeb460d948b02b8c308bc25f21e9b8e156053acfed2a4f5001812. Source manifest

sha256:9b784e52354bb2159ef4ae65e2d6e993400eaa1c2f06fad848b23cef7ab22726. Engine receipt

sha256:7228fa5be9be3cd341e17c494182a86aa8e4e0344042bd1b8d8839c53e50cccb at receipts/engine/model_admitted/SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001-7228fa5be9be3cd3.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MAGNETIC-SUSCEPTIBILITY-SRM
- NIST-PARAMAGNETIC-DIAMAGNETIC

Measured, observed, adverse and boundary records:

- MAGSC-001: 2 comparisons
- family sources: 12

- exact correspondence: True; omission rejected: True
- NIST-MAGNETIC-SUSCEPTIBILITY-SRM: specific susceptibility; magnetic moment
- NIST-PARAMAGNETIC-DIAMAGNETIC: paramagnetic or diamagnetic; parallel or antiparallel alignment

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:259f1f03c80c1f10ff15a45d1086008cb1955e9c499f4fb5ffd31a68999fbaeb; empirical validation
 sha256:07b8a385d02cc594a96f2a0066b12413063a5da5e7736fe6cf7d50a450f22855; external validation
 sha256:f20a1250375c918f1a9f87bcb591bdc7553cf8b8ebedf221ee49bed678fe1836; isolation
 sha256:508930ffaea1881266c64849de98a79d420250558fcb034cff62be9ca72ec8da; custody
 sha256:348284e5e92520355ba09ed587b6a73c14734fc2e2fb1e87100eeb1d465f3127.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

154. Diamagnetic response

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-002

Claim: SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002

Question, law and forced result. Diamagnetic response retains positive moment and field carriers with opposed orientation as a held label rather than a negative proof magnitude.

MAGSC-002 uniquely retains opposed-moment-field-susceptibility-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001

- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-002 axes before target release.

Boundary: Every positive finite MAGSC-002 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- opposed-moment-field-susceptibility-ledger
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-magnetic-carrier__opposed-moment-field-susceptibility-ledger__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.`

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:b43bc3383e47ddeddb2fdd595a78e3fa70ec23be950cb525ce5bb321b3410b4; generality`

`sha256:cbcfb2124ce70ce7df52f1611b99a16d00dd9fc39d73053a20dcbb7cba8f4ad3.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:32e2f8e53add19d831c8bfff086b81565c84fbef3ae0f3ccc335af8ba12166c4b.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:bb2f852703a66f301afe1e23cabd1d481a3950b07f0ce30f9d0ac46f55cbcd93.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:771d24072175086f78431b46cb5d816d922a4bda4ad54c441ccee684d2e81885.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:29e322f0cbc7853e5f1f867b37136e66d2c10f3d465c2b40452d53cbf52026ae.`

Independent reconstruction and engine admission. Independent implementation

`sha256:59d25b05eefb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0` regenerated the declared census and sole survivor; independent certificate

`sha256:8723b5f292bc8b29af8091835dcb6b697e820630760abbfc04a8568161c4be02.` Derivation seal

`sha256:71b5e1626bdfffa0a1a502510b0a2224c27ee557c4daf4186e0a20263917df61f.` Source manifest

`sha256:9e44a22b1574bdde09fbe1084fb58e6aa50c668632c6db8019249bb0b1912281.` Engine receipt

`sha256:cc74750ce36474675622217b4bacd5aee0bc00a2646cdf6db44f58677b98f8b5` at `receipts/eng`

ine/model_admitted/SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002-cc74750ce3647467.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-PARAMAGNETIC-DIAMAGNETIC
- NIST-MAGNETIC-SUSCEPTIBILITY-SRM

Measured, observed, adverse and boundary records:

- MAGSC-002: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-PARAMAGNETIC-DIAMAGNETIC: diamagnetic susceptibility; antiparallel alignment
- NIST-MAGNETIC-SUSCEPTIBILITY-SRM: pure platinum cylinder; specific susceptibility

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e23c0b5708e5b371bef0fb757eeb3588128e50162ba2927644eea289283843bc; empirical validation sha256:76356ae2cf99d992da5eaba9790fe8d3c9da76703f3f5860682b53059caf3120; external validation sha256:2734995e4f9a6b1b3240f6b4c4634e02172c8c01bc846a033b5613f52c06ba78; isolation sha256:47ef739c0a9fbdb1207c4bf2f76f0a5838077dd15f0e489a6cd8a804f37d28fb; custody sha256:d00c2571838327a414dec60b3d298439e440d6a92ed86160a171e2776e73e34a.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

155. Spin-glass freezing and history

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-003

Claim: SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003

Question, law and forced result. Spin-glass freezing is the first frozen state on a complete temperature and preparation-history word.

MAGSC-003 uniquely retains temperature-state-preparation-freezing-history with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-003 axes before target release.

Boundary: Every positive finite MAGSC-003 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- temperature-state-preparation-freezing-history
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__temperature-state-preparation-freezing-history__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:383dbfffb22c65708ab840a52484f72d6cb5dd3dbe26c7edb048c4f8ab38f7981; generality

sha256:ff58f76243e68e0e99e71ce55a736a3dc58685db7cfb5458aeb39febf21ef10c.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:7ad3be10242ec13e34084b665fbee76569a79669f75d4f44d404794dc2b7f764.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:244d94041de4cddb45b9d4102b6d3b8629c2481e311073e2a66fa2b9a374adde.

- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:84f0b3895a1890b0f5c14bdd6c4cbf1ba28d02612f05e649b3552b05693b53d8.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:8716a52cae14a04cba4fbec6a4f910f55367f9c5c31b0c2a6b26d1db37edb9a9.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eefb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate

sha256:17d51958fe88a41bb5d7dfaaf1622ca0e0f505948d875f7a12fcdfa7e8659f68. Derivation seal

sha256:cdbc8b32e3a182cab4dfaf76a18fa4315aa0678c0b3d21dfa841d33fba97cd7. Source manifest

sha256:36fd826f1c45a275b6afe652c94ca44813e5c124647dc20a38e2602e9a3cc00a. Engine receipt

sha256:9e7d4eccbc5c5cecc5bcf575caee702c4f66ff7f824bd0c341e4721bd6067621 at receipts/engine/model_admitted/SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003-9e7d4eccbc5c5cec.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MAGNETIZATION-LAB

Measured, observed, adverse and boundary records:

- MAGSC-003: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-MAGNETIZATION-LAB: spin freezing temperatures; magnetic state

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:85f5e7d00d63b6d4bc5ad61efeb34fc8684dd41ed3a4dc3fcdaf8aeda1f4be79; empirical validation

sha256:6cc27c1a741cfd4abcf263facb69b5b299188a7073eed59f9db92d4c9bd55cb8; external validation

sha256:13b15eb47c78b02abebcc9f20bb34b31c0378da45322576cc5b6326f61822d18; isolation

sha256:4a5fb507fefeb7b2b9ac7fffcdbd38de0669639f47453be58b0a8fb4f20fab3e36; custody

sha256:cc2c978f94b804eaa2d1522bf9e3310ffe5e6d925889015ad23e22934e0bd778.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

156. Magnetic domains and walls

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-004

Claim: SFT-MAT-MAGSC-DOMAINS-WALLS-004

Question, law and forced result. Magnetic domains retain exact domain and wall populations and the complete nucleation-growth-motion-disappearance path.

MAGSC-004 uniquely retains domain-wall-nucleation-growth-motion-disappearance with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-004 axes before target release.

Boundary: Every positive finite MAGSC-004 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- domain-wall-nucleation-growth-motion-disappearance
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__domain-wall-nucleation-growth-motion-disappearance__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:1c1efffb7f133367b8f2953e64f44bacfedad327aa346edeb6f02147ff6a1e10; generality

sha256:438fc4f232c8ae8e6a7b67efa0cf15c451d46bd9e61faf9d17b3e09475b62090.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:6fd71d2ae3a77f3f026cf68890dd24717a85eb3819850f98b96aff91eb5cf5f3.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:12945ca5c3e93c8c89a104dba8d61d95ce77056663c16ede21bc81f936ed5379.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:5f8cca2e1e554c90e3d4a1518d7088e5fd6a477dffbc3122c7dfbd4dd631fca1.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:8b57e7ad9120b325f4c817605d17ec28a965ea82819d3857e0c57325ceeb32e7.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eeb343c407cc23f50eb2c66f453eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate
sha256:ba3eda01cfa0b6f372d8396760458d2d978b03a1de22050e5936190021860a74. Derivation seal
sha256:975e80c8a5cf764cc9b9bb2c2b180f7cbbd0c40a44629ebd5577847f3b71b8ae. Source manifest
sha256:29904375e6295bb8b5ba1a7929afccd35d79f6a4f7e90b371b4051c3738309a9. Engine receipt
sha256:28ced43e9c0880b8e94dd7a0c29677e8fe89922a635b0f65a596ab7d63d39013 at
receipts/engine/model_admitted/SFT-MAT-MAGSC-DOMAINS-WALLS-004-28ced43e9c0880b8.json.
Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MAGNETIZATION-LAB
- NIST-MAGNETIC-IMAGING

Measured, observed, adverse and boundary records:

- MAGSC-004: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-MAGNETIZATION-LAB: magnetic domains while they are forming, growing, and disappearing; magnetization dynamics
- NIST-MAGNETIC-IMAGING: domain wall mobility; magnetic domains

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ca7ade7789196d085a07e175dbccfd938f888746f73a803b802bd6b56f57d913; empirical validation
sha256:37f18143338deaf55a4c8c650202151b619d97f858f5aeae63a975da7ae66533; external validation
sha256:8fb204f1f9bfb4e53b0e58d4d50b7a795700e9c443048e05b44532d47711ee9d; isolation
sha256:8caee4c6ef4b8647d04a394b98f3cfacd35a0676b1e969c38747c508d6393037; custody
sha256:ad0f04eb1b5ba1b561ce1eb8fc273727e353ffcafcdde8e93c2f440fc3fbdccf.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels

- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

157. Magnetic hysteresis loop ledger

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-005

Claim: SFT-MAT-MAGSC-HYSTERESIS-LOOP-005

Question, law and forced result. Magnetic hysteresis is a closed, ordered field-magnetization word with forward and reverse orientations retained at every point.

MAGSC-005 uniquely retains closed-forward-reverse-field-magnetization-word with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-DOMAINS-WALLS-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-005 axes before target release.

Boundary: Every positive finite MAGSC-005 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- closed-forward-reverse-field-magnetization-word
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held

- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__closed-forward-reverse-field-magnetization-word__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a7f4022d05a4d023a1e3e8273b08998ba2fa5109bd6262fa847b90583c043543; generality

sha256:5e1521d4252a786f429b62dbdf4922ba5b2188d5aa29f10fb40502a9e3b8910f.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:2f7a12954643c9ff45f2e24d16eb04e2cc52b425f57afe7dac42f30d824795f7.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:d224f960ea1c1b11a88ec17505853fec59728b4c20ccb1e498f40e9a814f6108.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:439cb71fc40fec0c15c1dc798c7f2c903d38bcdcdac0010928dbb0d5a01720a1.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ece61ea141097912ce5acedbf4f0ad4f533d04f898294b6db23028d7788d917b.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eefb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate

sha256:b19adeb0c728a15aab42d3e59129aade2f26a37435e13191bd45a7ad08db546d. Derivation seal

sha256:2e2f3d2536498c64e2360d7e5e720083a79b2798fbd4335c35765edc606e75b3. Source manifest

sha256:0975f79447caed6ddf8d5cfea2cd45fd5b43183e0d6afd66dab95ceb308afa59. Engine receipt

sha256:fd139ec657173ab8253dac9b667f604b6e7d1a74ba65b5b8b98969e7ff447cba at receipts/engine/model_admitted/SFT-MAT-MAGSC-HYSTERESIS-LOOP-005-fd139ec657173ab8.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MAGNETIC-IMAGING
- NIST-HYSTERESIS-STANDARD

Measured, observed, adverse and boundary records:

- MAGSC-005: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-MAGNETIC-IMAGING: magnetization reversal; field regimes
- NIST-HYSTERESIS-STANDARD: hysteresis loop; scaled geometry

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:f93d62d85cbc43dd68d3f71f8c6fd57ae2afca280f0d3ac7378d3d6db951aee6; empirical validation
 sha256:696e4c99a336d31cb3b5ca1621283c6d55b53c7e48f855325b36a913b5a06f68; external validation
 sha256:043f5585d4e1d1ddb02d783184b3bae52cbf6c6f4a93d1aef424e9fc610f923b; isolation
 sha256:7b42120d64e4360339545e5a5b179838de4fa523434100d06230adc505681be9; custody
 sha256:2b2ac5dba285604275d861349e9a00babb1d2a50baff8566e868f5a919ca7b8a.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

158. Magnetocrystalline anisotropy

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-006

Claim: SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006

Question, law and forced result. Magnetocrystalline anisotropy retains easy and hard positive response carriers, exact gap and crystal-axis identity.

MAGSC-006 uniquely retains easy-hard-crystal-axis-anisotropy-gap with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-HYSTERESIS-LOOP-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-006 axes before target release.

Boundary: Every positive finite MAGSC-006 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- easy-hard-crystal-axis-anisotropy-gap
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__easy-hard-crystal-axis-anisotropy-gap__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:230a8210bdc65a22cacac516c3a6e3727c70965efffa00f030ee9ad322b12cd3; generality

sha256:6777a294ae9eff338f77d10c75ba135c57bbfceb0895d4e66c979fc09a683000.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:59355266dd721087af61010e03062d52848c1779845c343c964b15535150dea9.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:6c6f17ab5a59820a761cd045a69f397f8533f0a0d63b81f4dd2ca14c5c7ea8c4.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:798f77cbc5713826703b7dc48b57aaf07337d26a85ddfdb974e4910bbc253c8c.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:cf0b4e412262754d3938c407fd21e4013b0ec188933319f62b811476928dce0c.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eefb343c407cc23f50eb2c66f453eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate

sha256:4729800a5714cd73025026bb1f3d42cf1f5a30d891f1317d6a8739c75f878eba. Derivation seal

sha256:656049dc03df358148469a55ebee6dcde82c9128ea6bc1fd208a3706bf371eb. Source manifest

sha256:37ebd3a90083154796a6e1100be42f4c5e01ab80f7f421ccda958b5368ff2b60. Engine receipt

sha256:490b9b5f874289adb99425c0825d9972735bc9c9c23a7a3c58dbd3d9d3e856f6 at receipts/engine/model_admitted/SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006-490b9b5f874289ad.json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.

Comparison passed: true.

Source identities:

- NIST-MAGNETIC-MATERIALS-METROLOGY
- NIST-MAGNETIZATION-LAB

Measured, observed, adverse and boundary records:

- MAGSC-006: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-MAGNETIC-MATERIALS-METROLOGY: magnetocrystalline anisotropy; magnetic anisotropy
- NIST-MAGNETIZATION-LAB: magnetic anisotropy; coercivity

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:79be3547e521ef79c0e75d486561451f0b8aad9a221b0b448f5606a0fd905407; empirical validation
 sha256:078b9cc15401ebdcb3cbea8a138f6daa12fd2bcb4f17bd9ffe77c44c4b5edfc4; external validation
 sha256:55d83cdb3ac63680ac143cfb6baef06440ec8fea6bdc7368c5855be84cdc1a70; isolation
 sha256:9450330be55bc81f451e7032baaed0663cce9253039088c1224a54daf96bf853; custody
 sha256:c6fe01ae1174e270e3029447dca26af257e18c33778c96c2acf120e556ecf2f1.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

159. Magnetoresistance and field-response relation

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-007

Claim: SFT-MAT-MAGSC-MAGNETORESISTANCE-007

Question, law and forced result. Magnetoresistance retains base and field-response carriers, exact rational change and held field orientation.

MAGSC-007 uniquely retains base-field-response-orientation-magnetoresistance with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001

- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-007 axes before target release.

Boundary: Every positive finite MAGSC-007 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- base-field-response-orientation-magnetoresistance
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__base-field-response-orientation-magnetoresistance__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:c28fbd4d4bcdcf110f9e56421b6d66f5058bdc70a72c1f1528a79b86e8b2c5f; generality

sha256:fd431d45846f45c6e323c8c1f81ea4b24651874429de6b7371780a29b4876866.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:2b8caa3c78b4890f20de38beb489e6b9231fe05c811a44fbf3cb17f1adca6ad5.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:1387418af7ae133ba56fdddb1c6f9f9561776829d19e044924f8ad7ae228e844.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:a35b5c0fb35a79b894579f88a659bb85c6706c35c41a50fae532f95756774635.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt

sha256:81baa4adb06df8df4cd5ff59752543529b55984064c16b3ad5ee100c2c90be71.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eefb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate

sha256:2c8ffccbc9f3a69eec7d7c5024cae698a3284f20696433edd8203ba18c16c18f. Derivation seal

sha256:2b33d8aa89f47a720e8278769153c57ea2917e04e4b99ee870e689e7aa575cfd. Source manifest

sha256:37c83145e9c01d167f9dbfcfb5f1f4dc2bb70434362cd80d34071c3f69412b9f. Engine receipt

sha256:ac5452c0a039eba0a778441d5fec468e9a7a6b65a25aa41b90e27dc443b7e4d3 at receipts/engine/model_admitted/SFT-MAT-MAGSC-MAGNETORESISTANCE-007-ac5452c0a039eba0.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SPIN-ORBIT
- NIST-MAGNETIC-IMAGING

Measured, observed, adverse and boundary records:

- MAGSC-007: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-SPIN-ORBIT: magnetoresistance; charge and spin current
- NIST-MAGNETIC-IMAGING: antisymmetric magnetoresistance; domain walls

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c25502fdd65ed8426d73bf03448f1dbf4504329461e165b40c4012c7a8249cec; empirical validation

sha256:530bf59f6dbb8811965896d58e12811212e3480be4193d790cfe84324861b99; external validation

sha256:b8186b62c7b0b18e9d0e812cc6779f48da8cfd702b0f6db7e571edccfa148eee; isolation

sha256:f750f673cf79d4ba358f61dfcfa1679219a4ab68ccb7befbfc41d03ba8234623; custody

sha256:326f90093caea8329886720e8ea4aab31ef9eb9ad2a81ed9e9e43d5fcc98c94c.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

160. Spin transport and relaxation

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-008

Claim: SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008

Question, law and forced result. Spin transport retains initial and surviving carriers, any lost distinction, counted relaxation steps and orientation.

MAGSC-008 uniquely retains initial-retained-lost-spin-relaxation-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-MAGNETORESISTANCE-007

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-008 axes before target release.

Boundary: Every positive finite MAGSC-008 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- initial-retained-lost-spin-relaxation-path
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__initial-retained-lost-spin-relaxation-path__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:782a05d71b878cab09064dcd117bbb42039e8a391781605872a69d57ef3d0282; generality

sha256:5ce8f6bf304144418998fb58e809fda55e6540f7db517da4e0380663e341bd22.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:a50f6c8d7f1dde78d4dbd7bdf7b2ad315fedd451a186ec91f0fc13a748233735.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:1cf3ad07982fc84592f6ee867cbe3a2edea94de4eb6fe2a5d28145c578c6b836.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:5dc046160d4362d83052bc2d6e35738baa00da1d255b1162448ed272894b8655.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:4c21bb5593183b8ba992634fada84eb22adcbc3fcdc11ce92abe4ed04d47ad73.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eefb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate

sha256:494c205b00bab1ccbb8735cf08096710d0cd3bdcb13e56081dff1d9db5d8473e. Derivation seal

sha256:14014d972897c37bb9d774fdbdlc277a4f16de6524fb8baa462d4e1ae1963a95. Source manifest

sha256:1122d7021801caf85d2b2555753cf98bf88833acb941acdd29e5bcfc1e8ee585. Engine receipt

sha256:ede3249d7281586061ae6f48c4a98ee7f162e7ec2b2fa9ffff24f88c1a813464 at receipts/engine/model_admitted/SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008-ede3249d72815860.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SPIN-TRANSPORT
- NIST-SPIN-SPECTROSCOPY

Measured, observed, adverse and boundary records:

- MAGSC-008: 2 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-SPIN-TRANSPORT: spin transport; OFF-state
- NIST-SPIN-SPECTROSCOPY: spin and magnon transport; energy dissipation

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:41073d9d28bfd4113403d24acaa8ffde9f97b01eb2c92d81b81cc099286678a2; empirical validation

sha256:3e98b9995745a37e768fee354ef40a33ac62e39b486ff978e98d76d1f3b8c0e9; external validation

sha256:8cc902907a1665290e58b5e4df39bf75c2f5887e00aa6396cdf53834d43dd563; isolation

sha256:a49a28ecf00a29f34295fb04955ca0d7730b538de99d2d95f88ee2c66b1d0b25; custody

sha256:a5ab6bfdbb8fedb0ed2f3498e3c0ce397437f508b6d75cc3db4077e826ea90f9.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained

- no failed attempt retires an obligation or changes protected authority

161. Superconducting critical-field organisation

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-009

Claim: SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009

Question, law and forced result. Superconducting critical fields are two ordered positive boundaries separating Meissner, mixed and normal state classes.

MAGSC-009 uniquely retains ordered-critical-fields-Meissner-mixed-normal with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-009 axes before target release.

Boundary: Every positive finite MAGSC-009 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- ordered-critical-fields-Meissner-mixed-normal
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing

- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__ordered-critical-fields-Meissner-mixed-normal__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:f462ff0518b6a0ce29b0fa10bb24dfae34251467a3f0004f43a04cca367c9d46; generality sha256:dbb74bc6a0c8885c0e095f2fd703091a5a315bf2b0d3b3259b4f3f224625d225.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:17de2656f7ba6d4db222a80dd4099abc71d1ad2c446e4c6b792fe0a5be6f2a7f.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:9a05754cfccd09e91a824d2696181c6828988a3b670423d9553239594a1c550e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:bec3b24ff8e35be9f097c834c9b47d5974f14ea13f74ac7f02158949003e4ba7.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:ed15e984017b7b67492baa6b59a225e786edf89d0c4e314d42177aefb0570cd1.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eefb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate

sha256:c57d2e6e935afe53d0359825efbfa6281e745f239ac7c7090e1330b57f66fccd. Derivation seal

sha256:ac2e2155573d31ee45f2189fc14234e0d9fdcbd6449d7782112cad94dfce3865. Source manifest

sha256:ce784ea5534ea0c70abe295bfc782f8d2186181d6603ca12b0eba6bf1782e464. Engine receipt

sha256:b36a35fac9671a641f6dadca15f42f257904ab2a56290630255acb5bf0dcb1d5 at receipts/engine/model_admitted/SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009-b36a35fac9671a64.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SC-CRITICAL-FIELDS

Measured, observed, adverse and boundary records:

- MAGSC-009: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-SC-CRITICAL-FIELDS: upper critical fields; Meissner state

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ebe4e11baae4baad2519295ededd66c4dfde119fe069e0f619157ddbac19c200; empirical validation

sha256:5e339be4fcbf9e19d3e5ea5b6f744ddff58f15d4411ecb1c42a699ac68b64b11; external validation

sha256:80b2e28785564a48ae08ec01340b1d1cf1e3c2e5040f596d849571e12b0b7ab7; isolation

sha256:711167fa150f9071c6c9cf9f25761287eb7e8a647e85ee771b43e11b43303bda; custody
sha256:bc8b12aa4055662ffe46c34f305e4f2e101c539c295b0140728f0e202889d4e1.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

162. Superconducting vortex matter and pinning

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-010

Claim: SFT-MAT-MAGSC-SC-VORTEX-PINNING-010

Question, law and forced result. Vortex matter is the exact partition of a complete vortex population into pinned and mobile carriers with lattice identity retained.

MAGSC-010 uniquely retains pinned-mobile-vortex-lattice-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-010 axes before target release.

Boundary: Every positive finite MAGSC-010 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- pinned-mobile-vortex-lattice-partition
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__pinned-mobile-vortex-lattice-partition__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:ad3b4b5c0e81d0244dc325b6fdaa4ffaccd4523292f68a238bfcc25c25169b92; generality

sha256:a483e0f8eeba218889f01f22eb5f998774f29aa8cc6ea82a9d33ebaacda2ce01.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:e9182b2eb9610fffbceaeff156a19b1383ecbd584225f54a566185751d4839ef6.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:7aed0ca8ff764183dbc6bcc7c616310836b74e1b54ac34b0523385c8e8e0bd47.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:4da7546519ebac3f2f82810bd6e2eb034322469cb400b2bea728ec7e9a2db158.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:086ae7c920b85ff2a0349770aab40d6362be5d4dd3ec749b119dcac9a9db0de9.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eeb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate

sha256:4839065682ebb901f9d3947f122f7eab290627bc20db72d3f8e89dc605a6b4ae. Derivation seal

sha256:427c15661cd78f2d92ad4456f38eca53d999f00893058f32d522f851a85ab2ed. Source manifest

sha256:ee6ee706a21a86110258e5e3aaee605903d46634563eaf08eace89f1291d0683. Engine receipt

sha256:298911f8bf33212f62692879d476eed5f5da9b31a636e259ac8b2146f435f1c3 at receipts/engine/model_admitted/SFT-MAT-MAGSC-SC-VORTEX-PINNING-010-298911f8bf33212f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SC-FLUX-LATTICE

Measured, observed, adverse and boundary records:

- MAGSC-010: 1 comparisons
- family sources: 12

- exact correspondence: True; omission rejected: True
- NIST-SC-FLUX-LATTICE: vortices; pin

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:f79b95835ee509b702adfc5cbee5f776e13d378ca4881d26e5e20c6948408b6a; empirical validation
 sha256:70d8284ed982dffa7848b295c69579ecc44c6cd3fd1a15b6b4a0b9868a897423; external validation
 sha256:ff48762cc7ee79f56316590b66e3304b10b54c058bbfd5cb87755987a3664856; isolation
 sha256:0165093ead15a14aeba1ef9cc828bc4868d68ad29e6b697ef84e4f20d9ea916e; custody
 sha256:0c8765eaae206c29b9a2bcf09b70e802b3c9f29d10629169a7fb9da310000a6.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

163. Superconducting coherence-length boundary

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-011

Claim: SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011

Question, law and forced result. Superconducting class retains exact coherence and penetration lengths, their rational relation and the type label.

MAGSC-011 uniquely retains coherence-penetration-type-boundary with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001

- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-SC-VORTEX-PINNING-010

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-011 axes before target release.

Boundary: Every positive finite MAGSC-011 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- coherence-penetration-type-boundary
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-magnetic-carrier__coherence-penetration-type-boundary__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:951f82eea137ca27d66b23d1d2e0615414a342ab664c2b9faa837c0948d4cc40; generality

sha256:07a82f7e9dc9febcb43dc05ed6f75b663856ee9be8efd85e743ac34c97e6411.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:a628169e36356207118e5ad47276f0a43fda19d8b1fb2f584593cffa2ce61f05.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:de8aaa571b207f98d4f080d28bbf4d37f429656de4c719b36725616f341457e4.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:02802d629d0b630058dd19b936f6ca36e64f17f8b3a640cbcafd87f3d590bf6e.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:4ce98c505b153b6ceabc9d1045a7324086667537da30b39050ffe8e4942d8535.

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eeb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate

sha256:3114f6cc61a7505f41796aa69ef3d5a7551fb27ac016a75516ab879863f18d90. Derivation seal

sha256:8fffb0312a70fc92ebbcdbbce535c3d7a8469eaa10fe447b3d77279ad85d9ebb. Source manifest

sha256:36ed71c57a46533eb01e04be35235fc9cab6b73fbbd792f0147a4c098dfa44a9. Engine receipt

sha256:9213d8e6499d8185965e0ecc3112214c416cc13414f39f895969cd989a849756 at receipts/engine/model_admitted/SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011-9213d8e6499d8185.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SC-CRITICAL-FIELDS

Measured, observed, adverse and boundary records:

- MAGSC-011: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-SC-CRITICAL-FIELDS: coherence length; penetration depth

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:7b3a4c864f8d298123f35fbfee3e6b045da183294bbccd796d70b8559c3649cf; empirical validation
 sha256:219ed64081efea68df168f16be8ea139152713a6ad995c0b7b7f11f179fab8c2; external validation
 sha256:16110461f5291a3a5632dd7dc1be5e85bb028e6e1c7aa37358547650a3e53181; isolation
 sha256:cfb6133575678b194570a32672fa23e4523bb60c20c066414f3fd81e2d3c2d41; custody
 sha256:8d54e96fa6b64bab83810689bc37d02e61dd9b4e32f4990a2636aebd13f92b08.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

164. Superfluid excitation and critical-flow ledger

Field: magnetism_spin_superconductivity_superfluidity

Obligation: SFT-MAT-OBL-MAGSC-012

Claim: SFT-MAT-MAGSC-SUPERFLUID-CRITICAL-FLOW-012

Question, law and forced result. Superfluid flow is an exact partition into persistent and excitation carriers under a retained critical-flow boundary.

MAGSC-012 uniquely retains persistent-excitation-critical-superfluid-flow with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001

- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight MAGSC-012 axes before target release.

Boundary: Every positive finite MAGSC-012 carrier with complete path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-magnetic-carrier
- persistent-excitation-critical-superfluid-flow
- complete-field-spin-vortex-flow-path
- specimen-method-field-temperature-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-magnetic-carrier__persistent-excitation-critical-superfluid-flow__complete-field-spin-vortex-flow-path__specimen-method-field-temperature-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.`

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:b9f2418c133370c732366af8146c9ee2075fafb6a99b08dbec797509dd63facf; generality sha256:dd14442f1f3ebf3640b6bd8f79d96042937277aad89a3dbd0c310502c94732e7.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:d71b9445bbb8b405632a7262667d5965cb84d5addbcc9f2f7ad36e73eae4cc8.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:cf810660242c8656c55430842c1249ce3b1b209af1cf7a86bb5eb9006f68589c.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:a2287a3feee39f07d3a53d1693511acb6d2b2eb6112f5c58b3f38845d9c9382f.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:66e2751a32ba329fdf3f2d470ec6c8b8d2beb2cddc536cf940645784577f473d.`

Independent reconstruction and engine admission. Independent implementation

sha256:59d25b05eeafb343c407cc23f50eb2c66f4533eadd8c37b59e72cbf6f00b628e0 regenerated the declared census and sole survivor; independent certificate
 sha256:9c81de7dee9c1156d6b82c4a1b9d77b307a78fe75c69125604229292b67d2847. Derivation seal
 sha256:17d399e74bfab27890b88ae985968f8b48e04a1af487d620e72a1c039f08f7ec. Source manifest
 sha256:b6419d777785c1f7c204ab0af275baf3b682a86b9f54f265b45b6b0ab697ff60. Engine receipt
 sha256:2e9849264a6199c9db15770d10fa9b5670e746881a1527f5b9c2d0e179a8a3ef at receipts/engine/model_admitted/SFT-MAT-MAGSC-SUPERFLUID-CRITICAL-FLOW-012-2e9849264a6199c9.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-SUPERFLUID-PERSISTENT-FLOW

Measured, observed, adverse and boundary records:

- MAGSC-012: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-SUPERFLUID-PERSISTENT-FLOW: persistent current; superfluidity

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:817dbfb2e023aec5ad3e1ea17fa54bda52d045b154c709d8cce89d5e3719b0a3; empirical validation
 sha256:667a9821caaa092a3e08df023dbf9a17cd5d90a7bc16f2e4eacdddb4dfcd478; external validation
 sha256:924a9c5d5dfd5295fe13f97e09513ada700ab68227b4cd6dca51677f2c247cc7; isolation
 sha256:44f4a3e85bbf62970eb0cdfcfcdc86226d9b2fb1811405880818a7485664b6c5; custody
 sha256:491dbf9d4f867ff1c8f821b9f2ac39c33e50d1fc2aa9ee0e3e9c5b6b2cc7dee4.

Prohibited inputs.

- no imported continuum magnetics, fitted response curve, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- opposed direction and absence remain held labels
- no external outcome selects a survivor
- all result classes retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Optics Photonics Polarization Electromagnetic Materials

165. Optical absorption and extinction

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-001

Claim: SFT-MAT-OPT-ABSORPTION-EXTINCTION-001

Question, law and forced result. Optical absorption and extinction retain the complete incident partition into transmitted, reflected, absorbed and scattered carriers with specimen identity and exact rational response parts.

OPT-001 uniquely retains incident-transmitted-reflected-absorbed-scattered-extinction-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-001 axes before target release.

Boundary: Every positive finite OPT-001 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- incident-transmitted-reflected-absorbed-scattered-extinction-partition
- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-optical-carrier__incident-transmitted-reflected-absorbed-scattered-extinction-partition__complete-optical-state-channel-path__specimen-method-geometry-spectrum-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d2f7e37999eafedfc5349a8094cda9d5eeec8fe465665d29d71cf8fe6a11c661; generality

sha256:6a10ed72c0dce521e8f75a9d109df95fc7abb6d224ad49bf78583c0d00545c77.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:066a3e2086e715859f467ea24dd478c6b1a3de0fb811b9356252bae0fab76807.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:e91c083b26373ba32248f5fb69096efbf32021c2e286382aa08ca38c6fa404bb.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:52009e44f81bf0d4970fae935d7d2b3495904aff150925710e23d82bfaadad78.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:41f8e12a9445cb481070f71039843e6e90f38637f478749a9296a1bd932a6978.

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate

sha256:ab9ac0dfbdb7ba09d112743b4bee6ac26358aead0305193314915af170aebdff. Derivation seal

sha256:b4481ce2dcfb4d6658cdf962665d85fdee3d531860b96d39a77e1b34ad5c2549. Source manifest

sha256:1ef42f09e4e0d6e533211f2841a97c8f3f070adbb6ab8e0df3ae23b35c6e0fdd. Engine receipt

sha256:78e7589d839869adfb80751b806c75f11edeeaa9633b1ccea0c35a467a384862 at receipts/engine/model_admitted/SFT-MAT-OPT-ABSORPTION-EXTINCTION-001-78e7589d839869ad.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SPECTROPHOTOMETRY

Measured, observed, adverse and boundary records:

- OPT-001: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-SPECTROPHOTOMETRY: transmittance or absorption properties; reflectance, transmittance, absorbance

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c10cfe53e14cccc55eb8555490eb267fd05d517261354ac9a5d8d774b6264a06; empirical validation

sha256:8d4a4718cb4371e81df7cb43c035a79aa999503004bfe2d61c9492ce90a8304c; external validation

sha256:a0576ad2ebac4a8253f6f60d509ca0aad8c9ef199d03d66be7474d08ef2535f6; isolation

sha256:9dd553760775571e408eab2c3212e4f0f2326fb5cf256d21077db4c394304f62; custody

sha256:0a9a09dcf1888115124c31c77bd885a7b4ea67af31df302a8d0187e906471cbb.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

166. Reflection and transmission ledger

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-002

Claim: SFT-MAT-OPT-REFLECTION-TRANSMISSION-002

Question, law and forced result. Reflection and transmission are exact positive parts of one incident carrier with retained specimen geometry and any internally retained optical support.

OPT-002 uniquely retains incident-reflection-transmission-retained-geometry-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-OPT-ABSORPTION-EXTINCTION-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-002 axes before target release.

Boundary: Every positive finite OPT-002 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- incident-reflection-transmission-retained-geometry-ledger
- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing

- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-optical-carrier__incident-reflection-transmission-retained-geometry-ledger__complete-optical-state-channel-path__specimen-method-geometry-spectrum-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:581126ad3fee90488fe1cd43bbf8f39b02ec6c963b2f089beb7c4ce1f09c809f; generality sha256:ab2a5f450a61287f24d37b22d1db69712e1ee47841b1ac48946858194c2b7157.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:68df4f875f11d0ecaf0ca470d4d7b8d136d578e6133e08b8a673e493cd51dd4e.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:2f9a8ad79b4ce09631c67d51c283ae144e3e2a2842a424886ee51a0f56539e7f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:cf88879096ea26702cabcc1bb4ef52f55cad2225e2ef2eb8469d96b78228fb5d.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:ee76e3ac7ba5a296fce05ceae49bf07730d2547eeb9bb4d0078d4a50e4b9540e.

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate sha256:d4df3fc997dac990913bd1209b65cdb6be32102a1405b66be9b3a4b474132358. Derivation seal sha256:fcbe1e00859e7eab0f8c746f4b27032731cf54ed2da055aa14fcd6b5b428fdf9. Source manifest sha256:4ba2d12c4c44855d995b8a095ed15747842bfd5208a1164e3233a68b0b868c82. Engine receipt sha256:e9433b2af8c835782e3ac6d25a87fc616a81b6f5d63ae2b327eb7090334ec361 at receipts/engine/model_admitted/SFT-MAT-OPT-REFLECTION-TRANSMISSION-002-e9433b2af8c83578.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SPECTROPHOTOMETRY
- NIST-OPTICAL-SCATTERING

Measured, observed, adverse and boundary records:

- OPT-002: 2 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-SPECTROPHOTOMETRY: quantitative measurement of the reflectance and transmittance; geometrical and spectral conditions of measurement
- NIST-OPTICAL-SCATTERING: Optical scattering from surfaces; Bidirectional Optical Scattering Facility

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:2d7d9988791abc9c9d20efd1e2e6b4d3e3f1d04f51374d6bb46d20208c348cea; empirical validation
 sha256:96b46d22f39f43c39ad5e9f563e5750f71bb46d55b66267c4ac2cac45364a418; external validation
 sha256:444b1c2d1a5031786346f219354bdcec6034c70e75737b1b23f1a5a8983dd030; isolation
 sha256:28cbb9a7c94d54ec35ae1c6a11f42f4a7ad221d10600baaf2c6d88ac306fc838; custody
 sha256:db0885713cad5ca3609e6e618e287ce8772b98283cb0569dcdcc3057588fe033.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

167. Luminescence and quantum-yield custody

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-003

Claim: SFT-MAT-OPT-LUMINESCENCE-YIELD-003

Question, law and forced result. Luminescence is an exact partition of absorbed excitations into emitted and nonradiative outcomes; quantum yield is the exact emitted part with excitation and emission identities retained.

OPT-003 uniquely retains absorbed-emitted-nonradiative-quantum-yield-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012

- SFT-MAT-OPT-REFLECTION-TRANSMISSION-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-003 axes before target release.

Boundary: Every positive finite OPT-003 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- absorbed-emitted-nonradiative-quantum-yield-partition
- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-optical-carrier__absorbed-emitted-nonradiative-quantum-yield-partition__complete-optical-state-channel-path__specimen-method-geometry-spectrum-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a7e649e847fb85592411cdf1f32e1494e1f99685487a24cdb88b32d527d76c4c; generality

sha256:ff934d3805cf44908e5fa86978206a20b5f76f73c47609529d7f1556e781c56f.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:d100e41ab1ca2d82cc5318ca7f26cce90900791ac260e3f41b16e85e5c09c70e.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:ffd205c34d70b37495621525581786e7b9ed0b26eb72eba13f4747acded51d49.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:2d081986cbae9408bce25007ba29ce402d2862c96c6615433d0ca809079af91f.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:9fb9e6ac04c27b7b21f3a349c48e8d262dd84526ebf6b7a7c17aa8ebb7ad67db.

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate

sha256:2e4b942ed10de0e3ec00f35455ca6877a8f3d04d9a511545be3f986e745281d0. Derivation seal

sha256:78703224ad8da91b0fde9a1b5fc8d6e5f1e04cb22688bfd69502c80080a78cf0. Source manifest

sha256:29fd7d1b5b87c69911bf94ee94391bd7d8a189a9027820229f56a8168d5aa5d8. Engine receipt

sha256:ab79045539d925089303124b8d0c6d3d7c7702e426c7f8b9395cc32e0f36b716 at receipts/engine/model_admitted/SFT-MAT-OPT-LUMINESCENCE-YIELD-003-ab79045539d92508.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FLUORESCENCE-RAMAN
- NIST-FLUORESCENCE-QUANTUM-YIELD-GUIDE

Measured, observed, adverse and boundary records:

- OPT-003: 2 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-FLUORESCENCE-RAMAN: Both fluorescence and Raman spectroscopy yield absolute signals; Fluorescence Intensity Correction Standards
- NIST-FLUORESCENCE-QUANTUM-YIELD-GUIDE: ratio of the number of molecules that fluoresce a photon to the number of molecules that absorb a photon from the excitation source; relative quantum yields are much more commonly measured than absolute

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:64923a8fef689b6209aa9b268633eba9ec95ed4d4fdcf90a3a474f04400f2f48; empirical validation
 sha256:4fc3f466969282147c46978c1bbc7954b249529a5a393a249f660fd07623bd9f; external validation
 sha256:c6b61bb5721da10764e35370f96563249ed69b4697c2c19fd2bd7d73b6f3dd93; isolation
 sha256:c4e3e493ac4a3bf3309dad6fcb8cd33c6c83bd58c5a518041cc0c60d38b5ff65; custody
 sha256:5eca4fff92f939fef8ee033ad947904607e9b20be7066c2cb205858e1f973016.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

168. Elastic and inelastic light scattering

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-004

Claim: SFT-MAT-OPT-LIGHT-SCATTERING-004

Question, law and forced result. Light scattering retains elastic, inelastic and unscattered channels as a complete incident partition with method and channel identity preserved.

OPT-004 uniquely retains elastic-inelastic-unscattered-channel-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-OPT-LUMINESCENCE-YIELD-003

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-004 axes before target release.

Boundary: Every positive finite OPT-004 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- elastic-inelastic-unscattered-channel-partition
- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-optical-carrier__elastic-inelastic-unscattered-channel-partition__complete-optical-state-channel-path__specimen-method-geometry-spectrum-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:c652233746d205e6ae19b9334a4fb894f49ac70b1afe639c69d6c5948f714c8e; generality

sha256:61484b56f5e535d01148353beb2377391c993fdee2f05b9df89e17489894865e.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:9e720ca6915f58477be1cb2e6a8608e6f03b8b5bc51d7252b881285bb1b9e0f3.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:e33d9d7143a09a36c52e91b32d1199e4bec18f7e8c9d312d14d8bf36a1738196.

- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:06d5b96b18d1fe7d37cbc5a2ea883153e75bac55a17637d075a83e62e3a23d3b.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:d2f4686a07f998cb7791d5cbdb8ec058066edf284549ea4bbb80970a1efc2b97.

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate
sha256:75b9b4e18c7a4e915d487868e38bfa1876295d0756cbae47112f7ec060268138. Derivation seal
sha256:9376763e743ed42ec22fdae0f4bad4077c2d73bda2e999ee00320ab2d72873c0. Source manifest
sha256:90d3f2f42be14335f9847377f37f875efc8434fef2ed644909ac6594bc84e334. Engine receipt
sha256:fda29881226f097c034019cc90dd00bcd1475e12806abad570a54fca70ec18b2 at receipts/engine/model_admitted/SFT-MAT-OPT-LIGHT-SCATTERING-004-fda29881226f097c.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-OPTICAL-SCATTERING
- NIST-FLUORESCENCE-RAMAN

Measured, observed, adverse and boundary records:

- OPT-004: 2 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-OPTICAL-SCATTERING: Optical scattering from surfaces; Optical Grating Scatterometry
- NIST-FLUORESCENCE-RAMAN: fluorescence and Raman spectroscopy; Raman Intensity Correction Standards

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:1e696cdb3c01d12c235776a687beb1990b10f17c1615e3300e12d1e3f4a45f63; empirical validation
sha256:f663d9de922f37b5df90ae5293d505be6c66e9c4de4007058f0756eb95f003ea; external validation
sha256:cf6a3d928394fa47db6c325087a14433d2d64eca4a1652b3580c3077224da2eb; isolation
sha256:f710dfc0582d838848523220684c822234eb74ea2a3c9e1c1e1ee7da3ad09a40; custody
sha256:9c189c93ced560c515f1a4ef7df03f0ce74a0351656358de64c4db5516ab66fd.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

169. Birefringence and optical anisotropy

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-005

Claim: SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005

Question, law and forced result. Optical anisotropy retains two positive response carriers, their distinct polarizations and axis, exact ratio and positive gap magnitude with ordering as a held label.

OPT-005 uniquely retains two-polarization-axis-ratio-positive-gap-anisotropy with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-OPT-LIGHT-SCATTERING-004

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-005 axes before target release.

Boundary: Every positive finite OPT-005 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- two-polarization-axis-ratio-positive-gap-anisotropy
- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-optical-carrier__two-polarization-axis-ratio-positive-gap-anisotropy__complete-optical-state-channel-path__specimen-method-geometry-spectrum

-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:bc418bc979e48f3ea7a113691cffffe1268a2531b0b314085a7bfe4d621bbc630; generality

sha256:19c2ee5ea697cd70bfe639b97de03274d243db479cd54c98780ff3d37de4c2b8.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:f32d19aafd2343879af9fd90a15310d55b177d00efb6e4ed2f4ecb2c1c972a8d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:0c8004fe6d5771b06c79a225a70e4203b568d9c6da68f4b41f29ec0e8da75646.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:74347828798bd97ba8f0c416867ea342a12e6f1733a8656dd44b0216b69c103a.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:a826d0b0b5d4053e88ff7a2bcd795674924d00281f7537a611d8432ed25af999.

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate

sha256:85f8bb1673476493fa0ecc55fa0943cefbec48208d6133abfbf9766bb1c75c00. Derivation seal

sha256:14f322eba22b223e29c361dd2bc4fe4b167c283c5cdbc17bf1dc26e0ae168224. Source manifest

sha256:113a5110d3f394918c6f6e47c911b724c120bcb18c1708f24d9d65a398d3e29. Engine receipt

sha256:77178a25bcf4f585bba258b8416f8b5e57b722e3b0b50b5ff545a44fc1d9f0d8 at receipts/engine/model_admitted/SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005-77178a25bcf4f585.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-BIREFRINGENCE

Measured, observed, adverse and boundary records:

- OPT-005: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-BIREFRINGENCE: crystal structure asymmetry; two polarizations

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:afa5520ac3fd70e1644423bd64ef4d5318648936651fdcf36c02affbed31318b; empirical validation

sha256:9a8538537ebcf9bfff4c5b108a6f892df93006f95555f3a5c86471682ecf5fa9e; external validation

sha256:8c770a7dc8e15ea37b9313f78dc52a51b091ad46a585a2ef87a17e42a41af13d; isolation

sha256:25b4d2ff59467cdcb6bf841a99ae1e67c41e8db49f7c01491911e2ee8733e1e7; custody

sha256:5aa91d0994c1aa2c89bbcfdd7b40132c0a83f57692bf2d9ac0ba7d2d0fe3829a.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude

- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

170. Nonlinear optical mixing response

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-006

Claim: SFT-MAT-OPT-NONLINEAR-MIXING-006

Question, law and forced result. Nonlinear optical mixing retains every positive input, output, polarization and complete sum, positive-difference or harmonic relation without importing a continuum field equation.

OPT-006 uniquely retains input-output-polarization-sum-difference-harmonic-mixing with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-006 axes before target release.

Boundary: Every positive finite OPT-006 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- input-output-polarization-sum-difference-harmonic-mixing

- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-optical-carrier__input-output-polarization-sum-difference-harmonic-mixing__complete-optical-state-channel-path__specimen-method-geometry-spectrum-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:62937a4441e0061543b0b51f9f763fdce95eeaa8cc4da4d16f8e5542aa38b295; generality
sha256:14882a61346d85851724cda82d62d6e30b77b9f260033b9ca603c193191d663e.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:fe3248aac7b835aed757d280593fb0e1737f0297623f57d5033cfaff5ff4a6db.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:396e1b964e4e1385f986008944dd66d6a016d60716ebcb9fb3f47491db946f73.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:3ad54280b26667a7fe17e14d711dd13ef1080ea279c68c56b52acf4d12f6710a.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:03ff14494eec7f83cfcd0e515f6918e10c0978ba93118249b284110f915133fa.

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate

sha256:bd1f01f94708d2216b8a75111c2ae395b7d575b7397326f3153f65b96f287b65. Derivation seal

sha256:3b3c603cd1b04cd62710fb403b42e953154207332b8691c4db79396d9e89b740. Source manifest

sha256:a03b75af135f54c97618bf135b11596686294f1392364a1f6c7ddffc1e6011f3. Engine receipt

sha256:8beecee8dcc03c64d1efe44d3a8224f0935eea80704812a0f76a2abc60f99684 at receipts/engine/model_admitted/SFT-MAT-OPT-NONLINEAR-MIXING-006-8beecee8dcc03c64.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-NONLINEAR-MIXING

Measured, observed, adverse and boundary records:

- OPT-006: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-NONLINEAR-MIXING: nonlinear optical mixing; three frequencies

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:8a00a2d775af26ab540ade42a666ca36b068e30e738596f81bb6abbd9f503013; empirical validation
 sha256:300ece65c0d49f0be87c5144f7e9fe5e89d9c0f674bf7ccaebd6725529580399; external validation
 sha256:656a14bbf6e50e8739d2b82dbf27496ae32138c329685f532c53478980749850; isolation
 sha256:c08b675f9e3d0ecb7ab6db4cc044f4a5026c7dfc10d5d38d93968339bf94050e; custody
 sha256:24ebc2a08aa130a9498e82a322a5b36fbf17c0a669fa9d8a89380664247d78ff.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

171. Waveguide confinement and loss

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-007

Claim: SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007

Question, law and forced result. Waveguide propagation is the exact partition of incident support into guided and lost carriers along a retained core-cladding path.

OPT-007 uniquely retains incident-guided-lost-core-cladding-path-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012

- SFT-MAT-OPT-NONLINEAR-MIXING-006

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-007 axes before target release.

Boundary: Every positive finite OPT-007 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- incident-guided-lost-core-cladding-path-partition
- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-optical-carrier__incident-guided-lost-core-cladding-path-partition__complete-optical-state-channel-path__specimen-method-geometry-spectrum-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:592d9fb47a5e67caabf500b6891d771dc0a4b7d002a123f890addc8e042d5025; generality

sha256:3570c9422763b36c23e0d3dde8f8a7dad08c698c7331d95f57a2c8a721c0c5a.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:558ab3d325387e41eee9088e7fac8b4c9fae5f0566493bb39fd44cab37378cb2.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:25664356c4eb0308cae2ba9deedbc4e45a358a04b04da5cc330572a04084e63.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:98d7fd8a2f479e1035e9a9abd5a626803da933738e6e027bcc681a2360575b7b.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:0ca5af5433f371e19f0170de3590de663833415c76fb1f775814273f5531d50f.

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate

sha256:feab5d632c85e0312e24c7ac5ccfbdaef226ec90516bd83cd2b073af14329f21. Derivation seal

sha256:262bffade27331d0d16ce173a681f36a8da151fffc7aab14454e5e34bc2b74f8. Source manifest

sha256:5853b382118d075f4e4d83e4398709f6b585a9ac250d0e3ecfba55ca405266e. Engine receipt

sha256:0612df0cfef508bcb99f1d101daf475c8e411d2927018fd33c68f4d3e9caf842 at receipts/engine/model_admitted/SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007-0612df0cfef508bc.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.

Comparison passed: true.

Source identities:

- NIST-WAVEGUIDE-LOSS

Measured, observed, adverse and boundary records:

- OPT-007: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-WAVEGUIDE-LOSS: single-mode waveguide; linear device losses

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0a5a1b09149c71fc62f335cfd8b35f40bd3aef87ccbea711d85a57837effaf94; empirical validation
 sha256:af0e714e44ef8cdecad4dc7001f551327ad5d7fbfc841aba5ac21f998d72b53b; external validation
 sha256:b4f20a4ae8ad8ed01a7ef2c4c46b8f2007a6038ba8c6fa73f45e84d372383b03; isolation
 sha256:a4baaa8988416f461d84c050c6cd9b8d9da931703956a3b7ea7fbafa8a345a8d; custody
 sha256:c2e5fc2eb0be4ed76d3b283c4318e437d505e8fbec54d7998d95d4a6d03d66a9.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

172. Photonic-band-gap and defect mode

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-008

Claim: SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008

Question, law and forced result. A photonic gap is an ordered positive excluded interval in periodic support; a retained defect generates an exact confined mode with both boundary margins preserved.

OPT-008 uniquely retains ordered-gap-periodic-support-confined-defect-mode with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003

- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-008 axes before target release.

Boundary: Every positive finite OPT-008 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- ordered-gap-periodic-support-confined-defect-mode
- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-optical-carrier__ordered-gap-periodic-support-confined-defect-mode__complete-optical-state-channel-path__specimen-method-geometry-spectrum-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:07d4e27de96604d3fc3b514c45965a718fc829dfdeb535a15c5357f9cd31afc4; generality sha256:b4524d82321f54c11d1175983be8fbf93dfa91480d8549b3710ae95cdbc38080.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:1e5a986077d76643566709d0a9c711591b7d07cc53a28d625594610b82c74c6a.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:e414e8bbe657cfa8c39d06c7ee15850c787f7faf7d2a40347ca57f16c0e8720.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:3faf8169658aad25c81ef9290e4273a1b9f0d971268ecec1edab7dcfc8ef546.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:21630055b1e717e07d08bacb074a8191b4b7553e0e8fa0285f5a1d248453f107.`

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate
 sha256:4f3be45c892e34cf84a9953a5bbf2a4e9088b0339eaa862d06295d10a2dc311. Derivation seal
 sha256:895776bea924bedfadb8aeb10cf5a904546907e2a26797bb04827af714f2f6f4. Source manifest
 sha256:d5e37a14863415e4b20d895df7de08d381e9e99392ebf7c6b24d617c31c98a56. Engine receipt
 sha256:a6461f40cf7c98b313a2375e2afe77729a9ed808be09e026934bade054a4ba5b at receipts/engine/model_admitted/SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008-a6461f40cf7c98b3.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-PHOTONIC-BANDGAP

Measured, observed, adverse and boundary records:

- OPT-008: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-PHOTONIC-BANDGAP: forbids light propagation at others; Photonic bandgap microcombs

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:b2fc9eaff3e6d4d71e2a6d9c3cde512d2cc587584827ec6f4887bf9fa5017c8b; empirical validation
 sha256:6eed9620e5f11cf8faad18e803423011d8b0c5f1477a2e4bb78155969227821e; external validation
 sha256:71e5f31eacb68fe6932e2810965ddb7da46fc3d741de6766fa42a3052390bba4; isolation
 sha256:8fd0c6c1667c0d045f63329e971afebf70d35c9be2a3f6454b9c271f6c525c9f; custody
 sha256:bbea1b34dd921c467cfab6056e8221c1abced1f0b1a3d58405d14027d220d9ec.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

173. Plasmonic collective response

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-009

Claim: SFT-MAT-OPT-PLASMONIC-RESPONSE-009

Question, law and forced result. Plasmonic response is the exact partition of interface excitation into a retained collective mode and dissipated support with mode and interface identity preserved.

OPT-009 uniquely retains interface-collective-dissipated-plasmonic-mode-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-009 axes before target release.

Boundary: Every positive finite OPT-009 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- interface-collective-dissipated-plasmonic-mode-partition
- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-optical-carrier__interface-collective-dissipated-plasmonic-mode-partition__complete-optical-state-channel-path__specimen-method-geometry-spectrum-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:e32bd6dd42640a522e911e9edc6fa60cc2d3bd7969ae68402dc3b422dcd42baf; generality sha256:aec2ec921517535b367f4612c3a6ff89bc5c9f93689a0bdc7cf91a2f9f3c2ad7.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:575b7e91867e84394f23f4fbc049f368f75b7582d5af3d1fdd5acf9f08779ab5.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:1825f63fe080da62ec19acaa474a344df1d1a2cc5a0657317f99a21fdc625e84.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:1878f60738bcc7f8dcaded0c69f61c37e71c79f198bdadf5af627a9b090602d7.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ec9de528b9a51846bda1c92fffdea492a4d372e25de6d0297ca9bea339e7b1e4.

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate

sha256:abde411e9da0935052f96a0ba50114f5a1a6205c5910f4bf50f43421ecbce8a9. Derivation seal

sha256:f21f8849654eb06aaf9e263ac267b8e750223a6c09db51168d5ad4749081204d. Source manifest

sha256:b8fb032502f4f7ecd8d1ea48ed452ced91eb2a511a47248ce819176e52b2039c. Engine receipt

sha256:1bdcbl8e974e7dae69c290e6242c01465f6c2dd8674443397237b45dcd22145d at receipts/engine/model_admitted/SFT-MAT-OPT-PLASMONIC-RESPONSE-009-1bdcbl8e974e7dae.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-PLASMONIC-MODES

Measured, observed, adverse and boundary records:

- OPT-009: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-PLASMONIC-MODES: collective oscillation of conduction electrons; nanoscale volumes of matter

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:91a751d808d395db9b92b34dfa934cbb9bef535f923e5a690e9a725ede761655; empirical validation

sha256:9b7531851813d007a9f849ef61b1066519a4419241ba08fc52c806ed8267dfbb; external validation

sha256:eeb40573e6dcf4f5fbb99c97a32774e5ec2600da8cf953a8066e7d79f6f66f11; isolation

sha256:787bad4f40c12ca6fa35d00de692354885af439fe083f304014b1d58dadbd5a5; custody

sha256:9643299e3541c332c65959a512cb0ca1aa8f01f4e0d5cc03e837b97550c254aa.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

174. Exciton generation, transport and recombination

Field: optics_photonics_polarization_electromagnetic_materials

Obligation: SFT-MAT-OBL-OPT-010

Claim: SFT-MAT-OPT-EXCITON-DYNAMICS-010

Question, law and forced result. Exciton dynamics retain generated, transported, recombined and surviving carriers and the complete generation-transport-terminal history.

OPT-010 uniquely retains generated-transported-recombined-retained-exciton-history with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-OPT-PLASMONIC-RESPONSE-009

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight OPT-010 axes before target release.

Boundary: Every positive finite OPT-010 carrier with complete state, channel, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-optical-carrier
- generated-transported-recombined-retained-exciton-history
- complete-optical-state-channel-path
- specimen-method-geometry-spectrum-polarization-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-optical-carrier__generated-transported-recombined-retained-exciton-history__complete-optical-state-channel-path__specimen-method-geometry-spectrum-polarization-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:091db1eacd84c3092de62e80573f037602335a03b4b936c72a0aa0c89a4f5b5f; generality
 sha256:d9e1a9d1224720e63d74a3f2d4ff6e32155daa1412ee220294ba8fe069dde436.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:149512407c5db6064d043c447fa117fa3db82fe60af1c8a9fc9fbea062efc737.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:f70772d1e8db12416765d5ccfcdclca90d487a333be22d20a55fd96b6f8a485a.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:08e4df87d43047f0dbd08dee0a9e762d4d8be16e63317f172473ed2d535975da.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:a0f9bae86f49418bbb66adc179b08d500b4d1c7fe377c04377ad92ad43349db3.

Independent reconstruction and engine admission. Independent implementation

sha256:5151cbabdec30ef6cbf4a76143df92fe72cbaba778a9b338dcabdb981532a04c regenerated the declared census and sole survivor; independent certificate
 sha256:3d915c72fb7e8b2c2814b7fa56f39d588e07e968f9135ad1159c02c8f08758a6. Derivation seal
 sha256:b4221805e77730c805731a85e4b386302fe0d0835335aed486cf01b15c7f31dd. Source manifest
 sha256:eede785e4e0cc0f8ce5d32e7fad772af657bee66cfa48eaf0739274abdaa4f12. Engine receipt
 sha256:03749083f3b44d3ca6eddddcfc97b0a75c07b96e0c3f003cde9b8e661bda217b at receipts/engine/model_admitted/SFT-MAT-OPT-EXCITON-DYNAMICS-010-03749083f3b44d3c.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-EXCITON-DYNAMICS
- NIST-CARRIER-DYNAMICS

Measured, observed, adverse and boundary records:

- OPT-010: 2 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-EXCITON-DYNAMICS: exciton population decay; radiative and non-radiative recombination
- NIST-CARRIER-DYNAMICS: initially generated excitons; charge recombination

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:d4401938e4a03a38c0b72023cd33778ba7ac7c565e77ad974e7dc6845e7b8441; empirical validation
 sha256:6e896857d9ffe5b1528bf2add3f479cc1035e6b45c58e3ba456cca68d0b51b30; external validation
 sha256:5ae2fff9b308f5f0e8f5bc9a38fed7b87c6880cc574374ba9b21db041c0ccd9f; isolation

sha256:c78fc0a7ad6982f901dfe3fffc666b8d2cf72b6ca65f134436eb7a90813bc216; custody
 sha256:22dfb3c6d8db0425752ed8bf48181a4ef4acfca3a4ab233c617541564a18585c.

Prohibited inputs.

- no imported continuum optical equation, fitted constitutive response, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- orientation, polarization, channel and structural absence remain held labels
- no external outcome selects a survivor
- all result classes and the initial source limitation remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Metals Alloys Ceramics Glasses Polymers Composites

175. Solid-solution and alloy phase organisation

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-001

Claim: SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001

Question, law and forced result. An alloy retains every constituent identity and the complete exact phase partition; solid-solution and compound phase labels are observational distinctions, never imported selectors.

CLASS-001 uniquely retains constituent-phase-label-exact-alloy-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001

- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-001 axes before target release.

Boundary: Every positive finite CLASS-001 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- constituent-phase-label-exact-alloy-partition
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__constituent-phase-label-exact-alloy-partition__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:22d7187f2874810934b5d218f56589823535204621f2b86ef75582a05bc4cc84; generality
sha256:f653275606b47b6ce9565971bb20f4a3ea3bcfaa03ee1353c6f864b7d917c5e7.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:bb1f45a79ee1be9dda782b8bd38da2af225af35737e8012598276c9e8b243f98.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:8049837f1534162f79ad5b89205db79c3722e74643a5faa52916f8de94d8246a.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:ca0e465e754dc189018f8031b6d1f5c9a8de5f5205d4e4416379b4a5ea1c899f.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:03946f9c99417b003fc2036fdbb7b7335d0edc1d14ad6eff277952ab429b5d1a.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:75bec1b77293e9984921734d8bd0acea7cc5244612fbbdccc623119378983a55. Derivation seal

sha256:743b46356bf2de516528ec1b3a865873431d438cf4f01ab64302278b3467eb5f. Source manifest

sha256:8f21d9162aaa4bb448b74ff403ad7084c9a4e1cfe3fa876fed77e727a1bdb6ad. Engine receipt

sha256:f4a93b8b79815bde21f22600d14a65da1a6105808d3cf945dcd5f59a5b13ddba at receipts/engine/model_admitted/SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001-f4a93b8b79815bde.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HEA-PHASE

Measured, observed, adverse and boundary records:

- CLASS-001: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-HEA-PHASE: ideal and non-ideal solid solution phases; solid solution phase and line compounds

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:da865a009513d477616a2ffc5566d14722d0b67bd4e88498df08098f84588dc3; empirical validation
sha256:76935a0358d3aacb0f0b0f42d837214f93573eb16613718574bdfe2d93f2575c; external validation
sha256:a50ee37dfbbb1eb0e476df4a5cda4aac7f4778c68f972cb41c14547f74b39dcb; isolation
sha256:93aa8fa2b44268729a7dde02699c0942a2e0edf4d906349639260eff43b78597; custody
sha256:0ceb14a20927b4bc73143cfe26698ffe22d5f1678ab430ba6e0474464db0e892.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

176. Ordered intermetallic compound organisation

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-002

Claim: SFT-MAT-CLASS-INTERMETALLIC-ORDER-002

Question, law and forced result. An ordered intermetallic retains compound identity, every distinguishable site, its occupant count and the exact site partition.

CLASS-002 uniquely retains ordered-site-occupancy-compound-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001

- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-002 axes before target release.

Boundary: Every positive finite CLASS-002 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- ordered-site-occupancy-compound-partition
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__ordered-site-occupancy-compound-partition__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a777fff849c1fd07e7cdf771a94061ef7bd75dbee661c3f2fd7ba0d1da1de291; generality

sha256:f197b72cae18d71928f237ab80a1eb8e1bc72ac2343d50833a7fdce874ad8fce.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:f1831c9bfff931837ee438999cf620af61657c8d556c9f2bd0fea819b05712aab.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:f8383da4f993b5bcd6dd4d750b11bd8240b38695e452d4fcd61df710b9c731d.

- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:796cd2a7652f335a221eb1a4dc29c9f029b3281703d466b052dd61dfcdc56a9f.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:25bb47d4b51abff56a004069136efba7eb908b68beeedd33033d90e82856c15b.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate
sha256:bdcb22b8d3f2978dd2b3f8f8688b1751d83974a299e69bff25f32eb19e01b661. Derivation seal
sha256:e5b5b89f96fffc9e228c72eb8cd05ae8a4b4e4f547df53834a560de5cfe52552e. Source manifest
sha256:c9c7f8d7af7eb9889c76d367e881ebf5723c0ed35d9bc553985e828150f4536b. Engine receipt
sha256:1f77f6063469a0c456146a5807612afb55c10ce801620968ec04cdcf6d32e28f at receipts/engine/model_admitted/SFT-MAT-CLASS-INTERMETALLIC-ORDER-002-1f77f6063469a0c4.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HEA-INTERMETALLIC
- NIST-HEA-PHASE

Measured, observed, adverse and boundary records:

- CLASS-002: 2 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-HEA-INTERMETALLIC: Single-Phase L10-Ordered High Entropy Thin Films; High Magnetic Anisotropy
- NIST-HEA-PHASE: solid solution phase and line compounds; multicomponent phase diagrams

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:3e1351c936c6d48a5777fa85df9ef59a60726d5ed18b171df368154b0720bf63; empirical validation
sha256:ad694c282b41074ce2e233ab223ea21da8d6269dd300081757c1ecf178c859d5; external validation
sha256:1687bbf62e9da26d9624a1ccb2e7a52de66b097903e26d4c3aa350f48ae984ad; isolation
sha256:e6a3062d2a3145428c026f520519ca1a87ac13e2112e11a330562ceaa0a40ab7; custody
sha256:b077d7a4c82b0c45488ce718b6ec872c4f51b140f59f410964d2626a8703a9b1.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

177. Compositionally complex alloy boundary

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-003

Claim: SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003

Question, law and forced result. A compositionally complex alloy retains every distinct component, its exact positive part and every resulting phase label; no named entropy threshold or selected phase is imported.

CLASS-003 uniquely retains complete-component-phase-complex-alloy-boundary with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-INTERMETALLIC-ORDER-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-003 axes before target release.

Boundary: Every positive finite CLASS-003 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- complete-component-phase-complex-alloy-boundary
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure

- no-extra-rule

Unique survivor: complete-positive-material-carrier__complete-component-phase-complex-alloy-boundary__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:6d49d00eb8baca980b83a782e794c0cf28ea188e76b58e85eef03d61944072f; generality

sha256:035d5c17f62cf914ed6375231e24d22bee612f1ce12ec06280165b87c2c04ab1.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:f4e51a31c3b480819d34b3beec78551e58ae564f0b39df391831e2d808ebde94.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:66de86ba078049b4d9587266254510630370fef35981d86f568161daed564b5e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:76b31813622b8f1f6e966c29bb937ea351ae0175e3816f3e7be839318e0cfb9f.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:93eb6a06b5f2ee3865fef64a7fc6d1a73df1a9dbbc786593db6346b2fe773653.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:08bbe8a71e56af425886749fae3ecab782833fbfb2a86ad2ee195ffd2d327a80. Derivation seal

sha256:34fd1f5e5583e5c925fffb4b5cef5285abfcddc274d161f25321cd11aacca0842f. Source manifest

sha256:1d2f3abe72015a482e5964ce065c603c3ca70a163587021a2ce7473b78e35ee5. Engine receipt

sha256:f6044a1580b64db186e3620881616f7177d0531ce8dfaa5254423642c16ce3a3 at receipts/engine/model_admitted/SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003-f6044a1580b64db1.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.

Comparison passed: true.

Source identities:

- NIST-HEA-PHASE
- NIST-HEA-INTERMETALLIC

Measured, observed, adverse and boundary records:

- CLASS-003: 2 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-HEA-PHASE: entropy of mixing; multicomponent phase diagrams
- NIST-HEA-INTERMETALLIC: High Entropy Thin Films; L10-Ordered

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c2912b760d75265b52625c2016eb98dd4993a63cd380c16b6e3d99c957a48d7a; empirical validation

sha256:fd8eac6e31681062774a7d46fe534a1c3a926810d61482c05c1911e3d77fff64; external validation

sha256:6b750d6069c125f16ff9ab0b0a1c824e62baa85eb93cf8b2eb3aa38f1edfc382; isolation

sha256:b836b51a06ae54bf230e9c628c587abfa4eda8071f9f784f16d6e67b62a00fb8; custody
 sha256:edd0729ae8a4633f07902576a222bfbe155b2265b4874d16ed317871a3fcbe2a.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

178. Refractory and ultra-high-temperature material class

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-004

Claim: SFT-MAT-CLASS-REFRACTORY-UHT-004

Question, law and forced result. A refractory or ultra-high-temperature class is a specimen-bound survival history retaining the service boundary, survived interval and every retained phase without a fitted temperature cutoff.

CLASS-004 uniquely retains service-survival-retained-phase-refractory-history with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001

- SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-004 axes before target release.

Boundary: Every positive finite CLASS-004 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- service-survival-retained-phase-refractory-history
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__service-survival-retained-phase-refractory-history__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:7bc0c1c0263857671a603f30ef875de66a00ad3d8de90017cd1cf4ac150c4c49; generality

sha256:9d92224da6887c66864dcfd1a7943d1b0a409abe74e87e5ae8245286f62d4f9e.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:4b787b424e3aeab99bcc61e45504470a2459a05344682dd71933ca77f548d197.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:c45e438018803c91f176a2e3fba330c43e906d0f6eeea0f213399c964ca624be.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:32604fccd6af3149be4215bcd2463fbc8e41e5d2f9d827741facbbf1daf4bd7.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:e0dc889f5e1dbc5f5d0ab07d57dac7451b6816d8bb4fced2c67a3d0e4ef8932f.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:831b83bdc5a4aabcdf8b2417292d8e55e7ab60583809e4ade899bfe480cccc3. Derivation seal

sha256:40f97f5e28558f83b0423ae84a785d1046224940ad3610349fad803059995677. Source manifest

sha256:2f1a3b6d38fb5953f51d4574e478ee640c31913e6f3450c85a0c4665bc7c9f1b. Engine receipt

sha256:ba549f60fab7b88361b73e97ca179007f76e86adb8339ecbe112925545a3d2ca at receipts/engine/model_admitted/SFT-MAT-CLASS-REFRACTORY-UHT-004-ba549f60fab7b883.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HIGH-TEMP-CERAMICS

Measured, observed, adverse and boundary records:

- CLASS-004: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-HIGH-TEMP-CERAMICS: high-temperature ceramics and cermets; elasticity and density at room temperature

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a12c10c80fcf423452aa3bc3b20d99d05ce8b8207a2c6a52b70cc6a7968e5b73; empirical validation
 sha256:dacaa577bd9521796ac3643f7876b847a133c35015397b152a8e599aa2458175; external validation
 sha256:ee17a35d35a98094e8a8936df195be5b124f4f21a4770ac0af49280027e34e0f; isolation
 sha256:3201667c529d263eb6959c73908e00c3c9307f24c8f913225b79da08023a6568; custody
 sha256:5f89b14d3fe8df611752342a3eb560df30332715f1aaf2f99f955e32ceb0eef0.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

179. Cementitious and concrete composite organisation

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-005

Claim: SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005

Question, law and forced result. A cementitious composite retains binder, aggregate, pore and hydration-product supports as exact positive parts of one specimen record.

CLASS-005 uniquely retains binder-aggregate-pore-hydration-composite-organization with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001

- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-REFRACTORY-UHT-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-005 axes before target release.

Boundary: Every positive finite CLASS-005 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- binder-aggregate-pore-hydration-composite-organization
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__binder-aggregate-pore-hydration-composite-organization__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:4bcebac53e28da40082ef8b0adb0dbdd60db62b6f3f12a1577c5697833d64aee; generality
sha256:563a4c2486d3eb3bc06e15ba4734f3c0b4c270ab854bbf49b8e8955df78d0068.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:d1bee91da978e0a61ceba300af643d07a4d4f36eed929d0a76401fa54c513062.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:1a8c4166426e690bc59f0a90ce155d3a2da57d9c5d8b0a25bafccd05a80212e3.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:8f679e62fda9c71f565512b99750ec44e57ee96c827402914fce3cc949ae91a3.

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:775c2c5ab16a4975b90b44b169e9fddcd51515d85d2779502f4f6763af3067f3.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:92bc59446a57d0a4c9171da98a9ec795815e4e6a23640aacd29f5468a64be213. Derivation seal

sha256:8d7672c188aa79145bd360669c37cd2e7bda4a1b72d4620f912dfd5db95b5683. Source manifest

sha256:c8edb2ce58a69f7c415ce6153ad480109df5b2154f4b38da62ddb0f83c210c84. Engine receipt

sha256:6e559584d473ceb17f1c5e0673555a5644dac42b49005f243e91d5d978f7d342 at receipts/engine/model_admitted/SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005-6e559584d473ceb1.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CEMENT-AM

Measured, observed, adverse and boundary records:

- CLASS-005: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-CEMENT-AM: fluid-to-solid transition; hydration products which act to link particles together creating a porous microstructure

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e4aec9fce792c5457f96e1d796289eec1b34d16ce9f737d365678cae90cde29a; empirical validation

sha256:e1a64e0fb408edd572a6b022cdcf21d73016545ec662468cdb881d9ebbb9bbd7; external validation

sha256:19f06366fc6178a71da4675ec6c66f868f50105859b1a264069ec49d81849f22; isolation

sha256:255cb58d57e1ebda75cb94c255030476d2431eac4e7287f15bd1b197ba73cc9e; custody

sha256:a944171531f8adc527db715073316be66d54e8ec2c487a7de15ca0956d9186cc.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

180. Fibre-reinforced composite load transfer

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-006

Claim: SFT-MAT-CLASS-FIBRE-REINFORCED-006

Question, law and forced result. Fibre reinforcement retains the applied load partition across fibre, matrix and interface together with fibre orientation.

CLASS-006 uniquely retains fibre-matrix-interface-load-orientation-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-006 axes before target release.

Boundary: Every positive finite CLASS-006 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- fibre-matrix-interface-load-orientation-partition
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-material-carrier__fibre-matrix-interface-load-orientation-partition__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.`

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:767baaacb6d0e54b420b096e203fedf5934e4cdf44487fffc63cc6fe5f416553; generality
 sha256:a284b6b34649fc21c1e8c2b5b581c2fb21f88e47edaf24ee86056cd6c6466ceb.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:acd552ecb23010d4fbb8e8f99e3591e276cbc4d9ff914b05c209b6c701ce89785.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:8e1b356e9f85fe5a0c263c9004e8160ac2c6c5d68a4cd2e38b26f6309c289c40.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:8234b948672762b92fcdc24d26f32e1b3a894f66dc607eaa31853d5486f043e.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:b91f6e476f75d5604a0847655452912df3b8edcf0a65699cf8f6ce8ac03441d8.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate
 sha256:fac9482fc3c7747cacd96fe79dbc35bd2d6579fe84e9fa9f5844da2b71d47b3e. Derivation seal
 sha256:fa04a3687ce728c87012635ddc00e4b2d4ef9664e44ecae78e324cfef72c651e. Source manifest
 sha256:5ab9da7c94375fbeb14ac1a37d631f1c0ad03d83561a3d36f821ad678276648. Engine receipt
 sha256:de902521804f45f5fe36e980d229e69a83923ed65294f07bf163a679b3160e62 at receipts/engine/model_admitted/SFT-MAT-CLASS-FIBRE-REINFORCED-006-de902521804f45f5.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-ADVANCED-COMPOSITES

Measured, observed, adverse and boundary records:

- CLASS-006: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-ADVANCED-COMPOSITES: transfer loads from the weaker polymer phase; stronger the interface, the greater the load transfer function

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:b062bbd3bc262873f47bb1972575284a73f6566a1e00c167a63214b745bc59d8; empirical validation
 sha256:3c2af25da479528e711279b7133af27707367446f9cf8a7ceea6500d350570; external validation
 sha256:6b1f1060b07ac717293bc49f7cc33760182169e86f588cc2d8813cb288ee4d24; isolation
 sha256:ccb15020dc8793be966cb452bdc6789ae44eeadcc4243cf82e3640b1795106e0; custody
 sha256:614fcd485bc92a1b7a1c2bdc191e72985969a19b5774a0df55f4d65229ee6906.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels

- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

181. Particle-reinforced composite load transfer

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-007

Claim: SFT-MAT-CLASS-PARTICLE-REINFORCED-007

Question, law and forced result. Particle reinforcement retains the applied load partition across particles, matrix and interface together with particle distribution.

CLASS-007 uniquely retains particle-matrix-interface-load-distribution-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-FIBRE-REINFORCED-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-007 axes before target release.

Boundary: Every positive finite CLASS-007 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- particle-matrix-interface-load-distribution-partition
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__particle-matrix-interface-load-distribution-partition__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:18179d709aca1dc129bf592d135b01a0cd91e46a64fa8b6314df20d0a1462b28; generality

sha256:c81fd2fe861e6c25d9802ab0c631deb16cad891818bf10ba99a96bc7096f0eb.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:df19ad80487439feb9499de85bb4499f5e1e33ee4d3f6711d34f082f3a8ede2d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:f3feb2e19f9efa42f289f923ccfd2484e2ee99b3323f83ff6e35a3f309a953cd.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:4616b17e16aedbca98f7dd35c7a99fff0f145df7a28843ac19e16da5d9bf24c8.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:e2d2d7ed7f68fa37ab2154272c08cf3212fc2ad79c92ccb365f4e7b76044143a.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:30e911c0f31401870492db7b9daadd7519ae177545dc79d3026afe026d2033b0. Derivation seal

sha256:2c299928ce8eb88c0c99d71d2f13509ddf1e7843f972e9df0d32e0003e37cea7. Source manifest

sha256:79c4df2305a7a3e0a7f5174787b0c69717efa441a13a552dc03ebe637d589b8f. Engine receipt

sha256:9e2c00016a582eef5e25f25c067c38be0d7bbd076631c2944a9b1e35b4697103 at receipts/engine/model_admitted/SFT-MAT-CLASS-PARTICLE-REINFORCED-007-9e2c00016a582eef.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-ADVANCED-COMPOSITES

Measured, observed, adverse and boundary records:

- CLASS-007: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True

- NIST-ADVANCED-COMPOSITES: interparticle interactions; filler surface

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:8db695670c0030e19d2ab8b44c2bbead0fffb6ce5a66b3d00c57b661c43634bb0; empirical validation
 sha256:85567e7c314f640d7d45fd362c8d9fea21691389f821b01056757f8380406a7f; external validation
 sha256:10f8abba2dd2102d9d67ec45d88a3e69cb9900cc5531c50930a6ec400c9ccf18; isolation
 sha256:5621281c635c9677440cdc624417d8f631a5541c997aeed04b393fa19afa51bf; custody
 sha256:c59aafc9502af83b01f14b6fa29cc1d785e000e25f58a68c98a28ff659b3f11a.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

182. Metallic-glass organisation

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-008

Claim: SFT-MAT-CLASS-METALLIC-GLASS-008

Question, law and forced result. A metallic glass retains composition, metastability and local chemical order while crystalline periodicity is held as structurally absent rather than numerical zero.

CLASS-008 uniquely retains local-order-nonperiodic-metastable-metallic-glass with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001

- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-PARTICLE-REINFORCED-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-008 axes before target release.

Boundary: Every positive finite CLASS-008 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- local-order-nonperiodic-metastable-metallic-glass
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__local-order-nonperiodic-metastable-metallic-glass__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:7c3c1d8e2cfc67aabab8bc45548b8a3ee6a10f887f07ba493cc3471d7a2696fb; generality

sha256:4ddc13d84022d522f5ba6dbf7e504e577eb0cf05d7144b05355022070d679903.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:39b3dc1733bc5f007dc348d75f76a5bfc9ef8eaf3d064debc33e48c255f82035.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:680efd5acf92750eac18805a0fe36e25d0a88f6e7062c01340b86d11a59e113a.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:714328c62a83630c10e1aca004a7858e719b8880dbf7fc96df618e88b494e677.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:0862c171adaa0d68fda520cc0a22e10294a8b40228bcd5c980b05d129bb9654d.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:6c8571b51299d94239884e9f752064734a49fcbdc64de3c038f97d4de028ecf1. Derivation seal

sha256:8f1a7641fec88b70a7d6b1c06f57a978ceaf95126964fe676656ee8d855e531. Source manifest sha256:20e56f768fdd9f8835659a0eb17627f12c0e4a0328c27ba73ae83a96ea2bff71. Engine receipt sha256:9718b54af7ca52008c6b43bb05c60a5d7d61c78870d87ccc83bda63ee479e8d1 at receipts/engine/model_admitted/SFT-MAT-CLASS-METALLIC-GLASS-008-9718b54af7ca5200.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-METALLIC-GLASS

Measured, observed, adverse and boundary records:

- CLASS-008: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-METALLIC-GLASS: noncrystalline homogeneous metastable phase; chemical ordering out to at least 12

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:799b45567f905f256f38b89802eb22eaa26120cd6a86e0901ef41e89b1af26b1; empirical validation sha256:7d3c600ab67814cbde24225805f79b8ade9216001d53998ef295c1f1b6a7bef2; external validation sha256:c152dacdf93de9508a200d3f95145f7c99d401d43f462723c194bd512622d9a9; isolation sha256:e222bfac6f973fd70b20acb9669cd891c7d446f8ce8beec4f6cf14b1839f6a2e2; custody sha256:c87cbab4770d566279d20f319d8c78a96a4d1036897a840c2590a77c50c82f8a.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

183. Structural and functional ceramic subclasses

Field:metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-009

Claim: SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009

Question, law and forced result. Ceramic subclasses retain composition and process while structural and functional response roles remain separately observable.

CLASS-009 uniquely retains structural-functional-composition-process-ceramic-subclasses with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-METALLIC-GLASS-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-009 axes before target release.

Boundary: Every positive finite CLASS-009 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- structural-functional-composition-process-ceramic-subclasses
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__structural-functional-composition-process-ceramic-subclasses__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d3dab070b5e343c6048144fa3c0785eab5ad78344e6c57e309278ceffd2e152f; generality

sha256:6e6c4333a124ea5d7463c069cda2d1e1639c4921bd2f3ad6b958e2f16c8c3d41.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:5572c81e7cd0a1e7159b47d7ec4410009e5ab673d168bd034dce71da930c3817.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:613028320d95ff5cf220b7db2033173de39593c09e93a23ad525bef62c3d3125.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:f94267b8d7e2b073bdbea9e3b32855a53c7b425d2847b6a67ce0432ee3b3c6e9.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:c36cbb842f92b776d5f07cd65597aa75e98529de7ef77558d05e0e1eb3e21420.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:ca3435e55105c52e7942c22f944056da9cea3ecbfb814fe8802fabdcbae49139. Derivation seal

sha256:84c1013a8e0642b0889783bfd2d3869817aed71f35ccfed7149e791509367652. Source manifest

sha256:cd61f1b2e96e1d7e0023ff29d0e8f1e9a325972b6cc38b3acc1543baf01bfd26. Engine receipt

sha256:dc16234399a1b13aaf6968f48602c508258b0e132e7904b58a8968af45c23b11 at receipts/engine/model_admitted/SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009-dc16234399a1b13a.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CERAMIC-AM
- NIST-HIGH-TEMP-CERAMICS

Measured, observed, adverse and boundary records:

- CLASS-009: 2 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-CERAMIC-AM: body armor, structural components, high-temperature systems; electronics, energy, and medical implants
- NIST-HIGH-TEMP-CERAMICS: high-temperature ceramics and cermets; elasticity and density

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:40a03429e4078fc37d2d65abe02a208362f3d0bd9cbe31edbc370711b026446c; empirical validation

sha256:848e8819d06b809aaaae498fa222f74e4038d9d37ede76246c7e4fe3e5070dde4; external validation

sha256:acbf11ffe2e4ec694ec2f24805f0e0e1987e68934519ba1f6064a6ccf6217fa2; isolation

sha256:569821405ada51aac71d533eba5adbcb35ba1836a4aef00535fcb1ebe122a6b; custody

sha256:e038d26fd3c6e4162f0309ae9842f75ff134b905866abae01caf458d3908bb.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained

- no failed attempt retires an obligation or changes protected authority

184. Thermoplastic, thermoset and elastomer distinction

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-010

Claim: SFT-MAT-CLASS-POLYMER-SUBCLASSES-010

Question, law and forced result. Thermoplastic reshaping, thermoset permanent crosslinking and elastomeric recovery are three separately retained response histories.

CLASS-010 uniquely retains thermoplastic-thermoset-elastomer-response-distinction with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-010 axes before target release.

Boundary: Every positive finite CLASS-010 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- thermoplastic-thermoset-elastomer-response-distinction

- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__thermoplastic-thermoset-elastomer-response-distinction__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:ed3c8f88f1140cf8bb86271da63f520ac861380ffd02f13c7e052fc25a8d0335; generality
sha256:aadf19ef1681882391e46fefba190c1fc3696dd60ad0344eab55fe47072faa9f.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:9ccd323cc852ed96d204689f3cd0503b8b95962250f8272c5f6fffe6f18f8711.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:10e05b84454e4fab8a957398773533975d5cf15df25c075d98876338c4a4cf7a.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:ec0e7698d431b44a64b2b33cc8a3d33c43c00835e2e440f1136cd58b27b2ab03.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:3e94eef3c6b36bd07a5bdfb9dc148308a0114f44bcd36cbe29dca70a6ba0e85a.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:7b09fc7c0640fec596cfdc1ba717a23dee37c31195310336b0f9576751ddbbee. Derivation seal

sha256:d0535dd836f4e4b78548d0f537111035f2f5a73f4b8b11816a0b19dd87960069. Source manifest

sha256:378fa76251fadf7c72740bb8f5f6d99a325ad31f536ddd466c32bc8bba46d997. Engine receipt

sha256:3377c28b097ac207e79b8530c9b73c5b43bf49f40ce8e9c73c7f33d84cc516fa at receipts/engine/model_admitted/SFT-MAT-CLASS-POLYMER-SUBCLASSES-010-3377c28b097ac207.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MACROMOLECULAR-ARCHITECTURES
- NIST-THERMOSET-GLOSSARY

Measured, observed, adverse and boundary records:

- CLASS-010: 2 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True

- NIST-MACROMOLECULAR-ARCHITECTURES: model thermoplastics, thermoplastic elastomers; polymer sequence, chemistry, and architectures
- NIST-THERMOSET-GLOSSARY: hardened into a permanent shape; not commonly subject to softening when heated

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:071fa752f1b7f7794143a5ff37443843af484f5b447bb6e005afd122c7424ff7; empirical validation
 sha256:2308075e8277048aa9e7a86cb4a458232f52185aaedbf19773a763ab365fa7cf; external validation
 sha256:ebfbb8b2d9ba35d33f7c5f01409e2fc87690dab7b0afc2633f17be28a7ed0c90; isolation
 sha256:6ce13b7dc7a2cd47d364ecc6a6bf96baa43dc0635d07c25bf2a6fdb309bf0aa2; custody
 sha256:66bd8b35b033397daee6f7917cde1a39b6bca67d71eadb5284dd3e3f04f98f54.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

185. Gradient and functionally graded materials

Field: metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-011

Claim: SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011

Question, law and forced result. A functionally graded material retains an ordered layer path, every exact positive property carrier and each adjacency from one endpoint to the other.

CLASS-011 uniquely retains ordered-layer-adjacency-property-gradient with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001

- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-POLYMER-SUBCLASSES-010

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-011 axes before target release.

Boundary: Every positive finite CLASS-011 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- ordered-layer-adjacency-property-gradient
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__ordered-layer-adjacency-property-gradient__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:deb30b3fc065049e7f68e92e1f78ac72fc6600005760ba67a0b9e13529073398; generality sha256:45b66105789fae274614f7cec18388a5f4462918936caa9a32cd81c7272eb041.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:78f5db6b4a3ae82ed88193134f68520556cf7280ac081019afe06d6aa39f3e6d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:c0fa31bde0cda47e583528e0633b6ffa34896ecf7dd2dd836dd60220975bb1e7.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:ec3d39e64e7e1c2165f6c84b13e167e663833d4e969979b9992620deb3bed723.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:3bc60b49b082463779b4564ed2a81ca8399f229e45d3ed6d3628535e29cbf46c.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:f0c0237154bbbb376071178f96b098cc49639ffc5748469f5663470a6708438b. Derivation seal
 sha256:6d783a34eaac257fa9021affb474a498c249f33ab7a8ce62eb603b468f30ca25. Source manifest
 sha256:74d5fefafe98126921dc2d5061aba73ec23053ec88918b8c93c1f45d48751130. Engine receipt
 sha256:3becd22d7a6345510be495ca931c777d4c7a075bc6c844d641ef4f3e20340818 at receipts/engine/model_admitted/SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011-3becd22d7a634551.json. Closure
 scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-AMMT-GRADED

Measured, observed, adverse and boundary records:

- CLASS-011: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-AMMT-GRADED: microstructure control, functionally graded materials; enabled by novel AM control techniques

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c76e204f0c00241b1a056e5e546d3cd8f072db1db86b1053a91b8ba6c5bb00ae; empirical validation
 sha256:f69edb882adfb354e3184f03b69d18e475a6d05658e0ea216e647afc57d46b73; external validation
 sha256:b51ffd06e8ea9509788318dd9c7efd25d9fa9bf39fbcbbdd95d96440fb929cd7a; isolation
 sha256:0326b09111171c169efabd34595f8324e515307cb4a5713ead2d0a01e8b1b765; custody
 sha256:99232f85d417e5ba2673d488e1fe4ac6257c5eabbabb9ef05b004901e314dbcd.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

186. Architected and cellular materials

Field:metals_alloys_ceramics_glasses_polymers_composites

Obligation: SFT-MAT-OBL-CLASS-012

Claim: SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012

Question, law and forced result. An architected cellular material retains nodes, links, cells, topology and response channels so architecture cannot be collapsed into bulk composition alone.

CLASS-012 uniquely retains node-link-cell-topology-response-architecture with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight CLASS-012 axes before target release.

Boundary: Every positive finite CLASS-012 material carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-material-carrier
- node-link-cell-topology-response-architecture
- complete-structure-state-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-material-carrier__node-link-cell-topology-response-architecture__complete-structure-state-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:4c9de97467c36fa63717f2c78d0c9af8135ab7088d1f5515a42d062a57d0cfae; generality

sha256:2ad3b3187db6f97415b2626f86aa382a9ea15d988e55ceab9a0a1f168348f46e.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:5942ff5232ba5062f11107727b1c6f46237868e891a79fdd6f33b838648fe6b5.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:6e6a624192b2ce5cea559cc86173ef638f9197592bafd9c23e8b88ab54843f77.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:ba153cc88c8fe9ed49f1d89287d01a8cc771ba4812c95d7c74fe051f0bb73e97.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:4ab553cfb8a87357c0d8f3d817d1a7ce6ad733152ca1305478cd9b3697ad.

Independent reconstruction and engine admission. Independent implementation

sha256:453e29faf5abf49b68e322aa171b2f8f0cb6c0a108dc6f79fa6da6e8c313f3a2 regenerated the declared census and sole survivor; independent certificate

sha256:9771b7b121794421e3a8dc670489e997a41fb9ec183f1fffb6ceda51a7bd91353. Derivation seal

sha256:65011966e50cfcc412cca4e3c876e57da3a8e5904ea412f91680ff15a8920017. Source manifest

sha256:38f917d47a8e76dd646b75d5bfddfd97beac466e1b233b6562d1878d5a282a8. Engine receipt

sha256:f7eec41372016f198c06805f29fc2f3fa759d6fe3e6ed83191ec861df994ffcb at receipts/engine/model_admitted/SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012-f7eec41372016f19.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-AUXETIC-ARCHITECTED

Measured, observed, adverse and boundary records:

- CLASS-012: 1 comparisons
- family sources: 11
- exact correspondence: True; omission rejected: True
- NIST-AUXETIC-ARCHITECTED: Auxetics defy common sense; widening when they're stretched and narrowing when they're compressed

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a3447f484180aa21a8fba614d473e7f7e0e01323d43d4570978f746325d80753; empirical validation

sha256:dd683f64e2a31b42ee4650ff3bd775b0c8e8a34b6798c105724983ec735ddd3f; external validation

sha256:b0d63bb95c0679ec9f825eaf1b250a4f48862254950d6bd0466ecdaa4c7ba1fa; isolation

sha256:6b8120fa65f61a7226538a540739eb2d208f25ac2da0d275c63c9fd979428efa; custody

sha256:f536def3c344e0c727171e51b1a32406efb20a79e7359b6e438be2785a4e90c5.

Prohibited inputs.

- no imported constitutive equation, fitted class threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence, class, phase, orientation, topology and response remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Soft Colloidal Liquid Crystalline Granular Materials

187. Colloidal stability and aggregation

Field: `soft_colloidal_liquid_crystalline_granular_materials`

Obligation: `SFT-MAT-OBL-SOFT-001`

Claim: `SFT-MAT-SOFT-COLLOID-AGGREGATION-001`

Question, law and forced result. A colloid retains the exact positive partition of free and aggregated particles together with interaction and condition labels.

SOFT-001 uniquely retains free-aggregated-interaction-condition-colloid-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MAT-MEAS-MATERIAL-001`
- `SFT-MAT-MEAS-SPECIMEN-001`
- `SFT-MAT-MEAS-PROPERTY-001`
- `SFT-MAT-MEAS-TRACEABILITY-001`
- `SFT-MAT-CLASS-POROUS-001`
- `SFT-MAT-CLASS-COMPOSITE-001`
- `SFT-MAT-MECH-RHEOLOGY-014`
- `SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012`

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight `SOFT-001` axes before target release.

Boundary: Every positive finite `SOFT-001` carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause `MAT-S009` applies. The full, unchanged clause is preserved in the shared-clause appendix.

- `complete-positive-soft-carrier`
- `free-aggregated-interaction-condition-colloid-partition`
- `complete-state-structure-response-path`
- `specimen-method-condition-scale-uncertainty-held`
- `complete-trace`
- `root-bound-forward-forcing`
- `positive-finite-successor-closure`
- `no-extra-rule`

Unique survivor: complete-positive-soft-carrier__free-aggregated-interaction-condition-colloid-partition__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:bd4d8f4b518240a0454c8abba961b8d4e24c42cc48c26203df21d8285ac3e3a6; generality
 sha256:d6c5dbe40bccal1c07cd885d9379312e7f0622ba877bcd92952450c8f2ceb.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:1cec915967970a8f3c6430b43a585fce07e6147b6ff2b1cfc59e2cf2a31f6a20.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:6f45b344c7b68bcd0dbd28880c2a05442b39ceec983017c42a6d0eb88c870e21.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:cf3d780b23a6347ceebf2c5c3aff6aa24bd9b2666a43640e175a97f32018671f.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:c0b522a1919c165b0113c3e4c5bdf528b48eb5614725790fdb2ece7fb7c7a598.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfefe90d4efca1cf7883413180fb87dc26 regenerated the declared census and sole survivor; independent certificate
 sha256:c9f6b2078a1c7b19cb41f649530f78b6010b4310979f9af7e8738e328bc9813c. Derivation seal
 sha256:92bbbe9576ff432e92c58aa1a6f9d4a226196041e8dabb6c1d0cc14abb8c5fde. Source manifest
 sha256:94b08759be6d782e976ead7b32f66dc6dfb6287c51fa3ebf5ab15988739d3eff. Engine receipt
 sha256:85d4ac6dba37c7352a3597f8080d2495312ca0aaa60d8a142326bd50905a0256 at receipts/engine/model_admitted/SFT-MAT-SOFT-COLLOID-AGGREGATION-001-85d4ac6dba37c735.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FLUID-SUSPENSIONS

Measured, observed, adverse and boundary records:

- SOFT-001: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-FLUID-SUSPENSIONS: self-association and aggregation; surface chemistry, surface adsorption, flow conditions, aggregate size

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:99a0da462b0b5873e4b4466913687bf47bd27a75088bb213d49e13a7a55ee883; empirical validation
 sha256:9895462ffb7c19b4ab37b0b0b8fdcf7e047104e766e308589d1fdbb4adb76f23; external validation
 sha256:6ealb6f8b7a958019e82cc4dc891b47192fd732f95ffb9a4f467c5a23cabd480; isolation
 sha256:9ea60c76f38672b97d6d94ecdfc40133980e570e5dbd99909ca853b84733c340; custody
 sha256:36954eab0de44ee0ea4151e05e40392470b93db9b35fb75dfaae5a53022639af.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

188. Gelation and percolated soft network

Field: `soft_colloidal_liquid_crystalline_granular_materials`

Obligation: `SFT-MAT-OBL-SOFT-002`

Claim: `SFT-MAT-SOFT-GEL-PERCOLATION-002`

Question, law and forced result. Gelation retains nodes, links, spanning path and mechanical response as one connected soft-network history.

SOFT-002 uniquely retains node-link-spanning-response-gel-network with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MAT-MEAS-MATERIAL-001`
- `SFT-MAT-MEAS-SPECIMEN-001`
- `SFT-MAT-MEAS-PROPERTY-001`
- `SFT-MAT-MEAS-TRACEABILITY-001`
- `SFT-MAT-CLASS-POROUS-001`
- `SFT-MAT-CLASS-COMPOSITE-001`
- `SFT-MAT-MECH-RHEOLOGY-014`
- `SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012`
- `SFT-MAT-SOFT-COLLOID-AGGREGATION-001`

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight SOFT-002 axes before target release.

Boundary: Every positive finite SOFT-002 carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- `complete-positive-soft-carrier`
- `node-link-spanning-response-gel-network`
- `complete-state-structure-response-path`

- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-soft-carrier__node-link-spanning-response-gel-network__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:b2ff4d0f8c19a1e8ba9dfa601b05acb2d023fa96ec9a3d0081ea5d83ece27fec; generality

sha256:463f0d7fa5e879ef69bc47fd45f3af2aadae5cc29831f896aa885a7dac1383e1.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:a99d62a26ac5864ec9fed86c0ef047ccbe3674b0ec8b11a43cd21c26d564b50.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:65a94b37db49bdd442e77753cc51d799d7b9adb61bb23a6ca04fad56e608f0cd.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:cdcde62abfb91a9f74cdbf42d5149b5c22911b583225d24efbf3780459233949.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:99cd6317e68454367f981667e7b4be49d7c55b942e9986c9da04982c31e5edd3.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfefe90d4efca1cf7883413180fb87dc26 regenerated the declared census and sole survivor; independent certificate

sha256:bb6225e3959b40597d2a399117acf362d755ee1c8e446d25d2883d872cbf71dd. Derivation seal

sha256:31ad1acb74b7e08b26cd6ece02416bd564e5c226ed6cf9eaedb57b6fb9c9947c. Source manifest

sha256:39ac3dd4f1070f3cf66bf6430a877e0da9b063b1c068f627bb61fa05d1429606. Engine receipt

sha256:aa1dc8180c723c0ce409d49ef2938fd857c1ea1bc9c73dfd8859bed9f917ba09 at receipts/engine/model_admitted/SFT-MAT-SOFT-GEL-PERCOLATION-002-aa1dc8180c723c0c.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-DYNAMIC-ARREST-GEL

Measured, observed, adverse and boundary records:

- SOFT-002: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-DYNAMIC-ARREST-GEL: physical gelation and rigidity percolation; particle dynamics and rheology

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:dd4ca0047ead77aad47861561a3097a9ff7e971eb5e0aee86c91c2b2cddb8c88; empirical validation
 sha256:ffecfa81b7a6582b2818725a630ceeb44c1fad9d27f0ceeac47890f81111b456; external validation
 sha256:eae81774fef2a4467982972f9332ac65a83111ab30e501eaf732a2a98e800c68; isolation
 sha256:42fecf1032c9c53cabcb9c1e6c0c65dddbf6cc552628a209478afa3ab7cea2bc; custody
 sha256:538f5f048e23994c84553d23be657ed1e0cd554fbe1f63edb3cc98a6ca0f84ef.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

189. Foam cell and drainage organisation

Field: soft_colloidal_liquid_crystalline_granular_materials

Obligation: SFT-MAT-OBL-SOFT-003

Claim: SFT-MAT-SOFT-FOAM-DRAINAGE-003

Question, law and forced result. A foam retains cells, gas identity and the exact liquid partition between retained and drained carriers.

SOFT-003 uniquely retains cell-gas-retained-drained-foam-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-SOFT-GEL-PERCOLATION-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SOFT-003 axes before target release.

Boundary: Every positive finite SOFT-003 carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-soft-carrier
- cell-gas-retained-drained-foam-ledger
- complete-state-structure-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-soft-carrier__cell-gas-retained-drained-foam-ledger__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:4721652b4c81b3e56fe1f80692a054bd0150873e5e1853caab1afe96b8cc12a6; generality

sha256:4670c7277be2344b79b506cd49f690ba3b1e2070e04f6994f8afde3a4c70b942.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:772c21f62b11ef9c7eaead8e99c2c9e225c261b20a6382ed0e214cd517625ddf.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:7dc65a639f422e8517156278a3f6275d4a756a126f990ba4afe6f1c3451f84dd.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:5cc6fbd6791674330c5aaf68e24ef51c0c3e790ab403dcda6886010d6e81b9d5.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:65e802710e4a8f44c36ff6e9ba4721671d406da49093f5bf19f8b66eaf9ba102.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfefe90d4efca1cf7883413180fb87dc26 regenerated the declared census and sole survivor; independent certificate

sha256:934ef66648c72f7e458ddf553aff4d5306d723ba700f4e0e51ea4de04a81e70d. Derivation seal

sha256:6acb78983ecb8ac5e2561028eeb0f3de90b57809aaabffbc9e495a813c66a3e1. Source manifest

sha256:60902e3dc1d62847f535a354d2f5394748fafc809b7c773efb64c82f0970ddf4. Engine receipt

sha256:872101dd10667aabe677ec1e25d3ce53f0b9bf998691193f289a7317f2d00926 at

receipts/engine/model_admitted/SFT-MAT-SOFT-FOAM-DRAINAGE-003-872101dd10667aab.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FOAM-DRAINAGE

Measured, observed, adverse and boundary records:

- SOFT-003: 1 comparisons

- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-FOAM-DRAINAGE: foam drainage; surfactants

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:69f466c9c6b99ce8499a28039444c0a73d221779c3312d6982724537d3674540; empirical validation
 sha256:9effc92fc2a7e595209d584f195632f9f917e0929784d2ec8f95e25077129458; external validation
 sha256:ccc74e3c66ef17b3c9dab8ed7fa210d58c13f58c21f728952a8cef15ec6d9f9b; isolation
 sha256:e12586028d503ea921382255721d89793ab5816b7b2805ec5fb189fb5ba3304a; custody
 sha256:37374646e7e16b64f5ca4ebcd0a9f8ba226f17445693ef2cd2a0a09c2a1d7150.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

190. Liquid-crystal orientational order

Field: soft_colloidal_liquid_crystalline_granular_materials

Obligation: SFT-MAT-OBL-SOFT-004

Claim: SFT-MAT-SOFT-LIQUID-CRYSTAL-ORDER-004

Question, law and forced result. Liquid-crystal order retains every orientation label and exact positive part, phase and defect record.

SFT-004 uniquely retains orientation-part-phase-defect-liquid-crystal-order with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-MECH-RHEOLOGY-014

- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-SOFT-FOAM-DRAINAGE-003

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SOFT-004 axes before target release.

Boundary: Every positive finite SOFT-004 carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-soft-carrier
- orientation-part-phase-defect-liquid-crystal-order
- complete-state-structure-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-soft-carrier__orientation-part-phase-defect-liquid-crystal-order__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d1c2e2fe3bc3df44b343200b758197389669d64555766c8fd6946b05014bbebcb; generality
sha256:2bf2c8aa9d802db15a2704f90bbb25ff24d24a6d9b50bbb58dacc1ed8960ab2d.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:4c37472f7c8a2d14fecdl782831ac70f1e484fff3e7c287178b6acf4bd60ab67.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:8effe230af06510f19dceb14eb05eec99199e2379199cd52bfcee1a31d71f30a.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:063b45be5fd70735e23d7b1d29a7d5e98431d0e20830a60c2b7d5ccce7e85b85.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:9c908513004dd928043dde764a3b249d1639724e14b3a97644104c19438c323b.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfef90d4efca1cf7883413180fb87dc26 regenerated the declared census and sole survivor; independent certificate

sha256:e9c9118d842c6741f41228441f6c8111029ff6b72997e9a142ba0936b7625758. Derivation seal

sha256:c600df0c2a274b7d66c32e9108e540f24a91274584f77d9b9d2f0ddfec7cc560. Source manifest

sha256:38df87fe83bc19fba084ab2ab6a14f130864dc5156dc5c44799c3ba943398db4. Engine receipt

sha256:572d0c1d48995f676100ff7bb7e8620ba36f5effaaa5f72bcf7b9f40f2618ca0 at receipts/engine/model_admitted/SFT-MAT-SOFT-LIQUID-CRYSTAL-ORDER-004-572d0c1d48995f67.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-LIQUID-CRYSTAL

Measured, observed, adverse and boundary records:

- SOFT-004: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-LIQUID-CRYSTAL: liquid crystal orientational order parameter; droplet shape, domain size, and liquid crystal defect density

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:2ce7ba2652687d388f5899449b268fab3503d55a5c631c29384b4e916c775d3a; empirical validation
sha256:8fbd1105f1e29a76e8c062d791e26f4a74bea8855053eb5af607d86b69385388; external validation
sha256:477f09bc58a6bce8269ab15a44320b929a73678cb8a1e31d70695a65ceafea1; isolation
sha256:df80c65f88892872867ac0b5fbc198e31293ec49de271e026dfa9440776c4411; custody
sha256:03b3818103f6ef69d02e5ac335fef7b0e17d326b77b18fb146f723960fbfef29.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

191. Emulsion and multiphase droplet organisation

Field: soft_colloidal_liquid_crystalline_granular_materials

Obligation: SFT-MAT-OBL-SOFT-005

Claim: SFT-MAT-SOFT-EMULSION-DROPLET-005

Question, law and forced result. An emulsion retains all droplet populations, phase identities, interface and complete shear history.

SFT-005 uniquely retains droplet-phase-interface-shear-emulsion-history with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001

- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-SOFT-LIQUID-CRYSTAL-ORDER-004

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SOFT-005 axes before target release.

Boundary: Every positive finite SOFT-005 carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-soft-carrier
- droplet-phase-interface-shear-emulsion-history
- complete-state-structure-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-soft-carrier__droplet-phase-interface-shear-emulsion-history__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:7b3b82c20a5dda5f5941aeecf162e656d9a796ea4106f8c4f8eec40a0392a4cd; generality

sha256:042ebcb8f0071384d0383c39670d02efe2a5bfabf1c50f143b55f0cba9b204a2.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:de401c3c7dd44f5516de91d4af8e82207b6abab6701df63b416d47546a5cf126.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:08c27c4c3c23499c674edf00b21b662d6f28d58e2183b13f684b55f701fe6bc2.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:7fcbca98fb9c7164d8ccd7650a205e26599dd2b10143d9c1025fb9e4e1500656.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:216b6928efa43cd73e7537cd31dd2c3d16a9bb76db3120c51458b984c8ef1ba4.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfeefe90d4efca1cf7883413180fb87dc26 regenerated the

declared census and sole survivor; independent certificate

sha256:e4f8d18f4d67cb7668d71c0c0c1e242b2ed9a3cf329b039d13b3c3950bdab095. Derivation seal
 sha256:2214ce772bc635a68501969b3ff858aa56ba05424ee7f00a72e9e9392368702c. Source manifest
 sha256:12cf3d5c72127345711805479502e35f0e0c72bb2752b582ee363d402635274d. Engine receipt
 sha256:69bf1637f23b87e943680c331d3db3c661d86710380382338e79f30ba10540b8 at receipts/engine/model_admitted/SFT-MAT-SOFT-EMULSION-DROPLET-005-69bf1637f23b87e9.json. Closure scope
 depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-EMULSION-DROPLETS

Measured, observed, adverse and boundary records:

- SOFT-005: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-EMULSION-DROPLETS: organisation of droplets in layers; interfacial stress effects overwhelm shear stress

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:acfd4955e887e9475159aaa40391e93c1f4cfbb26bfed32e1d3771e9505c002e; empirical validation
 sha256:3088000eae11a2b0a27a883626cf3ea10283186f06c93cc085c8dc5f82ce0c4; external validation
 sha256:4b3ad12d62dd25f742238c4c3601d16c93ffde65c3e70f498e66e6decc4ff73a; isolation
 sha256:3651cala1fae52f6210f160ed16833ef220299c54495485406298933c447db27; custody
 sha256:b97389aa8b0756b83664325c6a01423b333e20f9d0b250e9a9f5941c90d983c3.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

192. Membrane and thin-film soft matter

Field: soft_colloidal_liquid_crystalline_granular_materials

Obligation: SFT-MAT-OBL-SOFT-006

Claim: SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006

Question, law and forced result. A membrane retains its ordered layers and interface and the exact incident partition into transported and retained carriers.

SOFT-006 uniquely retains layer-interface-transport-retention-membrane-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-SOFT-EMULSION-DROPLET-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SOFT-006 axes before target release.

Boundary: Every positive finite SOFT-006 carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-soft-carrier
- layer-interface-transport-retention-membrane-ledger
- complete-state-structure-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-soft-carrier__layer-interface-transport-retention-membrane-ledger__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:92adf2f4ea02438b9525d280c3dc89a5b322b891d9b007ac1b90509edff291ac; generality
sha256:fe3f91887bafbc45726d5550cf4e5a10cc0c01ff83ec8756f407e40771613416.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:5d94d8fe2a26ec7979e2485c346813e013315b9733811bfe270e532155e6a29d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:a230fbfcfb9c4c9bd8e3051bd1f9d15110cc5ca0f150fbe0c251e1921c372f4d.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:cb31cecce06f562f831c272bc525bf26ec7cdfc4d793df123d930c74625d7754.

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:3d46dd367a6cb28c569f05bd97455cb4de0456c07d0fb3a67f1136ae99959253.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfefee90d4efca1cf7883413180fb87dc26 regenerated the declared census and sole survivor; independent certificate

sha256:0d9a82cfe5a8a59ee6c383ac321b616078025ab348e26e07696653b7adcd5b8e. Derivation seal

sha256:f899f0b489e2f83193dcb45f0985a5ffb43864e94b798fc6aaf60f090e953ede. Source manifest

sha256:d2aafacdb382a364eeec4675fbdbd52cc5133166beee6301ff7bb92bfff914940. Engine receipt

sha256:cc20019289aed6302bcd561f3f0a3333e641a7351847cb081d802da7dd6fb27 at receipts/engine/model_admitted/SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006-cc20019289aed630.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FUNCTIONAL-POLYMERS

Measured, observed, adverse and boundary records:

- SOFT-006: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-FUNCTIONAL-POLYMERS: thin films, interfaces, nanostructures, and membranes; thin film mechanics, adhesion

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:699f93adcaacd776d07f0d2a32fcef6d26e4031d178658b9933e7e6494676668; empirical validation

sha256:e6f48427246f2e6d843971249297e96cf7fb7f778cef9485a80949e5efb6311e; external validation

sha256:ee61f6de13d10d89f27a60959b3a501c6c76019a95489bd66082f7c040517d68; isolation

sha256:399bb50c2db9171bfe444bc330654be3bf3911e0894361adaf07f4943df8e99b; custody

sha256:9bd8728799f3ce6ce96ad3dae68c14da9c285fc02ac1ab1cef3c3292e86539af.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

193. Granular packing and force-chain support

Field: soft_colloidal_liquid_crystalline_granular_materials

Obligation: SFT-MAT-OBL-SOFT-007

Claim: SFT-MAT-SOFT-GRANULAR-FORCE-CHAIN-007

Question, law and forced result. Granular support retains grains, contacts, packing and every exact force-chain load part.

SOFT-007 uniquely retains grain-contact-packing-force-chain-support with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SOFT-007 axes before target release.

Boundary: Every positive finite SOFT-007 carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-soft-carrier
- grain-contact-packing-force-chain-support
- complete-state-structure-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-soft-carrier__grain-contact-packing-force-chain-support__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:7889dc4a4d7f80605ebaef2a52a7d7d38c6e397d7d45af3b9ed48ab6d557a8e0; generality sha256:0099f0620324eb2e5b42b09507b220c82a489debd466b6f9c0c7b551c67f4f53.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:b95c8bf89ef7abcdf0688fefbb98caebb23e5b014bab3549bb1265f35a61aac7.

- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:3786b8a9a0fd95cac4e11285f57e6a2363af6ec0a74051c6f2e87c00a60fea8e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:65a7f8e7af11003ac1d3ea9ac0cc036188bb586011b42c1d2054fd280a2f09b3.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:60f6d68fca33467c27fde3ccf6d8d643c57c5ebe172caa9e7e9bf5edd0359213.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfeefe90d4efca1cf7883413180fb87dc26 regenerated the declared census and sole survivor; independent certificate

sha256:238e17c5f6725a847f028bc3724efcd61136b01d7911ca6c2e3de66a255503be. Derivation seal

sha256:ce7fd22d89347a26b1117e78bddd0888a7f439bdbl1f7ae5f6c7b5ed88bc3a5a. Source manifest

sha256:bb440d784d8bc550ba9b721f7c7e1921a0a3e2f292fad6d08d6d7058f469fb02. Engine receipt

sha256:c9420e94b3f100db2e85a69f274253f95f8fa58e8b7e697e892f39c6986951b8 at receipts/engine/model_admitted/SFT-MAT-SOFT-GRANULAR-FORCE-CHAIN-007-c9420e94b3f100db.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-STRESS-CHAINS

Measured, observed, adverse and boundary records:

- SOFT-007: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-STRESS-CHAINS: Stress chain formation and evolution under shear; stress fluctuations and the suspension microstructure

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0832c7f614bb24d773e5fc22e376f84265178a343571e03de030aa95f0aaeab3; empirical validation

sha256:b53a848d38517e6736534e2e7002e4480cd8bfa9d2a216924c2611cef6d18b04; external validation

sha256:b5b7ac6b3d15b6c24dc4d93fdd1c2a21f7986e239797a158f52cdbe0a051afa7; isolation

sha256:9bfc236612da06d610117e3ac871be18ba520088c92c572f6d6cd1fcee2f0d44; custody

sha256:6f712ca027dd96251ed5151a89283b16076353b209b7592c291a01ec0ad8e7d1.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

194. Jamming and unjamming boundary

Field: soft_colloidal_liquid_crystalline_granular_materials

Obligation: SFT-MAT-OBL-SOFT-008

Claim: SFT-MAT-SOFT-JAMMING-BOUNDARY-008

Question, law and forced result. Jamming retains distinct before and after states, applied and resisted carriers and the complete transition path.

SOFT-008 uniquely retains before-after-applied-resisted-jamming-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-SOFT-GRANULAR-FORCE-CHAIN-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SOFT-008 axes before target release.

Boundary: Every positive finite SOFT-008 carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-soft-carrier
- before-after-applied-resisted-jamming-path
- complete-state-structure-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-soft-carrier__before-after-applied-resisted-jamming-path__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:408b699c930977edfdd00729687ab942c5b8ad2dff0dfc6b47303b99abbb7397; generality
 sha256:ce32e86c55a194093b4e12b1358d30a5385a5bf0d9293d29de8a37c27148af03.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:a2dec3bafc374711f790b308b1aa5e5bdee47d88cabd8b1fa717b3d288dbc36d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:f4225383fd84c05c77ebcd0048c7dceeebc61d0c588fe504b5ee16f5cccedb3e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:5ec9887be8eb6887bcf652b5130611adb01051025d66cb97e21ccc4197a14e20.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:dcbb705a5bcbbaad1510bf8c63a3cfb7c09dda43e0c108512d3c21cbeb5be574.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfefe90d4efca1cf7883413180fb87dc26 regenerated the declared census and sole survivor; independent certificate
 sha256:bd9a654aecedaa34062a11c9573d77cbf817d1230daa61fb90077acb89767ad8. Derivation seal
 sha256:c0247da271a85efd510a4aa1e8425520702c8c30e5ee451a41366dc912e71086. Source manifest
 sha256:e64ffflca40624303ee627c863cdb566bacdaa93d987619cbb70cc92658a98f8. Engine receipt
 sha256:2951fbb4b789a7dc97aa9425ea21ac58ed55904975180183865d15185a7229e5 at receipts/engine/model_admitted/SFT-MAT-SOFT-JAMMING-BOUNDARY-008-2951fbb4b789a7dc.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-JAMMING

Measured, observed, adverse and boundary records:

- SOFT-008: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-JAMMING: isolated, periodic and percolated aggregates; jammed fractal networks

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:3e3e94c465b6d6ce00dd045dc69c8b254552081b47d8d097bd767fd2b5d4a611; empirical validation
 sha256:e9b71eb7c6cbc54fe76f660baaae97656ee43d49b4f7b329a31907d08ca01564; external validation
 sha256:de98bcca2dd145de6f61f678163f9e0674e715fbc619cafe7e3a4438af80d18a; isolation
 sha256:d67297e5a3bb2916a5b2ad8fed4316f6213371b4ede665f8305e08f9d2db6ae3; custody
 sha256:614f7636152826b7aeaac8425e604f16628205311e690751066517f3123ee840.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels

- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

195. Responsive and stimuli-sensitive soft materials

Field: soft_colloidal_liquid_crystalline_granular_materials

Obligation: SFT-MAT-OBL-SOFT-009

Claim: SFT-MAT-SOFT-STIMULI-RESPONSIVE-009

Question, law and forced result. A responsive material retains stimulus, before and after states, exact response ratio and complete reversible history.

SOFT-009 uniquely retains stimulus-before-after-response-reversible-history with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-SOFT-JAMMING-BOUNDARY-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SOFT-009 axes before target release.

Boundary: Every positive finite SOFT-009 carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-soft-carrier
- stimulus-before-after-response-reversible-history
- complete-state-structure-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing

- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-soft-carrier__stimulus-before-after-response-reversible-history__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:1b260078ee7d532de055dd87f82fbf1097cb4f790983390d08fe3ce72f29bde6; generality sha256:7df11567a0d19011525771f01b10868a2163aa5054d58b73366773694af7d3db.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:31c0a8765d17f5887ce70470cbd83ba63b0104752073e39635c79dcaa10a3eae.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:92d8effa630df8e2150f994bbbe778b9dc363e19e7b7e974847d7c0931c9f854.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:3977982867f39418f30d97c98fe3ca31388609b2d1cbcbc33169ec57b729c975.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:cfd92c9753ae8b3770d0fe1db4468b108792eedef4aae518c0bc7239d6430bb4.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfefe90d4efca1cf7883413180fb87dc26 regenerated the declared census and sole survivor; independent certificate sha256:6e07aecc35ca43fd2dc0f1590ca9a5760bc4c7fbb797e07025cddb97e76b17c0. Derivation seal sha256:0e2e38d67de2dcff023966abb42ee542caa1939a1c7380342ab3fe24741aa686. Source manifest sha256:a5de963ef140b107a072cdf0d23ef6146c317de8fc491d38a2e4f7e3534e9fdc. Engine receipt sha256:dbabe6f570d78223954004a21806f541b639b186c7674d8482cf7bb995b5a53e at receipts/engine/model_admitted/SFT-MAT-SOFT-STIMULI-RESPONSIVE-009-dbabe6f570d78223.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-RESPONSIVE-POLYMER

Measured, observed, adverse and boundary records:

- SOFT-009: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-RESPONSIVE-POLYMER: stimuli-responsive chain cleavage; dynamic or reversible polymer chemistry

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:95b29dff367cc0ecde84caacaa4fff4f4333dd2025b2cc575111c3dde542940f; empirical validation sha256:0e996b713fcd7a2c08cbbf187e4cc8f14c1f9acd23a900c25682f2cce9f1999f; external validation sha256:f2cac700d0002286a66a4abdc2be6c546d0afcc752d5ae7366c059e233ef196c; isolation

sha256:0ddc557165413970b70e985bc4f82c8ec5e4337b3c5a00c00694d9852b349a1e; custody
sha256:cf295525b4a04a3f369e981d0ff623812387ebd9bf1101a83ef59a303ad1c6de.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

196. Active-material nonequilibrium organisation

Field: soft_colloidal_liquid_crystalline_granular_materials

Obligation: SFT-MAT-OBL-SOFT-010

Claim: SFT-MAT-SOFT-ACTIVE-NONEQUILIBRIUM-010

Question, law and forced result. Active matter retains agents and the exact input partition into organized motion and dissipation along the full nonequilibrium path.

SOFT-010 uniquely retains agent-input-motion-dissipation-nonequilibrium-history with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-SOFT-STIMULI-RESPONSIVE-009

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SOFT-010 axes before target release.

Boundary: Every positive finite SOFT-010 carrier with complete structure, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-soft-carrier

- agent-input-motion-dissipation-nonequilibrium-history
- complete-state-structure-response-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-soft-carrier__agent-input-motion-dissipation-nonequilibrium-history__complete-state-structure-response-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:db3542dbe49f146f9b01e1a83895115b5c750855c0a8eff5959d3f512910b186; generality sha256:3b25e940eb4a50134e2315309d8d49ea2274583271f2643d765df1f2aeeee604.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:ddf68b309b02f7aa9f903e9319b2f2d03e969fa00c45d04a04fdf6d3a887a918.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:b1adb9c695c1087c970f545269edda946a61aacfef51796ac9b00b379bdd5c51.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:8ac726d1823ef176960627178365d7076e5e38b9a50a6a8bfe799f08d3bc30f8.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:f7203116e4994f7d5fe915bd0f0f3f9322328ae6b2d4f0a3f38b5bc664e40baa.

Independent reconstruction and engine admission. Independent implementation

sha256:75bdf67067b137c517d58ede60194cfef90d4efca1cf7883413180fb87dc26 regenerated the declared census and sole survivor; independent certificate

sha256:58351d87824a8c25419b0e6bf08b48d76fe2d673dc578c4c8db915c6ae1b903c. Derivation seal

sha256:c981f8faf489e6bd9621374a8db73629919fc259182e37b76e411649c8336d61. Source manifest

sha256:28e34dde63992f7e5e038ef47dea4bce30c735931bb868f1e0589de6f9598f74. Engine receipt

sha256:aa19b4d8a60ac4e7fab62abab36c855e77df8c11914ee66649b1d0b0b13d5df8 at receipts/engine/model_admitted/SFT-MAT-SOFT-ACTIVE-NONEQUILIBRIUM-010-aa19b4d8a60ac4e7.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SOFT-NONEQUILIBRIUM

Measured, observed, adverse and boundary records:

- SOFT-010: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True

- NIST-SOFT-NONEQUILIBRIUM: far from the material's preferred state of equilibrium; sensitive to changes in pressure and temperature

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:2c7a4f5cb6212a68f5d9f0c3383ea57e2c84f53c1c4e172883871876e289137d; empirical validation
 sha256:4825cdc45c2d9c4a2a21f1c741af43d7383b656976d86eb0f13d59bbd1bf9a4b; external validation
 sha256:9042de152dd3fb957db74d65593858f21a78a17c3492426112c46e3a4f0fb99c; isolation
 sha256:4d84e4fa92146a30298cca69c326a51d8603c6aae6e62a3ca1877e91db1ae2c4; custody
 sha256:135487fc6aa637bb5b32ec429706334795172d79780c9be82c04e92fbd5b4332.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and all state, phase, topology and path distinctions remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Biomaterials Biologically Derived Materials

197. Biocompatibility as a material-interface boundary

Field:biomaterials_biologically_derived_materials

Obligation: SFT-MAT-OBL-BIO-001

Claim: SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001

Question, law and forced result. Biocompatibility is a specimen-, interface- and condition-bound exact partition of observed compatible and adverse responses, never an intrinsic unqualified label.

BIO-001 uniquely retains exposure-compatible-adverse-material-interface-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight BIO-001 axes before target release.

Boundary: Every positive finite BIO-001 carrier with complete material, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-biomaterial-carrier
- exposure-compatible-adverse-material-interface-ledger
- complete-material-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-biomaterial-carrier__exposure-compatible-adverse-material-interface-ledger__complete-material-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:79d1c3e95096542378cf3fef59ab0f16684e1c72cb3ccb2c469353898cf71580; generality

sha256:317b5bfc57f42620625a78412558550182860cde116b113d7d2125222deba97e.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:c7f7f90e1d199721f2d71558532197c251c4bc9cf32a73c69a846e86ebc6c2ff.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:4c9135013cdb73947184f2d05d31e55adf6de6627580494eded5201c7b46d27f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:450896d602c3307e5c0909a97c52acdaee91fc75cc014f4adfa917fe8bfc1499.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:213f30d1b5e459766e544fd81fdf8b37f3e14a86e140ac1a5347a513b887043d.

Independent reconstruction and engine admission. Independent implementation

sha256:0ffffc1bc958f8472c4f986d2585cc552b7490dc17de3cd831beb5f6077cc0dae regenerated the declared census and sole survivor; independent certificate

sha256:2e1810a2b879e5bbb067400e4766c76048ae84e6be300ff0325851b332ff1b14. Derivation seal

sha256:4490f8d3839b3650335dd7eb78af7f2c8b2f9f7f5616dec46c66356cdc2f2480. Source manifest

sha256:9949a66cc51fc3505c224b520c389dee7400135808c844c601aa3a0d3c9431cf. Engine receipt

sha256:a10aa7bcb625a83b2d3ebf9dd8e4ff28df0ae2d5d74ab13268cb4ef711b8bd2f at receipts/engine/model_admitted/SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001-a10aa7bcb625a83b.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-BIOMATERIAL-MEASUREMENTS

Measured, observed, adverse and boundary records:

- BIO-001: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-BIOMATERIAL-MEASUREMENTS: material designed to take a form that can direct; interactions with living systems

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:3ed08c927129a87f7a8880ae3fc07bfe15116af018d6da7109af0be416bf5ba1; empirical validation
 sha256:9f7c20f2f6d7224e822c249e382260ca6f991ce663e414e5d47a95f5b47d47cd; external validation
 sha256:216c7da3d79c726401bd9f7dd40eb2dde53ea3ea7f660dc1abd808612301162d; isolation
 sha256:052052bbe716a5fc02a96c3c8c8dbf6ee949f2cdf0289b60cfff231032d5c93b7; custody
 sha256:3b8c0ac4c4837b0c05a4570525fd9444dfde049dced071dcc821e1cb45be93a1.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, cell, interface, path and product distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

198. Bioresorption and degradation ledger

Field: biomaterials_biologically_derived_materials

Obligation: SFT-MAT-OBL-BIO-002

Claim: SFT-MAT-BIO-BIORESORPTION-DEGRADATION-002

Question, law and forced result. Bioresorption retains the exact partition of initial material into retained and resorbed carriers together with every product and time-ordered state.

BIO-002 uniquely retains initial-retained-resorbed-product-path-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-COMPOSITE-001

- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight BIO-002 axes before target release.

Boundary: Every positive finite BIO-002 carrier with complete material, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-biomaterial-carrier
- initial-retained-resorbed-product-path-ledger
- complete-material-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-biomaterial-carrier__initial-retained-resorbed-product-path-ledger__complete-material-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:f767ba673933edf5595b374cf8a937d541e685fbcc4318de5260bd5e4e79b6b1; generality`

`sha256:25cbc9c50aa430ca2e74e6a6d821023c17c6781bb98dfdb968289c7507c6843.`

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:9dbfd15eedaf5bdad1e4060d97fdc5ae0d054cb19209d2167a8ef27a16347401.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:e2a24a671d3b9dcf1ecffcd4b2568ca882e7a02f73ff9ec30f45ef46dc008292.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:0dcdfa5722cabdaee87f6a9b380fa84f24c1a647254e1eac705f7d8e922ce8ac.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:288feca742bee75d6990d71ff2f1ba9bb309b20430a465d592e6f293ee97f4c3.`

Independent reconstruction and engine admission. Independent implementation

`sha256:0fffc1bc958f8472c4f986d2585cc552b7490dc17de3cd831beb5f6077cc0dae` regenerated the declared census and sole survivor; independent certificate

`sha256:3f30f1026572822309ded35eba02c77a73034922a9f2eb20ca3375d24598af3d.` Derivation seal

`sha256:f39aad6e3a2a1eb3b42b57a47cfc4608e0b81c7c0e1815394d47a13d49ac6fc4.` Source manifest

`sha256:d3e9853158350a5273d74a5b9bef171a038b9ed9ea48efe684e2f15bb550ae4.` Engine receipt

`sha256:c09dea6fb0753c440817413d6c725e4766f0e60a5a0edd97b1ae4b07fdf61d99` at `receipts/eng`

ine/model_admitted/SFT-MAT-BIO-BIORESORPTION-DEGRADATION-002-c09dea6fb0753c44.json.
Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
Comparison passed: true.

Source identities:

- NIST-BIORESORPTION-RATE

Measured, observed, adverse and boundary records:

- BIO-002: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-BIORESORPTION-RATE: constant-composition dissolution system; factors that may control the resorption rate

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:f81236f20475d4f2af95ac6229aa8bffc6a419979d891f9e6402f6326e171da6; empirical validation
sha256:2f2edaaab21be8dad1cf7f09985f89572b2ec76784ead3b010d18d213f9ea5c1; external validation
sha256:958300d6552ad8ae63df54b0bcd015f3adeef1931d93858bd8514b56ec08346f; isolation
sha256:81649033dae466a0c373e698314eb6122d9da2d209d8368c3dcba839d2e7dc68; custody
sha256:53f3d301f096609328a578e7a12af221565c611d6b2092d9080b4eacf9d50072.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, cell, interface, path and product distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

199. Tissue-scaffold porosity and connectivity

Field:biomaterials_biologically_derived_materials

Obligation: SFT-MAT-OBL-BIO-003

Claim: SFT-MAT-BIO-SCAFFOLD-POROSITY-CONNECTIVITY-003

Question, law and forced result. A tissue scaffold retains pores, struts, links, pore classes and complete spanning connectivity as one measured organization.

BIO-003 uniquely retains pore-strut-link-class-spanning-scaffold with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001

- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-BIO-BIORESORPTION-DEGRADATION-002

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight BIO-003 axes before target release.

Boundary: Every positive finite BIO-003 carrier with complete material, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-biomaterial-carrier
- pore-strut-link-class-spanning-scaffold
- complete-material-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-biomaterial-carrier__pore-strut-link-class-spanning-scaffold__complete-material-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:3b86c2c8496ef725ed45129aa242d07deff00a09a68f5781394d54daa5863612; generality

sha256:14881b03c817f30f12db0797c7d2a01b8c831ee021fd2ce893303b02a6ff90fa.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:706a186ad91b8ecaf9a61e47138c5ac1966b1c6c6e959883b8cde75038ad915c.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:fe7020a3250f343f3cdf63abdc7112214bbf12340c21bd75d4b427836df3165.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:e3b351580e0cbe28feb845fda348e556031bafdaa3cfd855706dab356f44a589.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:46d94cf703beda38926aebe154b9a164207e04e231cb25d61ef9edb7b95834c9.

Independent reconstruction and engine admission. Independent implementation

sha256:0fffc1bc958f8472c4f986d2585cc552b7490dc17de3cd831beb5f6077cc0dae regenerated the

declared census and sole survivor; independent certificate

sha256:47d75cc50aabf01c165233eb6d9f01c9c95e7e59c73b4c3f08798134241f73e8. Derivation seal
 sha256:7ff4819baf9fc12a236940c6871f448ee97a538167f5f224d6a639c50b23e150. Source manifest
 sha256:fa49fe8db4be269d285a5e80c94a959794419a773bccf7a83edb927cf4449f42. Engine receipt
 sha256:1fe3a1e81c362902e3c4071d0e9dde9d04e9361616fa13f22c91314c8744f559 at receipts/engine/model_admitted/SFT-MAT-BIO-SCAFFOLD-POROSITY-CONNECTIVITY-003-1fe3a1e81c362902.json. Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-REFERENCE-SCAFFOLDS

Measured, observed, adverse and boundary records:

- BIO-003: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-REFERENCE-SCAFFOLDS: strut diameter, strut spacing and porosity; 47%, 60% and 69%

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:426e487f99ebf051582c43d8f7c79a85b1a48b090c9a519de7f73b4cab2039df; empirical validation
 sha256:3474e301e899fca9545f007fd337501afa71f7c17fa2ac430adec92e985c5f8b; external validation
 sha256:b1b50540ec36b011d06574a630e2367ca1542a1517dd2c9de492574aa7556670; isolation
 sha256:599f17fe396053f292d62928251b4bde6a1fdbf7786d7e761d4035c250205f8b; custody
 sha256:1329336e8f81b918d55a8a101a8547cf1d700446b4de75e45e491e0295b7dbb7.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, cell, interface, path and product distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

200. Cell-material adhesion handoff

Field: biomaterials_biologically_derived_materials

Obligation: SFT-MAT-OBL-BIO-004

Claim: SFT-MAT-BIO-CELL-MATERIAL-ADHESION-004

Question, law and forced result. Cell-material adhesion retains cell and material identities, interface conditions and the exact presented-cell partition into adherent and nonadherent outcomes.

BIO-004 uniquely retains presented-adherent-nonadherent-cell-material-interface with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-BIO-SCAFFOLD-POROSITY-CONNECTIVITY-003

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight BIO-004 axes before target release.

Boundary: Every positive finite BIO-004 carrier with complete material, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-biomaterial-carrier
- presented-adherent-nonadherent-cell-material-interface
- complete-material-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-biomaterial-carrier__presented-adherent-nonadherent-cell-material-interface__complete-material-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a8d2fe3a0a443d6537f7d651d230edf6f75bfb92459ddf3f7ed6ea4e4fd15f87; generality
sha256:591b4489c3647ef1d374002d4d959de5523b6c8dd7783fdc5588c0e0ae33d72d.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:e9bd3ba547ddc2b08edc890e28ed60832d664d36a0b713386c3da0f8551a57c3.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:3f060323321ab5d7e9ff4975112c2cce85e77b91f64a7a5107d7cf3a54ce8f6e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:52a48058658c6825b671604dce2d8d56c346894116e908842eb09e36a0b8da71.

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:7adb51f730ee20e744d68555647ee8fd4df19b4e6e907e833c8ae01d5f573b10.

Independent reconstruction and engine admission. Independent implementation

sha256:0fffc1bc958f8472c4f986d2585cc552b7490dc17de3cd831beb5f6077cc0dae regenerated the declared census and sole survivor; independent certificate

sha256:2657d10b6025c434786c26524c43fe5de6f55856bac8e572c899b41676e67073. Derivation seal

sha256:8d1397665b954c26c99582597ae44cb7d1f24082768332f4872759887d48d4d8. Source manifest

sha256:9b37a58602e3fff98f47fb45a85808eb5d6bde760bdf37e2c89bad3cc8b52e1e6. Engine receipt

sha256:9b7533748ea7aab0c1fb99c70effd14297d8563dbd71e9ddd7db7bcf1b8e4c3d at receipts/engine/model_admitted/SFT-MAT-BIO-CELL-MATERIAL-ADHESION-004-9b7533748ea7aab0.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CELL-MATRIX

Measured, observed, adverse and boundary records:

- BIO-004: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-CELL-MATRIX: Cells adhere to type I collagen; mechanical stiffness, and supramolecular organisation

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:744d65b453171a3d754534023cf319d13d6ca98badb9636d9f2e14e15076f7df; empirical validation

sha256:fbcb14c1cfcdce00ded18ee0612e8902dc66edc920a227e52fbb160711f7071148; external validation

sha256:42ebfc1febb2ffe519e89b44062be103374f58f6486be7034a0a61ff911d80ff; isolation

sha256:dbd0157796f29d570b158489330bfbc2d2458c260560d2d4e42de1a66359cada; custody

sha256:15d8ddff89d9fd2d0e088c0d55b8620c6924b3a2d612653b21b447e20e025273.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, cell, interface, path and product distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

201. Mechanical matching at a biological interface

Field:biomaterials_biologically_derived_materials

Obligation: SFT-MAT-OBL-BIO-005

Claim: SFT-MAT-BIO-MECHANICAL-MATCHING-005

Question, law and forced result. Mechanical matching retains the applied load partition across material, tissue and interface with specimen geometry held.

BIO-005 uniquely retains applied-material-tissue-interface-load-partition with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-BIO-CELL-MATERIAL-ADHESION-004

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight BIO-005 axes before target release.

Boundary: Every positive finite BIO-005 carrier with complete material, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-biomaterial-carrier
- applied-material-tissue-interface-load-partition
- complete-material-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-biomaterial-carrier__applied-material-tissue-interface-load-partition__complete-material-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:c3718d422b80969d684c61a520adefe3b3b7b47c818a295d58af2543d2e6cbc1; generality sha256:1bb17b50b337a824789c9ab93d261ed9cb68ad29527195d732bbc530cbcd1c29.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:9b769a12d252448b511c6fe7285fd8dc4bfc12746ec3efd980d3d0458fef0c.

- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:fcacf95b0bee8cb4e8d2eb39eabef6d434bbc42a37f2aad809e46508b60ad7f8b.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:870936546803b71cd6e123c533b1d4da7a45f66ca2321fff45125fc8acc81d88.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:e53e0cee9704849c1f2b9875a71e5f6d41a5ddc6ad37d6a877034860cda72e28.

Independent reconstruction and engine admission. Independent implementation

sha256:0fffc1bc958f8472c4f986d2585cc552b7490dc17de3cd831beb5f6077cc0dae regenerated the declared census and sole survivor; independent certificate

sha256:819ecfbfaf9f540066aa85fbd7f08e8cffb5fa6bd7ce23c76c6a536d97d1594c. Derivation seal

sha256:ab2dcc78052daf1332ff0faf96237a566a61d82c85c849bbb48539ce9b54b3a3. Source manifest

sha256:27fb84cef02cc8bf1b16ea83cd4d4d984c69554ac32a4d81961f1faa2fcf2584. Engine receipt

sha256:240cdfc025d172d1b3bf2421f5bf8a9f503e4b0ce16c22b4e959f56f6bf0429e at receipts/engine/model_admitted/SFT-MAT-BIO-MECHANICAL-MATCHING-005-240cdfc025d172d1.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-BONE-IMPLANT-LAYERS

Measured, observed, adverse and boundary records:

- BIO-005: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-BONE-IMPLANT-LAYERS: porosity needed for bone growth and the integrity required; strong, fiber-reinforced CPC paste

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:18fcddebe15c7913899ccaf3d930acb91f23d19f9a333db7c96bc979b0501e08; empirical validation

sha256:c9a7a8a88d7d22a4545c410a92fb922037841ac11e75a5c112ab177fc85d5f46; external validation

sha256:72f9de984170cf2bb4c07695b4c94a4f1fd325386218af02c9416234eaae0c05; isolation

sha256:1dbcc0ff0e04a7acdd2deca81f0327dbd2474a1bd2682cb92e42e3ded2182076; custody

sha256:0288c4d01d30be46a40221bbbab545a156652f752712d1114ebc75b6b46badae.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, cell, interface, path and product distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

202. Controlled-release material transport boundary

Field:biomaterials_biologically_derived_materials

Obligation: SFT-MAT-OBL-BIO-006

Claim: SFT-MAT-BIO-CONTROLLED-RELEASE-006

Question, law and forced result. Controlled release retains the exact loaded-carrier partition into released and retained quantities along the complete material path.

BIO-006 uniquely retains loaded-released-retained-carrier-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-BIO-MECHANICAL-MATCHING-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight BIO-006 axes before target release.

Boundary: Every positive finite BIO-006 carrier with complete material, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-biomaterial-carrier
- loaded-released-retained-carrier-path
- complete-material-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-biomaterial-carrier__loaded-released-retained-carrier-path__complete-material-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:1a3fa254c6ef509dd23e1a7a07b298632a5c89316a3ff1cb4401c163a920029c; generality sha256:cb5a197bbf8011f139ab07add43f779238cc7356370730f5b6dcd61cdc80865b.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:5616afe0d1d561434ba334957a64ddc3dc68bd958146304ddc490a74c82e4468.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:b1ced58e241c04543baba4d3641e27606f287782edf134a274750d7c2c71a81c.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:e854dd5f5ba9705b53ac7efe0002e0d441cfddde23e53fb8fa1b8d1fb4fe223e.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:e1f10e711a89fb01c6bc023a03e80cc3b80aee07b7a95db155d82e806b87eb04.

Independent reconstruction and engine admission. Independent implementation

sha256:0fffc1bc958f8472c4f986d2585cc552b7490dc17de3cd831beb5f6077cc0dae regenerated the declared census and sole survivor; independent certificate

sha256:4ff60efe32e6d3f2f087324d7bab698ef6599c6490be447e92b498e77bf790ec. Derivation seal

sha256:ad1392346b839103131341003c98c2ec5fbaf811a8bcb4a96640c318b6e6e42c. Source manifest

sha256:813ba03b7d15bb9af909407674026faa1bf69d6e613614eb59f85e3bd30f507a. Engine receipt

sha256:10c4443b381f4482ad2889e538708109711794e1c8c8aac77d6c7d66dac4d3a1 at receipts/engine/model_admitted/SFT-MAT-BIO-CONTROLLED-RELEASE-006-10c4443b381f4482.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CONTROLLED-PROTEIN-RELEASE

Measured, observed, adverse and boundary records:

- BIO-006: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-CONTROLLED-PROTEIN-RELEASE: Protein release was defined as mass of released protein/mass of total protein; control protein release via microstructural tailoring

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a5784dd1aefb3c9e040683b8d1f862da83e98d9b7a6976f7e1ef82ced551d061; empirical validation

sha256:27a945fb05965e968d5d1d90f3fe662eb9720fe418eb2ce52604813cbc8630f2; external validation

sha256:110dd37d78460241aa9e0648403e56b4184f3e6a67af5db502b2b91ed135f7e1; isolation

sha256:abb849a848abb1566320252601cc666e88cc81fccbac812e72f1611830e6123; custody

sha256:040de243ca88542bf1d5bf6007a0cd78d90ebaa5cd0349a623a58c1c51d52519.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, cell, interface, path and product distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained

- no failed attempt retires an obligation or changes protected authority

203. Mineralized biological material organisation

Field:biomaterials_biologically_derived_materials

Obligation: SFT-MAT-OBL-BIO-007

Claim: SFT-MAT-BIO-MINERALIZED-ORGANIZATION-007

Question, law and forced result. A mineralized biological material retains organic, mineral and pore supports, exact parts, mineral identity and structural organization.

BIO-007 uniquely retains organic-mineral-pore-identity-structure with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-BIO-CONTROLLED-RELEASE-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight BIO-007 axes before target release.

Boundary: Every positive finite BIO-007 carrier with complete material, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-biomaterial-carrier
- organic-mineral-pore-identity-structure
- complete-material-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-biomaterial-carrier__organic-mineral-pore-identity-structure__complete-material-interface-state-path__specimen-method-condition-scale-uncert

ainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:ab0f83e1dael65ac468609c2d22f055e90e864361e5ca7519caal8b10a87b6ce; generality

sha256:5c41755fcb74a5173ad061cd3ea2e24021b54e2fbaffd98aca0c3ea11956518b.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:2facc2f5136d016c4a46de5d8ef33525e9c9b86a88836a3eafe7d7fc93eb2c3c.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:01833520e08584b271531dedcb67405acaed1aa3ce9759903da8eb498c99385c.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:1f0a9e6eb443d096b28b07f99d377497a37bfe0d37e990dad8680cb92a5ecd20.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:82f0e6c5e367e59424b4db1d8da095000b93aac05854cb349021ccaf0f8483b2.

Independent reconstruction and engine admission. Independent implementation

sha256:0fffc1bc958f8472c4f986d2585cc552b7490dc17de3cd831beb5f6077cc0dae regenerated the declared census and sole survivor; independent certificate

sha256:23ae47317589c4cf2644da440beb4c3fd9f45e11bb57572e2d9cb90d5b49f32c. Derivation seal

sha256:95cb15f114ad12b8d2fef1aaf4c220923c67389f729e0a76454a71caedab7654. Source manifest

sha256:f044f0fe4ba943b32b03c62c2050486795449fbfde5c3749b0f9a3991f742226. Engine receipt

sha256:bd00986cccd114e52724965d3b2775686ee13a2353b6a88760f34404a1b4bd07 at receipts/engine/model_admitted/SFT-MAT-BIO-MINERALIZED-ORGANIZATION-007-bd00986cccd114e5.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HYDROXYAPATITE

Measured, observed, adverse and boundary records:

- BIO-007: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-HYDROXYAPATITE: calcium hydroxyapatite; crystal size and morphology, surface area, unit-cell parameters, crystallinity, and solubility

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:3a99d23e98f0c045e72b860179322a2e2fb983a38caaa3769120cd80282ba72f; empirical validation

sha256:24c36e7af0f200b9fd365c06a57642c3c85c59a37dd097de2ac77d32290ad1f4; external validation

sha256:cd39ea3d522975220e62ad48caffab12608d9163dfe13cecf9691dacc426cf7e; isolation

sha256:14b5e7c7fae3ee5fb3f8b3a511a369ab5e594bd910820afc1159095ffa832bca; custody

sha256:063d229b608a2e738844f6f7cb05dd8d1d841dd70bf43f21ce801f8572fd3067.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise

- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, cell, interface, path and product distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

204. Biologically derived and biofabricated material identity

Field: biomaterials_biotologically_derived_materials

Obligation: SFT-MAT-OBL-BIO-008

Claim: SFT-MAT-BIO-BIOFABRICATED-IDENTITY-008

Question, law and forced result. A biofabricated material retains biological source, cell state, complete process path and exact input-output custody.

BIO-008 uniquely retains biological-source-cell-process-input-output-custody with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-BIO-MINERALIZED-ORGANIZATION-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight BIO-008 axes before target release.

Boundary: Every positive finite BIO-008 carrier with complete material, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-biomaterial-carrier
- biological-source-cell-process-input-output-custody
- complete-material-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace

- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-biomaterial-carrier__biological-source-cell-process-input-output-custody__complete-material-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:140516fe72e7e62f23c4603dcdcd8fae6a99c7f3975ce16651e3efecc7bc47cd; generality

sha256:c65665d42714a5a211213bad18a2d60d75e21409f363769aacb43eca4261ed10.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:cb22542e1c3b8c5225574ccf7448ed945527f48d5845f983160fe631ec6a5fa1.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:26c8da4956e8c305b2d1009cbd48e48f8576eabae5a094edd887f277047da81e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:97060904d7f23491cf51295db4909f2f68101617b413b3e571ddce3d8b203334.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:50e2e66722ead31c87d8e610365d5d6a5b5ee4324ea7945eeab49ed9b6814825.

Independent reconstruction and engine admission. Independent implementation

sha256:0fffc1bc958f8472c4f986d2585cc552b7490dc17de3cd831beb5f6077cc0dae regenerated the declared census and sole survivor; independent certificate

sha256:5d1b03d373d6ba348219a440035efbb03cf606b3473ba7d8596f501436c16f36. Derivation seal

sha256:3aede996aff443053cb9410e870fb80a2841860c07ceeceb39715988ea8a74b5. Source manifest

sha256:71118e1503c11631292497416f1c47fbbba41614cc6d71280b4393438d1af668. Engine receipt

sha256:f39a1f35f33f68a03fb28adf80abbc665c69e85eae15894b03819db2fa269af4 at receipts/engine/model_admitted/SFT-MAT-BIO-BIOFABRICATED-IDENTITY-008-f39a1f35f33f68a0.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-BIOFABRICATION-METROLOGY

Measured, observed, adverse and boundary records:

- BIO-008: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-BIOFABRICATION-METROLOGY: structure of the constructs, cell viability in the constructs, and functional capacity; collection, validation, and standardization of reference data

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:d208305b8f0269af1ddd2eb588847cc919d38c75f92964f228276757a0c8c692; empirical validation

sha256:480ab72bf114ded61d577e075db8bbc990f3ba771849ac88a4c10d2d890d1be6; external validation

sha256:daba402f0088331a2a781fd44911b0a9534084e0171c29ec54032d76a7b3d9b0; isolation
 sha256:e8a153cb4e53289d846bfa2589e0e3202dff9713f5f42a33de10e9a127bcc7eb; custody
 sha256:09e4bb2d338609a5605d802f32963d20680b54b35778211d4d80d2c199a6dfe9.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, cell, interface, path and product distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Nanomaterials Two Dimensional Quantum Materials

205. Nanoparticle size and shape distribution

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-001

Claim: SFT-MAT-NANO-SIZE-SHAPE-DISTRIBUTION-001

Question, law and forced result. A nanoparticle population is an exact finite ledger of particle identities, three counted dimensions, held shape labels and the complete generated distribution; no mean or fitted continuum replaces its members.

NANO-001 uniquely retains particle-identity-three-dimension-shape-complete-distribution with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight NANO-001 axes before target release.

Boundary: Every positive finite NANO-001 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier
- particle-identity-three-dimension-shape-complete-distribution
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__particle-identity-three-dimension-shape-complete-distribution__complete-generated-discrete-organization__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:6a75bfl3b4dc5d53a0836e274b8ea30bd88623840049f5c2b336d6b218c7c86a; generality

sha256:40070048139af3d4a37a23afa055330ed221fa7d698795239429c90aff528288.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:fd550fbbcf0a9736a53cf673fbde48cf513963848566a7fa86670eec2c3d5072.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:ca799500f522df2ebe2e44f61be2cc35fb4f08adc3d3e51a73217b5188a70d6c.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:c7b00969b5bed57fc0dbfcc9b9e4bc8eaa099afbd60e05cdcfa8b012c2baec47.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:e9dfac62b80481ab81d592a0e2b357aa1655dc49ec98078ddc2b2bf451a7f14a.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the declared census and sole survivor; independent certificate

sha256:558aa8c4c5388ce4a29c7628cca73b3525d0af5aa7c14f2c0594d53a4c6c13d8. Derivation seal

sha256:d617bb105503a3de01f5f33a9f570235d2bd21d1087ef2dd7f58ce3e2c2d9ba9. Source manifest

sha256:f8182fdccd07397594a5056130fceb5999cd305e5e73ff78ab29caca8cd4d246. Engine receipt

sha256:c74d6d61481d890ee9570f1a2ae244149babdc2e4c76293cd4blae183775800a at receipts/engine/model_admitted/SFT-MAT-NANO-SIZE-SHAPE-DISTRIBUTION-001-c74d6d61481d890e.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-NANOPARTICLE-SIZE-SHAPE

Measured, observed, adverse and boundary records:

- NANO-001: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-NANOPARTICLE-SIZE-SHAPE: depend critically on their size; both SEM and AFM techniques; more accurate 3D shape

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:b0285f20cd98833a896f86d58308283d4f2fd7cf2ab2bd71fe7e2da3faed51fc; empirical validation
 sha256:3f9c82e021eaf11c4c892a7b57e45f0c46154d52dae07d69aa587121d09ac13b; external validation
 sha256:4481f5f07a14916fec77620e5c67b6b5a76549f5880d89765f82a156558affb; isolation
 sha256:0c34f6fb68104521271116558e4eb1fe144f057dc6fb1117f9a58de9a7432cae; custody
 sha256:dc38469b43b12bd56fc8a1d0cdb4867b48a716558886608f2b5336fcab1a1729.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

206. Nanowire and one-dimensional confinement

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-002

Claim: SFT-MAT-NANO-NANOWIRE-CONFINEMENT-002

Question, law and forced result. A nanowire retains one extended counted path, two finite transverse supports and both held terminal labels; one-dimensionality is a relation of supports rather than an imported continuum axis.

NANO-002 uniquely retains one-extended-two-finite-terminal-held-wire-support with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001

- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-NANO-SIZE-SHAPE-DISTRIBUTION-001

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOHING.

Complete candidate grammar. Complete literal product of eight NANO-002 axes before target release.

Boundary: Every positive finite NANO-002 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier
- one-extended-two-finite-terminal-held-wire-support
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__one-extended-two-finite-terminal-held-wire-support__complete-generated-discrete-organization__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:7aa564d62d5c690278a0f3a12b3b41a669db623f4bde5d061e1aeec781bab5b0; generality
 sha256:244e02595bf65e716dc3bfb26a589074fe9e42ac202672e78b8df5bd5e77e6a3.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:4d632347897c6e13b31efc849f6498f59f8fc415b0ad17818574f0046cb87a1c.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:4b0328274b9dd45f06a4b512ef971c75495d4dafc29fc56570aa568daa0f1a4c.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:a62103f329fa5f4dccf0023806a2ee2047d4370fc2d26771b50365526368c66d.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:b76664f4f872a3b9525d6c27bef9b939992ec9d94d1730680e04bfdcf459f0c8.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the declared census and sole survivor; independent certificate
 sha256:a3817bb235bb8d54d6481d510c4fc9aaa0f06232567d6ff578e783493746e46. Derivation seal
 sha256:bf6627eb93ccc7f91061be264a9ad7ddd63825bd29cb95ea5b423b54f5a73eec. Source manifest
 sha256:22635d6a6856ec157895aba42381c95007a213362de2ac9424b14b55376fb878. Engine receipt
 sha256:96ceb3232e229582f916a4da6fad2e9aba46d973d6b4f17e110e98743bc49fab at receipts/eng

ine/model_admitted/SFT-MAT-NANO-NANOWIRE-CONFINEMENT-002-96ceb3232e229582.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-GAN-NANOWIRE-GROWTH

Measured, observed, adverse and boundary records:

- NANO-002: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-GAN-NANOWIRE-GROWTH: control of nanowire location and diameter; quantum wells and quantum disks within the nanowires

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:32ef89e6284cf95c8bde1604974995737986061662e8592180135d96ee62ab26; empirical validation sha256:d73d5bc8344e019ebd4f50d5db0affed26e6d32ae511b48ba2921cdd1742e74c; external validation sha256:5e3ff900e21daaae089a16b53eb7e943860e15935a9777d7c9cb779c73c46207; isolation sha256:aa32de1bc9276d2c1fff9cb6a48c031a8758e70306d43bf2af8608a4c80c2b22; custody sha256:8723634e45d40a21ba5fc47a520c937d988ef226ab4af6c7c3ba6e2841c15309.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

207. Two-dimensional layer and stacking organisation

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-003

Claim: SFT-MAT-NANO-LAYER-STACKING-003

Question, law and forced result. A two-dimensional material stack is the exact ordered word of layer identities, adjacent interfaces and registry labels; no layer or interface may be erased.

NANO-003 uniquely retains ordered-layer-interface-registry-word with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-NANO-NANOWIRE-CONFINEMENT-002
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight NANO-003 axes before target release.

Boundary: Every positive finite NANO-003 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier
- ordered-layer-interface-registry-word
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__ordered-layer-interface-registry-word__complete-generated-discrete-organization__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:3a51d033558ba7621f1f9ef8ab86451af2e6b722520dd0189b7fc584247e31ab; generality sha256:7a37f5b33194032bef4fbe5687a8e35333c805672f22a1bee37b466318031cc3.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:d4aae6cdabf9ed07292da97d7d4909886280616137d634ed92b05dd25e7f38a9.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:413b980c9f652b378a7bc78942c3757648374ca460453c8c2063b673ca1a1a49.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:552f220d92cf344c3a069aca4054e48da5edcf69fe2e3bfea307e0f6e56552a9.

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ee0feccf5bc96528e66e40c4fd1738f43dab27b1c58b1cfe10c3312813aebef2.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the declared census and sole survivor; independent certificate
sha256:1bff5b7d1acf343c142e1427046363bd2a036e5996c1e8155ab817b179f831fa. Derivation seal
sha256:ce55f71db3066ea0921bd0da62d52f10e44fb06180c21c51df1228a0e6728611. Source manifest
sha256:ad48a7a291c63f1dd2532fbc95337a58e4dba9fe16f0836452ad904d0cf22fd8. Engine receipt
sha256:2e6d121c23ede76ba7490885b419daf8f70d23a15921228978606bb5b0fba16e at
receipts/engine/model_admitted/SFT-MAT-NANO-LAYER-STACKING-003-2e6d121c23ede76b.json.
Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
Comparison passed: true.

Source identities:

- NIST-QUANTUM-TRANSPORT-2D-STACKING

Measured, observed, adverse and boundary records:

- NANO-003: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-QUANTUM-TRANSPORT-2D-STACKING: novel two-dimensional materials; custom stacking station; twisted 2D materials

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:5d2c2e30b65287112d5ad146792a73b8cccfcfce92d4ad850ebb6cb341a96392; empirical validation
sha256:f674bf174ce0d12c7ff6b1131b239c034b1e7dedb578364f7be2ec9bbe4ac743; external validation
sha256:d7668acdfc5ad177ffa30592251617c818b15eaf5816d07186a5f5853f7d0e35; isolation
sha256:1ba13be60b3fcd17a46677c46e0fd590df133bc9c824004b97c8e01bbd33da0; custody
sha256:d00ac51a28385c6fb64c0e375ecba58961c20972a153fc4f5d6c62d03696b260.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

208. Quantum-dot finite confinement

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-004

Claim: SFT-MAT-NANO-QUANTUM-DOT-CONFINEMENT-004

Question, law and forced result. A quantum dot retains finite counted support, boundary support, carrier identity and every distinguished level label as one exact confinement record.

NANO-004 uniquely retains finite-site-boundary-carrier-level-confinement with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-NANO-LAYER-STACKING-003
- SFT-MAT-OPT-EXCITON-DYNAMICS-010

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight NANO-004 axes before target release.

Boundary: Every positive finite NANO-004 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier
- finite-site-boundary-carrier-level-confinement
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__finite-site-boundary-carrier-level-confinement__complete-generated-discrete-organization__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:80b49fb9dd043f3ff64a2218f3d9191e9499418d4e9c626e89faf79e147041b4; generality sha256:eaed12c1acf112549a74bb6acabfe50b396d6d31f3715d2f5f677d6d6dffff94.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:9ec76ebb90f553e1773f9ef22f2837b6be29bf0b8eb679d7fd29a1c1e7caeb79.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:75aa43798dfbae00b1d1f83adc509758aafe19d262c28479733f8f4f5f479559.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:2e340320e48797f5cf07c35317718c77d207ff6cd0415e78d7b709cbd54b3c18.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:183db9649f6dfe894de6549f03fb4d5b58d6b3a107b35039725c970df7e556ef.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the declared census and sole survivor; independent certificate

sha256:81bddeb5490f34a48ae2ae1e9d680dde6d5c5dd89978cb39778324a5e04d497. Derivation seal

sha256:7bc4a005ea9e180b126748f7b9ebdf52cc1502b10b822745a9dee50910058904. Source manifest

sha256:ac68d24890b67297a74331e2bb6a8eb77643107129a80e1aca58cee3419be2dd. Engine receipt

sha256:237de22de91f905cf55e4788c9fab3bf34a8a4b026cdcfbc46f817a87b783d9d at receipts/engine/model_admitted/SFT-MAT-NANO-QUANTUM-DOT-CONFINEMENT-004-237de22de91f905c.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-NANOWORLD-QUANTUM-DOTS

Measured, observed, adverse and boundary records:

- NANO-004: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-NANOWORLD-QUANTUM-DOTS: semiconductor quantum dots; electronic, optical and mechanical properties of ultrasmall structures

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:9c8043ae3c73a18b3ab5cbc8f4382936b77c3840de2d65b38bee60f78218843d; empirical validation

sha256:e864b149940fe541b8a771fc571c0e6928b029381f06c3650a58f8cfda48b662; external validation

sha256:95d5cfa232cdb9ab7e36e827cbaff8143cc3dc59031b5a04b38d97e1d4b22a28; isolation

sha256:dc1f8e9aa45f13692c0d2c5ed595a84332f4290600852d6cb4a0cce57aaa08b7; custody

sha256:40370335c359a4ae130d41196628280891bf6c3987b5c817a53cab743620d74.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

209. Surface-to-volume dominance relation

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-005

Claim: SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005

Question, law and forced result. For every generated cubic cellular support with positive interior, exact enumeration forces the boundary-cell part to fall under one-width succession; smaller supports therefore carry the larger boundary part without a fitted continuum formula.

NANO-005 uniquely retains counted-boundary-interior-successor-surface-part with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-NANO-QUANTUM-DOT-CONFINEMENT-004
- SFT-MAT-BULK-SIZE-SURFACE-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight NANO-005 axes before target release.

Boundary: Every positive finite NANO-005 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier
- counted-boundary-interior-successor-surface-part
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure

- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__counted-boundary-interior-success
or-surface-part__complete-generated-discrete-organization__specimen-method-condition-
scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-s
uccessor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d45712f7f25e8d2cd7371b02d35490201c01df56b4aa6f795648b7a192ddd7d1; generality

sha256:0eeef1265c12fe956d50af17c806dcdb076ae061d26e5ea3128f39b43e879bc.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:8200169d3cca2e2576a374522628f1e3a7eead14d91a2053aab0c2ac7dd517ee.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:eac6bdae85bb011691d89d17de3cf4a3cddc3666a0e5de4da78f1389555ac9ee.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:fe13bbbed80c59323958f70bd259b78660d54efd7a6f9ccba3e329dc3ffe4c9a2.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:7f15b8bdc28b4c560462745ba8f6ca371d4b667cc8fe078d1ec87eb23f2b97fe.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the
declared census and sole survivor; independent certificate

sha256:0d7443c00765995e0a558defc334e4050bf75c9d7ea8f4c0275e498cf82f8f9b. Derivation seal

sha256:2a7c9f655837de3a3b5ceb93212ed208ff74137bb4143dc52b9204758be6000d. Source manifest

sha256:bb3d1e34eb05b8263e1ea638309e89d079f12bf89d71b9e97322dfe9ba2bca5c. Engine receipt

sha256:57c22baf89837b4006bf20685ec4f7400457949c477ec442f2e4e8b523107293 at receipts/eng
ine/model_admitted/SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005-57c22baf89837b40.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
Comparison passed: true.

Source identities:

- NIST-NANOPARTICLE-SURFACE-AREA

Measured, observed, adverse and boundary records:

- NANO-005: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-NANOPARTICLE-SURFACE-AREA: mass, number of particles; surface area units; greater surface area

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:70a5ce3642bc689d586ffe80d7c09ab0934507b65c637fc0793b888cbda7398d; empirical validation

sha256:a0fc6d50260151bcf091840090dbb9fbf98327beea883703a897fadfde4156d8; external validation

sha256:9c424af8d346e5f9f8cd69e20b40357c5ac8e3d2b1d7c20fd29e4ca797b3b87f; isolation

sha256:ca6128e4be71640aa82e65c903ba628c83294e3b996c55cbe2e2ef95ad75704e; custody

sha256:2c6acf47f6b5223beb70af222bae9ec62c2064496ab8fcc6652e9fb24b0d43c6.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

210. Nanoscale phase and melting boundary

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-006

Claim: SFT-MAT-NANO-PHASE-MELTING-BOUNDARY-006

Question, law and forced result. A nanoscale phase boundary is an exact specimen-, size-, method- and condition-bound ordered observation separating two held phase identities; it is never promoted to a universal fitted temperature.

NANO-006 uniquely retains size-phase-method-conditioned-transition-boundary with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-PHASE-TRANSITION-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight NANO-006 axes before target release.

Boundary: Every positive finite NANO-006 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier

- size-phase-method-conditioned-transition-boundary
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__size-phase-method-conditioned-transition-boundary__complete-generated-discrete-organization__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:86e240a263ef2ab7f1de5aea5c40fd06a1c200f31da7cb69f8f1a9b42e7e4355; generality

sha256:94223ede8db129e1e5823abda7bcb6618f5485e780c27f8ec3098d61b0961920.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:3a0c0cffa8ddc1a81aace5423659992eda0c8d9a2034c8f7d2ff8463245c3e87.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:3a9fdee4ec9c311c30b469bd87e68493bc19e7b29617e035a26adc4f18122567.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:f00ab62d3745e3df8848b298cfc93e01ffa544d662ae69125045613891cfbace.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:e8594b7ea33eebc87084c84869534abd2b73830a99457d505b7a99faefbd5b8b.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the declared census and sole survivor; independent certificate

sha256:b0784e68bf28db0a689b43a26e42e829a4dc295e888e95c23104940c27de1046. Derivation seal

sha256:7aa153ccb5743e323051ebec9903be8c7339fd7968936969a0e7b141e7ac0c52. Source manifest

sha256:891365354cdb4ffca8fa92d82aa8706673f715af94ddf9cd0c852e0c2d6c6c3e. Engine receipt

sha256:d70a707f104b7d602449be424ef12ff523c1295ca6b76932e189ca72ec551393 at receipts/engine/model_admitted/SFT-MAT-NANO-PHASE-MELTING-BOUNDARY-006-d70a707f104b7d60.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-NANOCONFINED-FUSION

Measured, observed, adverse and boundary records:

- NANO-006: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True

- NIST-NANOCONFINED-FUSION: temperature of fusion T_{fus} of a nanoconfined liquid; T_{fus} can change significantly depending on the nanoparticle/nanopore size; nature of

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:6793123f5cb7e26985b72f5bc8c07b715284676b9e18281b5a91a0ae0aeda2dc; empirical validation
 sha256:52376332ca9a40de98f74a8a18809b593bc9c795ff43ea9957602afd27316417; external validation
 sha256:a66990b361523d41e055d2501a38266f1475eb99009fb26d52bd9a2b4f13d6f4; isolation
 sha256:e101e832b825df4c4dabb740285187fab4a97c605246819d30243038b083b83e; custody
 sha256:9c6bbba9137f78f82a850a9fb7bbc016744c6c265bc87374fba0c318bb822a80.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

211. Quantum-material collective-state classification

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-007

Claim: SFT-MAT-NANO-QUANTUM-COLLECTIVE-STATE-007

Question, law and forced result. A quantum-material collective state retains every local state, every correlation link, the joint-state identity and its topology label; the collective class cannot be reduced to disconnected local labels.

NANO-007 uniquely retains local-correlation-joint-topology-collective-state with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-NANO-PHASE-MELTING-BOUNDARY-006

- SFT-MAT-TOPO-INVARIANT-001
- SFT-MAT-TOPO-BULK-BOUNDARY-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight NANO-007 axes before target release.

Boundary: Every positive finite NANO-007 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier
- local-correlation-joint-topology-collective-state
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__local-correlation-joint-topology-collective-state__complete-generated-discrete-organization__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:bac50a7bdbc760f6a859742520e5dc2bea070c50ae86bfl1dbb0bb635c82e4aa1; generality
sha256:3e75b032d36054becb44c6f7d1cb27e2b241164e99165e8f6472054e1a104e45.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:996d92dae49c30749dffec15fea8b1d3d1e541a91b5724f50456b4510c440577.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:a9f0db88ff718c060fe486b359c30bd98c21a1ed2b5696ee7a36b031de5b1d6c.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:3bf447bbe0ce06b7c9be5f85b1e8bf5913c9a726d6e762793b2422790d619755.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:45e10822b2b9d0cad5ace7abd785e79e623bce0fd6c6643804f8708697e6a0cc.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the declared census and sole survivor; independent certificate

sha256:b4e1f86b8d02aca7655cbe1aa3549724fb1a347004867206e3416b395a59f936. Derivation seal

sha256:89364dd35b0ac482dd5fd292470d8789d6345ceb9cfbbb96a180899f6e8263fd. Source manifest

sha256:c3035ddc723256a1847276792ffeed8b0e78e47537a7380b28c4a7f528c3125d. Engine receipt

sha256:1a99613267e755337aa3ddb1966d3851081db5cbd2d294e7f0f28e6c10dd5d26 at receipts/engine/model_admitted/SFT-MAT-NANO-QUANTUM-COLLECTIVE-STATE-007-1a99613267e75533.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MOIRE-QUANTUM-PHASES

Measured, observed, adverse and boundary records:

- NANO-007: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-MOIRE-QUANTUM-PHASES: quantum phases resulting from the interplay between electron correlations and topology; superconductivity and correlated insulating states

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a59a5339a046347b8b9578d76ad09aa48ddedbeb086a7b2ea07ebd5c5888f644; empirical validation
sha256:45790ae53fc68e82b896c6d0257af464151d7512d1c923ff61889bcd95092c7e; external validation
sha256:414d68f687012ec0cc7fc833afacfa372f60a4368687cfad58a59b03de067000; isolation
sha256:0935c6102eeeee39d8ec1918368f6f1979f3a383c72cf1309029204a7016f7f8; custody
sha256:c73ee5c6defe1f4a1d7ef698990626e89cd6e16d10ccb91121e69a1fcdd10a3c.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

212. Moiré and twisted-layer superstructure

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-008

Claim: SFT-MAT-NANO-MOIRE-SUPERSTRUCTURE-008

Question, law and forced result. Two nonidentical layer recurrences force a least positive joint recurrence by complete enumeration, while layer identities and interlayer registry remain held as the moiré superstructure.

NANO-008 uniquely retains two-layer-least-positive-joint-recurrence with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001

- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-NANO-QUANTUM-COLLECTIVE-STATE-007

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight NANO-008 axes before target release.

Boundary: Every positive finite NANO-008 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier
- two-layer-least-positive-joint-recurrence
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__two-layer-least-positive-joint-recurrence__complete-generated-discrete-organization__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a63ac5d25ab64600eec6d0aef85572099142d4a912a34371caalc5e5a1d1186e; generality

sha256:e5a2b22e1d49c435d6ebadbe2a245a4e898dd8501930889f4c3698c7489d8073.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:22bdb96721ed1f633b308940b75dfc562195703b1333cd16f5f84ee9b4bb62b6.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:f7b766e96a109bf546efabee55cfc610e873426f12160fce5e07181601d2038f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:8087b4f2e0cabb3da4aac96c5cd3097dd25562e377514b588dd2a35246ad8d2a.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:bda0e8e7c3fc8f9180b0db284611e4c3223b321d8dfeea5cb48dfef249719815.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the declared census and sole survivor; independent certificate

sha256:ed7d4b5768b8e0ccbc4d7ec3da934a4df4bb261106b81b386f5a74212c329d33. Derivation seal

sha256:4e652bb177370caf5998fa142127829b729a5d08985ab49fe0144313cd12fcb7. Source manifest

sha256:3d73d2d4f3329b7e7aa733ade8b355a543f50dea5661b25d4a7685ef356a48ec. Engine receipt

sha256:d26bccebbdef31d38d5be6b79bffe940e6f1ca6b351580a7e2f8a9f9cf7547294 at receipts/engine/model_admitted/SFT-MAT-NANO-MOIRE-SUPERSTRUCTURE-008-d26bccebbdef31d38.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MOIRE-EXCITONS

Measured, observed, adverse and boundary records:

- NANO-008: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-MOIRE-EXCITONS: stacking two monolayer semiconductors; lattice mismatch or rotational misalignment introduces an in-plane moiré superlattice; confined within the moiré potential

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:19ec60e88d752041ccadb00b4498e3a59f831f45da104c65d8c689258d4b9396; empirical validation

sha256:cc3c5e2a68572ea9f954836046f3bb3f2949ddb12fe30c2d9fe9a8f70e4d9fc6; external validation

sha256:0a9b6e4b1d9b66a07d845a3a9efc4d97ab74d9a4e8300ad28616a3d618ead62d; isolation

sha256:214e961e7ddc8104fcc2378bf91c1c3ff8470fec0caf5632493256e7a728d166; custody

sha256:d8f47da39f3696a763c185c41bc8e8c2e4877fe79a3719b2128e7ce0173a3212.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

213. Nanocomposite interface density

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-009

Claim: SFT-MAT-NANO-NANOCOMPOSITE-INTERFACE-DENSITY-009

Question, law and forced result. A nanocomposite retains matrix units, inclusion units, their distinct identities and every counted interfacial contact, yielding an exact contact-per-unit fraction without fitted constitutive parameters.

NANO-009 uniquely retains matrix-inclusion-interface-contact-density-ledger with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-NANO-MOIRE-SUPERSTRUCTURE-008
- SFT-MAT-CLASS-COMPOSITE-001

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight NANO-009 axes before target release.

Boundary: Every positive finite NANO-009 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier
- matrix-inclusion-interface-contact-density-ledger
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__matrix-inclusion-interface-contact-density-ledger__complete-generated-discrete-organization__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:8d50d8f2bd275a9d8b327f2971be00e0d75418abbbdb6ae4bdcf2b8e6d63cf14; generality

sha256:88c92b53cf28834f485a43452cf197538e56a4a905180cc3a76fdb3ec737ef9.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:1b2cf3bd53e85f0aaa8d386c0cafdefe12b391c98058c4186aa89301bb946be6.

- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:29142d04c105906c808483b624f528ad27ee4112e03712f38d3cd3213f352b1e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:c8fcedcd305e1fd054e7bd09cbb9b2746d6644de4e8a76e90ea593d5993948dff.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:5443b7bd190438d0c48d47f8f689077ba79c096a84821fbb8880c99923c3b698.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the declared census and sole survivor; independent certificate
 sha256:59430f579b429b22c39180310a629d49ed2de7a597c642f1d56c6c6bbf66202e. Derivation seal
 sha256:f35cfd619516c0642eab875fe41044960de99a3b92a0a87300f73c769a52153c. Source manifest
 sha256:005cdd8a49db3d9a4d07d2e31c1974d62b07e86b5d268929f98d35d6f2d21f9d. Engine receipt
 sha256:675222df1d62c1af5aa1b5f5707fd2248cc9f568357492d0d0956e67a8afdf03 at receipts/engine/model_admitted/SFT-MAT-NANO-NANOCOMPOSITE-INTERFACE-DENSITY-009-675222df1d62c1af.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-NANOCOMPOSITE-INTERFACIAL-LAYER

Measured, observed, adverse and boundary records:

- NANO-009: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-NANOCOMPOSITE-INTERFACIAL-LAYER: thickness and density of the static layer; static interfacial layer of reduced polymer density; thickness of ca. 2 nm

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:f93c635b7cff3c96410c7179f036dda300812093ed73d1bf46771dc2b19aaf0b; empirical validation
 sha256:0af2b6a6944b8e3b2e6a60740fbc345966bbc908efc7251a95e3d41756ea4108; external validation
 sha256:35179e40ef0c8dff13360960711c857c21a906285e24cce586958b9efbe6ca61; isolation
 sha256:a3862a916e086a1fafd44a9158406a5ad9a3f60d393d1ea5b0c8405c36da85ad; custody
 sha256:ee6ac2fa8ace787bb3a9b31ca8da0614c95debc65f63fd3107de97927b877ffa.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

214. Nanomaterial aggregation and dispersion custody

Field: nanomaterials_two_dimensional_quantum_materials

Obligation: SFT-MAT-OBL-NANO-010

Claim: SFT-MAT-NANO-AGGREGATION-DISPERSION-CUSTODY-010

Question, law and forced result. A nanomaterial dispersion is the exact partition of held particle identities into clusters together with medium and condition path; aggregation changes organization without losing particle custody.

NANO-010 uniquely retains particle-cluster-medium-condition-complete-custody with complete carrier, organization, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-NANO-NANOCOMPOSITE-INTERFACE-DENSITY-009
- SFT-MAT-SOFT-COLLOID-AGGREGATION-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight NANO-010 axes before target release.

Boundary: Every positive finite NANO-010 carrier with complete particle, layer, boundary, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-nanomaterial-carrier
- particle-cluster-medium-condition-complete-custody
- complete-generated-discrete-organization
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-nanomaterial-carrier__particle-cluster-medium-condition-complete-custody__complete-generated-discrete-organization__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:5a61caa03a05820abf431b21f566b70769b2ac9911a5d77f43814a3b802400fc; generality
 sha256:e56942a76762661d2b9ab03e961fa43077ea46853b0bda32161aa00ec82059d5.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:e0e73b81d49ad17e4ce038b134796b1aa38335c68f2d2c7c3fdd5f5b95f1590d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:4cb9ea94fed8563e98b50c5da12b9ac3e887414688efb9d7c539b22b93462f1e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:36b038712e931bbddf441c68e8da53e11af36f9cb18d9bb4b8ed5807df7f21bc.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:7a089f0e57ff3347d66119a90f9305bde3a141b3d8d5047a7607ac6d6260036d.

Independent reconstruction and engine admission. Independent implementation

sha256:ea68af8e15532f9da76fe403778b36e539a15315a87d87940626a0c62ad90fe2 regenerated the declared census and sole survivor; independent certificate
 sha256:8c4c384cd63265f07e9e7b3edb1c00d0c71e8405c564ae05e098bf00a865eb54. Derivation seal
 sha256:56775b0be76746e41aef55fe89424d75f74c8a016830cd603d4c9b7c5f9c2220. Source manifest
 sha256:8194a94fa2d5f6992d8089d9899629707d461333a74df4bf4d40f3e815667f18. Engine receipt
 sha256:a2089b82b67287f189bc0e9d76a8145ab1916c4c24225ac9cb0d8c74e9f5d5f6 at receipts/engine/model_admitted/SFT-MAT-NANO-AGGREGATION-DISPERSION-CUSTODY-010-a2089b82b67287f1.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-NANOPARTICLE-AGGREGATION-DISPERSION

Measured, observed, adverse and boundary records:

- NANO-010: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-NANOPARTICLE-AGGREGATION-DISPERSION: real-time size, mass and concentration measurement; Differentiating aggregated NPs and non-aggregated states

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:254a40b06e46faa2605d93c8e0f72760b4086e7bec2d09c3b3231105b473cec6; empirical validation
 sha256:86fc5a6cf60bf6b11ba7ae85f21b3be23fea04129f22135795a6acef9de06eae; external validation
 sha256:a0fb92fa1f4cc0e8caf0f17c6819572695028f7f18eb3173dede1d48fb183930; isolation
 sha256:dcea0c5837b9f489529250be5030385b4654c3fca201de46b5fa47abb8655d24; custody
 sha256:bef695365bc7a6ba329b8fc990f1682840ba9316ce33db74542a93ccecca1e5e.

Prohibited inputs.

- no imported continuum constitutive equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude

- structural absence and every particle, layer, boundary, interface, cluster, state and path distinction remain held labels
- no external outcome selects a survivor
- all favourable, adverse, absent, unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Surfaces Coatings Adhesion Interfacial Response

215. Surface free-state and energy relation

Field: surfaces_coatings_adhesion_interfacial_response

Obligation: SFT-MAT-OBL-SURF-001

Claim: SFT-MAT-SURF-FREE-STATE-ENERGY-001

Question, law and forced result. A surface state retains distinct counted bulk and surface supports, every held state label and the observation condition; its exact excess support is a fraction, not an imported continuum energy model.

SURF-001 uniquely retains bulk-surface-state-condition-excess-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SURF-001 axes before target release.

Boundary: Every positive finite SURF-001 carrier with complete surface, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-surface-carrier
- bulk-surface-state-condition-excess-ledger
- complete-surface-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure

- no-extra-rule

Unique survivor: complete-positive-surface-carrier__bulk-surface-state-condition-excess-le
dger__complete-surface-interface-state-path__specimen-method-condition-scale-uncertai
nty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closu
re__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:41312bd68029f8d21eb875e2c7c3b191d22a81f0414572ff29a762095d6f638c; generality

sha256:b7dad558e1b65c224d5a1bf4abe996fba264a2e929fb6a6ca77d509fd8348a65.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:3bad46aefc05bf4e90c097a02191808ee42eb48e35d7ea8525d030e10e29da40.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:5bd63a0e8754a60bc15a5e1bbd2d251f50d56ecf583dd7ed923cbe418aa58664.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:f2e021a7d49d99b92b5500a06571494ea0784caa01b4f7aa95371aba175fbf24.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:54a4b1fbfa567327eee0b649b7e45a636f0301c9c99d6602f9979106f46d4dcb.

Independent reconstruction and engine admission. Independent implementation

sha256:49ce4ee7d3ec71dc320986b2be694f8fa4744c53d822a8a1fffbfbfb7898a5 regenerated the
declared census and sole survivor; independent certificate

sha256:a79bd1c6747c964db764f217fd574e6be2c8a158cb074b0b9d6f6dea4da19f37. Derivation seal

sha256:1cf25b518997ace251b28c1e6d3645711dedda99b791c3a9bd871e2d4509d7eb. Source manifest

sha256:3f50bec578740dad1082104329e6e0886d9bf740ca14031568003ad8b735f444. Engine receipt

sha256:dcde8fab4311e8d383a5fc687d9c3c15ef81a1662865ea82e0910a063308ca37 at receipts/eng
ine/model_admitted/SFT-MAT-SURF-FREE-STATE-ENERGY-001-dcde8fab4311e8d3.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
Comparison passed: true.

Source identities:

- NIST-POLYMER-SURFACE-INTERFACE

Measured, observed, adverse and boundary records:

- SURF-001: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-POLYMER-SURFACE-INTERFACE: surface and interface/ interphase play a major role; measuring
surface/interface properties

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:611adecbaba7ee267ab40b403be38f00801c6f23888d594f719cced01d13c0e6; empirical validation

sha256:8665495d1ea5b665e36685735c2e53400b91627b5d9d4398334d5bfa93371cb9; external validation

sha256:6706ff3c1b56490a84a099c054c223ed6f373b5083ea14d146f935d43451ef37; isolation

sha256:1af0d6f6876624ec1252b2dcccdf441d74645baeacb5b7c2cb959660ee1c67823; custody

sha256:34303514f567306ea519aff82c82e8609cd6d84b86f9366af7b79398b4e99028.

Prohibited inputs.

- no imported continuum surface equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every surface, layer, interface, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

216. Wetting and contact-angle custody

Field: surfaces_coatings_adhesion_interfacial_response

Obligation: SFT-MAT-OBL-SURF-002

Claim: SFT-MAT-SURF-WETTING-CONTACT-ANGLE-002

Question, law and forced result. Wetting retains liquid, surface, method, exact contact/noncontact partition and exact rational angle as one condition-bound observation.

SURF-002 uniquely retains liquid-surface-contact-noncontact-angle-method-custody with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-SURF-FREE-STATE-ENERGY-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SURF-002 axes before target release.

Boundary: Every positive finite SURF-002 carrier with complete surface, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-surface-carrier
- liquid-surface-contact-noncontact-angle-method-custody
- complete-surface-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace

- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-surface-carrier__liquid-surface-contact-noncontact-angle-method-custody__complete-surface-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:ec91eb89dac81c7289ccaebd28cc90f0416f9c28fe00de9510d20bbc2b87db0f; generality

sha256:650a741954afdd1a80ac86be401d6ad54d6db3f32c43738918da453308e04e4e.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:1341eb4239ba55a5b54160ede824fd6de3d6bbf613798bb464ca5150a33aed09.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:d686b60b2c5a80c383bdf9b396187f200212c4d3a9b17b02bf62e21717b931e4.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:04a2a3376d3908c439a6b427879bd5f265660ed7e4a49fcaa6a5297b19e19f87.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:a934050d61ed2432fb39a481a790781ece94be215a876ab71d71c6f282ed414f.

Independent reconstruction and engine admission. Independent implementation

sha256:49ce4ee7d3ec71dced320986b2be694f8fa4744c53d822a8a1fffbfbfb7898a5 regenerated the declared census and sole survivor; independent certificate

sha256:55bfd2bb302d38e15df3c028729151676e3c854f0ce32268c64754c7945a5644. Derivation seal

sha256:72ba917d55d59f796c7458e3e8e702565f5d3b0f29e30a79915865a4d9cb11f7. Source manifest

sha256:61effac3a73fba94cd6c031754f0cd9df8ffa64d6058c396ac2c3584e8622d01. Engine receipt

sha256:52ee8e2b2f42259b02e6094cf3801ebd59ac9f08ccc8264f0657f5f66a9aad06 at receipts/engine/model_admitted/SFT-MAT-SURF-WETTING-CONTACT-ANGLE-002-52ee8e2b2f42259b.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-WETTING-WRINKLED-SURFACES

Measured, observed, adverse and boundary records:

- SURF-002: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-WETTING-WRINKLED-SURFACES: systematic study of the wetting behaviour; anisotropic microstructure; simple sinusoidal profile

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:1a6c09281cd22fec214eca4f2f050c998c9ca205c7bb7ba208f90646fc8b484a; empirical validation

sha256:7e88c510b071ae03602e964e764280818f595c59b779df766b208413374f0219; external validation

sha256:364cd732dba03d0078320a6930eea6427e4434142391392b14d583227d903291; isolation
 sha256:0235543f059a1fa8d772d1c5314b6e00f34434631413c4cb139bdd1b5ddf9b73; custody
 sha256:bf7114b9b83b315a8d560a36875462b465551cfb37cb19f93e4cfd66bb93da55.

Prohibited inputs.

- no imported continuum surface equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every surface, layer, interface, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

217. Adhesion and work-of-separation ledger

Field: surfaces_coatings_adhesion_interfacial_response

Obligation: SFT-MAT-OBL-SURF-003

Claim: SFT-MAT-SURF-ADHESION-SEPARATION-003

Question, law and forced result. Adhesion retains the exact partition of interface links into retained and separated supports together with counted separation work and the complete path.

SURF-003 uniquely retains interface-link-separation-work-path-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-SURF-WETTING-CONTACT-ANGLE-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SURF-003 axes before target release.

Boundary: Every positive finite SURF-003 carrier with complete surface, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-surface-carrier
- interface-link-separation-work-path-ledger
- complete-surface-interface-state-path

- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-surface-carrier__interface-link-separation-work-path-ledger__complete-surface-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:25e46ab597ddc2fd468a50953b8807b3028f3eac894495a65313de09c1db23af; generality

sha256:f5e40bc93b1388b9b6c3bc2dcfe8f30ef4e40afb82fbadf0852fda5c512ce86e.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:5266dd4694ae1d6d22146cf4f27c7dd407f30d794dac49c4ccd98fa57489dd2d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:871dc210318ced54a476f5b2431f7c6647c687d13be7d4fdce4e0ae0b331e8b5.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:2fa0c2c0d84ed6838b35faedc83b2f92f126c09f39d0f1575b41677e75af7d95.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:dc0f08574d5892dd2e9eb7b329d9ac3d43dddbf35fe3b720ea10f5c64e9b17f8.

Independent reconstruction and engine admission. Independent implementation

sha256:49ce4ee7d3ec71dced320986b2be694f8fa4744c53d822a8a1fffbfb7898a5 regenerated the declared census and sole survivor; independent certificate

sha256:cec3e07fea44e6f5abaac5d78c0e2e0b7bc81b4c9e77854983df494e05b267bb. Derivation seal

sha256:4e98639e86d58b2fdbbc6c93d6dd9604c5dfa4b7a8f647aa089688724f544e664. Source manifest

sha256:4b5f1057529d11e70af62d4aae71e04720e47c1ffed6930b42693cf7312ee537. Engine receipt

sha256:fa0b6946cc725c0f96c2e7d20f4ab6187a483b55320e41deab1fb77a7a97078f at receipts/engine/model_admitted/SFT-MAT-SURF-ADHESION-SEPARATION-003-fa0b6946cc725c0f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-ADHESION-ENERGY-DISSIPATION

Measured, observed, adverse and boundary records:

- SURF-003: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-ADHESION-ENERGY-DISSIPATION: Adhesion plays a significant role; interfacial friction; energy dissipation

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:5bf92c74ea138f2e6146f6474a834607af14f4a920a2c12c083461368d5c7a9a; empirical validation
 sha256:2100aa347db2f88a5da7e835ce9a3a49570568639c2c7626be6853eb3d229a93; external validation
 sha256:fb61bf5e06ef9f8bcb3f72aea30c7cdca8a8500803d0dd0baa7dd96e4cef949f; isolation
 sha256:7270bd4444c5bc028a80b2d156d49b4dbe272bdb8c911169a1063a664e97bb57; custody
 sha256:340410f059999a7f474a81ac32181c3b5a9e673b17a0b79487b68066a089a2e2.

Prohibited inputs.

- no imported continuum surface equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every surface, layer, interface, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

218. Coating layer and substrate organisation

Field: surfaces_coatings_adhesion_interfacial_response

Obligation: SFT-MAT-OBL-SURF-004

Claim: SFT-MAT-SURF-COATING-SUBSTRATE-004

Question, law and forced result. A coating system is the exact ordered word of substrate, layers, interfaces and process states; no effective single layer may erase its organization.

SURF-004 uniquely retains substrate-layer-interface-process-word with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-SURF-ADHESION-SEPARATION-003
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SURF-004 axes before target release.

Boundary: Every positive finite SURF-004 carrier with complete surface, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-surface-carrier
- substrate-layer-interface-process-word
- complete-surface-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-surface-carrier__substrate-layer-interface-process-word__complete-surface-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:c87741c3a4ac515cce964c439172744372dd189266b00dd741b2757d4b8d0902; generality

sha256:3b09f55ad37d703abc20d7669f3029b0f82269f5dbac29010d554ca46ba05aeb.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:a4f05fd333e774e7d7c4fe65ce98449da53625174ffa17f6816fcbc47ffde816.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:16b8e492fa402b424b3240ab11a907403fda829083790ce537fa3550d5e8d516.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:1e9261008870006465737700f27cf8391bf68544e9c3d70aedef94fd3cb2819c6.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:4b8fc164654bd827b1cb3e20c177caf037b861fe34ec698552a88963a17d1180.

Independent reconstruction and engine admission. Independent implementation

sha256:49ce4ee7d3ec71dced320986b2be694f8fa4744c53d822a8a1fffbfbfb7898a5 regenerated the declared census and sole survivor; independent certificate

sha256:1f130b3bbe864118e03e60ec054f96f13ec5869b5ca867c7c5bb8b7d3c18f9a1. Derivation seal

sha256:a7f9e6378b11f97107a09ec2ddae8d6654d7c6f4167c926f2e500432c9802e52. Source manifest

sha256:900bc42a5880e889f284354f1f215c6f1dacfff4f7eb81c6cf4cb254394c3923. Engine receipt

sha256:2227007ad00495a284644907a691daabadbbddf9f1ef27e4635654eede3a5f1b at receipts/engine/model_admitted/SFT-MAT-SURF-COATING-SUBSTRATE-004-2227007ad00495a2.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-COATING-MICROSTRUCTURE

Measured, observed, adverse and boundary records:

- SURF-004: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True

- NIST-COATING-MICROSTRUCTURE: relationship between the microstructure and the appearance of coated materials; confocal laser microscopy

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a0dfcf96b944102d02220ab18e5854146faafd3ff69f82930b559d7d96d502e9; empirical validation
 sha256:af4b2d92745780b4caaf4ceb2eade622417a02718dd71d05fc2a1b9cac8df377; external validation
 sha256:ae733d09a64a07c1b2f09a01884fdf96eee28f275c82678952ebe5878d16f2a3; isolation
 sha256:0167c7d8b6fe4bb0810c066fb6baf98f7c2f60daee8ff93ea6631463794b6d3c; custody
 sha256:3f4e18386b342ce27c51245c327a8fbce916b7518e618e5fd7b2ee03dfd6f761.

Prohibited inputs.

- no imported continuum surface equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every surface, layer, interface, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

219. Surface roughness and scale boundary

Field: surfaces_coatings_adhesion_interfacial_response

Obligation: SFT-MAT-OBL-SURF-005

Claim: SFT-MAT-SURF-ROUGHNESS-SCALE-005

Question, law and forced result. Surface roughness is an exact finite height word bound to lateral scale and method, retaining every profile point and its counted range.

SURF-005 uniquely retains height-word-lateral-scale-method-range with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-SURF-COATING-SUBSTRATE-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SURF-005 axes before target release.

Boundary: Every positive finite SURF-005 carrier with complete surface, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-surface-carrier
- height-word-lateral-scale-method-range
- complete-surface-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-surface-carrier__height-word-lateral-scale-method-range__complete-surface-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:9836c61f3c4bd01ca7a608ce4c5839ae6aace52070e23753374904937d9f0dda; generality

sha256:621617d7a3ecd6c9fc1a84b7668618ffea02b749265c75cda6146992bfb4d352.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:b8dc875172382c4f74d1b12bf7246f93c5d88c2cd26f29d7a7745062b05b006a.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:b906caafc7fd46710cdcfd6eae4410231ce77587b4e7d0767dc14f4d6ded27e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:9b6bb84f3ba917a667853ef0987f37135eed7582f63d33f989c8fd71c43d4000.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:84d43d8073dd45b03dae5afbdece93712f889fffa75035b9869fe59f202d47bd.

Independent reconstruction and engine admission. Independent implementation

sha256:49ce4ee7d3ec71dc320986b2be694f8fa4744c53d822a8a1fffbfbfb7898a5 regenerated the declared census and sole survivor; independent certificate

sha256:de5b1ef65862e7faabd166d9cd0739b80f32084a817211df63db9691fc986bbf. Derivation seal

sha256:3f20fc2f20cc88e10b2a722ef9272beb670c502cd7d9ed8b28e28e7e266ba8d3. Source manifest

sha256:38082b7eedfcf7c7038de8c48d502e665a15bd49333b5b157a6d04572228ea23. Engine receipt

sha256:1ad9133ffa04881642cba041955c66a4edb1d1ccff397831018cdbab53d5c2e3 at receipts/engine/model_admitted/SFT-MAT-SURF-ROUGHNESS-SCALE-005-1ad9133ffa048816.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-COATING-SURFACE-ROUGHNESS

Measured, observed, adverse and boundary records:

- SURF-005: 1 comparisons

- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-COATING-SURFACE-ROUGHNESS: surface-roughness; residual stresses; interface between the ceramic top coat

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0cf66f57cee96ec46cce786055539719b45c52f21a0ae870decf59357f2f70f8; empirical validation
 sha256:49292959166bab1ff08b5d2c759eab28ccc127527f89904a538e0f963a0dbec0; external validation
 sha256:fa63aec4bf80f7c74fdb658baa75a8a0eb591c32a3c0a7725c658eea726abef4; isolation
 sha256:a9f0ee0f34f7087d29ad8add4a46d4acc66a8e2812841abe8e215be228bc073f; custody
 sha256:3d2c35105c4e0d20fdd232e3d2fdd3eba9d08ea3926465dcaac3d34364d5f537.

Prohibited inputs.

- no imported continuum surface equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every surface, layer, interface, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

220. Surface reaction and catalysis handoff

Field: surfaces_coatings_adhesion_interfacial_response

Obligation: SFT-MAT-OBL-SURF-006

Claim: SFT-MAT-SURF-REACTION-CATALYSIS-HANDOFF-006

Question, law and forced result. A surface reaction record retains surface sites, reactant and product identities, the full state path and unchanged catalytic surface identity while handing chemical identity and rate consequences to Chemistry.

SURF-006 uniquely retains surface-site-reactant-product-path-catalyst-handoff with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-SURF-ROUGHNESS-SCALE-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SURF-006 axes before target release.

Boundary: Every positive finite SURF-006 carrier with complete surface, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-surface-carrier
- surface-site-reactant-product-path-catalyst-handoff
- complete-surface-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-surface-carrier__surface-site-reactant-product-path-catalyst-handoff__complete-surface-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:152006a9bfff097e42a470d977da590ab1e517064674692343991521e38a95c96; generality

sha256:663ff365826b4fcc40ae9e4e8adf5e5ad6b64876ff5562285c34a53387789742.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:49637468e2acfb0e730cbbfef5b9800304155e57b01c8d9a274ae530b1cae694.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:a0c9061f9b0ffdfdf123efc731610557ed92fc9f2085e6a734930a24a64999d5.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:0d9e86b9c3876afd6401603c21a3e9750b554fed208ba83710324ac6b77255b7.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:691ebc918dceffa5762cc3bab523e155cfd711f048fd96ac6ee69afed380055b.

Independent reconstruction and engine admission. Independent implementation

sha256:49ce4ee7d3ec71dced320986b2be694f8fa4744c53d822a8a1fffbfbfb7898a5 regenerated the declared census and sole survivor; independent certificate

sha256:9f98701224e3ef736d268b7f387dff07c8960b5cc021b356e9812c3d5ddd4d29. Derivation seal

sha256:49a6534e402927dd9975d23d018eb5b4fa74b444cc4174521477a9f35a8af149. Source manifest

sha256:13957f037793454781546cec10d871921847bee044c47d0ae9ada10422881942. Engine receipt

sha256:bd13a0ba76ceb3706a1f1d6c4bbd9ab2664a093d9d4c276d921158bce33f6105 at receipts/engine/model_admitted/SFT-MAT-SURF-REACTION-CATALYSIS-HANDOFF-006-bd13a0ba76ceb370.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-POLYMER-SURFACE-CHEMISTRY

Measured, observed, adverse and boundary records:

- SURF-006: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-POLYMER-SURFACE-CHEMISTRY: chemical spectroscopy; chemical and morphological microstructure; polymer interfaces

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:02aa20af2a212c401cbd70fdaecd8fab8dbbc0d7b367f617fb164b965120529d; empirical validation
 sha256:fa9690dcdee52c021fda8ade1bf7041cd4f55c8d2069583b7cd9bd88d340f54e; external validation
 sha256:e309b3df4476928b4a2e46ffae4fbc3e6bfa9dcad447e034187099d2153b8709; isolation
 sha256:636a480103e10454f181d29b7b6bedea4c730bbba227649c4f2c095f4f4ffc6b; custody
 sha256:2051f57de9185ba04c4efebcf733897031561d84180afc7f0ded5ee625f32348.

Prohibited inputs.

- no imported continuum surface equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every surface, layer, interface, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

221. Tribofilm formation and retention

Field: surfaces_coatings_adhesion_interfacial_response

Obligation: SFT-MAT-OBL-SURF-007

Claim: SFT-MAT-SURF-TRIBOFILM-RETENTION-007

Question, law and forced result. A tribofilm retains film and substrate identities, complete formation path and the exact surface-site partition into covered and uncovered supports.

SURF-007 uniquely retains film-substrate-covered-uncovered-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-SURF-REACTION-CATALYSIS-HANDOFF-006

- SFT-MAT-MECH-FRICTION-CONTACT-012
- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SURF-007 axes before target release.

Boundary: Every positive finite SURF-007 carrier with complete surface, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-surface-carrier
- film-substrate-covered-uncovered-path
- complete-surface-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-surface-carrier__film-substrate-covered-uncovered-path__complete-surface-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:be5d76f5dfdbafb336300c5c086ca09edd93a7deb72f0db7a77e9a995da3ded2; generality sha256:c864fa8d8188624099473a3970428abe8782d223187691a85ce2f9b37fdcef0c.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:b6eacfd149d9e7a4bc8694e84b6333fd7933ee57eae05939440c47579c2b5ea2.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:a56392c95fad5330e51b5f1a5aa08e711df3df2f52742fd290c8495e2f1ee927.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:7d9bd7e3862b2f21ce02e441f94e01414d0b12af2b86c3208aa7a5baec5050a9.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:553ef3b9f0c18c201e06d1d54597bda32d54d75e07151257220b017991003db5.

Independent reconstruction and engine admission. Independent implementation

sha256:49ce4ee7d3ec71dc320986b2be694f8fa4744c53d822a8a1fffbfb7898a5 regenerated the declared census and sole survivor; independent certificate sha256:83a0d8574b7b6158bbb4c9bfeb1289dea9b362e970340aa378461caeddcfffb14. Derivation seal sha256:ffc695ef56a1b5032228f77b040678c886f885b42be114aa4408dcd217c8fa98. Source manifest sha256:b7bd304970381638297dd13967b551316eb72af6fee9a9e08bc36b6a36f7e0811. Engine receipt sha256:aea10406336c2c7ba505903bb8910588f5c56eabf7144124b4b409d3bb1c0d4d at receipts/engine/model_admitted/SFT-MAT-SURF-TRIBOFILM-RETENTION-007-aea10406336c2c7b.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-TRIBOLOGICAL-COATINGS

Measured, observed, adverse and boundary records:

- SURF-007: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-TRIBOLOGICAL-COATINGS: nanostructured coatings; tribological applications

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:068148fd0098c5e319e6d7b39b710cf8ff03999a99503d798e9d46383248af06; empirical validation
sha256:19b775260f1d379a7aa90428c182f699c485d0096cf43f66f85cf05b286562b2; external validation
sha256:fda36038de395e4053db2d7db39f76ba02ad83690ca6ef5419ac49220a5fe63a; isolation
sha256:32357ed6f97e08e087776c4a9f0e8fd7f80b9757f833c2e81f426d89ca6daecb; custody
sha256:59d4b12880b34115b4ebc625267db809fd9e59c50e9e3e2ed01714544b9fb3ed.

Prohibited inputs.

- no imported continuum surface equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every surface, layer, interface, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

222. Interface fracture and delamination

Field: surfaces_coatings_adhesion_interfacial_response

Obligation: SFT-MAT-OBL-SURF-008

Claim: SFT-MAT-SURF-DELAMINATION-008

Question, law and forced result. Interface fracture retains layer, substrate, intact and separated link supports and the complete delamination-front path as one exact ledger.

SURF-008 uniquely retains layer-substrate-intact-separated-front-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001

- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-SURF-TRIBOFILM-RETENTION-007
- SFT-MAT-MECH-FRACTURE-ENERGY-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SURF-008 axes before target release.

Boundary: Every positive finite SURF-008 carrier with complete surface, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-surface-carrier
- layer-substrate-intact-separated-front-path
- complete-surface-interface-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-surface-carrier__layer-substrate-intact-separated-front-path__complete-surface-interface-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:1e92266a063f4313bd747d8b68a23c72c951f93d161b9a48049d0f33b94472f4; generality

sha256:137511766e7f5fb39b55193dd23679abc2b286c3b4e2e0219e7255e4315b19de.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:398f4caa9338febe7a1c8b88839dbb990ac43403d3e76d2c5bc5be8dfef02547.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:0b62f3ccbbcc0881cba189fb37be96cd0422bce37d15778ae32b1259ee8b6dcbe.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:8dc102cdadf2cafb786a2be0939e64413cbc3bc33cf44ce79f27399dae9f78b8.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:5eb2aebf39a972fcc10db66a0a70344a7101ba8d6b51c4a93b37a05d098c3c3f.

Independent reconstruction and engine admission. Independent implementation

sha256:49ce4ee7d3ec71dced320986b2be694f8fa4744c53d822a8a1fffbfb7898a5 regenerated the declared census and sole survivor; independent certificate

sha256:88236e46f53fccd4c45fb081d9192c6797450cb08d58f716dc7d592a46ad8752. Derivation seal

sha256:1ab5968c73f8bf5d1d635bdaa72da1f95df7b3228b9923c6992dd62980b0a961. Source manifest

sha256:c7783024d9caf85763b5cf4b9b0f227a8b863380e9a17bd2bf3de4664a9a8f2c. Engine receipt

sha256:14e6d55a48f7c856bbb681901e5d3f582ff4bf88da0c03a77b6263273e230610 at
 receipts/engine/model_admitted/SFT-MAT-SURF-DELAMINATION-008-14e6d55a48f7c856.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-EDGE-DELAMINATION

Measured, observed, adverse and boundary records:

- SURF-008: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-EDGE-DELAMINATION: edge delamination test; failure of adhesion; temperature and film thickness

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c56cb6edb95196b2a205cf53ef89a20fd9d8eec41748a351eebbfbf9c3a7fb9a; empirical validation
 sha256:316e48c9cb42bc87bee71b0a5fb33e20aa79e47c82a0976961a160bde0f62f33; external validation
 sha256:7753fda8900ec3b7434c430f1b09410e81026fda3856ee9dab00ac586ba6ae4e; isolation
 sha256:4c7009d858441b5fff8d741326b2e931892eadfbdbfb6d6f5f8ca42c5c9a9826; custody
 sha256:3364a5d5ac1e0c46183c634d8973b10e7ee73d3b0f65e8a7dc831cdb2a8c02b5.

Prohibited inputs.

- no imported continuum surface equation, fitted threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every surface, layer, interface, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Corrosion Oxidation Wear Radiation Ageing

223. Oxidation scale growth and transport

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-001

Claim: SFT-MAT-DEGR-OXIDATION-SCALE-001

Question, law and forced result. Oxidation retains the exact partition of initial material into substrate and scale, transported support and every ordered state.

DEGR-001 uniquely retains initial-substrate-scale-transport-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-001 axes before target release.

Boundary: Every positive finite DEGR-001 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- initial-substrate-scale-transport-path
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__initial-substrate-scale-transport-path__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:8eedb8e3d61aa75176bed04c34ee59fce6ee467d6431ee187cd1395e7e85fab1; generality
sha256:c4d061b666b98933d6a48cb846f0a64f9fc1dd59a976a8e0ae5e5a8ffd4342fc.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:8cf0983a118dba7af3a8b5c11fbdb6f8bdc8ef0316f331738918c312006a4eba.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:e5240c0e01a41835243d3f81e5d7d8c8b52ee98b6a7b4ed082f54598d17ce8b7.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:49a849cd13192bf3cbf8dab8540295cc21b119931032cd97b02d718eb56f02cd.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:e2bded4eec2b0361dc973d24f569a6cbf1c2697836b75272fc9d9864369129db.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the

declared census and sole survivor; independent certificate

sha256:b45cc94143be4013d54f165ea9528aec5ea3a68636ac7e45e9ff90bfb8b2789a. Derivation seal
 sha256:623fe90ea92c56c957e7ba46dd7dfb2f42111435217ab4b9223e9c6518f539b6. Source manifest
 sha256:ece0993da254f1938928d3b43dd59aa6c01fd9bb7f3bf689ce605a446e4c6b86. Engine receipt
 sha256:947c31475711cd7e1f7e394a2115b1e7b9a672215ad17189bc8e8350d909d116 at receipts/engine/model_admitted/SFT-MAT-DEGR-OXIDATION-SCALE-001-947c31475711cd7e.json. Closure scope
 depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-OXIDE-GROWTH

Measured, observed, adverse and boundary records:

- DEGR-001: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-OXIDE-GROWTH: unidirectional growth mechanisms; size, shape

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e1a8162503013c3b1ce3e8ef8df48723fdf169701e67e34e533ac82a7a2b4d9b; empirical validation
 sha256:047ce2651752c16d162a702d408091415e452a9f82206f87d9fe969a2d562284; external validation
 sha256:6120132697167d45a40817859bbba511ad39f461326a3f769a26a51591726e2c; isolation
 sha256:e380ed6edd367b035cb81e038733e33b05d4204d3eb59c3899d9870e453d5bf7; custody
 sha256:061f66237128403f902d44e8ee20498982d6a88aa7933665d8cdc63eda7defb6.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

224. Corrosion-rate and electrochemical-path ledger

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-002

Claim: SFT-MAT-DEGR-CORROSION-PATH-002

Question, law and forced result. Corrosion retains exposed, retained and released carriers together with medium and complete electrochemical path; no fitted universal rate replaces the observation ledger.

DEGR-002 uniquely retains exposed-retained-released-medium-electrochemical-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-DEGR-OXIDATION-SCALE-001

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-002 axes before target release.

Boundary: Every positive finite DEGR-002 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- exposed-retained-released-medium-electrochemical-path
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__exposed-retained-released-medium-electrochemical-path__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f0f8b03ec32076310feb7b338f19cc7858469facd89e98f45347a99dabe46ee3; generality

sha256:2b79b1bb850fe649d77226fcd97cef62d62ae38f0f9dc91879847777e9fc2ee.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:125f8b5c322ba30b61e555d58aeb65df2c85467a148adb3d4af876d2a9e05af4.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:8d8b6509c0a6070c74f035847bc67a3855a85e615a06c6e1f6844df33a6b204f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:24e6d0b3dc8c59616107596594b73cb5d6c33b89b7c1a503c723a4820f78e2b3.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ee186f58255e98a99aac0694b07bbf3970a7303b6c65fd829e889583cab73bbb.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the declared census and sole survivor; independent certificate
 sha256:dd3d9f6c2ec512f81419a9382b3a2baff93249655bb6bb0f39c03debe32faac3. Derivation seal
 sha256:acce2c784ec13ee8f9f97ddb89bcfa44121a8fde60a3bb90aebfbb121da633. Source manifest
 sha256:4fae3b1e10111f15b467e3720d4bc7ab6b029087146a1e54f8e5ce4b5f8fc195. Engine receipt
 sha256:063ae7ade3f8e9e018cfb6234f834fa07b29f0e95a37e16a828c32a1545c603b at
 receipts/engine/model_admitted/SFT-MAT-DEGR-CORROSION-PATH-002-063ae7ade3f8e9e0.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-ELECTROCHEMICAL-CORROSION

Measured, observed, adverse and boundary records:

- DEGR-002: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-ELECTROCHEMICAL-CORROSION: electrochemical polarization; immersion corrosion

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ad4e8b271c0f5a8d48783e1bb366597d9cb0d8782f037dfdd0adbf6ff0f33428; empirical validation
 sha256:767ea60e4741db1e044a4c615e9ccd92db848a58f71ecf184ff138d8cf51d31a; external validation
 sha256:ef60106ec3f00d6b276d56e7a5c30c60c0df7c80d2380bdc12b4f4fc88e0a742; isolation
 sha256:1e8f347d1c8eed553f7fb7ebc9f0c6450ccb86b3b11bba01d7b8a22a05d5ea48; custody
 sha256:8a45dfde55dada94338f721db7f04302ef30ef431d3a291c7bd2d28164197d30.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

225. Passivation and film-breakdown boundary

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-003

Claim: SFT-MAT-DEGR-PASSIVATION-BREAKDOWN-003

Question, law and forced result. Passivation retains protected and broken site supports, film identity and exact conditions as one boundary record.

DEGR-003 uniquely retains site-protected-broken-film-condition-boundary with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-DEGR-CORROSION-PATH-002

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-003 axes before target release.

Boundary: Every positive finite DEGR-003 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- site-protected-broken-film-condition-boundary
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__site-protected-broken-film-condition-boundary__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:3c9b332f102c2956ab58f513a9abfa6fa32795560eb0eaf431718cb7c9b38db6; generality
sha256:73831c00269d7c2272c9f659e8aeb906bb91391512f6c8fe12e25024e8a433a1.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:dd411ddeaa134c118fe4a475cf7f8d5b061a9d202b46e80a55ceb52b154aa9cd.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:707c7510ad991ce7e56769eeeb4848c87097f205c6e94cf512e57abcb7d6f195.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:dc0b23153622964cdc1dae0ea31dbd37229a343f0e342b44af3f478766ce2300.

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:84411b04e22cec6c2ec1a81777a517132cb0736bf185c642a9fa4fc873d9bc2e.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the declared census and sole survivor; independent certificate

sha256:8d5c5a7899f1e226622c08a3a05e5e9612c8242951571e7826083fbb089d4bc6. Derivation seal

sha256:6479f36337e191ea3ffbc82fac58f36912e218e15c150491d08e9d4e65466234. Source manifest

sha256:585d9d8dd4db65934989f962a3b73d3d41e37a96f7b3a0c3a012702e1c6ab1b9. Engine receipt

sha256:26287756ed1abe3b51b51a1843d3b7145221786dac8dac086c1fb3f66f119f24 at receipts/engine/model_admitted/SFT-MAT-DEGR-PASSIVATION-BREAKDOWN-003-26287756ed1abe3b.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-LOCALIZED-CORROSION-RESISTANCE

Measured, observed, adverse and boundary records:

- DEGR-003: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-LOCALIZED-CORROSION-RESISTANCE: localized corrosion resistance; aggressive aqueous

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:9ae572b71e520fb8a8aca9fb0ba45174785c114fd2eee8007756517b850eefe0; empirical validation

sha256:d76240725f4390be9c6e998a3dbc18963309ddcf98a0dd5d3ade9198b0ddd120; external validation

sha256:261ba1713a879777eba5796425f0b4f275fb2c1fa41aa6be304db34320ccdc3; isolation

sha256:05fe79948b367dd34cf54772f79250463b1935068622642c680bedc49189046c; custody

sha256:7d9bdd97fd4c561bd7aba7ad846ccc493ed4910e49bafa8b1012b3b7b54692ac.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

226. Stress-corrosion cracking

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-004

Claim: SFT-MAT-DEGR-STRESS-CORROSION-004

Question, law and forced result. Stress-corrosion cracking retains load, medium, intact/cracked link partition and the complete crack path.

DEGR-004 uniquely retains load-medium-intact-cracked-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-DEGR-PASSIVATION-BREAKDOWN-003
- SFT-MAT-MECH-CRACK-GROWTH-008

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-004 axes before target release.

Boundary: Every positive finite DEGR-004 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- load-medium-intact-cracked-path
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__load-medium-intact-cracked-path__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:9b2757d455610a7c110497a581cd6146a4d91d83c35dfcaba6695fe6c35a4fe3; generality sha256:35bb89f66431a71923cf0ef442a6f45ee0e663f49c74f99742bb69709bed09d2.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:d664e817bd8f31ebc8840b8af629a3318430d1e441cf66b20af78e2fde89247f.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:f66a0b1674d80aca21b94ba27197c40d1a08c2fe40f51b2e1e93e2bd3032bba3.

- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:2fd9be4605005a82990e089e24e37dd8c55dd1f7afe2f16a8448c5d598d1a5aa.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:1a8fc97e8bee93ed90c1c09d9bb20171ffeb6d48342daaf93751e7386ffd9f5e.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the declared census and sole survivor; independent certificate

sha256:b9319da3716253aa0dd7b05ca44cbeba547137f52f54115a2b7ac7c2156c68dd. Derivation seal

sha256:916985696f817c840de80324ac73db2d902016f58fc89ba82b66e05ebe3e317c. Source manifest

sha256:67fa4219332796080592ddf09246fee11a263d003762de42a5e6ffcca6877270. Engine receipt

sha256:a92186f2b4df0f75f0ba19cff36f22bf111df5011f42dfcfa996c20b22bb74dc at receipts/engine/model_admitted/SFT-MAT-DEGR-STRESS-CORROSION-004-a92186f2b4df0f75.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-STRESS-CORROSION-CRACK

Measured, observed, adverse and boundary records:

- DEGR-004: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-STRESS-CORROSION-CRACK: Stress corrosion cracks typically nucleate; propagate away from the surface

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:74118af4686d5e3f597258d40738f840770135319c86fb26a4e6c543485c4abd; empirical validation

sha256:74c877e208dfc8c6263ed751d983e3fb16569564a8f3fdb6de57782aad815b4c; external validation

sha256:4b5134851c18c428c4f60dec3e816f06f92d3b6e12ffccd5181144de1879b02a; isolation

sha256:6efa3cd3b9b5b08627e1734219bb6b3fa629f7cc88b9489a6a284ad327d9634e; custody

sha256:02508f9224d64c7f89dc767fa2d118d6807c4f510d23dc1f5d288961325463c8.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

227. Hydrogen uptake and embrittlement

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-005

Claim: SFT-MAT-DEGR-HYDROGEN-EMBRITTELEMENT-005

Question, law and forced result. Hydrogen uptake retains presented, absorbed and rejected carriers and the affected/unaffected partition along the complete path.

DEGR-005 uniquely retains presented-absorbed-rejected-affected-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-DEGR-STRESS-CORROSION-004

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-005 axes before target release.

Boundary: Every positive finite DEGR-005 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- presented-absorbed-rejected-affected-path
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__presented-absorbed-rejected-affected-path__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:3af07b40dea574311529f510690623651a3aa22b246aa7101451d12ce6bf4cf4; generality
sha256:91b2bdc42cbab1e8317f92e0b14b9067698ee14997dc159012a2542ebfb3ef80.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:5ff78b3b01439885b6e49f5c4cf350f00d60c5aa5dafa40fc05f62facba3c0aa.

- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:e056073e758e7650051778f5991cd64724117f16b043105adc9a71910d965179.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:b0fb80bbb2e547fc710cdaef66347859b16adde485e12f3db8829fbb20e331ea.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:f0bb2db96e0066bf3e3b732cc846092d6819265f9a1e21db42b859787fb47c7c.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the declared census and sole survivor; independent certificate

sha256:07390814f2189c79000af7b73f89cabb64b5ff2ffa4d51443ac7d1bb2f9e4ea2. Derivation seal

sha256:c3c6a516c7b588efa8fc6f2fe81db0bbf395507bf0c59195b6731cca3f18ae8a. Source manifest

sha256:844864d95eeaaa40b6dfe514712621fa60b3eef467ae14cf832f62db327bca57. Engine receipt

sha256:79fe6b792a07b688c72d746dd58adf3e184f676f7c1beddf2aeed8327c6301b2 at receipts/engine/model_admitted/SFT-MAT-DEGR-HYDROGEN-EMBRITTLEMENT-005-79fe6b792a07b688.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HYDROGEN-UPTAKE-EMBRITTLEMENT

Measured, observed, adverse and boundary records:

- DEGR-005: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-HYDROGEN-UPTAKE-EMBRITTLEMENT: sensitive to hydrogen embrittlement; Hydrogen Uptake

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:6aa6ebdc31e634ebd2b7ef3a76cca8cce300deb1dd7974ffd18ad4f2f600b7e0; empirical validation

sha256:f02ce4e92efe911c0d0fac208249d63cd313b5d0095be2b3ba23134eddc6babd; external validation

sha256:4e0a1a92799b68db36a31dbaa56b5b9141a41dccde100e5256e0586f66191b03; isolation

sha256:07340c53a9494177a5e36c5022e5dd71c31ad148c20f185fcf367247580bcd0e; custody

sha256:25763f42a65e2fe5d94aad37a6830869ae8d5a4fd4b36c00fa46351d21d70d6b.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

228. Abrasive, adhesive and erosive wear distinction

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-006

Claim: SFT-MAT-DEGR-WEAR-MODE-DISTINCTION-006

Question, law and forced result. Wear retains abrasive, adhesive, erosive and retained carriers as four distinct exact parts under held surface and path conditions.

DEGR-006 uniquely retains abrasive-adhesive-erosive-retained-mode-ledger with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-DEGR-HYDROGEN-EMBRITTELEMENT-005
- SFT-MAT-MECH-FRICTION-CONTACT-012

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-006 axes before target release.

Boundary: Every positive finite DEGR-006 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- abrasive-adhesive-erosive-retained-mode-ledger
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__abrasive-adhesive-erosive-retained-mode-ledger__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:becd9d597bd7aecb112341d089842d7e6ffd8b25fc15e067e35f4ed25aaf761d; generality sha256:78c6d89e796eed20f563c150181f4cc036b56cecc4e3d60a1b6049ae6e4bba26.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:a2ae66db3780eb6060e1f3aa37e7c12da082b61c6a0cd9b08177173be5bb10b4.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:18c371a0015fb704f1c85ecd0bc787a415049db9c9d3000ca55a7a0cd4014c3.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:1a3f25163f9cc70b5247176d38ba78cd1f6ca874ccbf5381f533fb33fdc905b5.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:9aa1a24e074dd45aa0890003d87b237cfe90d61baabc368dbe7c700e799d3f9d.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the declared census and sole survivor; independent certificate

sha256:1070c65f2bda53acc234fae3d43472d9286e9762a09695ad2970a340650c9561. Derivation seal

sha256:306b033a4976a3c768707cf3c7b8bd2e3ba2773934b8b596373b1e2d179a073c. Source manifest

sha256:9a0edd7253e88ace78b644337855a9c6cdb7e56b12d2ca62d0b96ea42f7368ca. Engine receipt

sha256:cffc36d8c727d6e0d1a3d9fdcc174ef4e52a0e559797d92f8259ae3022b44dda at receipts/engine/model_admitted/SFT-MAT-DEGR-WEAR-MODE-DISTINCTION-006-cffc36d8c727d6e0.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MULTIAXIAL-WEAR

Measured, observed, adverse and boundary records:

- DEGR-006: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-MULTIAXIAL-WEAR: versatile wear tester; designed and built

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:644808cb9b13f458a4c9def35fcb762b90675e5d86679b231eba08323206bd7b; empirical validation

sha256:a6f21442f0be5bd63ffd5a027fd73164db0c415930e06b90204bbd023a14c9a5; external validation

sha256:09c7ca44f4c8b0f50510cc7d5502ed908614c0ce2fb933dab3dc43512e71254e; isolation

sha256:7ffb61b875c99b341f4781087d1c93ce0028c4c3bca40623b36a0dfb512684c2; custody

sha256:fbd8ea7cdc6a5b915311546bc59b930c39307bb3578bb23392f1a5072bbe8582.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

229. Radiation-defect accumulation and recovery

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-007

Claim: SFT-MAT-DEGR-RADIATION-DEFECT-RECOVERY-007

Question, law and forced result. Radiation damage retains created, recovered and retained defect carriers, every defect class, exposure and recovery path.

DEGR-007 uniquely retains created-retained-recovered-defect-exposure-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-DEGR-WEAR-MODE-DISTINCTION-006

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-007 axes before target release.

Boundary: Every positive finite DEGR-007 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- created-retained-recovered-defect-exposure-path
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__created-retained-recovered-defect-exposure-path__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:322c76efc0c4d888c718724f666d964da386441cee836f7c485e5e7bd7b63744; generality

sha256:6968a12a6441521931b59d46feb119661125a83556b5d1ad039784795d14b4ee.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:d88f426f8d93e942cd157c494fe644295ffe0323caf878ec938c28ea9f29cd67.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:108ff5b3b201da767bf505c36f4b9aa99b7538fdd051ed9c2d152b1e7fe70c5b.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:c8e5732700a17ba713fe92773365f9b351e5b265ab0b6ff912aee13f5fc1a18e.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:8c497b8f266f393aecb62dd3ddacce562b0cf391d78087031b6fe9f21266374c.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the declared census and sole survivor; independent certificate
sha256:f61d18c42fecf8fce3b584c2b73357e07fb807374adb6a219d46f9a973a2b852. Derivation seal
sha256:0f42e58136ae2ab6ff3fb5c5730d652116e8a650470801ba36235d454103171f. Source manifest
sha256:6705a469be5c483e9d5050636daa66793d4b290a5a93151e53c27427e5108f93. Engine receipt
sha256:58d0099c0a28fd3d4305be6797698fafb636dd31092304be72e4e006355868e3 at receipts/engine/model_admitted/SFT-MAT-DEGR-RADIATION-DEFECT-RECOVERY-007-58d0099c0a28fd3d.json.
Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-UV-RADIATION-DAMAGE

Measured, observed, adverse and boundary records:

- DEGR-007: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-UV-RADIATION-DAMAGE: direct measurements; UV-damaged silicon photodiodes

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:d678430545ba76ec3d3238fe5c490a015c3ad9dcfbf42bac466eab9919337c28d; empirical validation
sha256:8ca1a3a9abe5d2db5701521e622e25c177a39a4531ce8416c42f16e9a3fa36fd; external validation
sha256:a9f2eb65185d09e8ead8a73cba1c1f2833a112420e95b7dd0f328f5da357da8d; isolation
sha256:c4ec651055e420f0d892422c2535335941cf95d438705dc62be82dc73ddbc370; custody
sha256:98f12a7a6cb963e73a68b59dd56f50b76d3d5f8ad1d48bbbdad6ab42275f53e7.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

230. Physical ageing and property drift

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-008

Claim: SFT-MAT-DEGR-PHYSICAL-AGEING-008

Question, law and forced result. Physical ageing is an exact condition-bound ordered property record whose rational drift and direction remain held without a fitted continuum law.

DEGR-008 uniquely retains property-condition-time-direction-rational-drift with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-DEGR-RADIATION-DEFECT-RECOVERY-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-008 axes before target release.

Boundary: Every positive finite DEGR-008 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- property-condition-time-direction-rational-drift
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__property-condition-time-direction-rational-drift__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:d30074ddc5489344ea64d1258ea0ea4beea4249cab921f3df61d81f062ca8ea9; generality
 sha256:f42fbfda1d87e11343f069bbd7084907ca0c7d77b4443032c5cd820c01618b37.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:be62436e5603a4eb594b69aabdc9586ec1b745753d0e9a6f367da2a25cbb10c7.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:572e03783c2e191c7a21cf04829726539d784411eb2ac4802baefc528e18764f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:ff3691e4478a5b3a0c5a17b1182f369f8899e7e2243e99302a99864f7dc962d5.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:74e1b1b1126c9d6c0dbb04fcd410203503bcd943c7f8f530a2dd0a1198112319.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the declared census and sole survivor; independent certificate
 sha256:fd862a22abc63ba998aaf2062a4f6b63f13868815bfa95e84810f4c722dc003. Derivation seal
 sha256:b06caaa8a33346347eafc19be7e67fad2baf1cefafacfe18b164a438ab654e12. Source manifest
 sha256:d91137dba091e47890e46b139f573f77b54ec8ce4e7b63ec0f21896960e8efbc. Engine receipt
 sha256:b3c2d79260adfe051f83e284fe36363388d044927b59ebb4f66b2b8680ae5531 at receipts/engine/model_admitted/SFT-MAT-DEGR-PHYSICAL-AGEING-008-b3c2d79260adfe05.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-PHYSICAL-AGEING-RECOVERY

Measured, observed, adverse and boundary records:

- DEGR-008: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-PHYSICAL-AGEING-RECOVERY: hygrothermal effects; Physical Aging and Structural Recovery

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:abf4af6305be2dca2208d917442ecc30cd5c24af04571bbbec78ea60346e7402; empirical validation
 sha256:fe6fd7174bce031292cbb6adc69e8db042c28e711282237a541e6b468fb95859; external validation
 sha256:892c12802bce6ce2e5bf8ce329733006d5a33a7f62b863537874c0eae51d2502; isolation
 sha256:d2f0767a79053ee2ea5868ccd5e07d9c522669bf15d6363e00b6b10fedbae951; custody
 sha256:116dd511d18ec740e9120390d3fa92340a6e3b7e167bc8549de5d0d3de465889.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels

- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

231. Environmental attack and weathering

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-009

Claim: SFT-MAT-DEGR-WEATHERING-009

Question, law and forced result. Weathering retains every factor, condition state and the exact partition of exposed carriers into retained and changed outcomes.

DEGR-009 uniquely retains exposure-retained-changed-factor-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-DEGR-PHYSICAL-AGEING-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-009 axes before target release.

Boundary: Every positive finite DEGR-009 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- exposure-retained-changed-factor-path
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__exposure-retained-changed-factor-path__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held

d__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:9b5014b536ca6c4fed4a6affd9fad25447a00cb98c34490a566f087a54080335; generality

sha256:4d77ce6198ff1c177415ebf90452d94b28c90f17d09753bfeb9d6309ea2bfff3c.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:63b4720b4a8bf3e257f3f21e3551fb5b3b34284b819540d5c3ed1fd23b3d8a05.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:07a48c3b6146f26ed23f99454f9508a2f9e13a917cfc175cc963964edb69e513.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:9e2324d72ab634dc773a69abae82520e87871c930dff9e914e5152334fafd304.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:530e62b047b3fe2b6ad3f6bd1f29d1002314dfc71ba6c6b9e762dd2787b20e44.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the declared census and sole survivor; independent certificate

sha256:08d9e6456852da647ddb1ca5674dc1bf815c7c6b3efbf3d227c624087fe45317. Derivation seal

sha256:4d67c91930ffc4fb9317842f726159890ed08295bd2bf99c6f8c13c4b11a4af6. Source manifest

sha256:8a9622415236a8bfc7f28264b109e47dc344ff0fd06907e8e3c3250032b728d5. Engine receipt

sha256:c42fcb92ad214f709df497f726ebbdaf784842512437bbc215b8dbecb88c4545 at

receipts/engine/model_admitted/SFT-MAT-DEGR-WEATHERING-009-c42fcb92ad214f70.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-WEATHER-MATERIAL-RESPONSE

Measured, observed, adverse and boundary records:

- DEGR-009: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-WEATHER-MATERIAL-RESPONSE: fully instrumented weather station; outdoor test site

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:1d26c4ac0228e8a45bf16199cceb1f64699dc73f66c314741e891cfd95b1d25e; empirical validation

sha256:a088a7abd434e1ccce69a44b6664b73dda092a6ee11a6c48ef5507490c478fca; external validation

sha256:a09af49ccd6453269de24adee50183ca666dc04d7522714df8f3f45a233a8a14; isolation

sha256:23f317e95eae9ab5ec074f5ee94f69ce57108a30eb927479264b732729aa111; custody

sha256:37b39312ada21527ff3f425258faeaba489ea7ea04e9ff332db7ea9e2d030eba.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude

- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

232. Service-life and failure-time evidence boundary

Field: corrosion_oxidation_wear_radiation_ageing

Obligation: SFT-MAT-OBL-DEGR-010

Claim: SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010

Question, law and forced result. Service-life evidence retains every specimen, positive observation time, condition, failed/retained outcome and censoring rule; no universal extrapolation is admitted as observation.

DEGR-010 uniquely retains specimen-failure-retention-time-condition-censoring with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-DEGR-WEATHERING-009

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight DEGR-010 axes before target release.

Boundary: Every positive finite DEGR-010 carrier with complete material, defect, interface, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-degradation-carrier
- specimen-failure-retention-time-condition-censoring
- complete-degradation-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-degradation-carrier__specimen-failure-retention-time-condition-censoring__complete-degradation-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:b70a11df315f60cf3e656f83f4f7797c41b336461a968526e582fe34999a4c33; generality sha256:40d162ddda75ba149416dfaccf0226bbb5683dc9c4d14f0bef43578d6d6814f5.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:0b01722fcf8e1cb80f758aca459394abd9dfdaca10311b92cd19f67247bd3167.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:a962991ced308c1a14c3a94082bab97301c976beb9f4a597706450fec7e4dd8c.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:e965ed0c8a66a0092b7f25749573f3bc913cd99271ca3cd4ce2cc25f7d0db499.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:d228748878724c24d43f614c137c22060ea7a6b6e44a63f3464618e2d5f20f13.

Independent reconstruction and engine admission. Independent implementation

sha256:aaa66aeae0161d52b042c9374c79fc940e325a268e707390d248bf317e949b7 regenerated the declared census and sole survivor; independent certificate sha256:2d05682d8549e7ae87d4576dd38f7c97898a162660d7d7e69c5a0e3c505c6d2a. Derivation seal sha256:dfc56ff1c57581035bb884fc3bf10857a22f33061df55afddca4fae754f7f4b. Source manifest sha256:00032573caa5a0a40178c7a136353f1648c519ee94677d88bf3a729b7b687c88. Engine receipt sha256:cce897586e6bcd409ad936e2c6284691fcc6770e1156b471e3fff70e45a5fcfe at receipts/engine/model_admitted/SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010-cce897586e6bcd40.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SERVICE-LIFE-BOUNDARY

Measured, observed, adverse and boundary records:

- DEGR-010: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-SERVICE-LIFE-BOUNDARY: persistent problems; century of work

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a7819cae920ee0dfd924fb0ba4e71fcae55cb53658ce9000bba29e8ef66ea441; empirical validation sha256:92fdf6b6542f75350b81811073d858feba373a9fb79dcd07c572a35e37a0660c; external validation sha256:688b4e3a12b421c55a6ccb3c0ff1b71b54d40019b2063eaa66405ae6d1f47cd8; isolation sha256:80349213b3ad28e9fada193d0f5a40fc4c792139e75f87e9470a72750fd7c003; custody sha256:6eaca00e57056b32cca0ffcd4536fd001e1ecf0d8f4f354939f7e8a6fc1b3a81.

Prohibited inputs.

- no imported continuum degradation equation, fitted rate, threshold, named mechanism or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, defect, interface, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Processing Solidification Sintering Heat Treatment Additive

233. Casting and mould-filling material history

Field: processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-001

Claim: SFT-MAT-PROC-CASTING-HISTORY-001

Question, law and forced result. Casting retains charge, filled and retained carriers, every mould region and the complete filling/solidification history.

PROC-001 uniquely retains charge-filled-retained-mould-region-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight PROC-001 axes before target release.

Boundary: Every positive finite PROC-001 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- charge-filled-retained-mould-region-path
- complete-processing-state-path

- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-processing-carrier__charge-filled-retained-mould-region-path__complete-processing-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:b8fcc6c34cd6973c733519f014127d895988a9c6a31a0c186ad81c45d6b88564; generality

sha256:684ba520e01fc16681ff184853a8cd1b348cc5a4c50ac1f90a803673c6157b89.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:543386049aa8f45934759122332408393e7052bab2b8a4423edd1def80f7564d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:d8da4a90a9c906c3b1f724e5f81ee6c11f13ba48a654bc777b40924d5149e0d0.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:4e5d0b58b6e70d1629d373b5e6ec7af39a0a6d9e36ca637ec7c520501d52b8ce.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:82f7ac3d1542861b2498f19f574f0ac6909793ca22ee9729a9d95a2097dfebe1.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the declared census and sole survivor; independent certificate

sha256:0396700b29e5c2587e0b495b8380cd97b3ff1b7d82eef2fb159043a415262b9c. Derivation seal

sha256:f0cacc860fc6eb657c3bb92b3b85fdf297a007a1bf60d725c840b9642c0c3f4c. Source manifest

sha256:3f8a2d36e4b10d927d2d14b0bb04819e03f1c15079681effe6ec9d123bd2d192. Engine receipt

sha256:91c3072765ec33f4d5a0e2d1f2ee177e2a2df1c875ce5952b45e2f0eb8d06d03 at receipts/engine/model_admitted/SFT-MAT-PROC-CASTING-HISTORY-001-91c3072765ec33f4.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MOULD-FILLING

Measured, observed, adverse and boundary records:

- PROC-001: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-MOULD-FILLING: Reliable mold filling; most critical issue

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:6006d2cc235480e50dcd1930e0d39ea1664b52b3f506b5d7cb4028ba634fe44; empirical validation
 sha256:d99c295d0b10903b0d64239d9e397ecdf8e894980576f53127d1525898d5d041; external validation
 sha256:282301cdad683abd8d13094f8e57a2a7df353357e70c53eac245d9cf93624e77; isolation
 sha256:72e57b43a8f6c5d2deb7f681811c33193c87771dfbe9eb4368485c5c330fd153; custody
 sha256:4a26d3e4d8ae349193bdb0bd788e2e14c348c5e1c48a533eb9d9878abd6139c1.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

234. Thermomechanical forming and texture

Field: processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-002

Claim: SFT-MAT-PROC-FORMING-TEXTURE-002

Question, law and forced result. Thermomechanical forming retains every material element, before/after orientation words and the complete load and temperature paths.

PROC-002 uniquely retains element-before-after-load-temperature-texture with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-PROC-CASTING-HISTORY-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight PROC-002 axes before target release.

Boundary: Every positive finite PROC-002 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- element-before-after-load-temperature-texture
- complete-processing-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-processing-carrier__element-before-after-load-temperature-texture__complete-processing-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:36bb581dec236fe7ba7ea874bbdbe9747946ad15a54d92271626c6563212688b; generality

sha256:2254ed937fe7fbdec36a88b963afbb00100e9bcb161890016bbebf926f873c1f.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:31627a073343ae2d034dfbe8cd7fc27bbc653a70945152a57c11a8c4ea739a6a.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:30c0077d305c512cae2e619f708bc7dad7f2ce7ce331e74ba1338cb1633e7541.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:2723158f9e413a70b8309047bba391d16b81c9e41f427a8718e5863162ee5eca.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:9ae8d31125729cc4920f26afb339f20d5cb8543cc298ff3f21523d3d062e344.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the declared census and sole survivor; independent certificate

sha256:b1b166dd25e1d1ec893f6263229a6d1002e428600c41f5469d9a55a29a7fdf63. Derivation seal

sha256:cec7226610fd4a44823917dc19c245f81bda4b137b1a957bb946e6d24feb3dd9. Source manifest

sha256:5362f1ececbb37c5e9e9929a225446a0999c0fbaeca495a5355cdff789f17cb0. Engine receipt

sha256:82aa2ea4e778763837d0ffaaac35e0383964a6c79a26692c98e6e3a57640c2bd at receipts/engine/model_admitted/SFT-MAT-PROC-FORMING-TEXTURE-002-82aa2ea4e7787638.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CRYSTALLOGRAPHIC-TEXTURE

Measured, observed, adverse and boundary records:

- PROC-002: 1 comparisons
- family sources: 10

- exact correspondence: True; omission rejected: True
- NIST-CRYSTALLOGRAPHIC-TEXTURE: Crystallographic texture; methodologies for the analysis

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:4c59d7e94328f6b050198f52b51c8322a8fe776094e5d8e97909f8f5aa6d498c; empirical validation
 sha256:d75e0189eaae7e2a7a26a97ae9367b49dff5e72f3de4acde0ffbc62e1fcada2d; external validation
 sha256:ea01a69c54c098557b5d937d463cbc9b54a5e88b9a12700578e2bd2f004f1147; isolation
 sha256:0379147b83f64723cda435a0d9e32604e8b0279089f23c53100da7b936d2d0d5; custody
 sha256:f54b61c19087a29a5a25cfff6134ee853b7a92faa52d78d8d6e6a1d69cb82d33.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

235. Machining-induced surface and damage state

Field: processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-003

Claim: SFT-MAT-PROC-MACHINING-DAMAGE-003

Question, law and forced result. Machining retains operation, tool, complete path and the exact surface-site partition into intact and damaged support.

PROC-003 uniquely retains surface-intact-damaged-operation-tool-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-PROC-FORMING-TEXTURE-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight PROC-003 axes before target release.

Boundary: Every positive finite PROC-003 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- surface-intact-damaged-operation-tool-path
- complete-processing-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-processing-carrier__surface-intact-damaged-operation-to
ol-path__complete-processing-state-path__specimen-method-condition-scale-uncertainty-
held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__
no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:edab33a9b316fe1240ab14747d8ca2bb7c477289709e0d084fd54a958f6ba0fa; generality

sha256:5adbf2fdcffed86d6d628e80594eab62db8ad3eb16c5a75b80ff8117a7e4a531.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:9e20b85ed018b38b09c951f2afe20ba7a2c25b122c433b25f43a93bf6431e23a.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity;
receipt sha256:df431ad505b2dabba113ac3b5f434fcd43c57ee41ff8204808401b3e5032829f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing,
duplicate or additional survivor; receipt
sha256:83f0d8d9eab8e51a088d4c357d3dc50112e81762276530ad224420367a8877c4.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values,
target feedback and extra laws; receipt
sha256:2444f341e4f0d6e0b58403be8b4afef844fba7fbe16900427a1d818fa8410977.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the
declared census and sole survivor; independent certificate

sha256:adelcf17708963ff68044836e6ee1e4691df9a62a1fc5f43cea1836b46b12773. Derivation seal

sha256:507683988be0ae537a2182284475ee4f2fa5a46d5ef27e5595178c1d59d8cacc. Source manifest

sha256:761014921cbe90b467c0eac2dd91ff08feaa9e1aa00000b08ff34356da8cdb9d. Engine receipt

sha256:997502316d6b4dc58060c367cbdca5f1e02a50ca3393f10c9bc05f73e1a2711b at receipts/eng
ine/model_admitted/SFT-MAT-PROC-MACHINING-DAMAGE-003-997502316d6b4dc5.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
Comparison passed: true.

Source identities:

- NIST-SUBSURFACE-DAMAGE

Measured, observed, adverse and boundary records:

- PROC-003: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-SUBSURFACE-DAMAGE: characterization techniques; surface and subsurface damage

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:d23fbd0ea614cc88dbe356ac4cb66a794b7bec56c70fd3788c861232a5d497d; empirical validation
 sha256:c50219612fa9782d1030f81731b5626a9712062d53d6a136574a30bae2d73e5d; external validation
 sha256:057202cc1bcf6f51ee8921f080ee3eb84a240c558283326696e10eecab3b95c2; isolation
 sha256:3503537c6701a7eb5a02f89dd7359b27155acb7ae37d0943f01986232f33026d; custody
 sha256:eecee021a2b8bce5cb8c684c31773a6f96e2185933b535dc7cf30c85ee6928a1.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

236. Additive layer-build and melt-pool history

Field: processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-004

Claim: SFT-MAT-PROC-ADDITIVE-BUILD-004

Question, law and forced result. An additive build is the exact ordered word of layer, melt-pool and powder-batch identities together with the complete process path.

PROC-004 uniquely retains layer-meltpool-powderbatch-build-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004

- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-PROC-MACHINING-DAMAGE-003

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight PROC-004 axes before target release.

Boundary: Every positive finite PROC-004 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- layer-meltpool-powderbatch-build-path
- complete-processing-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-processing-carrier__layer-meltpool-powderbatch-build-path__complete-processing-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e5c22d84482a735bd68e2acf0e60c35ce1b4d6afcd3c2b6cfb059de7e7e50f9; generality
sha256:0551158b3f37e2fd47dae19bd8bbe352c3120930811dc6bce1314fbd0eecb04.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:aa9b82243812ec46f8c0898970c41be092bd19ec403248c503d08566205f2e42.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:59375d9f7ef7e37cfff5c3024da677fc78399229a79c5dc261cc127c7e2581306.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:09570a8414becdc0e8eadde3bc29ace0c5e70692fb4290cd5e26c702d191612b.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:2efee92d96138ebaed01c45a64dfc54637740d94a1c519c80f0ea61f61f1cd1.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the declared census and sole survivor; independent certificate

sha256:df316408f3b388317459a9cc1d84744872eedda56332f10ca3fda75d37c7d874. Derivation seal

sha256:335c46c9cad6f4b1868f20d873aae90f19670ec8ea149286d899685a834d42f5. Source manifest

sha256:52afe21303fc861ebdd22f00345845314572205dad98d893e87416eae7ccadf1. Engine receipt

sha256:e7d4f7875ee957bedb5077f32565db67a385cff2c55ec22edff2d739c7bd4b3d at

receipts/engine/model_admitted/SFT-MAT-PROC-ADDITIVE-BUILD-004-e7d4f7875ee957be.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MELT-POOL-COOLING

Measured, observed, adverse and boundary records:

- PROC-004: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-MELT-POOL-COOLING: Cooling rate is a decisive index; melt pool solidification

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ec5ab0d65a8e94f33a3d0eac6f5615607aa8cc21ace7214acaf7d0cbf4145873; empirical validation
 sha256:b35d50f044e3e8de2e5da2766060ff18c9181377a9ae50818bf16fe515285fba; external validation
 sha256:6b452f2500b0cb7fbd6cc1b1f7b1f869bfc792c46e514ecd52d4638e86469fa1; isolation
 sha256:f8ff4c8d6a1904d90a64f971e1a31b84e308f2f7cd9b367e8de09fdf9fcaa4a8; custody
 sha256:2ee97b547030fa5a9bf70c23f239f285cd8815e760009f0fd233b9db4aab1742.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

237. Thin-film deposition and growth

Field: processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-005

Claim: SFT-MAT-PROC-THIN-FILM-GROWTH-005

Question, law and forced result. Thin-film processing retains substrate, ordered layers, interfaces, growth states and deposition method as one exact history.

PROC-005 uniquely retains substrate-layer-interface-growthstate-method with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001

- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-PROC-ADDITIVE-BUILD-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight PROC-005 axes before target release.

Boundary: Every positive finite PROC-005 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- substrate-layer-interface-growthstate-method
- complete-processing-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-processing-carrier__substrate-layer-interface-growthstate-method__complete-processing-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:fbac6647e6be88923ce9ccd240b882aa9f442a3bcd3e6e7ee42c8dea2e17742; generality

sha256:2f55a8d47524b68b0082228dd00069aaf6e6ee5d98fcb2770c42c4c986caf0cb.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:bdafacea1950d435c934f85c4f345ba0d2c8e3fec7ec0fc56a957c9aef1cb9544.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:bbf9353fb27a29849f3764b04516af954c609b64704573ffdc0fa40214a64bd0.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:f8785954c49b1ab8676f5af28ddc4f17d5dc743c4f300f550ea928ff248b40f5.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:649e85b766b34246a70f0251a286a26f95beef83b40f9510bc3505465e815577.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the declared census and sole survivor; independent certificate

sha256:6df02f566f3940b7b22834475132148fb538ecbf181d12c1030fec71a16dbc2b. Derivation seal

sha256:54b2cf4b0dd6570da455459ef65ed0b839bd97a4498c696788a4f2214fc87a34. Source manifest

sha256:d0e33285341a0d2af8ad5315bec093654ad99b0d16946dcb30f189b642793eb8. Engine receipt sha256:7815bd4368b4df46fe44298998ceb8ec9db851a8f141edd185f437cfcd780bfd at receipts/engine/model_admitted/SFT-MAT-PROC-THIN-FILM-GROWTH-005-7815bd4368b4df46.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-PULSED-LASER-DEPOSITION

Measured, observed, adverse and boundary records:

- PROC-005: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-PULSED-LASER-DEPOSITION: Pulsed Laser Deposition; Dielectric Thin Films

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:3eaec1c3abb5bdb44d7f5a7f82777273624c55f2bb6b9a42ed1e7f6cdcf09f92; empirical validation sha256:6c73302f5666365df8c61520940439844db0803c641dd820e48bce67e7644da1; external validation sha256:95cb039c6e8af8d229a0b1268ff3118702f404cb0a8a919ec5df6503c1300116; isolation sha256:8db32c059bc4dd7f9c2769f85c0c2259cc247c31c50dd1738f4f82e596b2a8db; custody sha256:ed701a00eadde815ad8f1d510169c50db96e8ae2e3ff5ab70530cb95b43b7435.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

238. Epitaxial growth and lattice matching

Field:processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-006

Claim: SFT-MAT-PROC-EPITAXY-MATCHING-006

Question, law and forced result. Epitaxy retains film/substrate identities and orientation while complete positive recurrence enumeration forces their least joint lattice repeat.

PROC-006 uniquely retains film-substrate-orientation-least-joint-recurrence with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-PROC-THIN-FILM-GROWTH-005

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight PROC-006 axes before target release.

Boundary: Every positive finite PROC-006 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- film-substrate-orientation-least-joint-recurrence
- complete-processing-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-processing-carrier__film-substrate-orientation-least-joint-recurrence__complete-processing-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:115c48a77aa02241a6f82dd185c67e562ef5c9d992a2ce9cc6800354d9cbabb3; generality sha256:f5ad2dc267ecef2e63b26a3ba26af035b92507f6cacd36ac30722b4ebd38bae1.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:5a4e260fe1761de1a85d20f8f269d7a8b0d781c4c01bf50d3d3909e4214f1f0f.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:667532db4c363d8b17b725a2b74c0367c46642110fb1dfcf57b24ebba468c66f.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:e824eebf26f17471c561b6e1b5484d09d84c62dcefdbca89c37a9d23f91db4f0.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:97e24e9bcaf461ae56a26b1066da2c6a9028fb91398646bb00464728e31a3ba2.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the declared census and sole survivor; independent certificate
 sha256:27e53a617e100e6519b742641ae98a08a589b6a654de8286a91f0e33875eab44. Derivation seal
 sha256:e1b169f565b36108ffd8f14de9555df5af0bb2f09ec1cd4d769dbddd7aeabdc0. Source manifest
 sha256:6fb206b3a96dc9a76e15b91a9effc6b5ec91963547cc84757d4a7524ed6e86f1. Engine receipt
 sha256:218ceb30fdd566cc5fd54f49afe4ec745fb1c1a2f0b9b4b75745a35e8df65c0b at receipts/engine/model_admitted/SFT-MAT-PROC-EPITAXY-MATCHING-006-218ceb30fdd566cc.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-NANOWIRE-LATTICE-MATCH

Measured, observed, adverse and boundary records:

- PROC-006: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-NANOWIRE-LATTICE-MATCH: surface-directed growth of nanowires; number of semiconductors

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0fb5de03a800c47b07feb49671a583a216747c8b5ec6346c9ce3eb8e7edebf54; empirical validation
 sha256:d7f13ea20df9ee8a8f2b9cb077768b3024707f56c4d34a3e0df7c213010d634e; external validation
 sha256:fe200ef1e7f40efbfc43a54cc25a7b1c99f77f916d1c84739f44f705b455ccd; isolation
 sha256:0239623113602dfac13f61c032354ac21f8421adfd02bedbc02db9024b1f156d; custody
 sha256:5a600c1dc31425e298fb6caf04bd2708cd984b9184a1b7a6609506aeb16d4ba7.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

239. Welding, brazing and joining interface

Field: processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-007

Claim: SFT-MAT-PROC-JOINING-INTERFACE-007

Question, law and forced result. Joining retains component identities, process path and the exact partition of interface links into intact and broken support.

PROC-007 uniquely retains component-interface-intact-broken-process-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-PROC-EPITAXY-MATCHING-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight PROC-007 axes before target release.

Boundary: Every positive finite PROC-007 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- component-interface-intact-broken-process-path
- complete-processing-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-processing-carrier__component-interface-intact-broken-process-path__complete-processing-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:cc20bef236d57a2f89249411706d47e0e25489f66f4467c719d022e1342b043a; generality

sha256:6493c1bc405615e8f37ce561e6f8c715a5f75593fe75d144d48d0f8893d63efc.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:a692ceca76f429f6f3521c103f846979b7fc5b07a9b3126cc899687fe2b0456a.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:c4402ad345b80a3045db1b2b3f72bcbe279fb80466e56f2f9b52aa90a7137b2b.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt

sha256:c8e54cceb16e14c05437f6fa453da19b0084c2c4ace470bb7947749dc6229142.

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:91b20db38efaf657d1b8e914ede0f11c67d836e9d4ae51afc2e926f77996c5b8.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the declared census and sole survivor; independent certificate

sha256:aa8b75d384d0498262b65ac0c417507ee80d339f42d05aec0d38d29eae9a1e6f. Derivation seal

sha256:29c623968f4d59c6efee76572a662084e72c094694eb2e38e6950928a1873af7. Source manifest

sha256:3ffc00369654b877e9593a08f2fa8ce19b6f0c7ae0a563faf5191eebfa5a9d54. Engine receipt

sha256:abdac12922fc77dc89ad444cbb393a2db849d7fedbce5524dee039018d0c7a35 at receipts/engine/model_admitted/SFT-MAT-PROC-JOINING-INTERFACE-007-abdac12922fc77dc.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-WELD-MONITORING

Measured, observed, adverse and boundary records:

- PROC-007: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-WELD-MONITORING: on-line monitoring system; welding current and voltage

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:b3272950367d6a0a313e82adb5dc09e0672d79ce3e4d71db0cbcd6d20103fd00; empirical validation

sha256:f83f362bf3a742e063ede569204c185696983b60238a599cc87e36e8ff39008f; external validation

sha256:9785ecb2095c87c176bbc9c1e59189480032446dd0adaad670d343f1fc319b96; isolation

sha256:a1a83aae3135941b5e1628e709199bd70111f0556687663d128c012e2dc2faff; custody

sha256:1e638f96249a18540d8bdb9311743e6d41c493b79c6d5a490d1ccae4441fdf0a.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

240. Polymer processing and orientation history

Field: processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-008

Claim: SFT-MAT-PROC-POLYMER-ORIENTATION-008

Question, law and forced result. Polymer processing retains each chain orientation before and after processing together with complete process and thermal paths.

PROC-008 uniquely retains chain-before-after-process-thermal-orientation with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-PROC-JOINING-INTERFACE-007

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight PROC-008 axes before target release.

Boundary: Every positive finite PROC-008 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- chain-before-after-process-thermal-orientation
- complete-processing-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-processing-carrier__chain-before-after-process-thermal-orientation__complete-processing-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:5d4ef32563993e3d0449643680a7bf5e950a96a05ef49413eb117559bfc0e6aa; generality`

`sha256:6ba01e2d6d3345cc517364d2c25984a5cc6f4b8ea1535a7cc9344227b9cbb0b5.`

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:77d22e8fffb1261a6bca94993773c7ce5300e41c80957b2ee1cdd71e3dcabefbd.`

- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:d5202919cc9d51b80270f557d127e7264fd6bb9e4ea588b0f74f289d3eb8674d.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:e9e041ee8f454c25ed3b03e4b8669207dfd6bf35c5f1176b0618398f6dbdb1a6.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:0e3ae47522ae9b0354e626011031e132b7aada1cd6abd77046a22f31e8ab70ad.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the declared census and sole survivor; independent certificate

sha256:af238c5a06e8586d256438d1e4b975915d63d74ab8017847ecb2a9f5ede666e. Derivation seal

sha256:fca5b43d8ba6af0b9913c0dccc3a995ca3538a832544d8ce57a08de162b6bf. Source manifest

sha256:ac74db125d4acc14121ed136060ac3a7bd9bfb3bc2c5663ae51a573d9cf2d7d0. Engine receipt

sha256:06ff2fe86a03a7ea13581eb586e697af6c7c75cd71aca2229af4fdc4176a8695 at receipts/engine/model_admitted/SFT-MAT-PROC-POLYMER-ORIENTATION-008-06ff2fe86a03a7ea.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-POLYMER-ORIENTATION

Measured, observed, adverse and boundary records:

- PROC-008: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-POLYMER-ORIENTATION: 3D molecular orientations; shear deformed region

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:82efd38359a1b7b2eb21f0e899f03427a8d347cba40fdc501bd1bdd46fa6b6bc; empirical validation

sha256:1e4a6aa4b1151b02495935eacc9346b2dcb64ad44c67652065434d1fdcb6c698; external validation

sha256:977c115e4220a2af13cb608fa3118acfa68a4a5a1d3ff541602d382338dda39c; isolation

sha256:251ba27fb0d37fc2d742d15e894cfd767d727073cabd25b656ea71768c6dd613; custody

sha256:ab6aeea2f261a2bb00dda3b976f2d0448d2e8818eed4b8afba279c12e9ed53e0.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

241. Powder processing and compaction

Field: processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-009

Claim: SFT-MAT-PROC-POWDER-COMPACTION-009

Question, law and forced result. Powder processing retains every particle, compacted/uncompacted partition, counted pressure and complete state path.

PROC-009 uniquely retains particle-compacted-uncompacted-pressure-path with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-PROC-POLYMER-ORIENTATION-008

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight PROC-009 axes before target release.

Boundary: Every positive finite PROC-009 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- particle-compacted-uncompacted-pressure-path
- complete-processing-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-processing-carrier__particle-compacted-uncompacted-pressure-path__complete-processing-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:235e7ff688fe39c646b6dd380f89234e36b0a7205ac2c632c654c531cafb21c0; generality sha256:82ea4d0790776fd5f8f50708031f3c003168c2a53c523422b98266f00a89702a.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:626515cc4d26893a558b73c4946ac340d975d2719165999b2e043a5498d1fa2b.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:7727e6bc6ce9055e9dcbf2a4706bcb5b1f5d2f3d0f715a7c94296700261df055.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:8fb163a682312227264bd700ecf36ab911acb95beea6a3cfd6c6ef9e568e1a4a.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:b4e00da0d94ef2a6d40f92a8a9c8112e1dffc49b9fd823c8ea3f2b0e0159a34f.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the declared census and sole survivor; independent certificate

sha256:ba149c0f4997d4a67e7b4f0ae158273627380760764f6091374fe6a4faae19fe. Derivation seal

sha256:6ecb60c9f0d6a29dfdf01382d764611bd52ee2c1df6d1034cead7a81ae181903. Source manifest

sha256:bac141397eb41b3cbe84ef3c5bb0ccd1e92a7725925737a400a41c7c16ae431. Engine receipt

sha256:791d6c4a1d40f5e1f204718863833dc59f1ebe757f5511f318aaea9b7fbdb9e1 at receipts/engine/model_admitted/SFT-MAT-PROC-POWDER-COMPACTION-009-791d6c4a1d40f5e1.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-POWDER-COMPACTION

Measured, observed, adverse and boundary records:

- PROC-009: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-POWDER-COMPACTION: processing of nanosize ceramic powders; sintering them at low temperatures

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:aa8eeee13c336a81301013aa564c31a6b0f175a0b3c92eb8ee972b2c0fc37619; empirical validation

sha256:3c40b5c5815964a0ecb4c4b3bb1f546c0cb650752a935cfc2ac07bf2b1f0043e; external validation

sha256:6dd48d746d5880a8be58a18d268cfe03e18202d5a30f04f71b522d332583d45d; isolation

sha256:37f8827bd0134af8229cc31185254b4596469562af6cb0456997d9e494018abd; custody

sha256:3c9cfba838db3e6e45091379fd9a7ae9134e08fde0b71bf10e8ca3a38c75e360.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels
- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

242. Process-window provenance and reproducibility ledger

Field: processing_solidification_sintering_heat_treatment_additive

Obligation: SFT-MAT-OBL-PROC-010

Claim: SFT-MAT-PROC-WINDOW-PROVENANCE-010

Question, law and forced result. A process window is the complete trial ledger of exact conditions, outcomes and provenance; reproducibility is a retained repeated class, never a selected success.

PROC-010 uniquely retains trial-condition-outcome-provenance-repeated-class with complete carrier, path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-PROC-POWDER-COMPACTION-009

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight PROC-010 axes before target release.

Boundary: Every positive finite PROC-010 carrier with complete material, layer, interface, orientation, state, path and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-processing-carrier
- trial-condition-outcome-provenance-repeated-class
- complete-processing-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-processing-carrier__trial-condition-outcome-provenance-repeated-class__complete-processing-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:110cf71f11ca11c90aabe1129814902354403b0af457b27d948a6f025748b2bf; generality
 sha256:6ab62abda767ec9d184f782738d9383f7474a1d79050fedbd9896d4046fc60c3.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:48f34a58165f3135ee0d885ec9425e2735920581fc123bf4950c9c5978af6f36.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:98bc48dd7da8fbac906da6098359079ed08eb0443250a51709cf23c1afda0247.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:1ac5c151ddc31f875aedd60788262b68305b81eccc0648c5efe9d06cbfe79cfe.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:bf3b35b3b9a92ed9b5b28dc5ed8579794ebb6ee0c0e23d8ec393fead9f3a95dc.

Independent reconstruction and engine admission. Independent implementation

sha256:55f69621490196265bd06bdc46ff01b4ee86071568d75ee3e30a7066c194750d regenerated the declared census and sole survivor; independent certificate
 sha256:f5398ea64098d18b0fa62c23415b9f0a2341113b0b9e696c38c10881d7f09a4b. Derivation seal
 sha256:0497f6dac35d3c4c648856bac089fc191a1924c3f239a632d6373fd129a1aeefe. Source manifest
 sha256:e7146057be38e66b834768e196cd547c1e377bd99e87ae75922c58b7b2ffcc6b. Engine receipt
 sha256:0ad7b92e5c8d24cdfc2ee2617314ee031fbf333bf5e0b18f35e9809fa34d261f at receipts/engine/model_admitted/SFT-MAT-PROC-WINDOW-PROVENANCE-010-0ad7b92e5c8d24cd.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-REPRODUCIBLE-PROCESS-MONITORING

Measured, observed, adverse and boundary records:

- PROC-010: 1 comparisons
- family sources: 10
- exact correspondence: True; omission rejected: True
- NIST-REPRODUCIBLE-PROCESS-MONITORING: monitoring systems; print quality; additive manufacturing

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0f4139a4586a6f0b713517985f748e2bc9a715db10a4c73e2fde23e8e977d75b; empirical validation
 sha256:00a5ed1f75d933584be66dd9c133e944e634c87eaf8e3215cb18aa2df11435d4; external validation
 sha256:085c11716b7068769039eb7daa5c235efbb96471f05add1a954d1999b43c356c; isolation
 sha256:45a72bde71ec3af263c98b1421481cae75b90f4b301f86fc78ed7b7360a77645; custody
 sha256:315ddd9804b4f2ed22a84fb5dde0aa837889369c9509d9ba9047135139168601.

Prohibited inputs.

- no imported continuum process equation, fitted window, engineering choice or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every carrier, layer, interface, orientation, condition, state and path distinction remain held labels

- no external outcome selects a survivor
- all result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Computational Materials Structure Property Inference

243. Exact material-structure data representation

Field: `computational_materials_structure_property_inference`

Obligation: `SFT-MAT-OBL-COMP-001`

Claim: `SFT-MAT-COMP-DATA-REPRESENTATION-001`

Question, law and forced result. Material data is a finite set of unique identities binding exact structure, property and provenance words.

COMP-001 uniquely retains identity-structure-property-provenance-record with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-COMP-FORM-STATE-TRANSITION-001`
- `SFT-COMP-FORM-COMPOSITION-001`
- `SFT-COMP-ALG-NUMERICAL-001`
- `SFT-COMP-ALG-SYMBOLIC-001`
- `SFT-COMP-SEM-VERIFICATION-001`
- `SFT-MAT-MEAS-TRACEABILITY-001`
- `SFT-MAT-PROC-WINDOW-PROVENANCE-010`

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Complete literal product of eight COMP-001 axes before target release.

Boundary: Every positive finite COMP-001 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- `complete-positive-computational-material-carrier`
- `identity-structure-property-provenance-record`
- `complete-computational-trace`
- `specimen-method-condition-scale-uncertainty-held`
- `complete-trace`
- `root-bound-forward-forcing`
- `positive-finite-successor-closure`

- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__identity-structure-property-provenance-record__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:c8f5407eca0a7698f67dd6d37c4131fc31ba715cdc7a865149708d482b9eb5e9; generality

sha256:a9811eaf606d18fff8f015b396ceea2fa9e1b95c3d96b6d872fe8b799578a5aa.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:f1688f9e44512e6d3874acde906ade91c1387648fd7d74ad9eb2d1dae0aa7d16.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:300737596227db9242063c2d834645b4d3260e40676dc2040580f1a33f8babf1.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:583ecd8d392f071a2f2a8a63d6b7bad799167488d177b187830cec2bd6863ae0.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:e2aaa607250b69ca3b55f6779e6aef4e9b6fb11a0d05de871463012f476d29bb.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate

sha256:20ead48bfe111757953df31cf1f60f28f81257ba5e3d2229aa508649b262bd2d. Derivation seal

sha256:67f34aeb0030cdf96f4fc28a252b64d4c85af8a967134fd197d414a3550d4095. Source manifest

sha256:d0fa3a04ad8c8369350d7445dab57ca813bbcf72dda71f6cb080337e151958d0. Engine receipt

sha256:5cb145faa284f47b049c493613a77ec34d81ce7663ab2936f6646fd7c894fc92 at receipts/engine/model_admitted/SFT-MAT-COMP-DATA-REPRESENTATION-001-5cb145faa284f47b.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-RELIABLE-MATERIALS-DATA

Measured, observed, adverse and boundary records:

- COMP-001: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-RELIABLE-MATERIALS-DATA: Reliable Materials Data; Data Evaluation

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:d216f2f8d6547c75e656965bc88c5d637d80d625e4a82e78eb59107414e5a5d2; empirical validation

sha256:89e8a156cd1b1af75ec23af62ea627a82a025df3b058e10d158725883c143b51; external validation

sha256:ed4eb253d53647444616b3f18af301635964353512deed95d3284a919776a53f; isolation

sha256:658a200063a3c7cd30ab072fdbcebae75b8fbb78e8f58cbea118bc09e830ef29; custody

sha256:f457a688c04dc42d8e5655ddb6496cd61bde47fe4d263c6aa42abda92f94681a.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

244. Structure-property computation boundary

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-002

Claim: SFT-MAT-COMP-STRUCTURE-PROPERTY-002

Question, law and forced result. A structure-property computation retains complete structure input, property output, method and scope without promoting correlation to a universal law.

COMP-002 uniquely retains structure-property-scope-method-boundary with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-DATA-REPRESENTATION-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-002 axes before target release.

Boundary: Every positive finite COMP-002 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- structure-property-scope-method-boundary
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held

- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__structure-property-scope-method-boundary__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:6d0e12e3e04113debe257dcc96245112d5b072f2699f29ab5ba4df1511308ed7; generality

sha256:26b2712169f25923ada2368dedaa354c6f4be9c9107a939c2f49633e1230a383.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:d8d206523cb9a666c718f449cd3de6217b21c313d3ce3f00fcfccd6f70f22e4f.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:daafa11e12c65c99b48fe67333f8ac331728f97a62ce4f02f07e143f88ff32e4.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:3ab15b9f0ad09d2a03a0e70754988b19384537b6db63262c0db3078c2e39156d.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ccf5b9205ee39d8760a7bd3634f4cadef97adde0e8be77b5e6666ef8947d103c.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate

sha256:c65b7f4f9773d83d81fcd88960a2109cfc17eee45050e112183280c454819430. Derivation seal

sha256:7ddb05ea2380e6bf22c063be72a5e663d202cf56f4eecd8b82886b0360701a2. Source manifest

sha256:184e12840958aa71513b08ad3ebbf3094fcaa5d544d5cf639db904698faf3b09. Engine receipt

sha256:119129d06d51dbe1a5ca49a20a356dacb43a10a74faea9feda46b0205c446805 at receipts/engine/model_admitted/SFT-MAT-COMP-STRUCTURE-PROPERTY-002-119129d06d51dbe1.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-STRUCTURE-PROPERTY-MAPPING

Measured, observed, adverse and boundary records:

- COMP-002: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-STRUCTURE-PROPERTY-MAPPING: systematic mappings; materials microstructures; materials properties

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:b134152e96a49bd62c057452efe7bdb159ef0ee92e37159766c1129fd06647f7; empirical validation

sha256:6c2aa3416a9846e9121953786f027d6a5406f38574dc9a5f1cfa10919dfd087b; external validation
 sha256:88a54d022616e26c88f067a19f80c542656ed7a74580ed6ed8adad5d73d7dc50; isolation
 sha256:64592867b598620821f554ec1dd18ceca2acb93d94a17bd5a9f9ac32b39a8354; custody
 sha256:1ebb4e5e3303b7680f9bf0be584781e50f9045acbbb23d94946c23f8b0d7ce18.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

245. Finite numerical material simulation

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-003

Claim: SFT-MAT-COMP-FINITE-SIMULATION-003

Question, law and forced result. A finite material simulation is the complete exact state and transition trace under a held boundary.

COMP-003 uniquely retains initial-state-transition-finite-trace with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-STRUCTURE-PROPERTY-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-003 axes before target release.

Boundary: Every positive finite COMP-003 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier

- initial-state-transition-finite-trace
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__initial-state-transition-finite-trace__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:9889a2836fd6adcc7efef37ae40929de09c138f82d5858ce99439209bbdd2b3b; generality

sha256:b0fc1d30cd5bc422b091002da99873e1a522242e5d3979430e257a81bf2d2cc1.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:eaaf840f5eeba4fedf251c7ce436cda94ec4ba4f6e9422c2f543b6687e46960d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:6f6fdba50e19399fb13832068c50328140b1bba3814db5d6b115c8ccd2e849dd.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:65ba1b01f2c890389ddc79a4793a58eeee7cb392fa2cabd874aa9e06088c3c36.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ca81fec04840a06e000dc886e069dddaceca4dd8a0b3770a0c0e1329d7066931.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate

sha256:906f87bd7177db97908c0b243ff6378e33660bddc8bdc8636705eda1213ecb8b. Derivation seal

sha256:b3b8c731a33fd9f0952ad605be161b452ca52c938ae7f562df8d7700fad4b48a. Source manifest

sha256:8f984c8dbc2e3b27215f0efddfa9dde0929fa617ac31d2feb46beb83f9e3deb6. Engine receipt

sha256:f87320573329f585ad42f15acd55ac03c449594922b20a79986553d9086acbafe at receipts/engine/model_admitted/SFT-MAT-COMP-FINITE-SIMULATION-003-f87320573329f585.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-OOF3D

Measured, observed, adverse and boundary records:

- COMP-003: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-OOF3D: Image-Based Finite Element Solver; Materials Science

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e0674dcabd5cb591a0c2aa05c8c42fff9f02e31796751cfdc49e7c4f106182f9; empirical validation
 sha256:d6750c6c736dd3c389344a1b09bd6408da414d74dff5c3e45ac62c2c65fdc523; external validation
 sha256:7df92b7a2c116636b799c7325ad58862fb0f0329b9d514fc91d8a6ef0a8f10f7; isolation
 sha256:813831e45499ec4af48f47425ce735bb46317d0f93938b9a4493925b3a2943cc; custody
 sha256:5345ca6a9f96b7382b76befef73d09c151c29edb76880da6baa4dd3d6d40c154.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

246. Multiscale model composition

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-004

Claim: SFT-MAT-COMP-MULTISCALE-COMPOSITION-004

Question, law and forced result. Multiscale computation retains every scale, model and adjacent handoff map; composition cannot erase intermediate distinctions.

COMP-004 uniquely retains scale-model-handoff-composition with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-FINITE-SIMULATION-003

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-004 axes before target release.

Boundary: Every positive finite COMP-004 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- scale-model-handoff-composition
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__scale-model-handoff-composition__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f661a190aecbaa420d249451426870ba4b4b15e32eeeb9cef9387b0908265e46; generality

sha256:87e6580f2e65b6fd7d71cb62b93cf26c70e3ee7d56b0279b4a4f18669e1a6664.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:bea79bbb92058f6ba245baa2c883a084f15bd17f2e9909088cda6f1170e8ae2e.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:6af0dc66fa8bf72e2058fde45b3ce607a64f8a85188a0341271a25e352022a71.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:3364fc803b693fb3a621a54430317beba065b853f1432afab2d6ed099aa5b11a.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:7fa26897276a64a4c9f93f9a0c01f06414f0efe8f765fe7a9fbb2eb6781b3dfe.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate

sha256:38f1ec15933bc1740ac6c89c8828589dba5224721eccd9004ee1feb98402866b. Derivation seal

sha256:bee2d661d6b259c62865fe27827da704a58fbd161357455666f201c44b84f241. Source manifest

sha256:e96743418cb23958fd5aaf42f6075f32a0bd05c022a893ae3264c0ebe98da66f. Engine receipt

sha256:9eaf30305b42d6083076be451c45d63bc720441ebf1ac310a03aedfa68ae35b9 at receipts/engine/model_admitted/SFT-MAT-COMP-MULTISCALE-COMPOSITION-004-9eaf30305b42d608.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MULTISCALE-GREENS

Measured, observed, adverse and boundary records:

- COMP-004: 1 comparisons
- family sources: 12

- exact correspondence: True; omission rejected: True
- NIST-MULTISCALE-GREENS: multiscale Green's function method; two-dimensional nanomaterials

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:b7b733e023a2312d2c94921da2085bf7c190002433db760ba415603d526f86e6; empirical validation
 sha256:625b94b00948686a832fb5c57f6828fef77483a9d4558f76cd1bf617e76d2781; external validation
 sha256:3c7fea51f3b03ca3206360cee8ae43fb287f43c6a89e3d786e1725927ef074b1; isolation
 sha256:d092cdce9059a965d28ebe7ce2002603a628c6d48dbf6cf459deee01fc68a894; custody
 sha256:a1b32a7c48f805a702a02b5cfe2f79448e52f19c95882023ad0f4c9d6fd4c7.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

247. Numerical stability and error propagation in materials

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-005

Claim: SFT-MAT-COMP-ERROR-PROPAGATION-005

Question, law and forced result. Computational uncertainty is an exact positive component ledger whose declared total, scope and propagation custody must close.

COMP-005 uniquely retains positive-component-total-scope-error-ledger with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-MULTISCALE-COMPOSITION-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-005 axes before target release.

Boundary: Every positive finite COMP-005 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- positive-component-total-scope-error-ledger
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__positive-component-total-scope-error-ledger__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:1586d8c23db210bb34cd47175fe913797345b1e4baf4905bd6620748dc62610d; generality

sha256:41d7d984fb0394304851661e2a503f6ba813aa3d86909e8525639c7099dfefec.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:7826b8dbed343a3d981c8afa0c72fba8f44819f1f54ab3560a9bcf9c6dbc91a7.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:88795e1edefd47a5e02fb671c046ec244236c68be8ddc6fa16e4d5e468aa5a28.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:6cacf14847333f2c992bd628ce3701a9fb13b7580843340dd2cfb4529541490f.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:0b738a45c072f1b9597476cf9b505f3c9003f2424884bc937106d8c70119037a.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate

sha256:4d64e0f4525ea839929c7c3d98507c38c98538c6e05f617b7115c17e07497788. Derivation seal

sha256:6043e657a7498c7dd8cd5a4af0f9ab9ab7074fa995232e35ae795cfc8256de24. Source manifest

sha256:c4890d9ed8d0193e8a74ef50afaacb00bd3783685eef05baca8f546f50f52b70. Engine receipt

sha256:0e666132b5a7ea08e840c4c151af4249d039012740824d80a4a221df4f638b02 at receipts/engine/model_admitted/SFT-MAT-COMP-ERROR-PROPAGATION-005-0e666132b5a7ea08.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-UNCERTAINTY-PROPAGATION

Measured, observed, adverse and boundary records:

- COMP-005: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-UNCERTAINTY-PROPAGATION: propagating errors and uncertainties; first-order Taylor series expansion

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c6ae4612fbf0d0bfff28eecf61beb43750ae588668b41eaf9bbdaf994b9d42fa7; empirical validation
 sha256:af6e94cbc7751f322dacfd177e599717819e0e0ab02be6c6ed71bd2a99db2974; external validation
 sha256:67b4b9872d3f343e60d28749002b6e979b1b557cf05920d82c64e3b59024ca7a; isolation
 sha256:3908edee45c501a5bc5ef59b11e86fe73500945c2fba514c9f4eaaef9378a259; custody
 sha256:9e9ae17ec2691d003b9189a706d0c7dc90c8a092e893794d01779749b26c463c.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

248. Inverse materials problem

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-006

Claim: SFT-MAT-COMP-INVERSE-PROBLEM-006

Question, law and forced result. An inverse material result is admitted only when complete forward enumeration leaves exactly one candidate matching the held observation.

COMP-006 uniquely retains candidate-forward-target-unique-inverse-enumeration with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010

- SFT-MAT-COMP-ERROR-PROPAGATION-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-006 axes before target release.

Boundary: Every positive finite COMP-006 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- candidate-forward-target-unique-inverse-enumeration
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__candidate-forward-target-unique-inverse-enumeration__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:8257aa9335a5aba6885b7bdba03c31ba3c7672d82ee9beb37e2c973a8dda5683; generality

sha256:76b86ba7511a79469161543a745d39a94632ea31d3a58dd19cd048c312edfceb.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:f7b57b9bbfd815a36f5a6de582de5f045b7f5125462679f6517b45a822587a70.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:60e4043275057d76305879be5f20ef714ed843abdfbe3eccc8d5d406a8bfe6b2.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:29ccce2ac74767b9e3ae0c7e267b39710bc011d6c914bd61d19ab970ba5eac24.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:e25b313b946206b6217fb2dd8092f29a78ac33bed4f806d69b5af4d81da4b638.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate

sha256:6403eb28de57a22d1ecbacd6d8c6fd4da4c46ee99ed24827ef4d1aa27037f565. Derivation seal

sha256:8e6079936d01834f4b858b271ecb077f5ff494ec84e907bca8e60f1e080c2fa1. Source manifest

sha256:c77abb760a79fefeb617f5580b405e3d6526ad75e9283fe492b0e1911a4b315c. Engine receipt

sha256:c13fcf6f8de3af8b86bc8de025abd71956b434ca3ae754301e90d8829321bec6 at receipts/engine/model_admitted/SFT-MAT-COMP-INVERSE-PROBLEM-006-c13fcf6f8de3af8b.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-INVERSE-HEAT-PLACEMENT

Measured, observed, adverse and boundary records:

- COMP-006: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-INVERSE-HEAT-PLACEMENT: energy beam; powder bed surface; consolidate the material

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:9d2a74581f89d2674cbca4eece38a11d6d3ae6994310d144b4bbf95f1e5614c7; empirical validation
 sha256:fd74cc1013788be3c299899ecb724acce190625d22b27742a909ed8c169400fa; external validation
 sha256:da7c1fcd1b89d41844e2e4973c0c129c312db3506911f093dd11999611129e67; isolation
 sha256:fe54297160e5bfe244a474e888d19aa6dac4d7918aa7b24e8ba48f163981bbab; custody
 sha256:00cb225f92109a33e71906f379a3757eda1c2118994d20c3bd746b6d7d8d9f1f.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

249. Machine-learning materials inference boundary

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-007

Claim: SFT-MAT-COMP-LEARNING-BOUNDARY-007

Question, law and forced result. Learning retains training, test, predictions, method and every correct and incorrect row; an opaque score cannot select a law.

COMP-007 uniquely retains training-test-prediction-method-all-outcomes with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001

- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-INVERSE-PROBLEM-006

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-007 axes before target release.

Boundary: Every positive finite COMP-007 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- training-test-prediction-method-all-outcomes
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-computational-material-carrier__training-test-prediction-method-all-outcomes__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:fb44e15460c3fc643dfc142dfa3b0b9921d834f6d2f2ca1cd37344ed8e71b07d; generality`

`sha256:a00031df13a5fe0d1d1c5ec5bd73a540ebb14d4bae1672f63459821edeb26741.`

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:e7d73777eb3153e298b3c4766cfcebc137e5862f62072105f163077d6087792f.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:12e1233c5c4780a701916c5f46aad346be863a52790c059ba7c233380274c7ad.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:95a2f0936e0d7755d88b9483bce966a277833fdaddf8634f6e534b27434a0306.`
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
`sha256:7559e78eca0a90e95cbbdfb7b56680dc5d2176ffd402f9bf1d6b9dd2a07b736f.`

Independent reconstruction and engine admission. Independent implementation

`sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c` regenerated the declared census and sole survivor; independent certificate

`sha256:0e06ef83caa03116c4bc2787435e0de033eb85a6a6c3a6fd1799ad2b5051d07b.` Derivation seal

`sha256:e6b09603ab67ac2bd25992c46b3a70c21d42048d7c360d36be825947fad4c2f1.` Source manifest

`sha256:c2e694189289fb37ec85c0a9b8304934bf83368822346d7fd6b369fda69f04e8.` Engine receipt

`sha256:7b230e12ee2be28170850c668a9ae42e26541908878b91a938a1f1c649f07da9` at `receipts/eng`

ine/model_admitted/SFT-MAT-COMP-LEARNING-BOUNDARY-007-7b230e12ee2be281.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-ML-MATERIALS-ROBUSTNESS

Measured, observed, adverse and boundary records:

- COMP-007: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-ML-MATERIALS-ROBUSTNESS: robustness and generalizability; material database benchmarks; excellent benchmark score

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:1f392e500c7210d12e10ef2e424227e11b20bd68f2d60931f7b6d58767be5e9f; empirical validation
 sha256:2c7f334d94688ce7969470a21328168e42d386200d3031280c622e28f0f256bb; external validation
 sha256:4f15ea4556548eb4824549631866a8a2b19f415e1132b7c079fad4f48da1ccd9; isolation
 sha256:ad7f8c7b4592bdaa09f26a0031318f2f50efcf5d1f902325fb903d6ac19046aa; custody
 sha256:c323e71b5cab61f830357e73b188ba50a75a291bf4d754c1ce9b40c275040aaf.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

250. Materials database identity and provenance

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-008

Claim: SFT-MAT-COMP-DATABASE-PROVENANCE-008

Question, law and forced result. A materials database is the complete versioned set of unique record identities, payloads and sources.

COMP-008 uniquely retains identity-payload-source-version-database with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-LEARNING-BOUNDARY-007

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-008 axes before target release.

Boundary: Every positive finite COMP-008 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- identity-payload-source-version-database
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__identity-payload-source-version-database__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:2bbb54b3180737d17dedca851d00375331cd57d4e2e0fcf6e51070cff4b93680; generality

sha256:a90eb20915da157ae38aa64f011bca05b8c6c2ed4a3fe07ef0362f7e51889618.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:ce28a1d0fe7ff5da2836d289dd4b7d9e169bcea1bb797ab73192354ddc3dda9d.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:d0bf493d4c925d2024113cdab23adee0159ba6bad007e4835bd323773b3ca258.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:cf759782f24c2e8ca2d4d25ded01d3d3309f4443077711dfa135fd83a746b4d0.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:3f00b2929d2fdcfb899ed3bb8699d3267cee3ae0b93cd4da216e5be23a26cf7.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the

declared census and sole survivor; independent certificate

sha256:3b22146f3828d844d947b822c42489cd00ef9664cad644f7cbcdffdc93325e2. Derivation seal
 sha256:9b0501a23b87c19e5f02281efabb7112d5113731edd8806d9a1b7e0663ba424a. Source manifest
 sha256:0329c28ba7f943c8265b4095ca97adb6bec6628daa9a136346154bdfac2c5166. Engine receipt
 sha256:ba44476128acc010777e8bbc793004d9c167b221bb93a6f808a8c2d3bfe61839 at receipts/engine/model_admitted/SFT-MAT-COMP-DATABASE-PROVENANCE-008-ba44476128acc010.json. Closure
 scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-MATERIALS-DATABASES

Measured, observed, adverse and boundary records:

- COMP-008: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-MATERIALS-DATABASES: development and mining of materials databases; fatigue and corrosion experiments

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:5f4ebecc4077380f4313aad43a44fa9d301abbc0ba0167a4c5ab6cbea8bd9dbd; empirical validation
 sha256:94c3b43adfc838d56f71970c006096cbf352a7445d9272bfff8cbd815c9f1211; external validation
 sha256:f9ccef585dc89c923c6bd8cec7f762a9c82449eb4f66ec7d2ee9e5ed46d1e199; isolation
 sha256:3d8da2f9a888837eca75df9f330dbafdl145950388312ae7de8d9dcc7ca708d59; custody
 sha256:bc170a00abbb6f0b2e9e9b4461d9052a0dbbaf688f2e9bd46ae4343095b5295d.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

251. Phase-field computational correspondence

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-009

Claim: SFT-MAT-COMP-PHASE-FIELD-009

Question, law and forced result. Phase-field computation corresponds to an exact finite cell-state word and complete update trace under a held boundary, not an imported continuum premise.

COMP-009 uniquely retains cell-state-update-boundary-phasefield with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-DATABASE-PROVENANCE-008

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-009 axes before target release.

Boundary: Every positive finite COMP-009 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- cell-state-update-boundary-phasefield
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__cell-state-update-boundary-phasefield__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e891b80ffd0417e911a497a5c3304b4cfa2eec187045d3d669c201fb01699449; generality

sha256:5bab8e71b26ced4e09d1c5bd11befe219d143ecaf6ec66c77c42591faeb4d4b4.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:700b119a101a7ff0ba9e1f83ae26b0ddfa4488395aa3eef0f726f1333e51d294.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:9f70a6b6832d30153d6b661e84126433601a2ce50a4cf4b9474db49defbe5694.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:ba8f9c0b9bbd9d24a0c7c2accba07451b801d4d522285b64ba108c71a815028.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt

sha256:5b9f2ae1386f2fc6f0de546df01afed77d116c71c6068400e31635a86219fbb6.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate

sha256:e67182315c59f69c706845dfa818bcefe956a01bd2ea3eac5d460558c167d7e6. Derivation seal

sha256:0439b7a7b65c87e8a8b0eea4fbd99397ea35bfecd9db8cc988e2823eebffd1ad. Source manifest

sha256:4b3528d3530d027ff7d6f0457e850a3f7b4e20adc0ddcf466312683552c4205e. Engine receipt

sha256:4f46b3aba70bc6f456cbb7151de65fd380d8efa5a2eb386c0be4eb5096c2990f at

receipts/engine/model_admitted/SFT-MAT-COMP-PHASE-FIELD-009-4f46b3aba70bc6f4.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-PHASE-FIELD-BENCHMARK

Measured, observed, adverse and boundary records:

- COMP-009: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-PHASE-FIELD-BENCHMARK: benchmark problems; phase field models

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:97b0465b9459feba4049347f1c458d7a5583efb31bcf19d3932d829824cc3d47; empirical validation

sha256:405d69ee85e362a61a0517c162de3725351697d3f25f1e6faddf7d8a01d9a8d2; external validation

sha256:148f9d2c9b3a9488653469d78a5d8fa3525c7d8da90a8dbda4f3cae8bc69bfdf; isolation

sha256:bc202dc23d33c0a3a75679de4613e532cd4a49c02a4cff5d5047d338b62bb4a2; custody

sha256:32e53635565d34e6dd1df6ea490971934a9915394f7e2e48ffba79204f4b89d2.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

252. Molecular-dynamics computational correspondence

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-010

Claim: SFT-MAT-COMP-MOLECULAR-DYNAMICS-010

Question, law and forced result. Molecular dynamics corresponds to held particle identities and the complete exact state-transition trace under declared boundaries.

COMP-010 uniquely retains particle-state-transition-boundary-dynamics with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-PHASE-FIELD-009

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-010 axes before target release.

Boundary: Every positive finite COMP-010 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- particle-state-transition-boundary-dynamics
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: `complete-positive-computational-material-carrier__particle-state-transition-boundary-dynamics__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:d3da213cffeefbb6a93b4e395d24c8e5ecad147878f1ba5ab85f725ca3759d92; generality`

`sha256:f6c15db2a38805a6072ae9fc7f5e36c942ab8758c04e0a750eb308afbc02a15c.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:51ecad0f42a3a48060581d87f65d0d3ae6e35e07e78c8486bc542fb742fd5ed8.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:ab83ed6bdc3ec60738225f68363433eb250afdec9e3b1dae1273cc5a6acd5931.`

- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:9ffbb15ce9cdd01e168ee93025949c8cf90529c57a13bd74bddc0893cd748833.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:f383f428a4844e37e27e64b109afc58a539fd9927b1b8be22d5331e5f99391b0.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate
sha256:1458f0f1503ac16533f10a2a1bd79cec61ab885f390108f0eaa119eccc4db836. Derivation seal
sha256:9f71aca8df95a7524b2d657172dd99cefeb8bee199e665e1f8a0d83c36cea8c7. Source manifest
sha256:b1888f8758f91d1e18696fdd3b249fa5b76a3448191a4bad8a96c351cb9d840c. Engine receipt
sha256:75edb43db5151fd1f75084e2d7af29940612577167b1aba0bfb552d0715d5333 at receipts/engine/model_admitted/SFT-MAT-COMP-MOLECULAR-DYNAMICS-010-75edb43db5151fd1.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MOLECULAR-DYNAMICS

Measured, observed, adverse and boundary records:

- COMP-010: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-MOLECULAR-DYNAMICS: molecular dynamics simulation program; single sequential process

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:4aecc31bc853327ac8d4c84f99937deaa0c7bef7510ec6ea2a0ec1b71d6fc871; empirical validation
sha256:6d3a3878b9463def7123dfbb1cee60fe255de3a3e7201b32b8915af920627de6; external validation
sha256:17663a3af715de639ac5d54d09d1852fc4987efe6ab856066fa38cb998ae9048; isolation
sha256:044be213c4ca201b702230380dd803a3083fe19a8f63dee2bf2347e3986d6cb6; custody
sha256:90c28901e8aa2efc7883b855cd9ec5a937313d25258f78270f608c79a740dfb6.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

253. Electronic-structure computational correspondence

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-011

Claim: SFT-MAT-COMP-ELECTRONIC-STRUCTURE-011

Question, law and forced result. Electronic-structure computation retains sites, orbitals, positive occupations and method as one exact correspondence record.

COMP-011 uniquely retains site-orbital-occupation-method-electronic with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-MOLECULAR-DYNAMICS-010

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-011 axes before target release.

Boundary: Every positive finite COMP-011 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- site-orbital-occupation-method-electronic
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__site-orbital-occupation-method-electronic__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:f66d42e690a78c8c9b60868d1372c884028ad74e61efc209598b1ada5e8447de; generality sha256:b2667ea54abb1c1ee915dc34370f8cc6aad9e6622073f6f1bcc2924eb6ec5aa7.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt

sha256:80aa653cb434aa74f4b0ef1d8c3861e6aa38f096af0b2cbd823be5c314475d98.

- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:86271f2d78b15eb6c204d5daf60786f0654014a127270ded7bde33b38d2d8fa3.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:6b1a5254ffca0690bedd807eb6cf56e80070c0a1a337c8f402c210d16db09b51.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:15582fc1b507b32dbe2ef2cf0234a89c5954296e427a8eac23b3918c91be8e5a.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate

sha256:ddd4187506e8d4bd262dc5a8808f83b53a17fb5279f84cfb345999170da5045f. Derivation seal

sha256:8b782cb0732c11b0554ab8cc28a74dd105c80f2dddefd7050afa56340f44cb61. Source manifest

sha256:fa3351f3cbf92c2f3796c7762a46d54de35998e0a131bf140a0d81f81328fa6f. Engine receipt

sha256:636d9371895a27cf6fae2e5f7a2b050441799b66b3c9d65b42116ffc0af89c5b at receipts/engine/model_admitted/SFT-MAT-COMP-ELECTRONIC-STRUCTURE-011-636d9371895a27cf.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-COMPUTATIONAL-MATERIALS

Measured, observed, adverse and boundary records:

- COMP-011: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-COMPUTATIONAL-MATERIALS: Computational Materials Science; design, processing and performance

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e04fea0cf81461aa3ad389ac509a3bb933997aba3fbae7c404f91a664ef7ccaf; empirical validation

sha256:aa04f1d9b0c0dc86060d03ef4878e171d5ffbaa6110aa605188e63c858e49a4c; external validation

sha256:b3fb1a943023bfae4d665bd3628e4a842057143d5c8c193f704fa1e4e02a28b1; isolation

sha256:6213f779bc7772da78ecc469d533e346526a94e0af6bc195bffc43bac237d71e; custody

sha256:62198b7143bef6aef19096640d37f88f5b6bc09e0db36eeafc64e5d07210d3c2.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

254. Simulation-to-experiment validation ledger

Field: computational_materials_structure_property_inference

Obligation: SFT-MAT-OBL-COMP-012

Claim: SFT-MAT-COMP-SIMULATION-EXPERIMENT-012

Question, law and forced result. Simulation validation retains every predicted and observed value, unit, uncertainty, source and favourable or adverse result.

COMP-012 uniquely retains prediction-observation-unit-uncertainty-source-all-results with complete carrier, trace, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-ALG-NUMERICAL-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-ELECTRONIC-STRUCTURE-011

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight COMP-012 axes before target release.

Boundary: Every positive finite COMP-012 carrier with complete data, state, trace, provenance and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-computational-material-carrier
- prediction-observation-unit-uncertainty-source-all-results
- complete-computational-trace
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-computational-material-carrier__prediction-observation-unit-uncertainty-source-all-results__complete-computational-trace__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:9174d8cf24f1590b9bae05963f094d50739cb83e3f4c7ebdbc429d910bd66102; generality

sha256:b9f1ff0a3f56b366a5b200a5beaea65666bc787a88ed41d6c0acbd5c588b71dd.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:457b27605224a6a1219d5de969ce11387fa9288650c0beab1fc9b81ae9c0468f.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:8827851952bbf297c61473c8cac44d1be370ebb123ff972cd9b76a7a056c5556.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:327418b1a313c15edbd25a6444dd47c8fe3a2a476acdbd5f80a14c39b4eeb73e.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:253c0a68c58f1e6790a21802f1067f7c49c7c7f756583a5012bf735ee95cd340.

Independent reconstruction and engine admission. Independent implementation

sha256:d599131bbd5b12840a1dd37303152a8bcb68efc7a748e59e75456174d849d87c regenerated the declared census and sole survivor; independent certificate

sha256:70e68ed63673896e812b74830959d91229e83218e28fb6e21a4039726acd938b. Derivation seal

sha256:90294227e8d5a4779ee311ad6282f37314c43b663e1e0eb98bcf246d8ef94956. Source manifest

sha256:0088670d50b040dcf2315c0a0f17152903a7420c0aeebb89cca5a481e899c746. Engine receipt

sha256:eaf13f8ed5c6bc5816bfc32ebadb35bf75f26a473a6faa77dc11f344afffbca at receipts/engine/model_admitted/SFT-MAT-COMP-SIMULATION-EXPERIMENT-012-eaf13f8ed5c6bc58.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SIMULATION-VALIDATION

Measured, observed, adverse and boundary records:

- COMP-012: 1 comparisons
- family sources: 12
- exact correspondence: True; omission rejected: True
- NIST-SIMULATION-VALIDATION: complete and correct; before jobs are released

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:04f2e64e15e85a4110b53a5c23ac0adb3e52d47cb9c8002eb264692576b716b4; empirical validation

sha256:549a5c2bd3059d110c5b4d9a4f6a7143234bb3bd5db82a3a0c67fdbdc47d8d0f; external validation

sha256:4b49578279ad9bc2b5d01fb5ebee4d6cfb948f7138368e5c0dc1b275ae7d51c; isolation

sha256:1009ff62a0dc966f16601bc80d2b011456181140a366a3ae161bff1c9736e1d7; custody

sha256:27ccd0b46b7455b354ea5cde237fd43f3bd88f29d183a905cbc23d1a76d4b9d6.

Prohibited inputs.

- no imported continuum simulation equation, fitted predictor, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every data, state, scale, error, provenance, condition and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Materials Under Extreme Conditions

255. High-pressure material state

Field: materials_under_extreme_conditions

Obligation: SFT-MAT-OBL-EXT-001

Claim: SFT-MAT-EXT-HIGH-PRESSURE-STATE-001

Question, law and forced result. A high-pressure material state is the complete specimen, positive pressure-step, state-transition, method and uncertainty path; pressure cannot erase phase history.

EXT-001 uniquely retains specimen-pressure-state-transition-method-uncertainty-path with complete carrier, condition path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight EXT-001 axes before target release.

Boundary: Every positive finite EXT-001 carrier with complete specimen, condition, state, path, method, uncertainty and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-extreme-material-carrier
- specimen-pressure-state-transition-method-uncertainty-path
- complete-extreme-condition-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-extreme-material-carrier__specimen-pressure-state-transition-method-uncertainty-path__complete-extreme-condition-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:0871328a8c2fae06b6dd6029884996ef7a62164fc63b37a69ada3384d67e1b45; generality

sha256:65b24dced41616d55905fc8ce25515e3eb076fd55a182b03acb98d9e6de23490.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:9fc24e8815f805ea5b7dc9109bc0e3df7b2fffc7e501ba3c497dd456084275f27.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:f3ba318f75d82c087cbc08be2934425b32731c658360a04794e09ad80fca6449.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:29da351f9523d60eda2ef3123fbe51ebe4155ecb2a98fe8755215fd1cdf7c337.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:905bb9ef7a972bafdb8b704e2e60e4f33af9d8a4ddab5b5f3a6026d8a3948ae89.

Independent reconstruction and engine admission. Independent implementation

sha256:c042b1a3bc9c64bb13d24cfa912a70eb2e8265912a74d3665332c1cf01e63736 regenerated the declared census and sole survivor; independent certificate

sha256:21cf104e788ab4425180ea4a48c9871f3e14ee2eba1f5500379ba837f713cddc. Derivation seal

sha256:26adcad6b205f27e49bbf722e69798c570e24391d19c2e59ed5ed5ec7f2de6c5. Source manifest

sha256:5031100944a185e2dd15462e9a55a82aab3bad63e4ceda1afc6b2ff88d617588. Engine receipt

sha256:656183d7e8ecf4a9c8d09efd8e40eea8821392f36595186f9d934599149cb4bb at receipts/engine/model_admitted/SFT-MAT-EXT-HIGH-PRESSURE-STATE-001-656183d7e8ecf4a9.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HIGH-PRESSURE-MATERIAL-TESTING

Measured, observed, adverse and boundary records:

- EXT-001: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-HIGH-PRESSURE-MATERIAL-TESTING: mechanical properties of structural materials; high-pressure test chamber; repeatability

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:adfc898e0e86c2b01ac1e6531b90d81cc6ab93bc56c9f6e8f3d77d4ec4f1bd17; empirical validation

sha256:9eabccd91af0dec363f799effa092b8028a007bf8cc77d31ed0a5e34ca30c37; external validation

sha256:d6f0f7393adf0aaeca799e4ac303334bab69b2679cf40476f35ca0ec23b1ea57; isolation

sha256:6590cf5041395f74d53ccd8dcd16328260fcd28002899489f2f786f397266545; custody

sha256:dbc66327fe1bc9e44465b59269bcd9bf39f043030a19d17b903685c93cd93ed3.

Prohibited inputs.

- no imported continuum response equation, fitted constitutive law, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every specimen, condition, state, path, method, scale, uncertainty and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained

- no failed attempt retires an obligation or changes protected authority

256. High-temperature material state

Field: materials_under_extreme_conditions

Obligation: SFT-MAT-OBL-EXT-002

Claim: SFT-MAT-EXT-HIGH-TEMPERATURE-STATE-002

Question, law and forced result. A high-temperature material state is the complete specimen, positive temperature-step, state-transition, method and uncertainty path; a terminal property cannot replace its thermal history.

EXT-002 uniquely retains specimen-temperature-state-transition-method-uncertainty-path with complete carrier, condition path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-HIGH-PRESSURE-STATE-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight EXT-002 axes before target release.

Boundary: Every positive finite EXT-002 carrier with complete specimen, condition, state, path, method, uncertainty and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-extreme-material-carrier
- specimen-temperature-state-transition-method-uncertainty-path
- complete-extreme-condition-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-extreme-material-carrier__specimen-temperature-state-transition-method-uncertainty-path__complete-extreme-condition-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: 4f2f359d997ab0d734ddfa664ef0bd14959523e898b8be422a22ade72a0d3b38; generality

sha256: 905e2c762219f3258cefb08fb264db30e12a56cc1e03bf3ab70aefc00994dd14.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:42a3aca55eacea41b9ec9467e3f6292adcdc7f30a8a42403b32726f4971fa69b.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:23eab13a4adbc1b7cee586ce5c43b553059f647b3a80273dfab37547d0392d06.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:69580f04d4d3891d21c8472ed143617e74db766e4c1377ba88eb741172659d59.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:725dd5aea791d122df127869ed7b2bc12930e2a466d1929e0dff6d237177bc16.

Independent reconstruction and engine admission. Independent implementation

sha256:c042b1a3bc9c64bb13d24cfa912a70eb2e8265912a74d3665332c1cf01e63736 regenerated the declared census and sole survivor; independent certificate

sha256:e0e0bfa22b1e15d531d1a95e09ebc93988d4ad13a8e3ca395ed010bab088e7b6. Derivation seal

sha256:ffe8fb9a458bdb31880a351999cd980cb875760581759cd29ee5559a8308d4d5. Source manifest

sha256:69a568684f676b3259c4c92686b3d98ee99ab0820560b72df3a2c4e0870d48d9. Engine receipt

sha256:b4e46b616c93e8ba397d849a5ab3a5b82a6f219b6b65c206f8d9e0b6029c004b at receipts/engine/model_admitted/SFT-MAT-EXT-HIGH-TEMPERATURE-STATE-002-b4e46b616c93e8ba.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HIGH-TEMPERATURE-THERMOELECTRIC

Measured, observed, adverse and boundary records:

- EXT-002: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-HIGH-TEMPERATURE-THERMOELECTRIC: High Temperature Thermoelectric Properties Instrument; 295; 900 K; Seebeck coefficient

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ed31fb0d139387d844ee5dbcfba644a878cd44d086ef0c890f4039e09a2672a; empirical validation

sha256:ae06486cadd7b23e52997a8b77c27c6237012b1ffdf7ef7efcf6edd6cdef0c76f; external validation

sha256:88dfeec1dbb6b967c98062871fe8ee52c0ff8b54649ccfd056251bea9282ca91; isolation

sha256:135485db776d53f6b9ed3c2473193d9d44cc2b31bf9ca4fa11c6e10a5133bd00; custody

sha256:e4cb03785472b7f23f0473f52f0e80fdd4cd753342c9acfa2c7c76bbb3d7b207.

Prohibited inputs.

- no imported continuum response equation, fitted constitutive law, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every specimen, condition, state, path, method, scale, uncertainty and outcome distinction remain held labels
- no external outcome selects a survivor

- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

257. Cryogenic material response

Field: materials_under_extreme_conditions

Obligation: SFT-MAT-OBL-EXT-003

Claim: SFT-MAT-EXT-CRYOGENIC-RESPONSE-003

Question, law and forced result. Cryogenic response retains the exact descending positive temperature order and every material-state transition, method and uncertainty distinction.

EXT-003 uniquely retains descending-positive-temperature-state-transition-path with complete carrier, condition path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-HIGH-TEMPERATURE-STATE-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight EXT-003 axes before target release.

Boundary: Every positive finite EXT-003 carrier with complete specimen, condition, state, path, method, uncertainty and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-extreme-material-carrier
- descending-positive-temperature-state-transition-path
- complete-extreme-condition-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-extreme-material-carrier__descending-positive-temperature-state-transition-path__complete-extreme-condition-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:2be5ff3602f4c6634e71240c03b92040bea3e60298105dd1335f2cc96cf25f88; generality
 sha256:57166cb780b94307a92b72c579dada09c42708f584fd23730b19dc0d955edbb4.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:645f625037a08fa7cfbab27e62f0a7be534776dbc748048e7b6692ef459417c9.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:6d6c30e8c7063416d573f7e3fa820f52e5a67ac4bedcc482f42af85633f0f9e0.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:d1a04f19ac20216718b7ff235cb2df8c9010e7c0d9f2e42b633f9eb8228132fb.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:480d0bdeb9091139b9a821bf5399050d3bca9e3f3f67a207f2b6a0d43930232e.

Independent reconstruction and engine admission. Independent implementation

sha256:c042b1a3bc9c64bb13d24cfa912a70eb2e8265912a74d3665332c1cf01e63736 regenerated the declared census and sole survivor; independent certificate
 sha256:5c169f62a39604509a30cb65ce4f3cdd50f5a1c22b811c7d83e178ba1f523659. Derivation seal
 sha256:cba974039992158ee4b29f141cbe06e4cbb105e824e0a21f1d415f456e83a874. Source manifest
 sha256:dc8b61881abceaf4fd1b15cc96b7e6cc9bb7aecdd6b0d2cf852843ec1c254f1f8. Engine receipt
 sha256:861c495f2b5e062f82a7ba415d4e1b1a8fe02e9864298b7d09d86b2fa416ce2f at receipts/engine/model_admitted/SFT-MAT-EXT-CRYOGENIC-RESPONSE-003-861c495f2b5e062f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CRYOGENIC-MATERIAL-PROPERTIES

Measured, observed, adverse and boundary records:

- EXT-003: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-CRYOGENIC-MATERIAL-PROPERTIES: Properties of Selected Materials at Cryogenic Temperatures; 4 K to 300 K; thermal conductivity

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:3c1c7e322b31cbd4518a47b7f2b62ad02c8d0b1decc65372ce77ebdd3f9560b3; empirical validation
 sha256:62b55c46d58ac37cc6eff9134ed298bcd224f40ff959bf10a18c20dd8c91d815; external validation
 sha256:10e3f01f36d413f2425fac75ceb7989b34a2658a53bf6c486a3a43fe79dae355; isolation
 sha256:ff8ec9b9df6fb73a80cdee4fe85e45d0337281644bd4fd1e2296d74b7a86bdf6; custody
 sha256:550bd436ed1809819f1bf448d650bb2040e0126dc35ba12042e6d6b2025bbc2e.

Prohibited inputs.

- no imported continuum response equation, fitted constitutive law, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude

- structural absence and every specimen, condition, state, path, method, scale, uncertainty and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

258. High-electric-field material response

Field: materials_under_extreme_conditions

Obligation: SFT-MAT-OBL-EXT-004

Claim: SFT-MAT-EXT-ELECTRIC-FIELD-RESPONSE-004

Question, law and forced result. Electric-field response retains every positive field step and material-state transition with specimen, method and uncertainty custody.

EXT-004 uniquely retains positive-electric-field-state-transition-path with complete carrier, condition path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-CRYOGENIC-RESPONSE-003
- SFT-MAT-ELEC-DIELECTRIC-LOSS-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight EXT-004 axes before target release.

Boundary: Every positive finite EXT-004 carrier with complete specimen, condition, state, path, method, uncertainty and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-extreme-material-carrier
- positive-electric-field-state-transition-path
- complete-extreme-condition-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-extreme-material-carrier__positive-electric-field-state-transition-path__complete-extreme-condition-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:77f1cbb30b7bb862f96962acf5cebbbc933f6201d9207489402e7a1ca4c3f4fce; generality
 sha256:d11bde1980cd37b814502720e59b01b7040a83d22dd19c14a934610e74b67f63.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:3338ae26b99ece623853924254caefd729652213b4e1dc5a38ed16c790cd1650.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:3727c4642bd18399e351238e8483cad08e0b831d992fa2d6ee540eae46f1784d.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:f4ac9e99ce9e4dc7948f277b82f6f0eb2e2c3dc00766cf457dcccfc8c7e79f714.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:2f4f547daebe12ce2da223170c7c84c57301c3f2dbccd8ce027d92afffcfed37.

Independent reconstruction and engine admission. Independent implementation

sha256:c042b1a3bc9c64bb13d24cfa912a70eb2e8265912a74d3665332c1cf01e63736 regenerated the declared census and sole survivor; independent certificate
 sha256:0a7ced99df5838e1d227b484baa8c122ba1f51b7a672a7ec1d08eb25d14e1831. Derivation seal
 sha256:5a10e289937c92e6f1e39d0f661341b716c827ebfd82110c757ddc7ed0f1351a. Source manifest
 sha256:241a3c93e9eff2f019ba37641a2a077b0bb97624604d234c30f2035592d4c3aa. Engine receipt
 sha256:32f74faa03625c03d3d7d740b1a4a3fa629b35eb9ccca3bc9a407b7aee11f02c at receipts/engine/model_admitted/SFT-MAT-EXT-ELECTRIC-FIELD-RESPONSE-004-32f74faa03625c03.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-ELECTRIC-FIELD-RESPONSE

Measured, observed, adverse and boundary records:

- EXT-004: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-ELECTRIC-FIELD-RESPONSE: Electric Field Response; polarizability; Born effective charges

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:74b4a3286a12293a2c97228a6ddc01a3d8abd08bf73c426e7dd32b1db4c26fea; empirical validation
 sha256:3ea6c40e2d357de8587eb04f225ec29bbe665205041c0ee8ac4016657b7ba3fc; external validation
 sha256:c12a022d711efa8491b2d872a89709a0eeade9dc5e6d4411c2e3042ef64be525; isolation
 sha256:4e5ee5eef32594966978d526a638cd46576e67fca69d0af253f72050a64b2620; custody
 sha256:fca1ba7d1abbe3fdf27b0e1c89adaad2f0527dd4f04ce0d9e5d34e279106b414.

Prohibited inputs.

- no imported continuum response equation, fitted constitutive law, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every specimen, condition, state, path, method, scale, uncertainty and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

259. High-magnetic-field material response

Field: materials_under_extreme_conditions

Obligation: SFT-MAT-OBL-EXT-005

Claim: SFT-MAT-EXT-MAGNETIC-FIELD-RESPONSE-005

Question, law and forced result. Magnetic-field response retains every positive field step and material-state transition with specimen, method and uncertainty custody.

EXT-005 uniquely retains positive-magnetic-field-state-transition-path with complete carrier, condition path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-ELECTRIC-FIELD-RESPONSE-004
- SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight EXT-005 axes before target release.

Boundary: Every positive finite EXT-005 carrier with complete specimen, condition, state, path, method, uncertainty and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-extreme-material-carrier
- positive-magnetic-field-state-transition-path
- complete-extreme-condition-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure

- no-extra-rule

Unique survivor: complete-positive-extreme-material-carrier__positive-magnetic-field-state-transition-path__complete-extreme-condition-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:6f94fc3a2e6e3eb892db367cecda3f74e17e04052fc540687b2613c088a42be2; generality

sha256:5b92589d3e83edc9e955cc3f2dec8833ad21665e269189d491f3a6bb70ff004d.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:8ea5c162e117fee43afd60421b5fb410e7defcd91367342340b583dab722810c.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:2d5496ea44bb538c4d859eea26700d0a12e58e8994c4db70500fe9fcf4ed9af8.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:f955c2c7d5db30acfd5e33fe535257ae0e92a356f87bfa804d888bc21a451cf2.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:514de1dc0d618299a2f09e3f8017d327d3897ce752d6f4c80d357ef865d0cb71.

Independent reconstruction and engine admission. Independent implementation

sha256:c042b1a3bc9c64bb13d24cfa912a70eb2e8265912a74d3665332c1cf01e63736 regenerated the declared census and sole survivor; independent certificate

sha256:03e29ace587ad002e1f179ec037dd5526b8b999d7249e9bf65b2302a4cbdab18. Derivation seal

sha256:6a680e75250ad39af219953af52cc4016cfb9287be7ebc62a6ebe7114a7a3829. Source manifest

sha256:092cec5a8441b95d3f0f6bc143dec031c3f24c919ae9641bf56320b8b116e01a. Engine receipt

sha256:6c065f9b43a2507ea5416c4a6cb705da77d0cbf2e636ca1a3f679297e3915a0c at receipts/engine/model_admitted/SFT-MAT-EXT-MAGNETIC-FIELD-RESPONSE-005-6c065f9b43a2507e.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-HIGH-MAGNETIC-FIELD

Measured, observed, adverse and boundary records:

- EXT-005: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-HIGH-MAGNETIC-FIELD: SANS magnetic field capabilities; 9T Horizontal Magnet; 9.0

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c12a82736e4b83c556b60a48feb6e6d45d5223633cf872150cab10d9548ce2e; empirical validation

sha256:e6b97a6cdce7e787f0d2faedf0e31f43798d12b5989f9aef4b23d65717cd2056; external validation

sha256:5dbc56829d2a37cfce70fccc470b85e72af44d2bccba536f3e264769b329d7ec; isolation

sha256:7d34330f8f12dfd4fcbe8085e43a3e9e867359888a1e340d8a526c0a9fed683e; custody

sha256:1a845600074f2da50c9f8054773d9822f13ab0cafb3a11e7e0adb2a1f99bd37d.

Prohibited inputs.

- no imported continuum response equation, fitted constitutive law, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every specimen, condition, state, path, method, scale, uncertainty and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

260. High-strain-rate and shock response

Field: materials_under_extreme_conditions

Obligation: SFT-MAT-OBL-EXT-006

Claim: SFT-MAT-EXT-SHOCK-RESPONSE-006

Question, law and forced result. Shock response is the complete positive rate sequence, impact path and material-state transition trace; endpoint damage cannot erase loading history.

EXT-006 uniquely retains positive-rate-impact-state-transition-path with complete carrier, condition path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-MAGNETIC-FIELD-RESPONSE-005
- SFT-MAT-MECH-IMPACT-011

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight EXT-006 axes before target release.

Boundary: Every positive finite EXT-006 carrier with complete specimen, condition, state, path, method, uncertainty and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-extreme-material-carrier
- positive-rate-impact-state-transition-path
- complete-extreme-condition-state-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing

- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-extreme-material-carrier__positive-rate-impact-state-transition-path__complete-extreme-condition-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: fddde573ff6772d11cc7db78c9b2c0ebee41107c6557aeb2f7a5e124d5bfe749; generality sha256: 0883150de90b6ffb744e665cfdc127dd4413cbbb7d9eb439dbc6c1fcb9a3121c.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256: 5715b630601fb062932886275d4a2d47893691451a3725a49f0c2dcab7f2e829.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256: 4dd67340ff422cf26c7ad31798bfdbc0f89eb0681e334fd2ef077b3ebbb1dd0.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256: d0ad5d48c15d8a2fbae001f61cbf1d76c4c6321166d4354b83773285cae2ed3d.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256: 02c3f1c302e2e7cf9f83f4f1e48f9cbb527e0ae37df87cde5ad4682af57e2762.

Independent reconstruction and engine admission. Independent implementation

sha256: c042b1a3bc9c64bb13d24cfa912a70eb2e8265912a74d3665332c1cf01e63736 regenerated the declared census and sole survivor; independent certificate sha256: 81ea75b98d043294188b4eb7daf0d2f61d7d7a47023e2ad8e6aba02c2c3ef702. Derivation seal sha256: ada7107702bf680f72953a0f4d04fe91bfdf18b643e4362977171d53841be848. Source manifest sha256: fa380f03139bca364cb3e868e58bf62528a69aaa4c99f683434c1c4541a17e65. Engine receipt sha256: c3d8d24832ea1d7bab078386b61d78dcb11bcde6e53a05775e4e700b5b97705 at receipts/engine/model_admitted/SFT-MAT-EXT-SHOCK-RESPONSE-006-c3d8d24832ea1d7b.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SHOCKWAVE-MATERIAL-RESPONSE

Measured, observed, adverse and boundary records:

- EXT-006: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-SHOCKWAVE-MATERIAL-RESPONSE: visualizing shockwaves; high-velocity impacts; respond to extreme conditions

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256: caeb0558991b7c7d36dd541130747f1c9fcab9c58fcab60a03d07d82ad346e41; empirical validation sha256: 07aa9cc83ed7b3907f4d6f4f84680d4d296cd7abd33f36ede698eb443fd4bbb4; external validation sha256: b058f1d9a9409d807b5d5f42674a09f2b1da832a61b2de99396c2f2ab93b246c; isolation

sha256:6a23023a135704b0bd6be49571d278db258ab901f2232fb557bdb83112063e23; custody
 sha256:753cc0ed493e55bd70c98929b2a981fffccca876d10c466f94212d6c76e1a75e.

Prohibited inputs.

- no imported continuum response equation, fitted constitutive law, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every specimen, condition, state, path, method, scale, uncertainty and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

261. Extreme-radiation material response

Field: materials_under_extreme_conditions

Obligation: SFT-MAT-OBL-EXT-007

Claim: SFT-MAT-EXT-RADIATION-RESPONSE-007

Question, law and forced result. Extreme-radiation response retains event order and every identified before-and-after defect record alongside the material-state path.

EXT-007 uniquely retains radiation-event-defect-state-transition-path with complete carrier, condition path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-SHOCK-RESPONSE-006
- SFT-MAT-DEGR-RADIATION-DEFECT-RECOVERY-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight EXT-007 axes before target release.

Boundary: Every positive finite EXT-007 carrier with complete specimen, condition, state, path, method, uncertainty and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-extreme-material-carrier
- radiation-event-defect-state-transition-path
- complete-extreme-condition-state-path
- specimen-method-condition-scale-uncertainty-held

- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-extreme-material-carrier__radiation-event-defect-state-transition-path__complete-extreme-condition-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a3ba687e99f065821b67e17b8444fed7965c48ba5c82aaf2fdaf9733aa82d5bc; generality
sha256:ffbf66dd31bb86d6fab1e297d9ffed7a81eea7fb0a388bd5756e50b5eaa2c090.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:51a5f07e47214b1506f8db88f017c86bdd5d25867d62f599c4b37fc502ceb3e2.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:896eb51b2591e5c6fb677a658be940353d2adc95916d969113ecd9a4ae9b5291.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:ec1c8485399ee76f021ac5a33abe29de6a614c023bba297d1cab214fbe24cf59.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:4dec881961aa4866d2b37e1c8afdb696b768b3e89a22e9b4e99a0bc21068629c.

Independent reconstruction and engine admission. Independent implementation

sha256:c042b1a3bc9c64bb13d24cfa912a70eb2e8265912a74d3665332c1cf01e63736 regenerated the declared census and sole survivor; independent certificate

sha256:d71e5aa52a4114d3204624c4e49978c4aad2906cfa2a7ad46026e5007fbfac48. Derivation seal

sha256:c2ec7083a85dfc619a56d79ae5eb6adfe05d9233517ae4126480b14e82d8bca7. Source manifest

sha256:c2870911f1b229e987f04823f6bd0730b463e0d2a8a75a0c0abb5478e85704fd. Engine receipt

sha256:9378b3b6d4e9a2a2413e9d856f063be97bfc02e2cf7c289cfb9660efed86305d at receipts/engine/model_admitted/SFT-MAT-EXT-RADIATION-RESPONSE-007-9378b3b6d4e9a2a2.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-RADIATION-DAMAGE-MEASUREMENT

Measured, observed, adverse and boundary records:

- EXT-007: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-RADIATION-DAMAGE-MEASUREMENT: direct measurements; damaging effects; gamma rays; cosmic rays

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:75a4deb98834f22690aea54d2255bc8c2f7c493d08667815af5f74e295cc4bfe; empirical validation

sha256:8b843670f14c690b1e3acd821d34bea296f63f4d2dc012190afd84845ac83c4e; external validation
 sha256:5409ded1c2a6bf81fc2fd7112fefca88179d0d500df6a23879d52d69807c5f79; isolation
 sha256:c7855a18dab5d01a1ca0f83b4c159d2ad8df253b8dd8714ff08c11e7f71cb8ff; custody
 sha256:9cf3a96ad7b79a1707802c7ebad5326f01863debfe09831b02488624957b41bd.

Prohibited inputs.

- no imported continuum response equation, fitted constitutive law, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every specimen, condition, state, path, method, scale, uncertainty and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

262. Combined-extreme condition and path custody

Field: materials_under_extreme_conditions

Obligation: SFT-MAT-OBL-EXT-008

Claim: SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008

Question, law and forced result. A combined-extreme result retains at least two named conditions jointly at every ordered state; no single-condition projection may stand for the joint path.

EXT-008 uniquely retains joint-named-condition-state-transition-path with complete carrier, condition path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-RADIATION-RESPONSE-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight EXT-008 axes before target release.

Boundary: Every positive finite EXT-008 carrier with complete specimen, condition, state, path, method, uncertainty and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-extreme-material-carrier
- joint-named-condition-state-transition-path
- complete-extreme-condition-state-path
- specimen-method-condition-scale-uncertainty-held

- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-extreme-material-carrier__joint-named-condition-state-transition-path__complete-extreme-condition-state-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:3326a80a8f822a122fad6a751a616c0495fb00a8902ba4b54672bd3ead2193a3; generality
sha256:6ddb0631aea8d66a08df0f5914c8ae115145d8a57f2994469c880fcfeacaa6d.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:a7cda30f2b00f576a5811c5e8cd2ed4fadedf0724b1d55668fe55863a4d603c64.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:dd03ffb395e01f8ae02e0ee3534c01be89a98c03a4b7d7c37316e611238cd00b.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:e6d13acec7d4f56f75a08293a4402cf2d8d0ef176971656469fa2281c55cdb44.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:fb79b89b0a18688fecf25f87d8b49f6b1da829bbebcb2556011ac2660405c6a71.

Independent reconstruction and engine admission. Independent implementation

sha256:c042b1a3bc9c64bb13d24cfa912a70eb2e8265912a74d3665332c1cf01e63736 regenerated the declared census and sole survivor; independent certificate
sha256:b2dba9ffd872632a577bd36e96b9199e44823ca61f4fcd6f0ad9aac536d01214. Derivation seal
sha256:6ec520e4acd30f9d5da88a650dba721413177151c599215b6f97be5e4955b8f0. Source manifest
sha256:78264644b4d6814d69c4a1fb1cdc4406bb8e798cb153f2ca7834979ab70d3b3a. Engine receipt
sha256:f297de783de640f02297dbff2e2ce71955f5f557e873e78b5d89ce99f68f10e5 at receipts/engine/model_admitted/SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008-f297de783de640f0.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-COMBINED-EXTREME-KOLSKY

Measured, observed, adverse and boundary records:

- EXT-008: 1 comparisons
- family sources: 8
- exact correspondence: True; omission rejected: True
- NIST-COMBINED-EXTREME-KOLSKY: two extreme conditions; rapidly heating; applying force at the same time

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:1b2586d44738104e690e0aa6a36123a7768cb6e6e89a5cd74a0f89117cf6da28; empirical validation

sha256:63577ed7ed0156f9a042efe9e6f0686016ba6a732fc512ce38d8f9b6e87ab15f; external validation
 sha256:33f6eedba6c724e38688784f7895fdec4e6d2d8c75e1b50e9a00b155d2ad03da; isolation
 sha256:5ac757f0de2b424386ab698ac49ba343ce9842a9d66c331468530b24b3cc6f1f; custody
 sha256:020a323fbbbebbdef96d557f265599d3340323b1731b1e37661fc8b93a166a2.

Prohibited inputs.

- no imported continuum response equation, fitted constitutive law, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every specimen, condition, state, path, method, scale, uncertainty and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Sustainable Material Cycles Lifecycle

263. Embodied material and energy ledger

Field: sustainable_material_cycles_lifecycle

Obligation: SFT-MAT-OBL-SUST-001

Claim: SFT-MAT-SUST-EMBODIED-LEDGER-001

Question, law and forced result. An embodied ledger is the complete positive material-and-energy record with unique identities, source and declared scope.

SUST-001 uniquely retains identity-kind-positive-amount-source-scope-ledger with complete carrier, lifecycle path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SUST-001 axes before target release.

Boundary: Every positive finite SUST-001 carrier with complete material, amount, identity, path, scope, source and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-sustainable-material-carrier
- identity-kind-positive-amount-source-scope-ledger

- complete-material-lifecycle-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-sustainable-material-carrier__identity-kind-positive-amount-source-scope-ledger__complete-material-lifecycle-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e399e552550a15f509d568337fc2df9a81122e7ee247fdf86edba449f35893f1; generality
sha256:181c0869aeab05c2f89f44a0e88e929c2d161c0dec212461757b36fbf952febd.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:afc52ac1ffe8e8215aa322aa49fd8800271ce25daef05db44bdb6b295672cd35.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:8602541006f4694f559918bd6b81f4ba309d0773835e1050c759ca8771897de9.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:87fca7cc9cf93d9372a6895653742b8ced84e073fdaaaa827524e30970ddb0fd.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:b866a3d8983493e5e3c9fd9eaefdc8d2757ddef4eed73e1b4275befbdfc7bf3.

Independent reconstruction and engine admission. Independent implementation

sha256:90096a10407d33dfe96874384e4a80f867585849fbc029014cfb77f4e4e49bc7 regenerated the declared census and sole survivor; independent certificate

sha256:448e0d00e29d7180045dbd4f9d999cfbdd7135abb107028f185610a122556d69. Derivation seal

sha256:2f7a78c3037a5239f1c54973eac792e6d73874d1db2a70f21772212eb18819f6. Source manifest

sha256:7f9d5b8e4d22733f65a8ce4c254da95564cf019f1a7df4016620f68eae8e8. Engine receipt

sha256:37e497bdc37c1ca2dc454761261dff4f68f4d667889276415617f44d9295c187 at receipts/engine/model_admitted/SFT-MAT-SUST-EMBODIED-LEDGER-001-37e497bdc37c1ca2.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-LIFE-CYCLE-ASSESSMENT

Measured, observed, adverse and boundary records:

- SUST-001: 1 comparisons
- family sources: 9
- exact correspondence: True; omission rejected: True
- NIST-LIFE-CYCLE-ASSESSMENT: Life Cycle Assessment; environmental trade-offs; production and recycling methods

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ede0924410619802cf84967e7eb54d15b75a8d6e8cd3d652343260801663efde; empirical validation
 sha256:71d9ed1fb59dea90e8c6c560e7a0dc2368bfd2d6df8960a5c386593e2f1c5b65; external validation
 sha256:01cfdefe4bf2c374fc5c1500a82a3d8e2c83b18d6bb0102c507f68e3842fb325; isolation
 sha256:75292c7cb35cb505311d0eaea125403c22933f7875bc74f005e8aa7f71922084; custody
 sha256:b7c82f460538c6b77e150acac9e7a0ed0436aad667a8c8ef6867c0e94d9f15fd.

Prohibited inputs.

- no imported lifecycle equation, fitted score, economic selector, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, amount, identity, path, scope, source, residual and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

264. Critical-material availability boundary

Field: sustainable_material_cycles_lifecycle

Obligation: SFT-MAT-OBL-SUST-002

Claim: SFT-MAT-SUST-AVAILABILITY-BOUNDARY-002

Question, law and forced result. Availability compares positive held supply and requirement counts for one material, period and source without converting absence or shortfall into a negative magnitude.

SUST-002 uniquely retains material-available-required-period-source-boundary with complete carrier, lifecycle path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008
- SFT-MAT-SUST-EMBODIED-LEDGER-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SUST-002 axes before target release.

Boundary: Every positive finite SUST-002 carrier with complete material, amount, identity, path, scope, source and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-sustainable-material-carrier

- material-available-required-period-source-boundary
- complete-material-lifecycle-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-sustainable-material-carrier__material-available-required-period-source-boundary__complete-material-lifecycle-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:0faaa8196d4b4eddd555dde2404f5ff3f0ae4827662d6e22151d58b81485a987; generality

sha256:0e7a1100602af0a0d3e02e12b2d8fdcc2e38df74f55ec99318c6d101ab709cd7.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:c137e39aa75555bf3d70c06db00b653b7d706b27d4ea68f9c005d68ac45cb4f5.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:4590d3b03570d251ec54e4865be1fbf1c84257935258cbf93238c16587e31afd.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:bbe8e144095ad63ee47852a516dcd4fa9ec06bce13f04055f6cfab65cd70536f.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:372d8ff369ed026a0428ac4cc84d349be04198198bcc6405a95499181d586b9f.

Independent reconstruction and engine admission. Independent implementation

sha256:90096a10407d33dfe96874384e4a80f867585849fbc029014cfb77f4e4e49bc7 regenerated the declared census and sole survivor; independent certificate

sha256:60214922f56071c6b693ac66418de7082e49a6c6ded9c508d3cde6ca0d113e3b. Derivation seal

sha256:0c8c2eb100c6e7360a0c1db1c07e706a7ee1e9726368ee94264a034ca34386d6. Source manifest

sha256:5445546afbecfa486df0eb90c314d2bf546597a3368a3ff145a7a7569808f0a4. Engine receipt

sha256:1620043d743b49ff48e8468332379828b793c8ce100fbd796947cdd52e71790e at receipts/engine/model_admitted/SFT-MAT-SUST-AVAILABILITY-BOUNDARY-002-1620043d743b49ff.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CRITICAL-MINERALS-MATERIALS

Measured, observed, adverse and boundary records:

- SUST-002: 1 comparisons
- family sources: 9
- exact correspondence: True; omission rejected: True
- NIST-CRITICAL-MINERALS-MATERIALS: Critical minerals and materials; supply chain vulnerabilities; stable supply

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a7e747d6579a80e4741bc254cf9d73fc22bd14799b6e8c12a83ab5644ab71473; empirical validation
 sha256:161ca9a593978462f65bcb1b3988ccd5c52f9a064b73805d7299ae15f489ca6b; external validation
 sha256:d214af8f528fd988631393bfa4374897f0784b998f6c8fb224a679ad96d0bcf5; isolation
 sha256:d3164e1330615da2a9c6223854daf58b6e26ae4dbce0fc9e951dd4485c8eeb1d; custody
 sha256:992023a3dee125df372bf5a0ffd0342199c2bb6cee97c075e01a6ba3c4007a3.

Prohibited inputs.

- no imported lifecycle equation, fitted score, economic selector, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, amount, identity, path, scope, source, residual and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

265. Material reuse and remanufacture

Field: sustainable_material_cycles_lifecycle

Obligation: SFT-MAT-OBL-SUST-003

Claim: SFT-MAT-SUST-REUSE-REMANUFACTURE-003

Question, law and forced result. Reuse and remanufacture retain item identity through every state, operation and inspection.

SUST-003 uniquely retains identity-state-operation-inspection-reuse-path with complete carrier, lifecycle path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008
- SFT-MAT-SUST-AVAILABILITY-BOUNDARY-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SUST-003 axes before target release.

Boundary: Every positive finite SUST-003 carrier with complete material, amount, identity, path, scope, source and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-sustainable-material-carrier

- identity-state-operation-inspection-reuse-path
- complete-material-lifecycle-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-sustainable-material-carrier__identity-state-operation-inspection-reuse-path__complete-material-lifecycle-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:1d2e34a5934e0196af88a821b7815859777ee3a2a62102b1b25def74ce871ddd; generality sha256:d3def4669a033e44ad3224fb19dc306ad9e89443213240918c66b9e51a413e9e.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:5f1238bf976798b867b988e927d33145b3220f55aa5acfeaa046cd35dc7b3eb9.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:d85f62834ab295b24aa849770e4eec038f5fa8dff93479f70497e011654e56dd.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:6e14166e7cd7231c84270df4e7c233118eb1768941c0eadd8a4fea957d60c680.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:bea3d4376b3810c581c468d998fa0dda5db6ba308bc2e62f8729146566d42c09.

Independent reconstruction and engine admission. Independent implementation

sha256:90096a10407d33dfe96874384e4a80f867585849fbc029014cfb77f4e4e49bc7 regenerated the declared census and sole survivor; independent certificate

sha256:7a84069753fd1cf944b8b9f71c3d832acbf5659b174e8fea5a2c70d8722b99fc. Derivation seal

sha256:4a027306b60afd3866029178dbd514c0201ac661e88c08fd4de723f5f9986214. Source manifest

sha256:fa02171c0b3b4d1708bea0f9cf416ca380c6cc2bd738cd7032164dba8c118b92. Engine receipt

sha256:1d820f6b69450e79f26b2c5b0d75cde77f118713828dd2eb8b3f2eb61c380c52 at receipts/engine/model_admitted/SFT-MAT-SUST-REUSE-REMANUFACTURE-003-1d820f6b69450e79.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-REGENERATIVE-MANUFACTURING

Measured, observed, adverse and boundary records:

- SUST-003: 1 comparisons
- family sources: 9
- exact correspondence: True; omission rejected: True
- NIST-REGENERATIVE-MANUFACTURING: reuse, remanufacture, recycle; entire product life cycles; recovery

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:83ecc6c3ffe89920f854c67c31d51b8d22f240997d505109091557aaaa48c23; empirical validation
 sha256:8260739e2842c5c9092f4da96b3d65dd08c8a42899dd699b0d366246cf4b4ade; external validation
 sha256:7d624f014dc9aca7cfd30eb3321bc831d2ac092425c543f6b52ac3eccaac416f; isolation
 sha256:349f638c9756cde1d26d637ecad864d4fd77ad694e579c03bd5c1bdd1a33ab94; custody
 sha256:5de3deb2d7ece0680f577d45a09428df97a52fa5169e7a14d0721847a4593e55.

Prohibited inputs.

- no imported lifecycle equation, fitted score, economic selector, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, amount, identity, path, scope, source, residual and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

266. Recycling separation and recovery yield

Field: sustainable_material_cycles_lifecycle

Obligation: SFT-MAT-OBL-SUST-004

Claim: SFT-MAT-SUST-RECOVERY-YIELD-004

Question, law and forced result. Recovery yield is the exact positive feed partition into recovered and residual parts under a held method and scope.

SUST-004 uniquely retains positive-feed-recovered-residual-method-scope-partition with complete carrier, lifecycle path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008
- SFT-MAT-SUST-REUSE-REMANUFACTURE-003

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SUST-004 axes before target release.

Boundary: Every positive finite SUST-004 carrier with complete material, amount, identity, path, scope, source and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-sustainable-material-carrier
- positive-feed-recovered-residual-method-scope-partition
- complete-material-lifecycle-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-sustainable-material-carrier__positive-feed-recovered-residual-method-scope-partition__complete-material-lifecycle-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e931385ea253cd1f43126dda3c56a1e795a67b09d08f6e0fed100f31ae96fa23; generality

sha256:ffd5416ad29207334cbdc78ff7850b168cc9fc19d05cfd7b2c3a1374ea57bfa8.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:f54c84c4916c6c1d0b7769a2db3fa371a0b384be7859f6a947d78dbe4b79be11.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:309efa9d9134c0246fbf53f6629e0e6a9107ee33cd8797d97d81a3157d5a90ac.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:131f32978c4025b703274beb72cdad0e03413ff60f2c89582d596d556bba6bba.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:d3803a1810cd7cf84e4c503a15d16a49b2b5d9238c96adc7ec48bd002783404e.

Independent reconstruction and engine admission. Independent implementation

sha256:90096a10407d33dfe96874384e4a80f867585849fbc029014cfb77f4e4e49bc7 regenerated the declared census and sole survivor; independent certificate

sha256:b8ba8c531afd0d5cb27b2af413bdb2774a45b078b92b8e85eeaf31802274b555. Derivation seal

sha256:eb109dc8aba24440a3cc6e8194ea7f0f3c83eef48ad476539ce38f73517e5633. Source manifest

sha256:84a56333734b4bf59e61ce2487b8349303e87da1d017f709e53cc8e6c53d2fca. Engine receipt

sha256:4047e724adcc58e2a359e7a7a144564ded1f9bd34b050ad8409191613492def9 at

receipts/engine/model_admitted/SFT-MAT-SUST-RECOVERY-YIELD-004-4047e724adcc58e2.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.

Comparison passed: true.

Source identities:

- NIST-CRITICAL-MATERIAL-RECOVERY

Measured, observed, adverse and boundary records:

- SUST-004: 1 comparisons
- family sources: 9
- exact correspondence: True; omission rejected: True

- NIST-CRITICAL-MATERIAL-RECOVERY: metrics, indicators, models; critical materials recovery; maximize product value

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a4177cddcd1a73b748b31c671aa4fd68a5dcd1057cce394efea695c73e136ad; empirical validation
 sha256:f7330ddaa21dbe48c7f43e8198b509b1f864d291579de62fb3dfe95be0e0f88b; external validation
 sha256:8d2f3bf1016082f6adcc6cb1628a3e8298dab7092c6f591f5384012f11009c5d; isolation
 sha256:c5816c5c22ba28494896f05277ee8e36128d2a8a99d8a7d2210f27aeceb1df95; custody
 sha256:5f78a619834e600f17afefe401e63f22dda102e2c907cd9347f3ac061c50e8a6.

Prohibited inputs.

- no imported lifecycle equation, fitted score, economic selector, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, amount, identity, path, scope, source, residual and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

267. Circular material-flow organisation

Field: sustainable_material_cycles_lifecycle

Obligation: SFT-MAT-OBL-SUST-005

Claim: SFT-MAT-SUST-CIRCULAR-FLOW-005

Question, law and forced result. Circular flow is an exact directed network whose every positive transfer retains material identity and system boundary.

SUST-005 uniquely retains node-transfer-amount-material-boundary-network with complete carrier, lifecycle path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008
- SFT-MAT-SUST-RECOVERY-YIELD-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SUST-005 axes before target release.

Boundary: Every positive finite SUST-005 carrier with complete material, amount, identity, path, scope, source and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-sustainable-material-carrier
- node-transfer-amount-material-boundary-network
- complete-material-lifecycle-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-sustainable-material-carrier__node-transfer-amount-material-boundary-network__complete-material-lifecycle-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a0203d9ee6781edb66c44a3da6c771400b8323f0cf56e9f23ablece2997fdc5c; generality

sha256:040ba4fdf4485bfd03b25b1e618da38c28fa5f431fcb2b3d4008281ed148428a.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:0832c57dd03585aa070b4a73a735f1b848e82243da42d28ff90db9a4e51e30ff.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:76c2832f8a8fb7266ff6c4c05f7ebbd498528d373eb225c70ac884c3994431f3.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:915d18aa9c0960cb7cbdfbc3c2ca15806f19fd3db26e830f799101352efb6f0c.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:d6237e95657b8a930a3b62c647d7897ea28ee8ff71203ff636552e2abbb505ed.

Independent reconstruction and engine admission. Independent implementation

sha256:90096a10407d33dfe96874384e4a80f867585849fbc029014cfb77f4e4e49bc7 regenerated the declared census and sole survivor; independent certificate

sha256:e989358c7982830084ca05f9b953eccacbadb1ea1356a5c4879584d229c200f6. Derivation seal

sha256:e8081282891134218e8d0c8b6441c7df66c3594639df4f0fee526e4ae3a6475a. Source manifest

sha256:585de3a87cdee2fe4939ba57e2ba0bcbal0258f3d1f2c0a19f9bdf934556bde. Engine receipt

sha256:3be057d6b5c1faf6d8c88f7fe6f6fc11eab06366f270d65e12d7ea40bd3c8f03 at

receipts/engine/model_admitted/SFT-MAT-SUST-CIRCULAR-FLOW-005-3be057d6b5c1faf6.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MATERIAL-FLOW-CIRCULARITY

Measured, observed, adverse and boundary records:

- SUST-005: 1 comparisons
- family sources: 9

- exact correspondence: True; omission rejected: True
- NIST-MATERIAL-FLOW-CIRCULARITY: Material Flow Analysis; material stocks and flows; recovered CMM feedstocks

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:07cad8db25407f551171e4439322ad53f4f98af7b7fbff1a3331f84f096323be; empirical validation
 sha256:c04764d6094eeb882f72ce1166984bb672e73d97d803d1d167cb584f5b792a7e; external validation
 sha256:feee8c502f6ccf9b58a7668fae2818e75bf68e619351324f81e201bff5c2af4d; isolation
 sha256:dc41ffe09aa68baddbe6dc2b7eb8865b0d48a36d461e380ece60c33d61d81309; custody
 sha256:e9180e92787e3f29564fbf85f91f15e1f7f81b8edbce2855964124ba45bcefc7.

Prohibited inputs.

- no imported lifecycle equation, fitted score, economic selector, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, amount, identity, path, scope, source, residual and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

268. Durability and life-extension relation

Field: sustainable_material_cycles_lifecycle

Obligation: SFT-MAT-OBL-SUST-006

Claim: SFT-MAT-SUST-DURABILITY-EXTENSION-006

Question, law and forced result. Life extension retains the same item identity, positive baseline and extended service counts, interventions and evidence.

SUST-006 uniquely retains identity-baseline-extension-intervention-evidence with complete carrier, lifecycle path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008
- SFT-MAT-SUST-CIRCULAR-FLOW-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SUST-006 axes before target release.

Boundary: Every positive finite SUST-006 carrier with complete material, amount, identity, path, scope, source and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-sustainable-material-carrier
- identity-baseline-extension-intervention-evidence
- complete-material-lifecycle-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-sustainable-material-carrier__identity-baseline-extension-intervention-evidence__complete-material-lifecycle-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:129618a57b2047fe7267c7cc47ec85299346977dc76ce7300a5fa35d8c6330e0; generality

sha256:99e63992d9b2c31750096c1a1cd97c401ea2952bd05624ce74f81e8545d5560a.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:7374cb638494651a5480a36e021e29de9f0fb9ae986cd0f1771976d89f5fa6db.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:73362e265138240c0d2bdcd76d28291382396a183b37e4cd6be5793bf1c9f971.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:b077c86ef8d8d23ed81fce0a5f7ca0336c3fa44f7a39ede5c66bc06945c38286.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:688461e48fc3f7b815a4a05ba7a6c765329fe325285e0f67d231323aaeeeb332.

Independent reconstruction and engine admission. Independent implementation

sha256:90096a10407d33dfe96874384e4a80f867585849fbc029014cfb77f4e4e49bc7 regenerated the declared census and sole survivor; independent certificate

sha256:54fc2384bf41ec7847ce4ea4798f557aae70a83e61a88aad9b8ad6e78091577e. Derivation seal

sha256:5acdfe358b5d3225c3fd043629721e9788dccb5f42013dd8c32b3bf68c6065a. Source manifest

sha256:5667afdc1d8bccf7abf65081e1f212e7c2f44e87c15d6720abc45f30ef7cc808. Engine receipt

sha256:b31769a1805cc3dbebbd6bbebcb83225dff21a6a7249cce49dc8bcdd6181e5934 at receipts/engine/model_admitted/SFT-MAT-SUST-DURABILITY-EXTENSION-006-b31769a1805cc3db.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-MATERIALS-RESILIENCE

Measured, observed, adverse and boundary records:

- SUST-006: 1 comparisons
- family sources: 9
- exact correspondence: True; omission rejected: True
- NIST-MATERIALS-RESILIENCE: changes in material and structural performance; service life; long-term durability

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:8e7b6adb622783b232e660267cf85f935c8a3a546789112cd299404caf165536; empirical validation
 sha256:fd764b8d47a526742436315a78caa3ca8a9b5dcf0bd61d95ae2fe67c80b8ef79; external validation
 sha256:b488f56db326f48494690f753d761b3435ff6649ea7ae6ae733290f97a290c97; isolation
 sha256:3cbd3c713d4efe8303b13cd774baf723c2128a595cb37e172af4e1242e637103; custody
 sha256:d01b508f8811a162bec6b2ce12743146d81847508b3a9fdf835a87d06882f865.

Prohibited inputs.

- no imported lifecycle equation, fitted score, economic selector, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, amount, identity, path, scope, source, residual and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

269. Material toxicity and health handoff

Field: sustainable_material_cycles_lifecycle

Obligation: SFT-MAT-OBL-SUST-007

Claim: SFT-MAT-SUST-TOXICITY-HANDOFF-007

Question, law and forced result. Materials owns composition, exposure and observed specimen response; health consequence is exported explicitly to Medicine without erasing the evidence path.

SUST-007 uniquely retains material-exposure-observation-scope-health-owner-handoff with complete carrier, lifecycle path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008
- SFT-MAT-SUST-DURABILITY-EXTENSION-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SUST-007 axes before target release.

Boundary: Every positive finite SUST-007 carrier with complete material, amount, identity, path, scope, source and observation distinctions.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-sustainable-material-carrier
- material-exposure-observation-scope-health-owner-handoff
- complete-material-lifecycle-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-sustainable-material-carrier__material-exposure-observation-scope-health-owner-handoff__complete-material-lifecycle-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:b4ad5a35943fca41fdd4c894840b562220e1f4b37586358920b5bedbe184d1f; generality

sha256:996707b0c91809bdc688746751782978e73fff22f104ae9c9a66ea68cb310d8d.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:07b84cbdca6fda61c82d90bc25e07bce98d2e958c9e253b6155e88a0b4b2af40.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:398e90c2e662d703187a3fddb1a72743763410ebf5221902b78326c987f7803b.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:3bf3608b8172eca96425093aa6a46673ef3b626fcd4cf5c0a198560b06e8e7e6.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:ba147889799924a37b0f7f4bf5edc68493d8d7240619732784ee1e65d4ac3465.

Independent reconstruction and engine admission. Independent implementation

sha256:90096a10407d33dfe96874384e4a80f867585849fbc029014cfb77f4e4e49bc7 regenerated the declared census and sole survivor; independent certificate

sha256:2f5fcbc045c3f58976ff77bebf50564a20ab1fe9beb317eae9cc103ca233002f. Derivation seal

sha256:fd58eb201501f0cc80486b09cc90f80024b50506d5a35e6de148964b11113bef. Source manifest

sha256:2766953793e243170b2204288896080108fca9e6eacbde276ecb8369aa1f63ef. Engine receipt

sha256:4e27fad89061063b3f0d96fb80c982511ad0d38396c647ea04773e77156c91b0 at receipts/engine/model_admitted/SFT-MAT-SUST-TOXICITY-HANDOFF-007-4e27fad89061063b.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CIRCULAR-SAFETY

Measured, observed, adverse and boundary records:

- SUST-007: 1 comparisons
- family sources: 9
- exact correspondence: True; omission rejected: True
- NIST-CIRCULAR-SAFETY: material performance, processability, safety; landfills or the environment

Shared claim-record clause MAT-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:de4c6dfa83f5873256a496c49a35436f64932c6c7ac4d1865c0027779c9f45c8; empirical validation
 sha256:490af72b8a9043749ae640799723594d2c805cc07b8985ee85acc407fb3b3bf4; external validation
 sha256:afe73a6c6cee3916f7227812f84cd4a75cd06c1983e1f2f3c5f45db7fc5c88a8; isolation
 sha256:63ae9b08c70fcecde10cd65921cf730b8dfa9e535b2bfe88a518979acdf32f6e; custody
 sha256:6ed4723caba04e632fb357875eecd14ebb43e217d5a9fcbb7f997aa760e047d.

Prohibited inputs.

- no imported lifecycle equation, fitted score, economic selector, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, amount, identity, path, scope, source, residual and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

270. Material substitution and function preservation

Field: sustainable_material_cycles_lifecycle

Obligation: SFT-MAT-OBL-SUST-008

Claim: SFT-MAT-SUST-SUBSTITUTION-FUNCTION-008

Question, law and forced result. A substitution preserves every preregistered required function under the same held conditions while retaining distinct material identities.

SUST-008 uniquely retains distinct-material-required-function-condition-preservation with complete carrier, lifecycle path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008
- SFT-MAT-SUST-TOXICITY-HANDOFF-007

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SUST-008 axes before target release.

Boundary: Every positive finite SUST-008 carrier with complete material, amount, identity, path, scope, source and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-sustainable-material-carrier
- distinct-material-required-function-condition-preservation
- complete-material-lifecycle-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-sustainable-material-carrier__distinct-material-required-function-condition-preservation__complete-material-lifecycle-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:3973f3ed5bc20e72da4e3108ac8d02f5d8ed17ecd4625096c443e37a14dbaf02; generality

sha256:9bd606d252d2b2588cfdbb5cd38b7b89d4fc49e566fafc478266c6716e8144ab.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:adbc520695193e2e214a613fea99709dce0281439ad0900e0934f0ee82146346.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:e4450f6b3649d741537346b9bacb66b2e7954aec343a145c7410633db4bfbe47.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:7b27ab201c7250a781f4a56c09656b6170aa470a2d0ae91a9966e7448985e939.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:58c41da0ad3da8e151142be36dd00f210dcc34c09b702ca4c16cbb0fba896af1.

Independent reconstruction and engine admission. Independent implementation

sha256:90096a10407d33dfe96874384e4a80f867585849fbc029014cfb77f4e4e49bc7 regenerated the declared census and sole survivor; independent certificate

sha256:9262fee18133366ff0ef25677b3089341de6ad978107ae38f53edec4d7af157d. Derivation seal

sha256:47afeb734094331edcf09342e313d2b45e19f3b2f5dadf0cf7ecff6c63bacaa7. Source manifest

sha256:a4b738ab0bc7715b87a3f20af00ad0174213a8f42daff1ccd1a784f08980db30. Engine receipt

sha256:a76f6c73192aa6a94b613d47e653cf11d5be96453ae427e6a58fba1dae652fffb at receipts/engine/model_admitted/SFT-MAT-SUST-SUBSTITUTION-FUNCTION-008-a76f6c73192aa6a9.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CRITICAL-SUBSTITUTION

Measured, observed, adverse and boundary records:

- SUST-008: 1 comparisons
- family sources: 9
- exact correspondence: True; omission rejected: True
- NIST-CRITICAL-SUBSTITUTION: Substitution and Alternatives; alternative material systems; raw material shortages

Shared claim-record clause MAT-S014 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:31360d1367ab57575ff2e5d8dc918b56ceba1d85253f0a8aa1b32fd022de7e59; empirical validation
 sha256:6b06b5a0c9fcdf25c66931b4e20c7a74d9fdcff7b5f103f56a89b9d9d417e5ee; external validation
 sha256:2e6f45ddb12b57e72ca7799af15dc4314aa96f8b1b2a5f67a37a488eb66379f5; isolation
 sha256:354b3a2440b6ade7cf375cf2269ddbe685445610240f585c05f2b8b4679972ad; custody
 sha256:5f0517fe9530c8b8045d8ed3a419ec406a8493f439f5778ec83ef618d9e96882.

Prohibited inputs.

- no imported lifecycle equation, fitted score, economic selector, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, amount, identity, path, scope, source, residual and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

271. End-of-life fate and residual custody

Field: sustainable_material_cycles_lifecycle

Obligation: SFT-MAT-OBL-SUST-009

Claim: SFT-MAT-SUST-END-OF-LIFE-CUSTODY-009

Question, law and forced result. End-of-life evidence retains every material identity, fate, residual, source and system boundary; an unwanted sink cannot be erased.

SUST-009 uniquely retains identity-material-fate-residual-source-boundary-custody with complete carrier, lifecycle path, observation, proof, root provenance, successor closure and no extra rule.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008

- SFT-MAT-SUST-SUBSTITUTION-FUNCTION-008

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Complete literal product of eight SUST-009 axes before target release.

Boundary: Every positive finite SUST-009 carrier with complete material, amount, identity, path, scope, source and observation distinctions.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-positive-sustainable-material-carrier
- identity-material-fate-residual-source-boundary-custody
- complete-material-lifecycle-path
- specimen-method-condition-scale-uncertainty-held
- complete-trace
- root-bound-forward-forcing
- positive-finite-successor-closure
- no-extra-rule

Unique survivor: complete-positive-sustainable-material-carrier__identity-material-fate-residual-source-boundary-custody__complete-material-lifecycle-path__specimen-method-condition-scale-uncertainty-held__complete-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f7b538f04aec99a4b988447e64528c4b1c6f34ad580966fb712c6c82ef386f31; generality

sha256:f921d186182e7037a20a1e395f0274d541000eef9babffa21748a1c90ed444af.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:7a98e385872b61854f45b4ef9ca442be5df1100abb913b70bfb47972e8fdeb8f.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:e5db0fec05f3e8eda47a76166dd10e8c48d1c26bcc2adf4b777526a8a503aff1.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:5bc70e60aa26efd462bcf9cb623a7e750645e8d5b52fa93cd6aa3a5a244cdf29.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:2c5f24674555c3120a378bea5882e8d9366f9aea14d07d0dbceb9ff8a9d77abc.

Independent reconstruction and engine admission. Independent implementation

sha256:90096a10407d33dfe96874384e4a80f867585849fbc029014cfb77f4e4e49bc7 regenerated the declared census and sole survivor; independent certificate

sha256:94db5408bf0d6a2b864ad3e97e7a9015f5231616d6d4b274f49e69b142c64e79. Derivation seal

sha256:b5a7e0794a3c309fbb7c092d6d00c79c64c2006da507307e5bafa788abc5d08c. Source manifest

sha256:887d64d00f39911dea9b8912bdf4a801d89224ed4565964cb1fc694a526e0f59. Engine receipt

sha256:c722e8ed79cdea2dbbdf7dd45e64e9e33d6c0fe63c0d7b283a24e72f1b0f580 at receipts/engine/model_admitted/SFT-MAT-SUST-END-OF-LIFE-CUSTODY-009-c722e8ed79cdea2d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CIRCULAR-END-OF-USE

Measured, observed, adverse and boundary records:

- SUST-009: 1 comparisons
- family sources: 9
- exact correspondence: True; omission rejected: True
- NIST-CIRCULAR-END-OF-USE: end of their use; collecting and recirculating materials; unwanted sinks

Shared claim-record clause MAT-S014 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e03ddb6c804a20ca3979150bc9e4858b2227fdafdf055fcb41004b890f1c54f8; empirical validation
 sha256:3ba44dde5b1a6f0d75dbc11063a8e97663cff50764761e204877450b01e57b56; external validation
 sha256:96beedb7706c2d89c3fd7ab399cc4e9d74c862fc460a5bbbb45528fc721575d; isolation
 sha256:b54af20dfeba9901f3facbda3c7f4e6d8733a62f6d8e2836550482e1973f642b; custody
 sha256:2cc7e1fb0f5a6b1dc2e8c50b73999932529231475cd34448dd7d714aaf7bd678.

Prohibited inputs.

- no imported lifecycle equation, fitted score, economic selector, opaque oracle or prior proof as premise
- no numerical zero, negative, irrational, imaginary, floating, fitted or free proof magnitude
- structural absence and every material, amount, identity, path, scope, source, residual and outcome distinction remain held labels
- no external outcome selects a survivor
- all favourable adverse absent unavailable and unresolved result classes remain retained
- no failed attempt retires an obligation or changes protected authority

Complete-field family: Complete Materials External Validation

272. Crystallography and diffraction validation vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-001

Claim: SFT-MAT-VALIDATION-CRYSTALLOGRAPHY-DIFFRACTION-VECTOR-001

Question, law and forced result. VALID-001 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 8 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-001 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 8 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001
- SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007
- SFT-MAT-CRYS-PAIR-DISTRIBUTION-008
- SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004
- SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005

- SFT-MAT-CRYS-STRUCTURE-FACTOR-002
- SFT-MAT-CRYS-TEXTURE-ORIENTATION-003
- SFT-MAT-CRYS-TWIN-DOMAIN-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-001 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 8 value-free claim identities registered for Materials VALID-001, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-001-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: complete-receipt-bound-membership__complete-valid-001-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:863d03db998f5cde9ebf2b07156d711ec05aaae9cf417858923aab9e2cbd3fb7; generality
sha256:20b9b04ad17079b273ad189d08d3197daa52e441bb4e999c191b7b1f538d9684.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:c93d2a4782a152aa9b43a19a166b18a0a049554a584a71f2b4b7b10a6dbef743.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:164c6ec877981e052ddf2056a4baee2c7413173c269b241477b1a1be787204c.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:a08e47f9903ac11fc12f986975f3f5af0c054bf9e7d0ebe9b6eae7f6e7dc032b.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:df33bec7eeaf0fd97441ab4966fafd5a0315a6df7f54db728d3a4f30a43822d.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate

sha256:0c534461ae9bea7817e4dfc40da79ab0223baaede8ceeee79393980dd28d97b5. Derivation seal

sha256:02fdc4c8f68e55039cb478a24f1e9ad5b868aa51f1becb4bfccf326cfe89f232. Source manifest

sha256:b032c48049a932074573cce8872a0616970bcdb306f58a6be4caa9aaefef1cc9. Engine receipt

sha256:4ad9b7e3ae07359ffedfa94f3fb5489dee448ef95b63fb028df48f1097aa3428 at receipts/eng

ine/model_admitted/SFT-MAT-VALIDATION-CRYSTALLOGRAPHY-DIFFRACTION-VECTOR-001-4ad9b7e3ae07359f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-TOTAL-SCATTERING-PDF-2014
- IUCR-TWINNED-DIFFRACTION-DATA-2022
- IUCR-MODULATED-STRUCTURES-2009
- IUCR-INCOMMENSURATE-DICTIONARY-2026
- IUCR-MODULATION-WAVE-DIFFUSE-2015
- IUCR-STACKING-DIFFUSE-2023
- IUCR-STACKING-FAULT-LDH-2020
- NIST-SP846-POWDER-DIFFRACTION-1992
- NIST-NCAL-TEXTURE-PHASE-FRACTION-2026
- IUCR-TWIN-DICTIONARY-2026

Measured, observed, adverse and boundary records:

- VALID-001: complete registered membership count 8
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=6; explicit-status-lines=3
- SFT-MAT-CRYS-PAIR-DISTRIBUTION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-STRUCTURE-FACTOR-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-TEXTURE-ORIENTATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-TWIN-DOMAIN-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3

Shared claim-record clause MAT-S015 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:0573f61a6d9f508a2b22f9c2dff467e16a6d01e809e6d7ca4aa9f66973c46bc8; empirical validation
 sha256:a31d884e3ebd568d61244470f389b25d9c531a2911328ceea91e69df83019f6a; external validation
 sha256:a05d7478220c20f3415430dd31843e4c617cac9bc2322e73d8d23176b21215e6; isolation

sha256:7053e922962026e3c90cb77c26a9ba90c66aa7b485fcc1d15df0958179f5d0a8; custody
 sha256:e54a57ceeald542d0b144620911144c65a961a7ce4953fdc4781c1823cf6e3.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

273. Defect and microstructure validation vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-002

Claim: SFT-MAT-VALIDATION-DEFECT-MICROSTRUCTURE-VECTOR-002

Question, law and forced result. VALID-002 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 9 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-002 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 9 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005
- SFT-MAT-MICRO-COARSENING-TRANSFER-007
- SFT-MAT-MICRO-DEFECT-MIGRATION-002
- SFT-MAT-MICRO-DEFECT-POPULATION-001
- SFT-MAT-MICRO-DISLOCATION-REACTION-003
- SFT-MAT-MICRO-GRAIN-GROWTH-004
- SFT-MAT-MICRO-INTERFACE-MOBILITY-008
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006
- SFT-MAT-VALIDATION-CRYSTALLOGRAPHY-DIFFRACTION-VECTOR-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-002 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 9 value-free claim identities registered for Materials VALID-002, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-002-typed-evidence-vector

- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: complete-receipt-bound-membership__complete-valid-002-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:2af5a6cf8f91ae656b6adf80fc67764c598e1f200bf2d4145ec47419ab3bf5d6; generality
sha256:b3744f8be400908dc8abe4913fc6599d883e5818c4a72b2dd7f9de0ff8502f84.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:042e384099895082deb3873c8fdd86d0ed58231a08ffc02bf99bd9cdbe76a813.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:01f3b2df967456e2aebf69022bf544c601cbc8c9792482184e932e0818159bd8.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:16bfa54a0ce7120bec02385d033a31805c583e7fe99b6bc78d519704b1a99826.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:45eee85d3ddb818cef33430d59669904ee038b39fb59d011afc1d4dc27897d68.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate

sha256:4ba8da4274c90e79f505b21168cda542f812069d9200a6eb8824898e840ce1f2. Derivation seal

sha256:ae706d0d4508e249cfa1f9ff2e07614d643f2721a8b57908ea56d71849b4a59e. Source manifest

sha256:976b2d3a0e3a6f79fa4ed18ec2cdcbb70200e90270c5fc18b88f2fb90d830696. Engine receipt

sha256:07c5e25cffdc0077d4efbd0652068b1890a861cbffcfabc9826553b97d443dd3 at receipts/engine/model_admitted/SFT-MAT-VALIDATION-DEFECT-MICROSTRUCTURE-VECTOR-002-07c5e25cffdc0077.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SEGREGATION-PRECIPITATION-2021
- NIST-BENCHMARK-COARSENING-2017
- NIST-STRUCTURES-PRECIPITATION-HANDBOOK
- NIST-POINT-DEFECTS-2026
- NIST-DISLOCATION-DYNAMICS-2021
- NIST-DISLOCATION-CLIMB-MONOGRAPH-59

- NIST-SHARP-INTERFACE-GRAINS-2001
- NIST-UNIFIED-GRAIN-BOUNDARY-MOTION-2008
- NIST-MULTISCALE-MATERIALS-2026
- NIST-MICROSTRUCTURE-PROPERTY-TOOLS-2026

Measured, observed, adverse and boundary records:

- VALID-002: complete registered membership count 9
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=3
- SFT-MAT-MICRO-COARSENING-TRANSFER-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-MICRO-DEFECT-MIGRATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-DEFECT-POPULATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-MICRO-DISLOCATION-REACTION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-GRAIN-GROWTH-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=3
- SFT-MAT-MICRO-INTERFACE-MOBILITY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=4
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3

Shared claim-record clause MAT-S015 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:e2b6580b84f5b1e4debac2c31a75d9a0ff5140f2c0ae09ffc7d46ece538c3a77; empirical validation
 sha256:a2e0ac99018ee2cf3984cacf67e553a5077607b46fd9810e8499fe0651889703; external validation
 sha256:a7be423eac3da4ee14f8bba4f3dd62ad898680031cabbbfb78503348d9cfd1a4; isolation
 sha256:1829515d47f679649c2a49629f6047184422ab45bd475fced4d5e647c17d4161; custody
 sha256:2985710755f3acc70a58ed559613d625e9cc7bc11d4a935428108507087f81b3.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

274. Phase and transformation validation vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-003

Claim: SFT-MAT-VALIDATION-PHASE-TRANSFORMATION-VECTOR-003

Question, law and forced result. VALID-003 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 10 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-003 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 10 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-PHASE-COMPONENT-HANDOFF-003
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-PHASE-GLASS-ARREST-009
- SFT-MAT-PHASE-MARTENSITIC-006
- SFT-MAT-PHASE-METASTABLE-RETENTION-004
- SFT-MAT-PHASE-ORDER-DISORDER-008
- SFT-MAT-PHASE-RECONSTRUCTIVE-007
- SFT-MAT-PHASE-SPINODAL-INSTABILITY-005
- SFT-MAT-PHASE-TIE-LINE-LEVER-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-VALIDATION-DEFECT-MICROSTRUCTURE-VECTOR-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-003 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 10 value-free claim identities registered for Materials VALID-003, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-003-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: complete-receipt-bound-membership__complete-valid-003-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:64d3c9113168bb6d3ad85c72304c67e84c0af804792a44ad4f91e41bb7cf5ca5; generality
 sha256:6dac4a17fdde7a1c945a7e25a68279091281e9020923f97cce4916a6af85bbd4.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:2ed1b33eddd9ac5a974509dfb7968799ee79ac4aefc878abb1908f57e9332e68.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:53ad6286b369516288673febfeba3a94eda4d1f65d1c468f3f642ac25d8204e6.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:2e8116a65800a204c8b2a575f0b282d90fc286a4d7c050189f27157e651b8f8b.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:9b21a1015ac5441d6533d0c1d4497a86bcba6df90bbc5382497690b6c8dbe1c8.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate
 sha256:7a536370f64030c2acac60d9c22f7ca97565798045e8b259f9cf22010d436f3e. Derivation seal
 sha256:81408e24ee136b8cc59dd8a815dd6daff42876975cc421017bd2ff52b7e54b9b. Source manifest
 sha256:4ea74c18877f391f2df823052b77a5aaaa44ec383e6081978d3e153b3b1b9710. Engine receipt
 sha256:064539aa81cfee8f3d9a96eed1c554a7da11049ff07c78a34ac0bc42dba14c13 at receipts/engine/model_admitted/SFT-MAT-VALIDATION-PHASE-TRANSFORMATION-VECTOR-003-064539aa81cfee8f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-SAT-TMMC-COEXISTENCE
- NIST-TEXTURE-PHASE-FRACTION
- NIST-LEVER-RULE-SOLIDIFICATION
- NIST-GLASS-TRANSITION
- NIST-MARTENSITIC-MATERIALS-STUDY
- NIST-LIQUID-WATER-METASTABLE
- NIST-BINARY-HALIDE-TRANSFORMATIONS
- NIST-ORDER-DISORDER-SEPARATION
- NIST-LLE-TERNARY-TIE-LINES
- NIST-SOLIDIFICATION
- NIST-PHASE-TRANSITION-TEMPERATURES

Measured, observed, adverse and boundary records:

- VALID-003: complete registered membership count 10
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences

- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-PHASE-COMPONENT-HANDOFF-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-FRACTION-LEDGER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-GLASS-ARREST-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-MARTENSITIC-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-METASTABLE-RETENTION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-ORDER-DISORDER-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-RECONSTRUCTIVE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-SPINODAL-INSTABILITY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TIE-LINE-LEVER-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TIME-TEMPERATURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2

Shared claim-record clause MAT-S015 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:940b29f7ad3b32d75dfb8e17911dad3d91243f20ca3ab915ad19ccabbc8adad4; empirical validation
 sha256:332295ba9cbc5e686944a15b33b5fb805608b76e4a04a98c95be95c0bce5a223; external validation
 sha256:75019d5732e1057865cef139b00983748f58661fedd9756250f68c72b9b5ca03; isolation
 sha256:83e1a2a525a944710b62dd33246b9a3c9b1f1bf12bf75fc54514eb5fe8c0855e; custody
 sha256:b2eb762834a6de2794c5759737819c476ee3543cbfdadf140705039182073ee6.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

275. Mechanical and tribological validation vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-004

Claim: SFT-MAT-VALIDATION-MECHANICAL-TRIBOLOGICAL-VECTOR-004

Question, law and forced result. VALID-004 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 14 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-004 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 14 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-MECH-CRACK-GROWTH-008
- SFT-MAT-MECH-CREEP-RUPTURE-010
- SFT-MAT-MECH-FATIGUE-009
- SFT-MAT-MECH-FRACTURE-ENERGY-007
- SFT-MAT-MECH-FRICTION-CONTACT-012
- SFT-MAT-MECH-IMPACT-011
- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-MECH-TRANSVERSE-STRAIN-002
- SFT-MAT-MECH-VISCOELASTIC-MEMORY-003
- SFT-MAT-MECH-VISCOPLASTIC-FLOW-004
- SFT-MAT-MECH-WORK-HARDENING-006
- SFT-MAT-MECH-YIELD-PATH-005
- SFT-MAT-VALIDATION-PHASE-TRANSFORMATION-VECTOR-003

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-004 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 14 value-free claim identities registered for Materials VALID-004, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-004-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: `complete-receipt-bound-membership__complete-valid-004-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:09c68b2fdad85a79dcfbfbf0bcce842ed5e6e4f77464d0b7498d30129fe75fcdc; generality`

`sha256:43173457718c000d72039406daf82f829fffb90734100d8bbb94978b757d998c9.`

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:3eb9969e849e2cee1a9dd14d38c3e0968d5f64c6ad5a96f1bf24dea864e71fed.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:d9298dcd569c65c8a8f33a28b4025113e140aaabb24ff75a3d75cabcc83283ea.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:6d2653ba5f74184102e7bd33ff22e2822f3644b76ed4e02056824c1b9e68f686.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:bc00234d445547f43070f7c0c03844846438d70adfc33af6dac6682244a679ed.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate

sha256:e8ae411787bb257e9ee1bb27af36a14a3fd53e75370fd69e95bb3eeca2de7790. Derivation seal

sha256:c58c3022215187118cd64a88478d56f66a9a04b575412ee647045b496f162b36. Source manifest

sha256:aa541c3f015b028391b29d1dc761dc6d95434422f330017478c0fe6780d3073b. Engine receipt

sha256:2cecd395c1dafel1da88a1280beebbb9046503570b4dc994ccd98e7af625f7eca1 at receipts/engine/model_admitted/SFT-MAT-VALIDATION-MECHANICAL-TRIBOLOGICAL-VECTOR-004-2cecd395c1dafeld.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FAILURE-PROPERTY-TESTS
- NIST-FATIGUE-FRACTURE
- NIST-NANOTRIBOLOGY
- NIST-LUBRICATION-HANDBOOK
- NIST-LUBRICANT-FILM
- NIST-RHEOLOGY
- NIST-VISCOELASTIC-SEALANT
- NIST-MARTENSITIC-MATERIALS-STUDY

Measured, observed, adverse and boundary records:

- VALID-004: complete registered membership count 14
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-MECH-CRACK-GROWTH-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-CREEP-RUPTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-MECH-FATIGUE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-FRACTURE-ENERGY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-FRICTION-CONTACT-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-IMPACT-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-RHEOLOGY-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-TRANSVERSE-STRAIN-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-VISCOELASTIC-MEMORY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-VISCOPLASTIC-FLOW-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-WORK-HARDENING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-YIELD-PATH-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

Shared claim-record clause MAT-S015 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:bfa0f62e3232cc83ea7e01300ace07ef616017508df5cb52d6051d09c11a3522; empirical validation
 sha256:bc104f60468ec8c46ef5fd7d4179e80ae164d87091087fa928447b373ff37b83; external validation
 sha256:f12a1a3ecedc47b5e1708f8c5ae92187b81808e6a05b0a205826a6848b65256b; isolation
 sha256:ea65fb29ca106fa9f9a2ab3b8e198151c50ee983a7a2957722eb6b9dca143104; custody
 sha256:1ed4c2d8f8d50c55e3aacbac5b1d5d0cb7ca3565539fb43338aeb22fe76fdb11.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

276. Thermal and transport validation vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-005

Claim: SFT-MAT-VALIDATION-THERMAL-TRANSPORT-VECTOR-005

Question, law and forced result. VALID-005 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 7 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-005 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 7 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-THERM-BOUNDARY-RESISTANCE-002
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-THERM-PHASE-STORAGE-006
- SFT-MAT-THERM-PHONON-MEAN-PATH-003
- SFT-MAT-THERM-RADIATIVE-TRANSPORT-004
- SFT-MAT-THERM-SHOCK-FATIGUE-007
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-VALIDATION-MECHANICAL-TRIBOLOGICAL-VECTOR-004

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-005 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 7 value-free claim identities registered for Materials VALID-005, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-005-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: complete-receipt-bound-membership__complete-valid-005-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:6cc4e07ce837ef2a42cfe71eb22c6b057c463a0983b18506ab30c388ce39f459; generality
 sha256:e43c76cb202b2ccf532b74f4707514d6bc0452b2b75f797fd79cf1161816fc2c.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:e797e58ca8ab493a0a464c18449322051f9fc1c138cea7804d6827df973a863d.

- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:aefc40231855a0f817408a7a1e128bdf5ac332819b0c0b08999c57a7e7514159.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:6502405308b71a57630c898b6cb018eb0b946c0ab634e67e728b500d521068a8.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:7d8a8ef9fc3a68e266b9014bc2c3bdfaffbfc5e2e3abfcb8d477aafc1c9ccd5a.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate

sha256:325b2aa363f3ceb908ee697562c83cec900c266e95f991ba7a2ce995e13d89aa. Derivation seal

sha256:9c1cb0de78f1bddb4a95b9bd46a0eee132a2a947b08ce89a2af36fc185db4c0f. Source manifest

sha256:5617482093a5bf9f305c4b919921b7f8a95c352b6ab1ebfedd9e456071ac90e8. Engine receipt

sha256:f4f3ec66e67611e9c08643a7d846ac79c47b8964688ae2ab5ba05317a73c15f9 at receipts/engine/model_admitted/SFT-MAT-VALIDATION-THERMAL-TRANSPORT-VECTOR-005-f4f3ec66e67611e9.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-FDTR-TRANSPORT
- NIST-MATERIALS-DATA-GUIDE
- NIST-THERMAL-DIFFUSIVITY-AM
- NIST-PHASE-CHANGE-STORAGE
- NIST-NANOCALORIMETRY
- NIST-INTERFACE-SCATTERING
- NIST-PHONON-THERMAL-LIMITS
- NIST-INFRARED-OPTICAL-PROPERTIES
- NIST-FRACTOGRAPHY-THERMAL-SHOCK
- NIST-THERMAL-SHOCK-SILICON-NITRIDE
- NIST-THERMOELECTRIC-MEASUREMENTS
- NIST-TRANSPORT-THERMOELECTRIC

Measured, observed, adverse and boundary records:

- VALID-005: complete registered membership count 7
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-THERM-BOUNDARY-RESISTANCE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-THERM-DIFFUSIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

- SFT-MAT-THERM-PHASE-STORAGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-PHONON-MEAN-PATH-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-RADIATIVE-TRANSPORT-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-THERM-SHOCK-FATIGUE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

Shared claim-record clause MAT-S015 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:c754b94be5e45922cfd0cf66f587fd9531c4a04f2299d365e999bfbade26a84d; empirical validation
 sha256:e957e61711a0504b7d575568ea12366f26bd232076ef34c3bf00e4e6d5e69dd2; external validation
 sha256:653846c3a133eb5dfdb1ef4914b7e6930757bca6a019150beb3fce89f1f6a45d; isolation
 sha256:a69f345b43466dde885ea56748750e0c7726520ca8c347d9ec6b99365e8b4989; custody
 sha256:143534a42ba7d75434c7dfbe62caad8ba7275b2f3adcd039727a8a9d833a7824.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

277. Electronic, ionic and dielectric validation vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-006

Claim: SFT-MAT-VALIDATION-ELECTRONIC-IONIC-DIELECTRIC-VECTOR-006

Question, law and forced result. VALID-006 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 12 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-006 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 12 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-ELEC-BAND-ALIGNMENT-008
- SFT-MAT-ELEC-CARRIER-CONFINEMENT-009
- SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001
- SFT-MAT-ELEC-DEFECT-TRAP-STATES-010
- SFT-MAT-ELEC-DIELECTRIC-LOSS-004
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012

- SFT-MAT-ELEC-FINITE-BARRIER-TUNNELLING-007
- SFT-MAT-ELEC-HALL-RESPONSE-003
- SFT-MAT-ELEC-IONIC-TRANSFERENCE-005
- SFT-MAT-ELEC-MIXED-TRANSPORT-006
- SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002
- SFT-MAT-ELEC-SCREENING-DEPLETION-011
- SFT-MAT-VALIDATION-THERMAL-TRANSPORT-VECTOR-005

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-006 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 12 value-free claim identities registered for Materials VALID-006, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-006-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: complete-receipt-bound-membership__complete-valid-006-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:07862c01a5820db0764e5c91d7a9b11bea31fef7519bd5db261eae2d01d2f78; generality
 sha256:3bbc35e2f00522f1fe9edb771b5d11774ee74057c77061a0e9ff7d2a6663a453.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:3baa42396d16a23472c01b921e95e0f9627783df4d1474911afaa28044edcdcd.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:7e3becef53592e9fe6219a49051543b7ace512f99b06c37436056c4c47d4447e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:b651a50bb1f02c475f17589556bd4a2081e1e16015775fdceecdb6832a8e8600.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:825d19d266a97d57a86b0f791743ced9688ebda2ff7d34757f54e2d82d8300885.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate
 sha256:d3843d7e63928d592da5a293a9297e9493b81fcff8c856750790ebe6e20c5756. Derivation seal
 sha256:73f6d4d730b78b34a1ccd5b9ca2ba83dcee0dcff9353e1e2eecd3be015837db2. Source manifest
 sha256:47b5cfa1543a630a521263a4fbae176f3af84a4f3fe836a5c34b0de3bcc01e22. Engine receipt
 sha256:857f130a81d18da493390bd38c96dcc82521f45f01dd594259b36c9f2157fb1f at receipts/engine/model_admitted/SFT-MAT-VALIDATION-ELECTRONIC-IONIC-DIELECTRIC-VECTOR-006-857f130a81d18da4.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-TUNNEL-BAND-OFFSET
- NIST-OPTOELECTRONIC-BAND-METROLOGY
- NIST-HALL-EFFECT
- NIST-POINT-DEFECT-CHEMISTRY
- NIST-DIELECTRIC-PERMITTIVITY-LOSS
- NIST-OPERANDO-ELECTROCHEMICAL
- NIST-RESISTIVITY-HALL
- NIST-IONIC-CONDUCTIVITY
- NIST-MIXED-IONIC-ELECTRONIC
- NIST-DIELECTRIC-DEPLETION

Measured, observed, adverse and boundary records:

- VALID-006: complete registered membership count 12
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-ELEC-BAND-ALIGNMENT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-ELEC-CARRIER-CONFINEMENT-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-DEFECT-TRAP-STATES-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-DIELECTRIC-LOSS-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-FINITE-BARRIER-TUNNELLING-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-ELEC-HALL-RESPONSE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-IONIC-TRANSFERENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-MIXED-TRANSPORT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-SCREENING-DEPLETION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

Shared claim-record clause MAT-S015 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:d4b3ff84ed04422554e36c8c92569277a87d4b1b9b2791d4f58045d0893d590a; empirical validation
 sha256:035adcf9dda734e48c5a8d0ff24938c92d5fc41f69207f9d6451bb171aaa6b38; external validation
 sha256:34c5b87c3fbfe57d11cc33c03af2c8c2ea6ffcd4b4c23f6090c142e3fa86e12; isolation
 sha256:4b8f4bba8fb49a8bb33e3587de982cbb95ceeb3a4116c281d36d7b50cc67231a; custody
 sha256:0c94682e2dd3fec24ce99e1c531ce0c4733d0b6ae625545963b74bce1631c5e2.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

278. Magnetic, superconducting and topological validation vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-007

Claim: SFT-MAT-VALIDATION-MAGNETIC-SUPERCONDUCTING-TOPOLOGICAL-VECTOR-007

Question, law and forced result. VALID-007 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 12 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-007 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 12 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002
- SFT-MAT-MAGSC-DOMAINS-WALLS-004
- SFT-MAT-MAGSC-HYSTERESIS-LOOP-005
- SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006
- SFT-MAT-MAGSC-MAGNETORESISTANCE-007
- SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001

- SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011
- SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009
- SFT-MAT-MAGSC-SC-VORTEX-PINNING-010
- SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003
- SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008
- SFT-MAT-MAGSC-SUPERFLUID-CRITICAL-FLOW-012
- SFT-MAT-VALIDATION-ELECTRONIC-IONIC-DIELECTRIC-VECTOR-006

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-007 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 12 value-free claim identities registered for Materials VALID-007, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-007-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: complete-receipt-bound-membership__complete-valid-007-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a5a47677869aa2f2a443094104758092e2f49040d8981d00f0f5f4282a5ba192; generality
sha256:ad131e052c1f89b6af928212fa5d65c0f37ebd0298716c24ad2474e2e9c6f8bc.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:25019b49219b4c130f0ddcf0e2219909efa7dc909ea1f652ff2007186a5956a7.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:13119bb65c17edc43e6bd476be7067dec770bd94a9648a7517a11f7d432ecd8e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:76c9c22a155fed81c782785d686995a5ae86c41579bd7efe9f4bcf8e0a7c9b72.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:2da7f57de65a6829599861c51760bc192abab67e3d0fab24ce8e19b4129581b4.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate
 sha256:d3843d7e63928d592da5a293a9297e9493b81fcff8c856750790ebe6e20c5756. Derivation seal
 sha256:039fdea66772202ab00adde72dbbfa669e472bd2a2e6ca8829ccb436742885ad. Source manifest
 sha256:63e01ce915b630bceda0ad3bba9c0287233907b804b99363f466e9c349c546fb. Engine receipt
 sha256:136971745a3094bd8b155e164990a41b172b99d33bb7217ea211be34cdc06765 at receipts/engine/model_admitted/SFT-MAT-VALIDATION-MAGNETIC-SUPERCONDUCTING-TOPOLOGICAL-VECTOR-007-136971745a3094bd.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-PARAMAGNETIC-DIAMAGNETIC
- NIST-MAGNETIC-SUSCEPTIBILITY-SRM
- NIST-MAGNETIZATION-LAB
- NIST-MAGNETIC-IMAGING
- NIST-HYSTERESIS-STANDARD
- NIST-MAGNETIC-MATERIALS-METROLOGY
- NIST-SPIN-ORBIT
- NIST-SC-CRITICAL-FIELDS
- NIST-SC-FLUX-LATTICE
- NIST-SPIN-TRANSPORT
- NIST-SPIN-SPECTROSCOPY
- NIST-SUPERFLUID-PERSISTENT-FLOW

Measured, observed, adverse and boundary records:

- VALID-007: complete registered membership count 12
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-DOMAINS-WALLS-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-HYSTERESIS-LOOP-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-MAGNETORESISTANCE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

- SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SC-VORTEX-PINNING-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-SUPERFLUID-CRITICAL-FLOW-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

Shared claim-record clause MAT-S015 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:ffc01e9093eb1d067e019a34e2269a1e407dc9580a9f87e9dd436406de51bad5; empirical validation
 sha256:af750218f243a04ab994312faf76da2a263c538f764d5bf10f22c848f36d8dd7; external validation
 sha256:c716b90c8f0c36c5cb621b8bc8d742583599cf7b10b5c47d72a3f49c5164be08; isolation
 sha256:2db946900046d70fa62d75e63502203ee23e36bb5c29dca183d0fbefc3dc119e; custody
 sha256:a4d40ea0e1aec54236790baab766c5eabbef527f66498fb8fca676245c258669.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

279. Optical and photonic validation vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-008

Claim: SFT-MAT-VALIDATION-OPTICAL-PHOTONIC-VECTOR-008

Question, law and forced result. VALID-008 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 10 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-008 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 10 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-OPT-ABSORPTION-EXTINCTION-001
- SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005
- SFT-MAT-OPT-EXCITON-DYNAMICS-010
- SFT-MAT-OPT-LIGHT-SCATTERING-004
- SFT-MAT-OPT-LUMINESCENCE-YIELD-003

- SFT-MAT-OPT-NONLINEAR-MIXING-006
- SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008
- SFT-MAT-OPT-PLASMONIC-RESPONSE-009
- SFT-MAT-OPT-REFLECTION-TRANSMISSION-002
- SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007
- SFT-MAT-VALIDATION-MAGNETIC-SUPERCONDUCTING-TOPOLOGICAL-VECTOR-007

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-008 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 10 value-free claim identities registered for Materials VALID-008, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-008-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: complete-receipt-bound-membership__complete-valid-008-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:ad86a898d27978ed22b4178d81ca62ed388b80a067681a9a3f41ff5872fb9682; generality

sha256:9edf10d7ff8ab23871e8d2bee6f1c8f7c02a652fc2cb20d662c53cc68fd519a2.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:7cb4cf0c7de37ac7d1c7fbc94f6219632fbf4f90635c6236cd38ed3edda71b59.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:648fbb48814d019c9dc838eb2291ab1fd757f0fb7093f08963a06c6da8a5f892.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:bc4add7800a2191f47a8a1c66f717ebe056e016b538f90160f0d634108500b7.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:359aafd9544dc82a93a4434f0cdd1d04cd138eb9a8b13d92f680500bd66e4cc4.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the

declared census and sole survivor; independent certificate

sha256:7a536370f64030c2acac60d9c22f7ca97565798045e8b259f9cf22010d436f3e. Derivation seal
 sha256:0bc94864ec8ba4e016d8c116461dbe6d263a47de1e67302627b4b2452e7dbe3e. Source manifest
 sha256:f9e8d9ebf23820c0a3dd454dc94cc619fa7cd633c6e270faf8f9b002cbf4e439. Engine receipt
 sha256:20d2650407f790688fc8c8deae94315a17deb8235834956653bf5e970a755bbb at receipts/engine/model_admitted/SFT-MAT-VALIDATION-OPTICAL-PHOTONIC-VECTOR-008-20d2650407f79068.json. Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-SPECTROPHOTOMETRY
- NIST-BIREFRINGENCE
- NIST-EXCITON-DYNAMICS
- NIST-CARRIER-DYNAMICS
- NIST-OPTICAL-SCATTERING
- NIST-FLUORESCENCE-RAMAN
- NIST-FLUORESCENCE-QUANTUM-YIELD-GUIDE
- NIST-NONLINEAR-MIXING
- NIST-PHOTONIC-BANDGAP
- NIST-PLASMONIC-MODES
- NIST-WAVEGUIDE-LOSS

Measured, observed, adverse and boundary records:

- VALID-008: complete registered membership count 10
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-OPT-ABSORPTION-EXTINCTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-EXCITON-DYNAMICS-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-LIGHT-SCATTERING-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-LUMINESCENCE-YIELD-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-NONLINEAR-MIXING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-OPT-PLASMONIC-RESPONSE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-REFLECTION-TRANSMISSION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

Shared claim-record clause MAT-S015 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:791d347f9ffa8c3ae8fcfc7af46b64dc1a7d5ac5516629ef471e86aafd51fe4e; empirical validation
 sha256:e8c004a8c4738287657751328a8745954b8ea16d6ac066994461415a6ee7a092; external validation
 sha256:e431917af281e16cf3399911b0318ee61b040e7e340f14de966a36b7fce34b7f; isolation
 sha256:725e5016aa6747954ba9e60ca8e8a514339ae47a38eac28506a5465d0b2aed1f; custody
 sha256:3e0034f381c5a5e19a2d584ece0e87fbba67ddafca0b98f1e401857c3b9f220b.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

280. Material-class and processing validation vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-009

Claim: SFT-MAT-VALIDATION-MATERIAL-CLASS-PROCESSING-VECTOR-009

Question, law and forced result. VALID-009 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 22 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-009 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 22 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005
- SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009
- SFT-MAT-CLASS-FIBRE-REINFORCED-006
- SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011
- SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003
- SFT-MAT-CLASS-INTERMETALLIC-ORDER-002
- SFT-MAT-CLASS-METALLIC-GLASS-008
- SFT-MAT-CLASS-PARTICLE-REINFORCED-007
- SFT-MAT-CLASS-POLYMER-SUBCLASSES-010

- SFT-MAT-CLASS-REFRACTORY-UHT-004
- SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001
- SFT-MAT-PROC-ADDITIVE-BUILD-004
- SFT-MAT-PROC-CASTING-HISTORY-001
- SFT-MAT-PROC-EPITAXY-MATCHING-006
- SFT-MAT-PROC-FORMING-TEXTURE-002
- SFT-MAT-PROC-JOINING-INTERFACE-007
- SFT-MAT-PROC-MACHINING-DAMAGE-003
- SFT-MAT-PROC-POLYMER-ORIENTATION-008
- SFT-MAT-PROC-POWDER-COMPACTION-009
- SFT-MAT-PROC-THIN-FILM-GROWTH-005
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-VALIDATION-OPTICAL-PHOTONIC-VECTOR-008

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-009 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 22 value-free claim identities registered for Materials VALID-009, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-009-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: `complete-receipt-bound-membership__complete-valid-009-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof sha256:b1a31e82807f71e7451d8b2763b1d03e1d374014c2439d710a15102a0199c387; generality sha256:76436bf832d661b96f12df63f65ebf2a1bc323ebb942db5f1b76862f7973e6a1`.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt `sha256:e9falb5a9086a0f2402381f4f808fbedaaa6a946245c5cc39a540292b865e78b`.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt `sha256:27e1b452c442bf79bf3aca9b90dfb657aa06ee2080f5a02e648ad191d17e8889`.

- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:745a5fba3c7e4d894239459aff29e76e21a5df714c8fd12ed99b33c1cd4a67a7.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:3b588355dd95d73afecbb608a4a840c557222fbc1f00f77f3d13c7dd885f7e4.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate

sha256:f66198c68ed47bdbb1c557dfd0a076853688febc0d1ff35ff18f9a61bf1c7b2f. Derivation seal

sha256:0aac846e90d5c3569ad350159cedc409ca97280be783e1caa4d6885c083a72a8. Source manifest

sha256:d930e7c33b83de1d6547b9bd5e09d2dae246e2e1e14f692008a9f5b948ac078b. Engine receipt

sha256:f8fd9d1cd1bcf8cbefc282a747f11f0fdf688f47aa9a0e743b14d5ffda451b35 at receipts/engine/model_admitted/SFT-MAT-VALIDATION-MATERIAL-CLASS-PROCESSING-VECTOR-009-f8fd9d1cd1bcf8cb.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-AUXETIC-ARCHITECTED
- NIST-CEMENT-AM
- NIST-CERAMIC-AM
- NIST-HIGH-TEMP-CERAMICS
- NIST-ADVANCED-COMPOSITES
- NIST-AMMT-GRADED
- NIST-HEA-PHASE
- NIST-HEA-INTERMETALLIC
- NIST-METALLIC-GLASS
- NIST-MACROMOLECULAR-ARCHITECTURES
- NIST-THERMOSET-GLOSSARY
- NIST-MELT-POOL-COOLING
- NIST-MOULD-FILLING
- NIST-NANOWIRE-LATTICE-MATCH
- NIST-CRYSTALLOGRAPHIC-TEXTURE
- NIST-WELD-MONITORING
- NIST-SUBSURFACE-DAMAGE
- NIST-POLYMER-ORIENTATION
- NIST-POWDER-COMPACTION
- NIST-PULSED-LASER-DEPOSITION
- NIST-REPRODUCIBLE-PROCESS-MONITORING

Measured, observed, adverse and boundary records:

- VALID-009: complete registered membership count 22
- complete Materials base: 271 claims; 271 empirical; 0 formal-only

- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-FIBRE-REINFORCED-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-INTERMETALLIC-ORDER-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-METALLIC-GLASS-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-PARTICLE-REINFORCED-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-POLYMER-SUBCLASSES-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-REFRACTORY-UHT-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-ADDITIVE-BUILD-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-CASTING-HISTORY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-EPITAXY-MATCHING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-FORMING-TEXTURE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-JOINING-INTERFACE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-MACHINING-DAMAGE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-POLYMER-ORIENTATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-POWDER-COMPACTION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-THIN-FILM-GROWTH-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-PROC-WINDOW-PROVENANCE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

Shared claim-record clause MAT-S015 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:a8bf6654f0ecc9800b71c6aee8f19381fd0fdd70cf8a7c2ee5a2bce78a6c634c; empirical validation
 sha256:433ec6142744af5c080120f40442e40578b856234a08a3cfc89471d45ddd63bd; external validation
 sha256:37cbcd0c6d71a86c81daec4ea2886bae08b5def834b30b1b46e3f40e6461f597; isolation
 sha256:02eed5ccf08fa29ca46ca4c140e514458a2c77eaab87ed0f79bdfe174cd06ef2; custody
 sha256:bad23ef314353073ccb37d56ea86b4095a6fdd6298879b7e94fce9c065582225.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

281. Cross-source Materials reproducibility vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-010

Claim: SFT-MAT-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010

Question, law and forced result. VALID-010 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 62 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-010 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 62 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-BULK-CORRESPONDENCE-001
- SFT-MAT-BULK-SIZE-SURFACE-001
- SFT-MAT-BULK-WATER-RESPONSE-002
- SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009
- SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003
- SFT-MAT-CLASS-INTERMETALLIC-ORDER-002
- SFT-MAT-CLASS-POLYMER-SUBCLASSES-010
- SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001
- SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007
- SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004
- SFT-MAT-CRYS-STRUCTURE-FACTOR-002
- SFT-MAT-CRYS-TWIN-DOMAIN-006
- SFT-MAT-CRYST-BRAVAIS-001

- SFT-MAT-CRYST-PHONON-001
- SFT-MAT-DEGR-RADIATION-001
- SFT-MAT-ELEC-BAND-ALIGNMENT-008
- SFT-MAT-ELEC-CARRIER-CONFINEMENT-009
- SFT-MAT-ELEC-CARRIER-DUALITY-001
- SFT-MAT-ELEC-CONDUCTOR-CLASS-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002
- SFT-MAT-MAGSC-DOMAINS-WALLS-004
- SFT-MAT-MAGSC-HYSTERESIS-LOOP-005
- SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006
- SFT-MAT-MAGSC-MAGNETORESISTANCE-007
- SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001
- SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008
- SFT-MAT-MECH-CRACK-GROWTH-008
- SFT-MAT-MECH-FATIGUE-009
- SFT-MAT-MECH-FATIGUE-CREEP-001
- SFT-MAT-MECH-FRACTURE-001
- SFT-MAT-MECH-FRICTION-CONTACT-012
- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-MECH-STRESS-STRAIN-001
- SFT-MAT-MICRO-COARSENING-TRANSFER-007
- SFT-MAT-MICRO-DEFECT-MIGRATION-002
- SFT-MAT-MICRO-DISLOCATION-REACTION-003
- SFT-MAT-MICRO-INTERFACE-MOBILITY-008
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006
- SFT-MAT-OPT-EXCITON-DYNAMICS-010
- SFT-MAT-OPT-LIGHT-SCATTERING-004
- SFT-MAT-OPT-LUMINESCENCE-YIELD-003
- SFT-MAT-OPT-REFLECTION-TRANSMISSION-002
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-PHASE-METASTABLE-RETENTION-004
- SFT-MAT-PHASE-ORDER-DISORDER-008
- SFT-MAT-PHASE-SPINODAL-INSTABILITY-005
- SFT-MAT-PHASE-TIE-LINE-LEVER-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010

- SFT-MAT-PHASE-TRANSITION-001
- SFT-MAT-PROC-HEAT-TREATMENT-001
- SFT-MAT-PROC-PHASE-DIAGRAM-001
- SFT-MAT-SEMI-JUNCTION-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-SUST-LIFECYCLE-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-THERM-PHASE-STORAGE-006
- SFT-MAT-THERM-PHONON-MEAN-PATH-003
- SFT-MAT-THERM-SHOCK-FATIGUE-007
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-VALIDATION-MATERIAL-CLASS-PROCESSING-VECTOR-009

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-010 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 62 value-free claim identities registered for Materials VALID-010, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-010-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: complete-receipt-bound-membership__complete-valid-010-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:46d8ed9317be3c4a98ff4b30413a85b5b424cedecd5be8de792101f8b637c9c; generality sha256:6bf651acdded69a26c0acf9075a50d494770f4f8e683356f75038aac7d49dbf6.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:0e200a76a724b58d2eecfc5957f427b12b2fcac06361b0d8c7f5cd6623dd5d42.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:b64c616530f95d421df9db93e5f7361320c522875bfad5a8752bf973ea82fea9.

- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:dda3eb2e5ddaefef97ecd129658177e246b3a5f2db17812cb2a658e58da1c35f.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:5da0a196e302c1a0868bb1a4fb90b8a888704cc8ad5ea39626a2ba95f1be44ea.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate
sha256:ae4610ad2e012985645f587d2f8b5eb4bb620b1734cf5d37302dc98f17555f62. Derivation seal
sha256:ec9681d2d85b75a1dea067ce5a7c545e48633849e5596c328489d7a7c3baa9e7. Source manifest
sha256:78b850f6c4e290b1a220461a81b0fbcf41960efff82eab77463f1eee2c861141. Engine receipt
sha256:5fdf955b9220bffffd89fe0fce5754df31ff5b7db027b18c6c520233b4c5cc1c7 at receipts/engine/model_admitted/SFT-MAT-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010-5fdf955b9220bfff.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-POLYMER-COMPOSITES-2026-07-24
- NIST-MULTISCALE-MATERIALS-2026-07-24
- NIST-SURFACE-DAMAGE-2026-07-24
- NIST-NANO-PROTOCOLS-2026-07-24
- NIST-WATER-THERMOPHYSICAL-2026-07-27
- NIST-LIQUID-WATER-DENSITY-2026-07-27
- NIST-ICE-DENSITY-2026-07-27
- NIST-CERAMIC-AM
- NIST-HIGH-TEMP-CERAMICS
- NIST-HEA-PHASE
- NIST-HEA-INTERMETALLIC
- NIST-MACROMOLECULAR-ARCHITECTURES
- NIST-THERMOSET-GLOSSARY
- NIST-TOTAL-SCATTERING-PDF-2014
- IUCR-TWINNED-DIFFRACTION-DATA-2022
- IUCR-MODULATED-STRUCTURES-2009
- IUCR-INCOMMENSURATE-DICTIONARY-2026
- IUCR-MODULATION-WAVE-DIFFUSE-2015
- IUCR-STACKING-DIFFUSE-2023
- NIST-SP846-POWDER-DIFFRACTION-1992
- IUCR-TWIN-DICTIONARY-2026
- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24
- NIST-BRAVAIS-CLASSIFICATION-2026-07-24
- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24
- NIST-TUNNEL-BAND-OFFSET
- NIST-OPTOELECTRONIC-BAND-METROLOGY
- NIST-SEMICONDUCTOR-CARRIERS-2026-07-24
- NIST-OPTICAL-MATERIALS-2026-07-24
- NIST-TOPOLOGICAL-INSULATORS-2026-07-24
- NIST-SEMICONDUCTORS-2026-07-24
- NIST-PARAMAGNETIC-DIAMAGNETIC
- NIST-MAGNETIC-SUSCEPTIBILITY-SRM
- NIST-MAGNETIZATION-LAB
- NIST-MAGNETIC-IMAGING
- NIST-HYSTERESIS-STANDARD
- NIST-MAGNETIC-MATERIALS-METROLOGY
- NIST-SPIN-ORBIT
- NIST-SPIN-TRANSPORT
- NIST-SPIN-SPECTROSCOPY
- NIST-FAILURE-PROPERTY-TESTS
- NIST-FATIGUE-FRACTURE
- NIST-FATIGUE-FRACTURE-2026-07-24
- NIST-NANOTRIBOLOGY
- NIST-LUBRICATION-HANDBOOK
- NIST-LUBRICANT-FILM
- NIST-RHEOLOGY
- NIST-VISCOELASTIC-SEALANT
- NIST-SUPERCONDUCTING-GRAIN-BOUNDARIES-2026-07-24
- NIST-BENCHMARK-COARSENING-2017
- NIST-STRUCTURES-PRECIPITATION-HANDBOOK
- NIST-POINT-DEFECTS-2026
- NIST-DISLOCATION-DYNAMICS-2021
- NIST-DISLOCATION-CLIMB-MONOGRAPH-59
- NIST-SHARP-INTERFACE-GRAINS-2001
- NIST-UNIFIED-GRAIN-BOUNDARY-MOTION-2008
- NIST-MULTISCALE-MATERIALS-2026
- NIST-MICROSTRUCTURE-PROPERTY-TOOLS-2026
- NIST-SEGREGATION-PRECIPITATION-2021
- NIST-EXCITON-DYNAMICS
- NIST-CARRIER-DYNAMICS
- NIST-OPTICAL-SCATTERING

- NIST-FLUORESCENCE-RAMAN
- NIST-FLUORESCENCE-QUANTUM-YIELD-GUIDE
- NIST-SPECTROPHOTOMETRY
- NIST-TEXTURE-PHASE-FRACTION
- NIST-LEVER-RULE-SOLIDIFICATION
- NIST-LIQUID-WATER-METASTABLE
- NIST-BINARY-HALIDE-TRANSFORMATIONS
- NIST-ORDER-DISORDER-SEPARATION
- NIST-LLE-TERNARY-TIE-LINES
- NIST-SOLIDIFICATION
- NIST-PHASE-TRANSITION-TEMPERATURES
- NIST-SUPERCONDUCTING-FLUX-2026-07-24
- NIST-METAL-ADDITIVE-CORROSION-2026-07-24
- NIST-TEXTURE-PHASE-FRACTION-2026-07-24
- NIST-JOSEPHSON-STANDARD-2026-07-24
- NIST-MATERIALS-DIVISION-2026-07-24
- NIST-MATERIALS-DATA-GUIDE
- NIST-THERMAL-DIFFUSIVITY-AM
- NIST-PHASE-CHANGE-STORAGE
- NIST-NANOCALORIMETRY
- NIST-INTERFACE-SCATTERING
- NIST-PHONON-THERMAL-LIMITS
- NIST-FRACTOGRAPHY-THERMAL-SHOCK
- NIST-THERMAL-SHOCK-SILICON-NITRIDE
- NIST-THERMOELECTRIC-MEASUREMENTS
- NIST-TRANSPORT-THERMOELECTRIC

Measured, observed, adverse and boundary records:

- VALID-010: complete registered membership count 62
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-BULK-CORRESPONDENCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-BULK-SIZE-SURFACE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-BULK-WATER-RESPONSE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

- SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-INTERMETALLIC-ORDER-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-POLYMER-SUBCLASSES-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=6; explicit-status-lines=3
- SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-STRUCTURE-FACTOR-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-TWIN-DOMAIN-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-CRYST-BRAVAIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-PHONON-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEGR-RADIATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-BAND-ALIGNMENT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-ELEC-CARRIER-CONFINEMENT-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-ELEC-CARRIER-DUALITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-CONDUCTOR-CLASS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-BIOMATERIAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-DOMAINS-WALLS-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-HYSTERESIS-LOOP-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-MAGNETORESISTANCE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

- SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-CRACK-GROWTH-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-FATIGUE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-FATIGUE-CREEP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-FRACTURE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-FRICTION-CONTACT-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-RHEOLOGY-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-STRESS-STRAIN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MICRO-COARSENING-TRANSFER-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-MICRO-DEFECT-MIGRATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-DISLOCATION-REACTION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-INTERFACE-MOBILITY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=4
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-OPT-EXCITON-DYNAMICS-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-LIGHT-SCATTERING-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-LUMINESCENCE-YIELD-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-REFLECTION-TRANSMISSION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-PHASE-FRACTION-LEDGER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-METASTABLE-RETENTION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-ORDER-DISORDER-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2

- SFT-MAT-PHASE-SPINODAL-INSTABILITY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TIE-LINE-LEVER-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TIME-TEMPERATURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TRANSITION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-HEAT-TREATMENT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-PHASE-DIAGRAM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-JUNCTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-OPTICAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SUST-LIFECYCLE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-THERM-DIFFUSIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-PHASE-STORAGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-PHONON-MEAN-PATH-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-SHOCK-FATIGUE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

Shared claim-record clause MAT-S015 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:9d27c4820b583ae5ec40ff740b435ebcc3a59201a8645ec075bdb9057c554606; empirical validation
 sha256:7e6cf83ff14a8c26b1b4230e7da005dfcb3470491e30392e1ac4f91238d0b8dd; external validation
 sha256:f750bfc4a3eda3fa79a137fd97e0181d871f7d2d79e475f63a631164adad2962; isolation
 sha256:f16b36bdebfefa8d4dee795037a696576b51f6d7884f1976a7458307f44c7598; custody
 sha256:57a1b7fa58f412d84c0a18d49a7edb086b0e4629b28ecba80ece18231d8af439.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

282. Complete adverse, absent and out-of-bound Materials vector

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-011

Claim: SFT-MAT-VALIDATION-ADVERSE-ABSENT-OUT-OF-BOUND-VECTOR-011

Question, law and forced result. VALID-011 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 271 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-011 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 271 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001
- SFT-MAT-BIO-BIOFABRICATED-IDENTITY-008
- SFT-MAT-BIO-BIORESORPTION-DEGRADATION-002
- SFT-MAT-BIO-CELL-MATERIAL-ADHESION-004
- SFT-MAT-BIO-CONTROLLED-RELEASE-006
- SFT-MAT-BIO-MECHANICAL-MATCHING-005
- SFT-MAT-BIO-MINERALIZED-ORGANIZATION-007
- SFT-MAT-BIO-SCAFFOLD-POROSITY-CONNECTIVITY-003
- SFT-MAT-BULK-ANISOTROPY-001
- SFT-MAT-BULK-CORRESPONDENCE-001
- SFT-MAT-BULK-SIZE-SURFACE-001
- SFT-MAT-BULK-WATER-RESPONSE-002
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-FIBRE-REINFORCED-006
- SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003
- SFT-MAT-CLASS-INTERMETALLIC-ORDER-002
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-METALLIC-GLASS-008
- SFT-MAT-CLASS-PARTICLE-REINFORCED-007
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-POLYMER-SUBCLASSES-010
- SFT-MAT-CLASS-POROUS-001

- SFT-MAT-CLASS-REFRACTORY-UHT-004
- SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001
- SFT-MAT-COMP-DATA-REPRESENTATION-001
- SFT-MAT-COMP-DATABASE-PROVENANCE-008
- SFT-MAT-COMP-ELECTRONIC-STRUCTURE-011
- SFT-MAT-COMP-ERROR-PROPAGATION-005
- SFT-MAT-COMP-FINITE-SIMULATION-003
- SFT-MAT-COMP-INVERSE-PROBLEM-006
- SFT-MAT-COMP-LEARNING-BOUNDARY-007
- SFT-MAT-COMP-MOLECULAR-DYNAMICS-010
- SFT-MAT-COMP-MULTISCALE-COMPOSITION-004
- SFT-MAT-COMP-PHASE-FIELD-009
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-COMP-STRUCTURE-PROPERTY-002
- SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001
- SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007
- SFT-MAT-CRYS-PAIR-DISTRIBUTION-008
- SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004
- SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005
- SFT-MAT-CRYS-STRUCTURE-FACTOR-002
- SFT-MAT-CRYS-TEXTURE-ORIENTATION-003
- SFT-MAT-CRYS-TWIN-DOMAIN-006
- SFT-MAT-CRYST-BRAVAIS-001
- SFT-MAT-CRYST-CUBIC-COORDINATION-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-PHONON-001
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002
- SFT-MAT-CRYST-QUASICRYSTAL-001
- SFT-MAT-CRYST-QUASICRYSTAL-INFLATION-002
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-MAT-CRYST-ROTATION-RESTRICTION-001
- SFT-MAT-CRYST-SYSTEMS-001
- SFT-MAT-CRYST-TRANSLATION-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-DEFECT-DISLOCATION-001
- SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001
- SFT-MAT-DEFECT-POINT-001
- SFT-MAT-DEFECT-VACANCY-001

- SFT-MAT-DEGR-CORROSION-001
- SFT-MAT-DEGR-CORROSION-PATH-002
- SFT-MAT-DEGR-HYDROGEN-EMBRITTELEMENT-005
- SFT-MAT-DEGR-OXIDATION-SCALE-001
- SFT-MAT-DEGR-PASSIVATION-BREAKDOWN-003
- SFT-MAT-DEGR-PHYSICAL-AGEING-008
- SFT-MAT-DEGR-RADIATION-001
- SFT-MAT-DEGR-RADIATION-DEFECT-RECOVERY-007
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-DEGR-STRESS-CORROSION-004
- SFT-MAT-DEGR-WEAR-001
- SFT-MAT-DEGR-WEAR-MODE-DISTINCTION-006
- SFT-MAT-DEGR-WEATHERING-009
- SFT-MAT-DIEL-POLARIZATION-001
- SFT-MAT-ELEC-BAND-ALIGNMENT-008
- SFT-MAT-ELEC-BAND-GAP-001
- SFT-MAT-ELEC-CARRIER-CONFINEMENT-009
- SFT-MAT-ELEC-CARRIER-DUALITY-001
- SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001
- SFT-MAT-ELEC-CONDUCTOR-CLASS-001
- SFT-MAT-ELEC-DEFECT-TRAP-STATES-010
- SFT-MAT-ELEC-DIELECTRIC-LOSS-004
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-ELEC-FINITE-BARRIER-TUNNELLING-007
- SFT-MAT-ELEC-HALL-RESPONSE-003
- SFT-MAT-ELEC-IONIC-TRANSFERENCE-005
- SFT-MAT-ELEC-MIXED-TRANSPORT-006
- SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002
- SFT-MAT-ELEC-OCCUPATION-001
- SFT-MAT-ELEC-SCREENING-DEPLETION-011
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008
- SFT-MAT-EXT-CRYOGENIC-RESPONSE-003
- SFT-MAT-EXT-ELECTRIC-FIELD-RESPONSE-004
- SFT-MAT-EXT-HIGH-PRESSURE-STATE-001
- SFT-MAT-EXT-HIGH-TEMPERATURE-STATE-002
- SFT-MAT-EXT-MAGNETIC-FIELD-RESPONSE-005
- SFT-MAT-EXT-RADIATION-RESPONSE-007
- SFT-MAT-EXT-SHOCK-RESPONSE-006

- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-MAT-FUNC-FERROELECTRIC-001
- SFT-MAT-FUNC-MAGNETIC-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-FUNC-PIEZOELECTRIC-001
- SFT-MAT-FUNC-THERMOELECTRIC-001
- SFT-MAT-HALL-QUANTIZATION-002
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-MAG-FERRIMAGNETISM-002
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002
- SFT-MAT-MAGSC-DOMAINS-WALLS-004
- SFT-MAT-MAGSC-HYSTeresis-LOOP-005
- SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006
- SFT-MAT-MAGSC-MAGNETORESISTANCE-007
- SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001
- SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011
- SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009
- SFT-MAT-MAGSC-SC-VORTEX-PINNING-010
- SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003
- SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008
- SFT-MAT-MAGSC-SUPERFLUID-CRITICAL-FLOW-012
- SFT-MAT-MEAS-COMPOSITION-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MECH-CRACK-GROWTH-008
- SFT-MAT-MECH-CREEP-RUPTURE-010
- SFT-MAT-MECH-ELASTICITY-001
- SFT-MAT-MECH-FATIGUE-009
- SFT-MAT-MECH-FATIGUE-CREEP-001
- SFT-MAT-MECH-FRACTURE-001
- SFT-MAT-MECH-FRACTURE-ENERGY-007
- SFT-MAT-MECH-FRICTION-CONTACT-012

- SFT-MAT-MECH-IMPACT-011
- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013
- SFT-MAT-MECH-MODULUS-001
- SFT-MAT-MECH-PLASTICITY-001
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-MECH-SLIP-001
- SFT-MAT-MECH-STRENGTH-HARDNESS-001
- SFT-MAT-MECH-STRESS-STRAIN-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-MECH-TRANSVERSE-STRAIN-002
- SFT-MAT-MECH-VISCOELASTIC-MEMORY-003
- SFT-MAT-MECH-VISCOPLASTIC-FLOW-004
- SFT-MAT-MECH-WORK-HARDENING-006
- SFT-MAT-MECH-YIELD-PATH-005
- SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005
- SFT-MAT-MICRO-COARSENING-TRANSFER-007
- SFT-MAT-MICRO-DEFECT-MIGRATION-002
- SFT-MAT-MICRO-DEFECT-POPULATION-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-DISLOCATION-REACTION-003
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-GRAIN-GROWTH-004
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-INTERFACE-MOBILITY-008
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006
- SFT-MAT-NANO-AGGREGATION-DISPERSION-CUSTODY-010
- SFT-MAT-NANO-LAYER-STACKING-003
- SFT-MAT-NANO-MOIRE-SUPERSTRUCTURE-008
- SFT-MAT-NANO-NANOCOMPOSITE-INTERFACE-DENSITY-009
- SFT-MAT-NANO-NANOWIRE-CONFINEMENT-002
- SFT-MAT-NANO-PHASE-MELTING-BOUNDARY-006
- SFT-MAT-NANO-QUANTUM-COLLECTIVE-STATE-007
- SFT-MAT-NANO-QUANTUM-DOT-CONFINEMENT-004
- SFT-MAT-NANO-SIZE-SHAPE-DISTRIBUTION-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-OPT-ABSORPTION-EXTINCTION-001

- SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005
- SFT-MAT-OPT-EXCITON-DYNAMICS-010
- SFT-MAT-OPT-LIGHT-SCATTERING-004
- SFT-MAT-OPT-LUMINESCENCE-YIELD-003
- SFT-MAT-OPT-NONLINEAR-MIXING-006
- SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008
- SFT-MAT-OPT-PLASMONIC-RESPONSE-009
- SFT-MAT-OPT-REFLECTION-TRANSMISSION-002
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007
- SFT-MAT-PHASE-COMPONENT-HANDOFF-003
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-PHASE-GLASS-ARREST-009
- SFT-MAT-PHASE-MARTENSITIC-006
- SFT-MAT-PHASE-METASTABLE-RETENTION-004
- SFT-MAT-PHASE-ORDER-DISORDER-008
- SFT-MAT-PHASE-RECONSTRUCTIVE-007
- SFT-MAT-PHASE-SPINODAL-INSTABILITY-005
- SFT-MAT-PHASE-TIE-LINE-LEVER-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-PHASE-TRANSITION-001
- SFT-MAT-PROC-ADDITIVE-BUILD-004
- SFT-MAT-PROC-CASTING-HISTORY-001
- SFT-MAT-PROC-EPITAXY-MATCHING-006
- SFT-MAT-PROC-FORMING-TEXTURE-002
- SFT-MAT-PROC-HEAT-TREATMENT-001
- SFT-MAT-PROC-JOINING-INTERFACE-007
- SFT-MAT-PROC-MACHINING-DAMAGE-003
- SFT-MAT-PROC-PATH-001
- SFT-MAT-PROC-PHASE-DIAGRAM-001
- SFT-MAT-PROC-POLYMER-ORIENTATION-008
- SFT-MAT-PROC-POWDER-COMPACTION-009
- SFT-MAT-PROC-SINTERING-001
- SFT-MAT-PROC-SOLIDIFICATION-001
- SFT-MAT-PROC-THIN-FILM-GROWTH-005
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-SC-FLUX-QUANTIZATION-001
- SFT-MAT-SC-ISOTOPE-RESPONSE-002

- SFT-MAT-SC-JOSEPHSON-001
- SFT-MAT-SC-MEISSNER-001
- SFT-MAT-SC-PAIR-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SEMI-DOPING-001
- SFT-MAT-SEMI-JUNCTION-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-SEMI-PN-TYPE-001
- SFT-MAT-SEMI-RECTIFICATION-002
- SFT-MAT-SEMI-TRANSPORT-001
- SFT-MAT-SF-CIRCULATION-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-SOFT-ACTIVE-NONEQUILIBRIUM-010
- SFT-MAT-SOFT-COLLOID-AGGREGATION-001
- SFT-MAT-SOFT-EMULSION-DROPLET-005
- SFT-MAT-SOFT-FOAM-DRAINAGE-003
- SFT-MAT-SOFT-GEL-PERCOLATION-002
- SFT-MAT-SOFT-GRANULAR-FORCE-CHAIN-007
- SFT-MAT-SOFT-JAMMING-BOUNDARY-008
- SFT-MAT-SOFT-LIQUID-CRYSTAL-ORDER-004
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-SOFT-STIMULI-RESPONSIVE-009
- SFT-MAT-SURF-ADHESION-SEPARATION-003
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-SURF-FREE-STATE-ENERGY-001
- SFT-MAT-SURF-REACTION-CATALYSIS-HANDOFF-006
- SFT-MAT-SURF-ROUGHNESS-SCALE-005
- SFT-MAT-SURF-TRIBOFILM-RETENTION-007
- SFT-MAT-SURF-WETTING-CONTACT-ANGLE-002
- SFT-MAT-SUST-AVAILABILITY-BOUNDARY-002
- SFT-MAT-SUST-CIRCULAR-FLOW-005
- SFT-MAT-SUST-DURABILITY-EXTENSION-006
- SFT-MAT-SUST-EMBODIED-LEDGER-001
- SFT-MAT-SUST-END-OF-LIFE-CUSTODY-009
- SFT-MAT-SUST-LIFECYCLE-001
- SFT-MAT-SUST-RECOVERY-YIELD-004
- SFT-MAT-SUST-REUSE-REMANUFACTURE-003

- SFT-MAT-SUST-SUBSTITUTION-FUNCTION-008
- SFT-MAT-SUST-TOXICITY-HANDOFF-007
- SFT-MAT-THERM-BOUNDARY-RESISTANCE-002
- SFT-MAT-THERM-CONDUCTION-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-THERM-EXPANSION-001
- SFT-MAT-THERM-HEAT-CAPACITY-001
- SFT-MAT-THERM-PHASE-STORAGE-006
- SFT-MAT-THERM-PHONON-MEAN-PATH-003
- SFT-MAT-THERM-RADIATIVE-TRANSPORT-004
- SFT-MAT-THERM-SHOCK-FATIGUE-007
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-TOPO-BULK-BOUNDARY-001
- SFT-MAT-TOPO-EDGE-COUNT-002
- SFT-MAT-TOPO-INVARIANT-001
- SFT-MAT-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-011 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 271 value-free claim identities registered for Materials VALID-011, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-011-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: `complete-receipt-bound-membership__complete-valid-011-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof sha256:8b83c54330b01fda56814f669102210642fce5b45d76ad617ea9042ef71f9204; generality sha256:0ce5427fb4a0900e0a11297cbb26edad6719102d36ac0e4d09b3c2aacf784bf9`.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:6dbed068ea5a46e2e34a94040cdd821e9d61f5508d2320bffb4e9a2b05b256d5.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:8309bb350afe7d7d1d3188dd4088ba75f5511eb40db74fc947de3760afa2d562.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:d34f9a15cd821b658e30a8f85c248ff47d07e45385d87e195d01623bf21ce964.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:c4be233619fe55b01d23b0d44a52dd4b6593a53327fc18051db30013df7db2a8.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate

sha256:1037172b4357f24408b480b718c8348461993a5a83ef5bd6650e9febd4dfc43f. Derivation seal

sha256:48f5ae0b0afd5e955f3292fe72d82eefadf21d9d72dcc2d835d148fa7b8cd9f0. Source manifest

sha256:96d046380c48a4ab54c2658b4bebf4e5b067a66fd5dc6eec65ba9aee53ce86b1. Engine receipt

sha256:6c3a8fb1a8b0444d37a25fbb55244b4c5fb9130e991f8eb98d0cedf46e17c070 at receipts/engine/model_admitted/SFT-MAT-VALIDATION-ADVERSE-ABSENT-OUT-OF-BOUND-VECTOR-011-6c3a8fb1a8b0444d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-BIOMATERIAL-MEASUREMENTS
- NIST-BIOFABRICATION-METROLOGY
- NIST-BIORESORPTION-RATE
- NIST-CELL-MATRIX
- NIST-CONTROLLED-PROTEIN-RELEASE
- NIST-BONE-IMPLANT-LAYERS
- NIST-HYDROXYAPATITE
- NIST-REFERENCE-SCAFFOLDS
- NIST-TEXTURE-PHASE-FRACTION-2026-07-24
- NIST-POLYMER-COMPOSITES-2026-07-24
- NIST-MULTISCALE-MATERIALS-2026-07-24
- NIST-SURFACE-DAMAGE-2026-07-24
- NIST-NANO-PROTOCOLS-2026-07-24
- NIST-WATER-THERMOPHYSICAL-2026-07-27
- NIST-LIQUID-WATER-DENSITY-2026-07-27
- NIST-ICE-DENSITY-2026-07-27
- NIST-AUXETIC-ARCHITECTED
- NIST-CEMENT-AM
- NIST-CERAMIC-ADDITIVE-2026-07-24

- NIST-CERAMIC-AM
- NIST-HIGH-TEMP-CERAMICS
- NIST-ADVANCED-COMPOSITES
- NIST-AMMT-GRADED
- NIST-GLASS-TRANSITION-2026-07-24
- NIST-HEA-PHASE
- NIST-HEA-INTERMETALLIC
- NIST-SOLIDIFICATION-2026-07-24
- NIST-METALLIC-GLASS
- NIST-POLYMER-PROCESSING-2026-07-24
- NIST-MACROMOLECULAR-ARCHITECTURES
- NIST-THERMOSET-GLOSSARY
- NIST-RELIABLE-MATERIALS-DATA
- NIST-MATERIALS-DATABASES
- NIST-COMPUTATIONAL-MATERIALS
- NIST-UNCERTAINTY-PROPAGATION
- NIST-OOF3D
- NIST-INVERSE-HEAT-PLACEMENT
- NIST-ML-MATERIALS-ROBUSTNESS
- NIST-MOLECULAR-DYNAMICS
- NIST-MULTISCALE-GREENS
- NIST-PHASE-FIELD-BENCHMARK
- NIST-SIMULATION-VALIDATION
- NIST-STRUCTURE-PROPERTY-MAPPING
- NIST-TOTAL-SCATTERING-PDF-2014
- IUCR-TWINNED-DIFFRACTION-DATA-2022
- IUCR-MODULATED-STRUCTURES-2009
- IUCR-INCOMMENSURATE-DICTIONARY-2026
- IUCR-MODULATION-WAVE-DIFFUSE-2015
- IUCR-STACKING-DIFFUSE-2023
- IUCR-STACKING-FAULT-LDH-2020
- NIST-SP846-POWDER-DIFFRACTION-1992
- NIST-NCAL-TEXTURE-PHASE-FRACTION-2026
- IUCR-TWIN-DICTIONARY-2026
- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24
- NIST-BRAVAIS-CLASSIFICATION-2026-07-24
- NIST-SIX-NEIGHBOUR-MESH-2026-07-24
- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24
- NIST-PHONON-THERMAL-LIMITS-2026-07-27
- NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24
- NIST-QUASICRYSTAL-APERIODIC-2026-07-27
- NIST-SHECHTMAN-QUASICRYSTALS-2026-07-24
- NIST-SUPERCONDUCTING-GRAIN-BOUNDARIES-2026-07-24
- NIST-POINT-DEFECTS-2026-07-24
- NIST-METAL-ADDITIVE-CORROSION-2026-07-24
- NIST-ELECTROCHEMICAL-CORROSION
- NIST-HYDROGEN-UPTAKE-EMBRITTLEMENT
- NIST-OXIDE-GROWTH
- NIST-LOCALIZED-CORROSION-RESISTANCE
- NIST-PHYSICAL-AGEING-RECOVERY
- NIST-UV-RADIATION-DAMAGE
- NIST-SERVICE-LIFE-BOUNDARY
- NIST-STRESS-CORROSION-CRACK
- NIST-NANOTRIBOLOGY-2026-07-24
- NIST-MULTIAXIAL-WEAR
- NIST-WEATHER-MATERIAL-RESPONSE
- NIST-TUNNEL-BAND-OFFSET
- NIST-OPTOELECTRONIC-BAND-METROLOGY
- NIST-TOPOLOGICAL-INSULATORS-2026-07-24
- NIST-SEMICONDUCTOR-CARRIERS-2026-07-24
- NIST-OPTICAL-MATERIALS-2026-07-24
- NIST-HALL-EFFECT
- NIST-SEMICONDUCTORS-2026-07-24
- NIST-POINT-DEFECT-CHEMISTRY
- NIST-DIELECTRIC-PERMITTIVITY-LOSS
- NIST-OPERANDO-ELECTROCHEMICAL
- NIST-RESISTIVITY-HALL
- NIST-IONIC-CONDUCTIVITY
- NIST-MIXED-IONIC-ELECTRONIC
- NIST-DIELECTRIC-DEPLETION
- NIST-COMBINED-EXTREME-KOLSKY
- NIST-CRYOGENIC-MATERIAL-PROPERTIES
- NIST-ELECTRIC-FIELD-RESPONSE
- NIST-HIGH-PRESSURE-MATERIAL-TESTING
- NIST-HIGH-TEMPERATURE-THERMOELECTRIC

- NIST-HIGH-MAGNETIC-FIELD
- NIST-RADIATION-DAMAGE-MEASUREMENT
- NIST-SHOCKWAVE-MATERIAL-RESPONSE
- NIST-HALL-QUANTIZATION-2026-07-27
- NIST-ANTIFERROMAGNETIC-COUPPLING-2026-07-24
- NIST-FERRIMAGNETIC-SUBLATTICES-2026-07-27
- NIST-PARAMAGNETIC-DIAMAGNETIC
- NIST-MAGNETIC-SUSCEPTIBILITY-SRM
- NIST-MAGNETIZATION-LAB
- NIST-MAGNETIC-IMAGING
- NIST-HYSTERESIS-STANDARD
- NIST-MAGNETIC-MATERIALS-METROLOGY
- NIST-SPIN-ORBIT
- NIST-SC-CRITICAL-FIELDS
- NIST-SC-FLUX-LATTICE
- NIST-SPIN-TRANSPORT
- NIST-SPIN-SPECTROSCOPY
- NIST-SUPERFLUID-PERSISTENT-FLOW
- NIST-MATERIALS-DIVISION-2026-07-24
- NIST-REFERENCE-MATERIALS-2026-07-24
- BIPM-JCGM-VIM-TRACEABILITY-2026-07-24
- NIST-FAILURE-PROPERTY-TESTS
- NIST-FATIGUE-FRACTURE
- NIST-FATIGUE-FRACTURE-2026-07-24
- NIST-NANOTRIBOLOGY
- NIST-LUBRICATION-HANDBOOK
- NIST-LUBRICANT-FILM
- NIST-RHEOLOGY
- NIST-VISCOELASTIC-SEALANT
- NIST-MARTENSITIC-MATERIALS-STUDY
- NIST-SEGREGATION-PRECIPITATION-2021
- NIST-BENCHMARK-COARSENING-2017
- NIST-STRUCTURES-PRECIPITATION-HANDBOOK
- NIST-POINT-DEFECTS-2026
- NIST-DISLOCATION-DYNAMICS-2021
- NIST-DISLOCATION-CLIMB-MONOGRAPH-59
- NIST-SHARP-INTERFACE-GRAINS-2001
- NIST-POLYMER-INTERFACE-CONSORTIUM-2026-07-24

- NIST-UNIFIED-GRAIN-BOUNDARY-MOTION-2008
- NIST-MULTISCALE-MATERIALS-2026
- NIST-MICROSTRUCTURE-PROPERTY-TOOLS-2026
- NIST-NANOPARTICLE-AGGREGATION-DISPERSION
- NIST-QUANTUM-TRANSPORT-2D-STACKING
- NIST-MOIRE-EXCITONS
- NIST-NANOCOMPOSITE-INTERFACIAL-LAYER
- NIST-GAN-NANOWIRE-GROWTH
- NIST-NANOCONFINED-FUSION
- NIST-MOIRE-QUANTUM-PHASES
- NIST-NANOWORLD-QUANTUM-DOTS
- NIST-NANOPARTICLE-SIZE-SHAPE
- NIST-NANOPARTICLE-SURFACE-AREA
- NIST-SPECTROPHOTOMETRY
- NIST-BIREFRINGENCE
- NIST-EXCITON-DYNAMICS
- NIST-CARRIER-DYNAMICS
- NIST-OPTICAL-SCATTERING
- NIST-FLUORESCENCE-RAMAN
- NIST-FLUORESCENCE-QUANTUM-YIELD-GUIDE
- NIST-NONLINEAR-MIXING
- NIST-PHOTONIC-BANDGAP
- NIST-PLASMONIC-MODES
- NIST-WAVEGUIDE-LOSS
- NIST-SAT-TMMC-COEXISTENCE
- NIST-TEXTURE-PHASE-FRACTION
- NIST-LEVER-RULE-SOLIDIFICATION
- NIST-GLASS-TRANSITION
- NIST-LIQUID-WATER-METASTABLE
- NIST-BINARY-HALIDE-TRANSFORMATIONS
- NIST-ORDER-DISORDER-SEPARATION
- NIST-LLE-TERNARY-TIE-LINES
- NIST-SOLIDIFICATION
- NIST-PHASE-TRANSITION-TEMPERATURES
- NIST-SUPERCONDUCTING-FLUX-2026-07-24
- NIST-MELT-POOL-COOLING
- NIST-MOULD-FILLING
- NIST-NANOWIRE-LATTICE-MATCH

- NIST-CRYSTALLOGRAPHIC-TEXTURE
- NIST-WELD-MONITORING
- NIST-SUBSURFACE-DAMAGE
- NIST-POLYMER-ORIENTATION
- NIST-POWDER-COMPACTION
- NIST-PULSED-LASER-DEPOSITION
- NIST-REPRODUCIBLE-PROCESS-MONITORING
- NIST-SUPERCONDUCTING-ISOTOPE-2026-07-27
- NIST-JOSEPHSON-STANDARD-2026-07-24
- NIST-PN-RECTIFICATION-2026-07-27
- NIST-SUPERFLUID-CIRCULATION-2026-07-24
- NIST-SOFT-NONEQUILIBRIUM
- NIST-FLUID-SUSPENSIONS
- NIST-EMULSION-DROPLETS
- NIST-FOAM-DRAINAGE
- NIST-DYNAMIC-ARREST-GEL
- NIST-STRESS-CHAINS
- NIST-JAMMING
- NIST-LIQUID-CRYSTAL
- NIST-FUNCTIONAL-POLYMERS
- NIST-RESPONSIVE-POLYMER
- NIST-ADHESION-ENERGY-DISSIPATION
- NIST-COATING-MICROSTRUCTURE
- NIST-EDGE-DELAMINATION
- NIST-POLYMER-SURFACE-INTERFACE
- NIST-POLYMER-SURFACE-CHEMISTRY
- NIST-COATING-SURFACE-ROUGHNESS
- NIST-TRIBOLOGICAL-COATINGS
- NIST-WETTING-WRINKLED-SURFACES
- NIST-CRITICAL-MINERALS-MATERIALS
- NIST-MATERIAL-FLOW-CIRCULARITY
- NIST-MATERIALS-RESILIENCE
- NIST-LIFE-CYCLE-ASSESSMENT
- NIST-CIRCULAR-END-OF-USE
- NIST-CRITICAL-MATERIAL-RECOVERY
- NIST-REGENERATIVE-MANUFACTURING
- NIST-CRITICAL-SUBSTITUTION
- NIST-CIRCULAR-SAFETY

- NIST-FDTR-TRANSPORT
- NIST-MATERIALS-DATA-GUIDE
- NIST-THERMAL-DIFFUSIVITY-AM
- NIST-PHASE-CHANGE-STORAGE
- NIST-NANOCALORIMETRY
- NIST-INTERFACE-SCATTERING
- NIST-PHONON-THERMAL-LIMITS
- NIST-INFRARED-OPTICAL-PROPERTIES
- NIST-FRACTOGRAPHY-THERMAL-SHOCK
- NIST-THERMAL-SHOCK-SILICON-NITRIDE
- NIST-THERMOELECTRIC-MEASUREMENTS
- NIST-TRANSPORT-THERMOELECTRIC

Measured, observed, adverse and boundary records:

- VALID-011: complete registered membership count 271
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-BIOFABRICATED-IDENTITY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-BIORESORPTION-DEGRADATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-CELL-MATERIAL-ADHESION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-CONTROLLED-RELEASE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-MECHANICAL-MATCHING-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-MINERALIZED-ORGANIZATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-SCAFFOLD-POROSITY-CONNECTIVITY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BULK-ANISOTROPY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-BULK-CORRESPONDENCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-BULK-SIZE-SURFACE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-BULK-WATER-RESPONSE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=3; explicit-status-lines=1

- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-CERAMIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-COMPOSITE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-FIBRE-REINFORCED-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-GLASS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-INTERMETALLIC-ORDER-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-METAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-METALLIC-GLASS-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-PARTICLE-REINFORCED-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-POLYMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-POLYMER-SUBCLASSES-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-POROUS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-REFRACTORY-UHT-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-DATA-REPRESENTATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-DATABASE-PROVENANCE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-ELECTRONIC-STRUCTURE-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-ERROR-PROPAGATION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-COMP-FINITE-SIMULATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-INVERSE-PROBLEM-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-LEARNING-BOUNDARY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-MOLECULAR-DYNAMICS-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-MULTISCALE-COMPOSITION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-PHASE-FIELD-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-STRUCTURE-PROPERTY-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=6; explicit-status-lines=3
- SFT-MAT-CRYS-PAIR-DISTRIBUTION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-STRUCTURE-FACTOR-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-TEXTURE-ORIENTATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-TWIN-DOMAIN-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-CRYST-BRAVAIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-CUBIC-COORDINATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-LATTICE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-PHONON-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-CRYST-QUASICRYSTAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3

- SFT-MAT-CRYST-QUASICRYSTAL-INFLATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-CRYST-RECIPROCAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-ROTATION-RESTRICTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-SYSTEMS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-TRANSLATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-UNIT-CELL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEFECT-DISLOCATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEFECT-POINT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEFECT-VACANCY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEGR-CORROSION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEGR-CORROSION-PATH-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-HYDROGEN-EMBRITTLEMENT-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-OXIDATION-SCALE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-PASSIVATION-BREAKDOWN-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-PHYSICAL-AGEING-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-RADIATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEGR-RADIATION-DEFECT-RECOVERY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=1
- SFT-MAT-DEGR-STRESS-CORROSION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-WEAR-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEGR-WEAR-MODE-DISTINCTION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-DEGR-WEATHERING-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DIEL-POLARIZATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-BAND-ALIGNMENT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-ELEC-BAND-GAP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-CARRIER-CONFINEMENT-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-ELEC-CARRIER-DUALITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-CONDUCTOR-CLASS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-DEFECT-TRAP-STATES-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-DIELECTRIC-LOSS-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-FINITE-BARRIER-TUNNELLING-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-HALL-RESPONSE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-IONIC-TRANSFERENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-MIXED-TRANSPORT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-OCCUPATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-SCREENING-DEPLETION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-CRYOGENIC-RESPONSE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-ELECTRIC-FIELD-RESPONSE-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-HIGH-PRESSURE-STATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-EXT-HIGH-TEMPERATURE-STATE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-MAGNETIC-FIELD-RESPONSE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-RADIATION-RESPONSE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-SHOCK-RESPONSE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-FUNC-BIOMATERIAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-FERROELECTRIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-MAGNETIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-NANOMATERIAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-PHOTONIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-PIEZOELECTRIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-THERMOELECTRIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-HALL-QUANTIZATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MAG-FERRIMAGNETISM-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-MAG-FERROMAGNETISM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-DOMAINS-WALLS-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-HYSTeresis-LOOP-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-MAGNETORESISTANCE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SC-VORTEX-PINNING-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-SUPERFLUID-CRITICAL-FLOW-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MEAS-COMPOSITION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-MATERIAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-MICROSTRUCTURE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-PHASE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-PROPERTY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-SPECIMEN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-TRACEABILITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-CRACK-GROWTH-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-CREEP-RUPTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-ELASTICITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-FATIGUE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-FATIGUE-CREEP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-FRACTURE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-FRACTURE-ENERGY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-FRICTION-CONTACT-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-IMPACT-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

- SFT-MAT-MECH-MODULUS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-PLASTICITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-RHEOLOGY-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-SLIP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-STRENGTH-HARDNESS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-STRESS-STRAIN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-TRANSVERSE-STRAIN-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-VISCOELASTIC-MEMORY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-VISCOPLASTIC-FLOW-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-WORK-HARDENING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-YIELD-PATH-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=3
- SFT-MAT-MICRO-COARSENING-TRANSFER-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-MICRO-DEFECT-MIGRATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-DEFECT-POPULATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-MICRO-DIFFUSION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MICRO-DISLOCATION-REACTION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MICRO-GRAIN-GROWTH-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=3
- SFT-MAT-MICRO-INTERFACE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MICRO-INTERFACE-MOBILITY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=4

- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-NANO-AGGREGATION-DISPERSION-CUSTODY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-LAYER-STACKING-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-MOIRE-SUPERSTRUCTURE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=1
- SFT-MAT-NANO-NANOCOMPOSITE-INTERFACE-DENSITY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-NANOWIRE-CONFINEMENT-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-PHASE-MELTING-BOUNDARY-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-QUANTUM-COLLECTIVE-STATE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-QUANTUM-DOT-CONFINEMENT-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-SIZE-SHAPE-DISTRIBUTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-ABSORPTION-EXTINCTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-EXCITON-DYNAMICS-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-LIGHT-SCATTERING-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-LUMINESCENCE-YIELD-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-NONLINEAR-MIXING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-PLASMONIC-RESPONSE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-REFLECTION-TRANSMISSION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

- SFT-MAT-OPT-REFRACTIVE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PHASE-COMPONENT-HANDOFF-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-FRACTION-LEDGER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-GLASS-ARREST-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-MARTENSITIC-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-METASTABLE-RETENTION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-ORDER-DISORDER-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-RECONSTRUCTIVE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-SPINODAL-INSTABILITY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TIE-LINE-LEVER-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TIME-TEMPERATURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TRANSITION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-ADDITIVE-BUILD-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-CASTING-HISTORY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-EPITAXY-MATCHING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-FORMING-TEXTURE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-HEAT-TREATMENT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-JOINING-INTERFACE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-MACHINING-DAMAGE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-PATH-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-PHASE-DIAGRAM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3

- SFT-MAT-PROC-POLYMER-ORIENTATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-POWDER-COMPACTION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-SINTERING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-SOLIDIFICATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-THIN-FILM-GROWTH-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-WINDOW-PROVENANCE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SC-FLUX-QUANTIZATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SC-ISOTOPE-RESPONSE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-SC-JOSEPHSON-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SC-MEISSNER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SC-PAIR-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SC-ZERO-RESISTANCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-DOPING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-JUNCTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-OPTICAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-PN-TYPE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-RECTIFICATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-SEMI-TRANSPORT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SF-CIRCULATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SF-SUPERFLUID-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SOFT-ACTIVE-NONEQUILIBRIUM-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-COLLOID-AGGREGATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-SOFT-EMULSION-DROPLET-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-FOAM-DRAINAGE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-GEL-PERCOLATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-GRANULAR-FORCE-CHAIN-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-JAMMING-BOUNDARY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-LIQUID-CRYSTAL-ORDER-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-STIMULI-RESPONSIVE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-ADHESION-SEPARATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-COATING-SUBSTRATE-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-DELAMINATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-FREE-STATE-ENERGY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-REACTION-CATALYSIS-HANDOFF-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-ROUGHNESS-SCALE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-TRIBOFILM-RETENTION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-WETTING-CONTACT-ANGLE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-AVAILABILITY-BOUNDARY-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-CIRCULAR-FLOW-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-DURABILITY-EXTENSION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-EMBODIED-LEDGER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-END-OF-LIFE-CUSTODY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-LIFECYCLE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3

- SFT-MAT-SUST-RECOVERY-YIELD-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-REUSE-REMANUFACTURE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-SUBSTITUTION-FUNCTION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-TOXICITY-HANDOFF-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-THERM-BOUNDARY-RESISTANCE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-THERM-CONDUCTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-THERM-DIFFUSIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-EXPANSION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-THERM-HEAT-CAPACITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-THERM-PHASE-STORAGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-PHONON-MEAN-PATH-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-RADIATIVE-TRANSPORT-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-THERM-SHOCK-FATIGUE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-TOPO-BULK-BOUNDARY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-TOPO-EDGE-COUNT-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-TOPO-INVARIANT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3

Shared claim-record clause MAT-S015 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:b0ae9d2e9e6f62470e0116dd07e13f3af5020bfd54039320cf43d462e5833375; empirical validation
 sha256:79c09fb6047e6002004194496de13452d607176ea205dbf4138ead42e02a3bb9; external validation
 sha256:19d0a78fdfce4918fa358e8753be2bb9588cea463a7480f2b146a7a984e20747; isolation
 sha256:16fc3f02ef07c4f6516f91cb1ebfafad9ba0825e3adb0938191b69b9cfa9eea7; custody
 sha256:e1f04a441d3b5464ca6999fde04c1d30807d8e97cb6630f7d7c548f3346f76eb.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome

- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

283. Materials empirical Grand Lock

Field: complete_materials_external_validation

Obligation: SFT-MAT-OBL-VALID-012

Claim: SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012

Question, law and forced result. VALID-012 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 271 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-012 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 271 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001
- SFT-MAT-BIO-BIOFABRICATED-IDENTITY-008
- SFT-MAT-BIO-BIORESORPTION-DEGRADATION-002
- SFT-MAT-BIO-CELL-MATERIAL-ADHESION-004
- SFT-MAT-BIO-CONTROLLED-RELEASE-006
- SFT-MAT-BIO-MECHANICAL-MATCHING-005
- SFT-MAT-BIO-MINERALIZED-ORGANIZATION-007
- SFT-MAT-BIO-SCAFFOLD-POROSITY-CONNECTIVITY-003
- SFT-MAT-BULK-ANISOTROPY-001
- SFT-MAT-BULK-CORRESPONDENCE-001
- SFT-MAT-BULK-SIZE-SURFACE-001
- SFT-MAT-BULK-WATER-RESPONSE-002
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012
- SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005
- SFT-MAT-CLASS-CERAMIC-001
- SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009
- SFT-MAT-CLASS-COMPOSITE-001
- SFT-MAT-CLASS-FIBRE-REINFORCED-006
- SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011
- SFT-MAT-CLASS-GLASS-001
- SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003
- SFT-MAT-CLASS-INTERMETALLIC-ORDER-002
- SFT-MAT-CLASS-METAL-001
- SFT-MAT-CLASS-METALLIC-GLASS-008

- SFT-MAT-CLASS-PARTICLE-REINFORCED-007
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-CLASS-POLYMER-SUBCLASSES-010
- SFT-MAT-CLASS-POROUS-001
- SFT-MAT-CLASS-REFRACTORY-UHT-004
- SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001
- SFT-MAT-COMP-DATA-REPRESENTATION-001
- SFT-MAT-COMP-DATABASE-PROVENANCE-008
- SFT-MAT-COMP-ELECTRONIC-STRUCTURE-011
- SFT-MAT-COMP-ERROR-PROPAGATION-005
- SFT-MAT-COMP-FINITE-SIMULATION-003
- SFT-MAT-COMP-INVERSE-PROBLEM-006
- SFT-MAT-COMP-LEARNING-BOUNDARY-007
- SFT-MAT-COMP-MOLECULAR-DYNAMICS-010
- SFT-MAT-COMP-MULTISCALE-COMPOSITION-004
- SFT-MAT-COMP-PHASE-FIELD-009
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012
- SFT-MAT-COMP-STRUCTURE-PROPERTY-002
- SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001
- SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007
- SFT-MAT-CRYS-PAIR-DISTRIBUTION-008
- SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004
- SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005
- SFT-MAT-CRYS-STRUCTURE-FACTOR-002
- SFT-MAT-CRYS-TEXTURE-ORIENTATION-003
- SFT-MAT-CRYS-TWIN-DOMAIN-006
- SFT-MAT-CRYST-BRAVAIS-001
- SFT-MAT-CRYST-CUBIC-COORDINATION-001
- SFT-MAT-CRYST-LATTICE-001
- SFT-MAT-CRYST-PHONON-001
- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002
- SFT-MAT-CRYST-QUASICRYSTAL-001
- SFT-MAT-CRYST-QUASICRYSTAL-INFLATION-002
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-MAT-CRYST-ROTATION-RESTRICTION-001
- SFT-MAT-CRYST-SYSTEMS-001
- SFT-MAT-CRYST-TRANSLATION-001
- SFT-MAT-CRYST-UNIT-CELL-001

- SFT-MAT-DEFECT-DISLOCATION-001
- SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001
- SFT-MAT-DEFECT-POINT-001
- SFT-MAT-DEFECT-VACANCY-001
- SFT-MAT-DEGR-CORROSION-001
- SFT-MAT-DEGR-CORROSION-PATH-002
- SFT-MAT-DEGR-HYDROGEN-EMBRITTLEMENT-005
- SFT-MAT-DEGR-OXIDATION-SCALE-001
- SFT-MAT-DEGR-PASSIVATION-BREAKDOWN-003
- SFT-MAT-DEGR-PHYSICAL-AGEING-008
- SFT-MAT-DEGR-RADIATION-001
- SFT-MAT-DEGR-RADIATION-DEFECT-RECOVERY-007
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010
- SFT-MAT-DEGR-STRESS-CORROSION-004
- SFT-MAT-DEGR-WEAR-001
- SFT-MAT-DEGR-WEAR-MODE-DISTINCTION-006
- SFT-MAT-DEGR-WEATHERING-009
- SFT-MAT-DIEL-POLARIZATION-001
- SFT-MAT-ELEC-BAND-ALIGNMENT-008
- SFT-MAT-ELEC-BAND-GAP-001
- SFT-MAT-ELEC-CARRIER-CONFINEMENT-009
- SFT-MAT-ELEC-CARRIER-DUALITY-001
- SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001
- SFT-MAT-ELEC-CONDUCTOR-CLASS-001
- SFT-MAT-ELEC-DEFECT-TRAP-STATES-010
- SFT-MAT-ELEC-DIELECTRIC-LOSS-004
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012
- SFT-MAT-ELEC-FINITE-BARRIER-TUNNELLING-007
- SFT-MAT-ELEC-HALL-RESPONSE-003
- SFT-MAT-ELEC-IONIC-TRANSFERENCE-005
- SFT-MAT-ELEC-MIXED-TRANSPORT-006
- SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002
- SFT-MAT-ELEC-OCCUPATION-001
- SFT-MAT-ELEC-SCREENING-DEPLETION-011
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008
- SFT-MAT-EXT-CRYOGENIC-RESPONSE-003
- SFT-MAT-EXT-ELECTRIC-FIELD-RESPONSE-004
- SFT-MAT-EXT-HIGH-PRESSURE-STATE-001

- SFT-MAT-EXT-HIGH-TEMPERATURE-STATE-002
- SFT-MAT-EXT-MAGNETIC-FIELD-RESPONSE-005
- SFT-MAT-EXT-RADIATION-RESPONSE-007
- SFT-MAT-EXT-SHOCK-RESPONSE-006
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-MAT-FUNC-FERROELECTRIC-001
- SFT-MAT-FUNC-MAGNETIC-001
- SFT-MAT-FUNC-NANOMATERIAL-001
- SFT-MAT-FUNC-PHOTONIC-001
- SFT-MAT-FUNC-PIEZOELECTRIC-001
- SFT-MAT-FUNC-THERMOELECTRIC-001
- SFT-MAT-HALL-QUANTIZATION-002
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- SFT-MAT-MAG-FERRIMAGNETISM-002
- SFT-MAT-MAG-FERROMAGNETISM-001
- SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002
- SFT-MAT-MAGSC-DOMAINS-WALLS-004
- SFT-MAT-MAGSC-HYSTERESIS-LOOP-005
- SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006
- SFT-MAT-MAGSC-MAGNETORESISTANCE-007
- SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001
- SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011
- SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009
- SFT-MAT-MAGSC-SC-VORTEX-PINNING-010
- SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003
- SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008
- SFT-MAT-MAGSC-SUPERFLUID-CRITICAL-FLOW-012
- SFT-MAT-MEAS-COMPOSITION-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-MICROSTRUCTURE-001
- SFT-MAT-MEAS-PHASE-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-MEAS-SPECIMEN-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-MECH-CRACK-GROWTH-008
- SFT-MAT-MECH-CREEP-RUPTURE-010
- SFT-MAT-MECH-ELASTICITY-001
- SFT-MAT-MECH-FATIGUE-009

- SFT-MAT-MECH-FATIGUE-CREEP-001
- SFT-MAT-MECH-FRACTURE-001
- SFT-MAT-MECH-FRACTURE-ENERGY-007
- SFT-MAT-MECH-FRICTION-CONTACT-012
- SFT-MAT-MECH-IMPACT-011
- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013
- SFT-MAT-MECH-MODULUS-001
- SFT-MAT-MECH-PLASTICITY-001
- SFT-MAT-MECH-RHEOLOGY-014
- SFT-MAT-MECH-SLIP-001
- SFT-MAT-MECH-STRENGTH-HARDNESS-001
- SFT-MAT-MECH-STRESS-STRAIN-001
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001
- SFT-MAT-MECH-TRANSVERSE-STRAIN-002
- SFT-MAT-MECH-VISCOELASTIC-MEMORY-003
- SFT-MAT-MECH-VISCOPLASTIC-FLOW-004
- SFT-MAT-MECH-WORK-HARDENING-006
- SFT-MAT-MECH-YIELD-PATH-005
- SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005
- SFT-MAT-MICRO-COARSENING-TRANSFER-007
- SFT-MAT-MICRO-DEFECT-MIGRATION-002
- SFT-MAT-MICRO-DEFECT-POPULATION-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-MAT-MICRO-DISLOCATION-REACTION-003
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001
- SFT-MAT-MICRO-GRAIN-GROWTH-004
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-MICRO-INTERFACE-MOBILITY-008
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001
- SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006
- SFT-MAT-NANO-AGGREGATION-DISPERSION-CUSTODY-010
- SFT-MAT-NANO-LAYER-STACKING-003
- SFT-MAT-NANO-MOIRE-SUPERSTRUCTURE-008
- SFT-MAT-NANO-NANOCOMPOSITE-INTERFACE-DENSITY-009
- SFT-MAT-NANO-NANOWIRE-CONFINEMENT-002
- SFT-MAT-NANO-PHASE-MELTING-BOUNDARY-006
- SFT-MAT-NANO-QUANTUM-COLLECTIVE-STATE-007

- SFT-MAT-NANO-QUANTUM-DOT-CONFINEMENT-004
- SFT-MAT-NANO-SIZE-SHAPE-DISTRIBUTION-001
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005
- SFT-MAT-OPT-ABSORPTION-EXTINCTION-001
- SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005
- SFT-MAT-OPT-EXCITON-DYNAMICS-010
- SFT-MAT-OPT-LIGHT-SCATTERING-004
- SFT-MAT-OPT-LUMINESCENCE-YIELD-003
- SFT-MAT-OPT-NONLINEAR-MIXING-006
- SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008
- SFT-MAT-OPT-PLASMONIC-RESPONSE-009
- SFT-MAT-OPT-REFLECTION-TRANSMISSION-002
- SFT-MAT-OPT-REFRACTIVE-001
- SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007
- SFT-MAT-PHASE-COMPONENT-HANDOFF-003
- SFT-MAT-PHASE-FRACTION-LEDGER-001
- SFT-MAT-PHASE-GLASS-ARREST-009
- SFT-MAT-PHASE-MARTENSITIC-006
- SFT-MAT-PHASE-METASTABLE-RETENTION-004
- SFT-MAT-PHASE-ORDER-DISORDER-008
- SFT-MAT-PHASE-RECONSTRUCTIVE-007
- SFT-MAT-PHASE-SPINODAL-INSTABILITY-005
- SFT-MAT-PHASE-TIE-LINE-LEVER-002
- SFT-MAT-PHASE-TIME-TEMPERATURE-010
- SFT-MAT-PHASE-TRANSITION-001
- SFT-MAT-PROC-ADDITIVE-BUILD-004
- SFT-MAT-PROC-CASTING-HISTORY-001
- SFT-MAT-PROC-EPITAXY-MATCHING-006
- SFT-MAT-PROC-FORMING-TEXTURE-002
- SFT-MAT-PROC-HEAT-TREATMENT-001
- SFT-MAT-PROC-JOINING-INTERFACE-007
- SFT-MAT-PROC-MACHINING-DAMAGE-003
- SFT-MAT-PROC-PATH-001
- SFT-MAT-PROC-PHASE-DIAGRAM-001
- SFT-MAT-PROC-POLYMER-ORIENTATION-008
- SFT-MAT-PROC-POWDER-COMPACTION-009
- SFT-MAT-PROC-SINTERING-001
- SFT-MAT-PROC-SOLIDIFICATION-001

- SFT-MAT-PROC-THIN-FILM-GROWTH-005
- SFT-MAT-PROC-WINDOW-PROVENANCE-010
- SFT-MAT-SC-FLUX-QUANTIZATION-001
- SFT-MAT-SC-ISOTOPE-RESPONSE-002
- SFT-MAT-SC-JOSEPHSON-001
- SFT-MAT-SC-MEISSNER-001
- SFT-MAT-SC-PAIR-001
- SFT-MAT-SC-ZERO-RESISTANCE-001
- SFT-MAT-SEMI-DOPING-001
- SFT-MAT-SEMI-JUNCTION-001
- SFT-MAT-SEMI-OPTICAL-001
- SFT-MAT-SEMI-PN-TYPE-001
- SFT-MAT-SEMI-RECTIFICATION-002
- SFT-MAT-SEMI-TRANSPORT-001
- SFT-MAT-SF-CIRCULATION-001
- SFT-MAT-SF-SUPERFLUID-001
- SFT-MAT-SOFT-ACTIVE-NONEQUILIBRIUM-010
- SFT-MAT-SOFT-COLLOID-AGGREGATION-001
- SFT-MAT-SOFT-EMULSION-DROPLET-005
- SFT-MAT-SOFT-FOAM-DRAINAGE-003
- SFT-MAT-SOFT-GEL-PERCOLATION-002
- SFT-MAT-SOFT-GRANULAR-FORCE-CHAIN-007
- SFT-MAT-SOFT-JAMMING-BOUNDARY-008
- SFT-MAT-SOFT-LIQUID-CRYSTAL-ORDER-004
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006
- SFT-MAT-SOFT-STIMULI-RESPONSIVE-009
- SFT-MAT-SURF-ADHESION-SEPARATION-003
- SFT-MAT-SURF-COATING-SUBSTRATE-004
- SFT-MAT-SURF-DELAMINATION-008
- SFT-MAT-SURF-FREE-STATE-ENERGY-001
- SFT-MAT-SURF-REACTION-CATALYSIS-HANDOFF-006
- SFT-MAT-SURF-ROUGHNESS-SCALE-005
- SFT-MAT-SURF-TRIBOFILM-RETENTION-007
- SFT-MAT-SURF-WETTING-CONTACT-ANGLE-002
- SFT-MAT-SUST-AVAILABILITY-BOUNDARY-002
- SFT-MAT-SUST-CIRCULAR-FLOW-005
- SFT-MAT-SUST-DURABILITY-EXTENSION-006
- SFT-MAT-SUST-EMBODIED-LEDGER-001

- SFT-MAT-SUST-END-OF-LIFE-CUSTODY-009
- SFT-MAT-SUST-LIFECYCLE-001
- SFT-MAT-SUST-RECOVERY-YIELD-004
- SFT-MAT-SUST-REUSE-REMANUFACTURE-003
- SFT-MAT-SUST-SUBSTITUTION-FUNCTION-008
- SFT-MAT-SUST-TOXICITY-HANDOFF-007
- SFT-MAT-THERM-BOUNDARY-RESISTANCE-002
- SFT-MAT-THERM-CONDUCTION-001
- SFT-MAT-THERM-DIFFUSIVITY-001
- SFT-MAT-THERM-EXPANSION-001
- SFT-MAT-THERM-HEAT-CAPACITY-001
- SFT-MAT-THERM-PHASE-STORAGE-006
- SFT-MAT-THERM-PHONON-MEAN-PATH-003
- SFT-MAT-THERM-RADIATIVE-TRANSPORT-004
- SFT-MAT-THERM-SHOCK-FATIGUE-007
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005
- SFT-MAT-TOPO-BULK-BOUNDARY-001
- SFT-MAT-TOPO-EDGE-COUNT-002
- SFT-MAT-TOPO-INVARIANT-001
- SFT-MAT-VALIDATION-ADVERSE-ABSENT-OUT-OF-BOUND-VECTOR-011

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis VALID-012 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 271 value-free claim identities registered for Materials VALID-012, their current receipt-bound evidence and all result classes.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-receipt-bound-membership
- complete-valid-012-typed-evidence-vector
- sealed-law-then-capability-closed-release
- favourable-adverse-absent-unavailable-unresolved
- each-declared-closure-scope-held
- explicit-formal-standing-boundary
- omission-tamper-and-boundary-halt
- dated-complete-extension-open

Unique survivor: `complete-receipt-bound-membership__complete-valid-012-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:e2238c52917e730ac92c578064e12e376c56fc0af97e1f730fe5047c44778e73; generality
 sha256:2263a7f8d3a44871d4738e5b92feb5dafdaf5725a93a06878dc5a02dedbb4cee.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
 sha256:0bf419e47e3b09d77df583c35790e0a19b5392d9cf5b92a042564cfbbfdc643e.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
 sha256:feeedd55021f74cd0008a9b4a1f3b715198ddb784e0b461a57214583c829a4de.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
 sha256:8c26c445a0eca677362b75e3afff8e42d54c02425554b59d4f5b83e146149e0c.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
 sha256:ccab034c1a8bcf67eb7406384448cbf1c86af25a4a636226ddee7a80965b8401.

Independent reconstruction and engine admission. Independent implementation

sha256:22fb4003cfd6a55b6b421586a3aa4477fe57d4f24732f2a162be27885414d921 regenerated the declared census and sole survivor; independent certificate
 sha256:1037172b4357f24408b480b718c8348461993a5a83ef5bd6650e9febd4dfc43f. Derivation seal
 sha256:ceed08e577cfb1b879b39a9e4624942b8e743e52362c32f182fb2027a4045568. Source manifest
 sha256:5dd56b79b7e1c671f7042508879fdb4e441ffb952f73bbbaec7e391c0c65334. Engine receipt
 sha256:dbb75828d061c43824bfad454e100d0502d76b5ebc3ee74e9dd03cbcd8967268 at receipts/engine/model_admitted/SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012-dbb75828d061c438.json.
 Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true.
 Comparison passed: true.

Source identities:

- NIST-BIOMATERIAL-MEASUREMENTS
- NIST-BIOFABRICATION-METROLOGY
- NIST-BIORESORPTION-RATE
- NIST-CELL-MATRIX
- NIST-CONTROLLED-PROTEIN-RELEASE
- NIST-BONE-IMPLANT-LAYERS
- NIST-HYDROXYAPATITE
- NIST-REFERENCE-SCAFFOLDS
- NIST-TEXTURE-PHASE-FRACTION-2026-07-24
- NIST-POLYMER-COMPOSITES-2026-07-24
- NIST-MULTISCALE-MATERIALS-2026-07-24
- NIST-SURFACE-DAMAGE-2026-07-24
- NIST-NANO-PROTOCOLS-2026-07-24
- NIST-WATER-THERMOPHYSICAL-2026-07-27
- NIST-LIQUID-WATER-DENSITY-2026-07-27
- NIST-ICE-DENSITY-2026-07-27

- NIST-AUXETIC-ARCHITECTED
- NIST-CEMENT-AM
- NIST-CERAMIC-ADDITIVE-2026-07-24
- NIST-CERAMIC-AM
- NIST-HIGH-TEMP-CERAMICS
- NIST-ADVANCED-COMPOSITES
- NIST-AMMT-GRADED
- NIST-GLASS-TRANSITION-2026-07-24
- NIST-HEA-PHASE
- NIST-HEA-INTERMETALLIC
- NIST-SOLIDIFICATION-2026-07-24
- NIST-METALLIC-GLASS
- NIST-POLYMER-PROCESSING-2026-07-24
- NIST-MACROMOLECULAR-ARCHITECTURES
- NIST-THERMOSET-GLOSSARY
- NIST-RELIABLE-MATERIALS-DATA
- NIST-MATERIALS-DATABASES
- NIST-COMPUTATIONAL-MATERIALS
- NIST-UNCERTAINTY-PROPAGATION
- NIST-OOF3D
- NIST-INVERSE-HEAT-PLACEMENT
- NIST-ML-MATERIALS-ROBUSTNESS
- NIST-MOLECULAR-DYNAMICS
- NIST-MULTISCALE-GREENS
- NIST-PHASE-FIELD-BENCHMARK
- NIST-SIMULATION-VALIDATION
- NIST-STRUCTURE-PROPERTY-MAPPING
- NIST-TOTAL-SCATTERING-PDF-2014
- IUCR-TWINNED-DIFFRACTION-DATA-2022
- IUCR-MODULATED-STRUCTURES-2009
- IUCR-INCOMMENSURATE-DICTIONARY-2026
- IUCR-MODULATION-WAVE-DIFFUSE-2015
- IUCR-STACKING-DIFFUSE-2023
- IUCR-STACKING-FAULT-LDH-2020
- NIST-SP846-POWDER-DIFFRACTION-1992
- NIST-NCAL-TEXTURE-PHASE-FRACTION-2026
- IUCR-TWIN-DICTIONARY-2026
- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24

- NIST-BRAVAIS-CLASSIFICATION-2026-07-24
- NIST-SIX-NEIGHBOUR-MESH-2026-07-24
- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24
- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24
- NIST-PHONON-THERMAL-LIMITS-2026-07-27
- NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24
- NIST-QUASICRYSTAL-APERIODIC-2026-07-27
- NIST-SHECHTMAN-QUASICRYSTALS-2026-07-24
- NIST-SUPERCONDUCTING-GRAIN-BOUNDARIES-2026-07-24
- NIST-POINT-DEFECTS-2026-07-24
- NIST-METAL-ADDITIVE-CORROSION-2026-07-24
- NIST-ELECTROCHEMICAL-CORROSION
- NIST-HYDROGEN-UPTAKE-EMBRITTLEMENT
- NIST-OXIDE-GROWTH
- NIST-LOCALIZED-CORROSION-RESISTANCE
- NIST-PHYSICAL-AGEING-RECOVERY
- NIST-UV-RADIATION-DAMAGE
- NIST-SERVICE-LIFE-BOUNDARY
- NIST-STRESS-CORROSION-CRACK
- NIST-NANOTRIBOLOGY-2026-07-24
- NIST-MULTIAXIAL-WEAR
- NIST-WEATHER-MATERIAL-RESPONSE
- NIST-TUNNEL-BAND-OFFSET
- NIST-OPTOELECTRONIC-BAND-METROLOGY
- NIST-TOPOLOGICAL-INSULATORS-2026-07-24
- NIST-SEMICONDUCTOR-CARRIERS-2026-07-24
- NIST-OPTICAL-MATERIALS-2026-07-24
- NIST-HALL-EFFECT
- NIST-SEMICONDUCTORS-2026-07-24
- NIST-POINT-DEFECT-CHEMISTRY
- NIST-DIELECTRIC-PERMITTIVITY-LOSS
- NIST-OPERANDO-ELECTROCHEMICAL
- NIST-RESISTIVITY-HALL
- NIST-IONIC-CONDUCTIVITY
- NIST-MIXED-IONIC-ELECTRONIC
- NIST-DIELECTRIC-DEPLETION
- NIST-COMBINED-EXTREME-KOLSKY
- NIST-CRYOGENIC-MATERIAL-PROPERTIES

- NIST-ELECTRIC-FIELD-RESPONSE
- NIST-HIGH-PRESSURE-MATERIAL-TESTING
- NIST-HIGH-TEMPERATURE-THERMOELECTRIC
- NIST-HIGH-MAGNETIC-FIELD
- NIST-RADIATION-DAMAGE-MEASUREMENT
- NIST-SHOCKWAVE-MATERIAL-RESPONSE
- NIST-HALL-QUANTIZATION-2026-07-27
- NIST-ANTIFERROMAGNETIC-COUPPLING-2026-07-24
- NIST-FERRIMAGNETIC-SUBLATTICES-2026-07-27
- NIST-PARAMAGNETIC-DIAMAGNETIC
- NIST-MAGNETIC-SUSCEPTIBILITY-SRM
- NIST-MAGNETIZATION-LAB
- NIST-MAGNETIC-IMAGING
- NIST-HYSTERESIS-STANDARD
- NIST-MAGNETIC-MATERIALS-METROLOGY
- NIST-SPIN-ORBIT
- NIST-SC-CRITICAL-FIELDS
- NIST-SC-FLUX-LATTICE
- NIST-SPIN-TRANSPORT
- NIST-SPIN-SPECTROSCOPY
- NIST-SUPERFLUID-PERSISTENT-FLOW
- NIST-MATERIALS-DIVISION-2026-07-24
- NIST-REFERENCE-MATERIALS-2026-07-24
- BIPM-JCGM-VIM-TRACEABILITY-2026-07-24
- NIST-FAILURE-PROPERTY-TESTS
- NIST-FATIGUE-FRACTURE
- NIST-FATIGUE-FRACTURE-2026-07-24
- NIST-NANOTRIBOLOGY
- NIST-LUBRICATION-HANDBOOK
- NIST-LUBRICANT-FILM
- NIST-RHEOLOGY
- NIST-VISCOELASTIC-SEALANT
- NIST-MARTENSITIC-MATERIALS-STUDY
- NIST-SEGREGATION-PRECIPITATION-2021
- NIST-BENCHMARK-COARSENING-2017
- NIST-STRUCTURES-PRECIPITATION-HANDBOOK
- NIST-POINT-DEFECTS-2026
- NIST-DISLOCATION-DYNAMICS-2021

- NIST-DISLOCATION-CLIMB-MONOGRAPH-59
- NIST-SHARP-INTERFACE-GRAINS-2001
- NIST-POLYMER-INTERFACE-CONSORTIUM-2026-07-24
- NIST-UNIFIED-GRAIN-BOUNDARY-MOTION-2008
- NIST-MULTISCALE-MATERIALS-2026
- NIST-MICROSTRUCTURE-PROPERTY-TOOLS-2026
- NIST-NANOPARTICLE-AGGREGATION-DISPERSION
- NIST-QUANTUM-TRANSPORT-2D-STACKING
- NIST-MOIRE-EXCITONS
- NIST-NANOCOMPOSITE-INTERFACIAL-LAYER
- NIST-GAN-NANOWIRE-GROWTH
- NIST-NANOCONFINED-FUSION
- NIST-MOIRE-QUANTUM-PHASES
- NIST-NANOWORLD-QUANTUM-DOTS
- NIST-NANOPARTICLE-SIZE-SHAPE
- NIST-NANOPARTICLE-SURFACE-AREA
- NIST-SPECTROPHOTOMETRY
- NIST-BIREFRINGENCE
- NIST-EXCITON-DYNAMICS
- NIST-CARRIER-DYNAMICS
- NIST-OPTICAL-SCATTERING
- NIST-FLUORESCENCE-RAMAN
- NIST-FLUORESCENCE-QUANTUM-YIELD-GUIDE
- NIST-NONLINEAR-MIXING
- NIST-PHOTONIC-BANDGAP
- NIST-PLASMONIC-MODES
- NIST-WAVEGUIDE-LOSS
- NIST-SAT-TMMC-COEXISTENCE
- NIST-TEXTURE-PHASE-FRACTION
- NIST-LEVER-RULE-SOLIDIFICATION
- NIST-GLASS-TRANSITION
- NIST-LIQUID-WATER-METASTABLE
- NIST-BINARY-HALIDE-TRANSFORMATIONS
- NIST-ORDER-DISORDER-SEPARATION
- NIST-LLE-TERNARY-TIE-LINES
- NIST-SOLIDIFICATION
- NIST-PHASE-TRANSITION-TEMPERATURES
- NIST-SUPERCONDUCTING-FLUX-2026-07-24

- NIST-MELT-POOL-COOLING
- NIST-MOULD-FILLING
- NIST-NANOWIRE-LATTICE-MATCH
- NIST-CRYSTALLOGRAPHIC-TEXTURE
- NIST-WELD-MONITORING
- NIST-SUBSURFACE-DAMAGE
- NIST-POLYMER-ORIENTATION
- NIST-POWDER-COMPACTION
- NIST-PULSED-LASER-DEPOSITION
- NIST-REPRODUCIBLE-PROCESS-MONITORING
- NIST-SUPERCONDUCTING-ISOTOPE-2026-07-27
- NIST-JOSEPHSON-STANDARD-2026-07-24
- NIST-PN-RECTIFICATION-2026-07-27
- NIST-SUPERFLUID-CIRCULATION-2026-07-24
- NIST-SOFT-NONEQUILIBRIUM
- NIST-FLUID-SUSPENSIONS
- NIST-EMULSION-DROPLETS
- NIST-FOAM-DRAINAGE
- NIST-DYNAMIC-ARREST-GEL
- NIST-STRESS-CHAINS
- NIST-JAMMING
- NIST-LIQUID-CRYSTAL
- NIST-FUNCTIONAL-POLYMERS
- NIST-RESPONSIVE-POLYMER
- NIST-ADHESION-ENERGY-DISSIPATION
- NIST-COATING-MICROSTRUCTURE
- NIST-EDGE-DELAMINATION
- NIST-POLYMER-SURFACE-INTERFACE
- NIST-POLYMER-SURFACE-CHEMISTRY
- NIST-COATING-SURFACE-ROUGHNESS
- NIST-TRIBOLOGICAL-COATINGS
- NIST-WETTING-WRINKLED-SURFACES
- NIST-CRITICAL-MINERALS-MATERIALS
- NIST-MATERIAL-FLOW-CIRCULARITY
- NIST-MATERIALS-RESILIENCE
- NIST-LIFE-CYCLE-ASSESSMENT
- NIST-CIRCULAR-END-OF-USE
- NIST-CRITICAL-MATERIAL-RECOVERY

- NIST-REGENERATIVE-MANUFACTURING
- NIST-CRITICAL-SUBSTITUTION
- NIST-CIRCULAR-SAFETY
- NIST-FDTR-TRANSPORT
- NIST-MATERIALS-DATA-GUIDE
- NIST-THERMAL-DIFFUSIVITY-AM
- NIST-PHASE-CHANGE-STORAGE
- NIST-NANOCALORIMETRY
- NIST-INTERFACE-SCATTERING
- NIST-PHONON-THERMAL-LIMITS
- NIST-INFRARED-OPTICAL-PROPERTIES
- NIST-FRACTOGRAPHY-THERMAL-SHOCK
- NIST-THERMAL-SHOCK-SILICON-NITRIDE
- NIST-THERMOELECTRIC-MEASUREMENTS
- NIST-TRANSPORT-THERMOELECTRIC

Measured, observed, adverse and boundary records:

- VALID-012: complete registered membership count 271
- complete Materials base: 271 claims; 271 empirical; 0 formal-only
- retained evidence: 1040 measurement/evidence lines and 334 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-BIOFABRICATED-IDENTITY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-BIORESORPTION-DEGRADATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-CELL-MATERIAL-ADHESION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-CONTROLLED-RELEASE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-MECHANICAL-MATCHING-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-MINERALIZED-ORGANIZATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BIO-SCAFFOLD-POROSITY-CONNECTIVITY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-BULK-ANISOTROPY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-BULK-CORRESPONDENCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3

- SFT-MAT-BULK-SIZE-SURFACE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-BULK-WATER-RESPONSE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-CERAMIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-COMPOSITE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-FIBRE-REINFORCED-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-GLASS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-INTERMETALLIC-ORDER-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-METAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-METALLIC-GLASS-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-PARTICLE-REINFORCED-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-POLYMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-POLYMER-SUBCLASSES-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-CLASS-POROUS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CLASS-REFRACTORY-UHT-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-DATA-REPRESENTATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-DATABASE-PROVENANCE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-COMP-ELECTRONIC-STRUCTURE-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-ERROR-PROPAGATION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-FINITE-SIMULATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-INVERSE-PROBLEM-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-LEARNING-BOUNDARY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-MOLECULAR-DYNAMICS-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-MULTISCALE-COMPOSITION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-PHASE-FIELD-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-SIMULATION-EXPERIMENT-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-COMP-STRUCTURE-PROPERTY-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=6; explicit-status-lines=3
- SFT-MAT-CRYS-PAIR-DISTRIBUTION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-STRUCTURE-FACTOR-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-TEXTURE-ORIENTATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-CRYS-TWIN-DOMAIN-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-CRYST-BRAVAIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-CUBIC-COORDINATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-LATTICE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-PHONON-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3

- SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-CRYST-QUASICRYSTAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-QUASICRYSTAL-INFLATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-CRYST-RECIPROCAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-ROTATION-RESTRICTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-SYSTEMS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-TRANSLATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-CRYST-UNIT-CELL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEFECT-DISLOCATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEFECT-POINT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEFECT-VACANCY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEGR-CORROSION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEGR-CORROSION-PATH-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-HYDROGEN-EMBRITTLEMENT-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-OXIDATION-SCALE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-PASSIVATION-BREAKDOWN-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-PHYSICAL-AGEING-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-RADIATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEGR-RADIATION-DEFECT-RECOVERY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=1
- SFT-MAT-DEGR-STRESS-CORROSION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-DEGR-WEAR-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-DEGR-WEAR-MODE-DISTINCTION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DEGR-WEATHERING-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-DIEL-POLARIZATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-BAND-ALIGNMENT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-ELEC-BAND-GAP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-CARRIER-CONFINEMENT-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-ELEC-CARRIER-DUALITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-CONDUCTOR-CLASS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-DEFECT-TRAP-STATES-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-DIELECTRIC-LOSS-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-ELECTROCHEMICAL-INSERTION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-FINITE-BARRIER-TUNNELLING-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-HALL-RESPONSE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-IONIC-TRANSFERENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-MIXED-TRANSPORT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-ELEC-OCCUPATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-ELEC-SCREENING-DEPLETION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-CRYOGENIC-RESPONSE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-EXT-ELECTRIC-FIELD-RESPONSE-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-HIGH-PRESSURE-STATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-HIGH-TEMPERATURE-STATE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-MAGNETIC-FIELD-RESPONSE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-RADIATION-RESPONSE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-EXT-SHOCK-RESPONSE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-FUNC-BIOMATERIAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-FERROELECTRIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-MAGNETIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-NANOMATERIAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-PHOTONIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-PIEZOELECTRIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-FUNC-THERMOELECTRIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-HALL-QUANTIZATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-MAG-ANTIFERROMAGNETISM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MAG-FERRIMAGNETISM-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-MAG-FERROMAGNETISM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-DOMAINS-WALLS-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-HYSTeresis-LOOP-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-MAGNETORESISTANCE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

- SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SC-VORTEX-PINNING-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MAGSC-SUPERFLUID-CRITICAL-FLOW-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MEAS-COMPOSITION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-MATERIAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-MICROSTRUCTURE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-PHASE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-PROPERTY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-SPECIMEN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MEAS-TRACEABILITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-CRACK-GROWTH-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-CREEP-RUPTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-ELASTICITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-FATIGUE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-FATIGUE-CREEP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-FRACTURE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-FRACTURE-ENERGY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-FRICTION-CONTACT-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0

- SFT-MAT-MECH-IMPACT-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-MODULUS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-PLASTICITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-RHEOLOGY-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-MECH-SLIP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-STRENGTH-HARDNESS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-STRESS-STRAIN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-TRANSVERSE-STRAIN-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-VISCOELASTIC-MEMORY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-VISCOPLASTIC-FLOW-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-WORK-HARDENING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MECH-YIELD-PATH-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=3
- SFT-MAT-MICRO-COARSENING-TRANSFER-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-MICRO-DEFECT-MIGRATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-DEFECT-POPULATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-MICRO-DIFFUSION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MICRO-DISLOCATION-REACTION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-GRAIN-BOUNDARY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MICRO-GRAIN-GROWTH-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=3

- SFT-MAT-MICRO-INTERFACE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MICRO-INTERFACE-MOBILITY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=4
- SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-MICRO-NUCLEATION-GROWTH-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=3
- SFT-MAT-NANO-AGGREGATION-DISPERSION-CUSTODY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-LAYER-STACKING-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-MOIRE-SUPERSTRUCTURE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=1
- SFT-MAT-NANO-NANOCOMPOSITE-INTERFACE-DENSITY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-NANOWIRE-CONFINEMENT-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-PHASE-MELTING-BOUNDARY-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-QUANTUM-COLLECTIVE-STATE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-QUANTUM-DOT-CONFINEMENT-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-SIZE-SHAPE-DISTRIBUTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-NANO-SURFACE-VOLUME-DOMINANCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-ABSORPTION-EXTINCTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-EXCITON-DYNAMICS-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-LIGHT-SCATTERING-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-LUMINESCENCE-YIELD-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-NONLINEAR-MIXING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-OPT-PLASMONIC-RESPONSE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-OPT-REFLECTION-TRANSMISSION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-OPT-REFRACTIVE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PHASE-COMPONENT-HANDOFF-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-FRACTION-LEDGER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-GLASS-ARREST-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-MARTENSITIC-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-METASTABLE-RETENTION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-ORDER-DISORDER-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-RECONSTRUCTIVE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=2
- SFT-MAT-PHASE-SPINODAL-INSTABILITY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TIE-LINE-LEVER-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TIME-TEMPERATURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-MAT-PHASE-TRANSITION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-ADDITIVE-BUILD-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-CASTING-HISTORY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-EPITAXY-MATCHING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-FORMING-TEXTURE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-HEAT-TREATMENT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-JOINING-INTERFACE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-MACHINING-DAMAGE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-PROC-PATH-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-PHASE-DIAGRAM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-POLYMER-ORIENTATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-POWDER-COMPACTION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-SINTERING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-SOLIDIFICATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-PROC-THIN-FILM-GROWTH-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-PROC-WINDOW-PROVENANCE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SC-FLUX-QUANTIZATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SC-ISOTOPE-RESPONSE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-SC-JOSEPHSON-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SC-MEISSNER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SC-PAIR-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SC-ZERO-RESISTANCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-DOPING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-JUNCTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-OPTICAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-PN-TYPE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SEMI-RECTIFICATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-SEMI-TRANSPORT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SF-CIRCULATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SF-SUPERFLUID-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3

- SFT-MAT-SOFT-ACTIVE-NONEQUILIBRIUM-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-COLLOID-AGGREGATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-EMULSION-DROPLET-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-FOAM-DRAINAGE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-GEL-PERCOLATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-GRANULAR-FORCE-CHAIN-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-JAMMING-BOUNDARY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-LIQUID-CRYSTAL-ORDER-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SOFT-STIMULI-RESPONSIVE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-ADHESION-SEPARATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-COATING-SUBSTRATE-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-DELAMINATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-FREE-STATE-ENERGY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-REACTION-CATALYSIS-HANDOFF-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-ROUGHNESS-SCALE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-TRIBOFILM-RETENTION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SURF-WETTING-CONTACT-ANGLE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-AVAILABILITY-BOUNDARY-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-CIRCULAR-FLOW-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-DURABILITY-EXTENSION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-EMBODIED-LEDGER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0

- SFT-MAT-SUST-END-OF-LIFE-CUSTODY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-LIFECYCLE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-SUST-RECOVERY-YIELD-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-REUSE-REMANUFACTURE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-SUBSTITUTION-FUNCTION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-SUST-TOXICITY-HANDOFF-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-THERM-BOUNDARY-RESISTANCE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-THERM-CONDUCTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-THERM-DIFFUSIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-EXPANSION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-THERM-HEAT-CAPACITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-THERM-PHASE-STORAGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-PHONON-MEAN-PATH-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-RADIATIVE-TRANSPORT-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=4; explicit-status-lines=0
- SFT-MAT-THERM-SHOCK-FATIGUE-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=0
- SFT-MAT-TOPO-BULK-BOUNDARY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3
- SFT-MAT-TOPO-EDGE-COUNT-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=1
- SFT-MAT-TOPO-INVARIANT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3; explicit-status-lines=3

Shared claim-record clause MAT-S015 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt

sha256:88a650352c2bf25f431da12b579ceb2c2f39c3631fa24f6dd1781d7e876432f4; empirical validation
 sha256:c8571a2dbbbe1e736cde190e1b631d179000a41c9a897210d964344020492df4; external validation
 sha256:8551369d7074ded5eb00bd1bc730b3d2d14935ebac7498e983e4b8ddc436819e; isolation
 sha256:5fa656e3876427f67307e8bd0cd379c0d9e816d7a015c32a30f380a0b07e193e; custody
 sha256:798c865bd67df0116a7f86334f6f0bae991ddb7efad2a967a4e3f6403e1516f0.

Prohibited inputs.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favourable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Complete-field family: Cross Branch Handoffs

284. Materials-to-Engineering one-owner handoff

Field: `cross_branch_handoffs`

Obligation: `SFT-MAT-OBL-HAND-001`

Claim: `SFT-MAT-ENGINEERING-OWNERSHIP-HANDOFF-001`

Question, law and forced result. The Materials handoff to engineering_translation preserves complete paired admitted records and exactly one owner per coordinate: materials owns specimen organization, processing history and response while engineering owns designed use and implementation.

HAND-001 uniquely requires the complete receipt-bound pair in which materials owns specimen organization, processing history and response while engineering owns designed use and implementation; consumption is a directed dependency and never duplicate ownership.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- `SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012`
- `SFT-MAT-PROC-WINDOW-PROVENANCE-010`
- `SFT-ENG-PHYSICAL-HANDOFF-001`

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Generate the complete eight-axis HAND-001 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-001 owner identities, directed relations, paired receipt records, controls and extension boundary.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- `complete-owned-coordinate`
- `exactly-one-owner`
- `materials-to-engineering_translation-directed-one-owner-handoff`
- `directed-consumer-edge`
- `complete-paired-receipt-records`
- `value-free-registry-before-outcomes`
- `duplicate-missing-or-tampered-halts`
- `dated-complete-extension-open`

Unique survivor: `complete-owned-coordinate__exactly-one-owner__materials-to-engineering_translation-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt`

-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:1e929b8a734dacd591cbfa287a82460988266ab406424a7d97801ac17d0a8651; generality

sha256:7ee8cc1dfc53f033b325f983fef32400c541cf841669d6241b8eb5951a69929c.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:4535746b87e78e3d078dbedf5571acf2ada785749a29ce0c4d8c3b9926e9b62e.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:0b9e6a0ddb59bdab81cbec36009e73090bbd250eea7fbee6c0ed370fd5c9e73c.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:e4007293a8e8d22e5fd76a1e1931a94f18fabf523cc284ac24b984af4dafcfbc.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:f80ac37e9ff4bd752c57316926a11d9c6a37b3c0f24aef84596538ecd8b5f836.

Independent reconstruction and engine admission. Independent implementation

sha256:ce514b15f60d0a05712e120e17fc5eda23fbe18687c2c1ad30352dc5042431a5 regenerated the declared census and sole survivor; independent certificate

sha256:537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f. Derivation seal

sha256:e832769c55f75aae61e45aff33c3382a998f1b15d0a986bfaabbb7b45bb4a98. Source manifest

sha256:503b239b649e0f719e62816ad68e3159b462ea59e39975ae7c50bf99cea1b1d6. Engine receipt

sha256:be4f56a688538024ce7e0f47e3484316d97b382760f01ab93d795169959f5312 at receipts/engine/model_admitted/SFT-MAT-ENGINEERING-OWNERSHIP-HANDOFF-001-be4f56a688538024.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-REPRODUCIBLE-PROCESS-MONITORING
- CISA-SECURE-BY-DESIGN-001

Measured, observed, adverse and boundary records:

- HAND-001: value-free owner registry preceded outcome access
- frozen graph: 1681 claims, 28437 dependency edges, 19852 cross-branch uses
- paired external surface: 15 records, 28 source identities, 54 evidence lines
- all selected evidence rows preserved; omission control rejected True
- materials owns SFT-MAT-PROC-WINDOW-PROVENANCE-010; sources=1; evidence-lines=4; receipt=sha256:0ad7b92e5c8d24cdfc2ee2617314ee031fbf333bf5e0b18f35e9809fa34d261f
- engineering_translation owns SFT-ENG-PHYSICAL-HANDOFF-001; sources=1; evidence-lines=2; receipt=sha256:3c5e2bcfc0353b573de31a1f50e5b60e1983a348e9712b6ada307d889c5bbcf8

Falsification condition: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt

sha256:25d54e226d4508afb499cb40e01930979efb5ab5038592ba2744e5eae1f14c3; empirical validation

sha256:ce424d4169d58737d56105a3e4f58f61b5f75aedc7fabelac15fa7b23dd5c3e1; external validation

sha256:51e8ecc0fd77f9e8c2b07a4fc527eedfae35b7dd9b7e9ada7c4d9534cc476aed; isolation
 sha256:9a1c5fc81336c5697a80c31de6f8d37b0fcfa92d8326707d43d697fc21e0036c; custody
 sha256:e4b121fce65b98dcdb98a5a353c77fb99fff8ad13c09b728c13cde5bbcd0951c.

Prohibited inputs.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

285. Materials-to-Biology one-owner handoff

Field: cross_branch_handoffs

Obligation: SFT-MAT-OBL-HAND-002

Claim: SFT-MAT-BIOLOGY-OWNERSHIP-HANDOFF-002

Question, law and forced result. The Materials handoff to biology preserves complete paired admitted records and exactly one owner per coordinate: materials owns the material interface and response while biology owns living organization, adhesion and function.

HAND-002 uniquely requires the complete receipt-bound pair in which materials owns the material interface and response while biology owns living organization, adhesion and function; consumption is a directed dependency and never duplicate ownership.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001
- SFT-BIO-CELL-ADHESION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-MAT-ENGINEERING-OWNERSHIP-HANDOFF-001

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis HAND-002 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-002 owner identities, directed relations, paired receipt records, controls and extension boundary.

Shared claim-record clause MAT-S009 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complete-owned-coordinate
- exactly-one-owner
- materials-to-biology-directed-one-owner-handoff
- directed-consumer-edge
- complete-paired-receipt-records
- value-free-registry-before-outcomes

- duplicate-missing-or-tampered-halts
- dated-complete-extension-open

Unique survivor: complete-owned-coordinate__exactly-one-owner__materials-to-biology-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:52a371416a1d5cb003a6d8c3a2596ed26f02b6e48bc5ed73681ad7d15659622b; generality sha256:e7e06c9fc00730ae04379c1a049be5dd6a2d7d880606365fcf0d4c30d8de1cc6.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt sha256:9b571434a98245e4d3dbeb2bb21ee37e0c615f57da157e283e281f37603fbd7b.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:a6fba28cd1b335987f5e0e8ae9f0e2db931d998e7cb480394c4204a4f4a9c72e.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt sha256:4248fe2af54a1c49ed2f02100976624c299bd899efd7d29bbb426ab159a0ac91.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt sha256:b2cb9fcf84b8dcba201b86b871c88656d4b66c07cfd9d92ab3f0cfe07656da95.

Independent reconstruction and engine admission. Independent implementation

sha256:ce514b15f60d0a05712e120e17fc5eda23fbe18687c2c1ad30352dc5042431a5 regenerated the declared census and sole survivor; independent certificate sha256:537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f. Derivation seal sha256:4b0079efdeade3fbd235ec9e04bfb130737f0926b97c6ed3083533eee43084fc. Source manifest sha256:809b4aa7fb45bdd33e2fc0e2ab575adc9efba50c3a90b2c6b3b4d112b245a688. Engine receipt sha256:f24161d22e77e070e6bd4b052173fe4072996c01e3a29c7755c1dfa8d3252293 at receipts/engine/model_admitted/SFT-MAT-BIOLOGY-OWNERSHIP-HANDOFF-002-f24161d22e77e070.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-BIOMATERIAL-MEASUREMENTS
- SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1
- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010
- SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE
- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE

Measured, observed, adverse and boundary records:

- HAND-002: value-free owner registry preceded outcome access
- frozen graph: 1681 claims, 28437 dependency edges, 19852 cross-branch uses
- paired external surface: 15 records, 28 source identities, 54 evidence lines
- all selected evidence rows preserved; omission control rejected True

- materials owns SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001; sources=1; evidence-lines=4; receipt=sha256:a10aa7bcb625a83b2d3ebf9dd8e4ff28df0ae2d5d74ab13268cb4ef711b8bd2f
- biology owns SFT-BIO-CELL-ADHESION-001; sources=2; evidence-lines=4; receipt=sha256:0c854c871fd66f0b15e416e1da98cf127e8e562c49d095c2da11396e178f2e8c
- biology owns SFT-BIO-BIO-ASSAY-001; sources=2; evidence-lines=4; receipt=sha256:0f51212a550b8a30526dc82853d945b037c359b272dd8a66f96279ae3071df5c

Falsification condition: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt

sha256:2d42fdca4339052d1b5cd243858154fe6009adb602f916df1102b3117658cccd; empirical validation
 sha256:91a14cd6ff1a19bbd5dd240b459b2ef35b751b421db52af917132440a5c799f3; external validation
 sha256:b42e4f7c198e1cab61f24697fae4bd8f39f190f3b44f1e75a1ca04dc1f7b7cfa; isolation
 sha256:57b7ced135e45a3818be8d472b3fab2075f14e3c4308ba9c8f88a946a1e93ae0; custody
 sha256:7f463580f609d54d5281f54a72c6560407b44057bd49d553cc611abfd69ccabb.

Prohibited inputs.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

286. Materials-to-Medicine one-owner handoff

Field: cross_branch_handoffs

Obligation: SFT-MAT-OBL-HAND-003

Claim: SFT-MAT-MEDICINE-OWNERSHIP-HANDOFF-003

Question, law and forced result. The Materials handoff to medicine preserves complete paired admitted records and exactly one owner per coordinate: materials owns delivery-medium organization and release while medicine owns intervention, dose and health outcome.

HAND-003 uniquely requires the complete receipt-bound pair in which materials owns delivery-medium organization and release while medicine owns intervention, dose and health outcome; consumption is a directed dependency and never duplicate ownership.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-MAT-BIO-CONTROLLED-RELEASE-006
- SFT-MED-INTERVENTION-001
- SFT-MED-DOSE-001
- SFT-MED-RESPONSE-001
- SFT-MAT-BIOLOGY-OWNERSHIP-HANDOFF-002

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis HAND-003 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-003 owner identities, directed relations, paired receipt records, controls and extension boundary.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-owned-coordinate
- exactly-one-owner
- materials-to-medicine-directed-one-owner-handoff
- directed-consumer-edge
- complete-paired-receipt-records
- value-free-registry-before-outcomes
- duplicate-missing-or-tampered-halts
- dated-complete-extension-open

Unique survivor: complete-owned-coordinate__exactly-one-owner__materials-to-medicine-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:4968c79d3ea167840e3e35c7e324d435d57cd4fa1600b694d831e29cd826b070; generality

sha256:30ebc0528bbe2176294aa727a84cd234eb73414d1590d9f533ca255be9e3e5ac.

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:f2aca75da592b407ec0f438dbb46ee09213f90ff229009984153a97f0df9a5a9.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:9908c3556e06df6fb1e8c309cf08227323f5b98ca5ad5f865af9433f054299f5.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:25fa00504a903ac164c8bffa7e8a02f73c730eaaabdc053d868eb7a531d11202.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:bb61ea4bec174d3653bd80ab14c06747f4e66793d34e5c8b054df0b2c816e2ed.

Independent reconstruction and engine admission. Independent implementation

sha256:ce514b15f60d0a05712e120e17fc5eda23fbe18687c2c1ad30352dc5042431a5 regenerated the declared census and sole survivor; independent certificate

sha256:537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f. Derivation seal

sha256:a91e40fb59aacadf1240deab590ffc1c890e028a503e7ddc357294555405e79e. Source manifest

sha256:e9f1b6a71d4cdf47702f56f58e10b9b66da3cede2b7c7132cdcc06f86f812001. Engine receipt

sha256:cf60886a12c3826e6f3fbcfbfd12f4e46b86e24bd573d8696e203e967bd2409f3 at receipts/engine/model_admitted/SFT-MAT-MEDICINE-OWNERSHIP-HANDOFF-003-cf60886a12c3826e.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CONTROLLED-PROTEIN-RELEASE
- MED-ICH-E6R3-001
- MED-ICH-E9R1-001
- MED-FDA-EXPOSURE-RESPONSE-001
- MED-FDA-BIOMARKER-001

Measured, observed, adverse and boundary records:

- HAND-003: value-free owner registry preceded outcome access
- frozen graph: 1681 claims, 28437 dependency edges, 19852 cross-branch uses
- paired external surface: 15 records, 28 source identities, 54 evidence lines
- all selected evidence rows preserved; omission control rejected True
- materials owns SFT-MAT-BIO-CONTROLLED-RELEASE-006; sources=1; evidence-lines=4; receipt=sha256:10c4443b381f4482ad2889e538708109711794e1c8c8aac77d6c7d66dac4d3a1
- medicine owns SFT-MED-INTERVENTION-001; sources=2; evidence-lines=4; receipt=sha256:ef389261cfe2919237eefe5a95ed42263554fa6271eb87e13685b1d43ed1ddcc
- medicine owns SFT-MED-DOSE-001; sources=3; evidence-lines=4; receipt=sha256:ac1d83eb8670e428a755b801006e386ef850299d00310595b34b68fdbbac76a5
- medicine owns SFT-MED-RESPONSE-001; sources=3; evidence-lines=4; receipt=sha256:3a051896a773bd06407d83654f7e55a5f64aca695299eace57d1e33edef69367

Falsification condition: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt

sha256:3d251f537b4470226490c9f0797b34e2c42faf02b1e0a4a485f3fc27d9257fb; empirical validation
 sha256:1e278a85c13aa4c42a70251204a5bc39776505b7ee625efff815f32c1a40969e; external validation
 sha256:88f838ba45b9848165290628497fd8c8708ccfff36a400cd8b1fd735c27ccf80; isolation
 sha256:d7e1a7580de997de1723ea83338dec9a8614f316203fb0b8d863a783ec796783; custody
 sha256:aaa49b49747cb516f7fbfd5c9aa62bae88ca350f3e94445c16bf8f41809572b3.

Prohibited inputs.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

287. Materials-to-Earth and Environmental Science one-owner handoff

Field: cross_branch_handoffs

Obligation: SFT-MAT-OBL-HAND-004

Claim: SFT-MAT-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004

Question, law and forced result. The Materials handoff to earth_environment preserves complete paired admitted records and exactly one owner per coordinate: materials owns material lifecycle and end-of-life custody while Earth and Environmental Science owns environmental path and receptor context.

HAND-004 uniquely requires the complete receipt-bound pair in which materials owns material lifecycle and end-of-life custody while Earth and Environmental Science owns environmental path and receptor context; consumption is a directed dependency and never duplicate ownership.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-MAT-SUST-END-OF-LIFE-CUSTODY-009
- SFT-EARTH-POLLUTANT-IDENTITY-001
- SFT-EARTH-SOURCE-PATH-RECEPTOR-001
- SFT-MAT-MEDICINE-OWNERSHIP-HANDOFF-003

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis HAND-004 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-004 owner identities, directed relations, paired receipt records, controls and extension boundary.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-owned-coordinate
- exactly-one-owner
- materials-to-earth_environment-directed-one-owner-handoff
- directed-consumer-edge
- complete-paired-receipt-records
- value-free-registry-before-outcomes
- duplicate-missing-or-tampered-halts
- dated-complete-extension-open

Unique survivor: `complete-owned-coordinate__exactly-one-owner__materials-to-earth_environment-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open`.

Closure: `depth_independent; minimality true; named-shape uniqueness true; proof`

`sha256:89f596b29bf7358d39ddc498bd3dc702fd0784731a2b144a8405fef7c750d9ed; generality`

`sha256:d9db97c1aba84e87deee2eff8526e9b56817cfc328b775b5880ee01950931589.`

Shared claim-record clause MAT-S010 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
`sha256:0b6c565ba174a9f2d422f93df75dedf1001dde9070c4a0b974deec702b7b7dec.`
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
`sha256:493f1844d43705946bad1d05301d747c6749b397e0c8fb8bcc705b68d64e849e.`
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
`sha256:a3ae535e3d142ccc52f2862a3c69966229869521d20435d7af6648179d963638.`

- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:b6d97ecf64cff04cd006af951095eb1174f7b75f48107d3514258fd80b68cae5.

Independent reconstruction and engine admission. Independent implementation

sha256:ce514b15f60d0a05712e120e17fc5eda23fbb18687c2c1ad30352dc5042431a5 regenerated the declared census and sole survivor; independent certificate
sha256:537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f. Derivation seal
sha256:1483590bd4041ca0421d433f5b398ecf114002ea784feb2bbac749696f688e3e. Source manifest
sha256:abda0b0f8e997fb561cb7def1934eb3f6fafc551b3be8c85c1d96615e0642862. Engine receipt
sha256:cec7220113f95cacdefc5b3154df9194d625bc0f52c73000b227279ed5981cee at receipts/engine/model_admitted/SFT-MAT-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004-cec7220113f95cac.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-CIRCULAR-END-OF-USE
- EPA-AIRDATA-001
- WQP-WEB-SERVICES-001
- USGS-LANDSAT-ACCESS-001
- NEON-OPEN-DATA-001

Measured, observed, adverse and boundary records:

- HAND-004: value-free owner registry preceded outcome access
- frozen graph: 1681 claims, 28437 dependency edges, 19852 cross-branch uses
- paired external surface: 15 records, 28 source identities, 54 evidence lines
- all selected evidence rows preserved; omission control rejected True
- materials owns SFT-MAT-SUST-END-OF-LIFE-CUSTODY-009; sources=1; evidence-lines=4; receipt=sha256:c722e8ed79cdea2dbbdf7dd45e64e9e33d6c0fe63c0d7b283a24e72f1b0f580
- earth_environment owns SFT-EARTH-POLLUTANT-IDENTITY-001; sources=4; evidence-lines=4; receipt=sha256:f70ce5119a12f8ad926fd279bbbe6fcb49b650d666e604dff2f6e586b7d1f993
- earth_environment owns SFT-EARTH-SOURCE-PATH-RECEPTOR-001; sources=4; evidence-lines=4; receipt=sha256:29ff886be12a107ecbcb035eb5573a52bc661557faa4c0bfaf9ff7f28a8727

Falsification condition: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt

sha256:b1cc3e93791ef6c8329aab4393c8deb3548f6f0471aa35808027174e2d86fd6e; empirical validation
sha256:64b86bc0abbb10feaa3232eec803d1b2a5cdf2f0d6eec412fc55aef91466e95e; external validation
sha256:b8376f0b26c50337b0ee40dfe7f5118b0d584f10fd0fc60d4bcf43652c91b09d; isolation
sha256:10f525f2855d32d7427a50c4e700975239678f92026197b28fa57ee13c88d337; custody
sha256:dc0ddd5f92041c58e9f398589f8bace4dc34637f07b7814aeb7a4a59cbcd045.

Prohibited inputs.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored

- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

288. Materials-to-Astronomy one-owner handoff

Field: cross_branch_handoffs

Obligation: SFT-MAT-OBL-HAND-005

Claim: SFT-MAT-ASTRONOMY-OWNERSHIP-HANDOFF-005

Question, law and forced result. The Materials handoff to astronomy_cosmology preserves complete paired admitted records and exactly one owner per coordinate: materials owns radiation-conditioned response while astronomy owns the emitting source and cosmic observational context.

HAND-005 uniquely requires the complete receipt-bound pair in which materials owns radiation-conditioned response while astronomy owns the emitting source and cosmic observational context; consumption is a directed dependency and never duplicate ownership.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-MAT-EXT-RADIATION-RESPONSE-007
- SFT-ASTRO-SPECTRUM-001
- SFT-ASTRO-SOURCE-001
- SFT-MAT-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004

The complete owner/dependency graph in census/materials_hand_001_006_dependency_registry_v1.json independently proves this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Complete candidate grammar. Generate the complete eight-axis HAND-005 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-005 owner identities, directed relations, paired receipt records, controls and extension boundary.

Shared claim-record clause MAT-S009 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- complete-owned-coordinate
- exactly-one-owner
- materials-to-astronomy_cosmology-directed-one-owner-handoff
- directed-consumer-edge
- complete-paired-receipt-records
- value-free-registry-before-outcomes
- duplicate-missing-or-tampered-halts
- dated-complete-extension-open

Unique survivor: complete-owned-coordinate__exactly-one-owner__materials-to-astronomy_cosmology-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:eb01338bf3a22e57fd6a81e416bd5ccaf54ff1a502b2f373883531d2e3715b12; generality sha256:aef322af918bb6d2073f0cb39c9ce005772fa6ec06f86e7db8511e63323cb2a0.

Shared claim-record clause MAT-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:06d52d36fc30486c94ec32ba3ffa717fafd92152fe00b60fa90aba7861b52088.
- `tampered_source` - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt sha256:84701a02192e27ef13622ef3b79655d0856143e8c643728322af749c7cc28581.
- `tampered_artifact` - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:446c4ae09cfc9bdec657ae5f96821b856cb2ffd9ebcfaa410cb743e21bc2b595.
- `boundary` - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:c80ac44b0c2de05c9a719137e419a67a583cecb2e485e7abfdc514966a1bfdb2.

Independent reconstruction and engine admission. Independent implementation

sha256:ce514b15f60d0a05712e120e17fc5eda23fbe18687c2c1ad30352dc5042431a5 regenerated the declared census and sole survivor; independent certificate

sha256:537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f. Derivation seal

sha256:48f2a3bacb32748a449144ce569e3a6623ce2b4f4c4943055251644ea390d0a2. Source manifest

sha256:851a6b6c60efbb42fc7ba4d38abf9c34b2aa9ceb8b033c3da3b458daaa7a3f32. Engine receipt

sha256:d3ab4763a02984075b5d379e1ef1425b5de071c0bf584800a66465d52cc566b9 at receipts/engine/model_admitted/SFT-MAT-ASTRONOMY-OWNERSHIP-HANDOFF-005-d3ab4763a0298407.json.

Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-RADIATION-DAMAGE-MEASUREMENT
- ESA-GAIA-ARCHIVE-001

Measured, observed, adverse and boundary records:

- HAND-005: value-free owner registry preceded outcome access
- frozen graph: 1681 claims, 28437 dependency edges, 19852 cross-branch uses
- paired external surface: 15 records, 28 source identities, 54 evidence lines
- all selected evidence rows preserved; omission control rejected True
- materials owns SFT-MAT-EXT-RADIATION-RESPONSE-007; sources=1; evidence-lines=4; receipt=sha256:9378b3b6d4e9a2a2413e9d856f063be97bfc02e2cf7c289cfb9660efed86305d
- astronomy_cosmology owns SFT-ASTRO-SPECTRUM-001; sources=1; evidence-lines=2; receipt=sha256:3c2ae757898fd4ca1063639281403d361117113843e1a4bd69144ffa5565594
- astronomy_cosmology owns SFT-ASTRO-SOURCE-001; sources=1; evidence-lines=2; receipt=sha256:5473ed30f43a3ee1099bf327c08e196a9abcd888adb958b68d7c85d51681e958

Falsification condition: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt

sha256:9c6b0e6ba8d5be3cdaa6aeb8c9c623f6758ec9c703c928335c8b37ed39946a84; empirical validation

sha256:70758438cbf1323d9ed2e1d2fb690d17d1555586a4a2a5023e467689e75a1251; external validation

sha256:ebb37c06e3aedc032470cf875c1a0463525d7a390bc6e3b06acff4810c08d3bd; isolation

sha256:1b9bb8d9b4ea71e090c0bd95f1a2ed9e5c66541e59748357807e868ff5978d02; custody

sha256:eb9e8e0918ec6544bde26e05bc39ac3e9f671ab6b5df2125d11086c55974b176.

Prohibited inputs.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

289. Materials cross-branch one-owner completeness certificate

Field: `cross_branch_handoffs`

Obligation: `SFT-MAT-OBL-HAND-006`

Claim: `SFT-MAT-CROSS-BRANCH-ONE-OWNER-COMPLETENESS-006`

Question, law and forced result. Across the complete frozen graph of 1681 admitted claims and 28437 dependency edges, every claim has exactly one branch owner, every claim reaches the root theorem, and cross-branch use remains a directed dependency rather than duplicate ownership.

HAND-006 uniquely requires one owner for each of 1681 frozen claims, complete root reachability, and all 28437 directed dependency edges, with lawful extension adding rather than rewriting ownership.

Trace to the foundational theorem. The claim registration names no axioms and no free parameters. It depends only on these already admitted receipt identities:

- `SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012`
- `SFT-MAT-ENGINEERING-OWNERSHIP-HANDOFF-001`
- `SFT-MAT-BIOLOGY-OWNERSHIP-HANDOFF-002`
- `SFT-MAT-MEDICINE-OWNERSHIP-HANDOFF-003`
- `SFT-MAT-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004`
- `SFT-MAT-ASTRONOMY-OWNERSHIP-HANDOFF-005`

The complete owner/dependency graph in `census/materials_hand_001_006_dependency_registry_v1.json` independently proves this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Complete candidate grammar. Generate the complete eight-axis HAND-006 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-006 owner identities, directed relations, paired receipt records, controls and extension boundary.

Shared claim-record clause `MAT-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- `complete-owned-coordinate`
- `exactly-one-owner`
- `complete-one-owner-root-traced-consumer-graph`
- `directed-consumer-edge`
- `complete-paired-receipt-records`
- `value-free-registry-before-outcomes`
- `duplicate-missing-or-tampered-halts`
- `dated-complete-extension-open`

Unique survivor: complete-owned-coordinate__exactly-one-owner__complete-one-owner-root-traced-consumer-graph__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:38485f877dcda924e794fbdd34f3090d724467c9775a54b8e702e76daeb75b1c; generality

sha256:0ee5e96974699f2b1b8456271adb394e33375478c6f1d6247368c4011ef4bb75.

Shared claim-record clause MAT-S010 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject the form missing the first structural preservation; observed: reject the form missing the first structural preservation; receipt
sha256:9d10ce9cd51e8a27d3709d932e9d4fc85c2ce53b90f9e5a5b1783bca1a3fc841.
- tampered_source - passed; expected: reject a changed source identity; observed: reject a changed source identity; receipt
sha256:15f09720212512afe6d6913df1bd37bcaac8638426e079fa0cff40ca2b6da6a7.
- tampered_artifact - passed; expected: reject a missing, duplicate or additional survivor; observed: reject a missing, duplicate or additional survivor; receipt
sha256:e92484bb76354eadf991854f2f16556de9f568d6baf5cc14cfccf288b4f5b3c7.
- boundary - passed; expected: reject forbidden values, target feedback and extra laws; observed: reject forbidden values, target feedback and extra laws; receipt
sha256:dbfd8f7705bc4133868cf8c3ec1496e6e785e3a31585918326b03cf770c7d0ad.

Independent reconstruction and engine admission. Independent implementation

sha256:ce514b15f60d0a05712e120e17fc5eda23fbe18687c2c1ad30352dc5042431a5 regenerated the declared census and sole survivor; independent certificate

sha256:537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f. Derivation seal

sha256:37a22b384787da70130269616f39e8854af1218d223c951ae1b27d1efe5fab0b. Source manifest

sha256:7dced39cd8d2507bd7df089fdaccd2be4d766be02de6b5181a8c579e2b090f99. Engine receipt

sha256:7ca27841a01815e2ff3402430fcbb3c11115962de21e7c1d07d27005a60a5d7e at receipts/engine/model_admitted/SFT-MAT-CROSS-BRANCH-ONE-OWNER-COMPLETENESS-006-7ca27841a01815e2.j

son. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-seal external comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- NIST-REPRODUCIBLE-PROCESS-MONITORING
- CISA-SECURE-BY-DESIGN-001
- NIST-BIOMATERIAL-MEASUREMENTS
- SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1
- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010
- SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE
- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE
- NIST-CONTROLLED-PROTEIN-RELEASE
- MED-ICH-E6R3-001
- MED-ICH-E9R1-001
- MED-FDA-EXPOSURE-RESPONSE-001
- MED-FDA-BIOMARKER-001

- NIST-CIRCULAR-END-OF-USE
- EPA-AIRDATA-001
- WQP-WEB-SERVICES-001
- USGS-LANDSAT-ACCESS-001
- NEON-OPEN-DATA-001
- NIST-RADIATION-DAMAGE-MEASUREMENT
- ESA-GAIA-ARCHIVE-001

Measured, observed, adverse and boundary records:

- HAND-006: value-free owner registry preceded outcome access
- frozen graph: 1681 claims, 28437 dependency edges, 19852 cross-branch uses
- paired external surface: 15 records, 28 source identities, 54 evidence lines
- all selected evidence rows preserved; omission control rejected True
- complete owner graph: 1681 unique owners; 28437 dependency edges; 19852 cross-branch edges; all root-traced

Falsification condition: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt

sha256:29e544c72ad194d9d7badcc114909f03e6df7dc0dc1519afe49d718d41301090; empirical validation
 sha256:600ccdb64d3a2067df886f3a92f3e8dded86a030a031e9b762952ef87e61d32; external validation
 sha256:b05af999b2245937c1b98420702f89b33bfabf81bc9bd198546e2b2e0687537e; isolation
 sha256:cc72df20d2875ee60b5097f65032dc6e6745d091e53266728e8770f3f4b376cc; custody
 sha256:d33a1ea33ccc5af8c70638ce9ef216a5d62c3df4472a114a22620e44678b6437.

Prohibited inputs.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Reproducibility, falsification and extension boundary

The one-command repository validation remains the complete public route. This paper update additionally requires the 289/289 reconciliation identity

sha256:4d2a8f47c39d945fbe31a8de2c78d2dc66462db7c4773f1a785b8b139d2c148b, exact replay of the final HAND family, focused family tests, repository validation and both immutable authority seals. The final repository-wide heavy verification remains reserved for the final cross-branch Grand Lock; it is not redundantly rerun for this local paper preparation.

The Materials branch is complete to its dated current standard, not permanently locked. A lawful new observation may falsify an existing correspondence or create a new obligation. It must be preregistered, enumerated, independently reconstructed, passed through the same untouched engine, externally compared where possible and added by versioned extension. Neither popularity nor credentials can close or suppress it.

Conclusion

Materials Science is complete to its frozen 29 July 2026 census: **289/289 obligations, 73,984 candidate decisions, 289 unique survivors, 1,156 passed adverse controls, 289 implementation-distinct reconstructions and 289 receipt-bound empirical packages**. The result supplies a public, exact and machine-auditable path from the premise-free Fold theorem to each separately admitted law and its declared external evidence boundary.

The branch resolves a central conflation: a material law is not an attractive specimen, vendor value or context-free bulk number. Composition, phase, microstructure, preparation, direction, scale, method, condition and uncertainty remain attached to the observation that produced them. Exact structural classifications remain distinct from specimen-bound measured magnitudes, while mechanics, transport, response, processing, degradation and lifecycle relations retain their own carriers and falsifiers.

The Empirical Grand Lock preserves the complete pre-validation evidence surface, including favourable, adverse, absent, unavailable, unresolved and out-of-bound rows; it does not convert formal closure or test success into universal empirical confirmation. Historical records remain visible even when later evidence changes a current status. The one-owner handoff keeps biological function, clinical outcome, environmental fate and application engineering with their downstream branches.

The paper does not ask the reader to trust Maria Smith, Ernos Labs, an institution or an AI system. It exposes the grammars, exclusions, candidate censuses, one-for-one decisions, controls, source custody, receipts and falsifiers needed to reproduce or invalidate every claim. This is dated completion of the registered project-version investigation-not the end of Materials Science or science-and it remains open to lawful correction, falsification and discovery.

Publication and repository links

- Canonical repository: <https://github.com/MettaMazza/ernos-labs-sft-platform>
- Author: Maria Smith, Ernos Labs
- Contact: Maria.Smith.Sftoe@gmail.com
- Submissions and open review: <https://discord.gg/ucwGryVxGr>
- Paper license: CC BY 4.0; code license: Apache-2.0; authorship and creative rights retained.

Current authoritative statement and receipt reconciliation

The current model-admitted census is authoritative at claim level. The entries below expose exact current statement or receipt strings that the earlier successor prose omitted or paraphrased. Historical wording remains preserved elsewhere in the paper; this reconciliation does not retroactively rewrite a prediction, chronology, evidence class, result or receipt.

SFT-MAT-MEAS-MATERIAL-001

Earlier display gap: receipt hash.

Current exact statement: A material is the least complete constituent-and-relation carrier whose composition, bonding organization and declared observation boundary remain distinguishable.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:

`sha256:a603da455a0d64fb7ad615c083026bf63365752ae60f5ab151010705873f7a44`.

Receipt path:

`receipts/engine/model_admitted/SFT-MAT-MEAS-MATERIAL-001-a603da455a0d64fb.json`.

SFT-MAT-MEAS-SPECIMEN-001

Earlier display gap: receipt hash.

Current exact statement: A specimen is a source-bound finite part of a material with an explicit sampling map, retained preparation history and declared representativeness boundary.

Formal closure status: `depth_independent`.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:be89f08f9db698b344dbd6a7bd319ff4302aac70e120cc1531d07aba9e67cd7c.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MEAS-SPECIMEN-001-be89f08f9db698b3.json.

SFT-MAT-MEAS-COMPOSITION-001

Earlier display gap: receipt hash.

Current exact statement: Material composition is the exact part-to-whole record of every retained component on one declared amount or support basis.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:12f436cc0234aa79c73e3e13c80607ee40d2765efedab4566e0e1d3af6e789e3.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MEAS-COMPOSITION-001-12f436cc0234aa79.json.

SFT-MAT-MEAS-PHASE-001

Earlier display gap: receipt hash.

Current exact statement: A material phase is a maximal connected support whose constituents share one stable recurrence and symmetry observation class at declared conditions.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:89333be757443d46a610ebec1d82daecb409df0bd61811ebdfa9868f3fb0fc49.

Receipt path: receipts/engine/model_admitted/SFT-MAT-MEAS-PHASE-001-89333be757443d46.json.

SFT-MAT-MEAS-MICROSTRUCTURE-001

Earlier display gap: receipt hash.

Current exact statement: Microstructure is the complete declared-scale arrangement of phases, grains, interfaces and defects between constituent and bulk observation levels.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:41d5b521ac071b8c9388cc3fff563dc6c0b8e686e3a062f0657520cfd9d86352.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MEAS-MICROSTRUCTURE-001-41d5b521ac071b8c.json.

SFT-MAT-MEAS-PROPERTY-001

Earlier display gap: receipt hash.

Current exact statement: A material property is a reproducible relation between a declared material state and stimulus/response records, conditional on direction, scale, history and method.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:bf49f2ad1f00be8833e35eae3aad141cd5213b11094b232ae4fb632a3bad6535.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MEAS-PROPERTY-001-bf49f2ad1f00be88.json.

SFT-MAT-MEAS-TRACEABILITY-001

Earlier display gap: receipt hash.

Current exact statement: A materials measurement is admissible only with a complete result, uncertainty, method, specimen, condition, calibration and unbroken reference chain.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:da14ebc7c6205220ec0993a22cbb9b0fb41777bc23e4a17ffcd978c4b8f8ce53.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MEAS-TRACEABILITY-001-da14ebc7c6205220.json.

SFT-MAT-CRYST-LATTICE-001

Earlier display gap: receipt hash.

Current exact statement: A lattice is a completely generated site-and-adjacency network whose local word recurs under retained translations.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:060664b8cdcb7d7ab34749ac5f04012cc025b05d2fce3ed47cf5e011b39a4da1.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CRYST-LATTICE-001-060664b8cdcb7d7a.json.

SFT-MAT-CRYST-UNIT-CELL-001

Earlier display gap: receipt hash.

Current exact statement: A primitive unit cell is the least complete labelled cell whose generated translations reproduce every lattice site and relation once.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:7b95e56aaa1c2c51c79a84cfea8693aa695be4eb0f8fa0c9b948113547ecd68a.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CRYST-UNIT-CELL-001-7b95e56aaa1c2c51.json.

SFT-MAT-CRYST-TRANSLATION-001

Earlier display gap: receipt hash.

Current exact statement: Crystalline translational order is invariance of the complete local constituent word and adjacency class under a generated integer translation basis.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:be6c8d29c1753a134eb988fa34e523060c1e197246c7c9ca2dd6f835d2659c7f.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CRYST-TRANSLATION-001-be6c8d29c1753a13.json.

SFT-MAT-CRYST-CUBIC-COORDINATION-001

Earlier display gap: receipt hash.

Current exact statement: Three stable spatial generators with two held orientations per generator force six nearest neighbours for the simple-cubic adjacency class.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:b49f3d9cf57768f98f6fe15e0133c4df0cadcf3e7ccfc2a0c315ae31453b5ec0.

Receipt path: receipts/engine/model_admitted/SFT-MAT-CRYST-CUBIC-COORDINATION-001-b49f3d9cf57768f9.json.

SFT-MAT-CRYST-ROTATION-RESTRICTION-001

Earlier display gap: receipt hash.

Current exact statement: A finite rotation preserves a periodic lattice only for generated orders 1, 2, 3, 4 and 6; five is the least excluded positive order.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:a0e401100d13252d5f02d5be754a462b01b737b9b066ec9a1809671a6ac1a252.

Receipt path: receipts/engine/model_admitted/SFT-MAT-CRYST-ROTATION-RESTRICTION-001-a0e401100d13252d.json.

SFT-MAT-CRYST-SYSTEMS-001

Earlier display gap: receipt hash.

Current exact statement: Complete rank-three metric and crystallographic-rotation equivalence partitions periodic crystals into seven crystal-system classes.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:8eea622f2a962a3e3b995693f3ba38acf8c0b79242bd30c1804eff03952d7637.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CRYST-SYSTEMS-001-8eea622f2a962a3e.json.

SFT-MAT-CRYST-BRAVAIS-001

Earlier display gap: receipt hash.

Current exact statement: Complete rank-three translation lattices modulo basis and centering equivalence form fourteen Bravais classes.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:c04968088f5382d095822785a3b505081f9a9027b7a8d206438deafdb6f93a52.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CRYST-BRAVAIS-001-c04968088f5382d0.json.

SFT-MAT-CRYST-RECIPROCAL-001

Earlier display gap: receipt hash.

Current exact statement: Reciprocal support is the exact class of probe path labels whose phase returns coherently over the complete generated material organization; retained classes form diffraction peaks.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:257f4aeb601ec2c9cd2681faed0643db4d63f85dedc6ebdeba8bb87ec8c37f72.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CRYST-RECIPROCAL-001-257f4aeb601ec2c9.json.

SFT-MAT-CRYST-QUASICRYSTAL-001

Earlier display gap: receipt hash.

Current exact statement: Quasicrystalline order is long-range coherent recurrence with sharp reciprocal support and no finite translation period; fivefold order is thereby admitted only outside periodic-lattice closure.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:c397a36d1bb97f5bbab246e07e290b327c773ea18ec5c2b074d22794d7d1d094.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CRYST-QUASICRYSTAL-001-c397a36d1bb97f5b.json.

SFT-MAT-CRYST-PHONON-001

Earlier display gap: receipt hash.

Current exact statement: A rank-three lattice has exactly one longitudinal and two transverse displacement orientations, forcing three acoustic excitation branches.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:574d1650178ebf71cddebdd5128c2d2e9837d02ab67443c66d787cfc246e5b6c.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CRYST-PHONON-001-574d1650178ebf71.json.

SFT-MAT-DEFECT-POINT-001

Earlier display gap: receipt hash.

Current exact statement: A point defect is the least localized retained difference between an actual site record and its declared reference-lattice word.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:7fac1bb37f212cd0ea2c2fbff94a0c0e1d48392c90c532ef816fe3fee5de5de5.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-DEFECT-POINT-001-7fac1bb37f212cd0.json.

SFT-MAT-DEFECT-VACANCY-001

Earlier display gap: receipt hash.

Current exact statement: A vacancy is a retained reference site carrying structural occupant absence, never deletion of the site or a numerical-zero material amount.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:ec38624192bccb935013d71151437c8c90b43a6fae881c3ba54527cd41a85132.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-DEFECT-VACANCY-001-ec38624192bccb93.json.

SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001

Earlier display gap: receipt hash.

Current exact statement: Interstitial and substitutional defects are distinguished respectively by an added occupied site relation and a changed occupant label at a retained reference site.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:a0adb50fd640b1941a39bed7dc5e823c0b5a07a979f033e459c25160af154ba0.

Receipt path: receipts/engine/model_admitted/SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001-a0adb50fd640b194.json.

SFT-MAT-DEFECT-DISLOCATION-001

Earlier display gap: receipt hash.

Current exact statement: A dislocation is a line-supported lattice mismatch whose complete circuit retains a nonclosing held translation and supplies a lawful slip path.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:5b63a94c1368233f356123a5c3cf72d1e68f8cc92cbd72c5c1758642a4737821.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-DEFECT-DISLOCATION-001-5b63a94c1368233f.json.

SFT-MAT-MICRO-GRAIN-BOUNDARY-001

Earlier display gap: receipt hash.

Current exact statement: A grain is a maximal connected orientation-coherent crystal region; its boundary is the retained adjacency where two such orientation classes meet.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:97779d2048b92fda5a2968a339e3b2359a2bb5389563cda49f5cdda6aacc404e.

Receipt path: receipts/engine/model_admitted/SFT-MAT-MICRO-GRAIN-BOUNDARY-001-97779d2048b92fda.json.

SFT-MAT-MICRO-INTERFACE-001

Earlier display gap: receipt hash.

Current exact statement: A material interface is the complete boundary relation joining distinct material or phase carriers with explicit structural, content and transfer compatibility records.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:f8e65b2c9530d410a0a1b2dee9a225b829d130c02ccbd7d31dc52be63ad4e738.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MICRO-INTERFACE-001-f8e65b2c9530d410.json.

SFT-MAT-MICRO-DIFFUSION-001

Earlier display gap: receipt hash.

Current exact statement: Diffusion is counted adjacent-site redistribution of retained constituent labels with complete global content conservation and declared support/time recurrence.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:07a61b3ac976a385ccdfb2e48ca22dc8d09c675145423d0501d8b4dea7765500.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MICRO-DIFFUSION-001-07a61b3ac976a385.json.

SFT-MAT-MICRO-NUCLEATION-GROWTH-001

Earlier display gap: receipt hash.

Current exact statement: A nucleus is the least generated cluster whose interior recurrence persists against its complete boundary relation; growth appends compatible cells while retaining phase and transfer provenance.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:b314993b37eafa71cd6f643712590798c093ae7113db07b72d5fae73b9e6262c.

Receipt path: receipts/engine/model_admitted/SFT-MAT-MICRO-NUCLEATION-GROWTH-001-b314993b37eafa71.json.

SFT-MAT-ELEC-BAND-GAP-001

Earlier display gap: receipt hash.

Current exact statement: Electronic bands are complete phase-compatible recurrence classes on a material network; a band gap is a declared interval with no compatible generated state path.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:5fbc05f52167cabe089f6ac5ccb37bc8995972adce2580767b91009ff0d51404.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-ELEC-BAND-GAP-001-5fbc05f52167cabe.json.

SFT-MAT-ELEC-CONDUCTOR-CLASS-001

Earlier display gap: receipt hash.

Current exact statement: Conductor, semiconductor and insulator classes are forced by whether occupied carrier support connects to accessible states directly, across a finite activation gap, or not within the declared conditions.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:0851d0318085f1b7bdf0d9871bf74598b959240df24c4b8714e34f004cad981a.

Receipt path: receipts/engine/model_admitted/SFT-MAT-ELEC-CONDUCTOR-CLASS-001-0851d0318085f1b7.json.

SFT-MAT-ELEC-CARRIER-DUALITY-001

Earlier display gap: receipt hash.

Current exact statement: A binary exclusion-bounded state yields two transport descriptions: retained occupation and its held local absence, corresponding after sealing to electron and hole carriers.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:6f848016b57e0d09eae5c587efe10f4da04c6191319e870a6937958303a14727.

Receipt path: receipts/engine/model_admitted/SFT-MAT-ELEC-CARRIER-DUALITY-001-6f848016b57e0d09.json.

SFT-MAT-ELEC-OCCUPATION-001

Earlier display gap: receipt hash.

Current exact statement: An exclusion-labelled canonical material mode admits structural absence or one retained indistinguishable occupation; a duplicate occupation is rejected.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:a86b53942d003089e455938be92486df3b9ea73991cc106a4ab8c3ab2d446f3a.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-ELEC-OCCUPATION-001-a86b53942d003089.json.

SFT-MAT-SEMI-DOPING-001

Earlier display gap: receipt hash.

Current exact statement: Doping is a source-bound substitution or addition that retains host organization while introducing an exact carrier-support difference.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:c78add3668c0e8d5e00346b4de0be72933ce45fcb86adc5a3d18e2b1320b6f7.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SEMI-DOPING-001-c78add3668c0e8d.json.

SFT-MAT-SEMI-PN-TYPE-001

Earlier display gap: receipt hash.

Current exact statement: The two held orientations of a doped carrier imbalance force occupation-majority and absence-majority semiconductor classes.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:3ada2c3c62563f43e20c3857ca3ec0c9e8a79bfaf11b2387d938eb380900a96a.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SEMI-PN-TYPE-001-3ada2c3c62563f43.json.

SFT-MAT-SEMI-JUNCTION-001

Earlier display gap: receipt hash.

Current exact statement: Joining opposed doped regions forces interfacial carrier transfer until a retained depletion/space-charge organization balances transport and field response at declared conditions.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:ab3ac6d4d72771243c560c5ba63f56c21ecf46425253345f6c8e9a067b33a859.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SEMI-JUNCTION-001-ab3ac6d4d7277124.json.

SFT-MAT-SEMI-TRANSPORT-001

Earlier display gap: receipt hash.

Current exact statement: Semiconductor transport is the complete counted motion of occupied and held-absence carriers through accessible support under declared field, gradient and scattering conditions.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:b53d46b71c00244ede38457846ac2592276922367ced522052f68e98c03c3e18.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SEMI-TRANSPORT-001-b53d46b71c00244e.json.

SFT-MAT-SEMI-OPTICAL-001

Earlier display gap: receipt hash.

Current exact statement: An interband optical response exists only when a probe supplies a complete compatible transition between retained occupied and accessible band states under the material selection rules.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:c4341f21a28539567752ba56fae9c96581b507d5a41289a58c0996693d67d31c.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SEMI-OPTICAL-001-c4341f21a2853956.json.

SFT-MAT-SC-PAIR-001

Earlier display gap: receipt hash.

Current exact statement: The least exclusion-compatible collective carrier joins two opposed fermionic labels into one integer-composite recurrence class.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:37c8eddac13dfd4d0f0110117c6e85695a40f14bdbeb94ddfb21ccbc0c16d7b4.

Receipt path: receipts/engine/model_admitted/SFT-MAT-SC-PAIR-001-37c8eddac13dfd4d.json.

SFT-MAT-SC-ZERO-RESISTANCE-001

Earlier display gap: receipt hash.

Current exact statement: Superconducting zero-resistance transport is phase-locked pair recurrence on complete connected support with no retained label-compatible dissipative transition.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:eabc8076794eaf93d8139866bb2cb1b68f83fb9eff249cd9d3f785dab6c7f790.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SC-ZERO-RESISTANCE-001-eabc8076794eaf93.json.

SFT-MAT-SC-MEISSNER-001

Earlier display gap: receipt hash.

Current exact statement: A phase-locked charged carrier network generates a boundary response whose complete bulk recurrence closes an incompatible applied magnetic distinction, yielding flux exclusion apart from the declared penetration boundary.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:958f05ce5f5a809ccd23e229b67fee3c18dc732d43178f85843de9fc89ad4b01.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SC-MEISSNER-001-958f05ce5f5a809c.json.

SFT-MAT-SC-FLUX-QUANTIZATION-001

Earlier display gap: receipt hash.

Current exact statement: Single-valued return of a paired-carrier phase around a closed path forces whole winding classes and therefore discrete flux support.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:0ed97a6d6c7de89689c4aed43a9408b27fd7bddd9398b7efd9f9a0525499de4e.

Receipt path: receipts/engine/model_admitted/SFT-MAT-SC-FLUX-QUANTIZATION-001-0ed97a6d6c7de896.json.

SFT-MAT-SC-JOSEPHSON-001

Earlier display gap: receipt hash.

Current exact statement: A weak link between phase-locked pair supports carries a coherent transfer recurrence determined by their retained phase distinction without dissolving pair identity.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:aa8fb0a93ee97c5e9aa696f455108f5770b1882a050c6927f250d7a30b144426.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SC-JOSEPHSON-001-aa8fb0a93ee97c5e.json.

SFT-MAT-SF-SUPERFLUID-001

Earlier display gap: receipt hash.

Current exact statement: Superfluid flow is shared-phase recurrence of a neutral collective carrier when the complete accessible transition census contains no allowed dissipative path below its first retained excitation boundary.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:13bca5d4d8eefdaaae89f16793de97f2da0956b15f3c9090b7dd19a9864f7c4f.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SF-SUPERFLUID-001-13bca5d4d8eefdaa.json.

SFT-MAT-SF-CIRCULATION-001

Earlier display gap: receipt hash.

Current exact statement: Single-valued return of a superfluid phase around a closed path forces whole winding and discrete circulation classes with a retained vortex-core boundary.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:ea513957e5e43cc1e01e7a2187a08a7f2bfbd4d9a0d6561c8902174b3eecd6b9.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SF-CIRCULATION-001-ea513957e5e43cc1.json.

SFT-MAT-TOPO-INVARIANT-001

Earlier display gap: receipt hash.

Current exact statement: A material invariant is topological when its complete global path class is unchanged by every generated local deformation that retains connectivity and boundary records.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:fc5c012cceb34acee20825a1baf86f0c979fe88be20c0b06dae65074dca33361.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-TOPO-INVARIANT-001-fc5c012cceb34ace.json.

SFT-MAT-TOPO-BULK-BOUNDARY-001

Earlier display gap: receipt hash.

Current exact statement: A nontrivial bulk path class forces a boundary recurrence whose removal would require closing the retained bulk distinction; compatible local backscattering paths are therefore absent within the declared gap.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:d2a5beae9e015469d9a6ba517078466f3918892548d064025b696a2df8421dbc.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-TOPO-BULK-BOUNDARY-001-d2a5beae9e015469.json.

SFT-MAT-MECH-STRESS-STRAIN-001

Earlier display gap: receipt hash.

Current exact statement: Stress is oriented load transfer per generated boundary support; strain is the exact retained change of internal separation relative to the same reference material state.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:0445851d4b39830df5906e9cd54ca1b9c33557970627302ef3b5f1261d64e3e4.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MECH-STRESS-STRAIN-001-0445851d4b39830d.json.

SFT-MAT-MECH-ELASTICITY-001

Earlier display gap: receipt hash.

Current exact statement: Elastic response is a material transition path whose load release returns every retained constituent and adjacency relation to the original canonical state.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:b763df84f89c26c7f0250b07a2b29a053ece891b3635c136a6a66373fc9936ff.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MECH-ELASTICITY-001-b763df84f89c26c7.json.

SFT-MAT-MECH-PLASTICITY-001

Earlier display gap: receipt hash.

Current exact statement: Plastic response retains material content and phase identity while a defect-mediated transition changes the canonical adjacency organization after load release.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:acd1392cca2311cdf999b68957c53230344b1c9b8e31591ebc36802a08cecf2c.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MECH-PLASTICITY-001-acd1392cca2311cd.json.

SFT-MAT-MECH-SLIP-001

Earlier display gap: receipt hash.

Current exact statement: Slip is a counted whole-lattice translation propagated by a retained dislocation along a compatible plane/direction without requiring simultaneous rupture of the complete plane.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:b0bc6c6a3d4d3c3c8f16233b9d41b940ef1302f02c00ff2bb6f4c515e76a092d.

Receipt path: receipts/engine/model_admitted/SFT-MAT-MECH-SLIP-001-b0bc6c6a3d4d3c3c.json.

SFT-MAT-MECH-MODULUS-001

Earlier display gap: receipt hash.

Current exact statement: An elastic modulus is the exact stress/strain response relation for one declared material direction, condition, reference state and reversible response window.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:5f7a55c339d12573cad658697b86320f8d447eaa850d2c134ad20b95ef66a30b.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MECH-MODULUS-001-5f7a55c339d12573.json.

SFT-MAT-MECH-STRENGTH-HARDNESS-001

Earlier display gap: receipt hash.

Current exact statement: Strength and hardness are method-bounded thresholds at which a declared loading path first produces retained plastic, indentation or failure organization.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:4de60010c08d9fed3acf8c4728f2783ce2aeb1ef777f628f85195134380de79a.

Receipt path: receipts/engine/model_admitted/SFT-MAT-MECH-STRENGTH-HARDNESS-001-4de60010c08d9fed.json.

SFT-MAT-MECH-FRACTURE-001

Earlier display gap: receipt hash.

Current exact statement: Fracture is connected bond-path separation creating new retained boundary support; toughness is the declared resistance relation for crack initiation or advance with a complete work ledger.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:d47271030cd2fb14b7c385fec0736f231ee8a1dc4bed34ad8afd00a92bc262d5.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MECH-FRACTURE-001-d47271030cd2fb14.json.

SFT-MAT-MECH-FATIGUE-CREEP-001

Earlier display gap: receipt hash.

Current exact statement: Fatigue and creep are distinct history-dependent material processes: recurrence-counted damage under cyclic load and retained deformation under sustained load and time/temperature conditions.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:b94bc1254ab3d3025628c9435c5bc2cd1afa979be5035aebfb0b17dc1436f4d3.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MECH-FATIGUE-CREEP-001-b94bc1254ab3d302.json.

SFT-MAT-THERM-HEAT-CAPACITY-001

Earlier display gap: receipt hash.

Current exact statement: Material heat capacity is the exact relation between supplied thermal carrier change and the resulting accessible excitation-support change at declared conditions.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:6674f76f1ca6440a25798bd45bedccd1ea7e7b5df1bffe9674312f9fd0f409b5.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-THERM-HEAT-CAPACITY-001-6674f76f1ca6440a.json.

SFT-MAT-THERM-CONDUCTION-001

Earlier display gap: receipt hash.

Current exact statement: Thermal conduction is counted adjacent transfer of retained energy labels through connected material support with complete source, boundary and scattering provenance.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:902681fe9ffce6915387d8985f4b5cf95bc767a56094aa0b0b7f44a341150562.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-THERM-CONDUCTION-001-902681fe9ffce691.json.

SFT-MAT-THERM-EXPANSION-001

Earlier display gap: receipt hash.

Current exact statement: Thermal expansion is the condition-bounded change of recurrent constituent separation with excitation while material identity and complete adjacency provenance remain retained.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:5e9ca77f6756053d098868e4e031d724f44e4c42256357d74fa33a4d4567ac7d.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-THERM-EXPANSION-001-5e9ca77f6756053d.json.

SFT-MAT-MAG-FERROMAGNETISM-001

Earlier display gap: receipt hash.

Current exact statement: Ferromagnetism is connected recurrence of aligned local held-moment orientations producing a retained macroscopic orientation class.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:3860d6dec3e3056aa3eeb5268587b83a21373587f1e399dea61d3b9e840636cb.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-MAG-FERROMAGNETISM-001-3860d6dec3e3056a.json.

SFT-MAT-MAG-ANTIFERROMAGNETISM-001

Earlier display gap: receipt hash.

Current exact statement: Antiferromagnetism is stable opposed recurrence on distinguishable sublattices that retains local order while the complete macroscopic moment distinction closes.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:042d5e6093d1ae270db43da2ccb72f5299318e53632719ca3714ba648561dbbe.

Receipt path: receipts/engine/model_admitted/SFT-MAT-MAG-ANTIFERROMAGNETISM-001-042d5e6093d1ae27.json.

SFT-MAT-DIEL-POLARIZATION-001

Earlier display gap: receipt hash.

Current exact statement: Dielectric polarization is a retained field-conditioned redistribution or orientation of bound held-charge labels with complete conservation and relaxation records.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:46b063c128b7f353fb40d4b67a9fbf26f0266fd2cf441b9a103a8663ec260459.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-DIEL-POLARIZATION-001-46b063c128b7f353.json.

SFT-MAT-OPT-REFRACTIVE-001

Earlier display gap: receipt hash.

Current exact statement: Material refractive response is the source-bound phase/delay relation produced by coherent probe coupling to accessible material excitations while propagation, loss and dispersion records remain separate.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:3210b165450c76b959c9d5abf99c9b2487440238426780d044e5a6e810b2507e.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-OPT-REFRACTIVE-001-3210b165450c76b9.json.

SFT-MAT-PHASE-TRANSITION-001

Earlier display gap: receipt hash.

Current exact statement: A material phase transition is a provenance-retaining change between stable macro-recurrence and symmetry classes with explicit conditions, path and hysteresis boundary.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:7657d0f08ff975856f32bc1c00de5bd76507c0880b2ee959818a773e92a45208.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-PHASE-TRANSITION-001-7657d0f08ff97585.json.

SFT-MAT-CLASS-METAL-001

Earlier display gap: receipt hash.

Current exact statement: A metallic material is a connected atomic organization with delocalized carrier access across multiple constituent cells and no declared transport gap at its operating conditions.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:d319f9d2f4dc8869f881d394860212b924efe345b36bd8a11615a789ab14b81a.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CLASS-METAL-001-d319f9d2f4dc8869.json.

SFT-MAT-CLASS-CERAMIC-001

Earlier display gap: receipt hash.

Current exact statement: A ceramic is an inorganic nonmetallic material whose bulk carrier is a localized ionic/covalent network with retained phase, porosity and grain/interface organization.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:e985201b890a9c99983f9c3286047ccdf3bfe488da34cad3615f166d585a97cc.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CLASS-CERAMIC-001-e985201b890a9c99.json.

SFT-MAT-CLASS-POLYMER-001

Earlier display gap: receipt hash.

Current exact statement: A polymeric material is a population of covalently recurrent chain carriers with explicit repeat identity, sequence, length/branching distribution and interchain organization.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:afbfe9b832351be2b9532eb36f6b8d9b41a5423cf9d1bdfe3b32d9d94d910175.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CLASS-POLYMER-001-afbfe9b832351be2.json.

SFT-MAT-CLASS-COMPOSITE-001

Earlier display gap: receipt hash.

Current exact statement: A composite is a joint material whose distinguishable constituent carriers persist while interfaces and architecture compose their load, transport or barrier responses.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:e741199f029ca83879c34656b13c006bc8e54a12637e59a04a32d8f5c738b8d4.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CLASS-COMPOSITE-001-e741199f029ca838.json.

SFT-MAT-CLASS-GLASS-001

Earlier display gap: receipt hash.

Current exact statement: A glass is a connected material with retained local structural recurrence but no crystalline translation class, whose macrostate depends on preparation and observation-time relaxation.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:7a91b06f6cb50235c28be8058235baaa5b839f577f67b30a1ae3cd2b1e8042c2.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CLASS-GLASS-001-7a91b06f6cb50235.json.

SFT-MAT-CLASS-POROUS-001

Earlier display gap: receipt hash.

Current exact statement: A porous material is a joint solid/structural-absence support whose pore connectivity, interface and accessibility are retained at a declared resolution.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:7e6e7e7efc51d3e417adb6974dd16f8ec17d9d747971cd2456f92cc019ce0054.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-CLASS-POROUS-001-7e6e7e7efc51d3e4.json.

SFT-MAT-BULK-CORRESPONDENCE-001

Earlier display gap: receipt hash.

Current exact statement: A bulk material law is admitted only when the complete constituent/interaction network composes to a scale-stable observation while fluctuations, interfaces and preparation history remain explicitly bounded.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:d451b17d73ba7d892d5e79c947ca8b0979ee3e7806a0635afaa3285427ef29a5.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-BULK-CORRESPONDENCE-001-d451b17d73ba7d89.json.

SFT-MAT-BULK-ANISOTROPY-001

Earlier display gap: receipt hash.

Current exact statement: Anisotropy is inequality of response relations across retained material directions after specimen, condition and method identities are held fixed.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:4118df713bc0e9e8e34ac4bac952f7e9f774f5207bd40efcc4ed5adcb6524981.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-BULK-ANISOTROPY-001-4118df713bc0e9e8.json.

SFT-MAT-BULK-SIZE-SURFACE-001

Earlier display gap: receipt hash.

Current exact statement: A material size effect occurs when the exact boundary/interior support relation changes the accessible state, transfer or mechanical census; it is never a free scale correction.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:d3e923caf1c9a615cd149a43345002c9f5695b0b1573e4c82d834400a43fb09a.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-BULK-SIZE-SURFACE-001-d3e923caf1c9a615.json.

SFT-MAT-PROC-PATH-001

Earlier display gap: receipt hash.

Current exact statement: A materials process is a complete ordered path from input carrier through retained structural states to property relations; endpoint labels alone cannot erase the path.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:84b881964289b6369ff0a175b1415e110962c47669982eece1ebc77d7cbde690.

Receipt path: receipts/engine/model_admitted/SFT-MAT-PROC-PATH-001-84b881964289b636.json.

SFT-MAT-PROC-PHASE-DIAGRAM-001

Earlier display gap: receipt hash.

Current exact statement: A phase map partitions generated composition/condition support by the stable material recurrence class, retaining coexistence, fraction and transition boundaries without fitted extrapolation beyond its domain.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:10b42f4ba45befaa3cc9a5d8c854432e729109908f9feb9e1a3ef541f0a390bd.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-PROC-PHASE-DIAGRAM-001-10b42f4ba45befaa.json.

SFT-MAT-PROC-SOLIDIFICATION-001

Earlier display gap: receipt hash.

Current exact statement: Solidification is the ordered nucleation and growth path from mobile to position-retaining organization with complete energy, constituent rejection and interface provenance.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:47e7cf392984fba049ecd86ab40b7df92994589f140f31e58e4dfaebdb19ab7.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-PROC-SOLIDIFICATION-001-47e7cf392984fba0.json.

SFT-MAT-PROC-SINTERING-001

Earlier display gap: receipt hash.

Current exact statement: Sintering is interface-mediated constituent transport that joins particles and reorganizes structural absence while conserving material content and retaining the processing path.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:579b77a0e4d157c4027f3b362e50b0947b8ed266da098d1a86f1a7dfa3185c0b.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-PROC-SINTERING-001-579b77a0e4d157c4.json.

SFT-MAT-PROC-HEAT-TREATMENT-001

Earlier display gap: receipt hash.

Current exact statement: Heat treatment is a declared thermal/time path whose phase, defect and microstructure transitions are retained and whose property consequence is path-bound.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:173f913f1c66a6f72e38a5a5255fa572de13c997acd87bbda1ec5dc1495eb889.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-PROC-HEAT-TREATMENT-001-173f913f1c66a6f7.json.

SFT-MAT-DEGR-CORROSION-001

Earlier display gap: receipt hash.

Current exact statement: Corrosion is an interfacial reaction/transport process that transforms a material and environment jointly while retaining complete constituent, charge, product and boundary provenance.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:4542786b0ea6848bf34524495a0a40b678800eecd53efb77249282203f747945.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-DEGR-CORROSION-001-4542786b0ea6848b.json.

SFT-MAT-DEGR-WEAR-001

Earlier display gap: receipt hash.

Current exact statement: Wear is retained material transfer or surface reorganization produced by a declared contact/load/motion path; frictional response and material loss remain separately recorded.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:c59b96e47896844ac1b5ac4a3b204133efbb551e57bc8b3a48d26793af6577fb.

Receipt path: receipts/engine/model_admitted/SFT-MAT-DEGR-WEAR-001-c59b96e47896844a.json.

SFT-MAT-DEGR-RADIATION-001

Earlier display gap: receipt hash.

Current exact statement: Radiation damage is the complete source-bound path from probe energy transfer to retained displacement, excitation or transmutation defects and their subsequent migration/recovery.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:dbf66ce45a26ae89f38fdc829f0ecae959d30aff5c58d2d62890771ef2c84c7.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-DEGR-RADIATION-001-dbf66ce45a26ae89.json.

SFT-MAT-FUNC-PIEZOELECTRIC-001

Earlier display gap: receipt hash.

Current exact statement: Piezoelectricity is a reversible cross-relation between mechanical deformation and held electric polarization permitted only when the retained material symmetry does not close the orientation distinction.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:e334945fb5c5da1183b9dc19935a472a15ea0095fcfabf8cced2c319375c3bf9.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-FUNC-PIEZOELECTRIC-001-e334945fb5c5da11.json.

SFT-MAT-FUNC-FERROELECTRIC-001

Earlier display gap: receipt hash.

Current exact statement: Ferroelectricity is a material phase with stable opposed polarization recurrence classes and a retained field-driven switching path between them.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:668f7c77951470dd63ae4753fd69bd1d9806e623bbd8cd663f847c86ce7fe460.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-FUNC-FERROELECTRIC-001-668f7c77951470dd.json.

SFT-MAT-FUNC-THERMOELECTRIC-001

Earlier display gap: receipt hash.

Current exact statement: Thermoelectric response is the joint conserved relation between energy-gradient transport and held-charge transport on the same material network.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:a8cd3f6b52008aa2c5e86e4810d8312b4cd9071411cbb3015ce3240076d5405c.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-FUNC-THERMOELECTRIC-001-a8cd3f6b52008aa2.json.

SFT-MAT-FUNC-PHOTONIC-001

Earlier display gap: receipt hash.

Current exact statement: A photonic material organizes electromagnetic recurrence support so that allowed propagation classes and structural gaps follow from complete generated geometry and boundary relations.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:d90e52c8e5b4c5e0a92c05574a8fa2068d8b1bc5da6caa6ba76957c45c675a3a.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-FUNC-PHOTONIC-001-d90e52c8e5b4c5e0.json.

SFT-MAT-FUNC-MAGNETIC-001

Earlier display gap: receipt hash.

Current exact statement: A functional magnetic response is the complete relation among applied field, domain transition path, retained orientation, loss and remanence at declared frequency and temperature.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:c01cba9d83be4efcc037953fc82cc5162a40536f07d881413ad9ca9ada247e38.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-FUNC-MAGNETIC-001-c01cba9d83be4efc.json.

SFT-MAT-FUNC-NANOMATERIAL-001

Earlier display gap: receipt hash.

Current exact statement: A nanomaterial regime begins when boundary and finite-support distinctions remain in the material state/property census and cannot be closed by the bulk observation quotient.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:55317cf042b6531a6fe30ee1532ec73c758b2679f7aab742de64148ac5969b86.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-FUNC-NANOMATERIAL-001-55317cf042b6531a.json.

SFT-MAT-FUNC-BIOMATERIAL-001

Earlier display gap: receipt hash.

Current exact statement: A biomaterial claim is a joint material/biological interface relation with explicit exposure, transfer, response and function boundaries; biological outcomes cannot select the material law.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:99ee0945ba72d497738307051dbc5f9aa26a1f305b8f589e3dce5ccce07b2501.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-FUNC-BIOMATERIAL-001-99ee0945ba72d497.json.

SFT-MAT-SUST-LIFECYCLE-001

Earlier display gap: receipt hash.

Current exact statement: A material lifecycle is the complete source-to-processing-to-use-to-recovery flow ledger; recyclability requires an explicit identity-preserving recovery path and records every transformation, loss and quality boundary.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:be4a9a15df6225f1099d4f7d6c00291b3b6fd3dfd881617d9b565c3b34d16b32.

Receipt path:

receipts/engine/model_admitted/SFT-MAT-SUST-LIFECYCLE-001-be4a9a15df6225f1.json.

SFT-MAT-VALIDATION-CRYSTALLOGRAPHY-DIFFRACTION-VECTOR-001

Earlier display gap: exact statement.

Current exact statement: The complete frozen crystallography, reciprocal structure, diffraction and disorder vector retains every one of its 8 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:4ad9b7e3ae07359ffedfa94f3fb5489dee448ef95b63fb028df48f1097aa3428.

Receipt path: receipts/engine/model_admitted/SFT-MAT-VALIDATION-CRYSTALLOGRAPHY-DIFFRACTION-VECTOR-001-4ad9b7e3ae07359f.json.

SFT-MAT-VALIDATION-DEFECT-MICROSTRUCTURE-VECTOR-002

Earlier display gap: exact statement.

Current exact statement: The complete frozen defects, microstructure, interfaces and multiscale organization vector retains every one of its 9 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:07c5e25cffdc0077d4efbd0652068b1890a861cbffcfabc9826553b97d443dd3.

Receipt path: receipts/engine/model_admitted/SFT-MAT-VALIDATION-DEFECT-MICROSTRUCTURE-VECTOR-002-07c5e25cffdc0077.json.

SFT-MAT-VALIDATION-PHASE-TRANSFORMATION-VECTOR-003

Earlier display gap: exact statement.

Current exact statement: The complete frozen phase equilibria, transitions, kinetics and metastability vector retains every one of its 10 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:

`sha256:064539aa81cfee8f3d9a96eed1c554a7da11049ff07c78a34ac0bc42dba14c13`.

Receipt path: `receipts/engine/model_admitted/SFT-MAT-VALIDATION-PHASE-TRANSFORMATION-VECTOR-003-064539aa81cfee8f.json`.

SFT-MAT-VALIDATION-MECHANICAL-TRIBOLOGICAL-VECTOR-004

Earlier display gap: exact statement.

Current exact statement: The complete frozen mechanics, fracture, fatigue, creep, tribology and rheology vector retains every one of its 14 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:

`sha256:2cecd395c1dafel1da88a1280beebbb9046503570b4dc994ccd98e7af625f7eca1`.

Receipt path: `receipts/engine/model_admitted/SFT-MAT-VALIDATION-MECHANICAL-TRIBOLOGICAL-VECTOR-004-2cecd395c1dafel1d.json`.

SFT-MAT-VALIDATION-THERMAL-TRANSPORT-VECTOR-005

Earlier display gap: exact statement.

Current exact statement: The complete frozen thermal transport, capacity, expansion and shock response vector retains every one of its 7 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:

`sha256:f4f3ec66e67611e9c08643a7d846ac79c47b8964688ae2ab5ba05317a73c15f9`.

Receipt path: `receipts/engine/model_admitted/SFT-MAT-VALIDATION-THERMAL-TRANSPORT-VECTOR-005-f4f3ec66e67611e9.json`.

SFT-MAT-VALIDATION-ELECTRONIC-IONIC-DIELECTRIC-VECTOR-006

Earlier display gap: exact statement.

Current exact statement: The complete frozen electronic, ionic, dielectric and semiconductor transport vector retains every one of its 12 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:

`sha256:857f130a81d18da493390bd38c96dcc82521f45f01dd594259b36c9f2157fb1f`.

Receipt path: `receipts/engine/model_admitted/SFT-MAT-VALIDATION-ELECTRONIC-IONIC-DIELECTRIC-VECTOR-006-857f130a81d18da4.json`.

SFT-MAT-VALIDATION-MAGNETIC-SUPERCONDUCTING-TOPOLOGICAL-VECTOR-007

Earlier display gap: exact statement.

Current exact statement: The complete frozen magnetism, spin organization, superconductivity, superfluidity and topology vector retains every one of its 12 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:

`sha256:136971745a3094bd8b155e164990a41b172b99d33bb7217ea211be34cdc06765`.

Receipt path: `receipts/engine/model_admitted/SFT-MAT-VALIDATION-MAGNETIC-SUPERCONDUCTING-TOPOLOGICAL-VECTOR-007-136971745a3094bd.json`.

SFT-MAT-VALIDATION-OPTICAL-PHOTONIC-VECTOR-008

Earlier display gap: exact statement.

Current exact statement: The complete frozen optics, photonics, polarization and exciton response vector retains every one of its 10 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:

`sha256:20d2650407f790688fc8c8deae94315a17deb8235834956653bf5e970a755bbb`.

Receipt path: `receipts/engine/model_admitted/SFT-MAT-VALIDATION-OPTICAL-PHOTONIC-VECTOR-008-20d2650407f79068.json`.

SFT-MAT-VALIDATION-MATERIAL-CLASS-PROCESSING-VECTOR-009

Earlier display gap: exact statement.

Current exact statement: The complete frozen material classes, composites and complete processing paths vector retains every one of its 22 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:

`sha256:f8fd9d1cd1bcf8cbefc282a747f11f0fdf688f47aa9a0e743b14d5ffda451b35`.

Receipt path: `receipts/engine/model_admitted/SFT-MAT-VALIDATION-MATERIAL-CLASS-PROCESSING-VECTOR-009-f8fd9d1cd1bcf8cb.json`.

SFT-MAT-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010

Earlier display gap: exact statement.

Current exact statement: The complete frozen independently sourced and cross-source Materials correspondence vector retains every one of its 62 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:

sha256:5fdf955b9220bffffd89fe0fce5754df31ff5b7db027b18c6c520233b4c5cc1c7.

Receipt path: receipts/engine/model_admitted/SFT-MAT-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010-5fdf955b9220bfff.json.

SFT-MAT-VALIDATION-ADVERSE-ABSENT-OUT-OF-BOUND-VECTOR-011

Earlier display gap: exact statement.

Current exact statement: The complete frozen adverse, absent, unavailable, unresolved, tampered and scope-boundary evidence vector retains every one of its 271 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:6c3a8fb1a8b0444d37a25fbb55244b4c5fb9130e991f8eb98d0cedf46e17c070.

Receipt path: receipts/engine/model_admitted/SFT-MAT-VALIDATION-ADVERSE-ABSENT-OUT-OF-BOUND-VECTOR-011-6c3a8fb1a8b0444d.json.

SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012

Earlier display gap: exact statement.

Current exact statement: The complete frozen the complete pre-VALID Materials empirical surface vector retains every one of its 271 named Materials claim memberships, current receipts, controls, target-custody records, sources, measurements, adverse rows and explicit formal-only boundaries without aggregate scoring or omission.

Formal closure status: depth_independent.

Current external status: empirically_tested_and_independently_replicated.

Model-admitted receipt:

sha256:dbb75828d061c43824bfad454e100d0502d76b5ebc3ee74e9dd03cbcd8967268.

Receipt path: receipts/engine/model_admitted/SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012-dbb75828d061c438.json.

Data and code availability

The open repository is [`MettaMazza/ernos-labs-sft-platform`](#). The live model-admitted index is `census/claims.json`; the Materials Science field boundary is `census/materials_discipline_obligations.json`. Every represented claim has its own package under `claims/<claim-id>/` and its immutable model-admitted receipt under `receipts/engine/model_admitted/`.

The publication evidence map is `publications/successors/materials/evidence_map_v1_3.json` and the candidate manifest is `publications/successors/materials/manifest_v1_3.json`. Registered external records and source captures are retained under `experiments/external_sources/materials/` together with their source identities, transport outcomes and hashes. Files too large for ordinary Git remain checksum-bound and are distributed through the applicable existing Zenodo record rather than omitted or replaced.

Build, render, replay and verification programs are retained under `tools/`, with branch implementations under `sft/`, independent reconstructions under `generated/` and tests under `tests/`. The final approved Markdown, rendered PDF, evidence map, manifest, metadata and checksums must agree before deposit. No successful test is treated as empirical confirmation unless the claim-specific evidence protocol separately authorises that status.

Shared claim-record clauses

The clauses below previously appeared verbatim in multiple claim records. They are preserved once here to reduce mechanical repetition. Each in-text clause reference applies this exact wording at that claim location; no scientific statement, status, condition or qualification has been removed.

MAT-S001 - applied at 8 claim locations

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

MAT-S002 - applied at 84 claim locations

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

MAT-S003 - applied at 92 claim locations

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

MAT-S004 - applied at 84 claim locations

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

MAT-S005 - applied at 87 claim locations

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

MAT-S006 - applied at 13 claim locations

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

MAT-S007 - applied at 8 claim locations

Falsification boundary: The claim fails if any registered carrier, relation, organisation or observation feature is absent, any required external row fails, or any tampered row is accepted.

MAT-S008 - applied at 11 claim locations

Material classes arise from retained constituent and network organisation; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

MAT-S009 - applied at 197 claim locations

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

MAT-S010 - applied at 197 claim locations

What it means. This law retains every carrier, relation, organization, observation condition, record, provenance edge, positive-finite extension and explicit handoff required by its question. It does not turn a selected specimen, fitted magnitude, imported named model or application outcome into a law. Structural absence remains typed empty One; host numerical 0 is only its display notation.

MAT-S011 - applied at 8 claim locations

Falsification condition: Reject if any registered source identity, available fragment, unavailable row, specimen/method/condition boundary, prediction seal, custody identity, candidate, control or independent reconstruction is missing or changed.

MAT-S012 - applied at 9 claim locations

Falsification condition: Reject if any registered source identity, available fragment, unavailable row, failed route, specimen/method/condition boundary, prediction seal, custody identity, candidate, control or independent reconstruction is missing or changed.

MAT-S013 - applied at 10 claim locations

Falsification condition: Reject if any registered source identity, source byte, available fragment, condition boundary, phase distinction, prediction seal, custody identity, candidate, control or independent reconstruction is missing or changed.

MAT-S014 - applied at 152 claim locations

Falsification condition: Reject if any registered source, fragment, condition, path, seal, custody identity, candidate, control or independent reconstruction is missing or changed.

MAT-S015 - applied at 12 claim locations

Falsification condition: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

References

- Bureau International des Poids et Mesures / Joint Committee for Guides in Metrology, International Vocabulary of Metrology.
- National Institute of Standards and Technology records identified in the claim-level source ledgers.
- Smith, Maria. *From Nothing to Fold*. doi:10.5281/zenodo.21515629.
- Smith, Maria. *From Fold to Mathematics*. doi:10.5281/zenodo.21516146.
- Smith, Maria. *From Distinction to Information*. doi:10.5281/zenodo.21516916.
- Smith, Maria. *After Turing: The Fold Machine*. doi:10.5281/zenodo.21518311.
- Smith, Maria. *The Quantum Fold Machine*. doi:10.5281/zenodo.21518313.
- Smith, Maria. *From Fold to Physics*. doi:10.5281/zenodo.21520881.
- Smith, Maria. *From Fold to Chemistry*. doi:10.5281/zenodo.21531455.

Complete current-census successor supplement

The authoritative current scope of this paper is 289 live model-admitted claims (materials 289). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

Every current claim identifier for this paper's branch surface already occurs in the inherited scientific body. No claim-level prose delta is required.

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.3.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

This paper reports the complete dated Materials Science reconstruction: 289 of 289 registered obligations admitted through the untouched Smithian Fold Theory engine, while remaining open to lawful extension. The branch contains 73,984 exact generated candidates and decisions, 289 unique survivors, 1,156 passing adverse controls, 289 implementation-distinct reconstructions and 289 receipt-bound empirical packages. Every admitted law reaches the single premise-free foundational theorem. The complete-field extension adds quantitative crystallography and diffraction; defects and multiscale microstructure; phase transformation and metastability; fracture, fatigue, creep, tribology and rheology; thermal, electronic, ionic, magnetic, superconducting, optical and photonic response; complete material classes; soft, biological, nano and surface materials; degradation; processing; computational materials; extreme conditions; sustainability; complete empirical vectors; and

one-owner cross-branch handoffs. No axiom, free or fitted parameter, imported constitutive model, target-selected law, negative proof quantity, irrational or imaginary proof value, floating proof arithmetic or numerical-zero ontology is admitted.

Scope and ownership boundary

This paper owns the `materials` surface comprising 289 current claims. Ownership identifies responsibility for derivation and falsification; it does not prevent criticism or lawful cross-branch use. Application performance, consensus, credentials and Lean acceptance cannot transfer ownership or select a law.

Mathematical, computational and empirical constitutions

The mathematical constitution retains exact generated carriers, relations, candidate grammars, decisions, survivors and closure boundaries. The computational constitution retains executable identities, complete traces, controls, independent reconstructions and receipts. The empirical constitution retains source registration, transport, pre-seal and post-seal chronology, custody, measurement or observation, uncertainty, favourable and adverse rows, missingness, unresolved records and falsification conditions. None of these evidence classes substitutes for another.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Limitations and open frontier

The current result is formally complete only to the named dated grammar, census and artifact graph. Native Lean theoremhood is presently established for the operational root and generic gate implications; the other scientific claims are artifact checked rather than individually restated as Lean propositions. Empirical validation remains claim specific, and adverse, unavailable and unresolved evidence remains open where the current records say so. New discoveries, falsifications, stronger comparisons and native claim-level formalisations may enter only through a later version without rewriting this one.

Successor machine-identity appendix

- Preceding source:
publications/successors/materials/FROM_FOLD_TO_MATERIALS_PAPER_001_V1_3.md
- Preceding source SHA-256:
sha256:79fe7890cf31017536cfc5d76bf816783ab805f484de22b675e311e88d12c1da
- Lean report: generated/lean4_validation/reports/whole_model_validation.json
- Lean report SHA-256:
sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf
- Ordered census SHA-256:
sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71
- Execution manifest SHA-256:
sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41
- Protected engine seal:
sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification-authority seal:
sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8
- Publication authorised: false
- Remote actions performed: false

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is

`sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **289** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar:

`unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter.

Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors.

`#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted` gate only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.4.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 289 current claim(s) within `materials` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.